

Merged research report for the Surreal project

# **Omnific Integers and Omnific–Diophantine Geometry**

Retractions, rigidity, definability,  
and infinite families

Built from fifteen manuscripts dated September 22 and 23, 2026

### Abstract

The omnific integers  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  form a discretely ordered subring of the surreal field whose arithmetic is neither ordinary integer arithmetic nor a routine nonstandard enlargement of it. We organize their theory around the split ring retraction  $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$  that takes a normal form to its constant coefficient. This gives an exact transfer theorem for polynomial equations with ordinary integer coefficients, a classification of finite quotients, and a complete solution criterion for linear systems.

The Diophantine results distinguish existence from the structure of solutions. At a nonzero constant level, the omnific fiber of a product of affine linear forms is described exactly by a real common kernel; in particular binary forms with at least two distinct projective linear factors, Pell equations and norm forms of finite étale algebras have no new omnific or Gaussian omnific points. Support-preserving Euler derivations prove the same rigidity for every separated equation  $ax^m + by^n = c$  with complex  $abc \neq 0$  and  $m, n \geq 2$ , including  $y^2 = x^3 + 1$ , and for Fermat triples of exponent at least three with all coordinates nonzero that generate the unit ideal. In contrast, every represented nonzero level of a nondegenerate indefinite integral quadratic form in at least three variables carries explicit injective polynomial families through each integer point, and we prove an exhaustive quadratic-level dichotomy that includes degenerate forms. Pell rigidity combined with the three-square or the four-square theorem gives single quartic equations defining the ordinary integers inside  $\mathbf{Oz}$ . Further results concern non-atomic factorization, full-class common divisors, bounded semialgebraic rigidity, primitive and projective points, a failed existential induction instance, and failures of residue lifting.

The splitting  $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$  is itself Diophantine:  $x^2 = 2y^2$  defines  $\Pi$  with one witness, one order-free system defines  $\mathbb{Z}$  in  $\mathbf{Oz}$  and  $\mathbb{Z}[i]$  in  $\mathbf{Oz}[i]$ , and  $as = \Lambda(t)$ , for a classical intersective sextic  $\Lambda$ , is solvable exactly when  $\text{ct}(a) \neq 0$ . So the graph of  $\text{ct}$  is Diophantine and homomorphisms preserve constant terms. The multiplier ring of  $\Pi$  recovers  $\mathbb{R}$ ; the pure ring  $\mathbf{Oz}$  thus reconstructs the valued field  $\mathbf{No}$ , and its automorphisms fix  $\mathbb{R}$  pointwise, while a phase twist shows that  $\mathbf{Oz}[i]$  does not determine the real axis.

Fractions have an exact theory on the points with a unimodular presentation over the real nonnegative-support ring. There the constant coefficients decide whether the scalar multipliers of all omnific presentations form  $\mathbf{Oz}$  or  $\Pi$ . So such a point either has unique lowest terms and a principal denominator ideal, or no presentation of it has a gcd. This covers every tuple of real or complex rational functions of one purely infinite parameter. The fibers of the specialization are the orbits of the congruence kernel of  $\text{SL}_2(\mathbf{Oz})$ , and each of them is dense.

On proper varieties Euler derivations make global differentials vanish at points over the one-sided ring. So only the affine line among smooth affine curves has nonconstant such points, and every nonsingular  $y^2 = x^3 + ax + b$  has only ordinary omnific solutions. Logarithmic symmetric differentials on a normal-crossings compactification vanish in the same way, so a complement whose logarithmic forms separate tangent directions has only constant such points. A geometrically integral affine curve, singular or not, has nonconstant such points exactly when its normalization is the affine line. A monic polynomial over the one-sided ring whose discriminant is a nonzero constant is a purely infinite translate of its constant-term polynomial, and every unit of a finite étale algebra over that ring is algebraic over the coefficient field.

This report merges fifteen independently written manuscripts; it states which source proved each result, prints shared results once, and keeps every source's limitations. It is an AI-assisted draft: it has not been refereed, and independent review remains incomplete. Some elementary ring and arithmetic results have checked Lean proofs; the exact scope is recorded in the repository's `docs/FORMALIZATION.md`, not inferred from this manuscript.

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# 1 Scope, results, and conventions

Omnific integers are the surreal numbers with no negative exponents in Conway normal form and with an ordinary integer constant coefficient. The definition is classical, as are their discreteness and integer-part property [1, 2, 3]. This report develops their arithmetic with a particular emphasis on polynomial equations.

Three questions must be kept separate. Does an equation have any solution? Does it have a genuinely infinite solution? Does it have such a solution with a nonvanishing, positivity or primitivity condition? For example, every system of equations over  $\mathbb{Z}$  has an omnific solution exactly when it has an ordinary integer solution. Nevertheless, the equation  $X^n + Y^n = Z^n$  has positive infinite omnific solutions for every ordinary  $n \geq 2$ . There is no contradiction: the constant coefficients of these solutions are all zero. On the other hand, every omnific solution of

$$x^2 - 2y^2 = 1, \quad y^2 = x^3 + 17, \quad x^7 - y^5 = 3$$

has ordinary integer coordinates; these conclusions concern the entire solution classes, not just their constant terms.

The central decomposition is

$$\mathbf{Oz} = \mathbb{Z} \oplus \Pi, \quad \text{ct} : \mathbf{Oz} \longrightarrow \mathbb{Z}, \quad \text{ct}(a + b) = \text{ct}(a) + \text{ct}(b), \quad \text{ct}(ab) = \text{ct}(a) \text{ct}(b), \quad (1.1)$$

where  $\Pi$  is the ideal of normal forms supported on strictly positive exponents; coefficients of elements of  $\Pi$  may have either sign. The decomposition is a splitting of additive groups and a ring retraction, not a direct product decomposition of rings. In particular,  $\Pi$  has no identity.

## 1.1 Principal results

The structural results of Sections 2–4 explain why ordinary congruences see only the constant coefficient, although the fraction field of the full class  $\mathbf{Oz}$  is all of  $\mathbf{No}$ . The main Diophantine statements are the following. Bracketed numbers name the source manuscripts (section 1.2).

- (1) **Equational transfer** [01, 02, 05]. For every finite system over  $\mathbb{Z}$ ,  $V(\mathbf{Oz})$  is nonempty exactly when  $V(\mathbb{Z})$  is nonempty (theorem 5.1). This does not include inequations or order conditions (proposition 5.3).
- (2) **Constant-product rigidity** [01, 02, 05]. A nonzero constant product of normal forms with no negative exponents has only constant factors. At a nonzero constant level, the omnific points of a product of affine linear forms are exactly the integer points plus purely infinite vectors in the real common kernel of the linear parts (theorem 6.2). Binary forms with two distinct projective factors over  $\mathbb{C}$  at a level in  $\mathbb{C}^\times$  have only constant points in the complex nonnegative-support ring (theorem 6.7); Pell equations with  $D \neq 0$  and number-field norm forms are rigid at nonzero ordinary levels (corollary 6.9 and theorem 6.14). Source 13 of the sibling report adds the exact fibers over the Gaussian omnific integers  $\mathbf{Oz}[i]$  and the converse of the kernel criterion (theorem 6.3 and corollary 6.4); with source 07, norm rigidity at nonzero constant levels holds for finite étale algebras over any characteristic-zero base field, in the specified intermediate rings (theorem 6.15).
- (3) **Separated powers and unimodular Fermat** [05]. For complex  $a, b, c \neq 0$  and  $m, n \geq 2$ , every nonnegative-support solution of  $ax^m + by^n = c$  is constant (theorem 7.4; source 05 states the real case). For  $n \geq 3$ , a Fermat triple with all three coordinates nonzero that generates the unit ideal is constant (theorem 7.9); the threshold is sharp.

- (4) **Quadratic-level dichotomy** [01, 02]. For  $q \in \mathbb{Z}[X_1, \dots, X_n]$  quadratic homogeneous and  $0 \neq c \in \mathbb{Z}$ , all solutions are ordinary in the nondegenerate definite case and in nondegenerate dimension at most two. In every other case each integer point lifts to an injectively parametrized infinite family ([theorem 8.1](#)). For nondegenerate indefinite forms in at least three variables,  $(\Pi, +)$  embeds in the kernel of  $\mathrm{SO}(q, \mathbf{Oz}) \rightarrow \mathrm{SO}(q, \mathbb{Z})$  ([corollary 8.4](#)).

- (5) **Quartic definitions of  $\mathbb{Z}$**  [01, 05]. The equation

$$(U^2 - 2V^2 - 1)^2 + \left( U - \sum_{i=1}^n X_i^2 - W_1^2 - W_2^2 - W_3^2 \right)^2 = 0$$

with five existentially quantified auxiliary variables defines exactly  $\mathbb{Z}^n$  inside  $\mathbf{Oz}^n$  ([theorem 10.3](#)); a variant with six auxiliary variables and four squares defines  $\mathbb{Z}$  ([theorem 10.4](#)), and source 07 found the same variant independently. The graph of  $\mathrm{ct}$  and the ideal  $\Pi$  are first-order definable ([corollary 10.7](#)) and, by item (8), Diophantine; an explicit existential induction instance fails ([theorem 10.10](#)).

- (6) **Primitive and projective phenomena** [01, 02, 05]. There are infinite unimodular Pythagorean triples. A constant real projective direction has a primitive omnific representative precisely when it is rational ([proposition 14.4](#) and [theorem 14.5](#)).
- (7) **Bounded and homogeneous geometry** [01, 02]. Bounded semialgebraic sets have only integer omnific points ([theorem 9.2](#)); a homogeneous system has a nonzero omnific point exactly when it has a nonzero real point, and for integer coefficients this is decidable ([theorem 14.2](#)).

- (8) **Diophantine definitions** [06, 07]. In  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , and in the analogous rings over every ordered exponent group,  $\exists y (x^2 = 2y^2)$  defines the purely infinite ideal ([theorem 11.2](#)); one order-free system of three equations with five witnesses defines  $\mathbb{Z}$  in  $\mathbf{Oz}$  and  $\mathbb{Z}[i]$  in  $\mathbf{Oz}[i]$ , also in intermediate rings ([theorem 11.9](#)); the equation  $as = \Lambda(t)$  is solvable exactly when  $\mathrm{ct}(a) \neq 0$  ([theorem 12.2](#)), and  $\Pi$  is the largest ideal whose quotient has no root of  $\Lambda$  ([theorem 12.6](#)). The graph of  $\mathrm{ct}$  is Diophantine ([theorem 12.8](#)), in  $\mathbf{Oz}$  by one quartic with six witnesses ([corollary 12.9](#), added by the merge), and every unital homomorphism between such rings preserves  $\mathrm{ct}$  ([theorem 12.12](#)). The constant definition extends to subrings of number fields ([theorem 12.14](#)).

- (9) **Reconstruction and asymmetry** [06]. The multiplier ring of the ideal recovers the actual coefficient field ([theorem 13.2](#)); from the pure ring  $\mathbf{Oz}$  one reconstructs  $\mathbf{No}$ , its order, finite ring and standard part ([theorem 13.4](#)). Every automorphism of  $\mathbf{Oz}$  extends uniquely to  $\mathbf{No}$ , and this extension fixes  $\mathbb{R}$  pointwise ([theorem 13.6](#)); only  $\mathbb{Z}$  lies in  $\mathbf{Oz} \cap \mathbb{R}$ . A phase twist preserving  $\mathbf{Oz}[i]$  shows that neither  $\mathbf{No}$  nor conjugation is definable from  $\mathbf{Oz}[i]$  ([theorem 13.10](#)).

- (10) **Fractions and denominator ideals** [08, 09]. For a tuple that is unimodular over  $\mathcal{B}_{\mathbb{R}}$ , the omnific representations are its multiples by  $\mathbf{Oz}$  or by  $\Pi$ , according as its constant vector has rational direction or not ([theorem 15.11](#)). So every tuple of rational functions of one parameter  $0 \neq t \in \Pi$ , and every point of the region  $\mathcal{V} \subseteq \mathbb{P}^1(\mathbf{No})$  of points with a  $\mathcal{B}_{\mathbb{R}}$ -unimodular presentation, either has unique lowest terms and a principal denominator ideal, or has no presentation with a gcd and a denominator ideal generated by no set ([theorems 15.15](#) and [15.18](#)). The specialization fibers in  $\mathcal{V}$  are the orbits of the congruence kernel of  $\mathrm{SL}_2(\mathbf{Oz}) \rightarrow \mathrm{SL}_2(\mathbb{Z})$ , and all of them are dense ([theorem 15.32](#) and [corollary 15.36](#)). The section adds primitive coordinates on rational curves, a localization with residue field  $\mathbb{Q}$ , a nonflat integral scale extension with its  $\mathrm{Tor}_1$  witness, and the Gaussian form of [theorem 14.5](#) ([corollaries 15.28](#) and [15.55](#), [proposition 15.40](#), and [theorem 15.47](#)).

- (11) **Curve and differential rigidity** [C10–C16]. Over any field  $\mathbb{k}$  of characteristic zero and any ordered exponent group  $\Gamma$ , a smooth proper variety with globally generated cotangent sheaf, in particular a curve of positive genus or an abelian variety, has only constant points over the nonnegative-support Hahn ring  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  ([corollary 16.17](#)). If  $\Gamma \neq 0$ , a smooth geometrically integral affine curve has nonconstant points exactly when it is the affine line ([theorem 17.6](#)); for  $\Gamma = 0$ ,  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}$  and every point is constant. Every squarefree equation  $y^m = P(x)$  with  $m, \deg P \geq 2$  is rigid by an elementary certificate ([theorem 16.6](#)), so the nonsingular Weierstrass equations have only ordinary omnific solutions and the singular ones have polynomial parametrizations ([corollary 16.9](#) and [proposition 16.11](#)). For an affine model over  $\mathbb{Z}$  with smooth geometrically integral generic curve, a nonordinary omnific point exists exactly when its real fiber is  $\mathbb{A}_{\mathbb{R}}^1$  and there is an ordinary integer point. In that case every fiber of  $\text{ct}$  over an integer point is a copy of  $\Pi$  ([corollary 17.12](#)); this does not identify the integral model with  $\mathbb{A}_{\mathbb{Z}}^1$ . Semiabelian varieties are rigid. For a smooth connected commutative algebraic group the constant-term kernel is isomorphic, as a group, to  $\Pi_{\mathbb{k}}(\Gamma)^d$  after choosing vector coordinates; this is not an algebraic splitting of the group. On rigid projective targets only rational points have invertible coordinate ideals ([theorems 17.21, 18.4 and 18.6](#)). Global logarithmic symmetric differentials on a smooth proper compactification with normal-crossings boundary vanish on the Euler derivations at every point, so the complement is rigid when they separate tangent directions ([theorem 18.16](#) and [corollary 18.18](#)) [C15]. A geometrically integral affine curve, singular or not, has nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -points, for  $\Gamma \neq 0$ , exactly when its normalization is  $\mathbb{A}^1$ ; a conductor of the curve and Seidenberg’s identity give finite derivative-order certificates, and the rigid singular curves have only ordinary omnific points ([theorems 17.25, 17.28 and 17.41](#)) [C16].
- (12) **Constant discriminants and étale units** [C17]. Over any field  $\mathbb{k}$  of characteristic zero and any ordered exponent group, a monic polynomial of positive degree over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , or over  $\mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma)$  for a subring  $\mathfrak{o} \subseteq \mathbb{k}$ , whose discriminant is a nonzero constant is  $\overline{P}(X - h)$  for its constant-term polynomial  $\overline{P}$  and a unique purely infinite  $h$  ([theorem 19.12](#)); so a monic  $P \in \mathbf{Oz}[X]$  with  $\text{Disc}(P) \in \mathbb{Z} \setminus \{0\}$  has an omnific root exactly when  $\overline{P}$  has an integer root ([corollary 19.16](#)). Every unit of a finite étale  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -algebra is algebraic over  $\mathbb{k}$ , and monogenic ones are constant ([theorems 19.26 and 19.29](#)); a normal matrix over  $\mathcal{A}_{\mathbb{C}}(\Gamma)$  with nonzero constant characteristic discriminant is  $\text{ct } M + hI$  ([theorem 19.37](#)).

These statements are proved here, but proving a statement here is not a claim that nobody has previously proved it. The constant-term formalism is classical, and several consequences are natural applications of it. Each source names its own most distinctive contributions ([appendix B](#)); no exhaustive historical-priority claim is made.

The *universal set-sized quotient theorem*, that every unital ring homomorphism from the full class  $\mathbf{Oz}$  to a set-sized ring is  $x \mapsto \text{ct}(x)1$ , was proved by source 05 and is printed with its proof in the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers`. It is quoted here as [cited theorem 3.6](#), together with the elementary special cases proved by source 01.

## 1.2 Provenance of this report

This report is built from fifteen manuscripts written independently. Sources 01, 02 and 05, dated September 22, 2026, are items of the batch placed in commit `be06fc8` and pin repository commit `2cb9c0200af749fbbd27796c97d017e7f16fbf33` (107 commits before that placement). Sources 06 and 07, dated September 23, 2026, are items 01 and 07 of the batch placed in commit `cf350b1`; they are numbered 06 and 07 here to follow the earlier sources, as their

shipped file prefixes are. Both pin `71e9606`, which precedes the placement `be06fc8`, so neither saw this report or its sibling. Sources 08 and 09, also dated September 23, 2026, are items 01 and 02 of the batch placed in commit `a4dcb91`, numbered by their shipped file prefixes. Both pin `9a385d3`, 35 commits before that placement; they saw this report as assembled from sources 01, 02 and 05, before sources 06 and 07 were integrated. The tag [13] refers to manuscript 13 of the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` (item 03 of the batch placed in `cf350b1`, also pinned to `71e9606`); its Gaussian and étale fiber results are printed here once, and it is not counted among this report's sources. The tag [14] refers to manuscript 14 of the same report (item 04 of that batch, files prefixed `14-arithmetic-tensors-` there); its criterion for unimodular constant directions over  $\mathbb{Z}$  and  $\mathbb{Z}[i]$  is printed here once, in [corollary 15.55](#), and it is not counted among this report's sources either.

Sources C10–C14, all dated September 23, 2026, are items 02, 03, 04, 07 and 08 of the batch placed in commit `c6359e4` (their archives arrived in `5610500` and `21016dc`); their shipped files carry the prefixes `10-` to `14-`, and the letter C distinguishes their tags from the tags [13] and [14] above. Sources C10, C13 and C14 pin `4cdeaec`, source C11 pins `f6e031a` and source C12 pins `89bec38`; at all three pins this report had only sources 01, 02 and 05, and its questions were in Section 14. None of the five saw the others. Source C15, also dated September 23, 2026, is item 02 of the batch placed in commit `66d7e55` (its archive arrived in `65775fb`); its shipped files carry the prefix `15-`, and it is tagged like C10–C14. It pins `934810a`, 18 commits before `c6359e4`; at that pin this report had sources 01, 02, 05, 06 and 07, and source C15 saw none of C10–C14. Source C16, also dated September 23, 2026, is item 04 of the batch placed in commit `9d28e28` (its archive arrived in `39fe674`); its shipped files carry the prefix `16-`, and it is tagged like C10–C15. It pins `bcac55a`, 87 commits before `9d28e28`; at that pin sources C10–C14 were placed but not yet written into this report, source C15 was not yet placed, and source C16 saw none of C10–C15 except the audits of C13 and C14. Source C17, dated September 23, 2026, is manuscript 07 of batch 33, placed in commit `aa9c891` (its archive arrived in `73043eb`); its shipped files carry the prefix `17-discriminant-rigidity-`, and it is tagged like C10–C16. It pins `efc5446`, 70 commits before `aa9c891`; at that pin sources C10–C15 were written into this report and source C16 was placed but not yet written, and source C17 read this report's guide, not its article.

| Source | Manuscript                                                                                                            | Principal contribution to this report                                                                                                                                                                                                                                                                                                                                                    |
|--------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01     | <i>Omnific Integers and Omnific-Diophantine Geometry</i>                                                              | Base text and structure; exhaustive quadratic-level dichotomy; the five-auxiliary three-square guard; rational-direction theorem; no real point at infinity; symmetric-matrix rigidity; elementary uniqueness of ct.                                                                                                                                                                     |
| 02     | <i>Omnific Integers and Diophantine Geometry</i>                                                                      | Exact decomposable fibers; bounded semialgebraic rigidity; decidable homogeneous existence; quadratic-formula criterion; general pair without a gcd; explicit Lorentz matrix and the unipotent subgroup.                                                                                                                                                                                 |
| 05     | <i>Omnific Integers and Diophantine Rigidity</i>                                                                      | Binary rigidity over $\mathbb{C}$ ; Euler derivations; separated-power and unimodular Fermat rigidity; the four-square definition of $\mathbb{Z}$ ; failed existential induction; failed lifting; finite-support specialization; the universal set-sized quotient theorem (printed in the sibling report).                                                                               |
| 06     | <i>Definable Arithmetic and Coefficient Reconstruction in Omnific Integer Rings</i>                                   | Base of <a href="#">sections 11 to 13</a> ; the one-witness definition of $\Pi$ ; the Diophantine graph of ct; homomorphisms preserve ct; the multiplier identity and the reconstruction of $\mathbb{R}$ , $\mathbf{No}$ , order and standard part; automorphisms fix $\mathbb{R}$ ; the phase twist and the real-Gaussian asymmetry; c.e. sets; the quintic.                            |
| 07     | <i>Defining Arithmetic Inside Omnific Integers</i>                                                                    | The order-free system in intermediate rings (with 06); the augmentation root detector and the ideal test; support bounds and certificates; number-field coefficient rings; norm rigidity over any base field; recursive saturation; the quartic variant.                                                                                                                                 |
| 08     | <i>Fractions of Omnific Integers: Rational Specialization, Lowest Terms, Denominator Ideals, and Scale Extensions</i> | Base of <a href="#">section 15</a> : the multiplier theorem in any number of coordinates; the classification for rational-function tuples at any $t \in \Pi \setminus \{0\}$ ; real constants; rational curves; the localization at $\Pi$ and independent residues; polynomial models, the scale defect, nonflatness and $\text{Tor}_1$ ; monomial denominators; the Gaussian dichotomy. |
| 09     | <i>Omnific Fractions: Rationality, Denominator Ideals, and Dense Arithmetic Fibres</i>                                | The unimodular region $\mathcal{V}$ and its dichotomy; the calculus of denominator ideals and least denominators; fibers as congruence orbits; set-wise focusing, dense fibers and nowhere continuity; the pair version of the multiplier theorem and the one-scale case (special cases of 08).                                                                                          |

| Source | Manuscript                                                                                                                                                                           | Principal contribution to this report                                                                                                                                                                                                                                                                                                    |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C10    | <i>Curve Rigidity over the Omnific Integers: Elliptic equations, differential separation, and semiabelian varieties</i>                                                              | Exact fibers $\Pi \rightarrow \text{ct}^{-1}(\mathbf{n})$ on affine-line models; quasi-finite maps to rigid targets; nonsplit tori by faithful flatness; the general elliptic field point; the valuation-ring and verticality boundaries.                                                                                                |
| C11    | <i>Curve and Abelian Rigidity over the Omnific Integers: A complete smooth-affine-curve dichotomy, support-preserving derivations, and invariance over reduced coefficient rings</i> | Arithmetic fibers over any coefficient ring $\mathfrak{o}$ ; reduced coefficient algebras; the dual-number defect; the characteristic-two elliptic point; a discretely ordered ring with no omnific copy; proper classes of fiber points.                                                                                                |
| C12    | <i>Curve Rigidity over Omnific Integers: Logarithmic differentials, elliptic equations, and denominator ideals</i>                                                                   | The strict certificate proof; the logarithmic proof of the curve classification; descent for separated models over $\mathbb{Z}$ ; coordinate ideals (invertible, principal, rational); the singular Weierstrass dichotomy.                                                                                                               |
| C13    | <i>Differential Rigidity of Omnific Points: Smooth-curve classification, squarefree equations, and algebraic groups</i>                                                              | Annihilation of global tensors; symmetric differentials and ample cotangent bundles; general Weierstrass equations; smooth projective curves including $\mathbb{P}^1$ ; polynomial witnesses; the kernel of commutative groups; a characteristic- $p$ counterexample.                                                                    |
| C14    | <i>Differential Rigidity of Hahn Rings: Curves, Abelian Varieties, and Omnific Diophantine Points</i>                                                                                | Base of <a href="#">sections 16 to 18</a> : the two-ring principle and the curve classification over every field of characteristic zero and every exponent group; integer polynomial arcs; the congruence criterion; unimodular coordinates over $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ and rational projective points.        |
| C15    | <i>Geometric Rigidity over Omnific Integer Rings: Smooth curves, semiabelian varieties, and logarithmic differential certificates</i>                                                | Logarithmic symmetric-differential annihilation on normal-crossings compactifications and the tangent-separation criterion ( <a href="#">section 18.6</a> ); descent over any coefficient ring; finite products; independent proofs of the squarefree, curve, abelian and semiabelian results of C10–C14.                                |
| C16    | <i>Singular Curves over Omnific Integers: Conductor differentials, a complete Hahn-rigidity criterion, and polynomial witnesses at every surreal scale</i>                           | The criterion for singular geometrically integral affine curves ( <a href="#">section 17.6</a> ); Seidenberg’s identity and conductor certificates; the one-place obstruction in every characteristic; affine schemes of dimension at most one; repeated-root superelliptic equations; a pinched elliptic curve; the Gaussian dichotomy. |

| Source | Manuscript                                                                                                                                                  | Principal contribution to this report                                                                                                                                                                                                                                                                                                                                                                         |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C17    | <i>Constant Discriminants and Étale Unit Rigidity over Omnific Hahn Rings: Translation classification, arithmetic descent, and normal spectral rigidity</i> | Translation classification at a nonzero constant discriminant over $\mathcal{A}_{\mathbb{k}}(\Gamma)$ and $\mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma)$ , with a universal root-velocity certificate ( <a href="#">section 19.4</a> ); factors, Galois groups and resultants; units of finite étale algebras; monogenic and arithmetic descent; critical configurations; normal matrices; five boundary examples. |

Where the sources overlap, the statement with the weakest hypotheses is printed and the others are recorded as special cases. Results added by the merge carry the tag [merge] and have complete proofs. [Appendix A](#) records, section by section, which source proved what, where the merge had to choose, and every symbol renamed from a source.

**Current review and placement status.** Commit `bdd536` assembled sources 01, 02 and 05. The elementary algebra review, its preservation through that assembly, and later proof reviews of the original material in Sections 1–10 and [sections 14](#) and [19](#) are recorded in `RECONCILIATION.md`; assembly is not a review of every later proof, and later proofs and remaining imported foundations still require review. Sources 06 and 07, placed in `cf350b1`, were integrated concurrently with those reviews, as [sections 11](#) to [13](#), together with their material and that of source 13 in [section 6](#) and source 07’s quartic ([remark 10.5](#)). A subsequent pass reviews [sections 11](#) to [13](#), including the distinction between intermediate rings and full positive-support rings, the detector witnesses, coefficient reconstruction and logical consequences. A further pass reviews their Gaussian-fiber and étale-norm additions in [section 6](#), including support and coefficient-extension arguments and explicit boundary examples. The abstract étale norm proof also works in arbitrary characteristic; separability remains necessary. Source 07’s quartic in [section 10](#) now also has a proof review, including the ambient-ring argument and the integer witnesses for every intermediate ring. Sources 08 and 09, placed in `a4dcb91`, were integrated after those reviews as [section 15](#), together with additions to [section 20](#), one partly answered question and five new questions in [section 21](#). Further passes now review all of [section 15](#): denominator ideals, the multiplier theorem, rational-function and curve applications, specialization orbits and density, localization, scale defects, fixed workspaces and Gaussian fractions. The two residues can agree outside  $\mathbb{Q}$ , and the scale-defect module has infinite length despite finite real dimension; both points now have explicit proofs. The additions outside this section remain unreviewed. The insertion of [section 15](#) moved the three sections after it by one, from Sections 15–17 to 16–18. Sources C10–C14, placed in `c6359e4`, were integrated after that as [sections 16](#) to [18](#), with pointers in [sections 6](#), [7](#) and [14](#), additions to [section 20](#), and, in [section 21](#), [question 21.1](#) answered for smooth curves, two further questions answered in part and six new questions. A subsequent elementary pass reviews only [sections 16.2](#) and [16.3](#): the ring and valuation properties, Euler derivations, joint constants and relative algebraic closedness. The remaining curve and geometric material is outside the reviews. Its insertion moved the same three sections by three more, so that the former Sections 15–17 are now [sections 19](#) to [21](#); statement numbers in Sections 1–15 are unchanged. Source C15, placed in `66d7e55`, was integrated after that as [section 18.6](#), at the end of [section 18](#), with credits added to the results of [sections 16](#) to [18](#) that it reproves, a correction in [section 17.5](#) and re-scoped notes in [section 21](#); its elementary ring/Euler claims are included in the pass just described, while its other material remains unreviewed. No section or statement number changed. Source C16, placed in `9d28e28`, was integrated after that as

section 17.6, at the end of section 17, with credits added to the results it reproves, status notes in sections 17.5 and 21 and eight new questions; it is outside the reviews, and again no section or statement number changed. Nine earlier labels renamed or combined during assembly remain as aliases, so existing references resolve. Source C17, placed in aa9c891, was integrated after that as section 19.4, at the end of section 19, with credits added to the results it reproves, pointers in sections 5, 6, 9 and 20, status notes in sections 18, 19 and 21 and ten new questions; it is outside the reviews, and no section or statement number changed.

### 1.3 Class size and foundations

The full surreal field and  $\mathbf{Oz}$  are proper classes. We work in a customary class theory such as NBG, with the usual background for surreal normal forms; source 02 notes that one may equally read everything relative to universes, with “set-sized” replaced by “small in the specified lower universe”. Each individual surreal has a *set-sized* support. Every polynomial, system, matrix, or tuple in this report has ordinary finite size. Quantifiers over  $\mathbf{Oz}$  are class quantifiers expressed through their defining set-theoretic formulas, not an appeal to a set containing all surreals. Sources 06 and 07 state their results first for set-sized exponent groups, where they are ordinary first-order model theory, and read statements about  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  formula by formula (section 11.1).

Quotients by class ideals are interpreted through their quotient maps and congruence relations. Thus  $\mathbf{Oz}/\Pi \cong \mathbb{Z}$  uses the ordinary integer  $\text{ct}(x)$  as representative; for  $0 \neq n \in \mathbb{Z}$ ,  $\mathbf{Oz}/n\mathbf{Oz}$  uses the ordinary residue of  $\text{ct}(x)$  modulo  $n$ . Proper-class cosets are not elements of a set. Ideal generation uses finite sums, even for a class of generators. The notation  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  means representation by fractions inside  $\mathbf{No}$ , not a set of all pairs from  $\mathbf{Oz}$ .

Choice for sets is used to extend a rational linear functional on a set-sized exponent group (lemma 7.2). No proof printed in this report uses global choice. Source 05 works in NBG with global choice, which it uses only to pick a cardinal in the proof of the set-sized quotient theorem; the Hartogs ordinal of the target ring serves there without choice, since the pigeonhole step needs only that no injection from that ordinal into the target exists (remark 3.7).

Most proofs can be carried out in a fixed set-sized Hahn field. We identify the arguments that need the stronger fact that *every set* of surreal exponents has upper bounds and every set of positive surreal exponents has a strictly positive lower bound (section 4). One must not silently assume these properties for a fixed ordered exponent group (section 20).

## 1.4 Notation and the exponent convention

| Notation                                                                                                              | Meaning in this report                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\omega^\gamma$                                                                                                       | Conway's monomial; $\omega^\gamma \omega^\delta = \omega^{\gamma+\delta}$ with surreal addition of exponents.                                                                                                                                                  |
| $\Pi$                                                                                                                 | The ideal of normal forms supported on $\mathbf{No}^{>0}$ (purely infinite elements and 0).                                                                                                                                                                    |
| $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$                                                                                 | Omnific integers: nonnegative support, integer constant coefficient.                                                                                                                                                                                           |
| $\mathcal{B}_{\mathbb{R}} = \mathbb{R} \oplus \Pi$ ,<br>$\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}}[i]$      | Nonnegative-support rings with real or complex coefficients; $\Pi_{\mathbb{C}} = \Pi + i\Pi$ .                                                                                                                                                                 |
| $\mathcal{B}_{\mathbb{R},H}$ , $\mathcal{B}_{\mathbb{C},H}$                                                           | Their subrings with exponents in a set-sized divisible group $H \subseteq \mathbf{No}$ (lemma 2.6).                                                                                                                                                            |
| $\text{ct}(x)$                                                                                                        | Coefficient of $\omega^0$ ; not an order standard part.                                                                                                                                                                                                        |
| $\deg_\omega(x)$ , $\text{lc}(x)$                                                                                     | Largest exponent in $\text{supp}(x)$ and its coefficient; $\deg_\omega(0) = -\infty$ . Not an ordinal and not a birthday.                                                                                                                                      |
| $\text{Std}(t)$ , $\text{Std}_n(\mathbf{x})$                                                                          | The existential formulas of theorems 10.3 and 10.4 defining $\mathbb{Z}$ and $\mathbb{Z}^n$ .                                                                                                                                                                  |
| $\partial_\lambda$ , $W_\partial$                                                                                     | Local Euler derivation and two-term Wronskian (section 7); for an arbitrary exponent group and coefficient field, (16.1).                                                                                                                                      |
| $B$ , $q$ , $G$                                                                                                       | Symmetric bilinear form, its quadratic form $q(x) = B(x, x)$ , and the Lorentz Gram matrix $G = \text{diag}(1, -1, -1)$ .                                                                                                                                      |
| $R$ , $S$                                                                                                             | A generic commutative ring; a set-sized target ring.                                                                                                                                                                                                           |
| $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$                                                              | Gaussian omnific integers (section 6).                                                                                                                                                                                                                         |
| $\mathbb{k}$ , $\Gamma$ , $\mathfrak{o}$                                                                              | A coefficient field of characteristic zero, an ordered abelian exponent group, a unital subring of $\mathbb{k}$ (section 11.1).                                                                                                                                |
| $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , $\Pi_{\mathbb{k}}(\Gamma)$ ,<br>$\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ | Nonnegative-support series, its positive-support ideal, and $\mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma)$ ; an <i>intermediate ring</i> $A$ satisfies $\mathfrak{o} \subseteq A \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$ , $A \cap \mathbb{k} = \mathfrak{o}$ . |
| $\Lambda$ , $\Xi$ , $\Theta$ , $\text{Inf}$ , $\text{CT}$                                                             | The intersective polynomial, the order-free definition of $\mathbb{Z}$ and $\mathbb{Z}[i]$ , the detector of $\text{ct} \neq 0$ , the definition of $\Pi$ , the graph of $\text{ct}$ (sections 11 and 12).                                                     |
| $\text{st}$ , $\mathcal{O}_{\text{fin}}$ , $\mathfrak{m}$                                                             | Standard part, finite surreals, infinitesimals (theorem 13.4).                                                                                                                                                                                                 |
| $\mathfrak{D}$ , $\text{sp}$ , $\mathcal{V}$ , $\mathcal{U}$ , $\mathbf{K}$                                           | Denominator ideal, specialization, the $\mathcal{B}_{\mathbb{R}}$ - and $\mathbf{Oz}$ -unimodular regions of $\mathbb{P}^1(\mathbf{No})$ , the congruence kernel of $\text{SL}_2(\mathbf{Oz})$ (section 15.1).                                                 |
| $\mathbf{Oz}_{\Pi}$ , $\text{res}$ , $\text{Ex}$ , $\mathcal{P}[X]$                                                   | Localization at $\Pi$ , its rational residue, monic monomial exponents, $\mathbb{Z} + X\mathbb{R}[X]$ (section 15).                                                                                                                                            |
| $\mathbb{H}_{\mathbb{k}}(\Gamma)$ , $\mathcal{O}_{\mathbb{k}}(\Gamma)$ , $\mathfrak{m}_{\mathbb{k}}(\Gamma)$          | The Hahn field with exponents in $\Gamma$ , its valuation ring (support in $\Gamma^{\leq 0}$ ) and maximal ideal; opposite to $\mathcal{A}_{\mathbb{k}}(\Gamma)$ (section 16.1).                                                                               |
| $\mathcal{D}$ , $\mathbb{A}^n$ , $\mathbb{G}_{\mathbf{m}}$ , $\mathbb{G}_{\mathbf{a}}$ , $\langle \mathbf{a} \rangle$ | A family of derivations, affine space, the multiplicative and additive groups, the ideal generated by the coordinates of $\mathbf{a}$ (section 16).                                                                                                            |

The calligraphic  $\mathcal{B}_{\mathbb{R}}$ ,  $\mathcal{B}_{\mathbb{C}}$  name rings; the italic  $B$  is always a bilinear form. The capital

Greek letter  $\Pi$  always denotes the ideal; finite products are always written with the operator  $\prod$  and its index range. In the material of sources 06 and 07 the coefficient ring is written  $\mathfrak{o}$ , not  $D$ , because  $D$  is the Pell parameter; the coefficient field is written  $\mathbb{k}$ , because  $k$  is an ordinary index; and the exponent group is  $\Gamma$ , because  $G$  is the Lorentz Gram matrix. The sources' names are listed in [table 1](#). The constant-term polynomial  $\bar{P}$ , the discriminant  $\text{Disc}$ , the Hahn splitting field  $\mathbb{H}_{\mathbb{k}}(\Gamma_{\mathbb{Q}})$  with its valuation ring  $\mathcal{O}_{\mathbb{k}}(\Gamma_{\mathbb{Q}})$ , and a finite étale  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -algebra  $\mathbb{B}$  with its algebraic constants  $\mathbb{C}_{\mathbb{B}}$  are fixed in [section 19.4](#).

**Sign convention.** Exponents are those of Conway normal forms: supports are reverse well ordered, and  $\omega^\gamma$  is *infinite* for  $\gamma > 0$ , even for infinitesimal  $\gamma > 0$ . Thus

$$\Pi = \{x \in \mathbf{No} : \text{supp}(x) \subseteq \mathbf{No}^{>0}\} \quad (\omega\text{-exponents}).$$

In the Hahn convention  $t = \omega^{-1}$ , which is the convention of the trigonometry report (`docs/surcomplex/trigonometry`, equation `trigonometry:eq:split`), every exponent changes sign:  $\omega^\gamma = t^{-\gamma}$ , supports become well ordered, the valuation is  $v = -\deg_\omega$ , and the same class is

$$\Pi = \{x \in \mathbf{No} : \text{supp}_t(x) \subseteq \mathbf{No}^{<0}\} \quad (t\text{-exponents}).$$

So “strictly positive exponents” here and “negative  $t$ -exponents” there describe one class, and “no positive exponents” in the  $t$ -convention is “no negative exponents” here. The restriction concerns every supported exponent, not only the leading one:  $\mathcal{B}_{\mathbb{R}}$  consists of the forms whose entire  $t$ -support is nonpositive. It is not the set of elements with nonpositive valuation; for example  $\omega + \omega^{-1}$  has negative valuation but does not lie in  $\mathcal{B}_{\mathbb{R}}$ . Nor is  $\mathcal{B}_{\mathbb{R}}$  the valuation ring of finite elements. Using the wrong convention would invalidate the unit computation ([proposition 2.5](#)) and every rigidity proof below.

The repository’s foundations report defines  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  with the same letter  $\Pi$  (`docs/foundations-and-computation/foundations`, label `found:eq:omnific`); the notation here agrees with it. Other repository names for the same class are  $\mathbf{No}^\dagger$  and  $\mathbf{No}_{\text{PI}}$  in other reports. The sources used  $\mathcal{J}$  (01) and  $\mathcal{I}$  (02, 05); the complete per-source rename table is [table 1](#).

## 1.5 Relationship with the repository and literature

The repository distinguishes normal-form algebra, fixed Hahn workspaces, and the full surreal carrier. At the sources’ pin, the foundations report already defined  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  (`found:eq:omnific`), and the trigonometry report already used the omnific integer part, the floor formula and residue arithmetic (`trigonometry:eq:periodclass`); source 01 mentioned only the latter. At the pin there was no report devoted to  $\mathbf{Oz}$  and no Lean declaration about it. The collection now contains this report and the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers`. Commit `f879c1e` constructs the actual omnific ring and proves the constant-term package of [proposition 2.2](#) in Lean ([section 20](#) and [appendix D](#)). Sources 06 and 07 inspected the repository at the later pin `71e9606`, which still preceded both omnific reports; their statements that no explicit Diophantine definition of  $\mathbb{Z}$  in  $\mathbf{Oz}$  was found are stale, since [theorems 10.3](#) and [10.4](#) are in the collection ([appendix A](#)). Source 06 correctly notes that the surcomplex automorphism report (`docs/surcomplex/surcomplex-field-automorphisms`) already constructs phase twists that move the real axis while fixing complex coefficients.

The main external sources are Conway and Gonshor for the construction and normal forms [1, 2], L’Innocente–Mantova for modern factorization theory [3], Ehrlich–Kaplan for the initial integer-part context [4], and Jeřábek for the residue theory of  $\mathbb{Z}$ -groups [5]. MRDP, Tarski’s quantifier elimination, the three-square and four-square theorems, Shepherdson’s theorem, the Fermat theorem of Wiles and Taylor–Wiles, and Dirichlet’s theorem with quadratic reciprocity

are imported with explicit attribution at their points of use. The interseptive polynomial of [section 11](#) is classical [[22](#), [23](#), [24](#)]. The curve part ([sections 16 to 18](#)) imports standard algebraic geometry: Kähler differentials, the valuative criterion of properness for arbitrary valuation rings, smooth completion, Riemann–Roch and genus zero for curves, invariant differentials and the structure of commutative algebraic groups [[35](#), [38](#), [39](#), [41](#), [42](#), [48](#), [49](#)], and source C15 adds differentials with logarithmic poles along a normal-crossings divisor [[47](#)]; source C16 adds Seidenberg’s finite derivation identity [[53](#), [54](#)], finite normalization and Riemann–Hurwitz [[55](#), [57](#)]. Sources C10–C14 inspected the repository at commits where this report had only sources 01, 02 and 05, source C15 at one where it had sources 01, 02, 05, 06 and 07, and source C16 at one where it had the same sources and the placed files of C10–C14; their references to its numbering are translated in [appendix A](#). Source C17 imports étale ring maps, formal étaleness and the discriminant criterion [[61](#), [62](#), [63](#)] and the algebraic closedness of Hahn fields [[3](#), Fact 2.1.1]; it read this report’s guide at a commit where sources C10–C15 were written into it. Relations to neighbouring reports are listed in [section 21.7](#).

## 2 The normal-form rings and the integer part

### 2.1 Normal forms and three related rings

We use the standard normal-form theorem: each surreal has a unique expression

$$x = \sum_{\gamma \in S} r_\gamma \omega^\gamma, \quad r_\gamma \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

where  $S \subseteq \mathbf{No}$  is a reverse-well-ordered set: every nonempty subset has a greatest member. Conversely such a normal form represents a surreal. Addition is coefficientwise, multiplication is Hahn convolution (the sum of two reverse-well-ordered supports is reverse well ordered, and each exponent of a product receives finitely many contributions), and the sign of a nonzero normal form is the sign of its leading coefficient [[1](#), [2](#)]. The field  $\mathbf{No}$  is real closed.

**Definition 2.1.** Let

$$\begin{aligned} \Pi &= \{x \in \mathbf{No} : \text{supp}(x) \subseteq \mathbf{No}^{>0}\}, \\ \mathcal{B}_{\mathbb{R}} &= \mathbb{R} \oplus \Pi, \\ \mathbf{Oz} &= \mathbb{Z} \oplus \Pi, \\ \mathcal{B}_{\mathbb{C}} &= \mathcal{B}_{\mathbb{R}}[i] \subseteq \mathbf{No}[i]. \end{aligned}$$

The zero series belongs to  $\Pi$ . For an element of any of these rings,  $\text{ct}(x)$  denotes its coefficient at exponent zero. For  $x = a + ib \in \mathcal{B}_{\mathbb{C}}$ , this is  $\text{ct}(a) + i \text{ct}(b)$ . Put  $\Pi_{\mathbb{C}} = \Pi + i\Pi$ , so  $\mathcal{B}_{\mathbb{C}} = \mathbb{C} \oplus \Pi_{\mathbb{C}}$ .

Thus  $\sqrt{2}\omega$ ,  $\omega/17$ ,  $-\omega^\omega + 3$ ,  $\omega^{1/\omega}$  and  $\sum_{k \geq 1} \omega^{1/k}$  are omnific integers. The numbers  $\omega + \frac{1}{2}$  and  $\omega + \omega^{-1}$  are not. The word “integer” restricts the constant coefficient and excludes infinitesimal tails; it does not require all coefficients to lie in  $\mathbb{Z}$ . “Purely infinite” is a support condition, not a sign condition:  $-\omega \in \Pi$ , while  $\omega - 3 \notin \Pi$ . The normal-form definition of  $\mathbf{Oz}$  is equivalent to Conway’s simplicity characterization by the cut  $\{x - 1 \mid x + 1\}$  [[1](#), [2](#)]; no proof below uses that characterization.

For every positive surreal exponent  $\gamma$ , even an infinitesimal one, the Conway monomial  $\omega^\gamma$  exceeds every ordinary positive real number. This is a normal-form order fact about the omega-map; replacing that map by an unstated analytic expression  $\exp(\gamma \log \omega)$  would invalidate the support arguments below.

**Proposition 2.2** (Rings and retraction [01, 02, 05]).  $\mathcal{B}_{\mathbb{R}}$ ,  $\mathcal{B}_{\mathbb{C}}$ , and  $\mathbf{Oz}$  are integral domains, and  $\mathcal{B}_{\mathbb{R}}$ ,  $\mathbf{Oz}$  are ordered subrings of  $\mathbf{No}$ . The class  $\Pi$  is an ideal of  $\mathcal{B}_{\mathbb{R}}$  and  $\mathbf{Oz}$ , and is an  $\mathbb{R}$ -vector space;  $\Pi_{\mathbb{C}}$  is an ideal of  $\mathcal{B}_{\mathbb{C}}$ . Constant-coefficient extraction is a surjective ring homomorphism from  $\mathcal{B}_{\mathbb{R}}$  to  $\mathbb{R}$ , from  $\mathcal{B}_{\mathbb{C}}$  to  $\mathbb{C}$ , and from  $\mathbf{Oz}$  to  $\mathbb{Z}$ . In the last case its kernel is  $\Pi$ , the inclusion  $\mathbb{Z} \hookrightarrow \mathbf{Oz}$  is a section, and  $\mathbf{Oz}/\Pi \cong \mathbb{Z}$ .

*Proof.* Nonnegative exponents remain nonnegative under addition of exponents. A strictly positive exponent plus a nonnegative exponent is strictly positive. These observations and Hahn convolution prove the ring and ideal claims. In a product of two supported normal forms, exponent zero can arise only as  $0 + 0$ . Therefore  $\text{ct}(xy) = \text{ct}(x)\text{ct}(y)$ , and the constant coefficient of a product of elements of  $\mathbf{Oz}$  is an integer. Additivity is coefficientwise. Real scalar multiplication does not change a support except when the scalar is zero. Surjectivity and the section are supplied by constant series. All rings embed in a field, so have no zero divisors.  $\square$

*Remark 2.3* (Two misleading analogies). The map  $\text{ct}$  is not a ring homomorphism on all of  $\mathbf{No}$ :  $\text{ct}(\omega) = \text{ct}(\omega^{-1}) = 0$  but  $\text{ct}(1) = 1$ . The repository records the same warning for coefficient extraction at zero on  $\mathbf{No}$  (`diff:warn:ct` in the differential-equations report, and `docs/NOTATION.md`); the map  $\text{ct}$  of this report is the restriction of that extraction to  $\mathbf{Oz}$ , where it *is* multiplicative. Also  $\mathcal{B}_{\mathbb{R}}$  is not the ring of finite surreals. The latter permits negative exponents but no positive exponents;  $\mathcal{B}_{\mathbb{R}}$  has the opposite support restriction. Taking the standard part of a finite element and taking the constant coefficient of an omnific integer are different constructions, although both are multiplicative on their own domains. Finally,  $\text{ct}$  is not order preserving:  $\omega - 5 > 0$  but  $\text{ct}(\omega - 5) = -5$ .

## 2.2 Leading degree, units, and finite elements

For nonzero  $x \in \mathbf{No}[i]$ , combine real and imaginary coefficients and put

$$\deg_{\omega}(x) = \max \text{supp}(x), \quad \text{lc}(x) = [\omega^{\deg_{\omega}(x)}]x, \quad \deg_{\omega}(0) = -\infty.$$

On  $\mathcal{B}_{\mathbb{C}} \setminus \{0\}$  the degree lies in  $\mathbf{No}^{\geq 0}$ .

**Lemma 2.4** (Degree [01, 02, 05]). *For nonzero normal forms  $x, y$  with real or complex coefficients,*

$$\begin{aligned} \deg_{\omega}(xy) &= \deg_{\omega}(x) + \deg_{\omega}(y), & \text{lc}(xy) &= \text{lc}(x)\text{lc}(y), \\ \deg_{\omega}(x + y) &\leq \max\{\deg_{\omega}x, \deg_{\omega}y\}, \end{aligned}$$

*with equality in the last relation when  $\deg_{\omega}x \neq \deg_{\omega}y$ . An element of  $\mathcal{B}_{\mathbb{C}}$  of degree zero is a nonzero complex constant.*

*Proof.* If  $\gamma$  and  $\delta$  are the respective largest exponents, every contributing exponent of  $xy$  is at most  $\gamma + \delta$ . Equality requires both exponents to be largest: making either smaller makes their sum smaller. Thus the coefficient at  $\gamma + \delta$  is the nonzero product of leading coefficients. The statements about sums follow from cancellation at the greatest exponent. Finally, support in  $[0, 0]$  contains only zero.  $\square$

**Proposition 2.5** (Units and finite elements [01, 02, 05]). *The unit groups are*

$$\mathcal{B}_{\mathbb{R}}^{\times} = \mathbb{R}^{\times}, \quad \mathcal{B}_{\mathbb{C}}^{\times} = \mathbb{C}^{\times}, \quad \mathbf{Oz}^{\times} = \{1, -1\}.$$

*The omnific integers bounded in absolute value by some ordinary real number are exactly the ordinary integers; in particular  $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$ . Consequently  $\mathbf{Oz}$  is discretely ordered, with least positive element 1.*

*Proof.* If  $xy = 1$  in  $\mathcal{B}_{\mathbb{C}}$ , nonnegativity and additivity of degree force both degrees to be zero. This gives the first two unit statements; in  $\mathbf{Oz}$  the constant units are precisely the units of  $\mathbb{Z}$ .

A nonconstant element of  $\mathbf{Oz}$  has positive leading exponent, so its absolute value exceeds every positive real number by normal-form order. A constant element of  $\mathbf{Oz}$  is an integer. Hence no element lies strictly between 0 and 1, and every positive element is at least 1.  $\square$

Notice that  $\omega/2$  is an omnific integer even though  $1/2$  is not. This is no conflict with discreteness:  $\omega/2$  is infinite, and there are still no omnific integers in the interval  $(0, 1)$ .

The unit computation is stronger than saying that an infinite factor has large real absolute value: complex leading coefficients can cancel in sums, but they cannot cancel the unique leading coefficient in a product. It will do much of the Diophantine work.

### 2.3 Local workspaces

**Lemma 2.6** (Set-sized workspaces [05]). *Any set-sized family of normal forms belongs to a common Hahn field whose exponents lie in a set-sized ordered divisible group  $H \subseteq \mathbf{No}$ . One can take  $H$  to be the rational span of the union of the supports.*

*Proof.* The union of the supports is a set, and so is its rational span. Normal forms supported in  $H$  constitute the real or complex Hahn field with exponents in  $H$ . The Hahn field operations preserve this exponent group, including inverses of nonzero series.  $\square$

We write  $\mathcal{B}_{\mathbb{R},H}$ ,  $\mathcal{B}_{\mathbb{C},H}$  and  $\mathbf{Oz}_H$  for the intersections of  $\mathcal{B}_{\mathbb{R}}$ ,  $\mathcal{B}_{\mathbb{C}}$  and  $\mathbf{Oz}$ , respectively, with the workspace; in particular  $\mathbf{Oz}_H$  retains the integer constant-coefficient condition. Most proofs below, including all the main equation-rigidity theorems, take place in one such set-sized ring and need no proper-class derivation or basis. By contrast, a positive surreal smaller than every member of a specified set of positive exponents need not belong to a preassigned  $H$ ; the results of [section 4](#) genuinely use the full exponent class, and are marked there.

### 2.4 The exact omnific floor

Every surreal admits a unique decomposition

$$x = x_{>0} + r + \varepsilon, \quad x_{>0} \in \Pi, \quad r \in \mathbb{R}, \quad \text{supp}(\varepsilon) \subseteq \mathbf{No}^{<0}.$$

Here  $\varepsilon$  is infinitesimal, possibly zero.

**Theorem 2.7** (Integer-part formula [01, 02, 05]). *There is exactly one  $a \in \mathbf{Oz}$  such that  $a \leq x < a + 1$ . It is*

$$\lfloor x \rfloor_{\mathbf{Oz}} = \begin{cases} x_{>0} + \lfloor r \rfloor, & r \notin \mathbb{Z} \text{ or } \varepsilon \geq 0, \\ x_{>0} + r - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0. \end{cases} \quad (2.2)$$

*Proof.* For  $r \notin \mathbb{Z}$ , its distances from the two adjacent integers are strictly positive real numbers, and the infinitesimal  $\varepsilon$  cannot cross either boundary. If  $r$  is an integer and  $\varepsilon \geq 0$ , the remainder is  $\varepsilon \in [0, 1)$ . If  $r$  is an integer and  $\varepsilon < 0$ , the remainder after subtracting  $x_{>0} + r - 1$  is  $1 + \varepsilon \in (0, 1)$ . Two possible floors would differ by an omnific integer strictly between  $-1$  and  $1$ , so would coincide by discreteness.  $\square$

For example,

$$\lfloor \omega - \omega^{-1} \rfloor_{\mathbf{Oz}} = \omega - 1, \quad \lfloor \omega + \frac{3}{2} - \omega^{-1} \rfloor_{\mathbf{Oz}} = \omega + 1, \quad \lfloor -\omega^{-1} \rfloor_{\mathbf{Oz}} = -1.$$

Merely discarding negative exponents and flooring the constant gives the wrong answer at a negative infinitesimal displacement from an integer. The factorial floor profile of the holonomic-rigidity report (`hol:fh:thm:floor` of `docs/surcomplex/holonomic-rigidity-for-entire-hahn-functions`; batch 32) is the case of this formula with real part 0 or 1 and positive infinitesimal part. The omnific continued-fraction report re-derives [theorem 2.7](#), boundary correction included, as `ocf:lem:floor` of `docs/surreal/omnific-continued-fractions` (batch 33), and uses this floor as the digit map  $a_n = \lfloor x_n \rfloor_{\mathbf{Oz}}$  of its continued-fraction algorithm (`ocf:def:digits`).

*Provenance.* The integer-part assertion is classical. Ehrlich–Kaplan prove the stronger initiality characterization: for an initial subfield  $K$  of  $\mathbf{No}$ ,  $\mathbf{Oz} \cap K$  is its unique initial integer part [4, Lemma 11.1]. No alternative integer part is introduced. The formula agrees with the trigonometry report’s construction of the integer part, which takes “the appropriate side when  $f$  is infinitesimally adjacent to an integer” (text following `trigonometry:eq:split`).

**Proposition 2.8** (One-variable rigidity [02, 05; also C17]). *If  $P \in \mathbb{C}[T]$  is nonzero, every root of  $P$  in  $\mathcal{B}_{\mathbb{C}}$  belongs to  $\mathbb{C}$ . In particular, roots in  $\mathcal{B}_{\mathbb{R}}$  of a nonzero real polynomial are real, and the roots in  $\mathbf{Oz}$  of a nonzero  $P \in \mathbb{Z}[T]$  are exactly its ordinary integer roots.*

*Proof.* If a root  $x$  were nonconstant,  $\alpha = \deg_{\omega}(x) > 0$ . For nonzero coefficients  $a_j$ , the terms  $a_j x^j$  have distinct degrees  $j\alpha$ , so the highest-degree term cannot cancel. Thus  $P(x) \neq 0$ . A constant root in  $\mathbf{Oz}$  is an integer. The real case follows from  $\mathcal{B}_{\mathbb{R}} \cap \mathbb{C} = \mathbb{R}$ .  $\square$

Source 02 states the integer case and source 05 the real case. The same leading-degree argument proves the complex extension above.

Every infinite omnific integer is therefore transcendental over  $\mathbb{R}$ . This does not make a tuple of infinite omnific integers algebraically independent:  $(\omega, \omega^2)$  is a counterexample. Infinite coefficients change the problem:  $X^2 - \omega^2$  has the infinite roots  $\pm\omega$ .

### 3 Divisibility, residues, and quotients

#### 3.1 Division by ordinary integers

**Theorem 3.1** (Ordinary moduli [01, 02, 05]). *For every nonzero ordinary integer  $n$  and  $x \in \mathbf{Oz}$ ,*

$$n\mathbf{Oz} = \Pi + n\mathbb{Z}, \quad n \mid x \iff n \mid \text{ct}(x), \quad \mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}. \quad (3.1)$$

*In particular, for every ordinary prime  $p$ ,*

$$\bigcap_{n \geq 1} n\mathbf{Oz} = \Pi = \bigcap_{k \geq 1} p^k \mathbf{Oz}. \quad (3.2)$$

*If  $\text{ct}(x) \neq 0$ , the largest ordinary  $k$  with  $p^k \mid x$  is the  $p$ -adic order of  $\text{ct}(x)$ ; if  $\text{ct}(x) = 0$ , every power of every ordinary prime divides  $x$ .*

*Proof.* Necessity in the divisibility criterion follows by applying  $\text{ct}$  to  $x = ny$ . Multiplication by a nonzero real scalar preserves  $\Pi$  and is invertible on  $\Pi$ , so  $n\Pi = \Pi$ . If  $x = m + j$  with  $m \in \mathbb{Z}$ ,  $j \in \Pi$ , then  $x/n = m/n + j/n$  lies in  $\mathbf{Oz}$  exactly when  $m/n$  is an integer. Reduction of  $\text{ct}$  modulo  $n$  is therefore a surjection with kernel  $n\mathbf{Oz}$ . The only ordinary integer divisible by all positive integers, or by every power of one prime, is zero, which gives (3.2) and the statement on  $p$ -adic orders.  $\square$

This theorem solves every ordinary congruence. For instance,  $\sqrt{2}\omega + 14$  is even and is congruent to 2 modulo 3;  $\omega$  is divisible by every positive ordinary integer and  $\omega + 1$  is odd. The whole infinite part is invisible to these tests. The omnific-preserving-automorphisms report uses (3.2) as the sole structural input of automatic formal integration: every ring derivation  $\delta$  of  $\mathbf{Oz}$  or of  $\mathbf{Oz}[i]$  maps the ring into its purely infinite part, and the unique iterative Hasse–Schmidt derivation with first component  $\delta$  is  $\delta^n/n!$ , with values in the ring (opa:hs:thm:integration of docs/surreal/omnific-preserving-automorphisms; batch 33).

The statement does *not* extend to arbitrary omnific divisors. For example,  $\text{ct}(\omega + 2)$  divides  $\text{ct}(2)$  in  $\mathbb{Z}$ , but  $\omega + 2$  does not divide 2 in  $\mathbf{Oz}$ . More generally:

**Proposition 3.2** (Divisors of integers [05]). *If  $d \in \mathbf{Oz}$  divides a nonzero ordinary integer  $N$ , then  $d$  is an ordinary integer divisor of  $N$ .*

*Proof.* Write  $de = N$ . Neither factor is zero, so  $|e| \geq 1$  by discreteness and  $|d| \leq |N|$ . Hence  $d \in \mathbb{Z}$  by proposition 2.5; similarly  $e \in \mathbb{Z}$ .  $\square$

**Corollary 3.3** (Ordinary primes and mixed gcds [02, 05]). *Every positive ordinary prime  $p$  is a prime element of  $\mathbf{Oz}$ , and  $p\mathbf{Oz}$  is maximal. For  $0 \neq n \in \mathbb{Z}$  and  $x \in \mathbf{Oz}$ ,*

$$(n, x) = \gcd(n, \text{ct}(x)) \mathbf{Oz},$$

*so an ordinary gcd of  $n$  and  $\text{ct}(x)$  is a gcd of  $n$  and  $x$  in  $\mathbf{Oz}$ . The Chinese remainder theorem for pairwise coprime ordinary moduli holds in  $\mathbf{Oz}$  with exactly the ordinary residue rings.*

*Proof.* The quotient by  $p\mathbf{Oz}$  is the field  $\mathbb{Z}/p\mathbb{Z}$ . If  $x = m + j$  with  $j \in \Pi$ , then  $x - n(j/n) = m$ , so  $(n, x) = (n, m)$ , which an ordinary Bézout identity identifies. A generator of a two-generated principal ideal is a gcd of its generators in any domain. For coprime moduli, ordinary Bézout coefficients give comaximality, and the quotient descriptions give the Chinese remainder isomorphism.  $\square$

**Corollary 3.4** (Positive-characteristic quotients [01, 02, 05]). *Every proper ideal  $\mathfrak{a} \subset \mathbf{Oz}$  containing a nonzero ordinary integer is  $n\mathbf{Oz}$  for a unique ordinary  $n \geq 2$ ; it is prime exactly when  $n$  is an ordinary prime. Every unital homomorphism from  $\mathbf{Oz}$  to a finite ring kills  $\Pi$  and factors through  $\text{ct}$ , and every nonzero finite quotient of  $\mathbf{Oz}$  is a ring  $\mathbb{Z}/n\mathbb{Z}$ .*

*Proof.* If  $m \in \mathfrak{a}$  with  $0 \neq m \in \mathbb{Z}$ , then  $\Pi = m\Pi \subseteq \mathfrak{a}$ . Thus  $\mathfrak{a}$  is the inverse image of an ideal of  $\mathbf{Oz}/\Pi = \mathbb{Z}$ , namely  $n\mathbb{Z}$  with  $n > 0$ , and properness gives  $n \geq 2$ . Equation (3.1) identifies its inverse image with  $n\mathbf{Oz}$ ; primality is that of  $n\mathbb{Z}$ . If a finite target has characteristic  $m$ , then each  $j \in \Pi$  is  $m(j/m)$  and maps to zero. A finite nonzero unital ring has positive characteristic, so the last assertion follows.  $\square$

## 3.2 Completions

All finite-index ideals have just been classified, so the profinite completion, interpreted as the set-sized inverse limit over ordinary  $n \geq 1$ , is

$$\widehat{\mathbf{Oz}} \cong \varprojlim_n \mathbb{Z}/n\mathbb{Z} = \widehat{\mathbb{Z}}, \quad \varprojlim_k \mathbf{Oz}/p^k \mathbf{Oz} \cong \mathbb{Z}_p. \quad (3.3)$$

The completion maps are the composites of  $\text{ct}$  with the canonical maps of  $\mathbb{Z}$ , with kernel  $\Pi$  in both cases by (3.2). Thus these congruence topologies on  $\mathbf{Oz}$  are not Hausdorff and  $\mathbf{Oz}$  is not residually finite [01, 02, 05]. For example  $\omega$ ,  $\sqrt{2}\omega$  and 0 have identical images in every finite quotient. The separated quotient is  $\mathbf{Oz}/\Pi = \mathbb{Z}$ ; completion does not preserve the infinite

part. Status (batch 32): the failure of the Hausdorff property is forced. Every ring topology on  $\mathbf{Oz}$  in which each Hahn-summable family with sum in  $\mathbf{Oz}$  converges to its sum along finite partial sums is pulled back along  $\text{ct}$  from a ring topology on  $\mathbb{Z}$ , and none is Hausdorff (`opa:tc:thm:oztopology` of `docs/surreal/omnific-preserving-automorphisms`, which records the two topologies above as instances in `opa:tc:rem:congruence`).

The first inverse system is indexed by positive integers ordered by divisibility, with reduction maps  $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$  for  $m \mid n$ . The second uses the powers of a fixed ordinary prime  $p$ . These are set-sized diagrams of ordinary residue rings, not sets of class ideals or proper-class cosets.

### 3.3 Homomorphisms to $\mathbb{Z}$ and to set-sized rings

**Proposition 3.5** (Uniqueness of the retraction [01]). *The constant-term map is the unique unital ring homomorphism  $\mathbf{Oz} \rightarrow \mathbb{Z}$ . Every unital endomorphism  $\psi : \mathbf{Oz} \rightarrow \mathbf{Oz}$  preserves  $\Pi$  and satisfies  $\text{ct} \circ \psi = \text{ct}$ .*

*Proof.* For a homomorphism  $h : \mathbf{Oz} \rightarrow \mathbb{Z}$ , the integer  $h(j)$  is divisible by every positive integer whenever  $j \in \Pi$ . Hence  $h(j) = 0$ . Since  $h$  fixes ordinary integers,  $h = \text{ct}$ . An endomorphism preserves divisibility by each ordinary integer, so preserves their common divisibility kernel (3.2). It fixes  $\mathbb{Z}$  pointwise, which proves the final identity.  $\square$

This does not say that every endomorphism is the identity: the composite of  $\text{ct}$  with the inclusion  $\mathbb{Z} \hookrightarrow \mathbf{Oz}$  is a noninjective unital endomorphism [05].

*Provenance.* The divisible-kernel and residue-map perspective belongs to classical  $\mathbb{Z}$ -group theory; see [5, Sections 2 and 4]. Here it has a concrete ring-theoretic realization because  $\text{ct}$  is multiplicative. The finite-quotient and completion statements above are proved directly.

Source 05 proved a far-reaching generalization of [proposition 3.5](#) and [corollary 3.4](#). It is proved, in a more general coefficient-ring form and with the sharper theory of set-sized quotients assembled from several manuscripts, in the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers`, which also keeps source 05's route; we quote it without proof.

**Cited theorem 3.6** (Universal set-sized quotient; proved by source 05, printed in the sibling report). *Let  $S$  be a set-sized unital ring, not necessarily commutative, and let  $\varphi : \mathbf{Oz} \rightarrow S$  be a unital class ring homomorphism. Then  $\varphi(x) = \text{ct}(x) 1_S$  for every  $x \in \mathbf{Oz}$ . Consequently every set-sized quotient ring of  $\mathbf{Oz}$  is isomorphic to  $\mathbb{Z}$  (kernel  $\Pi$ ) or to  $\mathbb{Z}/n\mathbb{Z}$  (kernel  $n\mathbf{Oz}$ ) for an ordinary  $n \geq 1$ .*

Source 05's proof applies pigeonhole to the monomials  $\omega^{\delta/\omega^{\alpha+1}}$  for ordinals  $\alpha$  beyond the size of  $S$ , and kills a positive monomial  $\omega^\delta$  by the exact Hahn identity  $(\omega^a - \omega^b) \sum_{k \geq 0} \omega^{\delta-a-k(a-b)} = \omega^\delta$ , whose quotient lies in  $\Pi$  when  $0 < b < a \leq \delta/\omega$ ; it then uses [theorem 4.2](#) below to kill all of  $\Pi$ .

*Remark 3.7* (Foundations of the cited theorem). Source 05 chooses an infinite cardinal  $\kappa > |S|$ , using the global choice of its NBG setting. The Hartogs ordinal of  $S$ , the least ordinal admitting no injection into  $S$ , can be used instead: the pigeonhole step needs only that the family  $(\omega^{\delta/\omega^{\alpha+1}})_\alpha$  indexed by that ordinal does not inject into  $S$ . So neither global choice nor choice for sets is needed for [cited theorem 3.6](#).

[Proposition 3.5](#) is the case  $S = \mathbb{Z}$  of [cited theorem 3.6](#) together with its consequence for endomorphisms (apply it to  $\text{ct} \circ \psi$ ), and [corollary 3.4](#) is the case of finite  $S$ ; the elementary divisibility proofs above are independent second routes. The theorem does not apply to

class-sized targets such as  $\mathbf{Oz}$  itself, and it is a full-class statement: the subring  $\mathbb{Z} + \omega\mathbb{R}[\omega]$  of  $\mathbf{Oz}$  admits evaluation at many real values, and such maps do not extend to  $\mathbf{Oz}$  [05]. Source 05 also records that, by [cited theorem 3.6](#), the quotient  $\mathbf{Oz}/(f)$  by an infinite prime  $f$  with  $\text{ct}(f) = \pm 1$  (for example the prime of [3, Theorem B] in [section 4.2](#)) is not set-sized; that consequence is likewise printed in the sibling report.

## 4 Full-class support: common divisors, clearing, factorization

The next results use all surreal exponents, not merely the exponents in one fixed Archimedean or finite-rank group. This is an important source of arithmetic differences between full omnific integers and restricted series rings ([section 20](#)).

### 4.1 One monomial divides a whole set

**Lemma 4.1** (Set-sized exponent bounds [01, 02, 05]). *If  $E \subset \mathbf{No}^{>0}$  is a set, there is a surreal  $\delta > 0$  such that  $\delta < e$  for every  $e \in E$ . Every set of surreals also has a strict upper bound.*

*Proof.* A surreal separating the cut  $\{0 \mid E\}$  gives the first assertion, and a separator of  $\{E \mid \emptyset\}$  the second; their option collections are sets. The assertions also hold for the empty set. These are applications of the set-cut property, not an assertion of an infimum or supremum. No single  $\delta$  works for the entire proper class  $\mathbf{No}^{>0}$ .  $\square$

**Theorem 4.2** (Common monomial divisibility [01, 02, 05]). *For every set  $S \subseteq \Pi$  there is  $\delta > 0$  such that*

$$\omega^\delta \mid s \quad \text{and} \quad s/\omega^\delta \in \Pi \quad (s \in S).$$

*In particular, each nonzero element of  $\Pi$  is a product of two nonzero elements of  $\Pi$ , both nonunits of  $\mathbf{Oz}$ .*

*Proof.* The union of the supports of the elements of  $S$  is a set  $E$  of strictly positive surreal exponents. Choose  $\delta$  as in [lemma 4.1](#). Dividing by  $\omega^\delta$  shifts every exponent  $e$  to  $e - \delta > 0$ , preserving the reverse well-order of each support. The quotient is therefore in  $\Pi$ . For a nonzero  $s$ , both factors are nonzero elements of positive degree, and neither is a unit.  $\square$

Even the series  $\sum_{k \geq 1} \omega^{1/k}$  has such a divisor. The positive exponent  $\delta$  may be infinitesimal relative to every real  $1/k$ , for example  $\delta = 1/\omega$ . This is why the proof cannot be copied into a Hahn ring with exponent group  $\mathbb{R}$ .

**Corollary 4.3** (The purely infinite ideal [01, 02, 05]). *The ideal  $\Pi$  satisfies  $\Pi^2 = \Pi$ , and each element of  $\Pi$  is a single product of two elements of  $\Pi$ . It is not generated as an ideal by any set of elements; in particular it is not finitely generated. Consequently  $\Pi/\Pi^2 = 0$  although  $\Pi \neq 0$ ,  $\mathbf{Oz}/\Pi^k \cong \mathbb{Z}$  for every ordinary  $k \geq 1$ , and the  $\Pi$ -adic completion of  $\mathbf{Oz}$  is  $\mathbb{Z}$ . No nonzero element of  $\Pi$  is a finite product of irreducibles of  $\mathbf{Oz}$ .*

*Proof.* The inclusion  $\Pi^2 \subseteq \Pi$  follows from the ideal property. For  $j \neq 0$ , the factorization in [theorem 4.2](#) has both factors in  $\Pi$ , proving the reverse inclusion. Hence  $\Pi^k = \Pi$  for all  $k \geq 1$  and the inverse system  $\mathbf{Oz}/\Pi^k$  is constantly  $\mathbb{Z}$ .

Suppose a set  $S \subseteq \Pi$  generated  $\Pi$ . Choose  $\delta > 0$  strictly below every exponent in the supports of its elements. Each finite  $\mathbf{Oz}$ -linear combination of elements of  $S$  has support strictly above  $\delta$ : multiplication by an element of  $\mathbf{Oz}$  only adds nonnegative exponents. But  $\omega^{\delta/2} \in \Pi$  does not have such a support.

Finally, if  $j = p_1 \cdots p_k \neq 0$  with every  $p_i$  irreducible, then  $0 = \text{ct}(j) = \prod_{i=1}^k \text{ct}(p_i)$  would put some  $p_i$  in  $\Pi$ , and that factor would be reducible by [theorem 4.2](#).  $\square$

The factorization  $j = 2(j/2)$  already proves that a nonzero  $j \in \Pi$  is reducible, even in a fixed exponent workspace; the stronger content of [theorem 4.2](#) is a common *infinite monomial* divisor for an entire set, with all quotients still in  $\Pi$ . The vanishing of  $\Pi/\Pi^2$  does not say that the infinite part is small; it says that this particular quotient cannot detect it. The hypotheses of a finitely generated Nakayama argument are not present [02].

**Corollary 4.4** (Nilpotent tests [02]). *Let  $\phi : \mathbf{Oz} \rightarrow R$  be a ring homomorphism and  $\mathfrak{n} \subseteq R$  a nilpotent ideal with  $\phi(\Pi) \subseteq \mathfrak{n}$ . Then  $\phi(\Pi) = 0$ .*

*Proof.* Choose an ordinary  $N \geq 1$  with  $\mathfrak{n}^N = 0$ . By [corollary 4.3](#),  $\Pi^N = \Pi$ . The image of every finite sum of  $N$ -fold products from  $\Pi$  belongs to  $\mathfrak{n}^N = 0$ . This proves the claim without assuming that  $\phi(\Pi)$  is an ideal of all of  $R$ .  $\square$

## 4.2 Factorization

[Corollary 4.3](#) is an explicit failure of atomic factorization, not merely a failure of uniqueness of factorization. It leaves room for nonconstant prime elements having nonzero constant coefficient.

**Proposition 4.5** (Irreducibles [01, 02, 05]). *The constant irreducibles of  $\mathbf{Oz}$  are exactly the ordinary signed primes. Every nonconstant irreducible has constant coefficient 1 or  $-1$ . The latter condition is not sufficient for irreducibility. In particular,  $\mathbf{Oz}$  is neither atomic nor a unique factorization domain.*

*Proof.* Constant irreducibles are classified by [proposition 3.2](#). If a nonconstant element  $a$  has  $|\text{ct}(a)| > 1$ , choose an ordinary prime  $p$  dividing its constant coefficient. Then  $a = p(a/p)$  by [theorem 3.1](#) is a factorization into two nonunits: the second factor remains nonconstant. If  $\text{ct}(a) = 0$ , the factorization  $a = 2(a/2)$  has the same property. This leaves only constant coefficients  $\pm 1$ . Finally,  $(\omega + 1)^2$  has constant coefficient 1 and  $\omega^2 - 1 = (\omega - 1)(\omega + 1)$  has constant coefficient  $-1$ ; both are reducible.  $\square$

In particular  $\omega$  has no finite irreducible factorization, not for lack of factors: it is divisible by every ordinary positive integer and by every  $\omega^\delta$  with  $0 < \delta < 1$ . Genuine infinite primes exist. The series  $1 + \sum_{k \geq 1} \omega^{1/k}$  is prime by earlier results reviewed by L’Innocente and Mantova [3, Section 1.3] (irreducibility by Berarducci, primality by Pitteloud, and primality in  $\mathbf{Oz}$  by Biljaković–Kochetov–Kuhlmann, as credited there), and  $\omega^{\sqrt{2}} + \omega + 1$  is prime by their Theorem B [3, Theorem B], which answers a question of Gonshor. Their general theory uses substantially finer invariants than the degree employed here; no result of this report depends on these primality theorems.

## 4.3 Clearing arbitrary set-sized families of denominators

**Theorem 4.6** (Global monomial clearing [01, 02, 05]). *For every set  $S \subseteq \mathbf{No}$  there is  $h > 0$  such that*

$$\omega^h \in \Pi, \quad \omega^h x \in \Pi \quad (x \in S).$$

*Consequently  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ , where the equality means that every surreal is a quotient of two omnific integers with nonzero denominator.*

*Proof.* Take the set  $E$  obtained by uniting the supports of all elements of  $S$ . Choose  $h > 0$  strictly greater than every  $-e$  for  $e \in E$ . Multiplication by  $\omega^h$  shifts every exponent to  $h+e > 0$ , so all shifted normal forms belong to  $\Pi$ . Applying this to  $S = \{x\}$  gives  $x = (\omega^h x)/\omega^h$ . The reverse inclusion holds because  $\mathbf{Oz}$  is a subring of the field  $\mathbf{No}$ .  $\square$

A support unbounded below in a chosen real exponent group is still a set of surreal exponents and can be shifted above zero by a new, larger surreal exponent. The theorem does not assert that the required monomial lies in the original fixed Hahn workspace. The omnific-preserving-automorphisms report applies the theorem to the countable coefficient set of a formal power series, obtaining  $\text{Frac}(\mathbf{Oz}[[s]]) = \mathbf{No}((s))$  (`opa:hs:prop:formalfrac`); with formal square roots it shows that the parameter-free formula  $\exists a \exists b (b \neq 0 \wedge a^2 = (x^4 + 1)b^2)$ , which defines exactly  $\mathbf{Oz}$  in every finite polynomial ring over  $\mathbf{Oz}$ , holds for every element of  $\mathbf{Oz}[[s]]$  (`opa:hs:thm:collapse of docs/surreal/omnific-preserving-automorphisms; batch 33`). Its caveat on fixed workspaces is `opa:hs:rem:classscope`.

**Corollary 4.7** (Projective clearing and common multiples [01, 02, 05]). *Every point of  $\mathbb{P}^n(\mathbf{No})$  has homogeneous coordinates in  $\Pi^{n+1}$ ; more generally, any set-sized collection of coordinate tuples can be cleared by a single positive monomial, and a common zero of homogeneous polynomials remains a common zero after this rescaling. Every set  $S$  of nonzero omnific integers has a nonzero common multiple; in fact some nonzero  $M \in \mathbf{Oz}$  lies in  $a^k \mathbf{Oz}$  for every  $a \in S$  and ordinary  $k \geq 1$ .*

*Proof.* Apply [theorem 4.6](#) to the set of all coordinates and multiply all tuples by the resulting nonzero scalar; homogeneity preserves the equations. For the last claim apply [theorem 4.6](#) to the set  $\{a^{-k} : a \in S, k \geq 1\}$  and take  $M = \omega^h$ ; then  $M/a^k \in \Pi$ .  $\square$

This is projective *representability*, not primitivity: the coordinates of such a representative always have a common nonunit divisor, and [section 14](#) shows that some real projective directions have no primitive omnific representative at all. The common-multiple statement is an existence theorem, not an algorithm; nor is there one nonzero omnific integer divisible by every member of the proper class of nonzero omnific integers.

#### 4.4 Division with remainder is not a terminating Euclidean algorithm

**Proposition 4.8** (Ordered division [01, 02, 05]). *For  $a, b \in \mathbf{Oz}$  with  $b > 0$ , there are unique  $q, r \in \mathbf{Oz}$  such that  $a = bq + r$  and  $0 \leq r < b$ , namely  $q = \lfloor a/b \rfloor_{\mathbf{Oz}}$  and  $r = a - bq$ .*

*Proof.* The floor inequalities of [theorem 2.7](#), multiplied by  $b$ , give the remainder bounds. Conversely those bounds force  $q \leq a/b < q + 1$ , so floor uniqueness determines  $q$  and then  $r$ .  $\square$

Thus the familiar division step exists. Termination of repeated division is the missing ingredient: the positive order on  $\mathbf{Oz}$  is not well founded.

**Example 4.9** (A Euclidean algorithm that never stops [01, 02, 05]). Let  $\rho = \sqrt{2} - 1 \in (0, 1)$  and start with  $r_0 = \sqrt{2}\omega$ ,  $r_1 = \omega$ . The first quotient is 1, the remainder is  $r_2 = \rho\omega$ , and every subsequent quotient is 2:

$$r_k = \rho^{k-1}\omega \quad (k \geq 1), \quad r_{k-1} = 2r_k + r_{k+1} \quad (k \geq 2),$$

because  $1 - 2\rho = \rho^2$  and  $1/\rho = \sqrt{2} + 1 \in (2, 3)$ . Every remainder is positive and infinite, so the ordinary finite Euclidean procedure never terminates. (Source 05 starts at  $(\omega, \rho\omega)$  and sees the same tail.) A decreasing remainder is not a well-founded measure here.

**Theorem 4.10** (Pairs without a gcd [02], generalizing [01]). *Let  $0 \neq t \in \Pi$  and let  $r \in \mathbb{R} \setminus \mathbb{Q}$ . The pair  $(t, rt)$  has no greatest common divisor in  $\mathbf{Oz}$ , in the divisibility sense. In particular,  $\mathbf{Oz}$  is neither a GCD domain nor a Bézout domain.*

*Proof.* Take any common divisor  $d \in \mathbf{Oz}$ . Since  $t \neq 0$ , also  $d \neq 0$ , and there are nonzero  $u, v \in \mathbf{Oz}$  with  $t = du$  and  $rt = dv$ . Cancellation in  $\mathbf{No}$  gives  $v = ru$ . Taking constants in  $\mathcal{B}_{\mathbb{R}}$  yields  $\text{ct}(v) = r \text{ct}(u)$  with both constants ordinary integers, so irrationality of  $r$  forces both to be zero:  $u, v \in \Pi$ .

By [theorem 4.2](#) applied to  $\{u, v\}$ , choose  $\delta > 0$  with  $u/\omega^\delta, v/\omega^\delta \in \Pi$ . Then  $d' = d\omega^\delta$  is another common divisor of  $t, rt$ , and  $d \mid d'$ . But  $d' \nmid d$ , since the quotient would be  $\omega^{-\delta} \notin \mathbf{Oz}$ . Thus every common divisor can be strictly enlarged in the divisibility order, and none is a gcd. A Bézout domain would supply a gcd for each pair, namely a generator of the ideal the pair generates.  $\square$

*Remark 4.11* (A second route for the pair  $(\sqrt{2}\omega, \omega)$  [01]). Source 01 proves the instance  $t = \omega$ ,  $r = \sqrt{2}$  by degrees alone. Let  $d \neq 0$  be a common divisor,  $\theta = \deg_\omega(d)$ ; then  $0 \leq \theta \leq 1$ . If  $\theta = 1$ , both quotients are degree-zero omnific integers, hence nonzero ordinary integers, and their ratio would make  $\sqrt{2}$  rational. So  $\theta < 1$ ; with  $\eta = (\theta + 1)/2$ ,  $\omega^\eta$  is a common divisor (both quotients have exponent  $1 - \eta > 0$ ) that does not divide  $d$ , since its degree is larger. This route needs only the exponent  $(\theta + 1)/2$ , so it works in any workspace with a divisible exponent group, whereas the general [theorem 4.10](#) uses the full-class shift of [theorem 4.2](#).

[Example 4.9](#) shows at once why numerical descent is insufficient and why a division algorithm need not produce a gcd: its input has none.

## 4.5 Non-Noetherian and non-normal behavior

**Proposition 4.12** (Chains and integral closure [01, 02, 05]). *There is a strictly ascending chain of principal ideals*

$$(\omega) \subsetneq (\omega^{1/2}) \subsetneq (\omega^{1/4}) \subsetneq \cdots,$$

*so  $\mathbf{Oz}$  satisfies neither the ascending chain condition on principal ideals nor the Noetherian condition. The full ring  $\mathbf{Oz}$  is not integrally closed in its fraction field: both  $\sqrt{2}$  and  $\sqrt{\omega^2 + 1}$  are integral over  $\mathbf{Oz}$ , lie in  $\text{Frac}(\mathbf{Oz})$ , and are not in  $\mathbf{Oz}$ .*

*Proof.* Each generator is the square of the next, so the inclusions hold. They are strict because the reverse quotients  $\omega^{-1/2^{k+1}}$  have negative exponents. The number  $\sqrt{2} = (\sqrt{2}\omega)/\omega$  satisfies  $T^2 - 2 = 0$  and is not an integer [02, 05]. Since  $\mathbf{No}$  is real closed, let  $r = \sqrt{\omega^2 + 1} > 0$ ; it lies in  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  and satisfies  $T^2 - (\omega^2 + 1) = 0$ . If  $r \in \mathbf{Oz}$ , then  $(r - \omega)(r + \omega) = 1$  makes both factors units, hence elements of  $\{1, -1\}$  with equal signs; their difference  $2\omega$  would be 0, a contradiction [01].  $\square$

In particular the full omnific ring is not a UFD or a PID. Its factorization theory cannot be replaced by an ordinary Euclidean algorithm or by an unstated gcd assumption. The modern theory in [3] is substantially subtler. Conway's refinement question asks whether  $ab = cd$  in  $\mathbf{Oz}$  always admits  $a = ef$ ,  $b = gh$ ,  $c = eg$ ,  $d = fh$  with  $e, f, g, h \in \mathbf{Oz}$ ; this is a refinement property, not atomicity or the existence of gcDs, and none of the preceding arguments refutes it. [Appendix D](#) records why no current open-status claim about it is made.

## 5 Equational transfer and linear Diophantine equations

### 5.1 Existence is exactly ordinary existence

For a finite family of polynomials  $F_1, \dots, F_r$  over a ring  $R$ , write  $V_R(F_1, \dots, F_r)$  for its simultaneous zero set in  $R^n$ . All exponents of the variables are ordinary nonnegative integers and only finitely many monomials occur; equations with variable exponents, surreal exponentiation, or class-sized sums are not polynomial systems in this sense. Finite exponential sums  $\sum_j c_j \exp(\sum_k \lambda_{jk} x_k)$  in omnific-integer variables, with complex  $c_j$  and real  $\lambda_{jk}$ , are treated in the exponential-relations report (`docs/foundations-and-computation/exponential-relations-over-omnific-integers`; batch 33). Its ordinary-fiber criterion `exr:thm:fibers` is an analogue of [theorem 5.1](#) and of the fibers (5.2): over each ordinary integer tuple the fiber is a finite union of real-linear subspaces of  $\Pi^d$ . Ordered additive arithmetic on  $\mathbf{Oz}$  with the zero predicates of such sums, for algebraic coefficients and real algebraic slopes, has a decidable theory (`exr:thm:algdecision`); adjoining the single relation  $\exp(xy) = \exp(z)$ , the graph of multiplication, makes it undecidable (`exr:thm:undecidable`), a variant of [corollary 13.12](#).

**Theorem 5.1** (Retraction theorem for equations [01, 02, 05]). *Let  $F_1, \dots, F_r \in \mathbb{Z}[X_1, \dots, X_n]$ . Coordinatewise constant coefficient gives a retraction, with the inclusion as section,*

$$V_{\mathbf{Oz}}(F_1, \dots, F_r) \longrightarrow V_{\mathbb{Z}}(F_1, \dots, F_r).$$

In particular,

$$V_{\mathbf{Oz}}(F_1, \dots, F_r) \neq \emptyset \iff V_{\mathbb{Z}}(F_1, \dots, F_r) \neq \emptyset. \quad (5.1)$$

The same equivalence holds for positive existential formulas in the ring language, evaluated at ordinary integer parameters.

*Proof.* For every polynomial  $F$  over  $\mathbb{Z}$  and  $\mathbf{x} \in \mathbf{Oz}^n$ ,  $\text{ct}(F(\mathbf{x})) = F(\text{ct}(x_1), \dots, \text{ct}(x_n))$ , because  $\text{ct}$  is a ring homomorphism fixing all coefficients. Thus the constant tuple of an omnific solution is an integer solution. Conversely, an integer solution is already an omnific solution, and applying  $\text{ct}$  to it does nothing.

A positive existential formula is built from equalities using finite conjunction, finite disjunction, and existential quantification. Ring homomorphisms preserve such formulas, by induction on their formation, with witnesses mapped coordinatewise. Apply this to  $\text{ct}$  and to the inclusion of  $\mathbb{Z}$  in  $\mathbf{Oz}$ .  $\square$

The theorem answers a question about existence, not about cardinalities or complete solution sets. For  $\mathbf{a} \in V_{\mathbb{Z}}$ , its *constant-coefficient fiber* is

$$V_{\mathbf{a}}(\mathbf{Oz}) = \{\mathbf{a} + \mathbf{u} : \mathbf{u} \in \Pi^n, F_j(\mathbf{a} + \mathbf{u}) = 0 \text{ for all } j\}. \quad (5.2)$$

For  $Y = X^2$  the fiber above  $(0, 0)$  already contains  $(t, t^2)$  for every  $t \in \Pi$ . These are not infinitesimal deformations: every nonzero coordinate of  $\mathbf{u}$  is infinite. We call a system *rigid* when every fiber is its ordinary base point, that is, when  $V_{\mathbf{Oz}} = V_{\mathbb{Z}}$  [02].

**Corollary 5.2** (Hilbert's tenth problem [01, 02, 05]). *There is no algorithm deciding, from a finite list of ordinary integer polynomials, whether it has a common zero in  $\mathbf{Oz}$ . The yes-instances are exactly the ordinary-integer yes-instances of Hilbert's tenth problem, and form a computably enumerable set that is complete for computably enumerable sets under many-one reductions.*

*Proof.* Equation (5.1) identifies the two decision problems on exactly the same finite inputs. The undecidability and completeness of the ordinary problem are the Davis–Putnam–Robinson–Matiyasevich (MRDP) theorem and its standard consequences [7, 8, 9, 10]; one passes between the natural-number and integer formulations by the four-square representation of nonnegative integers, and conjunctions are combined as sums of squares. Enumerating ordinary integer tuples semidecides the omnific yes-instances; no enumeration of the proper class  $\mathbf{Oz}$  is involved.  $\square$

This corollary concerns *ordinary integer coefficients*. A decision problem allowing arbitrary omnific coefficients requires an explicit finite representation scheme and is a different question (section 19). The holonomic-rigidity report encodes the same undecidability into a rigid class of functional equations with integer data and a prescribed degree bound: no algorithm decides whether such an equation has a polynomial solution with coefficients in  $\mathbf{Oz}$  (`hol:pc:cor:undecidable` of `docs/surcomplex/holonomic-rigidity-for-entire-hahn-functions`; batch 31). Its proof passes through `ct` and MRDP, as here. For monic univariate equations with omnific coefficients and a discriminant in  $\mathbb{Z} \setminus \{0\}$ , in contrast, root existence reduces to the integer roots of one ordinary polynomial (corollary 19.16).

## 5.2 Why signs and nonvanishing do not transfer

Consider  $(X + 1)^2 - 2(Y + 1)^2 = 0$ . It has no solution in  $\mathbb{N}^2$ , since  $X + 1$  and  $Y + 1$  would be positive integers with ratio  $\sqrt{2}$ . However,  $X = \sqrt{2}\omega - 1$ ,  $Y = \omega - 1$  is a solution with both coordinates positive in  $\mathbf{Oz}$ . Its constant tuple is  $(-1, -1)$ , an ordinary solution of the equation without positivity constraints.

**Proposition 5.3** (Retraction closure [01, 02, 05]). *If  $D \subseteq \mathbf{Oz}^n$  is defined by a positive existential ring formula with ordinary integer parameters, then  $\mathbf{x} \in D$  implies  $\text{ct}(\mathbf{x}) \in D$ . Consequently none of*

$$\mathbf{Oz} \setminus \{0\}, \quad \mathbf{Oz}^{>0}, \quad \mathbf{Oz}^{\geq 0}$$

*is so definable.*

*Proof.* Apply `ct` to the free variables and to all existential witnesses, as in theorem 5.1. But  $\omega$  is nonzero and positive while  $\text{ct}(\omega) = 0$ ; and  $\omega - 1 > 0$  whereas  $\text{ct}(\omega - 1) = -1 < 0$ .  $\square$

Thus “nonzero” cannot be harmlessly added to an equational transfer statement, and the retraction is not an elementary embedding of ordered rings. Encoding nonvanishing by  $XY = 1$  would restrict  $X$  to  $\{1, -1\}$ , not to all nonzero omnific integers. The qualification on coefficients matters as well: a predicate with a fixed infinite parameter need not be preserved by a map that changes that parameter. For coefficients outside  $\mathbb{Z}$ , specialization gives a necessary residual system whose solvability need not lift (section 19).

## 5.3 A complete solution criterion for linear systems

**Theorem 5.4** (Smith normal form over the omnific integers [01, 02, 05]). *Let  $M \in \mathbb{Z}^{m \times n}$  and  $\mathbf{b} \in \mathbf{Oz}^m$ . Choose unimodular integer matrices  $U, V$  with*

$$UMV = \text{diag}(d_1, \dots, d_r, 0), \quad d_i > 0,$$

*in rectangular Smith normal form, and put  $\mathbf{c} = U\mathbf{b}$ . Then  $M\mathbf{x} = \mathbf{b}$  is solvable in  $\mathbf{Oz}^n$  if and only if*

$$d_i \mid \text{ct}(c_i) \quad (1 \leq i \leq r), \quad c_i = 0 \quad (r < i \leq m). \quad (5.3)$$

*All solutions are  $\mathbf{x} = V\mathbf{y}$ , where  $y_i = c_i/d_i$  for  $1 \leq i \leq r$  and  $y_{r+1}, \dots, y_n \in \mathbf{Oz}$  are arbitrary.*

*Proof.* Unimodular integer matrices and their inverses act on omnific tuples. The substitution  $\mathbf{x} = V\mathbf{y}$  transforms the system into  $\text{diag}(d_1, \dots, d_r, 0)\mathbf{y} = \mathbf{c}$ . For each nonzero diagonal entry, division is possible precisely under [theorem 3.1](#). Zero rows require the full equation  $c_i = 0$ , not merely its constant coefficient. The remaining coordinates are unconstrained.  $\square$

**Theorem 5.5** (Ordinary right-hand sides [05]). *Let  $M \in \mathbb{Z}^{m \times n}$  and  $\mathbf{b} \in \mathbb{Z}^m$ . All solutions in  $\mathbf{Oz}^n$  of  $M\mathbf{x} = \mathbf{b}$  are exactly*

$$\mathbf{x} = \mathbf{a} + \mathbf{u}, \quad \mathbf{a} \in \mathbb{Z}^n, \quad M\mathbf{a} = \mathbf{b}, \quad \mathbf{u} \in \Pi^n, \quad M\mathbf{u} = \mathbf{0}.$$

*If  $\mathbf{v}_1, \dots, \mathbf{v}_e \in \mathbb{Q}^n$  is a basis of  $\ker(M : \mathbb{Q}^n \rightarrow \mathbb{Q}^m)$ , the last condition is equivalent to  $\mathbf{u} = \sum_j h_j \mathbf{v}_j$  with uniquely determined  $h_j \in \Pi$ . In particular, if  $M$  has full column rank, all solutions are ordinary.*

*Proof.* Separate each coordinate into its integer constant and its member of  $\Pi$ . The integer and positive-support components of  $M\mathbf{x} - \mathbf{b}$  vanish separately. Gaussian elimination over  $\mathbb{Q}$  works with entries in the  $\mathbb{Q}$ -vector space  $\Pi$ , since nonzero rationals act invertibly on it; applying the same row operations and free-variable parametrization as over  $\mathbb{Q}$  gives the basis description and its uniqueness.  $\square$

For example, every solution of  $2x + 3y = 1$  is  $x = -1 + 3t$ ,  $y = 1 - 2t$  with  $t \in \mathbf{Oz}$ , and every solution of  $6x + 10y = 14$  is  $x = 4 + 5t$ ,  $y = -1 - 3t$  with  $t \in \mathbf{Oz}$ . The same formulas as over  $\mathbb{Z}$  work, but the parameter class is larger. The equation  $2x = \omega + 1$  has no solution, whereas  $2x = \omega + 2$  has the unique solution  $x = \omega/2 + 1$ . A zero row illustrates a different obstruction:  $0 \cdot x = \omega$  has no solution, although its constant-term equation is  $0 = 0$ .

## 6 Rigidity of constant products: binary forms, Pell, norms

### 6.1 The constant-product principle

**Lemma 6.1** (Constant products [01, 02, 05]). *If  $b_1, \dots, b_s \in \mathcal{B}_{\mathbb{C}}$  and  $b_1 \cdots b_s = c \in \mathbb{C}^\times$ , then every  $b_i$  belongs to  $\mathbb{C}^\times$ .*

*Proof.* Each  $b_i$  has inverse  $c^{-1} \prod_{j \neq i} b_j$  in  $\mathcal{B}_{\mathbb{C}}$ ; apply [proposition 2.5](#). Equivalently, their nonnegative degrees have sum zero, so every degree is zero.  $\square$

The importance of  $\mathcal{B}_{\mathbb{C}}$  is that linear factors may have irrational or complex coefficients. They need not themselves be omnific integers, but they have no negative exponents, so the unit argument applies.

**Theorem 6.2** (Exact fibers of decomposable equations [02]; case (b) [01]). *Let*

$$F(\mathbf{X}) = \prod_{j=1}^s L_j(\mathbf{X})^{e_j}, \quad L_j(\mathbf{X}) = \beta_j + \ell_j(\mathbf{X}),$$

*where  $\beta_j \in \mathbb{C}$ , each  $\ell_j$  is a complex linear form in  $n$  variables, and each  $e_j \geq 1$  is an ordinary integer. A nonzero scalar prefactor may instead be divided into the level. Fix  $c \in \mathbb{C}^\times$  and put*

$$\mathcal{K} = \{\mathbf{v} \in \mathbb{R}^n : \ell_j(\mathbf{v}) = 0 \text{ for every } j\}.$$

(a) The solutions of  $F(\mathbf{x}) = c$  in  $\mathbf{Oz}^n$  are exactly

$$\mathbf{x} = \mathbf{a} + \mathbf{u}, \quad \mathbf{a} \in \mathbb{Z}^n, \quad F(\mathbf{a}) = c, \quad \mathbf{u} \in \Pi^n, \quad \ell_j(\mathbf{u}) = 0 \text{ for all } j. \quad (6.1)$$

Equivalently, at each exponent the real coefficient vector of  $\mathbf{u}$  lies in  $\mathcal{K}$ ; if  $\mathbf{v}_1, \dots, \mathbf{v}_e$  is a real basis of  $\mathcal{K}$ , the fiber above  $\mathbf{a}$  is  $\mathbf{a} + \{\sum_k t_k \mathbf{v}_k : t_k \in \Pi\}$ . If  $\mathcal{K} = 0$ , all omnific solutions are ordinary.

(b) If the linear parts  $\ell_1, \dots, \ell_s$  have column rank  $n$  over  $\mathbb{C}$ , then every solution  $\mathbf{x} \in \mathcal{B}_{\mathbb{C}}^n$  of  $F(\mathbf{x}) = c$  lies in  $\mathbb{C}^n$ .

*Proof.* For  $\mathbf{x} \in \mathcal{B}_{\mathbb{C}}^n$ , every  $L_j(\mathbf{x})$  is in  $\mathcal{B}_{\mathbb{C}}$ , and if  $F(\mathbf{x}) = c \neq 0$ , [lemma 6.1](#) makes each  $L_j(\mathbf{x})$  constant.

(a) Write  $\mathbf{x} = \mathbf{a} + \mathbf{u}$  coordinatewise with  $\mathbf{a} \in \mathbb{Z}^n$  and  $\mathbf{u} \in \Pi^n$ . Then  $L_j(\mathbf{x}) = L_j(\mathbf{a}) + \ell_j(\mathbf{u})$ , where the first summand is constant and the second lies in  $\Pi_{\mathbb{C}}$ ; uniqueness of normal forms forces  $\ell_j(\mathbf{u}) = 0$ , and specialization gives  $F(\mathbf{a}) = c$ . Conversely those conditions give  $L_j(\mathbf{a} + \mathbf{u}) = L_j(\mathbf{a})$  and hence  $F(\mathbf{a} + \mathbf{u}) = c$ . At each exponent of the (set-sized, reverse-well-ordered) union of the coordinate supports, the equations  $\ell_j(\mathbf{u}) = 0$  say exactly that the real coefficient vector  $\mathbf{u}_{\gamma} \in \mathbb{R}^n$  at that exponent satisfies  $\ell_j(\mathbf{u}_{\gamma}) = 0$  for all  $j$ , that is, lies in  $\mathcal{K}$ . Expressing each coefficient vector in the basis  $\mathbf{v}_k$  produces  $t_k$  whose supports lie in the same union; if  $\mathcal{K} = 0$ , every coefficient vector vanishes.

(b) Select  $n$  of the  $\ell_j$  whose coefficient matrix is invertible over  $\mathbb{C}$ , subtract the constant terms and solve this ordinary linear system: all coordinates are complex constants.  $\square$

The kernel hypothesis in (a) is exactly what matters. A single-factor form can have infinite fibers,

$$(x - y)^2 = 1 \iff x = y \pm 1, \quad y \in \mathbf{Oz},$$

and the two-variable equation  $X^d = 1$  forces  $X$  to be constant but says nothing about  $Y$ . The real kernel in (a) need not equal the complex kernel in (b):  $L(X, Y) = X + iY$  has zero kernel on  $\mathbb{R}^2$ , so its level 1 has the unique omnific point  $(1, 0)$ . In  $\mathcal{B}_{\mathbb{C}}^2$  the same level contains  $(1 - it, t)$  for every  $t \in \mathcal{B}_{\mathbb{C}}$  (compare [remark 6.5](#)). On the zero level, even independent factors permit infinite points, since the unit argument cannot be applied to a product equal to zero:  $(\sqrt{2}\omega)^2 - 2\omega^2 = 0$ .

**Gaussian variables, the converse, and the kernel criterion.** Manuscript 13 of the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` (batch-25 item 03, written [13] here; see [section 1.2](#)) proves the same fiber theorem, with repeated factors in place of the exponents  $e_j$  (an equivalent formulation), for the pairs  $(\mathbb{Z}, \mathbb{R})$  and  $(\mathbb{Z}[i], \mathbb{C})$ . Its real case is [theorem 6.2\(a\)](#), printed once above. Its Gaussian case, its converse and its remark on real kernels are printed here, once; source 13's other results are printed in the sibling report. Write

$$\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}} = \{a + ib : a, b \in \mathbf{Oz}\}$$

for the *Gaussian omnific integers*. A finite union of reverse-well-ordered supports is reverse well ordered, so the two descriptions agree, and the constant coefficient of  $a + ib$  is  $\text{ct}(a) + i \text{ct}(b) \in \mathbb{Z}[i]$ . The name refers to this explicitly specified ring, not to another integer part of the surcomplex field [07].

**Theorem 6.3** (Exact fibers over the Gaussian omnific integers [13]). *Let  $F = \prod_j L_j^{e_j}$ ,  $L_j = \beta_j + \ell_j$  and  $c \in \mathbb{C}^{\times}$  be as in [theorem 6.2](#), and put*

$$\mathcal{K}_{\mathbb{C}} = \{\mathbf{v} \in \mathbb{C}^n : \ell_j(\mathbf{v}) = 0 \text{ for every } j\}.$$

The solutions of  $F(\mathbf{x}) = c$  in  $\mathbf{Oz}[i]^n$  are exactly

$$\mathbf{x} = \mathbf{a} + \mathbf{u}, \quad \mathbf{a} \in \mathbb{Z}[i]^n, \quad F(\mathbf{a}) = c, \quad \mathbf{u} \in \Pi_{\mathbb{C}}^n, \quad \ell_j(\mathbf{u}) = 0 \text{ for all } j. \quad (6.2)$$

Equivalently, at each exponent the complex coefficient vector of  $\mathbf{u}$  lies in  $\mathcal{K}_{\mathbb{C}}$ ; if  $\mathbf{v}_1, \dots, \mathbf{v}_e$  is a complex basis of  $\mathcal{K}_{\mathbb{C}}$ , the fiber above  $\mathbf{a}$  is  $\mathbf{a} + \{\sum_k t_k \mathbf{v}_k : t_k \in \Pi_{\mathbb{C}}\}$ . Fibers above distinct  $\mathbf{a}$  are disjoint. If  $\mathcal{K}_{\mathbb{C}} = 0$ , that is, if the linear parts have rank  $n$  over  $\mathbb{C}$ , every solution in  $\mathbf{Oz}[i]^n$  lies in  $\mathbb{Z}[i]^n$ .

*Proof.* Write  $\mathbf{x} = \mathbf{a} + \mathbf{u}$  with  $\mathbf{a} \in \mathbb{Z}[i]^n$  and  $\mathbf{u} \in \Pi_{\mathbb{C}}^n$ . If  $F(\mathbf{x}) = c$ , lemma 6.1 makes every  $L_j(\mathbf{x}) = L_j(\mathbf{a}) + \ell_j(\mathbf{u})$  a complex constant; since  $\ell_j(\mathbf{u}) \in \Pi_{\mathbb{C}}$ , uniqueness of normal forms gives  $\ell_j(\mathbf{u}) = 0$ , and applying ct gives  $F(\mathbf{a}) = c$ . Conversely these conditions give  $L_j(\mathbf{x}) = L_j(\mathbf{a})$  for every  $j$ , hence  $F(\mathbf{x}) = c$ . The coefficient of  $\omega^\gamma$  in  $\ell_j(\mathbf{u})$  is  $\ell_j(\mathbf{u}_\gamma)$ , where  $\mathbf{u}_\gamma \in \mathbb{C}^n$  is the coefficient vector of  $\mathbf{u}$  at  $\gamma$ ; so  $\ell_j(\mathbf{u}) = 0$  for all  $j$  says exactly that every  $\mathbf{u}_\gamma$  lies in  $\mathcal{K}_{\mathbb{C}}$ , and expanding each  $\mathbf{u}_\gamma$  in the basis gives series  $t_k$  supported in the (set-sized, reverse-well-ordered) union of the coordinate supports. More explicitly, the coordinate functionals of the basis are complex linear maps on  $\mathcal{K}_{\mathbb{C}}$ ; apply each to every coefficient vector to form  $t_k$ . Taking a subset of the common support preserves its admissibility. Conversely, arbitrary  $t_k \in \Pi_{\mathbb{C}}$  give a vector with coefficients in  $\mathcal{K}_{\mathbb{C}}$ , since only finitely many basis vectors are used. Independence of the  $\mathbf{v}_k$  makes these parameters unique coefficient by coefficient. This argument uses finite linear algebra and Hahn supports, not convergence of a sequence of partial sums. Disjointness holds because the constant coefficient of a normal form is unique. If  $\mathcal{K}_{\mathbb{C}} = 0$ , then  $\mathbf{u} = 0$ .  $\square$

**Corollary 6.4** (The kernel criterion is exact [13]). *Suppose the ordinary fiber  $\{\mathbf{a} \in \mathbb{Z}^n : F(\mathbf{a}) = c\}$  is nonempty. Then  $F(\mathbf{x}) = c$  has a solution in  $\mathbf{Oz}^n$  outside  $\mathbb{Z}^n$  if and only if the real kernel  $\mathcal{K}$  of theorem 6.2 is nonzero. Likewise, if  $F(\mathbf{a}) = c$  for some  $\mathbf{a} \in \mathbb{Z}[i]^n$ , it has a solution in  $\mathbf{Oz}[i]^n$  outside  $\mathbb{Z}[i]^n$  if and only if  $\mathcal{K}_{\mathbb{C}} \neq 0$ .*

*Proof.* If  $\mathbf{v} \neq 0$  lies in the relevant kernel and  $\mathbf{a}$  is an ordinary point, then  $\mathbf{a} + \mathbf{v}\omega$  is a point of the same fiber with an infinite coordinate. The other implications are theorem 6.2(a) and theorem 6.3.  $\square$

So, subject to one ordinary point, the linear-kernel criterion is not merely sufficient: it detects exactly whether a nonzero constant fiber acquires infinite directions [13]. The nonemptiness hypothesis cannot be omitted:  $F(X, Y) = X - Y$  and  $c = \sqrt{2}$  have kernel  $\mathbb{R}(1, 1)$ , but no omnific solution, because taking constants would give an integer difference equal to  $\sqrt{2}$ . In the Gaussian case the same example works because  $\sqrt{2} \notin \mathbb{Z}[i]$ . When an ordinary point  $\mathbf{a}$  and a nonzero kernel vector  $\mathbf{v}$  do exist, the map  $t \mapsto \mathbf{a} + t\mathbf{v}$  is injective on  $\Pi$  (and on  $\Pi_{\mathbb{C}}$  in the Gaussian case): a nonzero coordinate of  $\mathbf{v}$  allows cancellation. Thus every such fiber is a proper class, not merely an infinite set [merge].

*Remark 6.5* (Real and complex kernels [13]). For omnific variables the relevant kernel is  $\mathcal{K} = \ker_{\mathbb{C}} \cap \mathbb{R}^n$ , which can vanish although the complex rank of the linear parts is smaller than  $n$ ; the rank condition of theorem 6.2(b) is sufficient, not necessary, for real variables. Source 13's example is the single form  $X + iY$ , which is injective on  $\mathbb{R}^2$ : every omnific solution of  $x + iy = c$  is ordinary. For Gaussian variables the criterion is the complex kernel. For instance [merge],  $\mathcal{K}_{\mathbb{C}} = \mathbb{C}(1, i)$  for  $X + iY$ , and  $(1 + \omega, i\omega) \in \mathbf{Oz}[i]^2$  solves  $X + iY = 1$ , since  $(1 + \omega) + i \cdot i\omega = 1$ . For an explicit real rank test, write  $M$  for the complex matrix of the  $\ell_j$ . Then

$$\mathcal{K} = \ker_{\mathbb{R}} \begin{pmatrix} \operatorname{Re} M \\ \operatorname{Im} M \end{pmatrix} :$$

for a real vector,  $M\mathbf{v} = 0$  exactly when both its real and imaginary parts vanish. Full column rank of this real matrix is the necessary and sufficient kernel condition for real variables, whereas Gaussian variables use the rank of  $M$  over  $\mathbb{C}$ .

**Example 6.6** (A rank-deficient level [13]). Let  $F(X, Y, Z) = (X + Z + \sqrt{2}Y)(X + Z - \sqrt{2}Y) = (X + Z)^2 - 2Y^2$ . The common real kernel of the two linear parts is  $\mathcal{K} = \mathbb{R}(1, 0, -1)$ , so by [theorem 6.2\(a\)](#) the omnific solutions of  $F = 1$  are exactly

$$(x, y, z) = (a + u, b, c - u), \quad u \in \Pi, \quad a, b, c \in \mathbb{Z}, \quad (a + c)^2 - 2b^2 = 1.$$

For instance  $(3 - \omega, 2, \omega)$  is a nonordinary solution. It is the common kernel of *all* linear parts, not the nonzero level alone, that governs rigidity.

## 6.2 Binary forms, Pell equations, and conics

**Theorem 6.7** (Binary-form rigidity [05], generalizing [01, 02]). *Let  $F \in \mathbb{C}[X, Y]$  be homogeneous with at least two distinct projective linear factors over  $\mathbb{C}$ , and let  $c \in \mathbb{C}^\times$ . If  $x, y \in \mathcal{B}_{\mathbb{C}}$  satisfy  $F(x, y) = c$ , then  $x, y \in \mathbb{C}$ . In particular, for  $F \in \mathbb{Z}[X, Y]$  and  $c \in \mathbb{Z} \setminus \{0\}$ ,*

$$\{(x, y) \in \mathbf{Oz}^2 : F(x, y) = c\} = \{(x, y) \in \mathbb{Z}^2 : F(x, y) = c\}.$$

*Proof.* Factor  $F = \alpha \prod_{j=1}^d L_j$  over  $\mathbb{C}$  with  $\alpha \in \mathbb{C}^\times$  and nonzero linear forms repeated according to multiplicity. Two distinct projective factors have linearly independent coefficient vectors, so the linear parts have rank two; apply [theorem 6.2\(b\)](#). Constant omnific coordinates are integers.  $\square$

Sources 01 and 02 proved the statement for  $F \in \mathbb{Z}[X, Y]$ ,  $c \in \mathbb{Z}$  and  $\mathbf{Oz}$ -points; the complex form is 05's. The hypothesis excludes a pure power  $\alpha L^d$ , and that exclusion is necessary: fixing  $L(x, y)$  to a constant leaves a free infinite direction in the kernel of  $L$ . The requirement  $c \neq 0$  is also necessary, since zero levels contain lines through the origin. The hypothesis is weaker than squarefreeness in degree at least three:  $X^2Y$  qualifies.

**Corollary 6.8** (Thue levels and equal powers [01, 02, 05]). *If an integer binary form has at least two distinct projective factors, its nonzero ordinary constant levels have no additional omnific points. This applies to every irreducible binary form over  $\mathbb{Q}$  of degree at least two, hence to Thue equations, and to  $x^d - y^d = c$  with  $d \geq 2$  and  $c \in \mathbb{Z} \setminus \{0\}$ .*

*Proof.* In characteristic zero an irreducible binary form of degree at least two has distinct projective roots over  $\mathbb{C}$ , and  $X^d - Y^d$  has distinct linear factors. Apply [theorem 6.7](#).  $\square$

The conclusion identifies the omnific solution class with the classical integer solution set; it is not a new proof of ordinary finiteness, gives no bound for the ordinary solutions, and supplies no algorithm for them. No finiteness theorem of Thue or Baker is used.

**Corollary 6.9** (Pell rigidity [01, 02, 05]). *For ordinary integers  $D \neq 0$  and  $c \neq 0$ , every solution in  $\mathbf{Oz}^2$  of  $x^2 - Dy^2 = c$  is an ordinary integer solution. In particular*

$$\{(u, v) \in \mathbf{Oz}^2 : u^2 - 2v^2 = 1\} = \{(u, v) \in \mathbb{Z}^2 : u^2 - 2v^2 = 1\}.$$

*Proof.* Over  $\mathbb{C}$ , the factors  $X - \sqrt{D}Y$  and  $X + \sqrt{D}Y$  are distinct because  $D \neq 0$ . Apply [theorem 6.7](#).  $\square$

Source 02 states this corollary for  $D > 0$  only; its proof, like those of 01 and 05, uses only  $D \neq 0$ , and the conclusion holds whether or not  $D$  is a square. Source 13 states the corollary for positive nonsquare  $D$  (as  $X^2 - dY^2 = m$  with  $m \neq 0$ ); its own fiber theorem, like [theorem 6.2](#), covers every  $D \neq 0$ , and the statement for  $D \neq 0$  stands here. “Rigid” means no new points, not a finite number of old points: the classical Pell equation has ordinary solutions of arbitrarily large finite size, but no single infinite omnific solution, and there is no “Pell solution at index  $\omega$ ” obtained by substituting an omnific index into a finite recurrence. This is a first warning against treating  $\mathbf{Oz}$  as an elementary extension of ordinary integer arithmetic.

**Example 6.10** (The full class for  $D = 2$ ,  $c = 1$  [05]). Let  $X_k + Y_k\sqrt{2} = (3 + 2\sqrt{2})^k$  for ordinary  $k \geq 0$ . Every omnific solution of  $x^2 - 2y^2 = 1$  is  $(\pm X_k, \pm Y_k)$  with independent signs. Indeed, [corollary 6.9](#) first makes  $x, y$  ordinary. If  $y = 0$ , then  $x = \pm 1$ , giving  $k = 0$ . Otherwise, after changing signs suppose  $x, y > 0$ ; then  $y \geq 2$ , since  $y = 1$  would need  $x^2 = 3$ . Multiplication by  $3 - 2\sqrt{2}$  gives the integer Pell pair  $X = 3x - 4y$ ,  $Y = 3y - 2x$  with  $X > 0$  (as  $x > \sqrt{2}y > 4y/3$ ) and  $0 \leq Y < y$  (as  $x^2 = 2y^2 + 1 \leq 9y^2/4$  and  $x > y$ ). Ordinary descent on  $y$  reaches  $(1, 0)$ , and reversing it gives the asserted powers.

In the full surreal field the contrast is visible without solving a recurrence [01; also 07, 13]. For  $D > 0$ ,  $c \neq 0$  and  $s \in \mathbf{No}^\times$ , the pair  $x = (s + c/s)/2$ ,  $y = (s - c/s)/(2\sqrt{D})$  satisfies  $x^2 - Dy^2 = c$ . Taking  $s = \omega$  produces negative-exponent terms, so neither coordinate is an omnific integer, and discarding those terms changes the norm from  $c$  to zero. A surreal solution cannot be turned into an omnific solution by unverified truncation. Source 07 makes the same point for any nonconstant monomial of a Hahn field, and source 13 notes that the negative-exponent terms supply exactly the inverses missing from  $\mathcal{B}_{\mathbb{R}}$ : the rigidity theorems concern the nonnegative-support rings, not non-Archimedean fields in general.

**Proposition 6.11** (The zero Pell fiber [01; also 13]). *If  $D$  is a positive nonsquare ordinary integer, the solutions of  $x^2 - Dy^2 = 0$  in  $\mathbf{Oz}^2$  are exactly  $(\pm\sqrt{D}t, t)$  with  $t \in \Pi$ .*

*Proof.* Factor in  $\mathbf{No}$  to obtain  $x = \pm\sqrt{D}y$ . Write  $y = m + j$  with  $m \in \mathbb{Z}$  and  $j \in \Pi$ . The constant coefficient of  $x$  is  $\pm\sqrt{D}m$ , which is an integer only if  $m = 0$ . Conversely,  $\Pi$  is an  $\mathbb{R}$ -vector space, so all displayed pairs belong to  $\mathbf{Oz}^2$  and solve the equation.  $\square$

**Corollary 6.12** (Nonsingular central conics [05]). *Let  $F(X, Y) = q(X, Y) + \ell(X, Y) + d \in \mathbb{R}[X, Y]$ , where the quadratic form  $q$  has rank two, and let  $\mathbf{h} \in \mathbb{R}^2$  be the center, so that  $F(\mathbf{h} + \mathbf{u}) = q(\mathbf{u}) + F(\mathbf{h})$ . If  $F(\mathbf{h}) \neq 0$ , every solution of  $F = 0$  in  $\mathcal{B}_{\mathbb{R}}^2$  is constant; for integer coefficients, every omnific solution is ordinary.*

*Proof.* A rank-two binary quadratic form has two distinct projective factors over  $\mathbb{C}$ . The translated coordinates  $\mathbf{u} = (x, y) - \mathbf{h}$  lie in  $\mathcal{B}_{\mathbb{R}}^2$  and satisfy  $q(\mathbf{u}) = -F(\mathbf{h}) \neq 0$ . Apply [theorem 6.7](#) and translate back.  $\square$

This covers ellipses and hyperbolas with a nonzero centered level. Parabolas behave differently:  $Y = X^2$  has the infinite family  $(t, t^2)$ .

**Remark 6.13** (Power differences with a common factor [01]). For positive ordinary integers  $m, n$  with  $d = \gcd(m, n) > 1$  and  $c \in \mathbb{Z} \setminus \{0\}$ , every omnific solution of  $x^m - y^n = c$  is ordinary. Source 01’s argument applies [theorem 6.7](#) to  $X^d - Y^d$  with  $X = x^{m/d}$ ,  $Y = y^{n/d}$ . Both quotient exponents are positive. An element of  $\mathbf{Oz}$  whose positive power is an ordinary constant is ordinary: degree additivity proves this when that constant is nonzero, and the domain property proves it when it is zero. Positivity cannot be replaced by the gcd condition alone:  $\gcd(0, 2) = 2$ , but  $\omega^0 - 0^2 = 1$  has an infinite coordinate [merge]. Source 01 stated explicitly that this argument does not classify coprime cases such as  $x^2 - y^3 = 1$ . [Theorem 7.4](#)

of source 05 covers all  $m, n \geq 2$ , including coprime pairs; it does not include exponent one, where  $y = x^m - c$  supplies nonordinary solutions of  $x^m - y = c$ .

### 6.3 Number-field and étale norm equations

Here a norm form is first constructed as a polynomial over  $\mathbb{Q}$ , then evaluated on omnific coordinates. We do not embed the number field  $K$  into  $\mathbf{No}$ ; such an embedding need not exist. Equivalently, form the determinant of the multiplication matrix in a  $\mathbb{Q}$ -basis, with indeterminate coordinates. Its complex factorization is used only after this polynomial is formed. The same applies to the algebra  $E$  of [theorem 6.15](#), whose norm is factored over an algebraic closure of the coefficient field.

**Theorem 6.14** (Norm-form rigidity [01, 02]). *Let  $K/\mathbb{Q}$  be a number field of degree  $d$ , let  $e_1, \dots, e_d$  be a  $\mathbb{Q}$ -basis, and let  $N(\mathbf{X}) = N_{K/\mathbb{Q}}(\sum_{j=1}^d e_j X_j)$ . For every  $c \in \mathbb{Q}^\times$ , every solution  $\mathbf{x} \in \mathbf{Oz}^d$  of  $N(\mathbf{x}) = c$  has all coordinates in  $\mathbb{Z}$ . If the basis consists of algebraic integers (for example a power basis of an algebraic integer  $\theta$  generating  $K$ , or an integral basis), then  $N \in \mathbb{Z}[\mathbf{X}]$ , and the statement applies to the usual integer norm equations.*

*Proof.* The  $d$  embeddings  $\sigma_1, \dots, \sigma_d : K \rightarrow \mathbb{C}$  give

$$N(\mathbf{X}) = \prod_{i=1}^d \left( \sum_{j=1}^d \sigma_i(e_j) X_j \right).$$

The embedding matrix  $(\sigma_i(e_j))$  is invertible: for a power basis  $1, \theta, \dots, \theta^{d-1}$  of a primitive element of the separable extension  $K/\mathbb{Q}$  it is the Vandermonde matrix of the distinct conjugates of  $\theta$ , and a basis change over  $\mathbb{Q}$  preserves invertibility. The forms therefore have zero common kernel and [theorem 6.2](#) applies. If the  $e_j$  are algebraic integers, each coefficient of  $N$  is fixed by the Galois group of a normal closure, hence rational, and is an integer polynomial in conjugates of the  $e_j$ , hence an algebraic integer; so it is an ordinary integer [02].  $\square$

For example, consider the norm form for  $K = \mathbb{Q}(\sqrt[3]{2})$ :

$$N(X + Y\sqrt[3]{2} + Z\sqrt[3]{4}) = X^3 + 2Y^3 + 4Z^3 - 6XYZ.$$

For every  $c \in \mathbb{Z} \setminus \{0\}$ , every omnific solution of  $X^3 + 2Y^3 + 4Z^3 - 6XYZ = c$  is an ordinary integer triple. This is a three-variable rigidity statement on an unbounded real surface, unlike the indefinite quadratic forms of [section 8](#): dimension alone does not force infinite omnific points. The identity follows from the product over the three conjugates or from the determinant of multiplication by  $X + Y\theta + Z\theta^2$ ; the suites of 01 and 02 check it exactly. Source 13 computes the same determinant (its suite checks it too) and records the Gaussian analogue: every Gaussian omnific solution is a Gaussian integer triple, by [theorem 6.15\(b\)](#) below. No claim is made to enumerate or effectively bound the ordinary solutions of this cubic.

Sources 07 and 13 prove norm rigidity in two further directions: source 07 over an arbitrary base field and in intermediate rings, source 13 for finite étale algebras (finite products of number fields) and with Gaussian omnific variables. The mechanism is that of [theorem 6.14](#); the two generalizations are combined here. The statement uses the general nonnegative-support rings  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  and the intermediate rings  $A$  of [section 11.1](#): for a unital subring  $\mathfrak{o}$  of a field  $\mathbb{k}$  of characteristic zero,  $A$  is a ring with  $\mathfrak{o} \subseteq A \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$  and  $A \cap \mathbb{k} = \mathfrak{o}$ . Examples are  $\mathbf{Oz}$  (with  $\mathfrak{o} = \mathbb{Z}$ ,  $\mathbb{k} = \mathbb{R}$ ) and  $\mathbf{Oz}[i]$  (with  $\mathfrak{o} = \mathbb{Z}[i]$ ,  $\mathbb{k} = \mathbb{C}$ ).

**Theorem 6.15** (Étale norm rigidity in intermediate rings [07, 13] [merge]). *Let  $\mathbb{k}$  be a field of characteristic zero,  $K \subseteq \mathbb{k}$  a subfield,  $E$  a finite étale  $K$ -algebra of dimension  $r$ , that is, a finite product of finite separable field extensions of  $K$ , and  $e_1, \dots, e_r$  a  $K$ -basis of  $E$ . Put  $N(\mathbf{X}) = N_{E/K}(\sum_{j=1}^r e_j X_j) \in K[X_1, \dots, X_r]$ . Let  $\Gamma$  be an ordered abelian group and  $A$  an intermediate ring for  $\mathfrak{o} \subseteq \mathbb{k}$ . If  $\mathbf{x} \in A^r$  and  $N(\mathbf{x}) = a \in K^\times$ , then every  $x_j$  lies in  $\mathfrak{o}$ . In particular:*

- (a) *for a finite étale  $\mathbb{Q}$ -algebra, every solution in  $\mathbf{Oz}^r$  of  $N(\mathbf{x}) = a \neq 0$  lies in  $\mathbb{Z}^r$ ;*
- (b) *for a finite étale  $\mathbb{Q}$ - or  $\mathbb{Q}(i)$ -algebra, every solution in  $\mathbf{Oz}[i]^r$  of  $N(\mathbf{x}) = a \neq 0$  lies in  $\mathbb{Z}[i]^r$ .*

*Proof.* Let  $\bar{\mathbb{k}}$  be an algebraic closure of  $\mathbb{k}$ . Write  $E = \prod_{h=1}^s L_h$  as a product of finite separable extensions of  $K$  (the characterization of étale algebras over a field is [52, Lemma 10.143.4]). Choose a primitive element for each  $L_h$ . Its minimal polynomial splits over  $\bar{\mathbb{k}}$  into distinct linear factors, so the polynomial Chinese remainder theorem gives

$$L_h \otimes_K \bar{\mathbb{k}} \cong \prod_{\sigma: L_h \hookrightarrow \bar{\mathbb{k}}} \bar{\mathbb{k}}, \quad E \otimes_K \bar{\mathbb{k}} \cong \bar{\mathbb{k}}^r.$$

Each unital  $K$ -algebra map from  $E$  to a field selects one factor  $L_h$ : exactly one of its orthogonal idempotents maps to 1, and the others map to 0. Thus there are exactly  $r$  such maps  $\sigma_\ell$ , and they give the displayed isomorphism. In the basis  $e_j \otimes 1$ , its matrix is  $M = (\sigma_\ell(e_j))$ , so  $M$  is invertible. These maps need not be injective on the product algebra  $E$ ; only their combined map is injective. In the split coordinates, multiplication by  $\sum_j e_j X_j$  is diagonal. The determinant is unchanged by this constant change of basis, giving the polynomial identity

$$N(\mathbf{X}) = \prod_{\ell=1}^r \left( \sum_{j=1}^r \sigma_\ell(e_j) X_j \right) \quad \text{in } \bar{\mathbb{k}}[X_1, \dots, X_r].$$

Equivalently, the trace matrix is  $M^\top M$ , because its  $(i, j)$  entry is  $\sum_\ell \sigma_\ell(e_i) \sigma_\ell(e_j)$ ; hence its determinant is  $(\det M)^2 \neq 0$ . This is a bilinear trace pairing, with transpose, not conjugate transpose. Source 13 argues with power bases: for a field the matrix is a Vandermonde matrix of distinct conjugates, for a product it is block diagonal, and a change of basis over  $K$  preserves invertibility. Coefficient extension embeds  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  in  $\mathcal{A}_{\bar{\mathbb{k}}}(\Gamma)$ : injectivity of the coefficient-field map preserves every nonzero coefficient and thus every support, and finite coefficient convolutions commute with this map. Each factor  $\sum_j \sigma_\ell(e_j) x_j$  lies in  $\mathcal{A}_{\bar{\mathbb{k}}}(\Gamma)$ , and their product is the nonzero constant  $a$ , so every factor is constant by lemma 11.1. For each positive exponent  $\gamma$ , their coefficient vector is  $M \mathbf{x}_\gamma = 0$ . Invertibility gives  $\mathbf{x}_\gamma = 0$ , so no  $x_j$  has positive support. Its remaining coefficient was already in  $\mathbb{k}$ , so  $x_j \in \mathbb{k}$ , hence in  $A \cap \mathbb{k} = \mathfrak{o}$ . For (a) and (b) note that  $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$  and  $\mathbf{Oz}[i] \cap \mathbb{C} = \mathbb{Z}[i]$ . No closure of  $A$  under constant extraction is needed: the argument first makes the entire coordinate constant and only then uses  $A \cap \mathbb{k}$ .  $\square$

Theorem 6.14 is the case  $K = \mathbb{Q}$ ,  $E$  a number field,  $A = \mathbf{Oz}$ . Source 07 proved the theorem for a finite separable field extension of an arbitrary subfield  $K \subseteq \mathbb{k}$  and intermediate rings; source 13 for finite étale  $\mathbb{Q}$ -algebras with  $\mathfrak{o} = \mathbb{Z}$  and  $\mathfrak{o} = \mathbb{Z}[i]$ , and noted that the same proof applies to étale  $\mathbb{Q}(i)$ -algebras with Gaussian omnific variables. The combined statement, with its proof above, is the merge's; each ingredient is in one of the two sources. Two cautions from source 07 apply. When some coefficients of  $N$  in  $K$  do not lie in  $A$ , the equation is read in the ambient series ring; to use it literally as a formula with parameters in  $\mathfrak{o}$ , one assumes that its coefficients and right-hand side lie in  $\mathfrak{o}$ . The nonzero right-hand side and the étale

hypothesis are essential to this uniform assertion; this is not a theorem that every algebraic variety has only constant points. The norm on the left is evaluation of a finite polynomial, not a newly postulated norm on a class-sized algebra [13]. Pell rigidity in intermediate rings (corollary 11.5) is the case  $E = \mathbb{Q}(\sqrt{2})$ .

**Boundary examples and characteristic.** The restrictions can be seen in small polynomial rings [merge]. For the étale algebra  $E = \mathbb{Q} \times \mathbb{Q}$  with its product basis, the norm is  $N(X, Y) = XY$ ; its zero level contains  $(0, t)$  for every  $t \in \Pi$ . If étaleness is dropped, take the dual-number algebra  $E = \mathbb{Q}[\epsilon]/(\epsilon^2)$  with basis  $(1, \epsilon)$ . Multiplication by  $X + Y\epsilon$  has matrix  $\begin{pmatrix} X & 0 \\ Y & X \end{pmatrix}$  and norm  $X^2$ . Thus the nonzero level  $N = 1$  contains  $(1, t)$  for every  $t \in \Pi$ . A nonzero determinant makes this algebra element a unit; it does not force its nilpotent coordinate to be constant. Source C17 recovers theorem 6.15 in characteristic zero as the constant case of its unit theorem for finite étale algebras over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , and its finite free reduced algebra is a second way the étale hypothesis can fail (remark 19.28 and example 19.45).

Characteristic zero, in contrast, is unnecessary for the abstract étale norm argument [merge]. For any field  $\mathbb{k}$ , form the same reverse-well-ordered nonnegative-support Hahn ring and take  $\mathfrak{o} \subseteq A \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$  with  $A \cap \mathbb{k} = \mathfrak{o}$ . The conclusion of theorem 6.15 still holds for finite étale  $E/K$ ,  $K \subseteq \mathbb{k}$ . The splitting argument uses separability only. The units argument also works in any characteristic: in a product of nonzero series, the coefficient at the sum of the largest exponents is the product of the nonzero leading coefficients. Nonnegative degrees whose sum is zero all vanish. The remaining matrix argument is unchanged. This is a statement about these abstract Hahn rings; it does not replace the real or complex coefficient fields of the actual omnific rings.

Separability cannot be removed from this extension. For instance let  $K = \mathbb{F}_2(s)$  with  $s$  transcendental, let  $E = K(\theta)$  with  $\theta^2 = s$ , and use  $\mathbb{k} = E$  as the coefficient field. The polynomial  $X^2 - s$  is irreducible over  $K$  (the order at  $s = 0$  of a rational square is even), so  $E/K$  has degree two and is purely inseparable. Multiplication in the basis  $(1, \theta)$  has matrix  $\begin{pmatrix} X & sY \\ Y & X \end{pmatrix}$ , so its norm is  $N(X, Y) = X^2 - sY^2 = X^2 + sY^2$ . In  $A = \mathbb{k}[T] \cong \mathcal{A}_{\mathbb{k}}(\mathbb{Z})$ , the nonconstant pair  $(1 + \theta T, T)$  has norm 1, since  $2 = 0$  and  $\theta^2 = s$ . Here  $\mathfrak{o} = \mathbb{k}$ , so the claimed constant-coordinate conclusion would fail. The example locates the obstruction in the failure of separable splitting, even though  $E$  itself is a field.

## 6.4 An algebraic-geometric formulation

Let  $V$  be an affine complex variety whose coordinate ring  $\mathbb{C}[V]$  is generated, as a  $\mathbb{C}$ -algebra, by units. Every  $\mathcal{B}_{\mathbb{C}}$ -point of  $V$  is then a constant  $\mathbb{C}$ -point [01, 02]. Indeed, a  $\mathbb{C}$ -algebra map  $\mathbb{C}[V] \rightarrow \mathcal{B}_{\mathbb{C}}$  sends each generating unit to an element of  $\mathcal{B}_{\mathbb{C}}^{\times} = \mathbb{C}^{\times}$ , so its entire image lies in  $\mathbb{C}$ . This includes closed subvarieties of an algebraic torus  $(\mathbb{C}^{\times})^r$  in their very-affine coordinate presentation. The norm argument is a concrete linear-coordinate instance: a nonzero product relation forces its factors into a constant torus. It is not a claim that every affine variety is constant over  $\mathcal{B}_{\mathbb{C}}$ ; the affine line has the full class  $\mathcal{B}_{\mathbb{C}}$  of points. For smooth geometrically integral affine curves the affine line is the only exception: the unit argument gives the genus-zero case of the classification of theorem 17.6 (lemma 17.4), and curves of positive genus are rigid by a differential argument (theorem 17.2).

## 7 Euler derivations: separated powers and unimodular Fermat

Binary forms can be factored into linear forms, but equations such as  $Y^2 = X^3 + 17$  cannot be handled that way. The following local tool, due to source 05, gives short arguments without

any finite-support hypothesis.

## 7.1 Support-preserving Euler derivations

Fix a set-sized ordered divisible group  $H \subseteq \mathbf{No}$  and a  $\mathbb{Q}$ -linear map  $\lambda : H \rightarrow \mathbb{R}$ . On the real or complex Hahn field with exponents in  $H$ , define

$$\partial_\lambda \left( \sum_{h \in H} a_h \omega^h \right) = \sum_{h \in H} \lambda(h) a_h \omega^h. \quad (7.1)$$

This is a local, freely chosen Euler derivation. It is not a canonical surreal derivation and not the Berarducci–Mantova derivation. Under  $t = \omega^{-1}$  its rule is  $\partial_\lambda(t^\gamma) = -\lambda(\gamma)t^\gamma$ ; the minus sign comes from reversing exponents. It must not be identified with the differential equations report's  $\partial_\tau = -\omega\partial$ , whose real-exponent rule is  $\partial_\tau(t^\gamma) = \gamma t^\gamma$ .

**Lemma 7.1** (Euler derivations [05]). *The map  $\partial_\lambda$  is a derivation, kills every real or complex constant, and never introduces a new supported exponent; so  $\deg_\omega(\partial_\lambda f) \leq \deg_\omega(f)$  whenever  $\partial_\lambda f \neq 0$ . It restricts to derivations of  $\mathcal{B}_{\mathbb{R},H}$  and  $\mathcal{B}_{\mathbb{C},H}$ .*

*Proof.* The new support is a subset of the old one and is thus reverse well ordered. Additivity is immediate, and  $\lambda(0) = 0$  kills constants. For a fixed exponent  $h$ , the coefficient convolution is finite and  $\lambda(a+b) = \lambda(a) + \lambda(b)$ , so

$$[\omega^h] \partial_\lambda(fg) = \sum_{a+b=h} (\lambda(a) + \lambda(b)) f_a g_b = [\omega^h] ((\partial_\lambda f)g + f(\partial_\lambda g)).$$

This proves the Leibniz rule coefficientwise; no interchange of conditionally convergent sums is involved. The degree and restriction statements follow from support containment.  $\square$

**Lemma 7.2** (Detection of nonconstants [05]). *Every nonconstant element of a Hahn field with exponent group  $H$  is detected by some  $\partial_\lambda$ . If  $f \in \mathcal{B}_{\mathbb{C},H}$  has positive leading exponent  $\alpha$ , one may choose  $\lambda(\alpha) = 1$ ; then  $\deg_\omega(\partial_\lambda f) = \alpha$ .*

*Proof.* Choose a nonzero supported exponent  $h$ . A  $\mathbb{Q}$ -linear functional on  $\mathbb{Q}h$  taking  $h$  to 1 extends to  $H$  by extending a rational basis;  $H$  is a set, so choice for sets suffices. The derivative has nonzero coefficient at  $h$ . Choosing the leading exponent gives the final assertion.  $\square$

For  $\partial = \partial_\lambda$  define the two-term Wronskian  $W_\partial(f, g) = f \partial g - g \partial f$ . For  $f \neq 0$  it vanishes exactly when  $\partial(g/f) = 0$ ; it need not detect whether  $g/f$  is an absolute constant for one fixed  $\partial$ , which is why lemma 7.2 lets the derivation be chosen for the given ratio. For example, on  $H = \mathbb{Q} + \mathbb{Q}\sqrt{2}$  the functional  $\lambda(a + b\sqrt{2}) = a$  kills the nonconstant  $\omega^{\sqrt{2}}$ . Detection uses the family of functionals, not a claim that each individual derivation has only scalar constants.

**Lemma 7.3** (A strict Wronskian bound [05]). *For nonzero  $f, g \in \mathcal{B}_{\mathbb{C},H}$  with  $W_\partial(f, g) \neq 0$ ,  $\deg_\omega W_\partial(f, g) \leq \deg_\omega f + \deg_\omega g$ . If  $\deg_\omega f = \deg_\omega g = \alpha$ , the inequality is strict:  $\deg_\omega W_\partial(f, g) < 2\alpha$ .*

*Proof.* The weak bound follows from lemmas 2.4 and 7.1. If the leading coefficients of  $f, g$  are  $a, b$  at exponent  $\alpha$ , the two possible contributions at  $2\alpha$  are  $a\lambda(\alpha)b$  and  $b\lambda(\alpha)a$ , which cancel; every other contribution has smaller exponent. A nonzero Wronskian has a greatest supported exponent, by reverse well-ordering, so its degree is strictly below  $2\alpha$ ; one does not replace that maximum by an unattained supremum.  $\square$

## 7.2 Separated powers

**Theorem 7.4** (Separated-power rigidity [05]). *Let  $a, b, c \in \mathbb{C} \setminus \{0\}$  and let  $m, n \geq 2$  be ordinary integers. Every solution in  $\mathcal{B}_{\mathbb{C}}^2$  of*

$$ax^m + by^n = c \quad (7.2)$$

*has  $x, y \in \mathbb{C}$ . In particular, every solution in  $\mathcal{B}_{\mathbb{R}}^2$  is real. Consequently, if  $a, b, c$  are nonzero ordinary integers, every solution in  $\mathbf{Oz}^2$  is an ordinary integer solution.*

*Proof.* If one coordinate is constant, the other is constant by [proposition 2.8](#). Suppose both are nonconstant, with  $\alpha = \deg_{\omega} x > 0$  and  $\beta = \deg_{\omega} y > 0$ . Cancellation of the positive leading degree in (7.2) forces  $m\alpha = n\beta$ . Take a common set-sized workspace  $H$  for  $x, y$  ([lemma 2.6](#)) and choose  $\partial = \partial_{\lambda}$  with  $\lambda(\alpha) = 1$ , so  $\partial x \neq 0$ . Differentiating gives

$$amx^{m-1}\partial x + bny^{n-1}\partial y = 0, \quad (7.3)$$

and since  $x \neq 0$  also  $\partial y \neq 0$ .

We work in  $\mathcal{B}_{\mathbb{C},H}$ , in which  $a, b, c, m, n$  are units. Modulo the principal ideal  $(x^{m-1})$ , (7.2) says  $by^n = c$ , so the image of  $y$  is a unit; reducing (7.3) then shows that  $\partial y$  vanishes in the quotient. Equivalently  $\partial y = x^{m-1}q$  with  $q \in \mathcal{B}_{\mathbb{C},H}$ , and  $q \neq 0$  because  $\partial y \neq 0$ . Nonnegative degree and [lemma 7.1](#) give

$$(m-1)\alpha \leq \deg_{\omega}(\partial y) \leq \beta = \frac{m}{n}\alpha.$$

Since  $\alpha > 0$ , this gives  $mn - m - n \leq 0$ , that is  $(m-1)(n-1) \leq 1$ , which for  $m, n \geq 2$  forces  $m = n = 2$ . In that case  $aX^2 + bY^2$  has two distinct projective linear factors over  $\mathbb{C}$  because  $ab \neq 0$ , and [theorem 6.7](#) contradicts the assumed nonconstancy. Constant omnific coordinates are integers.  $\square$

Source 05 states the real case. The proof above gives its complex extension: it uses nonnegative support degrees and nonzero scalar units, not an order on the coefficients. The real and omnific conclusions follow by intersection with  $\mathcal{B}_{\mathbb{R}}$  and  $\mathbf{Oz}$ . The theorem is the case  $P(X) = (c - aX^m)/b$  of the squarefree theorem [theorem 16.6](#) of sources C10–C14, which holds over every field of characteristic zero and every exponent group and also covers  $y^2 = x^3 + ax + b$  with  $a \neq 0$ .

The quotient-ring step replaces any appeal to unique factorization: if  $y$  becomes a unit modulo  $x^{m-1}$ , a product of  $\partial y$  with a unit can vanish there only if  $\partial y$  itself does. The argument is valid in a non-Noetherian ring and for supports of arbitrary set-sized order type.

*Remark 7.5* (A direct divisibility identity [merge]). The divisibility  $x^{m-1} \mid \partial y$  also follows from an explicit identity. Put  $f = ax^m + by^n$ . For every derivation  $\partial$  that kills  $a$  and  $b$ ,

$$ax^{m-1}(nx\partial y - my\partial x) = n f \partial y - y \partial f,$$

as one sees by expanding  $\partial f = amx^{m-1}\partial x + bny^{n-1}\partial y$ : the right side is  $anx^m\partial y + bny^n\partial y - amx^{m-1}y\partial x - bny^n\partial y$ . Without that hypothesis, the right side has the additional term  $-y((\partial a)x^m + (\partial b)y^n)$ . The Euler derivations kill all scalar coefficients, including  $c$ . On a solution of (7.2), therefore,  $f = c$  and  $\partial f = 0$ , so  $nc\partial y = x^{m-1} \cdot a(nx\partial y - my\partial x)$ , and since  $nc$  is a unit of  $\mathcal{B}_{\mathbb{C},H}$ ,  $x^{m-1}$  divides  $\partial y$ . This route avoids the quotient ring. The suite of 05 checks a related identity only for  $a = 1, b = -1$ ; the identity above was checked symbolically for general  $a, b$  and  $2 \leq m, n \leq 7$  when this report was assembled.

**Corollary 7.6** (Mordell and Fermat curves [05]). *The following equations have no infinite omnific solutions; their omnific solution classes equal their ordinary integer solution sets:*

$$\begin{aligned} x^m - y^n &= c \quad (m, n \geq 2, c \in \mathbb{Z} \setminus \{0\}), \\ y^2 &= x^3 + k \quad (k \in \mathbb{Z} \setminus \{0\}), \quad x^n + y^n = 1 \quad (n \geq 2). \end{aligned}$$

*Proof.* Use  $(a, b) = (1, -1)$  or  $(-1, 1)$  as appropriate in [theorem 7.4](#), and  $(a, b, c) = (1, 1, 1)$  for the last equation.  $\square$

In particular  $Y^2 = X^3 + 1$ , which source 01 listed as an open question ([section 21.1](#)), has only its ordinary integer points in  $\mathbf{Oz}$ . The theorem reduces the omnific problem to ordinary integers; it does not by itself bound or list the integer solutions. A finite computation can verify particular solutions of  $y^2 = x^3 + 1$ , such as  $(-1, 0)$ ,  $(0, \pm 1)$ ,  $(2, \pm 3)$ , but a finite search is not a proof that such a list is complete, and the suite of 05 checks only the displayed identities.

The condition  $c \neq 0$  cannot be removed:  $x = t^n$ ,  $y = t^m$  solves  $x^m - y^n = 0$  for every  $t \in \mathbf{Oz}$ . Nor can an exponent equal one:  $y = x^m - c$  is an infinite family. The theorem treats fixed ordinary exponents and says nothing about variable omnific exponents. A general elliptic equation  $y^2 = x^3 + ax + b$  with  $a \neq 0$  is outside it: the differentiated equation has additional terms, and the divisibility by a high power of  $x$  no longer follows. This is a boundary of the proof, not a reason to extend the statement to all elliptic curves. Sources C10–C14 cross that boundary by a different certificate, a Bézout identity between the cubic and its derivative: every nonsingular equation has only ordinary omnific solutions, and every singular one has polynomial families ([corollary 16.9](#) and [proposition 16.11](#)).

### 7.3 Unimodular Fermat rigidity

The homogeneous Fermat equation has abundant infinite solutions ([example 14.3](#)), none of them primitive. An explicit unit-ideal condition excludes nonconstant solutions for every exponent at least three. We use the following terms, which sources 01 and 05 define identically in substance.

**Definition 7.7** (Primitive and unimodular [01, 05]). A nonzero tuple  $(x_1, \dots, x_r)$  over a commutative ring  $R$  is *unimodular* if  $\sum_i u_i x_i = 1$  for some  $u_i \in R$ , that is,  $(x_1, \dots, x_r) = R$ . Over the domain  $\mathbf{Oz}$  it is *primitive* if its entries have no common nonunit divisor.

Unimodularity implies primitivity. The converse is not assumed in the non-PID ring  $\mathbf{Oz}$ , and no Bézout hypothesis is used; whenever a tuple is claimed unimodular, an explicit identity is given.

**Lemma 7.8** (Pairwise comaximality [05]). *Let  $R$  be a commutative ring and suppose  $x^n + y^n = z^n$  with  $n \geq 1$  and  $(x, y, z) = R$ . Then  $(x, y) = (y, z) = (z, x) = R$ , and powers of any of these pairs are again comaximal.*

*Proof.* In  $R/(x, y)$  the element  $z$  is nilpotent, as  $z^n = 0$ , and a unit, because a unit-ideal witness reduces to  $wz = 1$ . The quotient is therefore zero. The other pairs are similar. For ordinary integers  $r, s \geq 1$ , if  $au + bv = 1$ , the binomial expansion of  $(au + bv)^{r+s-1}$  lies in  $(u^r, v^s)$ : each term has either at least  $r$  factors  $u$  or at least  $s$  factors  $v$ .  $\square$

**Theorem 7.9** (Unimodular Fermat rigidity [05]). *Let  $n \geq 3$  be an ordinary integer. Suppose  $x, y, z \in \mathcal{B}_{\mathbb{C}}$  are nonzero,*

$$x^n + y^n = z^n, \quad (x, y, z) = \mathcal{B}_{\mathbb{C}}.$$

Then  $x, y, z \in \mathbb{C}$ . In particular, a unimodular Fermat triple in  $\mathbf{Oz}$  with all three coordinates nonzero has ordinary integer coordinates.

*Proof.* First suppose that all ratios among  $x, y, z$  are complex constants:  $x = \mu z$ ,  $y = \nu z$  with  $\mu, \nu \in \mathbb{C}^\times$ . A unit-ideal witness  $ux + vy + wz = 1$  gives  $z(\mu u + \nu v + w) = 1$ , so  $z$  is a unit of  $\mathcal{B}_{\mathbb{C}}$ , and  $z$ , hence  $x, y$ , are constants by [proposition 2.5](#).

It remains to rule out a nonconstant ratio. Choose a set-sized workspace  $H$  containing the supports of  $x, y, z$  and of a unit-ideal witness; the ratios  $u = x/z$  and  $v = y/z$  lie in the Hahn field over  $H$  and satisfy  $u^n + v^n = 1$ . These ratios may have negative exponents; only the original coordinates, their derivatives and the chosen unit-ideal witness are required to lie in  $\mathcal{B}_{\mathbb{C}, H}$ . In particular that witness establishes unimodularity inside this ring before any comaximality argument is applied. If one of  $u, v$  is constant, the other is algebraic over  $\mathbb{C}$ , hence equal to one of the complex roots of its polynomial, because that polynomial splits over  $\mathbb{C}$  and the Hahn field has no zero divisors. So both are nonconstant. By [lemma 7.2](#), choose  $\partial$  with  $\partial v \neq 0$ ; differentiating  $u^n + v^n = 1$  gives  $\partial u \neq 0$ .

All three Wronskians are nonzero. Indeed  $W_{\partial}(y, z) = -z^2 \partial v$  and  $W_{\partial}(z, x) = z^2 \partial u$ . If  $W_{\partial}(x, y) = 0$ , then  $\partial u/u = \partial v/v = h$ , say, and differentiating  $u^n + v^n = 1$  gives  $nh(u^n + v^n) = nh = 0$ , so  $h = 0$ , contrary to  $\partial v \neq 0$ .

Differentiating the Fermat equation gives  $x^{n-1} \partial x + y^{n-1} \partial y = z^{n-1} \partial z$ . Multiplying by  $z$  and using  $z^n = x^n + y^n$ , then multiplying by  $y$  and using  $y^n = z^n - x^n$ , gives the two relations

$$x^{n-1} W_{\partial}(z, x) = y^{n-1} W_{\partial}(y, z), \quad -x^{n-1} W_{\partial}(x, y) = z^{n-1} W_{\partial}(y, z).$$

By [lemma 7.8](#), applied in  $\mathcal{B}_{\mathbb{C}, H}$ , there are  $r, s$  with  $rx^{n-1} + sy^{n-1} = 1$ ; multiplying by  $W_{\partial}(y, z)$  and using the first relation gives  $W_{\partial}(y, z) = x^{n-1}(rW_{\partial}(y, z) + sW_{\partial}(z, x))$ . The same argument with the roles of the comaximal pairs exchanged, in both relations, gives

$$x^{n-1} \mid W_{\partial}(y, z), \quad y^{n-1} \mid W_{\partial}(z, x), \quad z^{n-1} \mid W_{\partial}(x, y) \tag{7.4}$$

in  $\mathcal{B}_{\mathbb{C}, H}$ . Each divisibility has a nonzero quotient, since its Wronskian is nonzero. That quotient has nonnegative degree by the support restriction, which is essential for the next lower bound; divisibility in the ambient field alone would give no such bound.

Let  $\alpha$  be the greatest of the three coordinate degrees and  $\beta, \gamma$  the other two. If  $\alpha = 0$ , all coordinates are constants. If  $\alpha > 0$ , the divisibility in (7.4) for the coordinate of degree  $\alpha$  and the weak Wronskian bound give

$$(n-1)\alpha \leq \beta + \gamma \leq 2\alpha.$$

For  $n > 3$  this is impossible. For  $n = 3$  it forces  $\beta = \gamma = \alpha$ , and the relevant Wronskian then has degree strictly less than  $2\alpha$  by [lemma 7.3](#), contradicting its divisibility by a nonzero element of degree  $2\alpha$ . Hence a nonconstant ratio is impossible. A unit-ideal witness in  $\mathbf{Oz}$  is also one in  $\mathcal{B}_{\mathbb{C}}$ , and constant omnific elements are ordinary integers.  $\square$

**Corollary 7.10** (An omnific form of Fermat's Last Theorem [05]). *For every ordinary integer  $n \geq 3$  there is no triple of nonzero omnific integers with  $x^n + y^n = z^n$  and  $ux + vy + wz = 1$  for some  $u, v, w \in \mathbf{Oz}$ .*

*Proof.* By [theorem 7.9](#) such a triple is ordinary, and the classical theorem of Wiles and Taylor–Wiles excludes it [15, 16]. Its positive-integer formulation also excludes nonzero signed triples: for even exponents take absolute values, and for odd exponents move negative terms to the other side.  $\square$

The rigidity theorem is the argument supplied by source 05; the final nonexistence imports the classical theorem. The unimodular Pythagorean family of [proposition 14.4](#) is infinite, so the threshold  $n \geq 3$  is sharp. Zero coordinates are excluded intentionally:  $(1, 0, 1)$  is a unimodular constant solution for every  $n$ . The theorem uses actual Bézout witnesses and actual quotient rings; it makes no assertion about triples that merely lack a common nonunit divisor ([question 21.2](#)).

## 8 The quadratic-level dichotomy

Binary Pell rigidity might suggest that a nonzero constant quadratic level never contains infinite points. This is false as soon as there is enough indefinite dimension. The following result gives a precise classification.

For a real quadratic form we use the polarization convention

$$B(x, y) = \frac{1}{2}(q(x + y) - q(x) - q(y)), \quad q(x) = B(x, x).$$

Thus a coefficient of  $X_i X_j$  for  $i \neq j$  is twice the corresponding matrix entry of  $B$ . An integral polynomial  $q$  can have half-integral bilinear matrix entries. We perform this linear algebra over  $\mathbb{R}$ , and extend the same finite coordinate expressions to **No**. This does not assume that 2 is a unit in **Oz** or its residue rings. The parameter  $t$  in the following families is a free scalar, unrelated to the infinitesimal monomial  $t = \omega^{-1}$  used for the change of exponent convention.

**Theorem 8.1** (Quadratic-level dichotomy [01]; case (iii) also [02]). *Let  $n \geq 1$ , let  $q \in \mathbb{Z}[X_1, \dots, X_n]$  be a homogeneous quadratic form, and let  $c \in \mathbb{Z} \setminus \{0\}$ . Write*

$$V_{\mathbf{Oz}} = \{\mathbf{x} \in \mathbf{Oz}^n : q(\mathbf{x}) = c\}, \quad V_{\mathbb{Z}} = \{\mathbf{a} \in \mathbb{Z}^n : q(\mathbf{a}) = c\}.$$

*If  $V_{\mathbb{Z}}$  is empty, so is  $V_{\mathbf{Oz}}$ . Otherwise the following alternatives are exhaustive.*

- (i) *If  $q$  is nondegenerate and is either definite or of dimension at most two, then  $V_{\mathbf{Oz}} = V_{\mathbb{Z}}$ .*
- (ii) *If  $q$  is degenerate, every  $\mathbf{a} \in V_{\mathbb{Z}}$  lies on an injective affine-linear family  $\mathbf{a} + t\mathbf{v}$ ,  $t \in \Pi$ , in  $V_{\mathbf{Oz}}$ , with every  $t \neq 0$  giving an infinite point.*
- (iii) *If  $q$  is nondegenerate, indefinite, and  $n \geq 3$ , every  $\mathbf{a} \in V_{\mathbb{Z}}$  lies on an injective quadratic polynomial family  $\mathbf{a} + t\mathbf{v} + t^2\mathbf{w}$ ,  $t \in \Pi$ , in  $V_{\mathbf{Oz}}$ , with every  $t \neq 0$  giving an infinite point. The family is injective on all of **No**, so every such fiber contains a proper class of points.*

*In alternatives (ii) and (iii), the constant tuple of every point in the family is  $\mathbf{a}$ .*

We first prove the geometric construction needed in the indefinite case. All vectors and bilinear forms in it are ordinary real objects; only the parameter is later allowed to be surreal.

**Lemma 8.2** (A nilpotent quadratic isometry [01, 02]). *Let  $B$  be a symmetric bilinear form on  $\mathbb{R}^n$ , with  $q(x) = B(x, x)$ . Suppose  $e, u \in \mathbb{R}^n$  satisfy*

$$e \neq 0, \quad q(e) = 0, \quad B(e, u) = 0, \quad k = q(u) \neq 0,$$

*and put  $N(x) = B(x, e)u - B(x, u)e$ . Then*

$$N^2(x) = -kB(x, e)e, \quad N^3 = 0, \quad B(Nx, y) + B(x, Ny) = 0. \quad (8.1)$$

*For a parameter  $t$  in any field extension of  $\mathbb{R}$ , the polynomial map*

$$U_t = \text{id} + tN + \frac{t^2}{2}N^2, \quad U_t(x) = x + t(B(x, e)u - B(x, u)e) - \frac{k}{2}t^2B(x, e)e, \quad (8.2)$$

satisfies

$$q(U_t(x)) = q(x), \quad \det U_t = 1, \quad U_s U_t = U_{s+t}, \quad U_t^{-1} = U_{-t}.$$

If  $t \in \Pi$ , all matrix entries of  $U_t$  lie in  $\mathbf{Oz}$  and its entrywise constant coefficient is the identity. If  $\alpha = B(a, e) \neq 0$ , the curve  $t \mapsto U_t(a)$  is injective.

*Proof.* From  $B(e, e) = B(e, u) = 0$  one has  $N(e) = 0$  and  $N(u) = -ke$ , which gives the first two identities of (8.1). Directly,  $B(Nx, y) = B(x, e)B(u, y) - B(x, u)B(e, y) = -B(x, Ny)$ .

For preservation, put  $\alpha = B(x, e)$  and  $\beta = B(x, u)$ . In (8.2) the added vector is  $t\alpha u - (t\beta + \frac{k}{2}t^2\alpha)e$ . Its scalar product with  $x$ , multiplied by two, is  $-kt^2\alpha^2$ , because the two linear terms cancel. Its own quadratic value is  $kt^2\alpha^2$ , since  $e$  is isotropic and orthogonal to  $u$ . Their sum is zero.

Multiplying these polynomials in  $N$  and using  $N^3 = 0$  gives  $U_s U_t = U_{s+t}$  and hence  $U_t U_{-t} = \text{id}$  over  $\mathbb{R}[t]$ . The determinant is therefore a unit of  $\mathbb{R}[t]$ , a nonzero real constant, and evaluation at  $t = 0$  makes it 1. All identities remain valid after substituting a surreal  $t$ . Every entry of  $U_t$  is an entry of  $\text{id}$  plus a real combination of  $t$  and  $t^2$ ; for  $t \in \Pi$  that difference lies in  $\Pi$ .

Finally, for  $\alpha = B(a, e) \neq 0$  and  $\beta = B(a, u)$ ,

$$U_t(a) = a + tv + t^2w, \quad v = \alpha u - \beta e, \quad w = -\frac{k\alpha}{2}e.$$

The real vectors  $u, e$  are independent (as  $q(u) \neq 0 = q(e)$ ), so  $v, w$  are independent:  $w$  is a nonzero multiple of  $e$  and  $v$  has nonzero  $u$ -component. Their independence persists over a field extension because some  $2 \times 2$  real minor is nonzero. If  $U_s(a) = U_t(a)$  with  $s \neq t$ , then  $v + (s + t)w = 0$ , a contradiction.  $\square$

Source 01 wrote this map as a transvection  $T_t$  and source 02 as the nilpotent operator  $U_T$  with  $\text{id} + TN + \frac{T^2}{2}N^2$ ; they are the same map. No infinite exponential series is used: the expression is a finite polynomial because  $N^3 = 0$ .

**Lemma 8.3** (An isotropic direction visible from a nonzero point [01, 02]). *If  $q$  is a nondegenerate indefinite real quadratic form of dimension at least three and  $a \in \mathbb{R}^n$  is nonzero, there exist  $e, u$  as in lemma 8.2 with  $B(a, e) \neq 0$ .*

*Proof.* The real isotropic vectors span  $\mathbb{R}^n$ . Diagonalize and rescale to an orthogonal real basis with  $q(p_i) = 1$  and  $q(n_j) = -1$ ; both types exist by indefiniteness. Since  $B(p_i, n_j) = 0$ , each  $p_i \pm n_j$  is isotropic. Fixing one  $n_j$  recovers every  $p_i$  as a half-sum, and fixing one  $p_i$  recovers every  $n_j$  as a half-difference. Since the nonzero functional  $B(a, -)$  cannot vanish on a spanning set, some isotropic  $e$  has  $B(a, e) \neq 0$ .

Nondegeneracy gives  $f_0$  with  $B(e, f_0) = 1$ . Replacing it by  $f = f_0 - \frac{1}{2}q(f_0)e$  makes  $q(f) = 0$  while retaining  $B(e, f) = 1$ . The plane spanned by  $e, f$  has Gram matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , so it is nondegenerate. Explicitly,  $x - B(x, f)e - B(x, e)f$  is orthogonal to both  $e$  and  $f$ , which gives the direct sum with their orthogonal complement. The complement is nondegenerate: a vector in its radical would be orthogonal to the plane as well, hence to all of  $\mathbb{R}^n$ , and would vanish. Its dimension is  $n - 2 \geq 1$ . If  $q$  vanished identically there, polarization would make its bilinear form zero, contrary to nondegeneracy in positive dimension. It therefore contains  $u$  with  $q(u) \neq 0$ , and this  $u$  is orthogonal to  $e$ .  $\square$

Source 01 stated this lemma for  $q(a) \neq 0$ ; its proof, like source 02's, uses only  $a \neq 0$ .

*Proof of theorem 8.1.* The empty-set assertion is theorem 5.1.

Suppose first that  $q$  is definite and  $\mathbf{x} \in V_{\mathbf{Oz}}$  has an infinite coordinate. Let  $\gamma > 0$  be the largest leading exponent of its nonzero coordinates and  $r_i$  the coefficient of  $\omega^\gamma$  in  $x_i$  (zero when absent). The real vector  $\mathbf{r}$  is nonzero, and the coefficient of  $\omega^{2\gamma}$  in  $q(\mathbf{x})$  is exactly  $q(\mathbf{r}) \neq 0$ , contradicting  $q(\mathbf{x}) = c$ . Thus every coordinate is finite and hence an ordinary integer. In dimension two, nondegeneracy means that the binary quadratic has two distinct linear factors over  $\mathbb{C}$ , so theorem 6.7 applies. A nondegenerate one-dimensional form is definite. This proves (i).

For (ii), choose a nonzero real vector  $\mathbf{v} \in \text{rad}(B)$ . For every  $t$ ,  $q(\mathbf{a} + t\mathbf{v}) = q(\mathbf{a}) = c$ . If  $t \in \Pi$ , the coordinates of  $t\mathbf{v}$  lie in  $\Pi$ , so  $\mathbf{a} + t\mathbf{v} \in \mathbf{Oz}^n$  with constant tuple  $\mathbf{a}$ . The map is injective because  $\mathbf{v} \neq 0$ , and for  $t \neq 0$  some coordinate has positive degree.

For (iii),  $\mathbf{a} \neq 0$  since  $c \neq 0$ . Choose  $e, u$  by lemma 8.3 and use lemma 8.2. Each coordinate of  $U_t(\mathbf{a})$  is an integer constant plus real multiples of  $t$  and  $t^2$ , so it lies in  $\mathbf{Oz}$  with constant coefficient  $a_i$  for  $t \in \Pi$ . Injectivity was proved for all surreal  $t$ . If  $t \neq 0$ , its degree is positive, and in a coordinate where  $w_i \neq 0$  the quadratic term has degree  $2 \deg_\omega(t)$ , strictly larger than that of the linear term, so the point is infinite. Since  $\Pi$  contains  $\omega^\gamma$  for every surreal  $\gamma > 0$ , the fiber contains a proper class of points. The same proper-class conclusion holds for the injective affine family in (ii).  $\square$

The quadratic family in (iii) also preserves the ideal generated by the coordinates. For  $t \in \Pi$ , both  $U_t$  and  $U_{-t}$  have entries in  $\mathbf{Oz}$ . Writing  $\mathbf{x} = U_t \mathbf{a}$  gives  $(x_1, \dots, x_n) \subseteq (a_1, \dots, a_n)$ , and applying the inverse gives the reverse inclusion. For an ordinary starting tuple with gcd  $d > 0$ , that ideal is exactly  $d\mathbf{Oz}$ , by the ordinary integer Bézout identity. In particular, a primitive integer starting point gives an infinite family of unimodular omnific points. This conclusion uses the matrix orbit in (iii), not an arbitrary parametrized family.

## 8.1 Explicit families

For positive ordinary integers  $D, D'$  and  $t \in \Pi$ , set [01]

$$x = 1 + \frac{D'}{2}t^2, \quad y = \frac{D'}{2\sqrt{D}}t^2, \quad z = t. \quad (8.3)$$

All three coordinates are omnific integers, and  $x^2 - Dy^2 - D'z^2 = 1$ : the two quartic terms cancel, as do the two quadratic terms. For  $t \neq 0$  every coordinate is infinite. The tuple is unimodular, since

$$1 = x - \left(\frac{D'}{2}t\right)z, \quad \frac{D'}{2}t \in \mathbf{Oz}. \quad (8.4)$$

This illustrates a sharp contrast:

$$\begin{aligned} x^2 - Dy^2 = 1 & \text{ has only ordinary points,} \\ x^2 - Dy^2 - D'z^2 = 1 & \text{ has infinite unimodular points.} \end{aligned}$$

There is no appeal to convergence, analytic continuation, or infinite iteration; every expression is a finite polynomial.

For the Lorentz form  $q(x, y, z) = x^2 - y^2 - z^2$  with Gram matrix  $G = \text{diag}(1, -1, -1)$ , take  $e = (1, 1, 0)$  and  $u = (0, 0, 1)$  [02]. Then  $q(u) = -1$ ,  $N(x, y, z) = (z, z, x - y)$ , and

$$U_t = \begin{pmatrix} 1 + t^2/2 & -t^2/2 & t \\ t^2/2 & 1 - t^2/2 & t \\ t & -t & 1 \end{pmatrix}, \quad U_t^\top G U_t = G, \quad \det U_t = 1, \quad U_s U_t = U_{s+t}. \quad (8.5)$$

For  $t \in \Pi$  this is a one-parameter family of omnific Lorentz matrices with constant coefficient the identity. Acting on  $(1, 0, 0)$  gives  $(1 + t^2/2, t^2/2, t)$ , and replacing  $t$  by  $2s$  gives the denominator-free identity

$$(1 + 2s^2)^2 - (2s^2)^2 - (2s)^2 = 1,$$

valid for every  $s \in \mathbf{Oz}$ , with constant fiber  $(1, 0, 0)$  whenever  $s \in \Pi$ . For  $s = \omega$  the point is  $(1 + 2\omega^2, 2\omega^2, 2\omega)$ ; the squares contain terms of degrees four, two and zero, and the nonconstant terms cancel exactly. There is no negative-exponent tail. This is concretely how an extra negative square evades the Pell rigidity mechanism.

**Corollary 8.4** (An invisible unipotent subgroup [02]). *For every nondegenerate indefinite integral quadratic form  $q$  in at least three variables, there is an injective group homomorphism*

$$(\Pi, +) \longrightarrow \ker(\mathrm{SO}(q, \mathbf{Oz}) \xrightarrow{\mathrm{ct}} \mathrm{SO}(q, \mathbb{Z})), \quad t \longmapsto U_t.$$

*Proof.* Choose a nonzero real test vector  $a$  and use lemma 8.3 to construct  $e, u$ . Lemma 8.2 gives a homomorphism with entries in  $\mathbf{Oz}$ , determinant one, preservation of  $q$ , and constant coefficient id. If  $U_s = U_t$ , apply both to  $a$ ; injectivity of  $t \mapsto U_t(a)$  gives  $s = t$ . No ordinary point on a prescribed level is needed.  $\square$

Here  $\mathrm{SO}(q, \mathbf{Oz})$  means determinant-one matrices over  $\mathbf{Oz}$  preserving the integral quadratic polynomial  $q$ ; entrywise ct lands in  $\mathrm{SO}(q, \mathbb{Z})$ . This subgroup is invisible to every finite quotient of the coefficient ring (corollary 3.4), yet it is a proper class. It concerns indefinite forms only; for definite forms see corollary 9.4.

## 8.2 What the classification does and does not say

The classification concerns the presence of genuinely new points on a *represented nonzero level* [01, 02]. It does not decide whether the integer level is represented in the first place. Nor does it classify all infinite points or claim that every omnific point of a higher-dimensional quadric lies on the constructed orbit: a displayed curve is a rigorously constructed subset of the full fiber. Source 02's version covers nondegenerate forms only and excludes degenerate forms and the zero level; source 01's alternative (ii) covers degenerate forms.

For zero levels, homogeneity gives a different mechanism. An indefinite real quadratic form has real isotropic directions, and multiplying any such direction by an element of  $\Pi$  produces omnific points with zero constant tuple. Such points can exist even when there is no nonzero ordinary integer isotropic vector.

## 9 Bounded geometry, leading forms, and orthogonal matrices

Multiplicative rigidity is not the only mechanism. A different one comes from boundedness and the discreteness of  $\mathbf{Oz}$ .

### 9.1 Bounded sets

**Proposition 9.1** (Positive bounds [01, 05]). *If  $a_1, \dots, a_n$  are positive ordinary reals,  $m_1, \dots, m_n \geq 1$  are ordinary integers,  $x_1, \dots, x_n \in \mathbf{Oz}$ , and*

$$\sum_i a_i x_i^{2m_i} = C \in \mathbb{R},$$

then every  $x_i$  is an ordinary integer. In particular, an omnific point on a sphere of ordinary finite radius is an ordinary integer point.

*Proof.* Each summand is nonnegative and at most  $C$ . If  $C < 0$  there is no solution; otherwise  $|x_i| \leq (C/a_i)^{1/(2m_i)}$ , an ordinary nonnegative real bound. [Proposition 2.5](#) then applies. This also handles  $C = 0$ , when every coordinate is zero.  $\square$

**Theorem 9.2** (Bounded semialgebraic rigidity [02]). *Let  $X \subseteq \mathbb{R}^m$  be a bounded semialgebraic set defined by a finite formula in the language of ordered rings with real parameters, and let  $X(\mathbf{No})$  be the class defined by the same formula in  $\mathbf{No}$ . Then  $X(\mathbf{No}) \cap \mathbf{Oz}^m = X \cap \mathbb{Z}^m$ .*

*Proof.* First justify the passage to a set-sized field, including when the defining formula has quantifiers. Any set of surreal parameters is contained in a set-sized real closed subfield: generate a field and take all surreal elements algebraic over it. This is a set because each nonzero polynomial has finitely many roots, ordered by the inherited order. It is real closed because positive square roots and roots of odd-degree polynomials exist in  $\mathbf{No}$  and remain algebraic over the generated field.

For any set-sized real closed  $K \subseteq \mathbf{No}$ , a finite ordered-ring formula with parameters in  $K$  has the same truth value in  $K$  and in  $\mathbf{No}$ . Prove this simultaneously for all such  $K$  by induction on formulas. Atomic equalities and inequalities agree, and Boolean connectives preserve agreement. For an existential witness in  $\mathbf{No}$ , enlarge  $K$  to a set-sized real closed  $L$  containing the witness. The induction hypothesis transfers the subformula satisfied by that witness to  $L$ ; Tarski's quantifier elimination makes  $K \subseteq L$  elementary, so the existential assertion already holds in  $K$ . Conversely, a witness in  $K$  works in  $\mathbf{No}$  by induction. This uses model completeness only between set-sized fields [11, 31]; no set of all surreals is made into a model.

Now choose an ordinary real  $M > 0$  with  $|x_i| \leq M$  for every  $\mathbf{x} \in X$ . This bound is a first-order implication with real parameters. For a candidate point of  $X(\mathbf{No})$ , take such a  $K$  containing its coordinates and  $\mathbb{R}$ . Its defining formula holds in  $K$  by the preceding argument, and the bound holds there by elementarity of  $\mathbb{R} \subseteq K$ . Every omnific coordinate is therefore bounded by  $M$ , hence an ordinary integer by [proposition 2.5](#). An integer tuple satisfies the defining formula in  $K$  exactly when it does in  $\mathbb{R}$ , again by elementarity. This proves both inclusions, including when  $X$  is empty.  $\square$

**Corollary 9.3** (Definite levels [01, 02]). *For a positive or negative definite real quadratic form  $q$  and a real  $c$ , every omnific point of  $q(\mathbf{x}) = c$  is an ordinary integer tuple.*

*Proof.* In dimension zero there is only the empty tuple. In positive dimension, every real level set of a definite form is bounded, including an empty level. For a positive definite form, choose  $\delta > 0$  with  $q(\mathbf{r}) \geq \delta \sum_i r_i^2$  over  $\mathbb{R}$ , for example the least eigenvalue of its real symmetric matrix. A level  $c \geq 0$  then lies in the ball of radius  $\sqrt{c/\delta}$ , and a negative level is empty. For a negative definite form apply this to  $-q$ . Now [theorem 9.2](#) applies. Alternatively, the leading-coefficient argument in the proof of [theorem 8.1\(i\)](#) applies verbatim to any real level  $c$  [01].  $\square$

For instance,  $x^2 + y^2 = 1$  has exactly  $(\pm 1, 0), (0, \pm 1)$  as omnific solutions, and so has  $x^4 + y^4 = 1$ . These concern a fixed finite level; the homogeneous Pythagorean relation lets all coordinates grow together. Positivity does *not* imply being a sum of squares in  $\mathbf{Oz}$  ([remark 10.9](#)).

## 9.2 Orthogonal and symmetric matrices

**Corollary 9.4** (Euclidean orthogonal group [01, 02]). *For every ordinary  $n \geq 1$ ,  $O_n(\mathbf{Oz}) = O_n(\mathbb{Z})$  is the group of signed permutation matrices, of order  $2^n n!$ , and  $SO_n(\mathbf{Oz})$  consists of the signed permutation matrices of determinant one.*

*Proof.* For an orthogonal matrix, the sum of the squares in each row is 1, so every entry is ordinary by [proposition 9.1](#). An integer row of squared norm one has exactly one nonzero entry, equal to  $\pm 1$ , and orthogonality of different rows forces those entries into different columns. Conversely each signed permutation matrix is orthogonal.  $\square$

Requiring *surreal* entries gives the large rotation group of a real closed field, whereas *omnific* entries in standard Euclidean coordinates leave a finite group. This is not a claim that matrix groups over  $\mathbf{Oz}$  are finite:  $SL_2(\mathbf{Oz})$  contains all  $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ ,  $t \in \mathbf{Oz}$ , and indefinite forms have the unipotent subgroup of [corollary 8.4](#). The Euclidean group is rigid because of positivity. More generally, for any definite ordinary real form  $q$ , a matrix preserving  $q$  has its  $j$ th column on the fixed level  $q(e_j)$ , where  $e_j$  is the  $j$ th standard basis vector. [Corollary 9.3](#) makes all its omnific entries ordinary integers. Each such real level is bounded, so it contains only finitely many integer columns. Thus the omnific isometry group equals its finite integer isometry group for every definite  $q$ , although the signed-permutation description is specific to the standard sum of squares in the chosen coordinates.

**Corollary 9.5** (Symmetric matrix rigidity [01]). *Let  $M$  be a symmetric matrix over  $\mathbf{Oz}$ . If  $\text{tr}(M^2)$  is an ordinary real constant, every entry of  $M$  is an ordinary integer. In particular, this holds if its characteristic polynomial has ordinary integer coefficients.*

*Proof.* For symmetric  $M$ ,  $\text{tr}(M^2) = \sum_i m_{ii}^2 + 2 \sum_{i < j} m_{ij}^2$ , so [proposition 9.1](#) applies. When the size is at least two,  $\text{tr}(M^2) = (\text{tr } M)^2 - 2e_2(M)$ , where  $e_2(M)$ , the sum of the principal  $2 \times 2$  minors, is up to sign a coefficient of the characteristic polynomial, as is  $\text{tr } M$ ; for size one the assertion is immediate, and for size zero there are no entries to check.  $\square$

Symmetry is essential. The matrix  $\begin{pmatrix} 0 & \omega \\ 0 & 0 \end{pmatrix}$  has square zero and characteristic polynomial  $T^2$ , but has an infinite omnific entry. The trace bound works for symmetric matrices because it is a positive sum of entry squares, not because every matrix with fixed characteristic polynomial has bounded entries. Source C17 proves a normal-matrix analogue that allows a purely infinite trace: a normal matrix over  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ , in particular a symmetric one over  $\mathbf{Oz}$ , whose characteristic polynomial has nonzero constant discriminant is  $\text{ct } M + hI$  ([theorem 19.37](#) and [remark 19.40](#)).

## 9.3 No real point at infinity

**Theorem 9.6** (Leading-homogeneous obstruction [01]). *Let  $F_1, \dots, F_r \in \mathbb{Z}[X_1, \dots, X_n]$  be nonconstant polynomials, and let  $H_j$  be the top homogeneous part of  $F_j$ . If the  $H_j$  have no common nonzero real zero, then every common zero of the  $F_j$  in  $\mathbf{Oz}^n$  lies in  $\mathbb{Z}^n$ .*

*Proof.* Suppose a common zero  $\mathbf{x}$  has an infinite coordinate. Let  $\gamma > 0$  be the largest leading exponent of its coordinates and  $\mathbf{r} \neq 0$  the real vector of coefficients of  $\omega^\gamma$ . If  $d_j = \deg F_j$ , the coefficient at  $\omega^{d_j \gamma}$  in  $F_j(\mathbf{x})$  is  $H_j(\mathbf{r})$ : a product attains that exponent only by taking exponent  $\gamma$  in each of  $d_j$  factors, and terms of lower ordinary total degree have exponent strictly less than  $d_j \gamma$ . So every  $H_j(\mathbf{r}) = 0$ , contrary to the hypothesis. All coordinates are therefore finite and hence ordinary integers.  $\square$

The hypothesis is a sufficient absence-of-real-directions-at-infinity condition for the chosen system. Its converse is false: a Pell equation has real directions at infinity when  $D > 0$ , but its nonzero fibers are rigid by factorization. Initial-form obstructions and constant-product obstructions detect different mechanisms. The chosen generators also matter: the equations  $X - Y^2 = 0$  and  $X - Y^2 - Y = 0$  have only  $(0, 0)$ , since subtraction gives  $Y = 0$  and then  $X = 0$ . Their two top forms are both  $-Y^2$ , which vanish at the nonzero vector  $(1, 0)$ . Replacing them by the equivalent system  $X = Y = 0$  removes this direction. The theorem tests the displayed top forms, not every homogeneous consequence of the defining ideal.

## 10 Defining the ordinary integers

The retraction `ct` detects ordinary coefficients but is not a symbol of the ring language. Pell rigidity shows that the embedded copy of  $\mathbb{Z}$  is nevertheless detected by a single polynomial equation with existential auxiliary variables. Two versions were found independently by sources 01 and 05; source 07 found the second again ([remark 10.5](#)), and [section 11](#) gives an order-free definition that also works in the Gaussian ring.

### 10.1 Pell coordinates as finite bounds

Define ordinary Pell solutions by

$$X_k + Y_k\sqrt{2} = (3 + 2\sqrt{2})^k, \quad (X_{k+1}, Y_{k+1}) = (3X_k + 4Y_k, 2X_k + 3Y_k), \quad k \geq 0, \quad (10.1)$$

with  $(X_0, Y_0) = (1, 0)$ . The index  $k$  always belongs to the ordinary  $\mathbb{N}$ ; this recurrence does not define witnesses at infinite omnific indices.

**Lemma 10.1** (Pell witnesses [01, 05]). *For every ordinary  $k \geq 0$ ,  $X_k^2 - 2Y_k^2 = 1$ ; for  $k \geq 1$  both coordinates are positive and  $X_k \geq 3^k$ . Moreover  $X_{2j} \equiv 1$  and  $X_{2j+1} \equiv 3 \pmod{8}$ , so both parity subsequences are unbounded.*

*Proof.* Direct expansion gives  $(3X + 4Y)^2 - 2(2X + 3Y)^2 = X^2 - 2Y^2$ , and ordinary induction proves the identity, positivity and  $X_{k+1} \geq 3X_k$ . From  $X_k = \frac{1}{2}((3 + 2\sqrt{2})^k + (3 - 2\sqrt{2})^k)$  one gets the recurrence  $X_{k+2} = 6X_{k+1} - X_k$  with  $X_0 = 1, X_1 = 3$ ; modulo 8 it gives the alternating residues  $1, 3, 1, 3, \dots$  (as  $6 \cdot 3 - 1 \equiv 1$  and  $6 \cdot 1 - 3 \equiv 3$ ). Finally  $X_{2j} \geq 9^j$  and  $X_{2j+1} \geq 3 \cdot 9^j$ , so each residue class used below supplies witnesses above every ordinary integer bound.  $\square$

### 10.2 The five-auxiliary-variable guard

We use the Legendre–Gauss three-square theorem: a nonnegative ordinary integer  $N$  is a sum of three integer squares if and only if it is not of the form  $4^a(8b + 7)$  with  $a, b \geq 0$ . It is imported, not reproved; see the exposition [12] and the formal development in Isabelle’s Archive of Formal Proofs [13].

**Lemma 10.2** (A three-square gap below a Pell coordinate [01]). *For every ordinary nonnegative integer  $S$ , there are ordinary integers  $u, v, w_1, w_2, w_3$  such that*

$$u^2 - 2v^2 = 1, \quad u = S + w_1^2 + w_2^2 + w_3^2.$$

*Proof.* Choose the residue of  $u$  modulo 8 according to the following table; each choice is realized by an unbounded subsequence of the  $X_k$  ([lemma 10.1](#)).

|                    |   |   |   |   |   |   |   |   |
|--------------------|---|---|---|---|---|---|---|---|
| $S \bmod 8$        | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| chosen $u \bmod 8$ | 1 | 3 | 3 | 1 | 1 | 3 | 1 | 1 |
| $(u - S) \bmod 8$  | 1 | 2 | 1 | 6 | 5 | 6 | 3 | 2 |

Take  $u$  in that subsequence so large that  $N = u - S > 0$ . Its residue is one of 1, 2, 3, 5, 6, so it is neither divisible by 4 nor congruent to 7 modulo 8, and cannot be  $4^a(8b + 7)$ . The three-square theorem supplies the  $w_j$ , and the corresponding Pell coordinate supplies  $v$ .  $\square$

For each ordinary  $n \geq 1$ , define the integer polynomial

$$\Psi_n(\mathbf{X}, U, V, \mathbf{W}) = (U^2 - 2V^2 - 1)^2 + \left( U - \sum_{i=1}^n X_i^2 - W_1^2 - W_2^2 - W_3^2 \right)^2, \quad (10.2)$$

of total degree four in  $n + 5$  variables.

**Theorem 10.3** (Uniform quartic integrality guard [01]). *For every  $\mathbf{x} \in \mathbf{Oz}^n$ ,*

$$\mathbf{x} \in \mathbb{Z}^n \iff \exists u, v, w_1, w_2, w_3 \in \mathbf{Oz} \quad \Psi_n(\mathbf{x}, u, v, w_1, w_2, w_3) = 0. \quad (10.3)$$

*Every coordinate of every zero of  $\Psi_n$  in  $\mathbf{Oz}^{n+5}$  is an ordinary integer. The number of auxiliary variables is five, independently of  $n$ .*

*Proof.* Suppose  $\Psi_n = 0$  in  $\mathbf{Oz}$ . A sum of two squares in an ordered ring vanishes only when both squares vanish, so

$$u^2 - 2v^2 = 1, \quad u = \sum_i x_i^2 + w_1^2 + w_2^2 + w_3^2. \quad (10.4)$$

By Pell rigidity (corollary 6.9),  $u, v \in \mathbb{Z}$ . The second equation gives  $u \geq 0$  and bounds the square of each  $x_i$  and  $w_j$  by the ordinary number  $u$ . Explicitly, if  $z^2 \leq u$ , then  $|z| < u + 1$ ; otherwise  $z^2 \geq (u + 1)^2 > u$ . Thus each coordinate is finite and is an ordinary integer by proposition 2.5. Conversely, for  $\mathbf{x} \in \mathbb{Z}^n$  put  $S = \sum_i x_i^2$  and choose the witnesses of lemma 10.2.  $\square$

For  $x = 1$ , one witness is  $u = 3, v = 2, (w_1, w_2, w_3) = (1, 1, 0)$ ; for  $x = 2, u = 17, v = 12, (w_1, w_2, w_3) = (3, 2, 0)$ . The assertions for all integers rely on lemma 10.2, not on examples or a finite computation.

### 10.3 The four-square definition

**Theorem 10.4** (A quartic definition of  $\mathbb{Z}$  [05]). *For  $t \in \mathbf{Oz}$ , the following are equivalent:*

- (i)  $t \in \mathbb{Z}$ ;
- (ii) *there exist  $x, y, z_1, z_2, z_3, z_4 \in \mathbf{Oz}$  with*

$$x^2 - 2y^2 = 1, \quad x - t^2 = z_1^2 + z_2^2 + z_3^2 + z_4^2; \quad (10.5)$$

- (iii) *the quartic equation*

$$(x^2 - 2y^2 - 1)^2 + \left( x - t^2 - \sum_{j=1}^4 z_j^2 \right)^2 = 0 \quad (10.6)$$

*has a solution in its six existential variables over  $\mathbf{Oz}$ .*

Every witness to (10.5) has all six coordinates in  $\mathbb{Z}$ .

*Proof.* If (10.5) holds, Pell rigidity makes  $x, y$  ordinary, and  $0 \leq t^2 \leq x$  bounds  $t$ , so  $t \in \mathbb{Z}$ ; each  $z_j^2$  is bounded by the ordinary integer  $x - t^2$ , so every  $z_j$  is ordinary. Conversely, for  $t \in \mathbb{Z}$  choose  $X_k \geq t^2$  by lemma 10.1; the nonnegative integer  $X_k - t^2$  is a sum of four squares by Lagrange’s theorem, which is available in Mathlib as `Nat.sum_four_squares` [14]. In an ordered ring a sum of two squares vanishes exactly when both do, so (10.5) and (10.6) are equivalent. The polynomial has total degree four, integer coefficients and no infinite parameters.  $\square$

Write  $\text{Std}(t)$  for the existential formula (10.6) and  $\text{Std}_n(\mathbf{x})$  for (10.3). Replacing  $t^2$  by  $\sum_{i=1}^n t_i^2$  in (10.5) defines  $\mathbb{Z}^n$  with six auxiliary variables by the same proof. Here “standard” refers precisely to the embedded  $\mathbb{Z}$ ; it is not an extra predicate of the language.

*Remark 10.5* (Source 07’s quartic [07]). Source 07 found, independently, the quartic

$$F_4(x, u, v, \mathbf{a}) = (u^2 - 2v^2 - 1)^2 + \left(u^2 - x^2 - \sum_{j=1}^4 a_j^2\right)^2 \quad (10.7)$$

with the six existential variables  $u, v, a_1, \dots, a_4$ , and proved that it defines  $\mathbb{Z}$  in every ring  $A$  with  $\mathbb{Z} \subseteq A \subseteq \mathcal{A}_{\mathbb{R}}(\Gamma)$  and  $A \cap \mathbb{R} = \mathbb{Z}$  (notation of section 11.1), every zero in  $A^7$  having all seven coordinates in  $\mathbb{Z}$ . It differs from (10.6) only in comparing  $x^2$  with the square  $u^2$  of the Pell coordinate instead of  $t^2$  with the coordinate itself, and is printed once, here. Here is the argument for the stated intermediate ring, without assuming that it is closed under constant extraction. If  $F_4 = 0$ , the order on the ambient real Hahn field makes both squared expressions zero:

$$(u - \sqrt{2}v)(u + \sqrt{2}v) = 1, \quad x^2 + \sum_{j=1}^4 a_j^2 = u^2.$$

Both factors in the first equation belong to the ambient nonnegative-support ring, even when  $\sqrt{2} \notin A$ . The constant-product argument of lemma 11.1 makes them real constants. Adding and subtracting shows  $u, v \in \mathbb{R}$ , hence  $u, v \in A \cap \mathbb{R} = \mathbb{Z}$ . If any of  $x, a_1, \dots, a_4$  were nonconstant, let  $\delta > 0$  be the largest of their degrees. At exponent  $2\delta$ , the sum of their squares has coefficient equal to the sum of the squares of the nonzero leading coefficients of precisely those terms of degree  $\delta$ . That coefficient is positive, contradicting the constant right-hand side  $u^2$ . All five terms are therefore real constants, and membership in  $A$  makes them integers. This proves the assertion about the full zero set, including every witness.

Conversely, for  $x \in \mathbb{Z}$  choose an ordinary Pell pair  $(u, v) = (X_k, Y_k)$  with  $u \geq |x|$  by lemma 10.1. Lagrange’s theorem writes the nonnegative integer  $u^2 - x^2$  as four integer squares, including the case zero. These six integer witnesses belong to every allowed  $A$  and give  $F_4 = 0$ . Thus neither divisibility nor nontriviality of  $\Gamma$  is required; for  $\Gamma = 0$  the hypothesis already forces  $A = \mathbb{Z}$ . The polynomial has total degree exactly four: its  $x^4$  coefficient is 1. The order was used only in the proof, not as a symbol in the defining formula. Source 07 cites Lagrange’s theorem through [27, Section 6]. Over  $\mathbf{Oz}[i]$  the formula accepts every  $x$  (take  $u = 1, v = 0, a_1 = ix, a_2 = 1, a_3 = a_4 = 0$ ), so it is not asserted there; the order-free definition of  $\mathbb{Z}[i]$  in theorem 11.9.

*Remark 10.6* (Comparison of the two guards). Source 01’s guard is uniform in  $n$  with five auxiliary variables and needs the three-square theorem; source 05’s uses one more auxiliary variable and Lagrange’s four-square theorem, and source 01 had noted this simpler variant without writing it out. Neither source claims its count optimal (question 21.5). For formalization the difference matters: at the Mathlib revision pinned by this repository (81a5d257),

`Nat.sum_four_squares` is proved in `Mathlib/NumberTheory/SumFourSquares.lean`, while no three-square theorem is present. The four-square definitions of sources 05 and 07 therefore use an available Mathlib theorem; source 01's would need the three-square theorem imported or proved first. The four-square theorem is used only on an ordinary nonnegative integer; its analogue for positive omnific integers is false ([remark 10.9](#)). Source 07's quartic ([remark 10.5](#)) has the same counts as source 05's, and source 06's quintic ([remark 11.10](#)), with degree five and seven witnesses, is dominated by both guards.

## 10.4 Definability consequences

$\text{Std}(\mathbf{Oz}) = \mathbb{Z}$ : ordinary integer arithmetic is present not only as the quotient  $\mathbf{Oz}/\Pi$  but as an existentially definable subring of  $\mathbf{Oz}$ . There is no conflict with [proposition 5.3](#), because  $\mathbb{Z}$  is stable under  $\text{ct}$ , whereas the positive cone and the nonzero locus are not.

**Corollary 10.7** (Definability of  $\text{ct}$  and  $\Pi$  [01, 05]). *The graph of  $\text{ct}$  and the ideal  $\Pi$  are first-order definable in the pure ring language of  $\mathbf{Oz}$  without parameters:*

$$\begin{aligned} \text{ct}(a) = m &\iff \text{Std}(m) \wedge \forall k ((\text{Std}(k) \wedge k \neq 0) \Rightarrow \exists y (a - m = ky)), \\ j \in \Pi &\iff \forall k ((\text{Std}(k) \wedge k \neq 0) \Rightarrow \exists y (j = ky)). \end{aligned}$$

*Proof.* The right sides say that  $m \in \mathbb{Z}$  and that  $a - m$  (respectively  $j$ ) is divisible by every nonzero ordinary integer; by (3.2) this means  $a - m \in \Pi$  (respectively  $j \in \Pi$ ), which for  $m \in \mathbb{Z}$  is equivalent to  $\text{ct}(a) = m$ . For an explicit converse in the divisibility test, write  $j = p + n$  with  $p \in \Pi$  and  $n \in \mathbb{Z}$ . If  $n \neq 0$ , take the ordinary positive integer  $k = |n| + 1$ . The equation  $j = ky$  would imply  $n = k \text{ct}(y)$ , impossible since  $0 < |n| < k$ . If  $n = 0$ , division of every coefficient of  $p$  by any nonzero ordinary  $k$  preserves its positive support and gives the required  $y$ .  $\square$

All displayed quantifiers range over  $\mathbf{Oz}$ ;  $\text{Std}$  is an abbreviation for the finite existential formula already given. In particular, the universal divisibility test is one first-order formula, not an infinite conjunction indexed externally by the ordinary integers. The symbol  $\text{ct}$  occurs only on the left to identify the graph that the ring-language formula defines.

These definitions use a universal quantifier and an implication. Neither source 01 nor source 05 claims an existential (Diophantine) definition of  $\Pi$  or of the graph of  $\text{ct}$  ([question 21.4](#)). Source 06 gives both:  $\Pi = \{x : \exists y x^2 = 2y^2\}$  ([theorem 11.2](#)), and the graph of  $\text{ct}$  is Diophantine ([theorem 12.8](#)); source 07 adds a Diophantine definition of the complement  $\{\text{ct} \neq 0\}$  ([theorem 12.2](#)). The corollary above remains as the first-order route.

*Remark 10.8* (An abstract principle [01]). The proof of [theorem 10.3](#) needs less than all surreal numbers. Let  $R$  be any discretely ordered ring containing  $\mathbb{Z}$  in which every solution of  $U^2 - 2V^2 = 1$  is ordinary. Every element of  $R$  bounded by an ordinary integer is ordinary, by discreteness and a finite comparison with the integers in the bounding interval; so the same proof shows that  $\Psi_n$  defines  $\mathbb{Z}^n$  in  $R^n$ , and likewise for (10.6). The omnific rings considered here satisfy the Pell hypothesis by [corollary 6.9](#).

## 10.5 Sums of squares and induction

*Remark 10.9* (Squares and Peano arithmetic [01, 02, 05]). The positive omnific integer  $\omega - 1$  is not a sum of any finite number of squares in  $\mathbf{Oz}$ : applying  $\text{ct}$  to such an identity gives  $-1 = \sum_j m_j^2$  with  $m_j \in \mathbb{Z}$ . So the nonnegative cone of  $\mathbf{Oz}$  does not satisfy Peano arithmetic. A direct failed induction instance, independent of formalizing Lagrange's theorem in arithmetic, is given next. This is compatible with the classical fact that integer parts of real closed fields satisfy open induction [6, 3]; open induction is not full arithmetic.

**Theorem 10.10** (A failed existential induction instance [05]). *On the nonnegative part of  $\mathbf{Oz}$ , consider the formula*

$$\Phi(t) : \exists x, y \geq 0 (x^2 = 2y^2 + 1 \wedge x > t \wedge y > 0), \quad (10.8)$$

*existential with a quantifier-free ordered-ring matrix. Over  $\mathbf{Oz}_{\geq 0}$ ,*

$$\Phi(0), \quad \forall t \geq 0 (\Phi(t) \Rightarrow \Phi(t+1)), \quad \neg \Phi(\omega),$$

*and  $\Phi(t)$  holds exactly for the ordinary nonnegative integers  $t$ .*

*Proof.* The pair  $(3, 2)$  witnesses  $\Phi(0)$ . Given witnesses  $x, y$  for  $\Phi(t)$ , set  $x' = 3x + 4y$ ,  $y' = 2x + 3y$ . The identity of lemma 10.1 gives  $(x')^2 = 2(y')^2 + 1$ ; positivity and discreteness give  $x, y \geq 1$ , so  $x' > x + 1 > t + 1$  and  $y' > 0$ . This proves the successor step for every omnific  $t$ . Pell rigidity makes every witness ordinary, so  $x$  cannot exceed an infinite  $t$  such as  $\omega$ . Conversely an ordinary nonnegative  $t$  has a larger positive ordinary Pell witness by lemma 10.1. By proposition 2.5 these are the only two kinds of nonnegative omnific integers.  $\square$

By Shepherdson's theorem, an integer part of a real closed ordered field satisfies open induction on its nonnegative cone, that is, induction for quantifier-free ordered-ring formulas with parameters [6, 51]. In a divisible set-sized Hahn workspace  $H$ , the ring  $\mathbf{Oz}_H$  is an integer part of the real closed field of real Hahn series with exponents in  $H$ : the floor formula in theorem 2.7 only truncates support and changes the constant, so stays in this workspace.

Here is the passage to the full  $\mathbf{Oz}$  in detail. Fix a quantifier-free formula  $\theta(t, \mathbf{a})$  whose base case and successor implication hold in  $\mathbf{Oz}_{\geq 0}$ , and suppose it fails at some  $b \geq 0$ . Choose  $H$  to contain the supports of the finite parameter tuple  $\mathbf{a}$  and of  $b$ . Quantifier-free formulas agree between an ordered subring and its ambient ring: polynomial values, equality, order and Boolean operations are preserved. Thus the base case and successor implication hold in  $(\mathbf{Oz}_H)_{\geq 0}$ , whereas  $\theta(b, \mathbf{a})$  still fails there, contradicting Shepherdson's theorem. This proves each instance of the schema without treating the proper class  $\mathbf{Oz}$  as a set-sized model [05]. There is no conflict with theorem 10.10:  $\Phi$  is existential, not quantifier free. The sentence "every nonnegative  $t$  has a larger positive Pell solution" is true over  $\mathbb{Z}$  and false over  $\mathbf{Oz}$ ; thus  $\mathbf{Oz}$  is not an elementary extension of  $\mathbb{Z}$  as an ordered ring.

*Remark 10.11* (An induction failure in the pure ring language [merge]). Induction on the nonnegative cone also fails for an existential formula of the pure ring language, without order in its matrix: the existential formula  $\text{Std}(t)$  of theorem 10.4 satisfies  $\text{Std}(0)$  and  $\forall t (\text{Std}(t) \Rightarrow \text{Std}(t+1))$  over  $\mathbf{Oz}$  but not  $\text{Std}(\omega)$ , because it defines  $\mathbb{Z}$ , which contains 0 and is closed under  $t \mapsto t+1$ , and  $\omega \notin \mathbb{Z}$ . The induction variable is still restricted to the nonnegative cone; the displayed successor assertion is stronger, since it holds for every  $t \in \mathbf{Oz}$ .

One can also keep every existential witness inside that cone. In the language  $\{0, 1, +, \cdot\}$  of the nonnegative semiring, use

$$\exists x, y, z_1, z_2, z_3, z_4 \quad x^2 = 2y^2 + 1 \wedge x = t^2 + \sum_{j=1}^4 z_j^2.$$

For each ordinary  $t \geq 0$ , the Pell sequence and the four-square theorem supply nonnegative ordinary witnesses. Conversely, any witnesses in the nonnegative cone are also witnesses in  $\mathbf{Oz}$ , so force  $t \in \mathbb{Z}$ . This order-free existential formula therefore defines precisely  $\mathbb{N}$  inside  $\mathbf{Oz}_{\geq 0}$ , and has the same base case, successor step and failure at  $\omega$ . No subtraction or order predicate is needed in its matrix.

Separately, over the whole ring, the formula  $\text{Std}(t)$  also shows that  $\mathbf{Oz}$  is not an elementary extension of  $\mathbb{Z}$  in the ring language: the sentence  $\forall t \text{Std}(t)$  holds in  $\mathbb{Z}$  and fails in  $\mathbf{Oz}$ .

## 11 Diophantine arithmetic in omnific rings

This section and the next two are built from sources 06 and 07, written independently on September 23, 2026 (section 1.2). The two are twins: they share the order-free Gaussian system, the interjective polynomial and the Pell divisibility argument, and each has decisive results the other lacks. Source 06 is the base. It gives the Diophantine definition of  $\Pi$ , the Diophantine graph of  $\text{ct}$ , the homomorphism theorem and the reconstruction package; source 07 gives the constant-term detector, the ideal test, intermediate rings and number-field coefficients. Shared results are printed once with both sources named. Labels of this material carry the sub-prefix `odg:def:`.

### 11.1 General nonnegative-support rings

Let  $\mathbb{k}$  be a field of characteristic zero and  $\Gamma$  an ordered abelian group. For a set-sized  $\Gamma$  these constructions take place in an ordinary Hahn field. When  $\Gamma = \mathbf{No}$ , every series still has set-sized support; the same finite algebra is interpreted under the class convention below, rather than treating  $\mathbf{No}$  as a set. The notation  $\omega^\gamma$  is formal for an abstract group  $\Gamma$  and is a Conway monomial in the actual surreal instances. Consider series  $f = \sum_\gamma a_\gamma \omega^\gamma$  with  $a_\gamma \in \mathbb{k}$  and reverse-well-ordered support in  $\Gamma$ , added coefficientwise and multiplied by convolution: the Hahn field with the exponent order reversed, written in the  $\omega$ -convention of section 1.4 [06, 07]. Put

$$\begin{aligned} \mathcal{A}_{\mathbb{k}}(\Gamma) &= \{f : \text{supp}(f) \subseteq \Gamma^{\geq 0}\}, & \Pi_{\mathbb{k}}(\Gamma) &= \{f : \text{supp}(f) \subseteq \Gamma^{> 0}\}, \\ \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma) &= \mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma) & \text{for a unital subring } \mathfrak{o} \subseteq \mathbb{k}. \end{aligned}$$

As in proposition 2.2, the constant coefficient  $\text{ct}$  is a ring retraction of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  onto  $\mathbb{k}$  with kernel  $\Pi_{\mathbb{k}}(\Gamma)$ , restricting to a retraction of  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  onto  $\mathfrak{o}$ . Following source 07, an *intermediate ring* for  $\mathfrak{o} \subseteq \mathbb{k}$  is a ring  $A$  with

$$\mathfrak{o} \subseteq A \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma), \quad A \cap \mathbb{k} = \mathfrak{o}; \quad (11.1)$$

no inclusion  $\text{ct}(A) \subseteq \mathfrak{o}$  is required. This is weaker than being the full ring  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ . For example,  $A = \mathbb{Z}[\omega + \frac{1}{2}] \subseteq \mathbb{R}[\omega] = \mathcal{A}_{\mathbb{R}}(\mathbb{Z})$  has  $A \cap \mathbb{R} = \mathbb{Z}$ : a nonconstant polynomial in  $\omega + \frac{1}{2}$  still has positive degree. Nevertheless  $\text{ct}(\omega + \frac{1}{2}) = \frac{1}{2} \notin \mathbb{Z}$ . Thus the constant-term argument for  $\mathcal{R}_{\mathfrak{o}}$  must not be silently applied to every intermediate ring. The instances used in this report are

$$\begin{aligned} \mathcal{B}_{\mathbb{R}} &= \mathcal{A}_{\mathbb{R}}(\mathbf{No}), & \mathcal{B}_{\mathbb{C}} &= \mathcal{A}_{\mathbb{C}}(\mathbf{No}), & \Pi &= \Pi_{\mathbb{R}}(\mathbf{No}), & \Pi_{\mathbb{C}} &= \Pi_{\mathbb{C}}(\mathbf{No}), \\ \mathbf{Oz} &= \mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \mathbf{No}), & \mathbf{Oz}[i] &= \mathcal{R}_{\mathbb{Z}[i]}(\mathbb{C}, \mathbf{No}), \end{aligned}$$

and  $\mathbf{Oz}_{\Gamma} = \mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \Gamma)$  of section 20. For  $\Gamma = \mathbb{Z}$  every reverse-well-ordered subset of  $\mathbb{Z}^{\geq 0}$  is finite, so  $\mathcal{A}_{\mathbb{k}}(\mathbb{Z}) = \mathbb{k}[\omega]$  and  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \mathbb{Z}) = \mathfrak{o} + \omega\mathbb{k}[\omega]$  [06]. For  $\Gamma = \mathbb{R}$  supports such as  $\{1, \frac{1}{2}, \frac{1}{3}, \dots\}$  occur, and no proof below uses a least positive exponent. For  $\Gamma = \{0\}$  the ideal is zero and  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma) = \mathfrak{o}$ ; the statements in this section include that case.

**Lemma 11.1** (Degree, units and algebraic elements [06, 07]). *For nonzero  $f, g \in \mathcal{A}_{\mathbb{k}}(\Gamma)$ ,  $\deg_{\omega}(fg) = \deg_{\omega} f + \deg_{\omega} g$ , and  $\mathcal{A}_{\mathbb{k}}(\Gamma)^{\times} = \mathbb{k}^{\times}$ . If a finite product of elements of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is a nonzero constant, every factor is a nonzero constant. If  $Q \in \mathbb{k}[T]$  has degree  $n \geq 1$  and  $f \in \mathcal{A}_{\mathbb{k}}(\Gamma)$  is nonconstant, then  $\deg_{\omega} Q(f) = n \deg_{\omega} f > 0$ . Consequently every element of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  algebraic over  $\mathbb{k}$  lies in  $\mathbb{k}$ , and a polynomial over  $\mathbb{k}$  with no root in  $\mathfrak{o}$  has no root in any intermediate ring for  $\mathfrak{o} \subseteq \mathbb{k}$ .*

*Proof.* Write  $\alpha = \deg_{\omega} f$  and  $\beta = \deg_{\omega} g$ . Any exponents contributing to a coefficient of  $fg$  satisfy  $\gamma \leq \alpha$  and  $\delta \leq \beta$ . Equality  $\gamma + \delta = \alpha + \beta$  forces  $\gamma = \alpha$  and  $\delta = \beta$ , so the coefficient

there is the nonzero product of the two leading coefficients. No term of larger degree occurs. If  $fg = 1$ , the nonnegative degrees  $\alpha, \beta$  have sum zero, hence both vanish. A nonnegative-support form of degree zero is constant; conversely every nonzero constant has a constant inverse. The same degree argument applies to a finite product equal to a nonzero constant.

If  $f$  is nonconstant, then  $d = \deg_\omega f > 0$ . For  $Q(T) = a_n T^n + \cdots + a_0$  with  $a_n \neq 0$ , the leading coefficient of  $a_n f^n$  is  $a_n \text{lc}(f)^n \neq 0$  at exponent  $nd$ . Every nonzero term  $a_j f^j$  with  $j < n$  has degree  $jd < nd$ , so that coefficient cannot cancel. This proves  $\deg_\omega Q(f) = nd > 0$  without ordering the coefficient field. An algebraic nonconstant element would make such a nonzero polynomial vanish, a contradiction. A root of a nonconstant polynomial over  $\mathbb{k}$  in  $A$  must therefore lie in  $A \cap \mathbb{k} = \mathfrak{o}$ ; a nonzero constant polynomial has no root anywhere.  $\square$

In particular  $\mathbf{Oz}[i]^\times = \{\pm 1, \pm i\}$ , while the units of  $\mathcal{B}_\mathbb{C} = \mathbb{C} \oplus \Pi_\mathbb{C}$  are all of  $\mathbb{C}^\times$ ; the larger ring is the one in which the Pell factorization below takes place [06].

**Conventions.** A subset of  $A^n$  is *Diophantine* if it is defined by a formula  $\exists \mathbf{y} \bigwedge_{j=1}^m P_j(\mathbf{x}, \mathbf{y}) = 0$  with  $P_j \in \mathbb{Z}[\mathbf{X}, \mathbf{Y}]$  [06, 07]. This is the finite-system convention; single-polynomial presentations are stated explicitly, and negative atoms such as  $y \neq 0$  are not allowed. The numeral 2 is the ring term  $1 + 1$ . No formula below names an order, a monomial,  $\sqrt{2}$  or a coefficient map. For set-sized  $\Gamma$  the statements are ordinary first-order model theory. For  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  a formula is interpreted by relativizing its quantifiers to the class, formula by formula; no uniform satisfaction predicate for the class structure, no set of all surreals and no class of all class automorphisms is formed, and a statement about arbitrary automorphisms is a statement about each map individually [06, 07].

## 11.2 The purely infinite ideal is Diophantine

**Theorem 11.2** (A quadratic definition of the ideal [06]). *Let  $\mathfrak{o} \subseteq \mathbb{k}$  be a subring such that  $\sqrt{2} \in \mathbb{k}$  but 2 is not a square in  $\text{Frac}(\mathfrak{o})$ . For  $x \in \mathcal{R}_\mathfrak{o}(\mathbb{k}, \Gamma)$ ,*

$$x \in \Pi_\mathbb{k}(\Gamma) \iff \exists y \in \mathcal{R}_\mathfrak{o}(\mathbb{k}, \Gamma) \quad x^2 = 2y^2. \quad (11.2)$$

*In particular the parameter-free formula  $\text{Inf}(x) :\Leftrightarrow \exists y (x^2 = 2y^2)$  defines  $\Pi$  in  $\mathbf{Oz}$  and  $\Pi_\mathbb{C}$  in  $\mathbf{Oz}[i]$ , and the same formula works for every  $\Gamma$ .*

*Proof.* If  $x \in \Pi_\mathbb{k}(\Gamma)$ , then  $y = x/\sqrt{2}$  has the same support, so it lies in  $\Pi_\mathbb{k}(\Gamma)$  and witnesses the equation. Conversely, apply  $\text{ct}$  to a solution:  $a = \text{ct}(x)$  and  $b = \text{ct}(y)$  lie in  $\mathfrak{o}$  and  $a^2 = 2b^2$ . If  $b \neq 0$ , then  $(a/b)^2 = 2$  in  $\text{Frac}(\mathfrak{o})$ , which is excluded; so  $b = 0$  and then  $a = 0$ , that is,  $x \in \Pi_\mathbb{k}(\Gamma)$ . For  $\mathfrak{o} = \mathbb{Z}$  the hypothesis is the irrationality of  $\sqrt{2}$ ; for  $\mathfrak{o} = \mathbb{Z}[i]$ , a square root of 2 in  $\mathbb{Q}(i)$  would lie in  $\mathbb{Q}(i) \cap \mathbb{R} = \mathbb{Q}$ .  $\square$

The full positive-support part is needed for the forward implication: its closure under multiplication by  $1/\sqrt{2}$  supplies the witness. The conclusion does not hold for arbitrary intermediate rings. For example,  $\omega \in A \cap \Pi_\mathbb{R}(\mathbb{Z})$  for  $A = \mathbb{Z}[\omega]$ , but no  $y \in A$  satisfies  $\omega^2 = 2y^2$ : degree forces  $y$  to be linear and its leading integer coefficient would have square  $1/2$ . The nonsquare hypothesis also matters: in  $\mathcal{R}_{\mathbb{Z}[\sqrt{2}]}(\mathbb{R}, \Gamma)$  the nonzero constant  $x = \sqrt{2}$  satisfies the equation with  $y = 1$ , although it is not in the ideal.

For example  $\pi\omega^3 - \sqrt{3}\omega + 7$  does not satisfy  $\text{Inf}$ , because its constant term is 7, while  $\pi\omega^3 - \sqrt{3}\omega$  and  $\sum_{n \geq 1} \omega^{1/n}$  do; “purely infinite” includes 0 and negative elements such as  $-\omega$  [06]. The formula is existential, has no parameters and one witness, so it answers the first part

of [question 21.4](#) ([section 21.2](#)). It is compatible with [proposition 5.3](#):  $\Pi$  is stable under  $\text{ct}$ , since  $\text{ct}(\Pi) = \{0\} \subseteq \Pi$ , whereas the nonzero locus is not.

First-order definability of  $\mathbb{Z}$  in **Oz** is older. In a MathOverflow comment of 6 July 2018 the user *nombre* recognized ordinary integers by the absence of nonzero solutions of  $x^2 = 2y^2$  in a bounded interval, and observed the resulting failure of saturation [18]. Source 06 states that the equation of [theorem 11.2](#) belongs to the same obstruction and is not an unrelated discovery; the new point is its positive-existential form.

**Corollary 11.3** (Homomorphisms preserve the ideal [06]). *Let  $\mathbb{k}, \mathbb{k}'$  be fields of characteristic zero containing  $\sqrt{2}$ . Every unital ring homomorphism  $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma) \rightarrow \mathcal{R}_{\mathbb{Z}}(\mathbb{k}', \Gamma')$  maps  $\Pi_{\mathbb{k}}(\Gamma)$  into  $\Pi_{\mathbb{k}'}(\Gamma')$ , and the same holds between two rings  $\mathcal{R}_{\mathbb{Z}[i]}(\mathbb{k}, \Gamma) \rightarrow \mathcal{R}_{\mathbb{Z}[i]}(\mathbb{k}', \Gamma')$ . No injectivity or size condition on the target is needed.*

*Proof.* For  $x$  in the source ideal,  $x^2 = 2(x/\sqrt{2})^2$  with  $x/\sqrt{2}$  in the source ring. A unital homomorphism fixes the integer numeral 2, so its images satisfy the same equation. Apply [theorem 11.2](#) in the target. The homomorphism need not fix the chosen square root, the coefficient field or any monomial: the formula contains none of those parameters.  $\square$

*Remark 11.4* (Other radicands [06]). If a nonzero  $\delta \in \mathfrak{o}$  has a square root in  $\mathbb{k}$  but not in  $\text{Frac}(\mathfrak{o})$ , the same proof shows that  $\exists y (x^2 = \delta y^2)$  defines  $\Pi_{\mathbb{k}}(\Gamma)$  in  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ ; the formula is parameter-free when  $\delta$  is an integer numeral. The proof uses no ordering of  $\mathbb{k}$ , no divisibility of  $\Gamma$  and no real closedness.

### 11.3 Pell rigidity and divisible Pell coordinates

**Corollary 11.5** (Pell rigidity in intermediate rings [06, 07]). *Let  $A$  be an intermediate ring for  $\mathfrak{o} \subseteq \mathbb{k}$ . If  $u, v \in A$  satisfy  $u^2 - 2v^2 = 1$ , then  $u, v \in \mathfrak{o}$ .*

*Proof.* Extend the coefficients to  $\mathbb{k}(\sqrt{2})$ ; this does not change supports. There  $(u + \sqrt{2}v)(u - \sqrt{2}v) = 1$ , so both factors are constants by [lemma 11.1](#), and adding and subtracting them shows that  $u, v$  are constants. Their coefficients lie in  $\mathbb{k}$ , so  $u, v \in A \cap \mathbb{k} = \mathfrak{o}$ . Equivalently, apply [theorem 6.15](#) to  $E = \mathbb{Q}(\sqrt{2})$  with basis  $1, \sqrt{2}$ .  $\square$

Source 06 proves this in  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  with  $\sqrt{2} \in \mathbb{k}$ , source 07 in intermediate rings by coefficient extension; the argument is that of [corollary 6.9](#). It does not estimate  $u^2$  and  $2v^2$  separately, so cancellation between their leading terms is harmless, and  $\sqrt{2}$  occurs in the proof only, not as a parameter. For  $\mathfrak{o} = \mathbb{Z}[i]$  the conclusion is Gaussian constancy, not reality:  $(u, v) = (i, i)$  solves  $u^2 - 2v^2 = 1$ , and the Gaussian construction below uses only Gaussian constancy [06].

**Lemma 11.6** (Divisible Pell coordinates [06, 07]). *Let  $(X_k, Y_k)$  be the Pell sequence (10.1). For every ordinary  $m \geq 1$  there is  $k$  with  $1 \leq k \leq m^2$  such that  $X_k \equiv 1$  and  $Y_k \equiv 0 \pmod{m}$ ; then  $Y_k > 0$ .*

*Proof.* Let  $M = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}$ . Its inverse  $\begin{pmatrix} 3 & -4 \\ -2 & 3 \end{pmatrix}$  has integer entries, so  $M$  permutes  $(\mathbb{Z}/m\mathbb{Z})^2$ . Among the  $m^2 + 1$  points  $M^j(1, 0)$  for  $0 \leq j \leq m^2$ , two coincide, say at  $j < \ell$ . Applying  $M^{-j}$  gives  $M^{\ell-j}(1, 0) = (1, 0)$ , with  $1 \leq k = \ell - j \leq m^2$  [07]. In the ordinary integer recurrence  $X_0 = 1, Y_0 = 0$ , the first step gives  $(3, 2)$  and both coordinates remain positive, so  $Y_k > 0$ . This also covers  $m = 1$ , whose residue set has one point. Source 06 argues instead that  $3 + 2T$  is a unit of the ring  $(\mathbb{Z}/m\mathbb{Z})[T]/(T^2 - 2)$ , with inverse  $3 - 2T$ . Division by the monic polynomial  $T^2 - 2$  gives a unique representative  $a + bT$ , so the ring has  $m^2$  elements; some positive power at most  $m^2$  equals 1. Comparing the two coordinates gives the same congruences.  $\square$

No classification of Pell solutions, primitive-divisor theorem or distribution result is used [06]. Pell equations have a long history in Diophantine definability, from Davis's explicit work [20] and Denef's on polynomial rings [21] to Miller and Shlapentokh's on rings of functions [25]. Sources 06 and 07 use only the elementary fact that, after adjoining  $\sqrt{2}$ , every unit of the nonnegative-support ring is constant; source 07 stresses that no theorem about arbitrary Hahn supports is inferred from theorems about polynomial rings.

**Lemma 11.7** (Integers in principal ideals [07]). *Let  $\mathfrak{o}$  be a unital subring of a number field. For every nonzero  $a \in \mathfrak{o}$ , the ideal  $a\mathfrak{o}$  contains a nonzero ordinary integer; for a Gaussian integer  $a$  one may take  $a\bar{a}$ .*

*Proof.* Algebraicity supplies a nonzero integer polynomial vanishing at  $a$  after clearing the denominators of a rational relation. Choose one of least degree,  $A_n a^n + \cdots + A_1 a + A_0 = 0$ . If  $A_0 = 0$ , cancellation of  $a \neq 0$  gives a nonzero relation of smaller degree. Thus  $A_0 \neq 0$ , and  $A_0 = -a(A_1 + A_2 a + \cdots + A_n a^{n-1}) \in a\mathfrak{o}$ . The expression uses only integer coefficients and nonnegative powers of  $a$ , which belong to the unital subring  $\mathfrak{o}$ ; no algebraic-integrality hypothesis is needed. Changing its sign if necessary gives a positive ordinary integer in the same ideal.  $\square$

#### 11.4 An intersective polynomial

A polynomial in  $\mathbb{Z}[T]$  is *intersective* if it has a root modulo every positive integer. We use

$$\Lambda(T) = (T^2 - 13)(T^2 - 17)(T^2 - 221) = T^6 - 251T^4 + 6851T^2 - 48841. \quad (11.3)$$

It is a classical example: L   and Spencer list it in [23, Section 2, Example 4] and attribute it to Borevich and Shafarevich [22, p. 3]; Berend and Bilu study the general phenomenon of polynomials with roots modulo every integer [24]. Sources 06 and 07 both reprove the properties they need, because the definitions below use roots modulo *all* integers; the proofs are printed once.

**Proposition 11.8** (The polynomial  $\Lambda$  [06, 07]).

- (i) *For every ordinary  $N \geq 1$  there is  $r \in \mathbb{Z}$  with  $\Lambda(r) \equiv 0 \pmod{N}$ .*
- (ii)  *$\Lambda$  has no root in  $\mathbb{Q}(i)$ ; hence none in  $\mathbb{Z}$  or  $\mathbb{Z}[i]$ , and by [lemma 11.1](#) none in any intermediate ring for  $\mathbb{Z} \subseteq \mathbb{k}$  or  $\mathbb{Z}[i] \subseteq \mathbb{k}$ .*
- (iii) *For every nonzero  $v \in \mathbb{Z}[i]$  there are  $s \in \mathbb{Z}[i]$  and  $t \in \mathbb{Z}$  with  $vs = \Lambda(t)$ .*

*Proof.* (ii) A root in a field satisfies  $T^2 = d$  with  $d \in \{13, 17, 221\}$ . If  $(a + bi)^2 = d$  with  $a, b \in \mathbb{Q}$ , then  $2ab = 0$ ;  $a = 0$  gives  $d = -b^2 \leq 0$ , and  $b = 0$  would make  $d$  a rational square, which none of the three is.

(i) It suffices to find a root modulo each prime power  $p^a$  dividing  $N$  and to combine them by the Chinese remainder theorem; the chosen factor of  $\Lambda$  may depend on  $p$ . For  $N = 1$  every integer works. Let  $p$  be odd. If  $p \notin \{13, 17\}$ , then 13 and 17 are nonzero modulo  $p$ ; if neither is a square, their product 221 is, by multiplicativity of the Legendre symbol. For  $p = 13$  use  $17 \equiv 2^2$ , and for  $p = 17$  use  $13 \equiv 8^2$ . In each case some factor  $T^2 - d$  has a root  $r \not\equiv 0 \pmod{p}$ , and it lifts to every  $p^a$ : if  $r^2 \equiv d \pmod{p^j}$ , replacing  $r$  by  $r + hp^j$  changes  $(r^2 - d)/p^j$  by  $2rh$  modulo  $p$ , and  $2r$  is invertible modulo  $p$ . For  $p = 2$  use 17: the root 1 works modulo  $2^a$  for  $a \leq 3$ , since  $17 \equiv 1 \pmod{8}$ ; if an odd  $r$  satisfies  $r^2 \equiv 17 \pmod{2^a}$  with  $a \geq 3$ , then  $(r + 2^{a-1})^2 - r^2 = 2^a r + 2^{2a-2} \equiv 2^a \pmod{2^{a+1}}$ , so exactly one of  $r$  and  $r + 2^{a-1}$  is a

root modulo  $2^{a+1}$ . This step constructs a root at each power of 2; it need not preserve the previously chosen residue modulo  $2^a$ , and no compatible sequence of such choices is needed. For each prime-power divisor of  $N$ , choose one of the roots just constructed. The Chinese remainder theorem gives an integer congruent to all of them; its  $\Lambda$ -value is divisible by every such prime power, hence by their product  $N$ .

(iii) Put  $N = v\bar{v} \in \mathbb{Z}_{>0}$ , choose  $t$  by (i) with  $\Lambda(t) = Nq$ , and take  $s = q\bar{v}$ .  $\square$

The equation  $vs = \Lambda(t)$  is a positive-existential certificate of  $v \neq 0$  only when  $v$  is already known to be a Gaussian constant. It is not a definition of the nonzero elements: for  $v \in \Pi_{\mathbb{C}}$  it has no solution, since applying  $\text{ct}$  would give a root of  $\Lambda$  in  $\mathbb{Z}[i]$  [06].

### 11.5 One order-free system defining $\mathbb{Z}$ and $\mathbb{Z}[i]$

Let  $\mathcal{E}(x; u, v, w, s, t)$  be the system

$$x(u^2 - 2v^2 - 1) = 0, \quad x(v - xw) = 0, \quad x(vs - \Lambda(t)) = 0, \quad (11.4)$$

and put  $\Xi(x) :\Leftrightarrow \exists u, v, w, s, t \mathcal{E}(x; u, v, w, s, t)$ .

**Theorem 11.9** (One system for  $\mathbb{Z}$  and  $\mathbb{Z}[i]$  [06, 07]). *Let  $(\mathfrak{o}, \mathbb{k})$  be  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , let  $\Gamma$  be any ordered abelian group, and let  $A$  be an intermediate ring. Then  $\Xi$  defines  $\mathfrak{o}$  in  $A$ . In particular  $\Xi(\mathbf{Oz}) = \mathbb{Z}$  and  $\Xi(\mathbf{Oz}[i]) = \mathbb{Z}[i]$ : the same parameter-free system of three equations, of total degrees 3, 3 and 7, with five witnesses, defines both.*

*Proof.* Suppose  $\mathcal{E}(x; u, v, w, s, t)$  holds with  $x \neq 0$ . Cancelling  $x$  in the domain  $A$  gives

$$u^2 - 2v^2 = 1, \quad v = xw, \quad vs = \Lambda(t).$$

By [corollary 11.5](#),  $u, v \in \mathfrak{o}$ . If  $v = 0$ , then  $\Lambda(t) = 0$  with  $t \in A$ , which [proposition 11.8\(ii\)](#) excludes. So  $v$  is a nonzero constant, and  $xw = v$  forces  $x$  and  $w$  to be constants ([lemma 11.1](#)); hence  $x \in A \cap \mathbb{k} = \mathfrak{o}$ . If  $x = 0$  there is nothing to prove.

Conversely let  $x \in \mathfrak{o}$ . For  $x = 0$  take all witnesses zero. For  $x \neq 0$  let  $m = |x|$ , or  $m = x\bar{x}$  in the Gaussian case, so that  $m \in x\mathfrak{o}$ . By [lemma 11.6](#) choose a Pell pair  $(u, v) = (X_k, Y_k)$  with  $v > 0$  and  $m \mid v$ ; then  $w = v/x \in \mathfrak{o}$ . By intersectivity choose  $t \in \mathbb{Z}$  with  $v \mid \Lambda(t)$  and put  $s = \Lambda(t)/v \in \mathbb{Z}$ . These witnesses all lie in  $\mathfrak{o} \subseteq A$ :  $u, v, s, t$  are ordinary integers, while  $w$  may be a nonreal Gaussian integer. They satisfy the system.  $\square$

The proof uses the intermediate-ring condition, not a retraction  $A \rightarrow \mathfrak{o}$ . It therefore applies, for example, to  $\mathbb{Z}[\omega + \frac{1}{2}]$  above even though its constant coefficient need not be an integer. In fact the same proof works with any characteristic-zero coefficient field containing the indicated copy of  $\mathbb{Z}$  or  $\mathbb{Z}[i]$ : coefficient extension proves Pell rigidity, root exclusion uses  $A \cap \mathbb{k} = \mathfrak{o}$ , and the witnesses are already in  $\mathfrak{o}$ . The displayed theorem retains the two source pairs; the broader number-field version is stated in [theorem 12.14](#).

Source 07 states the theorem in this form, for both pairs and all intermediate rings. Source 06 proves the identical system (with witnesses  $h, r$  for  $s, t$ ) in  $\mathcal{R}_{\mathbb{Z}[i]}(\mathbb{C}, \Gamma)$  and  $\mathbf{Oz}[i]$ , noting that the root exclusion takes place after applying  $\text{ct}$ , so that algebraic closedness of the coefficient field is not needed. The factor  $x$  in each equation admits  $x = 0$  without a disjunction. For  $x = 1 + i$  a witness is  $u = 3, v = 2, w = 1 - i, t = 1, s = -21120$ , because  $\Lambda(1) = -42240$ ; for  $x = \omega + i$  there is none, since a nonzero Gaussian constant is not divisible by  $\omega + i$  in  $\mathcal{B}_{\mathbb{C}}$  [06, 07]. The formula contains no occurrence of  $i$  and no distinguished choice of it.

**Why the certificate is needed.** The relation  $v = xw$  alone would allow  $v = 0$  and hence every  $x$ . Writing  $v \neq 0$  would give an existential formula with an inequation, not a Diophantine one, and replacing it by  $vz = 1$  would force  $v$  to be a unit, which most Pell coordinates are not. The equation  $vs = \Lambda(t)$  excludes  $v = 0$  because  $\Lambda$  has no root in  $A$ , and accepts every nonzero ordinary  $v$  because  $\Lambda$  is interseptive [06, 07]. Over a ring containing  $i$  the three equations cannot be combined into one sum of squares ( $1^2 + i^2 = 0$ ), and no single-polynomial Gaussian definition is claimed. In the proof one may take the Pell index at most  $m^2$  and  $0 \leq t < v$ ; these are existence bounds, not claims of optimal or practical complexity [07]. The ring restriction is essential: in a field containing a root  $r$  of  $\Lambda$ ,  $\Xi(x)$  holds for every  $x$ , with  $u = 1$ ,  $v = w = s = 0$  and  $t = r$ ; so  $\Xi$  defines nothing in **No** or **No** $[i]$  [07].

## 11.6 Single real polynomials and the natural numbers

In the real case a conjunction of equations becomes one equation as a sum of squares. The report already prints the quartic guards of sources 01 and 05 (theorems 10.3 and 10.4), and source 07's quartic, which is source 05's up to the substitution described in remark 10.5. Source 06 proposed a quintic as its principal real formula; it is recorded here with the comparison.

*Remark 11.10* (Source 06's quintic, and a comparison [06]). Let  $\mathbb{k}$  be an ordered field containing  $\sqrt{2}$ . In  $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$ , and in particular in **Oz**,

$$x \in \mathbb{Z} \iff \exists u, v, w, s_1, \dots, s_4 \quad x \left[ (u^2 - 2v^2 - 1)^2 + (v - xw)^2 + \left( u - 2 - \sum_{j=1}^4 s_j^2 \right)^2 \right] = 0. \quad (11.5)$$

Indeed, for  $x \neq 0$  the three squares vanish; Pell rigidity makes  $u, v$  integers,  $u = 2 + \sum s_j^2 \geq 2$  excludes  $v = 0$ , and then  $xw = v \neq 0$  forces  $x \in \mathbb{Z}$ . Conversely, for  $0 \neq x \in \mathbb{Z}$  take a Pell pair with  $|x| \mid Y_k$  (lemma 11.6),  $w = Y_k/x$ , and four squares for  $X_k - 2 \geq 1$ . The polynomial has total degree exactly five (the monomial  $xu^4$ ) and seven witnesses. Source 06 tabulates complete ordinary witnesses for  $x \in \{1, 2, \pm 3, 5, 7\}$ , for instance  $u = 17$ ,  $v = 12$ ,  $w = 4$ ,  $(s_1, \dots, s_4) = (3, 2, 1, 1)$  for  $x = 3$ ; for  $x = \omega$  or  $\omega + 1$  there are none in **Oz**. It is dominated by theorem 10.3 (degree four, five witnesses, uniform in  $n$ ) and by theorem 10.4 and source 07's (10.7) (degree four, six witnesses); source 06's question on the least degree is thereby partly answered (section 21.2). Its mechanism, divisibility  $v = xw$  of a nonzero Pell coordinate, is the one that survives in the Gaussian system theorem 11.9. Source 06 restricts the quintic to the real case, correctly: over **Oz** $[i]$  it accepts  $x = \omega$  [merge], with  $u = 1$ ,  $v = 0$ ,  $w = 1$ ,  $s_1 = i$ ,  $s_2 = e^{-i\pi/4}\omega^{1/2}$ ,  $s_3 = s_4 = 0$ . Then  $u^2 - 2v^2 - 1 = 0$ ,  $v - xw = -\omega$  and  $u - 2 - s_1^2 - s_2^2 = -1 + 1 + i\omega = i\omega$ , and  $(-\omega)^2 + (i\omega)^2 = 0$ . Source 07 records the analogous collapse of its quartic (remark 10.5).

**Corollary 11.11** (The natural numbers [06, 07]). *Let Std be any of the Diophantine definitions of  $\mathbb{Z}$  in **Oz** (theorem 10.4, theorem 11.9, (11.5)). Then  $x \in \mathbb{N} \iff \text{Std}(x) \wedge \exists r_1, \dots, r_4 \ x = r_1^2 + r_2^2 + r_3^2 + r_4^2$ , where  $0 \in \mathbb{N}$ .*

*Proof.* The forward implication is Lagrange's theorem. Conversely  $\text{Std}(x)$  makes  $x$  an ordinary integer, and a sum of squares in the ordered ring **Oz** is nonnegative (alternatively, apply ct).  $\square$

Source 07 restricts the  $r_j$  by its constant-definition formula as well; in **Oz** this is harmless and unnecessary. Remark 10.11 gives a variant in the language  $\{0, 1, +, \cdot\}$  of the nonnegative semiring, with every witness in **Oz** $_{\geq 0}$ , that defines  $\mathbb{N}$  inside **Oz** $_{\geq 0}$ .

## 12 The constant term: detector, ideal test and graph

### 12.1 A root detector for augmentations

The following algebraic statement, due to source 07, isolates why a single polynomial can detect a nonzero constant term.

**Theorem 12.1** (Augmentation root detector [07]). *Let  $\mathcal{H}$  be a commutative unital  $\mathbb{k}$ -algebra with a  $\mathbb{k}$ -algebra homomorphism  $\varepsilon : \mathcal{H} \rightarrow \mathbb{k}$  that is the identity on  $\mathbb{k}$ . Let  $\mathfrak{o} \subseteq \mathbb{k}$  be a unital subring and  $A = \mathfrak{o} + \ker \varepsilon$ . Let  $f \in \mathfrak{o}[T]$  satisfy:*

- (a)  *$f$  has no root in  $\mathfrak{o}$ ;*
- (b)  *$f$  has a root  $r \in \mathbb{k}$ ;*
- (c) *for every nonzero  $c \in \mathfrak{o}$  there is  $b \in \mathfrak{o}$  with  $f(b) \in c\mathfrak{o}$ .*

Then for every  $a \in A$ ,

$$\varepsilon(a) \neq 0 \iff \exists s, t \in A \quad as = f(t). \quad (12.1)$$

No hypothesis that  $\mathcal{H}$  is a domain, reduced, complete or Henselian is needed.

*Proof.* Identify  $\mathbb{k}$  with its image in  $\mathcal{H}$ ; the retraction makes this embedding injective. Then  $A = \varepsilon^{-1}(\mathfrak{o})$ , since every  $z \in \mathcal{H}$  decomposes as  $\varepsilon(z) + (z - \varepsilon(z))$ , with the second summand in the kernel. If  $as = f(t)$  and  $\varepsilon(a) = 0$ , applying  $\varepsilon$  gives  $f(\varepsilon(t)) = 0$  with  $\varepsilon(t) \in \mathfrak{o}$ , contrary to (a). Conversely let  $c = \varepsilon(a) \neq 0$ , choose  $b$  by (c), and factor  $f(T) = (T - r)Q(T)$  in  $\mathbb{k}[T]$ . Put

$$\alpha = \frac{b - r}{c}, \quad t = r + \alpha a, \quad s = \alpha Q(t). \quad (12.2)$$

Since  $t - r = \alpha a$ , we get  $as = (t - r)Q(t) = f(t)$ . Moreover  $\varepsilon(t) = r + \alpha c = b \in \mathfrak{o}$  and  $c\varepsilon(s) = \varepsilon(as) = f(b) \in c\mathfrak{o}$ , so  $\varepsilon(s) \in \mathfrak{o}$ ; thus  $s, t \in \mathfrak{o} + \ker \varepsilon = A$ . Only the nonzero scalar  $c$  is inverted, in the field  $\mathbb{k}$ .  $\square$

The augmentation need not be part of the ring language: the conclusion is a ring-language formula that detects its kernel [07], with the coefficients of  $f$  as parameters. In the applications below  $f \in \mathbb{Z}[T]$ , so these are integer numerals and the formula is parameter-free.

For a concrete example with zero divisors, take  $\mathcal{H} = \mathbb{R} \times \mathbb{R}$ , embed  $\mathbb{R}$  diagonally, and let  $\varepsilon(u, v) = u$ . Then  $A = \mathbb{Z} \times \mathbb{R}$ . The zero divisor  $a = (1, 0)$  has nonzero augmentation and admits the certificate

$$t = (0, \sqrt{13}), \quad s = (-48841, 0), \quad as = (-48841, 0) = \Lambda(t).$$

Thus invertibility of  $a$  is not required; the cancellation in the proof takes place in  $\mathbb{k}$ , after applying  $\varepsilon$ .

**Theorem 12.2** (Detecting a nonzero constant term [07]). *Let  $(\mathfrak{o}, \mathbb{k})$  be  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$  and  $A = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ . For every  $a \in A$ ,*

$$\Theta(a) : \iff \exists s, t \quad (as = \Lambda(t)) \iff \text{ct}(a) \neq 0. \quad (12.3)$$

In particular this holds in  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ .

*Proof.* Apply [theorem 12.1](#) with  $\mathcal{H} = \mathcal{A}_{\mathbb{k}}(\Gamma)$ ,  $\varepsilon = \text{ct}$  and  $f = \Lambda$ . Hypothesis (a) is [proposition 11.8\(ii\)](#); (b) holds with  $r = \sqrt{13}$ . For (c), given  $c \neq 0$  in  $\mathfrak{o}$  choose a positive integer  $m \in c\mathfrak{o}$  ( $|c|$  or  $c\bar{c}$ ) and  $b \in \mathbb{Z}$  with  $\Lambda(b) \in m\mathbb{Z} \subseteq c\mathfrak{o}$ .  $\square$

With  $r = \sqrt{13}$  the witnesses (12.2) are explicit:  $t = r + \alpha a$  and  $s = \alpha(2r + \alpha a)(t^2 - 17)(t^2 - 221)$ , with  $\text{ct}(t) = b$  and  $\text{ct}(s) = \Lambda(b)/c$ . Neither  $r$  nor  $\alpha$  need lie in  $A$ ; the witnesses do, because their constant coefficients are right [07].

**Proposition 12.3** (Support control of the witnesses [07]). *Let  $S = \text{supp}(a)$  and  $S^* = S \cup \{0\}$ . The witnesses above satisfy  $\text{supp}(t) \subseteq S^*$  and  $\text{supp}(s) \subseteq \bigcup_{j=0}^5 jS^*$ , where  $jS^*$  is the  $j$ -fold sumset and  $0S^* = \{0\}$ . If  $a$  is nonconstant,  $\deg_\omega t = \deg_\omega a$  and  $\deg_\omega s = 5 \deg_\omega a$ .*

*Proof.* The witness  $t$  is affine in  $a$  and  $s$  is a polynomial of degree five in  $a$  with constant coefficients, so the support inclusions follow from finite convolution. Finite sumsets of reverse-well-ordered supports are again reverse well ordered, and each fixed exponent has only finitely many contributions in each such convolution. All exponents remain in  $\Gamma$ ; the scalars  $r, \alpha$  change coefficients only. Since  $b \in \mathfrak{o}$  and  $r \notin \mathfrak{o}$ ,  $\alpha \neq 0$ ; the leading coefficients of the two polynomials in  $a$  are  $\alpha$  and  $\alpha^6$ , and lemma 11.1 gives the degrees.  $\square$

The construction divides no series by  $a$ , expands no  $(1 + h)^{-1}$  and takes no limit; it applies unchanged to supports accumulating towards zero and to exponent groups of any rank [07].

The predicate  $\Theta$  detects a nonzero constant term: it rejects the nonzero element  $\omega$  and accepts the nonunit  $1 + \omega$ .

**Example 12.4** (Certificates [07]). For  $a = \omega$  (or  $a = \sum_{n \geq 1} \omega^{1/n}$ ) there is no certificate: a solution of  $as = \Lambda(t)$  would give  $\Lambda(\text{ct}(t)) = 0$  in  $\mathbb{Z}$ . For  $a = 1 + \omega$  take  $c = 1$ ,  $b = 0$ , so  $\alpha = -\sqrt{13}$  and

$$t = -\sqrt{13}\omega, \quad s = 13(\omega - 1)(13\omega^2 - 17)(13\omega^2 - 221), \quad (1 + \omega)s = \Lambda(t);$$

the constant coefficients are  $\text{ct}(t) = 0$  and  $\text{ct}(s) = -48841 = \Lambda(0)$ , and the irrational coefficient  $-\sqrt{13}$  of  $t$  is allowed. For  $a = 2 + \omega$  one may take  $b = 1$ , since  $\Lambda(1) = -42240$  is even: then  $\alpha = (1 - \sqrt{13})/2$ ,  $t = 1 + \alpha\omega$ , and  $\text{ct}(s) = -21120$ .

*Remark 12.5* (A root in the coefficient field is needed [07]). Let  $A = \mathbb{Z} + \omega\mathbb{Q}[\omega]$ , an intermediate ring for  $\mathbb{Z} \subseteq \mathbb{Q}$  with  $\Gamma = \mathbb{Z}$ . The proof of theorem 11.9 still shows that  $\Xi$  defines  $\mathbb{Z}$  in  $A$  (it needs only Pell rigidity and the absence of integer roots of  $\Lambda$ ). The detector fails: evaluation at  $\omega = -1$  is a surjective homomorphism  $A \rightarrow \mathbb{Q}$  (the element  $-q\omega$  maps to  $q$ ) whose kernel is  $(1 + \omega)A$ , because division by  $1 + \omega$  in  $\mathbb{Q}[\omega]$  leaves a quotient with the same constant coefficient. So  $A/(1 + \omega)A \cong \mathbb{Q}$  has no root of  $\Lambda$ , and  $(1 + \omega)s = \Lambda(t)$  has no solution although  $\text{ct}(1 + \omega) = 1$ . Here  $\mathbb{Q}$  contains no root of  $\Lambda$ , so hypothesis (b) of theorem 12.1 fails. The detector is not asserted for arbitrary intermediate rings, which need not contain the polynomial witnesses. For example, take  $A = \mathbb{Z}[\omega] \subseteq \mathbb{R}[\omega]$ . Here  $\text{ct}(A) \subseteq \mathbb{Z}$  and the ambient coefficient field contains  $\sqrt{13}$ , but a certificate  $(1 + \omega)s = \Lambda(t)$  in  $A$  would give  $\Lambda(t(-1)) = 0$  in  $\mathbb{Z}$  upon evaluation at  $\omega = -1$ . Thus the full pullback condition  $A = \mathfrak{o} + \ker \varepsilon$  also matters.

## 12.2 The largest ideal without a root

**Theorem 12.6** (One polynomial tests every ideal [07]). *Let  $(\mathfrak{o}, \mathbb{k})$  be  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ ,  $A = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  and  $I = \Pi_{\mathbb{k}}(\Gamma)$ . For every ideal  $J$  of  $A$ ,*

$$(\exists t \in A \quad \Lambda(t) \in J) \iff J \not\subseteq I. \quad (12.4)$$

*Thus  $\Lambda$  has no root in  $A/J$  exactly when  $J \subseteq I$ , and  $I$  is the unique largest ideal of  $A$  whose quotient has no root of  $\Lambda$ .*

*Proof.* If  $J \subseteq I$  and  $\Lambda(t) \in J$ , applying  $\text{ct}$  gives  $\Lambda(\text{ct}(t)) = 0$  in  $\mathfrak{o}$ , which is impossible. If  $J \not\subseteq I$ , choose  $a \in J$  with  $\text{ct}(a) \neq 0$ ; [theorem 12.2](#) gives  $as = \Lambda(t)$ , which lies in  $J$ . Since  $A/I \cong \mathfrak{o}$  has no root,  $I$  has the property and contains every ideal that has it.  $\square$

For a class ideal of a proper-class ring the displayed equivalence is the primary formulation; no set of quotient classes is formed. The case  $J = A$  is allowed: the zero ring has roots of every polynomial [07].

**Corollary 12.7** (A universal definition of the ideal [07]). *In the rings of [theorem 12.6](#),  $I = \{a : \forall s, t (as \neq \Lambda(t))\}$ , and  $\text{ct}(a) = \text{ct}(b)$  if and only if  $\forall s, t ((a - b)s \neq \Lambda(t))$ .*

*Proof.* Negate [theorem 12.2](#), and use additivity of  $\text{ct}$ .  $\square$

As a definition of  $I$  this universal formula is superseded by the existential one of [theorem 11.2](#); it is kept as the negation of the detector. Together, [theorems 11.2](#) and [12.2](#) make both  $I$  and its complement Diophantine. Source 07 also notes the external description  $I = \bigcap_{n \geq 1} nA$  for  $\mathfrak{o} = \mathbb{Z}, \mathbb{Z}[i]$ , of which [\(3.2\)](#) is the case of  $\mathbf{Oz}$ . Indeed, writing  $x = d + u$  with  $d \in \mathfrak{o}$  and  $u \in I$ , we have  $x \in nA$  exactly when  $d \in n\mathfrak{o}$ : necessity follows by applying  $\text{ct}$ , and sufficiency uses  $u/n \in I$ . An ordinary integer divisible by every positive integer is zero; applying this to both coordinates gives the same conclusion in  $\mathbb{Z}[i]$ . If instead  $\mathfrak{o}$  contains  $\mathbb{Q}$ , then  $nA = A$  for every  $n \geq 1$ , so this intersection does not detect  $I$ .

### 12.3 The graph of the constant term

**Theorem 12.8** (The graph of  $\text{ct}$  is Diophantine [06]). *Let  $(\mathfrak{o}, \mathbb{k})$  be  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$  and  $A = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ . The formula*

$$\text{CT}(x, n) : \Longleftrightarrow \Xi(n) \wedge \exists y ((x - n)^2 = 2y^2) \quad (12.5)$$

*defines the graph of  $\text{ct} : A \rightarrow \mathfrak{o}$ ; for each  $x$  there is exactly one such  $n$ . It is Diophantine, with six witnesses. In  $\mathbf{Oz}$ ,  $\Xi$  may be replaced by any Diophantine definition of  $\mathbb{Z}$ .*

*Proof.* The first clause says  $n \in \mathfrak{o}$  ([theorem 11.9](#)); by [theorem 11.2](#) the second says  $x - n \in I$ . In the decomposition  $A = \mathfrak{o} \oplus I$  the two conditions together say exactly  $n = \text{ct}(x)$ . Existence is explicit: take  $n = \text{ct}(x)$ , choose the five witnesses for  $\Xi(n)$ , and put  $y = (x - n)/\sqrt{2} \in I$ . Uniqueness concerns the output  $n$ , not the six existential witnesses: two possible outputs differ by an element of  $\mathfrak{o} \cap I = \{0\}$ .  $\square$

Source 06 proves the theorem with its own predicates, the quintic in the real case and the Gaussian system in the Gaussian case; its single real polynomial  $P(n, \dots)^2 + ((x - n)^2 - 2y^2)^2$ , with  $P$  the quintic of [\(11.5\)](#), has total degree ten and eight witnesses. Using the guard of [theorem 10.3](#) instead gives degree four:

**Corollary 12.9** (A quartic for the graph of  $\text{ct}$  [merge]). *Let  $\Psi_1(X, U, V, W_1, W_2, W_3)$  be the guard polynomial [\(10.2\)](#) with  $n = 1$ . For  $x, m \in \mathbf{Oz}$ ,*

$$m = \text{ct}(x) \iff \exists u, v, w_1, w_2, w_3, y \in \mathbf{Oz} \quad \Psi_1(m, u, v, w_1, w_2, w_3) + ((x - m)^2 - 2y^2)^2 = 0.$$

*The polynomial has integer coefficients, total degree four and six witnesses. With the six-witness quartics [\(10.6\)](#) or [\(10.7\)](#) in place of  $\Psi_1$  the same holds with seven witnesses.*

*Proof.* The polynomial is a sum of three squares of integer polynomials, since  $\Psi_1$  is itself a sum of two. In the ordered ring  $\mathbf{Oz}$  it vanishes exactly when the three squared polynomials vanish, that is, when  $\Psi_1(m, u, v, w_1, w_2, w_3) = 0$  and  $(x - m)^2 = 2y^2$ . By [theorem 10.3](#), witnesses for the first equation exist exactly when  $m \in \mathbb{Z}$ , and by [theorem 11.2](#), a witness  $y$  for the second exists exactly when  $x - m \in \Pi$ . Both together say  $m = \text{ct}(x)$ . Every term has degree at most four, and the coefficient of  $x^4$  is 1, since  $\Psi_1$  does not involve  $x$ ; hence the total degree is exactly four. The witnesses are  $u, v, w_1, w_2, w_3, y$ . The variant follows in the same way from [theorem 10.4](#) or [remark 10.5](#).  $\square$

By [remark 10.8](#) and [corollary 11.5](#) the corollary holds verbatim in  $\mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \Gamma)$  for every  $\Gamma$ . No single polynomial is claimed in the Gaussian case.

**Corollary 12.10** (The canonical splitting is definable [06]). *For the rings of [theorem 12.8](#), the exact sequence  $0 \rightarrow I \rightarrow A \xrightarrow{\text{ct}} \mathfrak{o} \rightarrow 0$  has a parameter-free definable retraction and a parameter-free definable section; its kernel and the graph of the retraction are Diophantine.*

*Proof.* The predicate  $\Xi$  defines the embedded copy of  $\mathfrak{o}$ . On that domain the section is the inclusion, with graph  $x = n$ . The retraction has graph  $\text{CT}$ , and its kernel is defined by  $\text{Inf}$ . These formulas have no parameters.  $\square$

The direct-sum decomposition is additive. In the coordinates  $x = n + u \leftrightarrow (n, u)$  its multiplication is

$$(n, u)(m, v) = (nm, nv + mu + uv),$$

so the splitting does not assert a direct product of rings. The existence of the sequence is elementary normal-form algebra; the content is the definability of the actual splitting, not merely an interpretation of  $A/I$  [06].

*Remark 12.11* (Source 07's prenex definitions; a second route [07]). Source 07 defines the graph by  $\text{CT}^\forall(a, c) : \Leftrightarrow \Xi(c) \wedge \forall s, t ((a - c)s \neq \Lambda(t))$ , with the two prenex forms

$$\exists \mathbf{u} \forall s, t [\mathcal{E}(c; \mathbf{u}) \wedge (a - c)s \neq \Lambda(t)], \quad \forall s, t \exists \mathbf{u} [\mathcal{E}(c; \mathbf{u}) \wedge (a - c)s \neq \Lambda(t)],$$

where  $\mathbf{u}$  is a quintuple of witnesses. In the first form the witnesses for  $\Xi(c)$  do not occur in the universal inequality; in the second, a single pair, for example  $(s, t) = (0, 0)$ , already produces witnesses for  $\Xi(c)$ , and conversely a fixed witness works for all  $(s, t)$ . Both are correct by [theorem 11.9](#) and [corollary 12.7](#). Source 07 claimed neither minimality nor the impossibility of an existential definition; source 06's [theorem 12.8](#) supplies one.

## 12.4 Homomorphisms preserve the constant term

**Theorem 12.12** (All homomorphisms respect the constant term [06]). *Let  $\mathbb{k}, \mathbb{k}'$  be fields of characteristic zero containing  $\sqrt{2}$ , and  $\Gamma, \Gamma'$  ordered abelian groups. Every unital ring homomorphism  $f : \mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma) \rightarrow \mathcal{R}_{\mathbb{Z}}(\mathbb{k}', \Gamma')$  satisfies  $\text{ct}(f(x)) = \text{ct}(x)$ . If  $i \in \mathbb{k} \cap \mathbb{k}'$  and  $f : \mathcal{R}_{\mathbb{Z}[i]}(\mathbb{k}, \Gamma) \rightarrow \mathcal{R}_{\mathbb{Z}[i]}(\mathbb{k}', \Gamma')$  is a unital ring homomorphism, then  $\text{ct}(f(x)) = \sigma(\text{ct}(x))$ , where  $\sigma$  is the identity or complex conjugation on  $\mathbb{Z}[i]$ , according as  $f(i) = i$  or  $f(i) = -i$ . No size restriction on the target is imposed.*

*Proof.* Write  $x = n + a$  with  $n \in \mathfrak{o}$  and  $a$  in the ideal. By [corollary 11.3](#),  $f(a)$  lies in the target ideal. A unital homomorphism fixes ordinary integers. In the Gaussian case  $f(i)^2 = -1$ , and by [lemma 11.1](#) a root of  $T^2 + 1$  in the target ring is a constant, hence  $\pm i$ . Taking constant terms of  $f(x) = f(n) + f(a)$  gives the claim.  $\square$

Source 06 states the theorem for the rings of its arithmetic sections; the proof uses only that both coefficient fields contain  $\sqrt{2}$ . It generalizes [proposition 3.5](#) (endomorphisms of  $\mathbf{Oz}$ ) and source 07's statement that ring automorphisms of  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  preserve  $\mathfrak{o}$ ,  $I$  and  $\text{ct}$  up to  $\sigma$ , which source 07 derives from the parameter-free definitions. It does not say that homomorphisms kill the ideal: the identity does not. Preservation of the ideal, annihilation of the ideal and pointwise fixation of the real coefficients are three different assertions [06]. For general homomorphisms the proof uses preservation of equations and their existential witnesses; parameter-free first-order definability alone would only ensure invariance under automorphisms. For set-sized  $\Gamma'$  it is consistent with [cited theorem 3.6](#): every unital homomorphism  $\mathbf{Oz} \rightarrow \mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \Gamma')$  is then  $x \mapsto \text{ct}(x)1$ . Source 07's version concerns ring automorphisms of the coefficient-restricted ring; it does not classify field automorphisms of  $\mathbf{No}$  or  $\mathbf{No}[i]$ , nor claim that an arbitrary field automorphism preserves  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$ .

## 12.5 Number-field coefficient rings

The polynomial  $\Lambda$  cannot serve every number field, since a field containing  $\sqrt{13}$  has a root of it. A tailored polynomial of the same shape exists for each fixed number field [07].

**Lemma 12.13** (A tailored intersective polynomial [07]). *For every number field  $K$  there are distinct primes  $p, q$  such that  $\Lambda_K(T) = (T^2 - p)(T^2 - q)(T^2 - pq)$  has no root in  $K$  and has a root modulo every positive integer.*

*Proof.* A number field has only finitely many quadratic subfields: it lies in a finite Galois extension of  $\mathbb{Q}$ , whose Galois group has finitely many subgroups. By Dirichlet's theorem choose a prime  $p \equiv 1 \pmod{8}$  with  $\mathbb{Q}(\sqrt{p}) \not\subseteq K$ , and then a prime  $q \equiv 1 \pmod{8p}$  with  $\mathbb{Q}(\sqrt{q}), \mathbb{Q}(\sqrt{pq}) \not\subseteq K$ ; only finitely many  $q$  are excluded, since distinct positive squarefree radicands give distinct quadratic fields. Both arithmetic progressions have residue 1 coprime to their modulus, so Dirichlet supplies infinitely many candidates; also  $q \equiv 1 \pmod{p}$  ensures  $q \neq p$ . Then none of  $p, q, pq$  is a square in  $K$ . For an odd prime  $\ell \notin \{p, q\}$ , one of  $p, q, pq$  is a nonzero square modulo  $\ell$ , as in [proposition 11.8](#). At  $\ell = p$  use  $q \equiv 1 \pmod{p}$ . At  $\ell = q$ , quadratic reciprocity and  $p \equiv 1 \pmod{4}$  give  $(p/q) = (q/p) = (1/p) = 1$ . The nonzero roots lift to all odd prime powers, and at 2 the number  $p \equiv 1 \pmod{8}$  has a root modulo every  $2^a$  by the lifting step of [proposition 11.8](#). The Chinese remainder theorem finishes the proof. Dirichlet's theorem and reciprocity are classical inputs [26].  $\square$

**Theorem 12.14** (Coefficient rings in number fields [07]). *Let  $\mathfrak{o}$  be a unital subring of a number field  $K$ , let  $\mathbb{k} \supseteq K$  be a field of characteristic zero, and let  $A$  be an intermediate ring for  $\mathfrak{o} \subseteq \mathbb{k}$ . Let  $\Xi_K$  be  $\Xi$  with  $\Lambda$  replaced by  $\Lambda_K$ . Then  $\Xi_K$  defines  $\mathfrak{o}$  in  $A$ . If moreover  $A = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  and  $\mathbb{k}$  contains a root of  $\Lambda_K$ , then  $\text{ct}(a) \neq 0 \Leftrightarrow \exists s, t \text{ (as } a = \Lambda_K(t))$ , and the ideal test, the universal definition of the ideal and both prenex definitions of the graph of  $\text{ct}$  hold with  $\Lambda_K$ .*

*Proof.* The proof of [theorem 11.9](#) used Pell rigidity ([corollary 11.5](#)), the absence of roots of the polynomial in  $\mathfrak{o}$  ([lemmas 11.1](#) and [12.13](#)), a nonzero integer in every  $x\mathfrak{o}$  ([lemma 11.7](#)) and roots modulo all integers. More explicitly, for  $x \neq 0$  the system forces  $u, v \in \mathfrak{o}$  by Pell rigidity and  $v \neq 0$  by root exclusion. Then  $xw = v$  forces  $x$  to have degree zero, hence to lie in  $A \cap \mathbb{k} = \mathfrak{o}$ . Conversely, if  $0 \neq x \in \mathfrak{o}$ , choose  $m > 0$  in  $x\mathfrak{o}$  and ordinary Pell coordinates  $u, v$  with  $0 < v \in m\mathbb{Z}$ . Put  $w = v/x \in \mathfrak{o}$ , choose  $t \in \mathbb{Z}$  with  $\Lambda_K(t) \in v\mathbb{Z}$ , and put  $s = \Lambda_K(t)/v \in \mathbb{Z}$ . These are witnesses already in  $\mathfrak{o}$ ;  $x = 0$  is admitted by the factors  $x$  in the defining equations. No integrality or finite-generation hypothesis on  $\mathfrak{o}$  is used. For the detector, given  $c \neq 0$  in  $\mathfrak{o}$  choose a positive integer  $m \in c\mathfrak{o}$  and  $b \in \mathbb{Z}$  with  $\Lambda_K(b) \in m\mathbb{Z}$ , and apply [theorem 12.1](#) with any root of  $\Lambda_K$  in  $\mathbb{k}$ . The ideal test and the graph use only the detector and the constant definition.  $\square$

The theorem covers subrings of a number field that are not finitely generated. It does not give one polynomial uniform in all number fields, and does not apply as stated to algebraic extensions of  $\mathbb{Q}$  of infinite degree: no polynomial of this kind is root-free in a field containing all algebraic numbers [07].

*Remark 12.15* (Existential definitions over number-field coefficients [merge]). The formula  $\exists y (x^2 = 2y^2)$  of [theorem 11.2](#) fails when  $\sqrt{2} \in \text{Frac}(\mathfrak{o})$ , for instance for  $\mathfrak{o} = \mathbb{Z}[\sqrt{2}]$ . In the setting of [theorem 12.14](#) choose a prime  $\delta$  with  $\mathbb{Q}(\sqrt{\delta}) \not\subseteq K$ ; it exists because  $K$  has finitely many quadratic subfields and distinct primes give distinct fields  $\mathbb{Q}(\sqrt{\delta})$ . Then  $\delta$  is not a square in  $K \supseteq \text{Frac}(\mathfrak{o})$ , since  $z^2 = \delta$  with  $z \in K$  would give  $\mathbb{Q}(\sqrt{\delta}) = \mathbb{Q}(z) \subseteq K$ . If  $\sqrt{\delta} \in \mathbb{k}$ , [remark 11.4](#) shows that  $\exists y (x^2 = \delta y^2)$  defines  $\Pi_{\mathbb{k}}(\Gamma)$  in  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ , and  $\Xi_K(n) \wedge \exists y ((x-n)^2 = \delta y^2)$  defines the graph of  $\text{ct}$ : the first clause says  $n \in \mathfrak{o}$ , by [theorem 12.14](#) applied to the intermediate ring  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ , and the second says  $x - n \in \Pi_{\mathbb{k}}(\Gamma)$ . Both definitions are Diophantine; they improve source 07's universal and prenex formulas whenever  $\sqrt{\delta} \in \mathbb{k}$ , and need no root of  $\Lambda_K$  in  $\mathbb{k}$ .

In particular, the detector hypothesis in [theorem 12.14](#) already suffices for these existential conclusions. If  $\Lambda_K(r) = 0$  in the field  $\mathbb{k}$ , one of its three factors vanishes, so  $r^2 = \delta$  for some  $\delta \in \{p, q, pq\}$ . Each of these radicands is nonsquare in  $K$ , by [lemma 12.13](#). The preceding argument therefore applies with this  $\delta$ , including the composite choice  $pq$ : primality was only a way to find a suitable radicand. Thus the ideal has a one-witness Diophantine definition, its complement has the two-witness detector, and the graph of  $\text{ct}$  has a six-witness Diophantine definition under exactly the stated detector hypotheses. This is a further consequence of combining sources 06 and 07. The integer  $\delta$  is chosen once for the fixed coefficient fields; the formulas contain its numeral, not a parameter naming  $r$ , and no uniform choice across all number fields is asserted.

## 13 Coefficient reconstruction and logical consequences

### 13.1 The multiplier ring of the ideal

Throughout this subsection  $\Gamma \neq 0$ ,  $\mathfrak{o} \subseteq \mathbb{k}$  is a unital subring,  $\mathcal{R} = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ ,  $I = \Pi_{\mathbb{k}}(\Gamma)$ ,  $\mathcal{A} = \mathcal{A}_{\mathbb{k}}(\Gamma)$ , and  $\mathcal{F} = \text{Frac}(\mathcal{R})$ , viewed inside the Hahn field of all series with reverse-well-ordered support in  $\Gamma$  [06]. Put  $(I : I)_{\mathcal{F}} = \{r \in \mathcal{F} : rI \subseteq I\}$ .

**Lemma 13.1** ([06]).  $\mathcal{A} \subseteq \mathcal{F}$ .

*Proof.* Choose  $g \in \Gamma^{>0}$ . For  $a \in \mathcal{A}$  both  $\omega^g$  and  $a\omega^g$  lie in  $I \subseteq \mathcal{R}$ , and  $\omega^g \neq 0$ , so  $a = (a\omega^g)/\omega^g \in \mathcal{F}$ . This also covers constants  $c \in \mathbb{k}$ : although  $c$  need not belong to  $\mathcal{R}$ , the fraction  $(c\omega^g)/\omega^g$  does. No common positive lower bound on the support of  $a$  is needed.  $\square$

**Theorem 13.2** (The multiplier identity [06]). *Inside  $\mathcal{F}$ ,*

$$(I : I)_{\mathcal{F}} = \mathcal{A}, \quad \mathbb{k} = \{0\} \cup ((I : I)_{\mathcal{F}})^{\times}. \quad (13.1)$$

*Proof.* A nonnegative and a strictly positive exponent add to a strictly positive exponent, so  $\mathcal{A}I \subseteq I$ . Conversely let  $r \in \mathcal{F} \setminus \mathcal{A}$  and choose a negative exponent  $h$  in  $\text{supp}(r)$ ; it need not be the least, and there need not be a least. Then  $\omega^{-h} \in I$ , and the coefficient of  $r\omega^{-h}$  at exponent zero is the nonzero coefficient of  $\omega^h$  in  $r$ , so  $r\omega^{-h} \notin I$ . Multiplication by a single monomial translates support injectively; no other coefficient contributes at zero and no cancellation can remove it. This witnesses the failure to be a multiplier. The second identity follows from  $\mathcal{A}^{\times} = \mathbb{k}^{\times}$  ([lemma 11.1](#)).  $\square$

An integral domain interprets its fraction field by pairs  $(a, b)$  with  $b \neq 0$ , where  $(a, b) \sim (c, d)$  means  $ad = bc$ . Addition and multiplication are represented by  $(ad + bc, bd)$  and  $(ac, bd)$ , and the original ring by  $a \mapsto (a, 1)$ . The denominator  $bd$  is nonzero because the ring is a domain. For the proper classes “quotient” means this definable presentation, with set-valued representatives supplied by the surreal quotient, and no proper class is used as an element of another [06]. With  $\text{Inf}$  from [theorem 11.2](#), put

$$\begin{aligned} \text{Mult}(a, b) &: \iff b \neq 0 \wedge \forall x (\text{Inf}(x) \Rightarrow \exists y (\text{Inf}(y) \wedge ax = by)), \\ \text{Coeff}(a, b) &: \iff b \neq 0 \wedge (a = 0 \vee (a \neq 0 \wedge \text{Mult}(a, b) \wedge \text{Mult}(b, a))). \end{aligned}$$

The first says that the fraction  $a/b$  multiplies  $I$  into  $I$ , and is invariant under change of representative. Indeed  $ax = by$  is equivalent in  $\mathcal{F}$  to  $(a/b)x = y$ ; all  $x, y$  in this formula range over the original ring, where  $\text{Inf}$  defines  $I$ . The second formula handles zero separately and otherwise requires both  $a/b$  and its inverse  $b/a$  to be multipliers, so [theorem 13.2](#) identifies it with  $a/b \in \mathbb{k}$ . It does not assert that every nonzero multiplier is a coefficient:  $\omega^g \in \mathcal{A}$  is not a unit there for  $g > 0$ .

**Corollary 13.3** (Internal reconstruction [06]). *Assume  $\Gamma \neq 0$  and the hypotheses of [theorem 11.2](#) (for instance in the real and Gaussian rings). Then  $\mathcal{A}$  and the actual embedded coefficient field  $\mathbb{k}$  are parameter-free first-order definable in the interpreted fraction field, and the coefficient map  $\mathcal{A} \rightarrow \mathbb{k}$  has the definable graph  $\{(a, c) : a \in \mathcal{A}, c \in \mathbb{k}, a - c \in I\}$ .*

*Proof.* Use  $\text{Mult}$  and  $\text{Coeff}$  in the interpreted fraction field. The last condition means that  $a - c$  is the image of an element of the original ring satisfying  $\text{Inf}$ . Every  $a \in \mathcal{A}$  has a unique decomposition  $a = c + u$  with  $c \in \mathbb{k}$  and  $u \in I$ , so the displayed relation is exactly the coefficient map. This recovers the embedded field, not just an abstract isomorphic copy.  $\square$

The same reconstruction works whenever  $I$  has a parameter-free definition in  $\mathcal{R}$ : the multiplier argument imposes no condition on  $\mathfrak{o}$  beyond being a unital subring of  $\mathbb{k}$ . In particular, for  $\Gamma \neq 0$  the integer-radical definitions in [remark 12.15](#) recover the coefficient field and map under the number-field detector hypotheses, even when 2 is a square in  $\text{Frac}(\mathfrak{o})$ . Substitute that ideal predicate for  $\text{Inf}$  in the two formulas above; no square root is named in them.

The formulas do not mention monomials; monomials are used only to prove that they define the right sets, and  $\sqrt{2}$  is not a hidden parameter. The multiplier formula contains a universal quantifier, so the reconstruction is first order and is not claimed positive existential [06]. Two boundaries are necessary. If  $\Gamma = 0$ , then  $I = 0$  and every fraction multiplies it, so the formula recovers  $\text{Frac}(\mathfrak{o})$  rather than  $\mathbb{k}$ . And  $\mathcal{F}$  need not be the whole Hahn field: for  $\Gamma = \mathbb{Z}$  it is the rational function field  $\mathbb{k}(\omega)$ ; the multiplier identity holds in  $\mathcal{F}$  without any such equality [06]. For example, in  $\mathbb{Z} + \omega\mathbb{R}[\omega]$  the ideal  $\omega\mathbb{R}[\omega]$  is defined by  $\text{Inf}$ , the fraction field is  $\mathbb{R}(\omega)$ , the multiplier ring is  $\mathbb{R}[\omega]$  and its units are  $\mathbb{R}^\times$ ; the reconstruction phenomenon is not an artifact of proper-class size. With  $\Gamma = \mathbb{R}$  and  $f = \sum_{n \geq 1} \omega^{1/n}$ , the ideal formula is witnessed by  $f/\sqrt{2}$ , and the multiplier proof uses one negative exponent, not a least one [06].

## 13.2 Reconstructing the surreal field

[Theorem 4.6](#) gives  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ , with a common monomial denominator for every set of surreals; applied to the set of real and imaginary parts it gives  $\text{Frac}(\mathbf{Oz}[i]) = \mathbf{No}[i]$ . Source 06 proves the same statement (its common-denominator lemma), printed once as [theorem 4.6](#); it uses bounds for sets of exponents only, and asserts no common denominator for a proper class of inputs.

**Theorem 13.4** (Reconstruction of the real structure [06]). *From the pure ring  $\mathbf{Oz}$  one can parameter-free interpret the field  $\mathbf{No}$  and define in it the canonical subfield  $\mathbb{R}$ , the order, the ring  $\mathcal{O}_{\text{fin}}$  of finite elements, the ideal  $\mathfrak{m}$  of infinitesimals and the graph of the standard part  $\text{st} : \mathcal{O}_{\text{fin}} \rightarrow \mathbb{R}$ .*

*Proof.* Interpret the fraction field, which is  $\mathbf{No}$ . By [corollary 13.3](#) the predicate  $\text{Coeff}$  defines its subfield  $\mathbb{R}$ . Since  $\mathbf{No}$  is real closed [1, 2],  $x > 0 \Leftrightarrow x \neq 0 \wedge \exists z (z^2 = x)$  defines the order, with  $x < y$  obtained by applying it to  $y - x$ . Here  $z$  ranges over the interpreted field, not just  $\mathbf{Oz}$ . Put  $\mathcal{O}_{\text{fin}}(x) :\Leftrightarrow \exists r \in \mathbb{R} (r > 0 \wedge -r \leq x \leq r)$  and  $\mathfrak{m}(x) :\Leftrightarrow \forall r \in \mathbb{R} (r > 0 \Rightarrow -r < x < r)$ . A nonzero surreal is bounded by a real number exactly when its leading exponent is nonpositive. For a finite surreal, removing the coefficient at zero leaves either zero or a remainder with a largest, strictly negative exponent; that remainder is therefore smaller in absolute value than every positive real. This proves existence of standard part, including zero by taking coefficient zero. It is unique because a nonzero real is not infinitesimal. So  $\text{ST}(x, r) :\Leftrightarrow \mathcal{O}_{\text{fin}}(x) \wedge r \in \mathbb{R} \wedge \mathfrak{m}(x - r)$  defines the standard part.  $\square$

The two constant-term operations must not be confused ([remark 2.3](#)):  $\text{ct}$  reads the constant term of a number with nonnegative exponents,  $\text{st}$  that of a finite number, with nonpositive exponents. For example  $\text{ct}(\omega + 7) = 7$  and  $\text{st}(7 + \omega^{-1}) = 7$ , but  $\omega + 7$  is not finite [06].

**Corollary 13.5** (The value group [06]). *The natural ordered value group of  $\mathbf{No}$  is interpretable from  $\mathbf{Oz}$  as  $\mathbf{No}^\times / \mathcal{O}_{\text{fin}}^\times$ , ordered by  $v(x) \geq v(y) \Leftrightarrow x/y \in \mathcal{O}_{\text{fin}}$ .*

*Proof.* For  $x \neq 0$ , membership in  $\mathcal{O}_{\text{fin}}^\times$  means that both  $x$  and  $x^{-1}$  are finite. Thus  $x \sim y$  iff  $x/y \in \mathcal{O}_{\text{fin}}^\times$  is definable, and multiplication descends to the quotient. If the leading exponents of  $x, y$  are  $g, h$ , then  $x/y$  has leading exponent  $g - h$ . It is finite exactly when  $g \leq h$ , and a finite unit exactly when  $g = h$ . Consequently the displayed relation is a well-defined total order on the quotient, compatible with multiplication; in additive valuation notation  $v(xy) = v(x) + v(y)$  it corresponds externally to  $v(x) = -g$ .  $\square$

This recovers the value group as an interpreted quotient, not a canonical cross-section of monomials, and does not show that the  $\omega$ -map, the simplicity order, birthdays or the surreal exponential are definable [06]. The external representative  $-g$  also explains the proper-class convention: it is a surreal number representing a valuation class, not a proper class used as an element. Its use to verify the quotient does not put a definable monomial cross-section into the interpreted language. The fraction-field construction can be phrased as a canonical bi-interpretation between the pure ring  $\mathbf{Oz}$  and the field  $\mathbf{No}$  with a predicate for  $\mathbf{Oz}$ ; the essential content is the reconstruction of the *canonical* coefficient field.

### 13.3 Automorphisms of $\mathbf{Oz}$

**Theorem 13.6** (Automorphisms fix the real coefficients [06]). *Every unital ring automorphism  $f$  of  $\mathbf{Oz}$  extends uniquely to a field automorphism  $\tilde{f}(a/b) = f(a)/f(b)$  of  $\mathbf{No}$ . It preserves the order,  $\mathbb{R}$ ,  $\mathcal{O}_{\text{fin}}$  and  $\mathfrak{m}$ , fixes every real number, and commutes with  $\text{st}$ .*

*Proof.* The extension is well defined by cross multiplication and unique because every element of  $\mathbf{No}$  is a fraction. It preserves the parameter-free definable sets of [theorem 13.4](#), so it restricts to an order-preserving automorphism of  $\mathbb{R}$  fixing  $\mathbb{Q}$ . If  $\tilde{f}(r) \neq r$ , a rational strictly between  $r$  and  $\tilde{f}(r)$  contradicts order preservation. The definable graph of  $\text{st}$ , with its real coordinate fixed, gives  $\text{st} \circ \tilde{f} = \text{st}$  on  $\mathcal{O}_{\text{fin}}$ .  $\square$

The statement concerns automorphisms. A universal multiplier formula need not be preserved by a nonsurjective embedding, so the argument does not apply unchanged to endomorphisms; [theorem 12.12](#) applies to all unital homomorphisms [06].

**Example 13.7** (Coefficient rigidity is not total rigidity [06]). The map  $S_2(\sum_g a_g \omega^g) = \sum_g a_g \omega^{2g}$  is a field automorphism of  $\mathbf{No}$  with inverse  $S_{1/2}$ , since doubling exponents is an order automorphism of  $(\mathbf{No}, +)$ ; it preserves supports and  $\mathbf{Oz}$ , fixes real coefficients and sends  $\omega$  to  $\omega^2$ . Nonnegative exponents remain nonnegative, and the coefficient at zero is unchanged, so the inverse also preserves  $\mathbf{Oz}$ . This is an exponent substitution, not the squaring map  $x \mapsto x^2$ ; for example  $S_2(1 + \omega) = 1 + \omega^2$ . Thus  $\text{Aut}(\mathbf{Oz})$  is not trivial.

The dilations  $S_a$  are studied in the repository's surcomplex automorphism report (`docs/surcomplex/surcomplex-field-automorphisms`), and the single-dilation report (`docs/surcomplex/single-dilation-hahn-support`) defines the coefficient field and the constant coefficient from one dilation of a field; [theorem 13.4](#) starts instead from the pure ring  $\mathbf{Oz}$  and its Diophantine ideal. [Theorem 13.6](#) and [example 13.7](#) are also relevant to the questions on automorphisms of  $\mathbf{Oz}$  studied in the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers`.

### 13.4 The Gaussian ring does not determine the real axis

The Gaussian system and the multiplier theorem recover  $\mathbf{No}[i] = \text{Frac}(\mathbf{Oz}[i])$  and its actual coefficient field  $\mathbb{C}$ , but not the real form  $\mathbf{No}$  [06].

**Proposition 13.8** (All coefficient automorphisms occur [06]). *Every field automorphism  $\sigma$  of  $\mathbb{C}$  extends coefficientwise to a field automorphism  $\hat{\sigma}(\sum_g c_g \omega^g) = \sum_g \sigma(c_g) \omega^g$  of  $\mathbf{No}[i]$  preserving  $\mathbf{Oz}[i]$ . Every automorphism of  $\mathbf{Oz}[i]$  extends to  $\mathbf{No}[i]$  and preserves  $\mathbb{C}$  setwise, so restriction to  $\mathbb{C}$  is surjective and  $\sigma \mapsto \hat{\sigma}$  is a composition-preserving section.*

*Proof.* A coefficient automorphism preserves supports and the finite sums and products defining Hahn convolution; its inverse is  $\widehat{\sigma^{-1}}$ . The coefficients of each product are finite sums, so no continuity or analytic regularity of  $\sigma$  is needed. It fixes  $\mathbb{Z}$  and sends  $i$  to  $\pm i$ , so it preserves  $\mathbb{Z}[i]$  and hence  $\mathbf{Oz}[i]$ . An automorphism of  $\mathbf{Oz}[i]$  extends to the fraction field and preserves the first-order coefficient predicate of [corollary 13.3](#). Finally  $\widehat{\sigma \circ \rho} = \hat{\sigma} \circ \hat{\rho}$  coefficientwise, and restriction of  $\hat{\sigma}$  to  $\mathbb{C}$  is  $\sigma$ , giving the stated section.  $\square$

For proper-class fields this is a statement about individual maps and their compositions, not a construction of a set of all class automorphisms.

Let  $\eta : \mathbf{No} \rightarrow \mathbb{R}$  send a surreal to the coefficient of  $\omega^0$  in its normal form. It is additive on  $\mathbf{No}$  but not multiplicative ( $\eta(\omega) = \eta(\omega^{-1}) = 0$ ,  $\eta(1) = 1$ ), and is used only as an additive map on the exponent group. Put  $\chi(g) = \exp(i\pi\eta(g)/2) \in \mathbb{C}^\times$ , the ordinary complex exponential of an ordinary real multiple of  $i$ ; then  $\chi(g+h) = \chi(g)\chi(h)$ ,  $\chi(0) = 1$  and  $\chi(1) = i$ .

**Lemma 13.9** (A phase twist preserving  $\mathbf{Oz}[i]$  [06]). *The map  $\tau(\sum_g c_g \omega^g) = \sum_g c_g \chi(g) \omega^g$  is an automorphism of  $\mathbf{No}[i]$  that fixes  $\mathbb{C}$  pointwise, preserves every support and  $\mathbf{Oz}[i]$ , and satisfies  $\tau(\omega) = i\omega$ .*

*Proof.* All values of  $\chi$  are nonzero, so supports are unchanged; additivity is clear. If  $a+b=g$ , the contribution of  $(a,b)$  to the coefficient at  $g$  is multiplied by  $\chi(a)\chi(b) = \chi(g)$ , so  $\tau$  commutes with convolution;  $\chi^{-1}$  gives the inverse. Constants sit at exponent zero and are fixed. A Gaussian omnific integer keeps its positive support and its Gaussian integer constant term, since  $\chi(0) = 1$ ; the inverse has the same property. These operations are coefficientwise on

set-supported normal forms, without an infinite analytic summation. The exponential in  $\chi$  is only the ordinary complex exponential of a real scalar.  $\square$

The character-twist construction and its use to move the real axis while fixing complex coefficients are already in the repository's surcomplex automorphism report (`docs/surcomplex/surcomplex-field-automorphisms`); source 06 credits it there. The additional fact needed here is that the twist preserves  $\mathbf{Oz}[i]$ .

**Theorem 13.10** (No definable real axis or conjugation [06]). *The subfield  $\mathbf{No} \subseteq \mathbf{No}[i]$  is not first-order definable in the structure  $(\mathbf{No}[i], \mathbf{Oz}[i])$ , even with every complex number named by a constant, and neither is conjugation. The subring  $\mathbf{Oz} \subseteq \mathbf{Oz}[i]$  is not definable in the pure ring  $\mathbf{Oz}[i]$ , even with parameters from  $\mathbb{Z}[i]$ .*

*Proof.* The automorphism  $\tau$  preserves the field, the predicate  $\mathbf{Oz}[i]$  and every complex constant, but sends  $\omega \in \mathbf{No}$  to  $i\omega \notin \mathbf{No}$ ; definable sets are invariant under automorphisms fixing the parameters. Conjugation does not commute with  $\tau$ :  $\tau(\bar{\omega}) = i\omega$  while  $\tau(\omega) = -i\omega$ . Finally the restriction of  $\tau$  to  $\mathbf{Oz}[i]$  fixes  $\mathbb{Z}[i]$  and moves  $\omega \in \mathbf{Oz}$  outside  $\mathbf{Oz}$ . Naming every element of  $\mathbb{C}$  does not affect the first obstruction:  $\tau$  fixes each of them, and every individual formula uses only finitely many names. For the restriction to  $\mathbf{Oz}[i]$ , the parameters  $\mathbb{Z}[i]$  already lie in that ring; arbitrary complex constants need not.  $\square$

There is no conflict with the recovery of  $\mathbb{C}$ : the twist fixes  $\mathbb{C}$  pointwise. The coefficient field and a distinguished real form are different structure. Naming a suitable involution would suffice, since its fixed ring can be taken, but that gives the real structure explicitly instead of reconstructing it [06].

### 13.5 Logical consequences

The equational transfer of [theorem 5.1](#) is proved again by source 06 (its retraction-transfer lemma), by source 07 (positive-existential conservation) and by source 13 (equation-only specialization and the Hilbert's-tenth-problem transfer), for all rings  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  with parameters in  $\mathfrak{o}$  and, in the Gaussian case, with  $\mathbb{Z}[i]$  in place of  $\mathbb{Z}$ . It is printed once, as [theorem 5.1](#) and [corollary 5.2](#); the proof is the same for every  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ . Transfer does not make solutions constant:  $y = x$  has many nonconstant solutions [07]. Both the parameters and the free tuple must be in the coefficient subring for this two-way transfer. Existential witnesses may range over the entire ring: applying ct supplies coefficient witnesses.

**Theorem 13.11** (Diophantine subsets of  $\mathbb{Z}^n$  [06]). *Let  $\mathbb{k}$  be an ordered field containing  $\sqrt{2}$ . A subset  $S \subseteq \mathbb{Z}^n$ , regarded as a subset of  $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)^n$  (in particular of  $\mathbf{Oz}^n$ ), is Diophantine there with integer coefficients if and only if it is computably enumerable.*

*Proof.* If  $S$  is defined by equations in  $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$ , its trace on  $\mathbb{Z}^n$ , which is  $S$ , is defined by the same equations over  $\mathbb{Z}$ , by transfer; enumerating integer witnesses shows that it is computably enumerable. Conversely, by the MRDP theorem [7] (in its integer form, by sign coding and sums of squares)  $S$  is defined over  $\mathbb{Z}$  by an integer system  $\Phi_S$ . In  $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$  impose  $\text{Std}(x_1) \wedge \cdots \wedge \text{Std}(x_n) \wedge \Phi_S(\mathbf{x})$ , with  $\text{Std}$  the quintic (11.5); once the free tuple is standard, transfer makes the remaining condition equivalent to  $\Phi_S$  over  $\mathbb{Z}$ . The guards exclude nonstandard free tuples; no guards on the auxiliary witnesses of  $\Phi_S$  are needed, since ct sends any such witnesses to integer ones. In the real ring the conjunction can be written as one sum of squares.  $\square$

Here  $S$  is required to consist of standard tuples. No computability classification of arbitrary subsets of the ring is claimed, and arbitrary surreal parameters are not finite inputs to an algorithm [06].

**Generality of the classification [merge].** With the finite-system convention for Diophantine definitions, the same equivalence holds in  $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$  for every characteristic-zero field  $\mathbb{k}$ , without an ordering or a square root of 2. Apply [theorem 12.14](#) with  $\mathfrak{o} = \mathbb{Z}$  and  $K = \mathbb{Q}$  to obtain the integer guard  $\Xi_{\mathbb{Q}}$ ; its first assertion needs no root of  $\Lambda_{\mathbb{Q}}$  in  $\mathbb{k}$ . Replace Std by this guard in the proof, and use the same retraction in both directions. All formula coefficients are integers. A single sum-of-squares equation is not claimed over an unordered field. This is a consequence of combining the order-free definition with source 06’s transfer argument.

**Corollary 13.12** (Standard arithmetic is definable [06, 07]). *The structures  $(\mathbb{Z}, +, \cdot, 0, 1)$  and  $(\mathbb{N}, +, \cdot, 0, 1)$  are parameter-free definable in the pure ring  $\mathbf{Oz}$ , and every arithmetic sentence has an effective translation obtained by relativizing its quantifiers to Std and to [corollary 11.11](#). For a set-sized intermediate ring  $A$  for  $\mathbb{Z} \subseteq \mathbb{R}$ , true arithmetic therefore reduces many-one to  $\text{Th}(A)$ ; so  $\text{Th}(A)$  is undecidable and, being complete, is not recursively axiomatizable.*

*Proof.* The translation replaces  $\exists x$  by  $\exists x(\text{Std}(x) \wedge \dots)$  and  $\forall x$  by  $\forall x(\text{Std}(x) \Rightarrow \dots)$ ; induction on formulas proves it correct. For  $A$  use  $\Xi$  ([theorem 11.9](#)) in place of Std, and the same four-square definition of  $\mathbb{N}$ , which is valid in  $A$  because sums of squares are nonnegative in the ordered ring  $\mathcal{A}_{\mathbb{R}}(\Gamma)$ . A recursive axiomatization of a complete theory would decide each sentence by searching for a proof of it or of its negation.  $\square$

For  $\mathbf{Oz}$  the same holds sentence by sentence under the class convention; no definable global satisfaction predicate for the class ring is asserted. Source 07 credits the non-axiomatizability of omnific arithmetic to Jeřábek’s 2011 MathOverflow answer [19], and source 06 cites the same answer for the logical limitations of  $\mathbf{Oz}$ ; the undecidability of integer-coefficient solvability over  $\mathbf{Oz}$  follows from transfer alone ([corollary 5.2](#)). None of these consequences is presented as new, and none is a new undecidability theorem about  $\mathbb{Q}$  or about number fields.

**Theorem 13.13** (What equations cannot define [06, 07]). *Let  $\Gamma \neq 0$ , let  $\mathfrak{o} \subseteq \mathbb{k}$  be a unital subring and  $\mathcal{R} = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ . No positive-existential formula with parameters in  $\mathfrak{o}$  defines a subset of  $\mathcal{R}$  that contains a nonzero element of  $\Pi_{\mathbb{k}}(\Gamma)$  and excludes 0. In particular  $\mathcal{R} \setminus \{0\}$ ,  $\mathcal{R} \setminus \mathfrak{o}$  and  $\Pi_{\mathbb{k}}(\Gamma) \setminus \{0\}$  are not so definable, and in the real case neither is the strictly positive cone.*

*Proof.* The composite  $\mathcal{R} \xrightarrow{\text{ct}} \mathfrak{o} \hookrightarrow \mathcal{R}$  is an endomorphism fixing  $\mathfrak{o}$ , and positive-existential formulas with parameters fixed by an endomorphism are preserved by it (as in [theorem 5.1](#)). If such a formula held at a nonzero  $a \in \Pi_{\mathbb{k}}(\Gamma)$ , it would hold at the image  $\text{ct}(a) = 0$ . Each displayed set contains  $\omega^g$  for  $g \in \Gamma^{>0}$ , which is positive in the real case, and excludes 0.  $\square$

The parameter restriction is essential to this endomorphism argument: for a fixed nonzero  $h \in I$ , the equation  $x = h$  defines  $\{h\}$  with parameter  $h$ , which is not fixed by ct. In the real case “strictly positive” means  $x > 0$ , so zero is excluded. This generalizes [proposition 5.3](#). It is consistent with the Diophantine definitions of  $\Pi$  and of  $\{\text{ct} \neq 0\}$ : the first contains 0, and the second contains no element of  $\Pi$ . The detector avoids the obstruction precisely because it defines  $\text{ct}(a) \neq 0$ , not  $a \neq 0$ :  $\omega$  is nonzero but divides no value  $\Lambda(t)$  [07]. The set  $\mathcal{R} \setminus \{0\}$  has the quantifier-free definition  $x \neq 0$ ; what fails is a positive-existential definition. Source 07 left open whether  $\Pi$  itself has one; [theorem 11.2](#) of source 06 shows that it has.

**Proposition 13.14** (A quantifier-free lower bound [06]). *In  $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$  with  $\Gamma \neq 0$ , neither  $\Pi_{\mathbb{k}}(\Gamma)$  nor  $\mathbb{Z}$  is definable by a quantifier-free ring formula in one variable, even with arbitrary parameters; the same holds for the ideal and  $\mathbb{Z}[i]$  in  $\mathcal{R}_{\mathbb{Z}[i]}(\mathbb{k}, \Gamma)$ . Hence the complete theory of such a set-sized ring does not eliminate quantifiers in the ring language.*

*Proof.* In an infinite domain a nonzero one-variable polynomial has finitely many roots. Each atomic equality is therefore true either everywhere (the zero polynomial) or at finitely many points; finite Boolean combinations of finite or cofinite sets remain finite or cofinite. This also covers negated equalities and any finite tuple of ring parameters. Fix  $g \in \Gamma^{>0}$ . The ideal contains the infinitely many distinct monomials  $\omega^{kg}$ ,  $k \geq 1$ , and its complement contains the nonzero ordinary integers. The coefficient subring is infinite and its complement contains those same monomials. Neither set is finite or cofinite. If the complete theory eliminated quantifiers, the coefficient-subring definition from [theorem 12.14](#), with  $K = \mathbb{Q}$  or  $\mathbb{Q}(i)$ , would be equivalent to a quantifier-free formula, contradicting what was just proved.  $\square$

This is a first lower bound on logical complexity only; it does not show that degree four, or five witnesses, is optimal [06].

**Theorem 13.15** (Omitted types [07; also 06]). *Let  $A$  be a set-sized intermediate ring for  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , or a set-sized ring as in [theorem 12.14](#), and let  $\Xi_A$  be the corresponding constant-definition formula. The computable parameter-free type*

$$\{\Xi_A(x)\} \cup \{Q(x) \neq 0 : 0 \neq Q \in \mathbb{Z}[X]\}$$

*is finitely satisfiable and omitted; so  $A$  is not recursively saturated. For the full rings  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  the same explicit type is finitely satisfiable and omitted.*

*Proof.* Use a fixed canonical syntax for integer polynomials, with signed integer coefficients and no trailing zero coefficient. Testing that a formula is  $\Xi_A(x)$  or a displayed inequality for a nonzero polynomial is decidable; this makes the type computable, not just countable. Its only coefficients are integer numerals, including those of the fixed guard  $\Xi_A$ . A finite part excludes the roots of finitely many nonzero polynomials. If the sum of their degrees is  $D$ , at most  $D$  of the ordinary integers  $0, 1, \dots, D$  are excluded; at least one satisfies every inequality and  $\Xi_A$ . A realization of the whole type would lie in  $\mathfrak{o}$ , hence be algebraic over  $\mathbb{Q}$ ; clearing denominators in an annihilating rational polynomial gives a nonzero integer polynomial vanishing at it, a contradiction. Recursive saturation would realize this computable finitely satisfiable type over no parameters.  $\square$

For  $\mathfrak{o} = \mathbb{Z}$  the simpler countable type  $\{\text{Std}(x)\} \cup \{x \neq n : n \in \mathbb{Z}\}$  already shows that  $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$  is not countably saturated, and the Gaussian analogue holds with all Gaussian constants [06]. For the full classes this is the class-relative formulation: no set-sized model carrier is assumed. Nonsaturation of  $\mathbf{Oz}$  has the 2018 precedent [18]. Even a very rich exponent order does not make these rings recursively saturated; the statement concerns the ring, not its order or a surrounding field [07].

*Remark 13.16* (Transfer without elementarity [06; also 13]). For  $\Gamma \neq 0$  the inclusion  $\mathbb{Z} \hookrightarrow \mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$  is not elementary, although it transfers positive-existential formulas: the sentence  $\exists x, y (x^2 = 2y^2 \wedge y \neq 0)$  is false in  $\mathbb{Z}$  and in  $\mathbb{Z}[i]$  and holds in the ring, with  $y = \omega^g$  and  $x = \sqrt{2}\omega^g$  for any  $g > 0$  (when  $\sqrt{2} \in \mathbb{k}$ ). This is a second route to the non-elementarity recorded after [theorem 10.10](#) and in [remark 10.11](#). The same non-elementarity conclusion for  $\mathbb{Z} \hookrightarrow \mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$  holds even when  $\mathbb{k}$  does not contain  $\sqrt{2}$ : the sentence  $\forall x \Xi_{\mathbb{Q}}(x)$ , using the guard above, holds in  $\mathbb{Z}$  but fails at  $\omega^g$  in the larger ring. The guard defines  $\mathbb{Z}$  in both structures by [theorem 12.14](#); this argument uses no ordering of  $\mathbb{k}$ . A formula that defines the actual integers

in this ring need not select the standard copy in an elementary extension, where it defines an elementary extension of the induced arithmetic; this is compatible with the omitted types above [06].

## 14 Infinite families, primitivity, and projective points

### 14.1 Polynomial curves through integer points

**Proposition 14.1** (Finite polynomial lifting [01, 02, 05]). *Let  $F_1, \dots, F_r \in \mathbb{R}[X_1, \dots, X_n]$ , and let  $P_1(T), \dots, P_n(T) \in \mathbb{R}[T]$  satisfy*

$$F_j(P_1(T), \dots, P_n(T)) = 0 \quad (1 \leq j \leq r), \quad P_i(0) \in \mathbb{Z} \quad (1 \leq i \leq n).$$

*Then  $\mathbf{P}(t) \in \mathbf{Oz}^n$  solves the system for every  $t \in \Pi$ , with constant tuple  $\mathbf{P}(0)$ . If some  $P_i$  is nonconstant, every  $t \in \Pi \setminus \{0\}$  produces an infinite point.*

*Proof.* Every nonconstant term of  $P_i(t)$  lies in  $\Pi$ , since  $\Pi$  is an ideal and a real vector space, so the constant coefficient is exactly  $P_i(0)$ . Polynomial identities hold under substitution into any field extension. If  $P_i$  has degree  $d > 0$  and  $\deg_\omega(t) = \gamma > 0$ , its top term has degree  $d\gamma$ , strictly larger than every other term's degree.  $\square$

No integrality is required of the nonconstant coefficients: for example  $(1 + t^2/2, t^2/2, t)$  is omnific for every  $t \in \Pi$  although  $1/2 \notin \mathbf{Oz}$ ; it satisfies  $X^2 - Y^2 - Z^2 = 1$  by expansion. For  $t < 0$ , leading degree still proves infinite size, though it need not preserve the eventual signs of the polynomials as  $T \rightarrow +\infty$ . The finiteness of the polynomials matters, however. Evaluating an infinite real power series at an infinite omnific argument is not licensed by this proposition; the Taylor expansion of  $\sqrt{1+T}$  is not an admissible way to assert that  $\sqrt{1+\omega}$  lies in  $\mathbf{Oz}$ . Polynomial maps supply a systematic sufficient condition for nontrivial fibers; they do not exhaust all normal-form constructions, and no such exhaustion is assumed (but see [theorem 14.7](#) for finite supports). On smooth affine curves they do decide existence. More precisely, for an affine  $\mathbb{Z}$ -model with smooth geometrically integral generic curve, a nonordinary omnific point exists exactly when the real fiber is isomorphic to  $\mathbb{A}_{\mathbb{R}}^1$  and an ordinary integer point exists. If  $\phi : \mathbb{A}_{\mathbb{R}}^1 \rightarrow C_{\mathbb{R}}$  is such an isomorphism and  $\mathbf{n}$  is that integer point, the entire fiber over  $\mathbf{n}$  is  $\phi(\phi^{-1}(\mathbf{n}) + \Pi)$ , bijectively ([corollaries 17.12](#) and [17.16](#)). The coordinates of  $\phi$  are finite real polynomials, not necessarily integer polynomials, and the model need not be the affine line over  $\mathbb{Z}$ . Taking a nonzero monomial parameter gives a nonordinary point of finite support with that constant term; arbitrary parameters in  $\Pi$  need not have finite support.

### 14.2 Infinite solutions of homogeneous equations

**Theorem 14.2** (Homogeneous cone criterion [01]; part (b) [02]). *Let  $F_1, \dots, F_r \in \mathbb{R}[X_1, \dots, X_n]$  be homogeneous of positive degrees.*

- (a) *Their common zero locus has a nonzero point in  $\mathbf{Oz}^n$  if and only if it has a nonzero point in  $\mathbb{R}^n$ .*
- (b) *If the  $F_j$  have ordinary integer coefficients, the equivalence in (a) persists with any fixed combination of coordinate sign and nonvanishing conditions preserved by positive scaling, and these existence problems are decidable.*

*Proof.* (a) A real nonzero common zero  $\mathbf{a}$  gives the omnific common zero  $\omega\mathbf{a} \in \Pi^n$  by homogeneity. Conversely, take a nonzero common zero  $\mathbf{x} \in \mathbf{Oz}^n$ . If every coordinate is constant, it is a real point. Otherwise let  $\gamma > 0$  be the maximum of the coordinates' leading degrees and  $r_i = [\omega^\gamma]x_i$ ; the vector  $\mathbf{r}$  is nonzero. For  $F_j$  homogeneous of degree  $d_j$ , the coefficient at  $\omega^{d_j\gamma}$  in  $F_j(\mathbf{x})$  is  $F_j(\mathbf{r})$ , since a product attains that exponent only by taking exponent  $\gamma$  in every factor. So  $\mathbf{r}$  is a nonzero real common zero. This route uses no quantifier elimination.

(b) A real solution  $\mathbf{r}$  gives  $\omega\mathbf{r}$ , which has the same signs and zero pattern. Conversely an omnific solution lies, with  $\mathbb{R}$ , in a set-sized real closed subfield of  $\mathbf{No}$  (generate a field and take its real closure inside  $\mathbf{No}$ ). Quantifier elimination makes the inclusion of  $\mathbb{R}$  elementary [11], so the existential statement with its finite sign conditions already holds over  $\mathbb{R}$ ; quantifier elimination is an effective decision procedure for this finite formula with integer coefficients.  $\square$

The leading-vector route of (a) does not preserve sign or nonvanishing conditions on coordinates of lower degree: for example  $(\omega^2, \omega)$  has both coordinates positive, but its highest-degree coefficient vector is  $(1, 0)$ . This is why (b) uses Tarski. Any homogeneous solution over  $\mathbf{No}$ , not only over  $\mathbb{R}$ , can be multiplied by a common positive monomial into  $\Pi^n$  preserving signs and zero patterns (corollary 4.7) [02, 05]. Neither statement decides affine equations: homogenizing an affine equation and requiring the new coordinate to be nonzero allows division in  $\mathbf{No}$ , not in  $\mathbf{Oz}$ . For example  $X^2 - 2Z^2 = 0$  has the omnific solution  $(X, Z) = (\sqrt{2}\omega, \omega)$  with  $Z \neq 0$ . Its affine coordinate  $X/Z = \sqrt{2}$  is not omnific, and  $x^2 = 2$  has no omnific solution by proposition 2.8.

**Example 14.3** (Fermat scaling [01, 02, 05]). For every ordinary integer  $n \geq 2$  and  $0 < t \in \Pi$ ,

$$x = t, \quad y = t, \quad z = 2^{1/n}t \quad (14.1)$$

is a positive infinite solution of  $x^n + y^n = z^n$ ; for  $t = \omega$  this is  $\omega^n + \omega^n = (2^{1/n}\omega)^n$ . The irrational coefficient is permitted by the definition of  $\mathbf{Oz}$ . All constant coefficients are zero, consistent with theorem 5.1, whose specialization is  $(0, 0, 0)$ ; it does not refute Fermat's Last Theorem about nonzero ordinary integers. The coordinates are divisible by every ordinary integer and have a common infinite monomial divisor (theorem 4.2), so these solutions are not primitive and must not be cited as resolving a primitive Fermat problem.

*Provenance.* Source 02 states that the failure of the nonzero analogue of Fermat's Last Theorem for omnific integers "is already known", citing comments by Emil Jeřábek (27 June 2018) and Timothy Chow (16 November 2021) under MathOverflow answer 303688 [17]. This attribution could not be verified when this report was assembled (the site was not reachable), and it is recorded as source 02's, unverified. No source claims novelty for the example.

### 14.3 Unimodular Pythagorean triples

**Proposition 14.4** (Infinite unimodular Pythagorean triples [01, 05]). *For every  $t \in \Pi$ , put  $x = t^2 - 1$ ,  $y = 2t$ ,  $z = t^2 + 1$ . Then  $x^2 + y^2 = z^2$  and the triple is unimodular. For  $t \neq 0$  all three coordinates are infinite, and for  $t > 0$  they are positive.*

*Proof.* The identity is the ordinary polynomial identity  $(t^2 - 1)^2 + 4t^2 = (t^2 + 1)^2$ . Two unit-ideal witnesses are

$$\frac{t^2}{2}z - \left(\frac{t^2}{2} + 1\right)x = 1 \text{ [01]}, \quad \frac{t}{2}y - x = 1 \text{ [05]}, \quad (14.2)$$

with  $t^2/2, t/2 \in \Pi \subseteq \mathbf{Oz}$ . Positive leading degree gives the assertion about infinite size.  $\square$

The constant tuple is  $(-1, 0, 1)$ , not the zero tuple; that is exactly where the common-divisor obstruction for  $\Pi^3$  stops applying. Infinite solutions therefore need not arise by scaling an ordinary tuple, and exponent two differs from exponents at least three ([theorem 7.9](#)). More generally  $(u^2 - v^2)^2 + (2uv)^2 = (u^2 + v^2)^2$  for all  $u, v \in \mathbf{Oz}$  [02].

#### 14.4 Which real projective directions have primitive representatives?

A point  $[r_0 : \cdots : r_n] \in \mathbb{P}^n(\mathbb{R})$  is *rational* if it has homogeneous coordinates in  $\mathbb{Q}$ , equivalently a nonzero integer coordinate tuple.

**Theorem 14.5** (Rationality of primitive constant directions [01]). *For a real projective point  $p = [r_0 : \cdots : r_n]$ , the following are equivalent:*

- (i)  $p$  is rational;
- (ii)  $p$  has a unimodular representative in  $\mathbf{Oz}^{n+1}$ ;
- (iii)  $p$  has a primitive representative in  $\mathbf{Oz}^{n+1}$ .

*Proof.* For (i) $\Rightarrow$ (ii), choose an integer representative with ordinary gcd one; the integer Bézout identity makes it unimodular over  $\mathbf{Oz}$ . (ii) $\Rightarrow$ (iii) is immediate. For (iii) $\Rightarrow$ (i), write a primitive representative as  $z_i = \lambda r_i$  with  $\lambda \in \mathbf{No}^\times$  and choose  $r_j \neq 0$ . Since  $z_j \in \mathbf{Oz} \subseteq \mathcal{B}_{\mathbb{R}}$  and division by the nonzero real  $r_j$  preserves  $\mathcal{B}_{\mathbb{R}}$ ,  $\lambda \in \mathcal{B}_{\mathbb{R}}$ . Put  $c = \text{ct}(\lambda)$ ; then  $\text{ct}(z_i) = cr_i \in \mathbb{Z}$  for all  $i$ . If  $c = 0$ , every  $z_i$  lies in  $\Pi$ , so 2 divides every coordinate, contradicting primitivity. Hence  $c \neq 0$  and  $(cr_0, \dots, cr_n)$  is a nonzero integer representative of  $p$ .  $\square$

In fact every primitive representative of a real projective point is an ordinary coprime integer tuple. To see this additional consequence, the theorem first makes the point rational. Fix an ordinary representative  $\mathbf{m}$  with gcd one and integers  $b_i$  such that  $\sum_i b_i m_i = 1$ . Every omnific representative has the form  $\mathbf{z} = s\mathbf{m}$  for some  $s \in \mathbf{No}^\times$ , and

$$s = \sum_i b_i z_i \in \mathbf{Oz}, \quad (z_0, \dots, z_n) = s\mathbf{Oz}.$$

Indeed each  $z_i$  is a multiple of  $s$ , and the displayed Bézout identity puts  $s$  in the coordinate ideal. Thus such a representative is primitive if and only if  $s$  is a unit, equivalently  $s = \pm 1$ , and then it is also unimodular. This equivalence is specific to representatives of real directions; no general equivalence of primitivity and unimodularity over  $\mathbf{Oz}$  is asserted.

For example,  $[1 : \sqrt{2}]$  has omnific representatives such as  $(\omega, \sqrt{2}\omega)$  but no primitive, hence no unimodular, representative; source 02 proved the unimodular half of this example directly. The theorem concerns real projective points; it is not a classification of all points of  $\mathbb{P}^n(\mathbf{No})$ , and neither monomial clearing nor the existence of a homogeneous solution answers the primitive-representative question. [Section 15](#), from sources 08 and 09, extends the classification to all points of  $\mathbb{P}^n(\mathbb{R}(t))$  and  $\mathbb{P}^n(\mathbb{C}(t))$  for  $0 \neq t \in \Pi$ , and to the region  $\mathcal{V}$  of  $\mathbb{P}^1(\mathbf{No})$  ([theorems 15.15](#), [15.18](#) and [15.54](#) and [corollary 15.28](#)); the Gaussian form of the theorem is [corollary 15.55](#). A classification of all points of  $\mathbb{P}^n(\mathbf{No})$  is still not claimed ([questions 21.12](#) and [21.13](#)). On projective varieties that are rigid over the one-sided rings, such as curves of positive genus, sources C12 and C14 show that only rational points have invertible, and in particular unimodular, omnific coordinates ([theorem 17.21](#) and [corollary 17.22](#)).

## 14.5 Finite supports specialize to polynomial arcs

The rigidity theorems above do not depend on finite supports. Finite-support expressions nevertheless admit an additional reduction [05].

**Lemma 14.6** (Finite ordered specialization [05]). *Let  $H$  be a finitely generated subgroup of an ordered abelian group and  $E \subseteq H$  finite. There is an additive map  $\ell : H \rightarrow \mathbb{Z}$  with  $\ell(a) > \ell(b)$  whenever  $a, b \in E$  and  $a > b$ .*

*Proof.* A finitely generated torsion-free abelian group is isomorphic to  $\mathbb{Z}^r$ . In such coordinates, form the finite set of vectors  $a - b$  for  $a, b \in E$  with  $a > b$ . If this set is empty,  $\ell = 0$  works. Otherwise enumerate it as  $v_1, \dots, v_s$ ; each vector represents a positive element in the original group order.

We first show that 0 is not in their ordinary real convex hull. If it were, choose a solution of

$$\sum_j c_j v_j = 0, \quad \sum_j c_j = 1, \quad c_j \geq 0$$

with the smallest possible number of positive coefficients. The augmented columns  $(v_j, 1)$  for those coefficients are linearly independent. Otherwise a nonzero dependence  $d_j$  has  $\sum_j d_j = 0$ , hence both positive and negative entries; moving the coefficient vector along this dependence until one positive coefficient reaches zero preserves all constraints and reduces its positive support. Independence gives a square nonsingular minor with integer entries, so the positive  $c_j$  are rational by solving that subsystem. Clearing denominators yields a nonempty sum of positive integer multiples of the  $v_j$  equal to zero, impossible in an ordered group.

Let  $C$  be their convex hull and choose  $p \in C$  of least Euclidean norm. Compactness supplies  $p$ , and  $p \neq 0$ . For  $v \in C$ , minimality along the segment  $p + u(v - p)$ ,  $0 \leq u \leq 1$ , gives  $p \cdot (v - p) \geq 0$ . Consequently  $p \cdot v_j \geq \|p\|^2 > 0$  for every  $j$ . A rational vector sufficiently close to  $p$  keeps these finitely many strict inequalities. Multiplying it by a positive common denominator gives an integer vector  $w$ , and  $\ell(h) = w \cdot h$  is the required additive map. All convexity here is in the ordinary finite-dimensional space  $\mathbb{R}^r$ , not in a surreal topological space.  $\square$

**Theorem 14.7** (Finite-support solutions specialize to arcs [05]). *Suppose  $\mathbf{x} \in \mathbf{Oz}^n$  has finite normal-form supports and satisfies a finite polynomial system over  $\mathbb{R}$ . There are real polynomials  $f_1(T), \dots, f_n(T)$  satisfying the same polynomial identities with  $f_i(0) = \text{ct}(x_i)$ , and every nonconstant  $x_i$  gives a nonconstant  $f_i$ . A prescribed finite list of nonzero real-coefficient polynomial expressions in  $\mathbf{x}$  can be kept nonzero, with the same eventual signs as  $T \rightarrow +\infty$  as their surreal signs.*

*Proof.* Let  $E$  contain zero, all exponents of the coordinates, and all exponents of the prescribed expressions after expansion; it is finite. Let  $H$  be its integer span and choose  $\ell$  by lemma 14.6. The group algebra  $\mathbb{R}[H]$  identifies with the finite-support real normal forms on  $H$ : distinct monomials are linearly independent, and their multiplication adds exponents. Thus the original equations hold in this algebra. The map  $\mathbb{R}[H] \rightarrow \mathbb{R}[T, T^{-1}]$ ,  $\omega^h \mapsto T^{\ell(h)}$ , is a ring homomorphism fixing every real coefficient. Positive coordinate exponents map to positive integers, so coordinate images are polynomials with the same constant coefficients; injectivity of  $\ell$  on  $E$  prevents cancellation of distinct supported terms, in particular of leading terms. Polynomial identities are preserved, and each prescribed expression keeps its leading coefficient and hence its eventual sign for large positive  $T$ .  $\square$

Source 05 states the system over  $\mathbb{Z}$ ; fixing real scalars in the same homomorphism proves the extension above. Coefficients of the additional expressions are specified as real for the same reason. The map need not be injective or order preserving on all of  $H$ : it preserves the finite list of comparisons used in its construction. For instance, on  $H = \mathbb{Z} + \mathbb{Z}\sqrt{2}$ , the map  $\ell(1) = 1$ ,  $\ell(\sqrt{2}) = 2$  preserves the order of  $E = \{0, 1, \sqrt{2}, 1 + \sqrt{2}\}$  but kills the nonzero group-algebra element  $\omega^{\sqrt{2}} - \omega^2$ . To preserve that expression as well, its supported exponent 2 must be included in  $E$  and a suitable map chosen again. Equations need no such extra condition: every ring homomorphism preserves them, even when it has a kernel.

With [proposition 14.1](#) (substituting  $t \in \Pi$  for  $T$ ), this gives an exact equivalence between the existence of a finite-support nonconstant solution and of a nonconstant real polynomial arc with integer constant coordinates. For the reverse implication choose  $t = \omega$ : the resulting tuple has finite supports, and each nonconstant polynomial has a nonconstant value. Positive  $t \in \Pi \setminus \{0\}$  also preserves the prescribed signs, because its positive leading term makes each polynomial's leading term determine its sign. A single specialization therefore preserves any finite list of polynomial equalities, disequalities and sign conditions. This does not turn infinite-support solutions into polynomial arcs. For smooth affine curves, however, every nonordinary point shares its constant term with a nonordinary point of finite support ([corollary 17.16](#)).

Finite-support Bézout certificates can be retained as well. If  $x_i, u_i \in \mathbf{Oz}$  all have finite supports and  $\sum_i u_i x_i = 1$ , apply the theorem to the combined tuple and include this equation. Its images satisfy  $\sum_i g_i(T) f_i(T) = 1$ , with integer constant coefficients in every  $f_i$  and  $g_i$ . Evaluating at any  $t \in \Pi$  gives an omnific Bézout certificate, so the resulting family is unimodular. This argument requires finite supports for the supplied witnesses  $u_i$ , not just for the coordinates  $x_i$ ; existence of arbitrary omnific witnesses does not supply that additional hypothesis.

For a finite-support implementation one may store an element as a map from exact exponents to nonzero exact real coefficients; the essential operations are coefficientwise addition, exponent addition in convolution, combination of equal exponents, and deciding integrality of the constant coefficient. Rational exponents with real algebraic coefficients give a practical decidable representation; general symbolic reals or arbitrary surreal exponents need further equality and order oracles. Useful exact certificates are polynomial identities, Bézout witnesses such as (14.2), standard-modulus quotients, and finite normal-form obstructions such as the first negative term in (19.2) [05].

## 15 Fractions, denominator ideals and unimodular presentations

This section is built from sources 08 and 09 ([section 1.2](#)), with source 08 as the base. [Theorem 4.6](#) makes every surreal a fraction of omnific integers, but it says nothing about lowest terms, and [theorems 4.10](#) and [14.5](#) treat only the constant pairs  $(t, rt)$  and the constant real directions. The central theorem below ([theorem 15.11](#)) removes the constancy. For every tuple that is unimodular over  $\mathcal{B}_{\mathbb{R}}$ , the constant vector alone decides whether the admissible scalar multipliers form  $\mathbf{Oz}$  or  $\Pi$ . This classifies every omnific representation of every point of  $\mathbb{P}^n(\mathbb{R}(t))$  for  $0 \neq t \in \Pi$  ([theorem 15.18](#)). It fills the gap stated after [theorem 14.5](#) and answers [question 21.3](#) in part ([corollary 15.30](#)). Source 09 adds the region  $\mathcal{V}$  of points with a  $\mathcal{B}_{\mathbb{R}}$ -unimodular presentation. It identifies the specialization fibers with the orbits of the congruence kernel of  $\mathrm{SL}_2(\mathbf{Oz}) \rightarrow \mathrm{SL}_2(\mathbb{Z})$  and proves that these fibers are dense. Source 08 adds a localization with residue field  $\mathbb{Q}$ , polynomial models, the exact denominator defect of a scale extension with its  $\mathrm{Tor}_1$  witness, and the Gaussian dichotomy. The Gaussian dichotomy contains the Gaussian unimodular-direction criterion of manuscript 14 of the sibling report ([corollary 15.55](#)). The proof review recorded in `RECONCILIATION.md` now covers this whole section, including the localization, scale-extension, fixed-workspace and Gaussian arguments. The later additions in

sections 20 and 21, other new material and remaining imported foundations still need review.

## 15.1 Conventions

Throughout,  $t$  denotes a nonzero element of  $\Pi$ , as in propositions 14.1 and 14.4. It is not the Hahn variable  $t = \omega^{-1}$  of section 1.4. Source 08 writes  $\tau$ , which here names the phase twist of lemma 13.9; source 09 uses only the monomials  $T = \omega^\alpha$  with  $\alpha > 0$ . Polynomials are written in an indeterminate  $T$ , and  $p(t)$  is evaluation at  $t$ .

**Coordinate order.** Numerators come first and the denominator last. A tuple  $(x_1, \dots, x_n)$  is the point  $[x_1 : \dots : x_n : 1]$ , and  $\mathbb{P}^1(\mathbf{No}) = \mathbf{No} \cup \{\infty\}$  through  $[P : Q] \mapsto P/Q$  for  $Q \neq 0$  and  $[P : 0] \mapsto \infty$ . Source 09 uses this order, while source 08 puts the denominator first. The specialized values, such as  $f(0)$ , are the same in both orders. A point of  $\mathbb{P}^1(\mathbf{No})$  other than  $\infty$  is called *affine*. Source 09 calls such points “finite”, a word reserved here for bounded surreals.

**Primitivity.** “Primitive” and “unimodular” are as in definition 7.7. Over another domain, a tuple is primitive when its only common divisors are units. Source 09’s “divisor-primitive” presentations and “Bézout fractions” are the primitive and the unimodular presentations here. Unimodularity implies primitivity; the converse is not assumed over  $\mathbf{Oz}$  and is proved below only for the classified representations.

**New symbols.** The symbols  $\mathfrak{D}$ ,  $\text{sp}$ ,  $\mathcal{V}$ ,  $\mathcal{U}$ ,  $\mathbf{K}$ ,  $\text{res}$ ,  $\text{Ex}$ ,  $\mathcal{P}$ ,  $F_{x,b}$  and  $\mathbf{N}_x$  are introduced below. Most of them replace source symbols that collide with symbols of this report (table 1). In this report  $\Gamma$  is an exponent group,  $\sigma$  an automorphism,  $\rho = \sqrt{2} - 1$ ,  $D$  the Pell parameter,  $G$  the Gram matrix,  $U_t$  the quadratic transvection,  $A$  an intermediate ring,  $\mathcal{A}_{\mathbf{k}}(\Gamma)$  and  $\mathcal{F}$  the rings of sections 11 and 13,  $B$  a bilinear form,  $\Delta$  a discriminant and  $V$  a zero set. The source symbols are therefore renamed. The calligraphic  $\mathcal{V}$  below is a class of projective points, not a zero set.

## 15.2 Denominator ideals

**Definition 15.1** (Denominator ideals [08, 09]). For surreals  $x_1, \dots, x_n$  put

$$\mathfrak{D}(x_1, \dots, x_n) = \{b \in \mathbf{Oz} : bx_j \in \mathbf{Oz} \text{ for every } j\}.$$

For a domain  $R'$  in a field containing the  $x_j$ ,  $\mathfrak{D}_{R'}$  is defined in the same way with  $R'$  in place of  $\mathbf{Oz}$ . A *representation* of  $(x_1, \dots, x_n)$  is a vector  $(bx_1, \dots, bx_n, b) \in \mathbf{Oz}^{n+1}$  with  $b \neq 0$ .

Zero belongs to  $\mathfrak{D}(x_1, \dots, x_n)$ . The actual common denominators are its nonzero members, and a common one exists by applying theorem 4.6 to the finite set of coordinates. Moreover  $\mathfrak{D}(x_1, \dots, x_n) = \bigcap_j \mathfrak{D}(x_j)$ , so it is an ideal as well. For a general domain  $R'$  in an ambient field, this ideal can instead be zero: a nonzero denominator forces every  $x_j$  to lie in  $\text{Frac}(R')$ . Conversely, if all coordinates lie in  $\text{Frac}(R')$ , the product of finitely many individual denominators is a nonzero common denominator. Ideals of  $\mathbf{Oz}$  here may be proper classes; their operations and generated ideals use finite sums and products, as in section 1.4. Source 09 writes  $D(x)$ ; the letter  $D$  is the Pell parameter here.

**Proposition 15.2** (Calculus of denominator ideals [09]). *For every  $x \in \mathbf{No}$ ,  $\mathfrak{D}(x)$  is a nonzero ideal of  $\mathbf{Oz}$ , with  $\mathfrak{D}(0) = \mathbf{Oz}$  and  $\mathfrak{D}(x) = \mathbf{Oz} \cap x^{-1}\mathbf{Oz}$  for  $x \neq 0$ . For  $z \in \mathbf{Oz}$ ,*

$$\mathfrak{D}(x + z) = \mathfrak{D}(x), \quad x \mathfrak{D}(x) = \mathfrak{D}(1/x) \quad (x \neq 0).$$

For  $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbf{Oz})$  and  $cx + d \neq 0$ ,

$$\mathfrak{D}\left(\frac{ax + b}{cx + d}\right) = (cx + d) \mathfrak{D}(x). \quad (15.1)$$

*Proof.* Closure under addition and under multiplication by  $\mathbf{Oz}$  is direct, and nonzero denominators exist by [theorem 4.6](#). For  $h \in \mathbf{Oz}$ ,  $h(x + z) \in \mathbf{Oz}$  is equivalent to  $hx \in \mathbf{Oz}$ , by subtracting  $hz$ ; this also covers  $x = 0$ . For  $x \neq 0$ , multiplication by  $x$  sends a denominator  $h$  of  $x$  to  $xh \in \mathbf{Oz}$ , with  $(xh)/x = h \in \mathbf{Oz}$ . Its inverse sends a denominator  $k$  of  $1/x$  to  $k/x$ . This proves the reciprocal identity. For (15.1) put  $y = (ax + b)/(cx + d)$ . If  $h \in \mathfrak{D}(x)$ , then  $k = (cx + d)h \in \mathbf{Oz}$  and  $yk = (ax + b)h \in \mathbf{Oz}$ . Conversely let  $k \in \mathfrak{D}(y)$  and write  $\delta_g = ad - bc \in \mathbf{Oz}^\times$  for the determinant. Since  $a - cy = \delta_g/(cx + d)$  and  $dy - b = \delta_g x/(cx + d)$ , the elements

$$h = \delta_g^{-1}(a - cy)k = \frac{k}{cx + d}, \quad xh = \delta_g^{-1}(dy - b)k$$

lie in  $\mathbf{Oz}$ : both  $k$  and  $yk$  do, and  $\delta_g^{-1} \in \mathbf{Oz}$ . Thus  $h \in \mathfrak{D}(x)$  and  $k = (cx + d)h$ . In particular the right side of (15.1) lies in  $\mathbf{Oz}$ , even though the scalar  $cx + d$  itself need not lie there.  $\square$

**Proposition 15.3** (Least denominators [09]). *A nonzero ideal  $\mathfrak{a} \subseteq \mathbf{Oz}$  is principal if and only if its positive members have a least element. Hence  $x \in \mathbf{No}$  has a least positive omnific denominator if and only if  $\mathfrak{D}(x)$  is principal. If  $\mathfrak{D}(x) = b_0 \mathbf{Oz}$  with  $b_0 > 0$ , the representations of  $x$  are exactly  $(a_0 h, b_0 h)$  with  $a_0 = b_0 x$  and  $0 \neq h \in \mathbf{Oz}$ , and  $(a_0, b_0)$  is primitive.*

*Proof.* For  $\mathfrak{a} = b_0 \mathbf{Oz}$  with  $b_0 > 0$ , the positive members are  $b_0 h$  with  $h \geq 1$  ([proposition 2.5](#)), so  $b_0$  is least. Conversely, if  $b_0$  is the least positive member and  $a \in \mathfrak{a}$ , ordered division ([proposition 4.8](#)) gives  $a = qb_0 + r$  with  $0 \leq r < b_0$  and  $r \in \mathfrak{a}$ , so  $r = 0$ . The description of the representations follows from the definition. If  $d$  divides  $a_0$  and  $b_0$ , then  $b_0/d \in \mathfrak{D}(x)$ , so  $b_0/d = b_0 h$  with  $h \in \mathbf{Oz}$ , and cancellation of  $b_0 \neq 0$  gives  $dh = 1$ , so  $d$  is a unit. This uses one ordered division for each  $a$ ; it does not assume termination of a Euclidean algorithm in  $\mathbf{Oz}$ . The argument proves primitivity; the stronger Bézout conclusion below comes from the multiplier theorem.  $\square$

**Lemma 15.4** (Unimodular fractions [08, 09]). *Let  $R'$  be a domain and  $(p_1, \dots, p_n, q) \in R'^{n+1}$  unimodular with  $q \neq 0$ . Then  $\mathfrak{D}_{R'}(p_1/q, \dots, p_n/q) = qR'$ . Every representation over  $R'$  is  $h(p_1, \dots, p_n, q)$  for a unique  $0 \neq h \in R'$ , and  $h$  is a gcd of it. For  $R' = \mathbf{Oz}$  and  $q > 0$ , the vector itself is the only primitive representation with positive denominator, and two unimodular representations differ by a sign.*

*Proof.* Take  $u_1 p_1 + \dots + u_n p_n + u_0 q = 1$  in  $R'$ . If  $b$  is a denominator, then

$$\frac{b}{q} = u_0 b + \sum_j u_j \frac{b p_j}{q} \in R'.$$

Thus  $b = qh$ , and the numerators are  $p_j h$ . Conversely every  $h \in R'$  gives a denominator, and  $q \neq 0$  makes the multiplier  $h = b/q$  unique and nonzero for an actual representation. A common divisor of  $h(p_1, \dots, p_n, q)$  divides the same linear combination, which is  $h$ , and  $h$  is a common divisor. For  $R' = \mathbf{Oz}$  a representation is primitive exactly when  $h$  is a unit, that is,  $h = \pm 1$ .  $\square$

**Proposition 15.5** (Ordinary denominators [08, 09]). *Inside  $\mathbf{No}$ ,*

$$\mathbf{Oz}[(\mathbb{Z} \setminus \{0\})^{-1}] = \mathbb{Q} \oplus \Pi, \quad 0 \longrightarrow \Pi \longrightarrow \mathbb{Q} \oplus \Pi \xrightarrow{\text{ct}} \mathbb{Q} \longrightarrow 0$$

with the sequence split exact. The ring  $\mathbb{Q} \oplus \Pi$  meets  $\mathbb{R}$  in  $\mathbb{Q}$ , has unit group  $\mathbb{Q}^\times$ , is not a field, and has  $\mathbb{Q}$  as its only subfield. If  $x = y + m/n$  with  $y \in \Pi$ ,  $n > 0$  and  $\gcd(m, n) = 1$ , then  $\mathfrak{D}(x) = n\mathbf{Oz}$ , and  $x = (ny + m)/n$  is a unimodular presentation with least positive denominator  $n$ : for integers  $u, v$  with  $um + vn = 1$ ,

$$u(ny + m) + (v - uy)n = 1, \quad v - uy \in \mathbf{Oz}. \quad (15.2)$$

*Proof.* If  $a = k + y \in \mathbf{Oz}$  with  $k \in \mathbb{Z}$ ,  $y \in \Pi$ , and  $0 \neq n \in \mathbb{Z}$ , then  $a/n = k/n + y/n \in \mathbb{Q} \oplus \Pi$ ; conversely  $y + m/n = (ny + m)/n$ . The constant-term map has kernel  $\Pi$  and is split by the inclusion of rational constants. The direct sum describes the additive decomposition; it is not a direct product of rings, since rational constants act on  $\Pi$  by scalar multiplication. A nonzero element of this ring has nonnegative leading degree. If it and its inverse both belong to the ring, additivity of leading degrees forces both degrees to be zero. Their supports are nonnegative, so both are rational constants. Conversely every nonzero rational is a unit. Also  $\omega$  lies in the ring while  $\omega^{-1}$  does not. For  $b \in \mathbf{Oz}$  the product  $bx$  has no negative exponent, and its constant coefficient is  $(m/n) \text{ct}(b)$ . So  $b \in \mathfrak{D}(x)$  exactly when  $n \mid \text{ct}(b)$ , which is  $b \in n\mathbf{Oz}$  by [theorem 3.1](#). The identity (15.2) is direct, and a positive element of  $n\mathbf{Oz}$  is at least  $n$ . Finally, a subfield contains  $\mathbb{Q}$ . A nonreal member of  $\mathbb{Q} \oplus \Pi$  is infinite, and its inverse is a nonzero infinitesimal outside the ring. So every member of a subfield is real, hence rational.  $\square$

An element of  $\mathbf{No}$  algebraic over  $\mathbb{Q}$  is a real algebraic number. Indeed, a nonzero polynomial in  $\mathbb{Q}[T]$  factors over the real algebraic numbers into linear factors and irreducible quadratics. After a nonzero scalar is removed, each quadratic has the form  $(T - a)^2 + b^2$  with  $b \neq 0$ , and has no root in any ordered extension. A surreal root must therefore annihilate a linear factor and be real algebraic. This uses only the real closedness of the real algebraic numbers, not a transfer theorem for classes. Hence the elements of  $\mathbb{Q} \oplus \Pi$  algebraic over  $\mathbb{Q}$  are exactly the rationals, and those of  $\mathbf{Oz}$  are the integers [09]. Fractions that produce  $\sqrt{2}$  need infinite omnific denominators. Also  $\mathcal{B}_{\mathbb{R}} \cap \mathcal{O}_{\text{fin}} = \mathbb{R}$  and  $(\mathbb{Q} \oplus \Pi) \cap \mathcal{O}_{\text{fin}} = \mathbb{Q}$ :  $\text{ct}$  discards the infinite terms and  $\text{st}$  the infinitesimal ones ([remark 2.3](#)) [09]. So there are four rings to keep apart [08]:

$$\mathbb{Q} \subsetneq \mathbb{Q} \oplus \Pi \subsetneq \mathbf{Oz}_{\Pi} \subsetneq \text{Frac}(\mathbf{Oz}) = \mathbf{No}.$$

The second admits only ordinary integer denominators. The third, the localization of  $\mathbf{Oz}$  at the prime ideal  $\Pi$ , admits every denominator with a nonzero constant coefficient; it is studied in [section 15.8](#). The strict inclusions have concrete witnesses:  $\omega$  is not rational;  $1/(1 + \omega)$  belongs to  $\mathbf{Oz}_{\Pi}$  but is a nonzero infinitesimal and hence is not in  $\mathbb{Q} \oplus \Pi$ ; and  $\sqrt{2}$  is an omnific fraction but cannot have a denominator with nonzero integer constant coefficient, by [theorem 15.7](#).

**Proposition 15.6** (No set of denominators [08]). *No set  $S \subseteq \mathbf{Oz} \setminus \{0\}$  supplies a denominator for every surreal: there is  $0 < x \in \mathbf{No}$  with  $bx \notin \mathbf{Oz}$  for every  $b \in S$ .*

*Proof.* The leading exponents of the members of  $S$  form a set. By [lemma 4.1](#) choose  $\eta > 0$  exceeding every  $\deg_{\omega}(b)$ , and put  $x = \omega^{-\eta}$ . Then  $bx \neq 0$  has negative leading exponent, so it is a nonzero infinitesimal and does not lie in  $\mathbf{Oz}$ .  $\square$

By contrast, the class  $\mathcal{M} = \{\omega^{\beta} : \beta \geq 0\}$  of monomials does serve:  $\mathcal{M}^{-1}\mathbf{Oz} = \mathbf{No}$ , with the localization read for the class of denominators. This is `osq:prop:fractions(iii)` of the sibling report, which source 09 proves again. A single monomial  $\omega^{\beta}$  fails at  $x = \omega^{-\beta-1}$  [09]. Even a set-sized subfield of  $\mathbf{No}$  has one common omnific denominator for all its elements ([theorem 4.6](#)), but the whole class does not [08].

**Theorem 15.7** (Real constants [08, 09]). *For  $r \in \mathbb{R}$ ,*

$$\mathfrak{D}(r) = \begin{cases} q \mathbf{Oz}, & r = p/q \in \mathbb{Q}, \gcd(p, q) = 1, q > 0, \\ \Pi, & r \notin \mathbb{Q}. \end{cases}$$

*For irrational  $r$ , every representation  $r = a/b$  has  $a, b \in \Pi$ , every nonzero ordinary integer divides both, and the pair has no gcd. For real constants each of the following is equivalent to  $r \in \mathbb{Q}$ : a primitive representation exists; a unimodular representation exists;  $\mathfrak{D}(r)$  is principal;  $\mathfrak{D}(r)$  is generated by a set; a least positive omnific denominator exists.*

*Proof.* Write  $b = k + y$  with  $k \in \mathbb{Z}$  and  $y \in \Pi$ . Multiplication by a real introduces no new exponents (and preserves support when the scalar is nonzero), so  $rb \in \mathbf{Oz}$  exactly when  $rk \in \mathbb{Z}$ . This includes  $r = 0$ , whose reduced denominator is  $q = 1$ . For  $r = p/q$  in lowest terms this means  $q \mid k$ , and since  $\Pi = q\Pi$  (theorem 3.1) the ideal is  $q\mathbf{Oz}$ . For irrational  $r$  it forces  $k = 0$ , giving  $\Pi$ .

For irrational  $r$ , then  $a = rb \in \Pi$  as well. Dividing their coefficients by any nonzero ordinary integer preserves positive support, so every such integer divides both. In particular the pair is not primitive. If  $d$  is a common divisor of  $a$  and  $b$ , then  $(a/d, b/d)$  is another representation of  $r$ , so both quotients again lie in  $\Pi$  and are divisible by 2. Hence  $2d$  is a common divisor; it does not divide  $d$ , because  $1/2 \notin \mathbf{Oz}$ . So no common divisor is greatest. Corollary 4.3 excludes a set of generators, and halving shows that there is no least positive denominator. In the rational case the ordinary pair  $(p, q)$  is unimodular, and lemma 15.4 gives the remaining assertions.  $\square$

*Provenance.* Source 09 proves  $\mathfrak{D}(r) = \Pi$  and the absence of a gcd for irrational  $r$  and credits the absence of a gcd to theorem 4.10; source 08 proves the whole theorem and credits the same result. The gcd part is theorem 4.10 for the pair  $(b, rb)$ . For irrational  $r$  the ideal  $\mathfrak{D}(r)$  is the dual  $\text{Hom}_{\mathbf{Oz}}(L(M), \mathbf{Oz})$  of the coefficient-lattice module  $L(M) = M + \Pi$  with  $M = \mathbb{Z} + \mathbb{Z}r$ , and the equality  $\mathfrak{D}(r) = \Pi$  is the rank-two case of `osq:tn:thm:duals` of the sibling report (manuscript 14 there). Hamkins discusses the failure of lowest terms and the omnific fractions of irrational reals [32], as source 08 records. The theorem does not erase the distinction between  $\mathbb{Q}$  and  $\mathbb{R} \setminus \mathbb{Q}$ , although  $\text{Frac}(\mathbf{Oz}) \cap \mathbb{R} = \mathbb{R}$  [08].

*Remark 15.8* (The halving route [merge]). Sources 08 and 09 show that no gcd exists by replacing a common divisor  $d$  by  $2d$ . This differs from the shift  $d \mapsto d\omega^\delta$  in the proof of theorem 4.10 and from the degree argument of remark 4.11. It needs no new exponent, and so the conclusion of theorem 4.10 holds in every fixed workspace. *Claim:* for every ordered abelian group  $\Gamma \neq 0$ , every  $0 \neq t \in \Pi_\Gamma$  and every  $r \in \mathbb{R} \setminus \mathbb{Q}$ , the pair  $(t, rt)$  has no gcd in the ring  $\mathbf{Oz}_\Gamma$  of section 20. Indeed, let  $d$  be a common divisor, with  $t = du$  and  $rt = dv$ . Then  $v = ru$ , and  $\text{ct}(v) = r \text{ct}(u)$  with both constants in  $\mathbb{Z}$ , so  $u, v \in \Pi_\Gamma$ . Dividing coefficients by 2 preserves supports, so  $u/2, v/2 \in \Pi_\Gamma$  and  $2d$  is a common divisor. It does not divide  $d$ , because  $1/2 \notin \mathbf{Oz}_\Gamma$ .

### 15.3 The multiplier theorem

We first isolate the algebraic mechanism. Throughout this subsection,  $R$  is a domain containing  $\mathbb{R}$  with a ring retraction  $c : R \rightarrow \mathbb{R}$  that fixes  $\mathbb{R}$ . Put  $I_R = \ker c$  and  $R_\mathbb{Z} = c^{-1}(\mathbb{Z}) = \mathbb{Z} \oplus I_R$ , and let  $K$  be any field containing  $R$ , possibly larger than  $\text{Frac}(R)$ . The main example is  $(R, c) = (\mathcal{B}_\mathbb{R}, \text{ct})$ , with  $R_\mathbb{Z} = \mathbf{Oz}$  and  $I_R = \Pi$ . Another is the polynomial ring  $R = \mathbb{R}[T]$  with evaluation at  $T = 0$ ; its image  $\mathbb{R}[t]$  under evaluation at the omnific parameter is treated in section 15.9. Every element of  $R$  decomposes uniquely as  $c(h) + (h - c(h)) \in \mathbb{R} \oplus I_R$ , and multiplication by a nonzero real preserves  $I_R$ . No nonzero-kernel hypothesis is needed for the algebraic theorem: when  $I_R = 0$ , the irrational case simply has no nonzero admissible scalar.

**Definition 15.9** (Rational directions [08]). A nonzero  $\mathbf{v} \in \mathbb{R}^{n+1}$  has *rational projective direction* if its line contains a nonzero vector of  $\mathbb{Q}^{n+1}$ , equivalently of  $\mathbb{Z}^{n+1}$ . Such a line has a primitive integral generator, unique up to sign.

**Lemma 15.10** (The scalar lattice of a real line [08]). For  $\mathbf{v} \neq 0$  put  $\mathcal{L}_{\mathbf{v}} = \{s \in \mathbb{R} : s\mathbf{v} \in \mathbb{Z}^{n+1}\}$ . If the direction of  $\mathbf{v}$  is not rational, then  $\mathcal{L}_{\mathbf{v}} = \{0\}$ . If  $\mathbf{v}$  is a primitive integral vector, then  $\mathcal{L}_{\mathbf{v}} = \mathbb{Z}$ .

*Proof.* A nonzero  $s$  in the first case would make the line rational. In the second case choose integers  $u_i$  with  $\sum_i u_i v_i = 1$ . If every  $sv_i$  is an integer, so is  $s = \sum_i u_i(sv_i)$ . Conversely every integer  $s$  works. More generally, if  $\lambda\mathbf{v}$  is primitive integral, then  $s\mathbf{v} \in \mathbb{Z}^{n+1}$  exactly when  $s/\lambda \in \mathbb{Z}$ , giving  $\mathcal{L}_{\mathbf{v}} = \lambda\mathbb{Z}$ . Clearing ordinary denominators and dividing by the positive integer gcd constructs this primitive generator; the same lattice calculation shows that it is unique up to sign.  $\square$

**Theorem 15.11** (All ambient scalar representations [08]; pairs [09]). Let  $\mathbf{P} = (P_0, \dots, P_n) \in R^{n+1}$  be unimodular over  $R$ . Its residue vector  $\mathbf{v} = (c(P_0), \dots, c(P_n))$  is nonzero, and

$$\{h \in K : hP_i \in R_{\mathbb{Z}} \text{ for all } i\} = \{h \in R : c(h)\mathbf{v} \in \mathbb{Z}^{n+1}\}. \quad (15.3)$$

- (a) If  $[\mathbf{v}] \in \mathbb{P}^n(\mathbb{Q})$ , multiply  $\mathbf{P}$  by a nonzero real scalar so that  $\mathbf{v}$  becomes primitive integral. The set (15.3) is then exactly  $R_{\mathbb{Z}}$ , and the normalized vector is unimodular over  $R_{\mathbb{Z}}$ .
- (b) If  $[\mathbf{v}] \notin \mathbb{P}^n(\mathbb{Q})$ , the set (15.3) is exactly  $I_R$ , and this does not change under nonzero real rescaling of  $\mathbf{P}$ .

*Proof.* Choose  $U_i \in R$  with  $\sum_i U_i P_i = 1$ . Applying  $c$  shows  $\mathbf{v} \neq 0$ . If every  $hP_i$  lies in  $R_{\mathbb{Z}} \subseteq R$ , then

$$h = \sum_i U_i(hP_i) \in R.$$

For  $h \in R$ , membership of  $hP_i$  in  $R_{\mathbb{Z}}$  is exactly  $c(h)v_i \in \mathbb{Z}$ . This proves (15.3), and lemma 15.10 gives the two scalar sets. It remains to show that the normalized vector in (a) is unimodular over  $R_{\mathbb{Z}}$ , not merely primitive. Here  $P_i$  denotes the normalized coordinate, and  $U_i$  is a Bézout row for that normalized vector: rescaling the original vector by  $\lambda$  rescales its row by  $\lambda^{-1}$ . The normalized coordinates  $P_i$  belong to  $R_{\mathbb{Z}}$  because their residues are integers. Choose  $u_i \in \mathbb{Z}$  with  $\sum_i u_i v_i = 1$ , put  $S = \sum_i u_i P_i$ , so that  $c(S) = 1$ , and define

$$V_i = u_i + (1 - S)U_i. \quad (15.4)$$

Each  $V_i$  has residue  $u_i \in \mathbb{Z}$ , so  $V_i \in R_{\mathbb{Z}}$ , and  $\sum_i V_i P_i = S + (1 - S)\sum_i U_i P_i = 1$ .  $\square$

The first identity of the proof forces  $h$  back into  $R$  even when  $h$  is allowed in a much larger field. For  $R = \mathcal{B}_{\mathbb{R}}$  this is why numerators and denominators with arbitrary infinite supports, or with scales outside the supports of  $\mathbf{P}$ , cannot evade the classification; formal reduction of a rational function would not show this [08, 09]. The scalar set includes zero; nonzero scalars give the actual projective representations. If  $\lambda\mathbf{v}$  is primitive integral, the scalar set for the *original* vector is  $\lambda R_{\mathbb{Z}}$ , since  $c(h) \in \lambda\mathbb{Z}$  and  $\lambda I_R = I_R$ ; it becomes  $R_{\mathbb{Z}}$  after normalization. Unimodularity over  $R$  is essential. For instance the pair  $(\omega, \omega)$  over  $\mathcal{B}_{\mathbb{R}}$  admits the scalar  $\omega^{-1} \notin \mathcal{B}_{\mathbb{R}}$ , which takes it to  $(1, 1)$ , but has no Bézout identity over  $\mathcal{B}_{\mathbb{R}}$ .

*Remark 15.12* (A second route to the lift [09]). Source 09 adjusts constants instead of using (15.4). For a pair  $(A_1, A_0)$  normalized to the primitive constant vector  $(m_1, m_0)$  with  $U_0 A_1 + V_0 A_0 = 1$  over  $R$ , the real solutions  $(y_1, y_0)$  of  $y_1 m_1 + y_0 m_0 = 1$  differ by multiples of

$(m_0, -m_1)$ . Choose integers  $u_1, u_0$  with  $u_1 m_1 + u_0 m_0 = 1$  and the real  $k$  with  $(u_1, u_0) = (c(U_0), c(V_0)) + k(m_0, -m_1)$ . Then  $U_0 + kA_0$  and  $V_0 - kA_1$  have constants  $u_1$  and  $u_0$  and still form a Bézout row. With  $n + 1$  coordinates and constant vector  $\mathbf{m}$ , let  $\mathbf{w} = \mathbf{u} - c(\mathbf{U})$  for an integer row  $\mathbf{u}$  with  $\mathbf{u} \cdot \mathbf{m} = 1$ , and choose a real row  $\mathbf{r}$  with  $\mathbf{r} \cdot \mathbf{m} = 1$ . Write  $\mathbf{A}$  for the normalized column and  $e_j$  for the  $j$ th standard row. Add to  $\mathbf{U}$  the row  $\sum_{i,j} r_i w_j (A_i e_j - A_j e_i)$ , summing over all ordered pairs. Each summand pairs to zero with  $\mathbf{A}$ , so the Bézout identity is unchanged. Its constant row is  $(\mathbf{r} \cdot \mathbf{m})\mathbf{w} - (\mathbf{w} \cdot \mathbf{m})\mathbf{r} = \mathbf{w}$ , since  $\mathbf{w} \cdot \mathbf{m} = 0$ . Source 08's formula (15.4) is shorter.

*Provenance.* The ring  $R_{\mathbb{Z}}$  is the pullback of  $c$  along  $\mathbb{Z} \hookrightarrow \mathbb{R}$ . Neither source claims the pullback construction or the use of coefficient retractions as new. Source 08 cites the retraction transfer lemma of L'Innocente–Mantova [3, Lemma 9.2.1] as a methodological precedent, and source 09 cites Kława [33] for background on  $D + M$  rings. The sibling report credits the  $D + M$  context to Dobbs–Papick, and its computation of scalar homomorphisms `osq:tn:thm:hom` (manuscript 14 there) uses the same step: a scalar  $q$  with  $qm_0 \in B$  satisfies  $q = (qm_0)/m_0 \in B$ , and then only the constant part matters. Source 09's pair version (its multiplier theorem, with multipliers in  $\Pi \oplus \{c \in \mathbb{R} : cp, cq \in \mathbb{Z}\}$ ) is the case  $n = 1$ ,  $R = \mathcal{B}_{\mathbb{R}}$  of theorem 15.11; its sketch for  $n + 1$  coordinates is contained in it.

## 15.4 The unimodular region and its specialization

**Definition 15.13** (The unimodular region [09]). Let  $\mathcal{V} \subseteq \mathbb{P}^1(\mathbf{No})$  be the class of points with a presentation  $[P : Q]$  such that  $P, Q \in \mathcal{B}_{\mathbb{R}}$  and  $UP + VQ = 1$  for some  $U, V \in \mathcal{B}_{\mathbb{R}}$ . For such a presentation put

$$\mathrm{sp}([P : Q]) = [\mathrm{ct}(P) : \mathrm{ct}(Q)] \in \mathbb{P}^1(\mathbb{R}).$$

Let  $\mathcal{U} \subseteq \mathcal{V}$  be the class of points with a presentation that is unimodular over  $\mathbf{Oz}$ .

**Lemma 15.14** (Well-defined specialization [09]). *Two  $\mathcal{B}_{\mathbb{R}}$ -unimodular presentations of the same point differ by a nonzero real scalar, so  $\mathrm{sp} : \mathcal{V} \rightarrow \mathbb{P}^1(\mathbb{R})$  is well defined. Moreover  $\mathbb{P}^1(\mathbb{R}) \subseteq \mathcal{V}$  with  $\mathrm{sp}(r) = r$ , and  $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{V}$  with  $\mathrm{sp}(x) = \mathrm{ct}(x)$ .*

*Proof.* Applying  $\mathrm{ct}$  to a Bézout identity shows that the two constants of a unimodular pair are not both zero. If  $(P', Q') = h(P, Q)$  with  $h \in \mathbf{No}^\times$ , the Bézout identity of  $(P, Q)$  gives  $h = UP' + VQ' \in \mathcal{B}_{\mathbb{R}}$ , and that of  $(P', Q')$  gives  $h^{-1} \in \mathcal{B}_{\mathbb{R}}$ . So  $h \in \mathcal{B}_{\mathbb{R}}^\times = \mathbb{R}^\times$  (proposition 2.5), and the constant vectors define the same point. Use the pairs  $(r, 1)$ ,  $(1, 0)$  and  $(x, 1)$  for the last assertions.  $\square$

**Theorem 15.15** (Denominator dichotomy on the unimodular region [09]; [08]). *Let  $x = P/Q$  be an affine point of  $\mathcal{V}$ , with  $(P, Q)$  unimodular over  $\mathcal{B}_{\mathbb{R}}$  and  $Q \neq 0$ .*

- (i) *If  $\mathrm{sp}(x) \in \mathbb{P}^1(\mathbb{Q})$ , choose  $\lambda \in \mathbb{R}^\times$  with  $(\lambda \mathrm{ct} P, \lambda \mathrm{ct} Q)$  a primitive integer vector and  $\lambda Q > 0$ . Then  $\mathfrak{D}(x) = \lambda Q \mathbf{Oz}$ . The presentation  $x = \lambda P / \lambda Q$  is unimodular over  $\mathbf{Oz}$  and is the only primitive presentation with positive denominator, and  $\lambda Q$  is the least positive element of  $\mathfrak{D}(x)$ .*
- (ii) *If  $\mathrm{sp}(x) \notin \mathbb{P}^1(\mathbb{Q})$ , then  $\mathfrak{D}(x) = Q \Pi$ . This ideal is not generated by any set. Every presentation has both entries in  $\Pi$ , none is primitive, none has a gcd, and every positive denominator can be halved.*

Consequently  $\mathcal{U} = \mathrm{sp}^{-1}(\mathbb{P}^1(\mathbb{Q}))$ .

*Proof.* Apply [theorem 15.11](#) with  $n = 1$ ,  $(R, c) = (\mathcal{B}_{\mathbb{R}}, \text{ct})$  and  $K = \mathbf{No}$ . The representations of  $x$  are the vectors  $h(P, Q)$  with  $h$  in the set [\(15.3\)](#), and their denominators are  $hQ$ . In case (i),  $(\lambda P, \lambda Q)$  is unimodular over  $\mathbf{Oz}$ , and [lemma 15.4](#) gives the rest. In case (ii) the admissible multipliers form  $\Pi$ , so  $\mathfrak{D}(x) = Q\Pi$ . Multiplication by  $Q$  is an  $\mathbf{Oz}$ -linear bijection  $\Pi \rightarrow Q\Pi$ , so [corollary 4.3](#) excludes a set of generators. Every representation is  $h(P, Q)$  with  $h \in \Pi$  and  $P, Q \in \mathcal{B}_{\mathbb{R}}$ , so both entries lie in  $\Pi$ . If  $d$  is a common divisor of a representation, the quotient is again a representation, so its entries lie in  $\Pi$  and  $2d$  is a common divisor that does not divide  $d$ . Halving  $h \in \Pi$  keeps it in  $\Pi$ . Finally, the constant vector of an  $\mathbf{Oz}$ -unimodular pair is a primitive integer vector (apply  $\text{ct}$  to its Bézout identity), so  $\mathcal{U}$  lies in the rational fibers. Conversely, (i) gives an  $\mathbf{Oz}$ -unimodular presentation of every affine point with rational specialization, and  $\infty = [1 : 0]$ .  $\square$

*Remark 15.16* (The exact strength of the dichotomy [\[09\]](#)). For affine points of  $\mathcal{V}$  the following coincide: rational specialization, unimodularity over  $\mathbf{Oz}$ , principality of the denominator ideal, a least positive denominator, and a primitive presentation. The equivalence is proved on  $\mathcal{V} \cap \mathbf{No}$  only. The point  $\infty$  also has rational specialization and the unimodular presentation  $[1 : 0]$ , but the affine denominator ideal and its least positive element are not defined there. No such equivalence is asserted outside  $\mathcal{V}$ , and neither source claims that every surreal lies in  $\mathcal{V}$  ([question 21.13](#)). Source 09 notes that L’Innocente–Mantova use “reduced” for a different, support-theoretic notion [\[3\]](#); the word is not used here for lowest terms. For  $n + 1$  coordinates the analogue is [theorem 15.11](#) itself.

## 15.5 Rational functions of one parameter

**Lemma 15.17** (Evaluation at an omnific parameter [\[08\]](#); monomials [\[09\]](#)). *Evaluation at  $t$  embeds  $\mathbb{R}[T]$  in  $\mathcal{B}_{\mathbb{R}}$  and extends to an embedding  $\mathbb{R}(T) \hookrightarrow \mathbf{No}$ . For every polynomial  $p$ ,  $\text{ct}(p(t)) = p(0)$ , and for  $p \neq 0$ ,  $\deg_{\omega}(p(t)) = (\deg p) \deg_{\omega}(t)$ .*

*Proof.* Every power of  $t$  has nonnegative support, so polynomial evaluation lands in  $\mathcal{B}_{\mathbb{R}}$ . Applying the retraction and using  $\text{ct}(t) = 0$  gives  $\text{ct}(p(t)) = p(0)$ . Put  $\delta = \deg_{\omega}(t) > 0$ ; this holds for either sign of  $t$ . The nonzero terms  $a_j t^j$  have distinct leading exponents  $j\delta$ , so the term of largest degree cannot cancel; this is the argument of [proposition 2.8](#). Thus evaluation is injective. In particular  $q(t) \neq 0$  for every nonzero  $q \in \mathbb{R}[T]$ , and the fraction-field extension is well defined and injective by cross multiplication. For later use, if  $0 \neq f = p/q \in \mathbb{R}(T)$ , multiplicativity of leading exponents gives

$$\deg_{\omega}(f(t)) = (\deg p - \deg q)\delta = (\deg f)\delta.$$

$\square$

**Theorem 15.18** (Representations of rational-function tuples [\[08\]](#)). *Let  $p_1, \dots, p_n, q \in \mathbb{R}[T]$  with  $q \neq 0$  and  $\gcd(p_1, \dots, p_n, q) = 1$ , put  $x_j = p_j(t)/q(t)$  and*

$$\text{sp}(\mathbf{x}) = [p_1(0) : \dots : p_n(0) : q(0)] \in \mathbb{P}^n(\mathbb{R}).$$

*This point is defined. Every representation of  $\mathbf{x} = (x_1, \dots, x_n)$  is uniquely of the form*

$$h(p_1(t), \dots, p_n(t), q(t)), \quad h \neq 0,$$

*and:*

- (a) *If  $\text{sp}(\mathbf{x}) \in \mathbb{P}^n(\mathbb{Q})$ , rescale the polynomials by a common nonzero real scalar so that their constant vector is primitive integral. Then  $h$  ranges over  $\mathbf{Oz} \setminus \{0\}$ ,  $\mathfrak{D}(\mathbf{x}) = q(t) \mathbf{Oz}$ , and the polynomial vector at  $t$  is unimodular over  $\mathbf{Oz}$ .*

- (b) If  $\text{sp}(\mathbf{x}) \notin \mathbb{P}^n(\mathbb{Q})$ , no normalization is needed. Then  $h$  ranges over  $\Pi \setminus \{0\}$ , and  $\mathfrak{D}(\mathbf{x}) = q(t)\Pi$ , an ideal with no set of generators.

*These statements cover representations with arbitrary surreal supports.*

*Proof.* The polynomial gcd gives a Bézout identity  $\sum_j U_j(T)p_j(T) + U_0(T)q(T) = 1$  in  $\mathbb{R}[T]$ . By lemma 15.17 its evaluation shows that the vector at  $t$  is unimodular over  $\mathcal{B}_{\mathbb{R}}$ , with residue vector the constant vector of the polynomials. Evaluating the same identity at 0 shows that this vector is nonzero. Equivalently, if all constants vanished then  $T$  would be a common polynomial factor. Apply theorem 15.11 with  $(R, c) = (\mathcal{B}_{\mathbb{R}}, \text{ct})$  and  $K = \mathbf{No}$ . Every representation has the unique multiplier  $h = b/q(t) \neq 0$ , and every admissible nonzero  $h$  gives a representation. In (a),  $q$  and the  $p_j$  henceforth mean the rescaled polynomials; for an original denominator  $q_{\text{old}}$  the ideal is  $\lambda q_{\text{old}}(t)\mathbf{Oz}$ , where  $\lambda$  is the common normalizing scalar. The ideals are the classes of last coordinates, including zero. In (b), multiplication by  $q(t) \neq 0$  is an  $\mathbf{Oz}$ -linear bijection  $\Pi \rightarrow q(t)\Pi$ . A set of generators for its image would pull back to a set of generators for  $\Pi$ , contrary to corollary 4.3.  $\square$

The projective specialization is independent of the chosen reduced polynomial vector. Indeed, two such evaluated vectors for the same tuple differ by a nonzero surreal scalar. Their two  $\mathcal{B}_{\mathbb{R}}$ -Bézout identities force both that scalar and its inverse into  $\mathcal{B}_{\mathbb{R}}$ , just as in lemma 15.14; hence the scalar is a nonzero real. Taking constants gives the same projective point. This argument applies to any number of coordinates and is compatible with  $\text{sp}$  on  $\mathcal{V}$ .

*Provenance.* Source 09 proves the case  $n = 1$ ,  $t = \omega^\alpha$  (its one-scale classification), including the formula  $\text{sp}(P(\omega^\alpha)/Q(\omega^\alpha)) = [P(0) : Q(0)]$  and the membership of these points in  $\mathcal{V}$ . It is the special case of theorem 15.18 and theorem 15.15 and is not printed separately. Source 08's theorem has any number of coordinates and any  $t \in \Pi \setminus \{0\}$ ; a nonzero  $t$  that is merely infinite, with a nonintegral constant, is not allowed [08]. The coprimality over  $\mathbb{R}[T]$  is necessary: an uncanceled pair such as  $(T(T+r), T(T+1))$  has constant vector zero [09].

**Corollary 15.19** (Lowest terms are all or nothing [08]; [09]). *In theorem 15.18(a) every representation has a gcd, namely its multiplier  $h$  up to sign. The only primitive representations are the two signs of the normalized polynomial vector, and both are unimodular. The least positive common denominator is  $|q(t)|$ . In case (b) no representation is primitive or has a gcd, every nonzero ordinary integer divides every coordinate of every representation, every common divisor can be replaced by a strictly larger one in the divisibility order, and there is no least positive common denominator.*

*Proof.* Case (a) is lemma 15.4 with  $\mathbf{Oz}^\times = \{\pm 1\}$ , and  $|h| \geq 1$  for  $0 \neq h \in \mathbf{Oz}$ . The denominator  $|q(t)|$  is attained by choosing  $h = 1$  or  $h = -1$  according to the sign of  $q(t)$ . In case (b) every coordinate is  $h$  times an element of  $\mathcal{B}_{\mathbb{R}}$  with  $h \in \Pi$ , hence lies in  $\Pi$ . If  $d$  is a common divisor, dividing by  $d$  gives another representation, whose multiplier  $h/d$  lies in  $\Pi$  by theorem 15.18; so  $2d$  is a common divisor not dividing  $d$ : otherwise cancellation of  $d \neq 0$  would put  $1/2$  in  $\mathbf{Oz}$ . Division by any nonzero ordinary integer preserves positive support, proving the asserted ordinary divisibility of all coordinates. Replacing  $h$  by  $h/2$  halves a positive denominator and preserves the whole representation.  $\square$

This strengthens theorem 4.10: there the pair is the constant point  $(t, rt)$ , while here the absence of a gcd holds for every representation of every point with irrational specialization, for example  $h(t+r, t+1)$ .

**Corollary 15.20** (Leading exponents of denominators [08]). *The leading exponents of the nonzero common denominators of  $\mathbf{x}$  in theorem 15.18 are exactly the surreals  $\deg_\omega q(t) + \eta$  with  $\eta \geq 0$  in case (a), and with  $\eta > 0$  in case (b).*

*Proof.* Use lemma 2.4 and the two ideal formulas. The nonzero elements of  $\mathbf{Oz}$  have every nonnegative leading exponent, and those of  $\Pi$  every positive one; ordinary integers and monomials  $\omega^\eta$  realize them.  $\square$

This measures the leading scale, not the support. In case (b) the endpoint is missing although denominators exist at every larger leading exponent [08].

**Corollary 15.21** (One rational function [08, 09]). *Let  $f = p/q \in \mathbb{R}(T)$  be reduced. Then  $f(t) \in \mathcal{V}$  with  $\text{sp}(f(t)) = f(0)$ , read in  $\mathbb{R} \cup \{\infty\}$ . It has a primitive omnific representation if and only if  $f(0) \in \mathbb{Q} \cup \{\infty\}$ ; lowest terms are then unique once the denominator is positive, and otherwise no representation has a gcd. The criterion does not depend on  $t$ .*

*Proof.* For a reduced pair  $p(0)$  and  $q(0)$  are not both zero, and  $[p(0) : q(0)]$  is rational exactly when  $f(0) \in \mathbb{Q} \cup \{\infty\}$ . Apply theorem 15.18 and corollary 15.19. In particular a pole at 0 means  $q(0) = 0$  and  $p(0) \neq 0$ , giving the rational projective point  $[1 : 0]$ ; it does not mean that  $q(t) = 0$ . The zero function uses the reduced pair  $(0, 1)$  and has least denominator 1.  $\square$

*Remark 15.22* (A formal specialization [08, 09]). The value  $f(0)$  is a substitution in an ordinary rational function before evaluation at  $t$ . It is not a surreal value of the infinite number  $t$ , nor the valuation residue at infinity. There is no operation of “setting  $\omega = 0$ ” on arbitrary normal forms, and no homomorphism  $\mathbf{No} \rightarrow \mathbb{R}$  extending the substitution: a unital field homomorphism extending  $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$  would send the invertible element  $\omega$  to 0. The criterion belongs to the rational-function subfield and the retraction on  $\mathcal{B}_\mathbb{R}$ . The map  $\text{sp}$  is not a field homomorphism either: its domain is only  $\mathcal{V}$ , its values are projective, and field operations can make both constants of a presentation vanish [09].

**Example 15.23** (Denominator ideals [08, 09]). Let  $t > 0$  in  $\Pi$  and let  $r$  be an irrational real. Generators are given up to sign.

| $x$                                | $\text{sp}(x)$ | $\mathfrak{D}(x)$     | Lowest terms | Source |
|------------------------------------|----------------|-----------------------|--------------|--------|
| $m/n$ , $\gcd(m, n) = 1$ , $n > 0$ | $m/n$          | $n\mathbf{Oz}$        | yes          | 08, 09 |
| $r$                                | $r$            | $\Pi$                 | no           | 08, 09 |
| $t + r$                            | $r$            | $\Pi$                 | no           | 08     |
| $rt + 3/5$                         | $3/5$          | $5\mathbf{Oz}$        | yes          | 09     |
| $1/t$                              | $\infty$       | $t\mathbf{Oz}$        | yes          | 08     |
| $r/t$                              | $\infty$       | $(t/r)\mathbf{Oz}$    | yes          | 08     |
| $(rt + 1)/(t + 1)$                 | 1              | $(t + 1)\mathbf{Oz}$  | yes          | 08     |
| $(2t + 3)/(5t + 7)$                | $3/7$          | $(5t + 7)\mathbf{Oz}$ | yes          | 09     |
| $(rt + 2)/(t + 3)$                 | $2/3$          | $(t + 3)\mathbf{Oz}$  | yes          | 09     |
| $(t + r)/(t + 1)$                  | $r$            | $(t + 1)\Pi$          | no           | 08, 09 |
| $(t^2 + 1)/(t^2 + r)$              | $1/r$          | $(t^2 + r)\Pi$        | no           | 09     |

The constant-denominator rows are coprime automatically. In the nonconstant rows involving  $r$ , a common root would force  $r = 1$ ,  $r = 2/3$  or  $r = 0$ , as applicable; irrationality excludes these possibilities. For  $(2T + 3, 5T + 7)$ , the combination  $5(2T + 3) - 2(5T + 7) = 1$  is already a Bézout identity. The denominator  $t/r$  in the row  $r/t$  comes from rescaling  $(r, T)$  by  $1/r$ , not from asserting that  $r \in \mathbf{Oz}$ . The function  $(rt + 1)/(t + 1)$  has irrational standard part  $r$  but

unique lowest terms, while  $(t + r)/(t + 1)$  has standard part 1 and none. So the standard part does not decide cancellation [08].

The class of surreals with a primitive omnific representation is not a subring [08]. Both  $t$  and  $\sqrt{2}/t$  have one, but their product  $\sqrt{2}$  does not. Both  $1/t$  and  $\sqrt{2} - 1/t = (\sqrt{2}t - 1)/t$  specialize to  $\infty$ , but their sum  $\sqrt{2}$  does not. These are statements about primitive representations, not about the existence of fractions.

**Example 15.24** (Nearly equal values, opposite denominators [09]). Let  $r$  be irrational and  $t > 0$  in  $\Pi$ , and put

$$x_+ = r + \frac{1}{t} = \frac{rt + 1}{t}, \quad x_- = r + \frac{1}{t + 1} = \frac{rt + r + 1}{t + 1}.$$

Then  $\text{sp}(x_+) = \infty$ , and  $t$  is the least positive denominator of  $x_+$ , with the certificate

$$(1 - rt)(1 + rt) + (r^2t)t = 1 \tag{15.5}$$

over **Oz**:  $1 - rt$  has integer constant 1, while  $r^2t \in \Pi$ , so the real coefficients do not obstruct membership of the witnesses. But  $\text{sp}(x_-) = r + 1 \notin \mathbb{Q}$ , so  $\mathfrak{D}(x_-) = (t + 1)\Pi$  and no presentation of  $x_-$  has a gcd. Yet  $x_+ - x_- = 1/(t(t + 1))$  and  $\text{st}(x_+) = \text{st}(x_-) = r$ . The standard part does not detect the obstruction, even at the next infinitesimal scale.

*Remark 15.25* (A certificate procedure [08, 09]). For  $f_1, \dots, f_n \in \mathbb{R}(T)$  the proofs give a finite procedure. (1) Choose a common denominator, divide out the polynomial gcd, and compute a polynomial Bézout identity. (2) Evaluate at 0 and decide whether the direction is rational. (3) If it is not, return  $\mathfrak{D} = q(t)\Pi$ . (4) If it is, normalize to a primitive integer vector, compute an integer Bézout identity, and apply (15.4) (or [remark 15.12](#)) to get an omnific Bézout certificate. To perform step (2), choose a nonzero constant coordinate  $v_k$ ; the direction is rational exactly when every  $v_j/v_k$  is rational. Clearing their denominators and then their integer gcd gives the primitive vector and the scalar used in step (4). (5) For the chosen parameter, choose the sign so that  $q(t) > 0$ :  $\text{sgn}(q(t)) = \text{sgn}(\text{lc}(q)) \text{sgn}(t)^{\deg q}$ . The polynomial certificate works for every nonzero purely infinite parameter; the positive-denominator sign may depend on that parameter. The ambient-scalar identity makes the classification apply to representations with arbitrary supports. The procedure is effective over coefficient fields with exact arithmetic, decidable equality and rationality, and the required sign tests, for example real algebraic number fields; it is not a uniform algorithm for reals given by approximation oracles, and no factorization in **Oz** is used [08, 09].

**Proposition 15.26** (Prescribed specialization near a given function [09]). *Give  $\mathbb{R}(T)$  its topology at infinity:  $g$  is  $N$ -close to  $f$  if  $\deg(g - f) < -N$ , where  $\deg$  is the degree of a rational function ( $\deg 0 = -\infty$ ). For every  $f \in \mathbb{R}(T)$ , every ordinary  $N > 0$  and every  $r \in \mathbb{R}$  there is  $g \in \mathbb{R}(T)$ , regular at 0, with  $g(0) = r$  and  $\deg(g - f) < -N$ .*

*Proof.* Choose an ordinary integer  $L > 0$  larger than the pole order of  $f$  at 0 (take that order to be 0 when  $f$  is regular), and for an ordinary integer  $M \geq 1$  put

$$H = (1 + T^L)^{-M}, \quad g = (1 - H)f + rH.$$

The numerator of  $1 - H$  is  $(1 + T^L)^M - 1$ , whose lowest nonzero term is  $MT^L$ , and its denominator is nonzero at 0. Thus  $(1 - H)f$  is regular and vanishes at 0, while  $H(0) = 1$ , proving  $g(0) = r$ . Moreover  $g - f = H(r - f)$ . If  $f = r$  this is 0; otherwise its degree is  $\deg(r - f) - LM$ . Choose  $M \geq 1$  with  $LM > N + \deg(r - f)$  to obtain the strict inequality. Degrees of rational functions add under multiplication, even if numerator and denominator factors cancel.  $\square$

Source 09 states the conclusion “unless  $g = f$  exactly”; the clause is not needed, since  $g = f$  gives degree  $-\infty$ . This topology is not the topology induced by the full surreal order. More explicitly [merge], fix  $t$  and put  $\delta = \deg_\omega(t) > 0$ . The set-cut bound supplies  $\eta > N\delta$  for every ordinary  $N \geq 0$ . Every nonzero  $f(t) \in \mathbb{R}(t)$  has leading exponent  $m\delta$  for an integer  $m$ , by [lemma 15.17](#), and hence  $|f(t)| > \omega^{-\eta}$ . The interval of radius  $\omega^{-\eta}$  around any member of  $\mathbb{R}(t)$  therefore meets  $\mathbb{R}(t)$  only there: the induced topology is discrete. In contrast, the degree topology has nonzero  $T^{-M}$  arbitrarily close to zero. [Theorem 15.35](#) achieves arbitrary surreal accuracy by choosing a new monomial after the target interval [09].

## 15.6 Simultaneous denominators and rational curves

**Example 15.27** (No least common denominator [08]). Let  $t > 0$  in  $\Pi$ . Each of  $x_1 = 1/t$  and  $x_2 = \sqrt{2}/t$  has unique lowest terms, with  $\mathfrak{D}(x_1) = t \mathbf{Oz}$  and  $\mathfrak{D}(x_2) = (t/\sqrt{2}) \mathbf{Oz}$ . But

$$\mathfrak{D}(x_1, x_2) = t \mathbf{Oz} \cap \frac{t}{\sqrt{2}} \mathbf{Oz} = t \Pi. \quad (15.6)$$

This ideal has no set of generators and no least positive member, so the two canonical denominators have no least common multiple in the divisibility order.

*Proof.* The common polynomial vector is  $(1, \sqrt{2}, T)$ , with specialization  $[1 : \sqrt{2} : 0]$ , which is not rational. [Theorem 15.18](#) gives (15.6). An lcm would generate the intersection of the two principal ideals: being a common multiple gives one inclusion, and dividing every common multiple gives the other. This would make  $t\Pi$  principal, contrary to its lack of a set of generators. The individual assertions are [corollary 15.21](#).  $\square$

The two fractions are already reduced; the obstruction is the irrational relation between their residues. Every finite tuple in  $\mathbb{R}(t)$  has either a principal common denominator ideal or a scalar copy of  $\Pi$ , with no intermediate case [08]. For monomial  $t = \omega^g$ , the identity (15.6) is the intersection formula  $X^g A \cap cX^g A = X^g \Pi$  of `osq:tn:thm:syzygy(iii)` in the sibling report, with  $c = 1/\sqrt{2}$ .

**Corollary 15.28** (Primitive coordinates on rational curves [08]). *Let  $p_0, \dots, p_n \in \mathbb{R}[T]$  have gcd one, and let  $\varphi = [p_0 : \dots : p_n]$ . For every  $0 \neq t \in \Pi$ , the point  $\varphi(t)$  has primitive omnific homogeneous coordinates if and only if  $\varphi(0) \in \mathbb{P}^n(\mathbb{Q})$ . Such coordinates are then unimodular and, after the common real normalization of the  $p_i$ , equal to  $\pm(p_0(t), \dots, p_n(t))$ . Otherwise no omnific coordinate vector of  $\varphi(t)$  has a gcd.*

*Proof.* The polynomial gcd gives a Bézout identity, so the vector at  $t$  is  $\mathcal{B}_{\mathbb{R}}$ -unimodular with residue vector  $(p_0(0), \dots, p_n(0))$ . Apply [theorem 15.11](#) and the argument of [corollary 15.19](#). No affine chart is needed.  $\square$

For constant  $p_i$  this is [theorem 14.5](#) of source 01. The corollary replaces a constant vector by any reduced polynomial parametrization; the criterion concerns the specialization at  $T = 0$ , not the direction at infinity [08]. Source 08’s illustration, the conic  $(1 - t^2, 2t, 1 + t^2)$  with certificate  $(1 + \frac{t^2}{2})(1 - t^2) + \frac{t^2}{2}(1 + t^2) = 1$  and specialization  $[1 : 0 : 1]$ , is [proposition 14.4](#) with source 01’s witness (14.2), up to the sign of the first coordinate; source 08 presents it as an illustration, not as a new construction.

**Example 15.29** (A curve without primitive coordinates [08]). The vector  $(1, \sqrt{2} + t, t^2)$  has specialization  $[1 : \sqrt{2} : 0]$ , which is not rational. Its entries are not all omnific, but  $h(1, \sqrt{2} + t, t^2)$  is omnific for every  $0 \neq h \in \Pi$ , and by [theorem 15.11](#) these are all the omnific

coordinate vectors of the point. None is primitive, although the first unscaled coordinate is 1: the unscaled vector does not lie in  $\mathbf{Oz}^3$ .

**Corollary 15.30** (Solutions on rational curves [08] [merge]). *Let  $F_1, \dots, F_r \in \mathbb{R}[X_0, \dots, X_n]$  be homogeneous, and let  $p_0, \dots, p_n \in \mathbb{R}[T]$  have gcd one, satisfy  $F_j(p_0(T), \dots, p_n(T)) = 0$  for all  $j$ , and have  $[p_0(0) : \dots : p_n(0)] \in \mathbb{P}^n(\mathbb{Q})$ . After the normalization of [corollary 15.28](#), for every  $0 \neq t \in \Pi$  the vector  $(p_0(t), \dots, p_n(t))$  is a unimodular omnific solution of the system, and it has an infinite coordinate when some  $p_i$  is nonconstant. If instead  $[p_0(0) : \dots : p_n(0)] \notin \mathbb{P}^n(\mathbb{Q})$ , no point  $\varphi(t)$  of the curve with  $0 \neq t \in \Pi$  has a primitive, in particular a unimodular, omnific coordinate vector.*

*Proof.* A common real rescaling by  $\lambda \neq 0$  preserves each equation: homogeneity gives  $F_j(\lambda p_0, \dots, \lambda p_n) = \lambda^{\deg F_j} F_j(p_0, \dots, p_n) = 0$  for nonzero  $F_j$ , and zero equations are automatic. Evaluation then preserves these identities. [Corollary 15.28](#) gives the unimodular vector in the rational case and excludes primitive coordinates in the irrational case. If  $p_i$  has positive degree, [lemma 15.17](#) gives  $\deg_\omega(p_i(t)) = (\deg p_i) \deg_\omega(t) > 0$ , so that coordinate is infinite in absolute value; rescaling by a nonzero real does not change its degree.  $\square$

This answers [question 21.3](#) in part: a homogeneous system whose projective zero locus contains such a rational curve has unimodular omnific solutions, and along the curve the points with primitive coordinates are exactly those of rational specialization. It says nothing about points off rational curves, nor about the dependence on an integral model asked there ([section 21.3](#)). The corollary is the merge’s formulation of source 08’s [corollary 15.28](#). [Proposition 14.1](#) asks for integer constants of the  $p_i$  and gives omnific solutions, not unimodularity; here only its projective direction needs to be rational, and the output after normalization is unimodular. The polynomial gcd condition is also essential for this stronger conclusion.

## 15.7 Specialization fibers, congruence orbits and density

Apply  $\text{ct}$  entrywise to matrices. It gives a split exact sequence

$$1 \longrightarrow K \longrightarrow \text{SL}_2(\mathbf{Oz}) \xrightarrow{\text{ct}} \text{SL}_2(\mathbb{Z}) \longrightarrow 1, \quad K = \ker(\text{ct} : \text{SL}_2(\mathbf{Oz}) \rightarrow \text{SL}_2(\mathbb{Z})),$$

split by the integer matrices. The kernel of  $\text{ct} : \text{SL}_2(\mathcal{B}_{\mathbb{R}}) \rightarrow \text{SL}_2(\mathbb{R})$  is the same group  $K$ : determinants are polynomial expressions, so taking constants commutes with determinants and matrix multiplication, and a matrix over  $\mathcal{B}_{\mathbb{R}}$  with constant matrix  $I$  has integer constants, hence entries in  $\mathbf{Oz}$  [09]. Source 09 writes  $\Gamma$  for  $K$ .

**Lemma 15.31** (Equivariance [09]). *The class  $\mathcal{V}$  is stable under  $\text{SL}_2(\mathcal{B}_{\mathbb{R}})$ , and  $\text{sp}(gx) = \text{ct}(g) \text{sp}(x)$ . In particular  $K$  preserves  $\text{sp}$ .*

*Proof.* An invertible matrix carries a unimodular column to a unimodular column: apply its inverse to transport a Bézout row. Explicitly, if a row  $u$  and a column  $v$  satisfy  $uv = 1$ , then  $(ug^{-1})(gv) = 1$  with entries still in  $\mathcal{B}_{\mathbb{R}}$ . Taking constants commutes with matrix multiplication, giving  $\text{ct}(gv) = \text{ct}(g) \text{ct}(v)$  and the asserted projective formula.  $\square$

**Theorem 15.32** (Specialization fibers are congruence orbits [09]). *For every  $r \in \mathbb{P}^1(\mathbb{R})$ ,  $\text{sp}^{-1}(r) = K \cdot r$ . Hence*

$$\mathcal{V} = \coprod_{r \in \mathbb{P}^1(\mathbb{R})} K \cdot r, \quad \mathcal{U} = \coprod_{r \in \mathbb{P}^1(\mathbb{Q})} K \cdot r,$$

*disjoint unions of classes, and every  $K$ -orbit in  $\mathcal{V}$  contains exactly one constant real projective point (including  $\infty$ ).*

*Proof.* One inclusion is [lemma 15.31](#). For the other let  $(P, Q)$  be a  $\mathcal{B}_{\mathbb{R}}$ -unimodular column representing  $x$ , with  $UP + VQ = 1$ , and put

$$M = \begin{pmatrix} P & -V \\ Q & U \end{pmatrix} \in \mathrm{SL}_2(\mathcal{B}_{\mathbb{R}}), \quad M_0 = \mathrm{ct}(M) \in \mathrm{SL}_2(\mathbb{R}), \quad g = MM_0^{-1}.$$

Then  $M\infty = x$ ,  $\mathrm{ct}(g) = I$ , so  $g \in \mathbf{K}$ , and  $g(M_0\infty) = x$  with  $M_0\infty = \mathrm{sp}(x)$ . Distinct real projective points have distinct specializations, and the description of  $\mathcal{U}$  follows from [theorem 15.15](#).  $\square$

**Corollary 15.33** (Modular equivalence [09]). *For  $r, s \in \mathbb{P}^1(\mathbb{R})$ ,  $s \in \mathrm{SL}_2(\mathbf{Oz}) \cdot r$  if and only if  $s \in \mathrm{SL}_2(\mathbb{Z}) \cdot r$ . Moreover  $\mathcal{U} = \mathrm{SL}_2(\mathbf{Oz}) \cdot \infty$ .*

*Proof.* If  $s = gr$ , then  $s = \mathrm{sp}(s) = \mathrm{ct}(g)r$  by [lemma 15.31](#); the converse uses integer matrices. An  $\mathbf{Oz}$ -unimodular column completes to a determinant-one matrix over  $\mathbf{Oz}$  as in the proof of [theorem 15.32](#).  $\square$

The orbits carry a canonical real projective label, and the rational labels select exactly the fibers whose affine points have lowest terms [09]. The fiber labeled  $\infty$  has affine points too, such as  $1/t$ .

Here and below the topology on  $\mathbb{P}^1(\mathbf{No})$  is the ordered-field topology in its two affine charts. Around a surreal point use order intervals. Around  $\infty$  use the reciprocal chart  $[P : Q] \mapsto Q/P$ : basic neighborhoods are  $\{\infty\} \cup \{z \in \mathbf{No} : |z| > 1/\varepsilon\}$  for  $\varepsilon > 0$ . In particular every nonempty projective open neighborhood contains an affine interval. These are class-defined neighborhoods; no set of all surreal open subsets is assumed.

**Lemma 15.34** (Focusing matrices [09]). *For  $x \in \mathbf{No}$  and  $b \in \Pi$  with  $bx, bx^2 \in \Pi$  put*

$$\mathbf{F}_{x,b} = \begin{pmatrix} 1 + bx & -bx^2 \\ b & 1 - bx \end{pmatrix} = I + b\mathbf{N}_x, \quad \mathbf{N}_x = \begin{pmatrix} x & -x^2 \\ 1 & -x \end{pmatrix}.$$

*Then  $\mathbf{N}_x^2 = 0$ ,  $\mathbf{F}_{x,b} \in \mathbf{K}$ ,  $\mathbf{F}_{x,b}$  fixes  $x$ ,  $\mathbf{F}_{x,b}\mathbf{F}_{x,c} = \mathbf{F}_{x,b+c}$  when both are defined,  $\mathbf{F}_{x,b}^{-1} = \mathbf{F}_{x,-b}$ , and*

$$\mathbf{F}_{x,b}(z) = x + \frac{z - x}{1 + b(z - x)}, \quad \mathbf{F}_{x,b}(\infty) = x + \frac{1}{b} \quad (b \neq 0), \quad (15.7)$$

*the first formula where its denominator is nonzero. For  $b = 0$  the matrix is the identity and fixes  $\infty$ . For  $b \neq 0$ , the pole  $x - 1/b$  is sent to  $\infty$ .*

*Proof.* The entries other than the constant diagonal lie in  $\Pi$ , and  $\det \mathbf{F}_{x,b} = (1 + bx)(1 - bx) + b^2x^2 = 1$ . The identities follow from  $\mathbf{N}_x^2 = 0$ . The hypotheses also hold for  $b + c$  and  $-b$  because  $\Pi$  is an additive group, which justifies the composition and inverse inside  $\mathbf{K}$ . The action on a column  $(z, 1)$  has denominator  $1 + b(z - x)$ ; subtracting  $x$  gives (15.7). The image of  $(1, 0)$  is  $(1 + bx, b)$ , giving the stated infinity formula when  $b \neq 0$  and the fixed point when  $b = 0$ . At  $z = x - 1/b$  the numerator is  $-1/b \neq 0$ , so the projective image is indeed  $\infty$ .  $\square$

Source 09 calls  $\mathbf{F}_{x,b}$  an omnific transvection. It is unrelated to the quadratic isometry of [lemma 8.2](#).

**Theorem 15.35** (Set-wise focusing [09]). *Let  $S \subseteq \mathbb{P}^1(\mathbf{No})$  be a set,  $x \in \mathbf{No}$  and  $\varepsilon > 0$ . There is a monomial  $b = \omega^\beta$  with  $\beta > 0$  such that  $\mathbf{F}_{x,b} \in \mathbf{K}$  and*

$$\mathbf{F}_{x,b}(S) \subseteq (x - \varepsilon, x + \varepsilon).$$

*In particular no point of  $S$  is sent to  $\infty$ .*

*Proof.* The set  $\{1, x, x^2\}$  can be cleared by one monomial ([theorem 4.6](#)), and any larger exponent also clears it. The positive surreals  $2/\varepsilon$  and  $2/|z - x|$ , for affine  $z \in S \setminus \{x\}$ , form a set, so by [lemma 4.1](#) we may also take  $\beta$  larger than all their leading exponents, as well as larger than the clearing exponent and zero. A positive monomial with strictly greater leading exponent dominates each specified positive bound regardless of its real leading coefficient. Then  $b, bx, bx^2 \in \Pi$ ,  $b > 2/\varepsilon$ , and  $b|z - x| > 2$  for those  $z$ . For such  $z$  put  $w = z - x$ ; then  $|1 + bw| \geq |bw| - 1 > \frac{1}{2}|bw| > 0$  and

$$|F_{x,b}(z) - x| = \frac{|w|}{|1 + bw|} < \frac{2}{b} < \varepsilon.$$

The point  $x$  is fixed, and  $\infty$  goes to  $x + 1/b$ , in the same interval.  $\square$

The conclusion holds for every set-sized family, not only for finite ones. It fails for the proper class  $\mathbb{P}^1(\mathbf{No})$ , since an invertible matrix acts bijectively; the restriction to sets is essential [09]. In this explicit family its pole  $x - 1/b$  always maps to infinity. The bound  $b|z - x| > 2$  ensures that this pole is outside the chosen set  $S$ .

**Corollary 15.36** (Dense fibers [09]). *Every  $K$ -orbit in  $\mathbb{P}^1(\mathbf{No})$  is dense in the topology of its ordered-field charts. In particular every fiber  $\mathrm{sp}^{-1}(r)$ ,  $r \in \mathbb{P}^1(\mathbb{R})$ , is dense, and  $\mathcal{V}$  contains continuum many pairwise disjoint dense fibers.*

*Proof.* Apply [theorem 15.35](#) to a singleton and a nonempty interval; every nonempty open set of  $\mathbb{P}^1(\mathbf{No})$  contains an interval. Then use [theorem 15.32](#).  $\square$

**Corollary 15.37** (Both denominator types in every interval [09]). *Every nonempty open interval of  $\mathbf{No}$  contains a fraction with a least positive omnific denominator and a unimodular presentation, and a fraction whose denominator ideal is not generated by any set and none of whose presentations has a gcd. The first can be taken in any prescribed rational fiber, the second in any prescribed irrational fiber.*

*Proof.* Combine [corollary 15.36](#) and [theorem 15.15](#).  $\square$

Explicitly [09], fix  $x \in \mathbf{No}$ ,  $\varepsilon > 0$  and an irrational real  $r$ . Apply [theorem 15.35](#) with  $S = \{\infty, r\}$ , choosing  $b$  with the bounds in its proof. The point with specialization  $\infty$  is

$$y = x + \frac{1}{b} = \frac{1 + bx}{b}, \quad (1 - bx)(1 + bx) + (bx^2)b = 1,$$

so  $b$  is the least positive denominator of  $y$ , and  $|y - x| < \varepsilon$ . For the already chosen irrational real  $r$ , the point  $y = F_{x,b}(r) = x + (r - x)/(1 + b(r - x))$  also satisfies  $|y - x| < \varepsilon$  and has  $\mathfrak{D}(y) = (1 + b(r - x))\Pi$ , since the transformed  $\mathcal{B}_{\mathbb{R}}$ -unimodular column has constant vector  $(r, 1)$ . Its second entry is  $1 + b(r - x)$ , and its first is  $r + bx(r - x)$ ; applying a determinant-one matrix preserves  $\mathcal{B}_{\mathbb{R}}$ -unimodularity. The bound in the focusing proof excludes a zero denominator. For  $r = x$  this is  $y = r$  and the denominator factor is 1.

**Corollary 15.38** (Nowhere continuity [09]). *The map  $\mathrm{sp} : \mathcal{V} \rightarrow \mathbb{P}^1(\mathbb{R})$  is continuous at no point, for the relative topology from the ordered-field charts on  $\mathbb{P}^1(\mathbf{No})$  and the usual topology on  $\mathbb{P}^1(\mathbb{R})$ .*

*Proof.* At  $y \in \mathcal{V}$  choose  $r \neq \mathrm{sp}(y)$  and a neighbourhood of  $\mathrm{sp}(y)$  omitting  $r$ . Every relative neighborhood of  $y$  is the intersection with  $\mathcal{V}$  of a projective open neighborhood, and the latter meets the dense fiber  $\mathrm{sp}^{-1}(r) \subseteq \mathcal{V}$ . Thus no such neighborhood maps into the chosen target neighborhood. The same argument uses the reciprocal chart when  $y = \infty$ .  $\square$

Denominator arithmetic is thus invisible to order approximation of any accuracy. This does not conflict with the continuity of the field operations. The dense fibers are invariant arithmetic classes; they are not open and not separated components [09].

*Remark 15.39* (Elementary matrices do not suffice [merge]). The matrices  $\mathbf{N}_x$  are the transposes of the matrices  $Q_\alpha$  of the non-elementary unipotents in `docs/surreal/omnific-groups-and-lattices` (`ogl:el:thm:none1`), at  $\alpha = x$ . That report proves that  $h \cdot \infty$  has rational standard part whenever  $h \in E_2(\mathbf{Oz})$  and  $h \cdot \infty$  is finite (`ogl:el:prop:cuspresidue`), where  $E_2$  is the group generated by elementary matrices. Two consequences follow.

- (i) If  $x$  is finite with  $\text{st}(x) \notin \mathbb{Q}$ , for example an irrational real, then  $F_{x,b} \notin E_2(\mathbf{Oz})$  for every  $b$  as in [lemma 15.34](#) with  $b \neq 0$ . Indeed  $F_{x,b}(\infty) = x + 1/b$  is finite with standard part  $\text{st}(x)$ .
- (ii) The orbit  $E_2(\mathbf{Oz}) \cdot \infty$  is not dense. It misses the interval  $(\sqrt{2} - \omega^{-1}, \sqrt{2} + \omega^{-1})$ , whose points are finite with standard part  $\sqrt{2}$ . But  $\mathbf{K} \cdot \infty = \text{sp}^{-1}(\infty)$  is dense ([corollary 15.36](#)), and  $\mathcal{U} = \text{SL}_2(\mathbf{Oz}) \cdot \infty$  strictly contains  $E_2(\mathbf{Oz}) \cdot \infty$ . For example  $\sqrt{2} + \omega^{-2}$  lies in that interval and in  $\mathcal{U}$ , with the unimodular pair  $(1 + \sqrt{2}\omega^2, \omega^2)$  and the certificate  $(1 - \sqrt{2}\omega^2)(1 + \sqrt{2}\omega^2) + 2\omega^2 \cdot \omega^2 = 1$ .

So the density of the orbit of  $\infty$  needs matrices outside  $E_2(\mathbf{Oz})$ , and source 09 rightly works with  $\mathbf{K}$  rather than with the elementary kernel of that report. Source 09 never claims  $\mathbf{K} \subseteq E_2(\mathbf{Oz})$ ; the remark adds the comparison.

## 15.8 The localization at the purely infinite ideal

The ideal  $\Pi$  is prime, since  $\mathbf{Oz}/\Pi \cong \mathbb{Z}$  is a domain. Let

$$\mathbf{Oz}_\Pi = (\mathbf{Oz} \setminus \Pi)^{-1} \mathbf{Oz} = \left\{ \frac{a}{b} : a, b \in \mathbf{Oz}, \text{ct}(b) \neq 0 \right\} \subseteq \mathbf{No},$$

the localization at  $\Pi$ , not the quotient  $\mathbf{Oz}/\Pi$ . Its denominators are closed under multiplication because their nonzero integer constant coefficients multiply. Fractions are realized inside  $\mathbf{No}$ , so this notation does not require a set containing all pairs of omnific integers. Source 08 writes  $\mathcal{A}$  and  $\rho$  for  $\mathbf{Oz}_\Pi$  and  $\text{res}$ .

**Proposition 15.40** (A local ring with residue field  $\mathbb{Q}$  [08]). *The map  $\text{res} : \mathbf{Oz}_\Pi \rightarrow \mathbb{Q}$ ,  $\text{res}(a/b) = \text{ct}(a)/\text{ct}(b)$ , is a well-defined surjective ring homomorphism. The ring  $\mathbf{Oz}_\Pi$  is local with maximal ideal  $\ker \text{res} = (\mathbf{Oz} \setminus \Pi)^{-1} \Pi$ , its real elements are exactly  $\mathbb{Q}$ , and it is neither a field nor a valuation ring of  $\mathbf{No}$ .*

*Proof.* If  $a/b = a'/b'$  with  $\text{ct}(b), \text{ct}(b') \neq 0$ , then  $ab' = a'b$ ; taking constant coefficients proves well-definedness, and the fraction rules give a surjective homomorphism: ordinary fractions already realize every rational value. Its kernel consists exactly of fractions with numerator in  $\Pi$ . If  $\text{res}(a/b) \neq 0$ , then  $\text{ct}(a) \neq 0$  and  $b/a \in \mathbf{Oz}_\Pi$ , so  $a/b$  is a unit; an element of  $\ker \text{res}$  is not a unit, because its image in the nonzero field  $\mathbb{Q}$  is zero. Thus all nonunits form this proper ideal. Every proper ideal contains no unit and hence lies in it, proving that it is the unique maximal ideal. For  $r \in \mathbb{R}$ , a representation with denominator outside  $\Pi$  exists exactly when  $\mathfrak{D}(r)$  has an element with nonzero constant, which by [theorem 15.7](#) happens exactly for  $r \in \mathbb{Q}$ . So neither  $\sqrt{2}$  nor  $1/\sqrt{2}$  lies in  $\mathbf{Oz}_\Pi$ . Its fraction field is  $\mathbf{No}$ , since  $\mathbf{Oz} \subseteq \mathbf{Oz}_\Pi \subseteq \mathbf{No}$  and  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ . A valuation ring of that field would contain  $\sqrt{2}$  or its inverse, so this ring is not one. Finally  $t \in \ker \text{res}$  is a nonzero nonunit.  $\square$

This is consistent with the sibling report's list of the prime ideals with set-sized quotient:  $\Pi$  is prime with  $\mathbf{Oz}/\Pi = \mathbb{Z}$  and fraction field  $\mathbb{Q}$  (`osq:cor:smallprimes`), and `res` restricts to `ct` on  $\mathbf{Oz}$ . Status (batch 34): that report also builds set-sized analogues of  $\mathbf{Oz}_\Pi$ . For every infinite cardinal  $\kappa$  there is a real closed field  $\mathbb{E} \subseteq \mathbf{No}$  of cardinality  $\kappa$  with integer part  $B_\mathbb{E} = \mathbb{E} \cap \mathbf{Oz}$  and  $\text{Frac } B_\mathbb{E} = \mathbb{E}$ , whose localization  $\mathcal{L}$  at  $B_\mathbb{E} \cap \Pi$  is local with residue field  $\mathbb{Q}$ , of cardinality  $\kappa$ , with maximal ideal  $\mathfrak{n}$  satisfying  $\mathfrak{n}^2 = \mathfrak{n} \neq 0$  (`osq:hb:cor:local`); the maximal ideal of  $\mathbf{Oz}_\Pi$  is idempotent as well, since  $\Pi^2 = \Pi$  ([example 19.5](#)). The completion of  $\mathcal{L}$  is  $\mathbb{Q}$ , while its henselization has exactly  $2^\kappa$  minimal primes (`osq:hb:thm:main`).

**Proposition 15.41** (Rational functions in the localization [08]). *Under  $\mathbb{R}(T) \cong \mathbb{R}(t)$ ,*

$$\mathbf{Oz}_\Pi \cap \mathbb{R}(t) = \{f(t) : f \in \mathbb{R}(T) \text{ regular at } 0, f(0) \in \mathbb{Q}\},$$

*and  $\text{res}(f(t)) = f(0)$  on this ring.*

*Proof.* Write  $f = p/q$  in reduced form. If  $f(0)$  is irrational, every denominator lies in  $q(t)\Pi \subseteq \Pi$  ([theorem 15.18](#)). If  $f$  has a pole at 0, then  $\mathfrak{D}(f(t)) = \lambda q(t)\mathbf{Oz}$  with  $q(0) = 0$ , and again all denominators lie in  $\Pi$ . In both cases  $f(t) \notin \mathbf{Oz}_\Pi$ . If  $f$  is regular at 0 with  $f(0) \in \mathbb{Q}$ , normalize  $(p(0), q(0))$  to a primitive integer pair; its second entry is nonzero, so the normalized denominator lies outside  $\Pi$ , and  $\text{res}(f(t)) = f(0)$ .  $\square$

**Theorem 15.42** (Independent real and rational residues [08]). *Let  $\mathbf{Oz}_\Pi^{\text{fin}} = \mathbf{Oz}_\Pi \cap \mathcal{O}_{\text{fin}}$ . The map  $(\text{st}, \text{res}) : \mathbf{Oz}_\Pi^{\text{fin}} \rightarrow \mathbb{R} \times \mathbb{Q}$  is surjective, its two kernels are comaximal, and  $\mathbf{Oz}_\Pi^{\text{fin}}/(\ker \text{st} \cap \ker \text{res}) \cong \mathbb{R} \times \mathbb{Q}$ . For  $r \in \mathbb{R}$ ,  $q \in \mathbb{Q}$  and  $t > 0$  in  $\Pi$ , the element*

$$x_{r,q} = \frac{rt + q}{t + 1} = r + \frac{q - r}{t + 1}$$

*lies in  $\mathbf{Oz}_\Pi^{\text{fin}}$  with  $\text{st}(x_{r,q}) = r$  and  $\text{res}(x_{r,q}) = q$ .*

*Proof.* With  $q = m/n$ ,  $x_{r,q} = (nrt + m)/(nt + n)$  has numerator and denominator in  $\mathbf{Oz}$  and denominator constant  $n \neq 0$ , so  $\text{res}(x_{r,q}) = m/n$ ; the second expression differs from  $r$  by an infinitesimal. The elements  $u = t/(t + 1)$  and  $v = 1/(t + 1)$  lie in  $\mathbf{Oz}_\Pi^{\text{fin}}$ , satisfy  $u + v = 1$ , and have  $(\text{st}, \text{res})(u) = (1, 0)$  and  $(\text{st}, \text{res})(v) = (0, 1)$ . So the kernels are comaximal, and the map to the product has kernel  $\ker \text{st} \cap \ker \text{res}$ . Thus two elements have the same pair of residues exactly when their difference lies in that intersection. This gives the asserted quotient isomorphism through ordinary pairs in  $\mathbb{R} \times \mathbb{Q}$  as representatives, without treating proper-class cosets as set elements.  $\square$

The standard part sees the behaviour at an infinite parameter and `res` the constant-coefficient specialization. Both fix  $\mathbb{Q}$  [08], but they can also agree on nonrational elements. In fact [`merge`], on their common domain,

$$\{x \in \mathbf{Oz}_\Pi^{\text{fin}} : \text{st}(x) = \text{res}(x)\} = \mathbb{Q} \oplus (\ker \text{st} \cap \ker \text{res})$$

as an additive decomposition. If the common value is  $q$ , it lies in  $\mathbb{Q}$  and  $x - q$  lies in both kernels; conversely every such sum has both residues  $q$ . The intersection meets  $\mathbb{Q}$  only in zero. It is nonzero: for  $t > 0$  in  $\Pi$ , the element  $t/(t + 1)^2$  has both residues zero and is a positive infinitesimal, hence is not rational. The earlier phrase “agree only on  $\mathbb{Q}$ ” was therefore too strong. Neither map is a retraction of  $\mathbf{No}$ , and their domains differ. The positive element  $u$  above has rational residue zero. To see that `res` even fails to preserve weak order, take  $r > 0 > q$ : then  $x_{r,q} > 0$  has rational residue  $q < 0$ . For nonzero reduced  $f = p/q$ , [lemma 15.17](#) makes  $f(t)$  finite exactly when  $\deg p \leq \deg q$ . With strict inequality it is infinitesimal. With

equal degrees, subtracting the ratio  $\text{lc}(p)/\text{lc}(q)$  lowers the numerator degree, so that ratio is its standard part. The zero function is finite with both residues zero. So  $\mathbf{Oz}_{\Pi}^{\text{fin}} \cap \mathbb{R}(t)$  consists of the rational functions regular at 0 and at  $\infty$  with a rational value at 0 and any real value at  $\infty$ . The surjection onto  $\mathbb{R} \times \mathbb{Q}$  is a set-sized quotient of the class ring  $\mathbf{Oz}_{\Pi}^{\text{fin}}$ . That ring meets  $\mathbf{Oz}$  in  $\mathbb{Z}$  and contains no nonzero element of  $\Pi$ , so the sibling report's `osq:prop:fractions(iv)`, about localizations in which an element of  $\Pi$  becomes a unit, does not apply. There is no contradiction.

## 15.9 Polynomial models and scale extensions

For an indeterminate  $X$  write  $\mathcal{P}[X] = \mathbb{Z} + X\mathbb{R}[X]$ ; source 08 writes  $A_{\tau}$ ,  $A_0$ ,  $A_m$  and source 09 writes  $O_{\mathbb{Z}}$ . These are set-sized rings.

**Proposition 15.43** (The omnific part of a one-parameter field [08]).  $\mathcal{B}_{\mathbb{R}} \cap \mathbb{R}(t) = \mathbb{R}[t]$  and  $\mathbf{Oz} \cap \mathbb{R}(t) = \mathbb{Z} + t\mathbb{R}[t]$ , the image  $\mathcal{P}_t$  of  $\mathcal{P}[T]$  under  $T \mapsto t$ . Moreover  $\text{Frac}(\mathcal{P}_t) = \mathbb{R}(t)$ .

*Proof.* Polynomial division writes  $f = p/q$  as  $a + b/q$ , with  $a, b, q \in \mathbb{R}[T]$  and either  $b = 0$  or  $\deg b < \deg q$ . By lemma 15.17, a nonzero  $b(t)/q(t)$  is infinitesimal. If  $f(t) \in \mathcal{B}_{\mathbb{R}}$ , then  $b(t)/q(t) = f(t) - a(t) \in \mathcal{B}_{\mathbb{R}}$  as well, since  $a(t) \in \mathcal{B}_{\mathbb{R}}$ . That ring has no nonzero infinitesimal, so  $b = 0$ . This proves the first intersection. For a polynomial  $a$ , the identity  $\text{ct}(a(t)) = a(0)$  gives the second. The fraction field contains  $t$  and every real  $r = (rt)/t$ , since both  $rt$  and  $t$  lie in  $\mathcal{P}_t$  and  $t \neq 0$ . Conversely it lies in  $\mathbb{R}(t)$ , proving equality.  $\square$

Real constants are absent from  $\mathcal{P}_t$  except for  $\mathbb{Z}$  but reappear in its fraction field. Transport evaluation at  $T = 0$  from  $\mathbb{R}[T]$  to  $\mathbb{R}[t]$  through the injective evaluation map  $T \mapsto t$ . With this retraction, theorem 15.11 gives, in the notation of theorem 15.18,

$$\mathfrak{D}_{\mathcal{P}_t}(\mathbf{x}) = q(t) \mathcal{P}_t \quad \text{or} \quad q(t) \cdot t\mathbb{R}[t], \quad (15.8)$$

according as  $\text{sp}(\mathbf{x})$  is rational or not, with the same normalization. The primitive or nonprimitive status is the same in the model and in  $\mathbf{Oz}$ , although the ideal of all denominators becomes much larger.

**Proposition 15.44** (Generator counts [08]). *The ideal  $t\mathbb{R}[t]$  of  $\mathcal{P}_t$  needs exactly  $\mathfrak{c} = |\mathbb{R}|$  generators, and so does every  $q(t)t\mathbb{R}[t]$  with  $q \neq 0$ . The corresponding ideal  $q(t)\Pi$  of  $\mathbf{Oz}$  has no set of generators.*

*Proof.* With  $I = t\mathbb{R}[t]$ , every product of two members has degree at least two, and  $t^2a(t) = t \cdot ta(t)$  for  $a \in \mathbb{R}[T]$ , so  $I^2 = t^2\mathbb{R}[t]$ . The coefficient of  $t$  gives  $I/I^2 \cong (\mathbb{R}, +)$ . The action of  $\mathcal{P}_t$  on this quotient factors through  $\mathcal{P}_t/I = \mathbb{Z}$ . If a family of cardinality  $\kappa < \mathfrak{c}$  generated  $I$ , its images would generate  $(\mathbb{R}, +)$  as an abelian group. But finite integer linear combinations of such a family number at most  $\max(\kappa, \aleph_0) < \mathfrak{c}$ , a contradiction; this uses no continuum hypothesis. Conversely  $\{rt : r \in \mathbb{R}\}$  generates  $I$ , because every monomial  $rt^j$ ,  $j \geq 1$ , is  $(rt)t^{j-1}$  with  $t^{j-1} \in \mathcal{P}_t$ , and polynomials have finite support. Multiplication by  $q(t) \neq 0$  is a module isomorphism onto the scaled ideal. Multiplication by the same scalar identifies  $\Pi$  with  $q(t)\Pi$ , so corollary 4.3 gives the last assertion even without a specified rational-function tuple.  $\square$

For an ideal  $J$  of  $\mathcal{P}_t$  let  $J\mathbf{Oz}$  be its extension to  $\mathbf{Oz}$ : finite sums of products  $ja$  with  $j \in J$  and  $a \in \mathbf{Oz}$ .

**Theorem 15.45** (Enlargement defect [08]). *In the notation of theorem 15.18,  $\mathfrak{D}_{\mathcal{P}_t}(\mathbf{x}) \mathbf{Oz} = q(t) \mathbf{Oz} = \mathfrak{D}(\mathbf{x})$  in the rational case, and  $\mathfrak{D}_{\mathcal{P}_t}(\mathbf{x}) \mathbf{Oz} = q(t)t\mathcal{B}_{\mathbb{R}} \subsetneq q(t)\Pi = \mathfrak{D}(\mathbf{x})$  otherwise.*

*Proof.* Only the second case needs an argument. The extension  $(t\mathbb{R}[t])\mathbf{Oz}$  contains  $rt$  for real  $r$  and  $tu$  for  $u \in \Pi$ , hence  $t\mathcal{B}_{\mathbb{R}}$ ; conversely a product of an element of  $t\mathbb{R}[t]$  and an omnific integer lies in  $t\mathcal{B}_{\mathbb{R}}$ . For strictness choose  $\varepsilon = \frac{1}{2} \deg_{\omega}(t) > 0$ : then  $\omega^{\varepsilon} \in \Pi$ , but  $\omega^{\varepsilon}/t$  has negative leading exponent, so it is a nonzero infinitesimal outside  $\mathcal{B}_{\mathbb{R}}$ . Multiplication by  $q(t)$  preserves the strict inclusion.  $\square$

Enlarging the available scales and recomputing a denominator ideal can thus reveal denominators not generated by those of the smaller model. The primitive or nonprimitive answer is stable, but the module of all denominators is not obtained by extending scalars [08].

In the rest of this subsection  $m \geq 2$ , the rings are  $\mathcal{P}[T] \subseteq \mathcal{P}[U]$  with  $T = U^m$ , embedded in  $\mathbf{Oz}$  by  $T = \omega$  and  $U = \omega^{1/m}$ , and  $I_T = T\mathbb{R}[T]$ ,  $I_U = U\mathbb{R}[U]$ .

**Theorem 15.46** (Scale-extension defect [08]). *Let  $p_1, \dots, p_n, q \in \mathbb{R}[T]$  have gcd one,  $q \neq 0$ ,  $\mathbf{f} = (p_j/q)_j$  and  $\mathbf{s} = [p_1(0) : \dots : p_n(0) : q(0)]$ . If  $\mathbf{s} \in \mathbb{P}^n(\mathbb{Q})$ , then  $\mathfrak{D}_{\mathcal{P}[U]}(\mathbf{f}(U^m)) = \mathfrak{D}_{\mathcal{P}[T]}(\mathbf{f}(T))\mathcal{P}[U]$ . If  $\mathbf{s} \notin \mathbb{P}^n(\mathbb{Q})$ , then*

$$\mathfrak{D}_{\mathcal{P}[U]}(\mathbf{f}(U^m)) = q(U^m)U\mathbb{R}[U], \quad \mathfrak{D}_{\mathcal{P}[T]}(\mathbf{f}(T))\mathcal{P}[U] = q(U^m)U^m\mathbb{R}[U],$$

*and division by  $q(U^m)$  induces an isomorphism of the quotient of the first by the second with  $U\mathbb{R}[U]/U^m\mathbb{R}[U]$ , which has the real basis  $U, U^2, \dots, U^{m-1}$  and real dimension  $m-1$ .*

*Proof.* A polynomial Bézout identity persists under  $T = U^m$ , and the constant vector is unchanged; apply [theorem 15.11](#) in both polynomial rings. In the rational case both ideals are the same principal ideal. Otherwise they are  $q(T)I_T$  and  $q(U^m)I_U$ , and  $I_T\mathcal{P}[U] = U^m\mathbb{R}[U]$ : the inclusion  $\subseteq$  is by degrees, and every  $rU^m$  comes from  $rT$ , while multiplication by  $U^k$  gives the higher monomials. Multiplication by the nonzero polynomial  $q(U^m)$  identifies  $U\mathbb{R}[U]$  with the new denominator ideal and  $U^m\mathbb{R}[U]$  with the extended old one. Its inverse therefore induces the stated quotient isomorphism, which is  $\mathcal{P}[U]$ -linear. Every class has a unique representative  $\sum_{j=1}^{m-1} r_j U^j$ , giving its real basis.  $\square$

The dimension counts real coefficient layers, not module length [08]. The  $\mathcal{P}[U]$ -action on the displayed quotient is multiplication followed by truncation modulo  $U^m$ . In fact this module has infinite length [merge]: on its top layer  $\mathbb{R}U^{m-1}$ , positive-degree terms of  $\mathcal{P}[U]$  act as zero and constants act as integers. Thus

$$\mathbb{Z}U^{m-1} \subsetneq \frac{1}{2}\mathbb{Z}U^{m-1} \subsetneq \frac{1}{4}\mathbb{Z}U^{m-1} \subsetneq \dots$$

is a strictly increasing chain of  $\mathcal{P}[U]$ -submodules. Finite real dimension here does not imply finite module length.

**Theorem 15.47** (An integral extension that is neither finite nor flat [08]). *For every  $m \geq 2$  the inclusion  $\mathcal{P}[T] \hookrightarrow \mathcal{P}[U]$  is integral, but  $\mathcal{P}[U]$  is neither finitely generated nor flat as a  $\mathcal{P}[T]$ -module.*

*Proof.* Each  $rU^k$  with  $k \geq 1$  is a root of the monic polynomial  $X^m - r^m T^k$  over  $\mathcal{P}[T]$ : its constant term belongs to  $I_T$  even when  $r$  is not an integer. An element of  $\mathcal{P}[U]$  is an integer plus a finite sum of such monomials. Adjoining these finitely many integral elements gives a finite module, spanned by products of bounded powers; the determinant argument then gives a monic equation for their sum. This proves integrality. If  $\mathcal{P}[U]$  were a finite  $\mathcal{P}[T]$ -module, then  $\mathcal{P}[U]/I_T\mathcal{P}[U] \cong \mathbb{Z} \oplus \bigoplus_{j=1}^{m-1} \mathbb{R}U^j$  would be finitely generated over  $\mathcal{P}[T]/I_T = \mathbb{Z}$ , which it is not: its projection onto the coefficient of  $U$  is the uncountable group  $(\mathbb{R}, +)$ .

For flatness put  $r = \sqrt{2}$  and consider the row  $(rT, T) : \mathcal{P}[T]^2 \rightarrow \mathcal{P}[T]$ . The equation  $rTa + Tb = 0$  implies  $b = -ra$  by cancellation of  $T \neq 0$ . For  $a \in \mathcal{P}[T]$ , both  $a$  and  $ra$  belong

to that ring exactly when its integer constant  $\text{ct}(a)$  is zero, since  $r$  is irrational. Thus the old kernel is  $L = \{(h, -rh) : h \in I_T\}$ . After substitution  $T = U^m$ , the identical argument gives the new kernel  $\{(h, -rh) : h \in I_U\}$ . The image of  $L \otimes_{\mathcal{P}[T]} \mathcal{P}[U]$  in  $\mathcal{P}[U]^2$  consists of finite sums of old relations multiplied by new scalars, hence is exactly  $\{(h, -rh) : h \in I_T \mathcal{P}[U] = U^m \mathbb{R}[U]\}$ . The relation  $(U, -rU)$  is in the new kernel but not in this image because  $m \geq 2$ . Flatness would make these two modules equal by preservation of exactness at the middle of the row sequence [34, Tag 00HK]. Therefore  $\mathcal{P}[U]$  is not flat.  $\square$

**Proposition 15.48** (The defect as a Tor group [08]). *Let  $J = (\sqrt{2}T, T) \subseteq \mathcal{P}[T]$  and  $C = \mathcal{P}[T]/J$ . Then  $\text{Tor}_1^{\mathcal{P}[T]}(C, \mathcal{P}[U]) \cong U\mathbb{R}[U]/U^m\mathbb{R}[U]$  as  $\mathcal{P}[U]$ -modules.*

*Proof.* Put  $A_0 = \mathcal{P}[T]$  and  $A_m = \mathcal{P}[U]$  for this proof, with the specified embedding  $T \mapsto U^m$ . From  $0 \rightarrow J \rightarrow A_0 \rightarrow C \rightarrow 0$ , the long exact sequence of Tor and  $\text{Tor}_1^{A_0}(A_0, A_m) = 0$  identify

$$\text{Tor}_1^{A_0}(C, A_m) \cong \ker(J \otimes_{A_0} A_m \rightarrow A_m)$$

(see [34, Section 10.75]). Let  $L = \ker((\sqrt{2}T, T) : A_0^2 \rightarrow J)$ . Right exactness of tensoring gives

$$J \otimes_{A_0} A_m \cong A_m^2 / \text{im}(L \otimes_{A_0} A_m \rightarrow A_m^2).$$

Under this presentation the map to  $A_m$  is induced by the row  $(\sqrt{2}U^m, U^m)$ . Its kernel is therefore the new relation module modulo the image of the old one, both computed in theorem 15.47. Taking the first coordinate identifies this quotient with  $I_U/(I_T A_m) = U\mathbb{R}[U]/U^m\mathbb{R}[U]$ ,  $A_m$ -linearly. In particular  $(U, -\sqrt{2}U)$  represents a nonzero class. This argument uses right exactness, which holds without flatness, and does not assume that  $L \otimes_{A_0} A_m \rightarrow A_m^2$  is injective.  $\square$

*Remark 15.49* (Scope of the scale computations [08]; [merge]). Integrality, nonflatness and non-finiteness of pullback extensions are familiar; the point is the exact denominator-sensitive defect realized by a finer surreal monomial, and the same  $m - 1$  layers witness the failure of flatness. All tensor products and Tor groups here are over set-sized rings [08]. The tower  $\mathcal{P}[T] \subseteq \mathcal{P}[T^{1/2}] \subseteq \mathcal{P}[T^{1/6}] \subseteq \dots$  (exponents  $1/m!$ ) has union the set-sized core  $A_1 \cong T_{\mathbb{Q}}$  of the sibling report (osq:hd:lem:cores), with  $(D, K) = (\mathbb{Z}, \mathbb{R})$ . The steps have exponent group  $\mathbb{Z}$  and are not among the cores  $T_{\Gamma}$  there, which need a divisible  $\Gamma$ ; their ideals  $I_T$  are not idempotent ( $I_T/I_T^2 \cong \mathbb{R}$ ), whereas  $\Pi_{\Gamma}^{\text{fin}}$  is. No statement of that report is contradicted.

## 15.10 Monomial denominators and fixed workspaces

For  $x \in \mathbf{No}$  let  $\text{Ex}(x) = \{\beta \in \mathbf{No} : \beta \geq 0, \omega^\beta x \in \mathbf{Oz}\}$ , the class of exponents of monic monomial denominators. The corresponding denominators  $\omega^\beta$  have coefficient 1; ordinary integers such as 2 are not of this form. Source 08 writes  $M(x)$  and source 09  $E(x)$ .

**Proposition 15.50** (Monomial denominators [08, 09]). *Let  $x = \sum_{\gamma \in S} r_\gamma \omega^\gamma \neq 0$  and  $r_\eta = 0$  for  $\eta \notin S$ . Then  $\beta \in \text{Ex}(x)$  if and only if  $\beta \geq 0$ ,  $\gamma + \beta \geq 0$  for all  $\gamma \in S$ , and  $r_{-\beta} \in \mathbb{Z}$ . The class  $\text{Ex}(x)$  is nonempty and upward closed, and  $\omega^{\beta'} x \in \Pi$  whenever  $\beta' > \beta \in \text{Ex}(x)$ . It has a least member exactly when  $x \in \mathbf{Oz}$  (least member 0) or  $S$  has a negative least element  $a$  with  $r_a \in \mathbb{Z}$  (least member  $-a$ ). Also  $\text{Ex}(0) = \mathbf{No}^{\geq 0}$ .*

*Proof.* Multiplication by  $\omega^\beta$  shifts supports by  $\beta$  and keeps coefficients, which gives the criterion; nonemptiness is theorem 4.6, and a strictly larger exponent makes every shifted exponent positive. Let  $\beta > 0$  be least. If the shifted support were entirely positive, a positive  $\varepsilon$  below  $\beta$  and below every shifted exponent, which exists by lemma 4.1, would make  $\beta - \varepsilon$

admissible. So 0 is in the shifted support and is its minimum; hence  $-\beta$  is a negative least element of  $S$  with integral coefficient. Conversely such an element makes  $-a$  admissible and forbids smaller exponents. The case  $\beta = 0$  is  $x \in \mathbf{Oz}$ .  $\square$

**Example 15.51** (Monomial and algebraic minimization differ [08, 09]). (i) The rational  $1/2$  has least omnific denominator 2, but  $\text{Ex}(1/2) = \{\beta : \beta > 0\}$  has no least member [08]. (ii) The number  $x = \frac{1}{2}\omega^{-1} = 1/(2\omega)$  has least positive denominator  $2\omega$ , by the unimodular pair  $(1, 2\omega)$ , but  $\text{Ex}(x) = \{\beta : \beta > 1\}$ , since at  $\beta = 1$  the numerator is  $1/2$  [09]. (iii) For irrational  $r$ ,  $\text{Ex}(r) = \{\beta : \beta > 0\}$ , and here even unrestricted denominators have no least member (theorem 15.7) [09]. (iv) The geometric series  $x = 1/(\omega - 1) = \sum_{n \geq 1} \omega^{-n}$  has least positive denominator  $\omega - 1$ , but  $\text{Ex}(x)$  is the class of  $\beta$  exceeding every ordinary integer, with no least member: if  $\beta$  exceeds every ordinary integer, so does  $\beta - 1$  [08]. The unrestricted least denominator follows from the unimodular pair  $(1, \omega - 1)$ . Straightforward monomial clearing therefore needs an exponent above every integer, where a two-term denominator of leading exponent 1 suffices; support bounds cannot replace denominator-ideal arithmetic, and a preferred monomial denominator need not be the arithmetically preferred one.

**Example 15.52** (One fixed workspace [08, 09]). Let  $K = \mathbb{R}((\omega^{-1}))$ , the Laurent series in  $\omega^{-1}$ . A reverse-well-ordered set of nonnegative integers is finite, so  $K \cap \mathbf{Oz} = \mathcal{P}_\omega = \mathbb{Z} + \omega\mathbb{R}[\omega]$ , and

$$\{\omega^n : n \in \mathbb{N}\}^{-1} \mathcal{P}_\omega = \mathbb{R}[\omega, \omega^{-1}] \subsetneq \text{Frac}(\mathcal{P}_\omega) = \mathbb{R}(\omega) \subsetneq K.$$

For the localization equality, multiplying any Laurent polynomial by a sufficiently high power of  $\omega$  makes every exponent positive, putting it in  $\omega\mathbb{R}[\omega]$  even if its coefficients are not integral. Conversely dividing an element of  $\mathcal{P}_\omega$  by a power of  $\omega$  gives a Laurent polynomial. The fraction-field equality follows from  $r = (r\omega)/\omega$  for every  $r \in \mathbb{R}$  and the reverse containment  $\mathcal{P}_\omega \subseteq \mathbb{R}(\omega)$ . The first inclusion is strict because  $1/(\omega - 1) = \sum_{n \geq 1} \omega^{-n}$  is not a Laurent polynomial [09]. The second is strict by either of two elements.

- (i)  $E = \sum_{n \geq 0} \omega^{-n}/n!$  [08]. If  $\sum_n z^n/n!$  were rational, a nonzero  $\sum_{j=0}^d q_j z^j$  would give  $\sum_j q_j/(n-j)! = 0$  for all large  $n$ ; multiplying by  $n!$  for  $n \geq d$  gives  $\sum_j q_j n(n-1) \cdots (n-j+1) = 0$  for infinitely many  $n$ . This polynomial is zero, and the falling factorials have distinct degrees and leading coefficient one, forcing every  $q_j = 0$ , a contradiction.
- (ii)  $F = \sum_{n \geq 1} \omega^{-n!}$  [09]. If  $F = A(u)/B(u)$  in  $u = \omega^{-1}$  with  $B(0) \neq 0$  after cancellation (the series is regular at zero), write  $B = \sum_{j=0}^d b_j u^j$  and  $F = \sum c_n u^n$ . Comparing coefficients in  $BF = A$  gives  $b_0 c_n = -\sum_{j=1}^d b_j c_{n-j}$  for  $n > \deg A$ , with  $n \geq d$ . Since  $b_0 \neq 0$ , a block of  $d$  consecutive zero coefficients beyond this bound forces all later ones to vanish by induction. The factorial gaps give such blocks followed by a coefficient 1, a contradiction. A constant  $B$  would make  $F$  a polynomial, also impossible.

In **No** the monomial  $\omega^\omega$  clears  $F$ , with exponents  $\omega - n! > 0$ , but  $\omega^\omega \notin K$ , since  $K$  has only integer exponents [09].

The example is consistent with source 06's non-claim that the fraction field of a fixed Hahn ring need not be the whole Hahn field ( $\Gamma = \mathbb{Z}$  gives  $\mathbb{k}(\omega)$ ), and with the cofinality criterion of L'Innocente–Mantova [3, Proposition 2.4.5]. The small common divisor of theorem 4.2 may likewise need an exponent outside a workspace. Even where the needed shifts exist, the dichotomy in a workspace requires a unimodular pair over its nonnegative-support ring; the polynomial Bézout certificate of theorem 15.18 supplies it, and a formal cancellation in an ambient field does not [09].

### 15.11 Gaussian omnific fractions

Here  $\mathcal{B}_{\mathbb{C}} = \mathbb{C} \oplus \Pi_{\mathbb{C}}$  and  $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$  (definition 2.1), with  $\text{ct} : \mathcal{B}_{\mathbb{C}} \rightarrow \mathbb{C}$  a retraction and  $\mathbf{Oz}[i] = \text{ct}^{-1}(\mathbb{Z}[i])$ .

**Proposition 15.53** (Gaussian fractions [08]; [09]).  $\mathbf{Oz}[i][(\mathbb{Z} \setminus \{0\})^{-1}] = \mathbb{Q}(i) \oplus \Pi_{\mathbb{C}}$ . The units of  $\mathbf{Oz}[i]$  are  $\pm 1, \pm i$ , and  $\Pi_{\mathbb{C}}$  is divisible by every nonzero ordinary integer and not generated by any set as an ideal of  $\mathbf{Oz}[i]$ .

*Proof.* Clearing the two rational constants gives the first identity. By lemma 2.4 a unit of  $\mathcal{B}_{\mathbb{C}}$  is a complex constant, and the units of  $\mathbb{Z}[i]$  are the four displayed. Division by a nonzero integer preserves positive support. For a set of proposed generators, the union of their real and imaginary supports is a set of positive exponents. Choose  $\varepsilon > 0$  strictly below every exponent in this union, using lemma 4.1. Multiplication by an element of  $\mathbf{Oz}[i]$  cannot introduce exponents below  $\varepsilon$ , since its exponents are nonnegative; neither can a finite sum. Thus  $\omega^{\varepsilon/2}$  lies outside the generated ideal, proving the last assertion.  $\square$

The identity  $\text{Frac}(\mathbf{Oz}[i]) = \mathbf{No}[i]$ , proved by sources 08 and 09 through clearing the real and imaginary parts with one monomial, is already stated with source 06 in section 13, and the units of  $\mathbf{Oz}[i]$  are recorded in section 11.1. Source 09 gives no complex lowest-term classification; source 08 does.

**Theorem 15.54** (Gaussian representation dichotomy [08]). Let  $t \in \Pi$  be nonzero and let  $p_1, \dots, p_n, q \in \mathbb{C}[T]$ ,  $q \neq 0$ , have gcd one. Theorem 15.18 holds with  $(\mathbb{R}, \mathbb{Z}, \mathbb{Q}, \mathcal{B}_{\mathbb{R}}, \mathbf{Oz}, \Pi)$  replaced by  $(\mathbb{C}, \mathbb{Z}[i], \mathbb{Q}(i), \mathcal{B}_{\mathbb{C}}, \mathbf{Oz}[i], \Pi_{\mathbb{C}})$ . In particular primitive homogeneous coordinates exist exactly when  $[p_1(0) : \dots : p_n(0) : q(0)] \in \mathbb{P}^n(\mathbb{Q}(i))$ ; they are then unimodular and unique up to a Gaussian unit. Otherwise no representation has a gcd, and the denominator ideal is  $q(t)\Pi_{\mathbb{C}}$ , with no set of generators.

*Proof.* Evaluation at  $t$  is injective on  $\mathbb{C}[T]$ : a nonzero polynomial of degree  $d$  has leading exponent  $d \deg_{\omega}(t)$  and nonzero leading complex coefficient. In particular  $q(t) \neq 0$ . The evaluated vector is unimodular over  $\mathcal{B}_{\mathbb{C}}$  by a complex polynomial Bézout identity, and the identity forces every ambient scalar giving Gaussian omnific coordinates into  $\mathcal{B}_{\mathbb{C}}$ . For a nonzero complex residue vector  $\mathbf{v}$ , the complex  $s$  with  $s\mathbf{v} \in \mathbb{Z}[i]^{n+1}$  are 0 unless the line is defined over  $\mathbb{Q}(i)$ . In that case choose  $\lambda \in \mathbb{C}^{\times}$  so that  $\lambda\mathbf{v}$  is a primitive Gaussian integral vector. Replace all polynomials by their  $\lambda$  multiples and the Bézout row by its  $\lambda^{-1}$  multiple. Since  $\mathbb{Z}[i]$  is Euclidean, the normalized constant vector has a Gaussian Bézout identity; as in lemma 15.10, its admissible constant scalars are exactly  $\mathbb{Z}[i]$ . This description is for the normalized vector; before normalization the set is  $\lambda\mathbb{Z}[i]$ . Formula (15.4) gives a Gaussian omnific Bézout certificate. The normalized scalar sets are therefore  $\mathbf{Oz}[i]$  or  $\Pi_{\mathbb{C}}$ . In the first case the lifted identity shows that a common divisor of the coordinates of a scalar multiple divides that scalar; primitive multiples are precisely unit multiples. The four units give the stated uniqueness. In the second case every representing tuple has all entries in  $\Pi_{\mathbb{C}}$ . If  $d$  is any common divisor, dividing by  $d$  gives another representing tuple, so its entries again lie in  $\Pi_{\mathbb{C}}$  by the scalar classification. They remain Gaussian omnific after division by two. Thus  $2d$  is still a common divisor, but  $2d \mid d$  would require  $1/2 \in \mathbf{Oz}[i]$ , a contradiction to the existence of a gcd. The generator assertion is proposition 15.53.  $\square$

**Corollary 15.55** (Constant directions over  $\mathbb{Z}$  and  $\mathbb{Z}[i]$  [08, 14]). Let  $(D, k, A_D)$  be  $(\mathbb{Z}, \mathbb{R}, \mathbf{Oz})$  or  $(\mathbb{Z}[i], \mathbb{C}, \mathbf{Oz}[i])$ . For a point  $[r_0 : \dots : r_n] \in \mathbb{P}^n(k)$  the following are equivalent:

- (i) it lies in  $\mathbb{P}^n(\text{Frac } D)$ ;

- (ii) it has a unimodular representative in  $A_D^{n+1}$ ;
- (iii) it has a primitive representative in  $A_D^{n+1}$ .

The primitive representatives are then the unit multiples of one primitive tuple in  $D^{n+1}$ .

*Proof.* Choose a nonzero coordinate and permute it into the denominator position. Apply [theorem 15.54](#) (or [theorem 15.18](#) for  $D = \mathbb{Z}$ ) to these constant polynomials: the chosen denominator is nonzero, and their polynomial gcd is one because at least one constant is nonzero. In the rational-direction case normalization gives a primitive tuple in  $D^{n+1}$ , and the classification gives exactly its unit multiples as primitive representatives. Otherwise no representative has a gcd. Undoing the permutation preserves every divisibility and Bézout property.  $\square$

*Provenance.* The real case is [theorem 14.5](#) of source 01. Manuscript 14 of the sibling report proves the criterion (i) $\Leftrightarrow$ (ii) for both pairs  $(D, k)$ , using only that  $D$  is a principal ideal domain: taking constants in a Bézout identity makes the constant multiplier nonzero, and a primitive tuple over  $D$  has a Bézout identity over  $D$ . That report records that the Gaussian and principal-ideal form is printed here with [theorem 14.5](#); this corollary is that statement. Manuscript 14 notes that the criterion concerns coordinate representatives, not all points of a projective scheme over a ring, and it credits only the example  $[1 : \sqrt{2}]$  of the present report, not [theorem 14.5](#).

For constants  $z \in \mathbb{C}$ , ordinary Gaussian rationality  $z \in \mathbb{Q}(i)$  is thus exactly the existence of a primitive Gaussian omnific fraction, although Gaussian omnific fractions produce every surcomplex number [08]. The localization of  $\mathbf{Oz}[i]$  at the elements with nonzero constant has residue field  $\mathbb{Q}(i)$ , and on its finite part (both real and imaginary coordinates finite), the complex standard part and Gaussian-rational residue are prescribed independently in  $\mathbb{C} \times \mathbb{Q}(i)$  by  $(zt + w)/(t + 1)$  with  $z \in \mathbb{C}$ ,  $w \in \mathbb{Q}(i)$  and  $0 < t \in \Pi$ . A common nonzero ordinary denominator for the two parts of  $w$  puts this fraction in the localization. Its constant-term residue is  $w$ , while subtracting  $z$  leaves  $(w - z)/(t + 1)$  with infinitesimal real and imaginary parts. This is the proof of [theorem 15.42](#) coordinatewise [08].

## 15.12 What is and is not classified

The slogan that all surreals are rational over the omnific integers is correct only for membership in the fraction field. It does not give ordinary rationality, lowest terms, or a rational specialization map [08]. The results classify every tuple in  $\mathbb{R}(t)$  and  $\mathbb{C}(t)$  for  $0 \neq t \in \Pi$ , and every point of  $\mathcal{V}$ . They do not classify  $\mathfrak{D}(x)$  for arbitrary  $x \in \mathbf{No}$ , and they do not show that  $\mathcal{V} = \mathbb{P}^1(\mathbf{No})$ ; density is weaker than equality, as the disjoint dense fibers show [09]. A multivariate polynomial gcd of one does not give a Bézout identity ( $T_1$  and  $T_2$  have gcd one in  $\mathbb{R}[T_1, T_2]$  but generate an ideal annihilated by evaluation at  $(0, 0)$ ), so the proofs do not extend directly to several parameters [08]. The scale computation is one explicit test case, not a general flatness or normalization theorem for Hahn integer parts [08]. The questions left by the two sources are [questions 21.12](#) to [21.16](#).

## 16 Differential rigidity over one-sided Hahn rings

This section and the next two form the part *Curve and differential rigidity*. They are built from five manuscripts written independently on September 23, 2026, tagged [C10]–[C14] after their

file prefixes 10– to 14– (section 1.2); the tags [13] and [14] without the letter keep their meaning as manuscripts of the sibling report. All five prove the same spine. A smooth geometrically integral affine curve with constant coefficients has nonconstant points over a one-sided Hahn ring with nonzero exponent group exactly when it is the affine line (theorem 17.6); in particular every omnific solution of a nonsingular Weierstrass equation is an ordinary integer solution (corollary 16.9). The elementary route gives a Bézout certificate for squarefree equations  $y^m = P(x)$ . The geometric *two-ring* argument treats more general targets: properness and the Euler derivations force the contractions of every global differential form at a point to vanish. Source C14 is the base because it has the weakest hypotheses on the shared theorems: any coefficient field of characteristic zero and any ordered exponent group. For the affine-curve classification, sources C10 and C12 use an algebraically closed field, C11 an algebraically closed field or  $\mathbb{R}$ , and C13 the fields  $\mathbb{R}$ ,  $\mathbb{C}$ . The abstract proper-rigidity results of C11 and C13 already allow every characteristic-zero coefficient field; C13 assumes a divisible exponent group. The results answer question 21.1 for smooth curves and its Weierstrass test completely, and they answer questions 21.3 and 21.8 in part (section 21.4). They contain theorem 7.4 and corollary 7.6 as special cases. Labels of this material carry the sub-prefix `odg:cr:`. The proof review in RECONCILIATION.md covers this section’s ring and Euler prerequisites, squarefree certificate and two-ring differential principle, through symmetric forms and inheritance. It also covers the curve classification in sections 17.1 and 17.2, including arbitrary coefficient descent and the logarithmic proof, and the arithmetic fibers, polynomial witnesses and separated-model descent in section 17.3. It also covers projective coordinates and their ideal obstruction in section 17.4. The review also covers section 17.5 and the singular-curve criterion through corollary 17.37. The repeated-root and later geometric applications remain outside it; imported foundations are identified separately. A sixth manuscript, source C15 (prefix 15–), reproves much of the part independently and extends the two-ring argument to logarithmic forms in every dimension; its new results form section 18.6, with the sub-prefix `odg:log:`, and its other results are credited where they are printed. A seventh, source C16 (prefix 16–), extends the curve classification to singular curves; its new results form section 17.6, with the sub-prefix `odg:sg:`, and its other results are credited where they are printed.

## 16.1 Conventions

Throughout the part,  $\mathbb{k}$  is a field of characteristic zero,  $\bar{\mathbb{k}}$  an algebraic closure, and  $\Gamma$  a set-sized ordered abelian group;  $\Gamma$  is *not* assumed divisible, and  $\Gamma_{\mathbb{Q}} = \Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$  is its ordered divisible hull, into which  $\Gamma$  embeds because it is torsion free. We use the  $\omega$ -convention and the rings of section 11.1: the series  $\sum_{\gamma} a_{\gamma} \omega^{\gamma}$  with  $a_{\gamma} \in \mathbb{k}$  and reverse-well-ordered support form the Hahn field  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ , and inside it

$$\begin{aligned} \mathcal{A}_{\mathbb{k}}(\Gamma) &= \{f : \text{supp } f \subseteq \Gamma^{\geq 0}\}, & \Pi_{\mathbb{k}}(\Gamma) &= \{f : \text{supp } f \subseteq \Gamma^{> 0}\}, \\ \mathcal{O}_{\mathbb{k}}(\Gamma) &= \{f : \text{supp } f \subseteq \Gamma^{\leq 0}\}, & \mathfrak{m}_{\mathbb{k}}(\Gamma) &= \{f : \text{supp } f \subseteq \Gamma^{< 0}\}. \end{aligned}$$

For  $0 \neq f \in \mathbb{H}_{\mathbb{k}}(\Gamma)$  the valuation is  $v(f) = -\deg_{\omega} f$ , with  $v(0) = +\infty$ ; so  $\mathcal{O}_{\mathbb{k}}(\Gamma)$  is its valuation ring and  $\mathfrak{m}_{\mathbb{k}}(\Gamma)$  the maximal ideal. For a subring  $\mathfrak{o} \subseteq \mathbb{k}$  the arithmetic ring is  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma) = \mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma)$ . In the surreal instances  $\Gamma \subseteq \mathbf{No}$ ,  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is the part of  $\mathcal{B}_{\mathbb{R}}$  or  $\mathcal{B}_{\mathbb{C}}$  supported in  $\Gamma$ ,  $\mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \Gamma) = \mathbf{Oz}_{\Gamma}$ . For  $\mathbb{k} = \mathbb{R}$ ,  $\mathcal{O}_{\mathbb{k}}(\Gamma)$  is the workspace form of the finite surreals  $\mathcal{O}_{\text{fin}}$ , with  $\mathfrak{m}_{\mathbb{k}}(\Gamma)$  its infinitesimals; for  $\mathbb{k} = \mathbb{C}$ , both real and imaginary coordinates must be finite, and both are infinitesimal in  $\mathfrak{m}_{\mathbb{k}}(\Gamma)$ . The two rings  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  and  $\mathcal{O}_{\mathbb{k}}(\Gamma)$  point in opposite support directions. For  $\Gamma \neq 0$  they are distinct:  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is not the valuation ring or the ring of finite elements (remark 2.3). For  $\Gamma = 0$  both reduce to the coefficient field.

**The  $t$ -convention of three sources.** Sources C11, C12 and C14 write  $\mathbb{k}((t^\Gamma))$  with well-ordered supports and  $t^\gamma = \omega^{-\gamma}$ , as in [section 1.4](#). Their ring  $B_\Gamma(k)$  of series supported on  $\Gamma_{\leq 0}$  is  $\mathcal{A}_\mathbb{k}(\Gamma)$  with exponents negated, their ideal of series supported on  $\Gamma_{< 0}$  ( $P_\Gamma(k)$  in C11,  $I$  in C12) is  $\Pi_\mathbb{k}(\Gamma)$ , and their valuation ring ( $\mathcal{O}$  in C11 and C14,  $V$  in C12) is  $\mathcal{O}_\mathbb{k}(\Gamma)$ . Sources C10 and C13 use the  $\omega$ -convention of this report. Every statement below is translated; no normalization changes.

**Points and the class convention.** For a  $\mathbb{k}$ -scheme  $X$  and a commutative  $\mathbb{k}$ -algebra  $S$ ,  $X(S) = \text{Hom}_\mathbb{k}(\text{Spec } S, X)$ . A point is *constant* if it comes from  $X(\mathbb{k})$ . The retraction  $\text{ct}$  acts on points functorially. All schemes, modules of differentials and valuation rings in this part are set-sized. For the class rings  $\mathcal{B}_\mathbb{R}$ ,  $\mathcal{B}_\mathbb{C}$ ,  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  we never form a spectrum: an affine point is a finite tuple satisfying the equations, and for a separated  $\mathbb{k}$ -scheme of finite type a point over  $\mathcal{B}_\mathbb{R}$  means a point over some  $\mathcal{A}_\mathbb{R}(\Gamma)$ ,  $\Gamma \subseteq \mathbf{No}$  set-sized, two such being identified after passage to a common larger workspace [C10, C12, C13]. A projective point over a ring is given by an invertible quotient; its finite presentation data, like a unit-ideal witness, fit into one workspace [C12]. By [lemma 2.6](#) every finite datum lies in some  $\mathcal{A}_\mathbb{R}(\Gamma)$  or  $\mathcal{A}_\mathbb{C}(\Gamma)$ , and the theorems below hold for arbitrary  $\Gamma$  subject to their stated hypotheses, so either the rational span of the supports [C10, C12, C13] or the group they generate [C11, C14] may be used.

**Letters.** In this part  $\partial_\lambda$  is an Euler derivation (the sources'  $D_\lambda$ ) and  $\mathcal{D}$  a family of derivations;  $C$  is a curve,  $\overline{C}$  its smooth projective completion,  $\Sigma = \overline{C} \setminus C$  its boundary and  $g$  the genus;  $A$  is an abelian variety (not an intermediate ring as in [section 11](#)),  $G$  an algebraic group,  $T$  a torus and  $U$  a unipotent group;  $S$  is a commutative  $\mathbb{k}$ -algebra of coefficients (as elsewhere, a set-sized ring);  $\langle \mathbf{a} \rangle$  is the ideal generated by the coordinates of a tuple  $\mathbf{a}$ . A calligraphic  $\mathcal{O}$  with a scheme as subscript is a structure sheaf, while  $\mathcal{O}_\mathbb{k}(\Gamma)$  is the valuation ring above. The sources write the valuation ring with a macro `\OO` (source C12 writes  $V$ ), affine space with `\A`, and use `\dd` for the degree (C10) or for the differential (C11, C13). In this report `\OO` is the orthogonal group  $O$  and `\V` a zero set  $V$ , and the omnific groups report uses `\A` for a ring; so the new material writes  $\mathcal{O}_\mathbb{k}(\Gamma)$ ,  $\mathbb{A}$ ,  $\deg_\omega$  and  $d$  instead. The complete list of renamed symbols is in [table 1](#).

## 16.2 Two opposite rings in one Hahn field

**Lemma 16.1** (Intersections, units and degree [C14; also C10–C13, C15–C17]). (a)  $\mathcal{A}_\mathbb{k}(\Gamma)$  is a  $\mathbb{k}$ -subalgebra of  $\mathbb{H}_\mathbb{k}(\Gamma)$ ,  $\Pi_\mathbb{k}(\Gamma)$  is an ideal of it,  $\mathcal{A}_\mathbb{k}(\Gamma) = \mathbb{k} \oplus \Pi_\mathbb{k}(\Gamma)$ , and  $\text{ct}$  is a ring retraction of  $\mathcal{A}_\mathbb{k}(\Gamma)$  onto  $\mathbb{k}$  with kernel  $\Pi_\mathbb{k}(\Gamma)$ . Likewise  $\mathcal{O}_\mathbb{k}(\Gamma) = \mathbb{k} \oplus \mathfrak{m}_\mathbb{k}(\Gamma)$  is the valuation ring of  $v$ , with fraction field  $\mathbb{H}_\mathbb{k}(\Gamma)$ , maximal ideal  $\mathfrak{m}_\mathbb{k}(\Gamma)$  and residue map  $\text{ct}$ .

(b)  $\mathcal{A}_\mathbb{k}(\Gamma) \cap \mathcal{O}_\mathbb{k}(\Gamma) = \mathbb{k}$ ,  $\mathcal{A}_\mathbb{k}(\Gamma) \cap \mathfrak{m}_\mathbb{k}(\Gamma) = \{0\}$  and  $\Pi_\mathbb{k}(\Gamma) \cap \mathcal{O}_\mathbb{k}(\Gamma) = \{0\}$ .

(c)  $\mathcal{A}_\mathbb{k}(\Gamma)^\times = \mathbb{k}^\times$ , and for a subring  $\mathfrak{o} \subseteq \mathbb{k}$ ,  $\mathcal{R}_\mathfrak{o}(\mathbb{k}, \Gamma) \cap \mathbb{k} = \mathfrak{o}$  and  $\mathcal{R}_\mathfrak{o}(\mathbb{k}, \Gamma)^\times = \mathfrak{o}^\times$ .

(d) If  $P \in \mathbb{k}[X]$  has degree  $d \geq 1$  and  $x \in \mathcal{A}_\mathbb{k}(\Gamma) \setminus \mathbb{k}$ , then  $P(x) \neq 0$  and  $\deg_\omega P(x) = d \deg_\omega x$ . Every element of  $\mathcal{A}_\mathbb{k}(\Gamma)$  algebraic over  $\mathbb{k}$  lies in  $\mathbb{k}$ .

*Proof.* Parts (a), (c) for  $\mathcal{A}_\mathbb{k}(\Gamma)$  and (d) are [lemma 11.1](#) and its proof. On  $\mathcal{O}_\mathbb{k}(\Gamma)$ , addition and multiplication preserve nonpositive support; in a product exponent zero can arise only from two zero exponents. Thus  $\text{ct}$  is a ring homomorphism onto  $\mathbb{k}$ , with kernel  $\mathfrak{m}_\mathbb{k}(\Gamma)$ . For a nonzero  $f \in \mathbb{H}_\mathbb{k}(\Gamma)$ ,  $v(f) \geq 0$  means  $f \in \mathcal{O}_\mathbb{k}(\Gamma)$ , while  $v(f) < 0$  implies  $v(f^{-1}) > 0$  and  $f^{-1} \in \mathfrak{m}_\mathbb{k}(\Gamma)$ . This proves the valuation-ring property and also  $\text{Frac } \mathcal{O}_\mathbb{k}(\Gamma) = \mathbb{H}_\mathbb{k}(\Gamma)$ : in the latter case  $f = 1/f^{-1}$ . A nonzero  $f \in \mathcal{O}_\mathbb{k}(\Gamma)$  is a unit exactly when  $v(f) = 0$ , since  $v(f^{-1}) = -v(f)$ . Its

nonunits are therefore precisely  $\mathfrak{m}_{\mathbb{k}}(\Gamma)$ , which is the maximal ideal, and  $\mathcal{O}_{\mathbb{k}}(\Gamma)/\mathfrak{m}_{\mathbb{k}}(\Gamma) \cong \mathbb{k}$  via ct. Part (b) compares supports: a support in  $\Gamma^{\geq 0} \cap \Gamma^{\leq 0}$  is contained in  $\{0\}$ , and one in  $\Gamma^{\geq 0} \cap \Gamma^{< 0}$  or  $\Gamma^{> 0} \cap \Gamma^{\leq 0}$  is empty. A unit of  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  is a unit of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , hence a nonzero constant, and a constant lies in  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  exactly when it lies in  $\mathfrak{o}$ ; this gives (c).  $\square$

For  $\Gamma \neq 0$ , the ring  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is not a valuation ring. To see this, choose  $\gamma > 0$  and put  $q = \omega^{\gamma}/(1 + \omega^{\gamma})$ . Both  $q$  and  $q^{-1} = 1 + \omega^{-\gamma}$  have degree zero and are nonconstant, whereas every degree-zero element of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is constant. Neither belongs to  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , violating the valuation-ring criterion. If  $\Gamma = 0$ , instead  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathcal{O}_{\mathbb{k}}(\Gamma) = \mathbb{H}_{\mathbb{k}}(\Gamma) = \mathbb{k}$  is a field and hence a valuation ring. Nothing below assumes that  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is Noetherian, normal, a gcd domain, or that  $\text{Frac } \mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{H}_{\mathbb{k}}(\Gamma)$  [C10, C11, C12, C13, C14]. It is not integrally closed in general (section 17.5). For  $\Gamma = \mathbb{Z}$  it is the polynomial ring  $\mathbb{k}[\omega]$  (section 11.1); for  $\Gamma = \mathbb{Q}$  it contains  $\sum_{n \geq 1} \omega^{1/n}$ , whose support has no least element, and for lexicographically ordered  $\mathbb{Q}^2$  it contains  $\sum_{n \geq 1} \omega^{(1, 1/n)}$ , whose fine structure lies below the first Archimedean scale [C11, C12, C13, C14]. No proof below replaces such a support by a finite truncation.

### 16.3 Euler derivations on both rings

For a  $\mathbb{Q}$ -linear map  $\lambda : \Gamma_{\mathbb{Q}} \rightarrow \mathbb{k}$ , equivalently an additive map  $\Gamma \rightarrow (\mathbb{k}, +)$ , put

$$\partial_{\lambda} \left( \sum_{\gamma \in \Gamma} a_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma \in \Gamma} \lambda(\gamma) a_{\gamma} \omega^{\gamma}. \quad (16.1)$$

For a divisible  $\Gamma = H \subseteq \mathbf{No}$  and a real  $\lambda$  this is (7.1); here neither divisibility nor real values are needed, and values in  $\mathbb{Q} \subseteq \mathbb{k}$  already suffice for detection [C11, C12, C13]. In the  $t$ -convention the rule reads  $\partial(t^{\gamma}) = \lambda(-\gamma)t^{\gamma}$ ; the sources C11, C12 and C14 define their family by  $t^{\gamma} \mapsto \lambda(\gamma)t^{\gamma}$ , which is the same family with  $\lambda$  replaced by  $-\lambda$ . As in section 7, these are local, freely chosen derivations; they are not a canonical surreal derivation, not the Berarducci–Mantova derivation, and not derivatives in the surreal order topology [C10, C11, C12, C13, C14].

**Lemma 16.2** (Euler derivations on the two rings [C12, C13; also C10, C11, C14, C15, C16, C17]). *Each  $\partial = \partial_{\lambda}$  is a  $\mathbb{k}$ -derivation of  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  with  $\text{supp } \partial f \subseteq \text{supp } f$ , hence  $v(\partial f) \geq v(f)$ , and*

$$\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) \subseteq \Pi_{\mathbb{k}}(\Gamma), \quad \partial(\mathcal{O}_{\mathbb{k}}(\Gamma)) \subseteq \mathfrak{m}_{\mathbb{k}}(\Gamma). \quad (16.2)$$

*For  $u \in \mathbb{H}_{\mathbb{k}}(\Gamma)^{\times}$  one has  $\partial u/u \in \mathcal{O}_{\mathbb{k}}(\Gamma)$ , and  $\partial u/u \in \mathfrak{m}_{\mathbb{k}}(\Gamma)$  when  $u \in \mathcal{O}_{\mathbb{k}}(\Gamma)^{\times}$ .*

*Proof.* A subset of a reverse-well-ordered support is reverse well ordered, so (16.1) defines a series, and additivity is coefficientwise. At a fixed exponent the product has finitely many contributions, and  $\lambda(\alpha + \beta) = \lambda(\alpha) + \lambda(\beta)$  gives the Leibniz rule coefficientwise, as in the proof of lemma 7.1;  $\lambda(0) = 0$  kills  $\mathbb{k}$ . Deleting the exponent 0 from a support in  $\Gamma^{\geq 0}$  leaves one in  $\Gamma^{> 0}$ , and from a support in  $\Gamma^{\leq 0}$  one in  $\Gamma^{< 0}$ ; this is (16.2). Finally  $v(\partial u/u) = v(\partial u) - v(u) \geq 0$ , and if  $v(u) = 0$  then  $\partial u \in \mathfrak{m}_{\mathbb{k}}(\Gamma)$  by (16.2), so  $\partial u/u \in \mathfrak{m}_{\mathbb{k}}(\Gamma)$  [C12].  $\square$

The stronger support inclusion  $\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) \subseteq \Pi_{\mathbb{k}}(\Gamma)$  is proved by sources C11, C12 and C13, and appears already in the collection, as a derivation  $\mathbf{Oz} \rightarrow \Pi$  in `osq:prop:classder` and as a map  $\mathcal{R}_{\mathfrak{o}}(k, \Gamma) \rightarrow \Pi_k(\Gamma)$  in `opa:par:prop:Euler`. Sources C10 and C14 use only  $\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$ . The inclusion  $\partial(\mathcal{O}_{\mathbb{k}}(\Gamma)) \subseteq \mathfrak{m}_{\mathbb{k}}(\Gamma)$  on the *opposite* ring is the new ingredient of all five sources; it is the bridge to properness.

These image inclusions need not be proper. For example, when  $\Gamma = \mathbb{Z}$  and  $\lambda(n) = n$ , the ring  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}[\omega]$  has  $\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) = \omega \mathbb{k}[\omega] = \Pi_{\mathbb{k}}(\Gamma)$ : the monomial  $b\omega^n$ ,  $n > 0$ , is the derivative

of  $(b/n)\omega^n$ . Likewise  $v(\partial f) \geq v(f)$  need not be a strict inequality. For any  $u \neq 0$ , write  $u = a\omega^\alpha(1+\varepsilon)$  with  $a \in \mathbb{k}^\times$ ,  $\alpha = \deg_\omega u$  and  $\varepsilon \in \mathfrak{m}_\mathbb{k}(\Gamma)$ . Leibniz gives the more precise formula

$$\frac{\partial_\lambda u}{u} = \lambda(\alpha) + \frac{\partial_\lambda \varepsilon}{1+\varepsilon}, \quad \text{ct}\left(\frac{\partial_\lambda u}{u}\right) = \lambda(\alpha).$$

The second summand belongs to  $\mathfrak{m}_\mathbb{k}(\Gamma)$ , since its numerator does and its denominator is a unit of  $\mathcal{O}_\mathbb{k}(\Gamma)$ . This explains both logarithmic-derivative clauses: an arbitrary monomial can give a nonzero constant, whereas a unit of  $\mathcal{O}_\mathbb{k}(\Gamma)$  has  $\alpha = 0$  [merge].

**Lemma 16.3** (Joint constants and leading detection [C10, C11, C12, C13, C14, C15, C16, C17]). *The common kernel of all Euler derivations of  $\mathbb{H}_\mathbb{k}(\Gamma)$  is  $\mathbb{k}$ . If  $x \in \mathcal{A}_\mathbb{k}(\Gamma) \setminus \mathbb{k}$  has  $\deg_\omega x = \alpha$ , there is  $\lambda$  with  $\lambda(\alpha) = 1$ , and then  $\partial_\lambda x \neq 0$  and  $\deg_\omega \partial_\lambda x = \alpha$ .*

*Proof.* If  $f$  has a nonzero coefficient at  $\gamma \neq 0$ , extend  $\{\gamma\}$  to a basis of the  $\mathbb{Q}$ -vector space  $\Gamma_\mathbb{Q}$  and let  $\lambda$  be the coordinate functional taking  $\gamma$  to  $1 \in \mathbb{Q} \subseteq \mathbb{k}$ ; then  $\partial_\lambda f$  has the same nonzero coefficient at  $\gamma$ . Only choice for a set-sized vector space is used. For the last assertion take  $\gamma = \alpha$ : no new exponent is created and the leading coefficient survives. This is lemma 7.2 for an arbitrary  $\Gamma$  and  $\mathbb{k}$ .  $\square$

As after lemma 7.2, a single Euler derivation can kill a nonconstant series ( $\lambda(a + b\sqrt{2}) = a$  kills  $\omega^{\sqrt{2}}$  on  $\mathbb{Q} + \mathbb{Q}\sqrt{2}$ ); every constancy argument below uses the whole family [C10, C11, C12, C13, C14, C15].

**Corollary 16.4** (Relatively algebraically closed constants [C10, C14, C15, C17]). *Every element of  $\mathbb{H}_\mathbb{k}(\Gamma)$  algebraic over  $\mathbb{k}$  lies in  $\mathbb{k}$ .*

*Proof.* Let  $p$  be the minimal polynomial of  $f$  over  $\mathbb{k}$ . In characteristic zero it is separable, so  $p'(f) \neq 0$ , and  $0 = \partial(p(f)) = p'(f)\partial f$  for every Euler derivation. More explicitly, coprimality of  $p$  and  $p'$  gives  $ap + bp' = 1$  in  $\mathbb{k}[X]$ ; evaluating at  $f$  makes  $p'(f)$  invertible. The chain rule therefore forces  $\partial f = 0$  for every member of the family. Thus  $f$  is a joint constant, and lemma 16.3 applies.  $\square$

## 16.4 The squarefree certificate

**Lemma 16.5** (Differential division [C14; also C10, C11, C12, C13, C15]). *Let  $m \geq 1$  be an ordinary integer, let  $P \in \mathbb{k}[X]$  be squarefree, and choose  $U, W \in \mathbb{k}[X]$  with  $UP + WP' = 1$ . In independent variables  $x, y, \xi, v$  put*

$$F = y^m - P(x), \quad G = my^{m-1}v - P'(x)\xi, \quad H = U(x)y\xi + mW(x)v.$$

*Then*

$$\xi - y^{m-1}H + U(x)\xi F + W(x)G = 0. \quad (16.3)$$

*Consequently, if  $x, y$  lie in a commutative  $\mathbb{k}$ -algebra with  $y^m = P(x)$  and  $\partial$  is a  $\mathbb{k}$ -derivation of it,*

$$\partial x = y^{m-1}(U(x)y\partial x + mW(x)\partial y). \quad (16.4)$$

*Proof.* Characteristic zero makes  $P$  coprime to  $P'$ , so such  $U, W$  exist. Expanding in  $\mathbb{k}[x, y, \xi, v]$ , the terms in  $y^m\xi$  and in  $y^{m-1}v$  cancel and  $\xi(1 - U(x)P(x) - W(x)P'(x)) = 0$  remains. Substituting  $\xi = \partial x$ ,  $v = \partial y$  makes  $F$  and  $G$  vanish, since the derivative of  $y^m = P(x)$  is  $G = 0$  [C12, C13].  $\square$

The identity is a finite certificate in the ideal generated by the equation and its derivative; it replaces the unique factorization or the quotient-ring step of [theorem 7.4](#) and needs no cancellation in the ring [C10, C11, C12, C13, C14].

**Theorem 16.6** (Squarefree superelliptic rigidity [C12, C13, C15; also C10, C11, C14]). *Let  $P \in \mathbb{k}[X]$  be squarefree of degree  $d \geq 2$  and  $m \geq 2$  an ordinary integer. Every solution of  $y^m = P(x)$  in  $\mathcal{A}_{\mathbb{k}}(\Gamma)^2$  lies in  $\mathbb{k}^2$ .*

*Proof.* If  $x \in \mathbb{k}$ , then  $y$  is algebraic over  $\mathbb{k}$  and lies in  $\mathbb{k}$  by [lemma 16.1\(d\)](#). Suppose  $\alpha = \deg_{\omega} x > 0$ . By [lemma 16.1\(d\)](#),  $P(x) \neq 0$  has degree  $d\alpha$ , so  $y \neq 0$  and  $\beta = \deg_{\omega} y$  satisfies  $m\beta = d\alpha$ ; in particular  $\beta > 0$ . Choose  $\partial$  with  $\deg_{\omega} \partial x = \alpha$  ([lemma 16.3](#)) and put  $h = U(x)y\partial x + mW(x)\partial y$ , so that  $\partial x = y^{m-1}h$  by (16.4). The factors  $\partial x$  and  $\partial y$  lie in the ideal  $\Pi_{\mathbb{k}}(\Gamma)$  ([lemma 16.2](#)), and their coefficients lie in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , so  $h \in \Pi_{\mathbb{k}}(\Gamma)$ ; and  $h \neq 0$  because  $\partial x \neq 0$ . A nonzero element of  $\Pi_{\mathbb{k}}(\Gamma)$  has positive degree, so

$$\alpha = \deg_{\omega} \partial x = (m-1)\beta + \deg_{\omega} h > (m-1)\beta.$$

Multiplying by  $m$  gives  $m\alpha > (m-1)d\alpha$ , that is  $(m - (m-1)d)\alpha > 0$ . But  $m - (m-1)d \leq 0$  for all ordinary  $m, d \geq 2$ , and  $\alpha > 0$ , a contradiction. Only multiplication of group elements by ordinary integers is used, so  $\Gamma$  need not be divisible.  $\square$

*Remark 16.7* (The unit route at the corner (2, 2) [C10, C11, C14]). Sources C10, C11 and C14 use only  $h \in \mathcal{A}_{\mathbb{k}}(\Gamma)$ . Then  $\deg_{\omega} h \geq 0$  and the same computation gives  $(m - (m-1)d)\alpha \geq 0$ , a contradiction whenever  $(m-1)d > m$ ; since  $(m-1)d - m = (m-1)(d-1) - 1$ , this excludes every pair except  $m = d = 2$ . That corner they treat by units: extend coefficients to  $\bar{\mathbb{k}}$ , write  $P(X) = c(X - r_1)(X - r_2)$  with  $r_1 \neq r_2$ , choose  $q^2 = c$  and put  $u = x - (r_1 + r_2)/2$ ; then  $(y - qu)(y + qu) = -c(r_1 - r_2)^2/4 \in \bar{\mathbb{k}}^{\times}$ , so both factors are units of  $\mathcal{A}_{\bar{\mathbb{k}}}(\Gamma)$ , hence constants, and so are  $x, y$ , whose coefficients lie in  $\mathbb{k}$ . This is the constant-product mechanism of [lemma 6.1](#) and [theorem 6.7](#). The support inclusion  $h \in \Pi_{\mathbb{k}}(\Gamma)$  of sources C12 and C13 removes the corner without it; it is printed as the primary proof, and this unit argument is kept as the second route.

The hypotheses are sharp [C10, C11, C12, C13, C14, C15]. For  $d = 1$  the curve is an affine line:  $y^m = ax + b$  with  $a \neq 0$  has the points  $((u^m - b)/a, u)$  for all  $u \in \mathcal{A}_{\mathbb{k}}(\Gamma)$ . For  $m = 1$  the graph  $y = P(x)$  has the evident families. Squarefreeness is needed: the cusp  $y^2 = x^3$  has  $(u^2, u^3)$  ([proposition 16.11](#)). Characteristic zero is needed ([section 18.3](#)).

[Theorem 16.6](#) contains [theorem 7.4](#): for  $a, b, c \in \mathbb{C}^{\times}$  and  $m, n \geq 2$ , the equation  $ax^m + by^n = c$  is  $y^n = P(x)$  with  $P(X) = (c - aX^m)/b$ , which is squarefree of degree  $m$  because its derivative vanishes only at 0, where  $P(0) = c/b \neq 0$ . So [theorem 16.6](#) with  $\mathbb{k} = \mathbb{C}$  and  $\Gamma \subseteq \mathbf{No}$  reproves [theorem 7.4](#) and [corollary 7.6](#) without the binary-form step, and the case  $d = 2$  includes the Pell-type equations  $y^2 = Dx^2 + c$  with  $Dc \neq 0$  ([corollary 6.9](#)) [C13]. The separated-power proof of [section 7](#) remains a valid second proof of its case.

**Proposition 16.8** (A cubic certificate [C10, C11, C12, C13, C14, C15]). *For  $P(X) = X^3 + aX + b$  and  $\Delta_0 = 4a^3 + 27b^2$ ,*

$$(27b - 18aX)P(X) + (6aX^2 - 9bX + 4a^2)P'(X) = \Delta_0 \quad (16.5)$$

*in  $\mathbb{Z}[a, b, X]$ . For  $a, b \in \mathbb{k}$ , the polynomial  $P$  is squarefree if and only if  $\Delta_0 \neq 0$ . In every commutative  $\mathbb{k}$ -algebra, on  $y^2 = P(x)$  and for a derivation  $\partial$  killing  $a, b$ ,*

$$\Delta_0 \partial x = y((27b - 18ax)y\partial x + 2(6ax^2 - 9bx + 4a^2)\partial y), \quad (16.6)$$

whether or not  $\Delta_0$  vanishes. If  $\Delta_0 \neq 0$ , set  $U = (27b - 18aX)/\Delta_0$  and  $W = (6aX^2 - 9bX + 4a^2)/\Delta_0$ . Then the differential  $dx/y$ , initially defined where  $y \neq 0$ , extends regularly to the whole affine curve by

$$\frac{dx}{y} = U(x)y dx + 2W(x) dy. \quad (16.7)$$

*Proof.* Expansion proves (16.5). If  $\Delta_0 \neq 0$ , dividing this identity by  $\Delta_0$  proves coprimality of  $P$  and  $P'$ . A repeated root satisfies  $3r^2 + a = 0$  and  $r^3 + ar + b = 0$ , whence  $b = 2r^3$  and  $\Delta_0 = 4(-3r^2)^3 + 27(2r^3)^2 = 0$ . Conversely, if  $\Delta_0 = 0$  and  $a = 0$ , then  $b = 0$  and  $r = 0$  is repeated. If  $a \neq 0$ , put  $r = -3b/(2a)$ ; then  $r^2 = -a/3$  and  $r^3 = b/2$ , so  $P(r) = P'(r) = 0$ .

To obtain (16.6) without dividing by  $\Delta_0$ , multiply (16.5) at  $x$  by  $\partial x$  and substitute  $P(x) = y^2$  and  $P'(x)\partial x = 2y\partial y$ . Now assume  $\Delta_0 \neq 0$  and write  $\eta = U(x)y dx + 2W(x)dy$ . This polynomial expression is a global regular differential. The relation  $2y dy = P'(x)dx$  gives  $y\eta = (U(x)P(x) + W(x)P'(x))dx = dx$ . Thus  $\eta = dx/y$  where  $y \neq 0$ . Where  $P'(x) \neq 0$ , substituting  $dx = 2y dy/P'(x)$  instead gives  $\eta = 2dy/P'(x)$ . These two open sets cover the curve because  $U(x)y^2 + W(x)P'(x) = 1$ ; in particular the second formula explains regularity at every point with  $y = 0$  [C12, C13].  $\square$

The right side of (16.7) is regular on the whole affine curve, including its points with  $y = 0$ , and the element  $h$  in the proof of theorem 16.6, for  $m = 2$  and these  $U, W$ , is exactly its contraction with  $\partial$ . The half-form  $dx/(2y)$  instead contracts to  $h/2$ ; this is the normalization used in the direct proof below. This is the bridge between the elementary certificate and the geometric argument below: the certificate exhibits one global regular differential of the curve [C12, C13]. The classical discriminant of  $P$  is  $-\Delta_0$  and the elliptic discriminant is  $-16\Delta_0$ ; only the nonvanishing matters [C10, C12].

**Corollary 16.9** (Nonsingular short Weierstrass equations [C10, C11, C12, C13, C14, C15]). *For  $a, b \in \mathbb{k}$  with  $4a^3 + 27b^2 \neq 0$ , every solution of  $y^2 = x^3 + ax + b$  in  $\mathcal{A}_{\mathbb{k}}(\Gamma)^2$  is constant. In particular, for  $a, b \in \mathbb{Z}$  with the same nonvanishing condition,*

$$\{(x, y) \in \mathbf{Oz}^2 : y^2 = x^3 + ax + b\} = \{(x, y) \in \mathbb{Z}^2 : y^2 = x^3 + ax + b\},$$

and for  $a, b \in \mathbb{Z}[i]$  satisfying that condition the solutions in  $\mathbf{Oz}[i]^2$  are those in  $\mathbb{Z}[i]^2$ .

*Proof.* The first assertion is theorem 16.6 with  $(m, d) = (2, 3)$  and proposition 16.8. Directly: if  $\deg_{\omega} x = \alpha > 0$  then  $2\deg_{\omega} y = 3\alpha$ , and with  $\deg_{\omega} \partial x = \alpha$  the element  $g = \frac{1}{2}U(x)y \partial x + W(x) \partial y \in \Pi_{\mathbb{k}}(\Gamma)$  satisfies  $2yg = \partial x$ . Here  $g \neq 0$  since  $\partial x \neq 0$ , so  $\deg_{\omega} g = \alpha - \frac{3}{2}\alpha < 0$  in  $\Gamma_{\mathbb{Q}}$ , impossible [C13, C14]. For omnific coordinates put both in one workspace  $\mathcal{A}_{\mathbb{C}}(\Gamma)$  (lemma 2.6); they become complex constants, and  $\mathbf{Oz} \cap \mathbb{C} = \mathbb{Z}$ ,  $\mathbf{Oz}[i] \cap \mathbb{C} = \mathbb{Z}[i]$ .  $\square$

So  $y^2 = x^3 - x + 1$  ( $\Delta_0 = 23$ ),  $y^2 = x^3 + 2x + 3$  ( $\Delta_0 = 275$ ) and, by theorem 16.6,  $y^2 = x^5 - x + 1$  (whose gcd with  $5x^4 - 1$  is one) have only their ordinary omnific points [C12, C13]; source C15 works out the certificate for  $\Delta_0 = 23$  and also names  $y^2 = x^5 + x + 1$ ,  $y^3 = x^4 + x + 1$  and  $y^5 = x^2 - 2$ , whose right sides are squarefree. No complete list of those ordinary points is claimed [C12, C13, C15].

**Corollary 16.10** (General Weierstrass equations [C13], for every  $\mathbb{k}$ ). *Let  $a_1, a_2, a_3, a_4, a_6 \in \mathbb{k}$  and suppose the projective cubic of*

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6 \quad (16.8)$$

*is nonsingular. Every solution in  $\mathcal{A}_{\mathbb{k}}(\Gamma)^2$  is constant. If the  $a_j$  lie in  $\mathbb{Z}$  (or  $\mathbb{Z}[i]$ ), every solution in  $\mathbf{Oz}^2$  (or  $\mathbf{Oz}[i]^2$ ) is ordinary.*

*Proof.* Put  $Y = 2y + a_1x + a_3$ . Then  $Y^2 = Q(x)$  with

$$Q(X) = 4X^3 + (a_1^2 + 4a_2)X^2 + (2a_1a_3 + 4a_4)X + (a_3^2 + 4a_6),$$

because  $Y^2 - Q(x)$  is four times the difference of the two sides of (16.8). The change of variables is invertible over  $\mathbb{k}$  and extends to the invertible projective linear change  $[x : y : z] \mapsto [x : 2y + a_1x + a_3z : z]$ . A repeated root  $r \in \bar{\mathbb{k}}$  of  $Q$  would give the affine point  $(r, 0)$  on  $Y^2 = Q(x)$  where both partial derivatives  $-Q'(r)$  and  $2Y$  vanish, so its tangent space has dimension two rather than one (the plane-curve Jacobian criterion [36, Section 53.9]). This contradicts nonsingularity of the projective cubic. Hence  $Q$  is squarefree of degree three. Theorem 16.6 makes  $x$  and  $Y$  constant, hence  $y = (Y - a_1x - a_3)/2$ . Division by 2 takes place in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ . For omnific coordinates use a common complex workspace and then intersect the constant coordinates with  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$ , as in corollary 16.9; division by 2 in the integer coefficient ring is not required [C13].  $\square$

Source C13 states this over  $\mathbb{Z}$  and  $\mathbb{Z}[i]$  with the fields  $\mathbb{R}$  and  $\mathbb{C}$ ; the proof uses only characteristic zero.

**Proposition 16.11** (Singular cubics [C10, C11, C12, C14, C15]). *If  $4a^3 + 27b^2 = 0$ , put  $r = -3b/(2a)$  when  $a \neq 0$  and  $r = 0$  when  $a = 0$ . Then  $a = -3r^2$ ,  $b = 2r^3$ , and*

$$x = s^2 - 2r, \quad y = s^3 - 3rs \quad (16.9)$$

*solves  $y^2 = x^3 + ax + b$  in every commutative  $\mathbb{k}$ -algebra. If  $a, b \in \mathbb{Z}$ , then  $r \in \mathbb{Z}$ , and (16.9) gives omnific solutions for every  $s \in \mathbf{Oz}$ , nonordinary when  $s$  is. Likewise the node  $y^2 = x^2(x + 1)$  carries  $(h^2 - 1, h(h^2 - 1))$  and the cusp  $y^2 = x^3$  carries  $(h^2, h^3)$  [C11].*

*Proof.* If  $a = 0$ , then  $b = 0$  and the claims hold with  $r = 0$ . If  $a \neq 0$ , the identity  $4a^3 = -27b^2$  gives  $r^2 = 9b^2/(4a^2) = -a/3$  and  $r^3 = r \cdot r^2 = b/2$ . In both cases  $X^3 - 3r^2X + 2r^3 = (X - r)^2(X + 2r)$ , and at  $x = s^2 - 2r$  the two factors are  $(s^2 - 3r)^2$  and  $s^2$ , whose product is  $y^2$ . For integers  $a, b$ , the number  $r$  is a rational root of the monic integer polynomial  $X^3 + aX + b$ , hence an integer [C12]; equivalently,  $r = p/q$  in lowest terms and  $-3p^2/q^2 = a \in \mathbb{Z}$  force  $q = 1$  [C14]. A nonconstant  $s$  gives  $\deg_{\omega} x = 2 \deg_{\omega} s > 0$ . The node and cusp identities are  $(h(h^2 - 1))^2 = (h^2 - 1)^2 h^2$  and  $(h^3)^2 = (h^2)^3$ .  $\square$

The case  $r = 0$  is the cusp and  $r = 1$  the node  $y^2 = x^3 - 3x + 2$  with  $(s^2 - 2, s^3 - 3s)$  [C12, C14]. For positive infinite  $s \in \mathbf{Oz}$  the map  $s \mapsto s^2 - 2r$  is injective, so every singular integral short Weierstrass cubic has a proper class of distinct omnific points, while a nonsingular one has only its set of ordinary points [C12, C14]. This is a polynomial family, not a claim that every point of a singular curve has a parameter in the same ring [C14]. Thus the test equation of question 21.1 has two sharply different outcomes: every nonsingular equation is rigid, and every singular one has polynomial families [C10, C12].

## 16.5 The two-ring contraction principle

**Definition 16.12** (Differential separation data [C14; also C10, C11, C13]). Let  $L/\mathbb{k}$  be a field extension,  $R \subseteq L$  a  $\mathbb{k}$ -subalgebra, and  $V \subseteq L$  a valuation ring containing  $\mathbb{k}$ , with fraction field  $L$  and maximal ideal  $\mathfrak{m}_V$ . A family  $\mathcal{D}$  of  $\mathbb{k}$ -derivations of  $L$  is *separating* if

$$\partial(R) \subseteq R, \quad \partial(V) \subseteq \mathfrak{m}_V \quad (\partial \in \mathcal{D}), \quad R \cap \mathfrak{m}_V = \{0\}, \quad \bigcap_{\partial \in \mathcal{D}} \ker \partial = \mathbb{k}. \quad (16.10)$$

No inclusion between  $R$  and  $V$  is required, and  $R$  is not assumed to be a valuation ring, normal, finitely generated or of finite transcendence degree;  $L$  may be larger than  $\text{Frac } R$  [C10, C11, C14]. By [lemmas 16.1 to 16.3](#),  $(L, R, V, \mathcal{D}) = (\mathbb{H}_{\mathbb{k}}(\Gamma), \mathcal{A}_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \{\partial_{\lambda}\})$  is separating data, with the stronger inclusion  $\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) \subseteq \Pi_{\mathbb{k}}(\Gamma)$ .

Here is the meaning of differentiating a point. For a  $\mathbb{k}$ -algebra  $S$ , an  $S$ -point  $p_S$  and a  $\mathbb{k}$ -derivation  $\partial : S \rightarrow M$ , where  $M$  is an  $S$ -module, compose the natural pullback map with the universal map:

$$p_S^* \Omega_{X/\mathbb{k}}^1 \longrightarrow \Omega_{S/\mathbb{k}}^1 \xrightarrow{\iota_{\partial}} M, \quad df \longmapsto \partial(f(p_S)).$$

For  $S = L$  and  $M = L$  this is an  $L$ -linear functional on the cotangent fiber, denoted  $\partial p_L$ . On an affine open with coordinate algebra  $B$ , the corresponding tangent derivation of  $B \otimes_{\mathbb{k}} L$  is  $f \otimes a \mapsto a \partial(f(p_L))$ : it annihilates the new scalar factor  $L$ . Thus it is a tangent vector to  $X_L$  at  $p_L$ , although the original map  $\partial : L \rightarrow L$  is only  $\mathbb{k}$ -linear.

**Lemma 16.13** (Equality from an ambient field [C10, C11, C12, C13, C14, C15]). *Let  $X$  be a separated scheme and  $\sigma_1, \sigma_2 \in X(S)$ . If  $S \rightarrow T_i$  are ring maps whose kernels have zero intersection, and the images of  $\sigma_1, \sigma_2$  in each  $X(T_i)$  agree, then  $\sigma_1 = \sigma_2$ . In particular  $X(S) \rightarrow X(T)$  is injective for an injective ring map  $S \hookrightarrow T$ , such as  $S \subseteq L$  with  $L$  a field;  $L = \text{Frac } S$  is not needed.*

*Proof.* The equalizer of  $\sigma_1, \sigma_2$  is the inverse image of the closed diagonal, a closed subscheme of  $\text{Spec } S$  with ideal  $J$ . Agreement over  $T_i$  puts  $J$  in the kernel of  $S \rightarrow T_i$  for every  $i$ , so  $J = 0$  [C11].  $\square$

**Theorem 16.14** (Two-ring annihilation [C13]; one-forms [C10, C11, C12, C14]). *Let  $L, R, V, \mathfrak{m}_V$  and  $\mathcal{D}$  satisfy the first three conditions of (16.10). Let  $X$  be proper over  $\mathbb{k}$  and  $p \in X(R)$ , with  $p_L \in X(L)$  its image. For every  $r \geq 1$ , every global covariant tensor  $s \in H^0(X, (\Omega_{X/\mathbb{k}}^1)^{\otimes r})$  and all  $\partial_1, \dots, \partial_r \in \mathcal{D}$ ,*

$$s_{p_L}(\partial_1 p_L, \dots, \partial_r p_L) = 0, \tag{16.11}$$

*and every global symmetric differential of positive degree  $r$  vanishes on the diagonal  $(\partial p_L, \dots, \partial p_L)$ . Smoothness of  $X$  is not needed.*

*Proof.* A  $\mathbb{k}$ -derivation  $\partial : S \rightarrow M$  into an  $S$ -module factors uniquely through an  $S$ -linear map  $\iota_{\partial} : \Omega_{S/\mathbb{k}}^1 \rightarrow M$ ,  $\sum a_j db_j \mapsto \sum a_j \partial b_j$ ; the sums are finite, and no finite generation of  $S$  is needed [35, Tag 00RM]. Pull  $s$  back along  $p$ , apply the natural maps to  $(\Omega_{R/\mathbb{k}}^1)^{\otimes r}$ , and contract with the restrictions  $\partial_j|_R : R \rightarrow R$ . The resulting global function on  $\text{Spec } R$  is an element  $a_R \in R$ . By the valuative criterion for properness, which holds for arbitrary valuation rings and not only discrete ones [38, Tag 0BX5],  $p_L$  extends to  $q \in X(V)$ . The restrictions  $\partial_j|_V$  take values in the  $V$ -module  $\mathfrak{m}_V$ ; pulling  $s$  back along  $q$  and contracting gives a finite sum of products of  $r$  elements of  $\mathfrak{m}_V$  with coefficients in  $V$ . More precisely,  $a_V \in \mathfrak{m}_V^r \subseteq \mathfrak{m}_V$ , since  $r \geq 1$ .

For either  $S = R$  or  $S = V$ , restriction of the contraction to  $L$  is the composition of  $p_L^* \Omega_{X/\mathbb{k}}^1 \rightarrow \Omega_{L/\mathbb{k}}^1$  with the same maps  $\iota_{\partial_j}$ . Indeed, the natural map  $\Omega_{S/\mathbb{k}}^1 \otimes_S L \rightarrow \Omega_{L/\mathbb{k}}^1$  takes  $db$  to  $db$ , and both contractions take it to  $\partial_j b$ . Tensor products and multiplication preserve this compatibility. No claim that this natural map is an isomorphism, or that  $L = \text{Frac } R$ , is needed. Hence  $a_R$  and  $a_V$  have the same image in  $L$ , even when  $p$  and  $q$  use different affine charts, and that image lies in  $R \cap \mathfrak{m}_V = 0$  [C13, C14].

For a symmetric differential, use one derivation in every factor. The map  $(\alpha_1, \dots, \alpha_r) \mapsto \prod_j \iota_{\partial}(\alpha_j)$  is symmetric and therefore descends to  $\text{Sym}^r \Omega^1$ . Its value over  $V$  is still in  $\mathfrak{m}_V^r$ .

This uses the quotient definition of the symmetric power; no lift of a global symmetric section to a global tensor section is assumed.  $\square$

The positive-degree restriction matters: in degree zero the global section 1 evaluates to 1, and  $\mathfrak{m}_V^0 = V$  gives no vanishing. The joint constant condition in (16.10) is used only in the later descent from zero tangent vectors to constant points.

The derivations are relative to  $\mathbb{k}$ , not to  $L$ : an  $L$ -linear derivation of  $L$  is zero and would give nothing. The forms are global and regular; a rational form with a pole at the centre of  $q$  need not contract into  $\mathfrak{m}_V$  [C14]. The argument compares two extensions of one field-valued point; it does not assert a ring map  $R \rightarrow V$  compatible with the embeddings in  $L$ , and such a map generally does not exist [C10]. For  $X \subseteq \mathbb{P}_{\mathbb{k}}^n$  the valuative step is concrete: write  $p_L = [a_0 : \cdots : a_n]$ , choose  $j$  with  $a_j \neq 0$  of least valuation among the nonzero coordinates, and use  $a_i/a_j \in V$ . The  $j$ th coordinate is 1, so these ratios define a  $V$ -point of projective space. The homogeneous equations of  $X$  vanish in  $L$  and hence in  $V$ , so the point lies in  $X$ . This uses only the minimum of a finite set of values, with no rank-one or discreteness condition [C13]. A regular differential thus has two descriptions at the point, one with no strictly negative exponent and one, after an Euler derivation, with only strictly negative exponents; properness supplies the second even when the affine coordinates of the point are infinite [C14]. Nothing here is an order-topological or transfinite-limit argument.

**Lemma 16.15** (From vanishing tangents to a constant point [C13, C14; also C10, C11, C12]). *Let  $(L, R, V, \mathcal{D})$  be separating data,  $X$  a separated  $\mathbb{k}$ -scheme of finite type and  $p \in X(R)$ . If  $\partial(f(p_L)) = 0$  for every  $\partial \in \mathcal{D}$  and every regular function  $f$  on some affine open  $W \ni p_L$ , then  $p$  comes from a unique point of  $X(\mathbb{k})$ .*

*Proof.* The algebra map  $\mathbb{k}[W] \rightarrow L$  of  $p_L$  has image in the joint constants  $\mathbb{k}$ , so it defines  $p_0 \in W(\mathbb{k}) \subseteq X(\mathbb{k})$  whose extension to  $L$  is  $p_L$ . The constant  $R$ -point of  $p_0$  and  $p$  agree over  $L$ , hence over  $R$  by lemma 16.13. Uniqueness follows from  $\mathbb{k} \hookrightarrow L$ . The point  $p$  need not factor through  $W$  before this is proved [C13].  $\square$

It is enough to test finitely many algebra generators of  $\mathbb{k}[W]$ : the derivations kill coefficients in  $\mathbb{k}$  and satisfy the sum and product rules. This finite test concerns the chosen affine chart; it does not require a finitely generated subalgebra of  $L$  stable under all the derivations.

**Theorem 16.16** (Proper differential rigidity [C14; also C10, C11, C12, C13]). *Let  $(L, R, V, \mathcal{D})$  be separating data and  $X$  a smooth proper  $\mathbb{k}$ -scheme whose cotangent sheaf  $\Omega_{X/\mathbb{k}}^1$  is generated by global sections. Then  $X(\mathbb{k}) \rightarrow X(R)$  is a bijection.*

*Proof.* Let  $p \in X(R)$  and  $\partial \in \mathcal{D}$ . Differentiation of  $p_L$  defines an  $L$ -linear map  $p_L^* \Omega_{X/\mathbb{k}}^1 \rightarrow L$ . Global generation makes  $H^0(X, \Omega_{X/\mathbb{k}}^1) \otimes_{\mathbb{k}} L \rightarrow p_L^* \Omega_{X/\mathbb{k}}^1$  surjective: pull back the surjective evaluation map  $H^0(X, \Omega_{X/\mathbb{k}}^1) \otimes_{\mathbb{k}} \mathcal{O}_X \rightarrow \Omega_{X/\mathbb{k}}^1$  along  $p_L$ . Pullback is right exact, even when  $p_L$  is not a closed point. By theorem 16.14, the tangent functional vanishes on the image of this evaluation map, hence on the whole cotangent fiber. So  $\partial(f(p_L)) = 0$  for every regular  $f$  on an affine open through  $p_L$ , and lemma 16.15 makes  $p$  constant. Injectivity follows from  $\mathbb{k} \hookrightarrow L$ .  $\square$

In fact this proof also works for a proper, possibly singular  $X$  whose Kähler differential sheaf is globally generated: neither its local freeness nor smoothness was used. Smoothness in the stated theorem provides the cotangent-bundle setting used in the applications below.

**Corollary 16.17** (Hahn realization [C14; also C10, C11, C12, C13]). *For every field  $\mathbb{k}$  of characteristic zero, every set-sized ordered abelian group  $\Gamma$  and every smooth proper  $\mathbb{k}$ -scheme  $X$  with globally generated cotangent sheaf,  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$ , and  $\text{ct}$  is inverse to the inclusion of constant points. For  $\mathbb{k} = \mathbb{R}$  or  $\mathbb{C}$  the same holds over  $\mathcal{B}_{\mathbb{R}}$  or  $\mathcal{B}_{\mathbb{C}}$  in the class convention of section 16.1.*

*Proof.* Apply theorem 16.16 to the data  $(\mathbb{H}_{\mathbb{k}}(\Gamma), \mathcal{A}_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \{\partial_{\lambda}\})$ . The retraction fixes constant points [C10]. A class point lies over some  $\mathcal{A}_{\mathbb{R}}(\Gamma)$  or  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ .  $\square$

The proof uses only that global forms span the cotangent space at the particular field-valued point  $p_L$  [C11, C12]. Hence if  $Z \subseteq X$  is the locus where global forms fail to generate, every nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point has its  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ -point in  $Z$ , and every Euler tangent direction of a point lies in the common annihilator of the global one-forms there [C10, C11]. This is an obstruction, not a classification of varieties with a nonempty failure locus, and it does not prove that surviving directions integrate to nonconstant Hahn points.

**Definition 16.18** (Tangent detection by symmetric differentials [C13]). Global symmetric differentials *detect tangent directions* on a smooth proper  $\mathbb{k}$ -scheme  $X$  if for every field extension  $L/\mathbb{k}$ , every  $x \in X(L)$  and every nonzero tangent vector  $u \in T_x X_L$  there are  $r \geq 1$  and  $s \in H^0(X, \text{Sym}^r \Omega_{X/\mathbb{k}}^1)$  with  $s_x(u, \dots, u) \neq 0$ .

The quantification over extension fields is needed because the point  $p_L$  is generally not classical. For  $\mathcal{F} = \text{Sym}^r \Omega_{X/\mathbb{k}}^1$ , flat base change gives  $H^0(X_L, \mathcal{F}_L) = H^0(X, \mathcal{F}) \otimes_{\mathbb{k}} L$  [37, Tag 02KH]; here  $(\Omega_{X/\mathbb{k}}^1)_L = \Omega_{X_L/L}^1$ , with the base field changed along with the scheme. Thus a detecting section after field extension is a finite  $L$ -linear combination of base-field sections; if its value is nonzero, at least one of those sections has nonzero value. This justifies using sections over  $\mathbb{k}$  in the definition [C13].

**Theorem 16.19** (Symmetric-differential rigidity [C13, C15]). *If  $X$  is smooth and proper over  $\mathbb{k}$  and its global symmetric differentials detect tangent directions, then  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$  for every  $\Gamma$ .*

*Proof.* If some  $\partial_{\lambda} p_L$  were nonzero, a detecting symmetric differential would have nonzero diagonal value on it, contrary to theorem 16.14. So all Euler tangent vectors vanish and lemma 16.15 applies.  $\square$

The same proof works for any separating data  $(L, R, V, \mathcal{D})$ : tangent detection at its field  $L$  gives  $X(R) = X(\mathbb{k})$ . The Hahn realization supplies one such family, rather than an additional geometric ingredient.

**Corollary 16.20** (A semiample tautological bundle [C13]; ample case also [C15]). *Let  $X$  be smooth and projective over  $\mathbb{k}$ , and let  $\mathcal{L} = \mathcal{O}_{\mathbb{P}(\Omega_{X/\mathbb{k}}^1)}(1)$ , with  $\mathbb{P}(E)$  parametrizing one-dimensional quotients of  $E$ . If some positive tensor power of  $\mathcal{L}$  is globally generated, then  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$ . This holds in particular when  $\mathcal{L}$  is ample, which is what “ $\Omega_{X/\mathbb{k}}^1$  is ample” means here, and when  $\text{Sym}^r \Omega_{X/\mathbb{k}}^1$  is globally generated for some  $r \geq 1$ .*

*Proof.* A nonzero tangent vector  $u$  at  $x \in X(L)$  is a nonzero  $L$ -linear map  $x^* \Omega_{X/\mathbb{k}}^1 \rightarrow L$ , hence a surjection. It defines a point  $[u] \in \mathbb{P}(\Omega^1)(L)$  over  $x$ . If  $\mathcal{L}^{\otimes r}$  is globally generated, its evaluation at  $[u]$  is surjective onto a one-dimensional  $L$ -space; at least one base-field section is nonzero there. Since  $\Omega^1$  is locally free, the projective-bundle identity is  $H^0(\mathbb{P}(\Omega^1), \mathcal{L}^{\otimes r}) = H^0(X, \text{Sym}^r \Omega^1)$  [45, Tag 01OA]. Under the quotient identification provided by  $u$ , the value of that section

is exactly  $s_x(u, \dots, u) \neq 0$ . An ample line bundle on a projective scheme has a globally generated power [46, Tag 01PR], and global generation of  $\mathrm{Sym}^r \Omega^1$  gives that of  $\mathcal{L}^{\otimes r}$  through the universal quotient. Apply [theorem 16.19](#).  $\square$

Conventions for semiample vector bundles vary, so the hypothesis is stated for the tautological line bundle. No implication from nefness, bigness or general type is asserted [C13].

For a smooth projective  $X$ , tangent detection is in fact equivalent to this tautological semiampleness condition. To see the converse, put  $P = \mathbb{P}(\Omega^1)$  and suppose detection holds. Applying it over the residue field of each point of  $P$  shows that the nonvanishing loci of the sections of all positive powers of  $\mathcal{L}$  cover  $P$ . Quasi-compactness gives a finite subcover, from sections  $s_i$  of degrees  $r_i \geq 1$ . For a common positive multiple  $N$  of the  $r_i$ , the sections  $s_i^{N/r_i}$  of  $\mathcal{L}^{\otimes N}$  still have no common zero, so they generate that line bundle. If  $P$  is empty the conclusion is vacuous. This equivalence concerns tangent detection; it does not assert that every rigid variety satisfies either condition.

**Proposition 16.21** (Inheritance [C13; also C10, C11, C14, C15]). *Let  $X$  be a separated  $\mathbb{k}$ -scheme of finite type with  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$ . Every locally closed subscheme  $Y \subseteq X$  has  $Y(\mathcal{A}_{\mathbb{k}}(\Gamma)) = Y(\mathbb{k})$ . For a morphism  $f : Z \rightarrow X$ , every  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point  $z$  of  $Z$  satisfies  $f(z) = f(\mathrm{ct} z)$ ; so if every fiber of  $f$  over a  $\mathbb{k}$ -point is rigid, then  $Z(\mathcal{A}_{\mathbb{k}}(\Gamma)) = Z(\mathbb{k})$ .*

*Proof.* For  $y \in Y(\mathcal{A}_{\mathbb{k}}(\Gamma))$  its image in  $X$  is constant, equal to the image of  $y_0 = \mathrm{ct}(y) \in Y(\mathbb{k})$ ; an immersion is a monomorphism, so  $y = y_0$ . For  $f$ , the point  $f(z)$  is constant with constant term  $x_0 = f(\mathrm{ct} z)$ , so the universal property of  $Z \times_X \mathrm{Spec} \mathbb{k}$ , formed using  $x_0$ , puts  $z$  in that fiber as a  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point. If the fiber is rigid, this point comes from a  $\mathbb{k}$ -point of the fiber and hence of  $Z$ . Conversely, constant points always extend, and the retraction makes their extension injective. The immersion argument also works for singular or nonreduced  $Y$ ; no smoothness is used here.  $\square$

Properness and global generation in [theorem 16.16](#) cannot in general be dropped [C10, C11, C12, C13, C14]. For the counterexamples choose  $\gamma > 0$  in  $\Gamma$  (when  $\Gamma = 0$ ,  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}$  already). The affine line  $\mathbb{G}_a$  has trivial cotangent sheaf, generated by  $dT$ , but is not proper, and  $\mathbb{G}_a(\mathcal{A}_{\mathbb{k}}(\Gamma)) = \mathcal{A}_{\mathbb{k}}(\Gamma)$ : the point  $T = \omega^\gamma$  does not extend to a  $\mathcal{O}_{\mathbb{k}}(\Gamma)$ -point of that chart. The projective line is proper but has no nonzero global one-form, and  $[\omega^\gamma : 1]$  is a nonconstant point of  $\mathbb{P}^1(\mathcal{A}_{\mathbb{k}}(\Gamma))$  for  $\gamma > 0$ . Properness supplies the valuation-ring point and global generation the detection; they are separate hypotheses. Nor is finite generation a shortcut: a finite tuple of Hahn series generates a finitely generated algebra, but an Euler derivation need not preserve it, so one cannot infer that the tuple comes from a polynomial arc; polynomial arcs appear only after the classification has forced the affine line ([corollary 17.16](#)) [C14].

For an explicit failure of stability, put  $t = \omega^\gamma$ , choose  $\partial t = t$ , and set  $f = t + t^2$ . Then  $\partial f = 2f - t$  does not belong to the finitely generated algebra  $\mathbb{k}[f]$ : otherwise  $t$  would belong to it, whereas every nonconstant polynomial in  $f$  has degree at least  $2\gamma > \gamma = \deg_\omega t$ . The same point  $[t : 1]$  also illustrates why regularity of the form is essential. On  $\mathbb{P}^1$  the rational form  $dT$  contracts to  $t$ , which is not in  $\mathfrak{m}_{\mathbb{k}}(\Gamma)$ ; at the valuation centre, with coordinate  $u = T^{-1}$ , it is  $-u^{-2}du$  and has a pole.

## 17 Smooth curves and their omnific points

### 17.1 Positive genus

**Lemma 17.1** (Canonical forms have no common zero [C13, C14; also C10, C11, C12]). *Let  $\overline{C}$  be a smooth proper geometrically integral curve of genus  $g \geq 1$  over  $\mathbb{k}$ . Then  $\Omega_{\overline{C}/\mathbb{k}}^1$  is generated by its global sections.*

*Proof.* Global sections of a coherent sheaf on a proper scheme commute with field extension [37], and surjectivity descends along the faithfully flat extension  $\mathbb{k} \subseteq \bar{\mathbb{k}}$ , so assume  $\mathbb{k}$  algebraically closed. Let  $P$  be a closed point, hence a  $\mathbb{k}$ -rational point. A nonconstant rational function with at most a simple pole at  $P$  would be a degree-one morphism  $\overline{C} \rightarrow \mathbb{P}^1$ , an isomorphism of smooth projective curves, contradicting  $g \geq 1$ ; so  $h^0(\overline{C}, \mathcal{O}_{\overline{C}}(P)) = 1$ . Riemann–Roch and duality [41] give  $h^0(\overline{C}, \Omega^1(-P)) = h^0(\mathcal{O}_{\overline{C}}(P)) + g - 2 = g - 1$ , while  $h^0(\overline{C}, \Omega^1) = g$ . So some global differential does not vanish at  $P$  and spans the one-dimensional cotangent fiber there: the kernel of evaluation is exactly  $H^0(\overline{C}, \Omega^1(-P))$ . Nakayama’s lemma makes the evaluation map surjective at every closed point. Its coherent cokernel is therefore zero, since a nonempty closed support on a finite-type scheme over a field has a closed point. This proves global generation on the whole curve.  $\square$

**Theorem 17.2** (Positive-genus curves [C11, C13, C14; also C10, C12]). *Let  $\overline{C}$  be a smooth proper geometrically integral curve of genus at least one over  $\mathbb{k}$ . Then  $\overline{C}(\mathcal{A}_{\mathbb{k}}(\Gamma)) = \overline{C}(\mathbb{k})$ , and every locally closed subscheme of  $\overline{C}$ , in particular every open subcurve, has only constant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -points.*

*Proof.* Apply lemma 17.1, corollary 16.17, and proposition 16.21.  $\square$

Sources C11, C13 and C14 prove this over every field of characteristic zero, source C11 by extending coefficients to  $\bar{\mathbb{k}}$  and comparing a point with its constant term through lemma 16.13; sources C10 and C12 assume  $\mathbb{k}$  algebraically closed. No rational point is selected in advance: a genus-one torsor without a  $\mathbb{k}$ -point has no  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point at all [C14]. Sources C10 and C15 note that the projective statement also follows from an Abel–Jacobi embedding and theorem 18.2; the proof above avoids that construction.

*Remark 17.3* (One nonzero form suffices for curves [C11, C12, C15]). For curves the base-point-free statement can be bypassed. Let  $p$  be a nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point of a smooth geometrically integral curve  $X$  of positive genus (proper, or an open subcurve of a proper one). If the image of the generic point of  $\text{Spec } \mathcal{A}_{\mathbb{k}}(\Gamma)$  were a closed point, its residue field would be finite over  $\mathbb{k}$  and embedded in  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ , hence equal to  $\mathbb{k}$  by corollary 16.4, and  $p$  would be constant by lemma 16.13. So  $p$  gives an embedding  $F = \mathbb{k}(X) \hookrightarrow \mathbb{H}_{\mathbb{k}}(\Gamma)$ . A nonzero global regular form  $\eta$  on the completion spans the one-dimensional  $F$ -space  $\Omega_{F/\mathbb{k}}^1$ , and its contraction with every  $\partial_{\lambda}$  vanishes by theorem 16.14, applied after composing  $p$  with the inclusion into the proper completion. All contractions are taken in  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  through the displayed embedding; the derivations need not preserve  $F$ . Writing  $df = h_f \eta$  gives  $\partial_{\lambda} f = h_f \eta(\partial_{\lambda}) = 0$  for all  $f \in F$  and all  $\lambda$ , so  $F \subseteq \mathbb{k}$  by lemma 16.3, contradicting  $\text{trdeg}_{\mathbb{k}} F = 1$ . Source C11 gives this for affine curves over algebraically closed  $\mathbb{k}$  and source C12 for projective ones; the closed-point step for a general  $\mathbb{k}$  uses corollary 16.4.

### 17.2 The smooth affine classification

**Lemma 17.4** (Two punctures force constancy [C10, C11, C13, C14, C15]). *Let  $\mathbb{k}$  be algebraically closed. If  $U = \mathbb{P}_{\mathbb{k}}^1 \setminus \Sigma$  for a finite set  $\Sigma$  of at least two closed points, then  $U(\mathcal{A}_{\mathbb{k}}(\Gamma)) = U(\mathbb{k})$ .*

More generally, an integral affine curve  $C$  over  $\mathbb{k}$  whose coordinate ring has a nonconstant unit has  $C(\mathcal{A}_{\mathbb{k}}(\Gamma)) = C(\mathbb{k})$ .

*Proof.* Let  $\varphi : \mathbb{k}[C] \rightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$  be a point and  $u$  a nonconstant unit. Then  $\varphi(u) \in \mathcal{A}_{\mathbb{k}}(\Gamma)^{\times} = \mathbb{k}^{\times}$ , say  $\varphi(u) = c$ . The kernel of  $\varphi$  contains  $u - c \neq 0$ ; it is a prime of the one-dimensional finitely generated domain  $\mathbb{k}[C]$ , hence maximal with residue field  $\mathbb{k}$  by the Nullstellensatz, and  $\varphi$  factors through  $\mathbb{k}$  [C13]. For  $U$ , a coordinate  $z$  with its zero and pole at two removed points is such a unit: both  $z$  and  $z^{-1}$  are regular on  $U$  [C10, C11, C14].  $\square$

This extends the torus remark of [section 6](#) (“An algebraic-geometric formulation”). There the coordinate ring is generated by units; here a single nonconstant unit suffices because the coordinate ring has dimension one. It need not be generated by units. In either argument units must specialize to nonzero constants.

**Lemma 17.5** (Forms of the affine line [C14; also C16]; real case [C10, C11, C13]). *If  $C$  is a smooth geometrically integral affine curve over  $\mathbb{k}$  with  $C_{\bar{\mathbb{k}}} \cong \mathbb{A}_{\bar{\mathbb{k}}}^1$ , then  $C \cong \mathbb{A}_{\mathbb{k}}^1$ .*

*Proof.* The smooth projective completion  $\bar{C}$  exists over the perfect field  $\mathbb{k}$  [40]; its base change is the smooth projective completion of  $C_{\bar{\mathbb{k}}}$ , hence  $\mathbb{P}^1$  by uniqueness [39]. Its reduced boundary is finite étale over  $\mathbb{k}$  and has exactly one geometric point. Its degree is therefore one, so it consists of a  $\mathbb{k}$ -rational point  $P$ . A smooth proper genus-zero curve with a rational point is  $\mathbb{P}_{\mathbb{k}}^1$  [42]: by Riemann–Roch,  $h^0(\mathcal{O}_{\bar{C}}(P)) = 2$ , and a function with a single simple pole at  $P$  is a degree-one map to  $\mathbb{P}_{\mathbb{k}}^1$  sending  $P$  to  $\infty$ . Removing  $P$  gives  $C \cong \mathbb{A}_{\mathbb{k}}^1$ .  $\square$

For  $\mathbb{k} = \mathbb{R}$  the unique geometric boundary point is fixed by complex conjugation; sources C10, C11 and C13 prove this case. The affine unit circle  $x^2 + y^2 = 1$  instead has two geometric boundary points,  $[1 : i : 0]$  and  $[1 : -i : 0]$  on  $x^2 + y^2 = z^2$ , exchanged by conjugation. Its complexification is a projective line minus two points, so its rigidity follows from [lemma 17.4](#) and the coefficient descent below. It is not a form of the affine line. Source C13 includes it in the full real-descent argument; the one-boundary-point step alone does not apply to it.

**Theorem 17.6** (Smooth affine curves [C14; also C10, C11, C12, C13, C15, C16]). *Let  $\Gamma \neq 0$  and let  $C$  be a smooth geometrically integral affine curve over a field  $\mathbb{k}$  of characteristic zero. Then*

$$C(\mathcal{A}_{\mathbb{k}}(\Gamma)) \neq C(\mathbb{k}) \iff C \cong \mathbb{A}_{\mathbb{k}}^1,$$

*and in that case an isomorphism identifies  $C(\mathcal{A}_{\mathbb{k}}(\Gamma))$  with  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ . The following are equivalent:  $C \cong \mathbb{A}_{\mathbb{k}}^1$ ;  $C_L \cong \mathbb{A}_L^1$  for some (or every) field extension  $L/\mathbb{k}$ ; there is a nonconstant morphism  $\mathbb{A}_{\mathbb{k}}^1 \rightarrow C$ .*

*Proof.* First let  $\mathbb{k}$  be algebraically closed. Let  $\bar{C}$  be the smooth projective completion and  $\Sigma = \bar{C} \setminus C$ ; it is finite and nonempty, because a positive-dimensional proper integral curve is not affine [40]. If  $\bar{C}$  has positive genus, use [theorem 17.2](#). Otherwise  $\bar{C} \cong \mathbb{P}_{\mathbb{k}}^1$  [42]; if  $|\Sigma| \geq 2$  use [lemma 17.4](#), and if  $|\Sigma| = 1$  then  $C \cong \mathbb{A}_{\mathbb{k}}^1$ .

For general  $\mathbb{k}$ , the coefficientwise inclusion  $\mathcal{A}_{\mathbb{k}}(\Gamma) \hookrightarrow \mathcal{A}_{\bar{\mathbb{k}}}(\Gamma)$  preserves supports. Unless  $C_{\bar{\mathbb{k}}} \cong \mathbb{A}_{\bar{\mathbb{k}}}^1$ , a point  $\mathbb{k}[C] \rightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$  becomes constant over  $\bar{\mathbb{k}}$ , so its image lies in  $\mathcal{A}_{\mathbb{k}}(\Gamma) \cap \bar{\mathbb{k}} = \mathbb{k}$  and it is a  $\mathbb{k}$ -point [C14]. If  $C_{\bar{\mathbb{k}}} \cong \mathbb{A}_{\bar{\mathbb{k}}}^1$ , then  $C \cong \mathbb{A}_{\mathbb{k}}^1$  by [lemma 17.5](#), whose points are the elements of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ ; for  $0 < \gamma \in \Gamma$  the element  $\omega^{\gamma}$  is nonconstant.

For an arbitrary field extension  $L/\mathbb{k}$ , if  $C_L \cong \mathbb{A}_L^1$ , then  $\bar{C}_L \cong \mathbb{P}_L^1$  and its reduced boundary has degree one. Genus and the degree of the finite étale boundary are unchanged by field extension. Thus  $\bar{C}$  has genus zero and a boundary consisting of one  $\mathbb{k}$ -rational point. The

Riemann–Roch argument in [lemma 17.5](#) gives  $C \cong \mathbb{A}_{\mathbb{k}}^1$ . This does not require an embedding of  $L$  into  $\bar{\mathbb{k}}$ . The converse holds by base change.

A nonconstant morphism  $\mathbb{A}^1 \rightarrow C$  is dominant, since its image cannot be contained in a closed point. It therefore induces an injection  $\mathbb{k}[C] \hookrightarrow \mathbb{k}[T]$ . Composing with  $\mathbb{k}[T] \hookrightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$ ,  $T \mapsto \omega^\gamma$ , gives a nonconstant point: distinct powers have distinct exponents, so substitution is injective. Conversely  $\mathbb{A}^1 \cong C$  is such a morphism [C12]. The assumption  $\Gamma \neq 0$  is needed for these nonconstant examples; when  $\Gamma = 0$ ,  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}$  and every point is constant.  $\square$

| Completion and boundary (over $\bar{\mathbb{k}}$ ) | Mechanism [C14]                                                           |
|----------------------------------------------------|---------------------------------------------------------------------------|
| Positive genus, any boundary                       | Global regular differentials force constancy.                             |
| Genus zero, at least two missing points            | An invertible coordinate forces constancy.                                |
| Genus zero, one missing point                      | The affine line takes every value in $\mathcal{A}_{\mathbb{k}}(\Gamma)$ . |

The criterion is intrinsic to the affine curve, not to a plane equation: the parabola  $Y = X^2$  is an affine line and has the points  $(u, u^2)$ , whereas the hyperbola  $XY = 1$  is  $\mathbb{P}^1$  minus two points. Its  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -points are precisely  $(c, c^{-1})$  for  $c \in \mathbb{k}^\times$ ; imposing real omnific integer coordinates leaves only  $(1, 1)$  and  $(-1, -1)$  [C10, C11, C12]. Boundedness is not the mechanism:  $x^2 + y^2 = 1$  is rigid already by [theorem 9.2](#), and over  $\mathcal{B}_{\mathbb{C}}$  by  $(x + iy)(x - iy) = 1$ , but  $y^2 = x^3 + x$  is unbounded in its real locus and is rigid by [proposition 16.8](#) or by its global differential [C14]. The classification uses no finite generation, rank or order-type hypothesis on supports [C10, C11, C12, C13, C14, C15].

The rigid direction has a second proof, by logarithmic differentials, due to source C12. We print it over an algebraically closed field, where its local computation takes place; [theorem 17.6](#) then descends as above.

**Lemma 17.7** (Logarithmic forms [C12, C15]). *Let  $\mathbb{k}$  be algebraically closed,  $\bar{C}$  a smooth projective integral curve of genus  $g$  and  $\Sigma = P_1 + \cdots + P_r$  a nonempty reduced divisor. Then  $h^0(\bar{C}, \Omega^1(\Sigma)) = g + r - 1$ , which vanishes only for  $g = 0$ ,  $r = 1$ , that is, for  $\bar{C} \setminus \Sigma \cong \mathbb{A}_{\mathbb{k}}^1$ . If  $u$  is a local parameter at a closed point  $Q$ , a rational differential with at most a simple pole at  $Q$  has the form  $c du/u$  with  $c \in \mathcal{O}_{\bar{C}, Q}$ .*

*Proof.* Riemann–Roch for  $K_{\bar{C}} + \Sigma$ , where  $K_{\bar{C}}$  is a canonical divisor of degree  $2g - 2$ , gives  $h^0(\Omega^1(\Sigma)) - h^0(\mathcal{O}(-\Sigma)) = g + r - 1$ , and  $h^0(\mathcal{O}(-\Sigma)) = 0$  because a regular function on the proper integral curve is constant and vanishes at a point of  $\Sigma$  only if it is zero. A curve of genus zero is  $\mathbb{P}_{\mathbb{k}}^1$ . The local form holds because  $\mathcal{O}_{\bar{C}, Q}$  is a discrete valuation ring and  $du$  generates its rank-one module of differentials: its class spans the cotangent fiber at the smooth  $\mathbb{k}$ -point  $Q$ , so Nakayama’s lemma applies. Writing  $\eta = f du$ , the simple-pole bound says  $uf \in \mathcal{O}_{\bar{C}, Q}$ , giving  $c = uf$ .  $\square$

**Theorem 17.8** (The logarithmic route [C12; also C15], closed centre [merge]). *Let  $\mathbb{k}$  be algebraically closed and  $C$  a smooth integral affine curve over  $\mathbb{k}$  with  $C \not\cong \mathbb{A}_{\mathbb{k}}^1$ . Then  $C(\mathcal{A}_{\mathbb{k}}(\Gamma)) = C(\mathbb{k})$ .*

*Proof.* Let  $p : \mathbb{k}[C] \rightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$  be a nonconstant point. Its kernel is a prime; a nonzero prime of the one-dimensional domain  $\mathbb{k}[C]$  is maximal with residue field  $\mathbb{k}$ , which would make  $p$  constant. So  $p$  is injective and extends to an embedding  $F = \mathbb{k}(C) \hookrightarrow \mathbb{H}_{\mathbb{k}}(\Gamma)$ . By [lemma 17.7](#) there is a nonzero  $\eta \in H^0(\bar{C}, \Omega^1(\Sigma))$ . Its restriction to the affine curve  $C$  is regular, a finite

sum  $\sum_j a_j db_j$  with  $a_j, b_j \in \mathbb{k}[C]$ , so for every Euler derivation  $\partial$

$$\eta(\partial) = \sum_j p(a_j) \partial(p(b_j)) \in \Pi_{\mathbb{k}}(\Gamma)$$

by [lemma 16.2](#). On the other hand the  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ -point of  $\overline{C}$  extends, by properness, to a  $\mathcal{O}_{\mathbb{k}}(\Gamma)$ -point with centre  $Q$ . The centre is a closed point: if it were the generic point, then  $F \subseteq \mathcal{O}_{\mathbb{k}}(\Gamma)$  and the residue map would give an injective  $\mathbb{k}$ -algebra map  $F \rightarrow \mathbb{k}$ , impossible since  $\text{trdeg}_{\mathbb{k}} F = 1$  [merge]. In fact  $Q$  lies in the boundary: if  $Q \in C$ , every element of  $\mathbb{k}[C]$  would map into  $\mathcal{O}_{\mathbb{k}}(\Gamma) \cap \mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}$ , contradicting nonconstancy. The map  $\mathcal{O}_{\overline{C}, Q} \rightarrow \mathcal{O}_{\mathbb{k}}(\Gamma)$  is local, so it sends a local parameter  $u$  to  $\mathfrak{m}_{\mathbb{k}}(\Gamma)$ ; its image is nonzero because the map is the restriction of the embedding  $F \hookrightarrow \mathbb{H}_{\mathbb{k}}(\Gamma)$ . Since  $\eta$  has at most a simple pole at  $Q$ , [lemma 17.7](#) and [lemma 16.2](#) give  $\eta(\partial) = c \partial u / u \in \mathcal{O}_{\mathbb{k}}(\Gamma)$ . As  $\Pi_{\mathbb{k}}(\Gamma) \cap \mathcal{O}_{\mathbb{k}}(\Gamma) = 0$ ,  $\eta(\partial) = 0$ . Here the logarithmic derivative need only lie in  $\mathcal{O}_{\mathbb{k}}(\Gamma)$ , not in its maximal ideal. Both computations evaluate the same rational form in  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  under the same embedding of  $F$ . Now  $\Omega_{F/\mathbb{k}}^1$  is one-dimensional over  $F$  and spanned by  $\eta$ , so  $df = h_f \eta$  gives  $\partial f = h_f \eta(\partial) = 0$  for all  $f \in F$  and all  $\partial$ , and  $F \subseteq \mathbb{k}$  by [lemma 16.3](#), contradicting  $\text{trdeg}_{\mathbb{k}} F = 1$ .  $\square$

Source C12 writes “let  $Q$  be its centre” without showing that  $Q$  is closed; the step marked [merge] supplies it. Source C15’s proof of the same statement avoids the step, because its local computation ([lemma 18.15](#)) works wherever the centre lies ([section 18.6](#)). This route needs the support inclusion  $\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) \subseteq \Pi_{\mathbb{k}}(\Gamma)$ : with  $\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$  alone one would only get  $\eta(\partial) \in \mathcal{A}_{\mathbb{k}}(\Gamma) \cap \mathcal{O}_{\mathbb{k}}(\Gamma) = \mathbb{k}$ . It also avoids the unit lemma:  $\mathbb{G}_m$  has the logarithmic form  $dx/x$ , a genus-one affine curve has regular forms even with one point removed, and the affine line has no nonzero logarithmic forms: every nonzero regular form  $P(x)dx$  has a pole of order  $\deg P + 2 \geq 2$  at infinity. These are three values of  $g + r - 1$  [C12].

**Corollary 17.9** (Curve subalgebras [C11, C12]). *Let  $\mathbb{k}$  be algebraically closed. A finitely generated  $\mathbb{k}$ -subalgebra  $A_0 \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$  that is a smooth (equivalently, normal) domain of transcendence degree one is  $\mathbb{k}[u]$  for some  $u \in \mathcal{A}_{\mathbb{k}}(\Gamma) \setminus \mathbb{k}$ .*

*Proof.* A one-dimensional Noetherian domain is normal exactly when it is regular; over the perfect field  $\mathbb{k}$  its curve is then smooth [40]. The inclusion is a nonconstant point of  $\text{Spec } A_0$ . Transcendence degree one excludes  $\Gamma = 0$ , which would give  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}$ . Thus  $\text{Spec } A_0 \cong \mathbb{A}_{\mathbb{k}}^1$  by [theorem 17.6](#), and  $u$  is the image of the coordinate.  $\square$

Smoothness of the algebra cannot be dropped: for  $\gamma > 0$ , put  $t = \omega^\gamma$ . The cusp algebra  $\mathbb{k}[t^2, t^3] \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$  is not normal:  $t = t^3/t^2$  lies in its fraction field and is integral over it, but it is not in the algebra, whose polynomial exponents belong to  $\{0, 2, 3, 4, \dots\}$ . The corollary does not say that every finitely generated subalgebra of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is a polynomial ring; higher transcendence degree is outside it [C11].

**Corollary 17.10** (All smooth curves [C13; also C15], for every  $\mathbb{k}$ ). *Let  $\Gamma \neq 0$  and let  $C$  be a smooth geometrically integral separated curve of finite type over  $\mathbb{k}$ . Then  $C$  has a nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point if and only if  $C \cong \mathbb{A}_{\mathbb{k}}^1$  or  $C \cong \mathbb{P}_{\mathbb{k}}^1$ .*

*Proof.* A separated integral finite-type curve that is not proper is affine [40], and [theorem 17.6](#) applies. A proper curve of positive genus is rigid by [theorem 17.2](#). A proper curve of genus zero with an  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point has the  $\mathbb{k}$ -point obtained by ct, hence is  $\mathbb{P}_{\mathbb{k}}^1$  [42]. Conversely, choose  $\gamma > 0$ ;  $\omega^\gamma$  on  $\mathbb{A}^1$  and  $[\omega^\gamma : 1]$  on  $\mathbb{P}^1$  are nonconstant points.  $\square$

Source C13 proves the corollary for  $\mathbb{k} = \mathbb{R}$  and  $\mathbb{C}$ ; its proof uses only the curve classification and the rational point supplied by ct, so the statement holds over every field of characteristic zero. It is the only one of sources C10–C14 to include the projective line; source C15 includes it too, over an algebraically closed field, and descends the rigid direction to every field of characteristic zero.

### 17.3 Arithmetic models and exact fibers

The geometric classification controls points in  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k} \oplus \Pi_{\mathbb{k}}(\Gamma)$ . For arithmetic points we use its subring  $R = \mathfrak{o} \oplus \Pi_{\mathbb{k}}(\Gamma)$ , where  $\mathfrak{o} \subseteq \mathbb{k}$ . The distinction matters: coefficients of a parametrization may lie anywhere in  $\mathbb{k}$ , while the constant coordinates of an  $R$ -point must lie in  $\mathfrak{o}$ . We use closed affine presentations, so a point is a tuple satisfying the defining polynomial equations.

**Theorem 17.11** (Exact fibers over an arbitrary coefficient ring [C11], for every  $\mathbb{k}$ ). *Let  $\mathfrak{o} \subseteq \mathbb{k}$  be a subring,  $\mathcal{U} \subseteq \mathbb{A}_{\mathfrak{o}}^n$  a closed subscheme such that  $U = \mathcal{U}_{\mathbb{k}}$  is a smooth geometrically integral curve, and  $R = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ .*

(i) *If  $U \not\cong \mathbb{A}_{\mathbb{k}}^1$ , then  $\mathcal{U}(R) = \mathcal{U}(\mathfrak{o})$ .*

(ii) *If  $\phi : \mathbb{A}_{\mathbb{k}}^1 \rightarrow U$  is an isomorphism, then for every  $\mathbf{n} \in \mathcal{U}(\mathfrak{o})$ , with  $c_{\mathbf{n}} = \phi^{-1}(\mathbf{n}) \in \mathbb{k}$ ,*

$$\Pi_{\mathbb{k}}(\Gamma) \longrightarrow \{\mathbf{x} \in \mathcal{U}(R) : \text{ct}(\mathbf{x}) = \mathbf{n}\}, \quad s \longmapsto \phi(c_{\mathbf{n}} + s),$$

*is a bijection, and  $\phi(c_{\mathbf{n}} + s)$  is constant only for  $s = 0$ .*

*Proof.* (i) When  $\Gamma = 0$ ,  $R = \mathfrak{o}$  and the assertion is immediate. Otherwise a point of  $\mathcal{U}(R)$  becomes a point of  $U(\mathcal{A}_{\mathbb{k}}(\Gamma))$ , constant by [theorem 17.6](#); its coordinates lie in  $R \cap \mathbb{k} = \mathfrak{o}$ . They still satisfy the defining equations, giving the claimed ordinary point. The reverse inclusion is induced by  $\mathfrak{o} \hookrightarrow R$ .

(ii) Write  $\mathcal{U} = \text{Spec}(\mathfrak{o}[X_1, \dots, X_n]/J)$  and  $\phi = (\phi_1, \dots, \phi_n)$  with  $\phi_j \in \mathbb{k}[T]$ . Each  $\phi_j(c_{\mathbf{n}} + s) - \phi_j(c_{\mathbf{n}})$  is a polynomial in  $s$  without constant term, hence lies in  $\Pi_{\mathbb{k}}(\Gamma)$ , and  $\phi_j(c_{\mathbf{n}}) = n_j \in \mathfrak{o}$ ; so the point lies in  $R^n$  and has constant term  $\mathbf{n}$ . For every  $F \in J$ ,  $F(\phi(T)) = 0$  in  $\mathbb{k}[T]$ . After substitution it vanishes in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , hence in  $R$ , since  $R \hookrightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$  is injective. This proves that the tuple is a point of the original model, even if that model is not flat over  $\mathfrak{o}$ . The inverse of  $\phi$  is a regular function on  $U$ , represented by a polynomial  $q$  with coefficients in  $\mathbb{k}$ , and evaluation commutes with ct. For  $\mathbf{x}$  in the fiber,  $t = q(\mathbf{x}) \in \mathcal{A}_{\mathbb{k}}(\Gamma)$  has  $\text{ct}(t) = q(\mathbf{n}) = c_{\mathbf{n}}$ , so  $t = c_{\mathbf{n}} + s$  with a unique  $s \in \Pi_{\mathbb{k}}(\Gamma)$ , and  $\mathbf{x} = \phi(t)$ . If  $\phi(c_{\mathbf{n}} + s)$  is constant, it equals its constant term  $\mathbf{n} = \phi(c_{\mathbf{n}})$ , and injectivity gives  $s = 0$  [C10].  $\square$

The parametrization need not be defined over  $\mathfrak{o}$ : every nonconstant term lies in  $\Pi_{\mathbb{k}}(\Gamma)$ , and only the constant term must lie in  $\mathfrak{o}$  [C11, C13]. Source C11 states the theorem for  $\mathbb{k}$  algebraically closed or  $\mathbb{k} = \mathbb{R}$ ; it holds for every  $\mathbb{k}$  of characteristic zero by [theorem 17.6](#). Only the geometric fiber over  $\mathbb{k}$  is assumed smooth: the model may have bad reduction at primes of  $\mathfrak{o}$  [C11, C13, C14]. No integer-part assertion is made for an arbitrary  $\mathfrak{o}$  or  $\Gamma$  [C11].

For example the model  $2Y = 0$  in  $\mathbb{A}_{\mathbb{Z}}^2$  has generic fiber an affine line and fiber  $\mathbb{A}_{\mathbb{F}_2}^2$  at the prime 2. Its coordinate ring has nonzero  $\mathbb{Z}$ -torsion, so it is not flat over  $\mathbb{Z}$ . Nevertheless its omnific points are exactly  $(x, 0)$ ,  $x \in \mathbf{O}_{\mathbb{Z}}$ , and the fiber over  $(n, 0)$  is  $\{(n + s, 0) : s \in \Pi\}$ . The theorem uses the generic geometry and the injective coefficient maps, not smoothness or flatness of the integral model.

**Corollary 17.12** (Omnific points on smooth affine curves [C10, C11, C13, C14]). *Let  $\mathcal{C} \subseteq \mathbb{A}_{\mathbb{Z}}^n$  be a closed subscheme whose generic fiber  $C = \mathcal{C}_{\mathbb{Q}}$  is a smooth geometrically integral curve (equivalently, whose real or complex fiber is one).*

- (a) *If  $C_{\mathbb{C}} \not\cong \mathbb{A}_{\mathbb{C}}^1$  (equivalently  $C \not\cong \mathbb{A}_{\mathbb{Q}}^1$ , or  $C_{\mathbb{R}} \not\cong \mathbb{A}_{\mathbb{R}}^1$ ), then  $\mathcal{C}(\mathbf{Oz}) = \mathcal{C}(\mathbb{Z})$  and  $\mathcal{C}(\mathbf{Oz}[i]) = \mathcal{C}(\mathbb{Z}[i])$ .*
- (b) *If  $C \cong \mathbb{A}_{\mathbb{Q}}^1$ , then  $\mathcal{C}(\mathbf{Oz}) \neq \mathcal{C}(\mathbb{Z})$  if and only if  $\mathcal{C}(\mathbb{Z}) \neq \emptyset$ . For an isomorphism  $\phi : \mathbb{A}_{\mathbb{R}}^1 \rightarrow C_{\mathbb{R}}$  and  $\mathbf{n} \in \mathcal{C}(\mathbb{Z})$ , the fiber of  $\text{ct}$  over  $\mathbf{n}$  is exactly  $\{\phi(\phi^{-1}(\mathbf{n}) + s) : s \in \Pi\}$ , parametrized bijectively; the Gaussian fiber is parametrized by  $\Pi_{\mathbb{C}}$ .*
- (c) *Accordingly, either  $\mathcal{C}(\mathbf{Oz})$  is empty, or it equals the set  $\mathcal{C}(\mathbb{Z})$ , or every nonempty fiber of  $\text{ct}$  on it is a proper class.*

*The same holds for a model over  $\mathbb{Z}[i]$  and its points in  $\mathbf{Oz}[i]$ , with the complex fiber in place of  $C$ .*

*Proof.* Smoothness and geometric integrality are invariant under extension of fields of characteristic zero, in both directions; the affine-line condition descends across arbitrary extensions by [theorem 17.6](#). A finite tuple of omnific integers lies in  $\mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \Gamma)$  for a set-sized  $\Gamma$  ([lemma 2.6](#)); apply [theorem 17.11](#) with  $(\mathfrak{o}, \mathbb{k}) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ . The formula of [theorem 17.11\(ii\)](#) is finite polynomial evaluation, so it makes sense for every  $s \in \Pi$ ; every given point lies in a workspace where its parameter exists uniquely. Since  $\text{ct}$  is a retraction, an omnific point has an ordinary constant term, so  $\mathcal{C}(\mathbb{Z}) = \emptyset$  forces  $\mathcal{C}(\mathbf{Oz}) = \emptyset$ . A nonempty fiber in case (b) contains the distinct points for  $s = \omega^{\alpha}$ ,  $\alpha$  ranging over the positive ordinals, so it is a proper class [C10, C11, C13]. This argument uses an injective class-indexed family, not a scheme over a proper-class ring. For Gaussian fibers choose a complex affine-line isomorphism and replace  $\Pi$  by  $\Pi_{\mathbb{C}}$ ; the same real monomials already witness proper-class size.  $\square$

Source C14 states (a) and (b) with the generic fiber over  $\mathbb{Q}$ , source C10 with the complex fiber, and sources C11 and C13 with the real fiber; the conditions agree by [lemma 17.5](#). In the rigid case the retraction  $\mathcal{C}(\mathbf{Oz}) \rightarrow \mathcal{C}(\mathbb{Z})$  is a bijection, not merely a surjection, and in the affine-line case it is surjective onto the integer points with explicitly parametrized fibers. The equality  $\mathcal{C}(\mathbf{Oz}) = \mathcal{C}(\mathbb{Z})$  is not an algorithm for the integer points and does not assert that there are finitely many [C10, C11, C13, C14]. The rigidity and fiber formulas also hold in one fixed workspace, with  $\Pi$  replaced by  $\Pi_{\mathbb{R}}(\Gamma)$  [C10, C13]. That fiber is a set: the proper-class conclusion uses all surreal supports. A nonzero fixed  $\Gamma$  still gives infinitely many distinct points, using  $s = \omega^{m\gamma}$  for  $m \geq 1$  and a fixed  $\gamma > 0$ ; for  $\Gamma = 0$  each fiber is a singleton.

**Example 17.13** (Complete fibers [C10, C11, C13]). On  $Y = X^2$  the fiber over  $(n, n^2)$  is  $\{(n + s, n^2 + 2ns + s^2) : s \in \Pi\}$ . On  $Y = X^2 - X + 3$  with  $\phi(T) = (2 + T, 5 + 3T + T^2)$ , the fiber over  $(2, 5)$  is  $\{(2 + u, 5 + 3u + u^2) : u \in \Pi\}$ ;  $u = \omega$  gives a point of finite support and  $u = \sum_{n \geq 1} \omega^{1/n}$  one of infinite support, both by the same formula.

**Lemma 17.14** (Integer polynomial arcs [C14]). *Let  $\mathcal{C} \subseteq \mathbb{A}_{\mathbb{Z}}^n$  be a closed subscheme with generic fiber isomorphic to  $\mathbb{A}_{\mathbb{Q}}^1$ . Through every  $\mathbf{n} \in \mathcal{C}(\mathbb{Z})$  there are polynomials  $q_i(S) \in \mathbb{Z}[S]$  with  $q_i(0) = n_i$  such that  $\mathbf{q} : R \rightarrow \mathcal{C}(R)$  is injective for every domain  $R \supseteq \mathbb{Z}$  of characteristic zero. Over  $\mathbf{Oz}$  every nonordinary parameter has a nonordinary image, and parameters in  $\Pi$  stay in the fiber of  $\text{ct}$  over  $\mathbf{n}$ .*

*Proof.* Write an isomorphism  $\mathbb{A}_{\mathbb{Q}}^1 \rightarrow \mathcal{C}_{\mathbb{Q}}$  as  $\mathbf{p}(T)$  with  $p_i \in \mathbb{Q}[T]$ , and its inverse as a polynomial  $u$  with rational coefficients, so that  $u(\mathbf{p}(T)) = T$ . Put  $t_0 = u(\mathbf{n}) \in \mathbb{Q}$ , expand  $p_i(t_0 + T) = \sum_j c_{ij}T^j$  with  $c_{i0} = n_i$ , and choose an ordinary  $N > 0$  clearing the denominators of all  $c_{ij}$

with  $j \geq 1$ . The coefficient of  $S^j$  is  $N^j c_{ij} = N^{j-1}(Nc_{ij}) \in \mathbb{Z}$ , while the constant coefficient is already  $n_i$ . Thus  $q_i(S) = p_i(t_0 + NS) \in \mathbb{Z}[S]$  and  $q_i(0) = n_i$ . The defining polynomials of  $\mathcal{C}$  vanish on  $\mathbf{q}(S)$  over  $\mathbb{Q}$ , hence in  $\mathbb{Z}[S]$ . If  $\mathbf{q}(s) = \mathbf{q}(s')$  in  $R$ , apply  $u$  in  $\text{Frac } R$  to get  $t_0 + Ns = t_0 + Ns'$ . Since  $N \neq 0$  in this characteristic-zero field,  $s = s'$ . Some  $q_i$  is nonconstant, since  $u(\mathbf{q}(S)) = t_0 + NS$ , and by [lemma 16.1\(d\)](#) its value at a nonordinary omnific parameter is nonordinary. For  $s \in \Pi$ ,  $\text{ct}(\mathbf{q}(s)) = \mathbf{q}(0) = \mathbf{n}$ .  $\square$

The nonemptiness condition is needed: the model  $2X = 1$  with a free coordinate  $Y$  has generic fiber  $\mathbb{A}_{\mathbb{Q}}^1$  but neither integer nor omnific points [C14].

The injection actually works for any  $\mathbb{Z}$ -torsion-free algebra  $R$ : if  $D > 0$  clears the coefficients of  $u$  and  $t_0$ , the inverse identity becomes the identity  $(Du)(\mathbf{q}(S)) = Dt_0 + DNS$  in  $\mathbb{Z}[S]$ . Equality of two images then gives  $DN(s - s') = 0$ , and torsion-freeness gives  $s = s'$ . Characteristic zero alone would not suffice for this argument. For the affine line and the allowed choice  $q(S) = 2S$ , the characteristic-zero ring  $\mathbb{Z} \times \mathbb{F}_2$  has distinct elements  $(0, 0)$  and  $(0, 1)$  with the same image. The original domain hypothesis excludes such torsion.

**Proposition 17.15** (A congruence criterion [C14]). *Keep  $\mathbf{p}$  and  $u$  from [lemma 17.14](#). Let  $N_1 > 0$  clear the denominators of the coefficients of  $u$ , and  $N_2 > 0$  satisfy  $P_i(V) = N_2 p_i(V/N_1) \in \mathbb{Z}[V]$  for all  $i$ . Then  $n \mapsto \mathbf{p}(n/N_1)$  is a bijection from*

$$\{n \in \mathbb{Z} : P_i(n) \equiv 0 \pmod{N_2} \text{ for all } i\}$$

*onto  $\mathcal{C}(\mathbb{Z})$ . This set is a union of residue classes modulo  $N_2$ , and whether it is empty is decided by checking  $0 \leq n < N_2$ .*

*Proof.* For  $\mathbf{x} \in \mathcal{C}(\mathbb{Z})$ ,  $u(\mathbf{x}) \in N_1^{-1}\mathbb{Z}$ ; write it  $n/N_1$ . Then  $x_i = p_i(n/N_1) = P_i(n)/N_2$  are integers, so  $n$  satisfies the congruences. Conversely the congruences make the coordinates integral, and the defining polynomials vanish identically on  $\mathbf{p}$ . The inverse identity gives the bijection, with inverse  $\mathbf{x} \mapsto N_1 u(\mathbf{x})$ . Finally,  $P_i(n + N_2 m) - P_i(n)$  is divisible by  $N_2$  for every integer  $m$ , so the congruences depend only on the residue class of  $n$ . The finite test includes  $N_2 = 1$ , when its sole residue is 0 and all parameters are admissible.  $\square$

This is an algorithm only after an affine-line isomorphism and its inverse are supplied; it is not an algorithm for integer points of arbitrary curves [C14]. For  $3Y = X^2 - X$  take  $p_1 = T$ ,  $p_2 = (T^2 - T)/3$  and  $u = X$ : then  $N_1 = 1$ ,  $N_2 = 3$ , the admissible residues are 0 and 1 modulo 3, and the arcs through  $(0, 0)$  and  $(1, 0)$  are  $(3S, 3S^2 - S)$  and  $(1 + 3S, 3S^2 + S)$ , valid for every omnific parameter. In the letters above,  $N$ ,  $N_1$ ,  $N_2$  are the sources'  $M$ ,  $D$ ,  $L$  ([table 1](#)).

Thus, once  $\mathbf{p}$  and  $u$  are supplied, the ordinary point set is either empty or countably infinite: every admissible residue class contains infinitely many integers, and the displayed parametrization is injective. This statement concerns the affine-line branch only. Moreover the constructed integer arc  $\mathbf{q}$  parametrizes the entire omnific fiber over  $\mathbf{n}$  when its parameter ranges over  $\Pi$ : multiplication by the nonzero ordinary integer  $N$  is a bijection of  $\Pi$ . In the exact fiber formula,  $\mathbf{p}(t_0 + s) = \mathbf{q}(s/N)$ , and  $s/N \in \Pi$ . Thus denominator clearing retains the full fiber, including its infinite-support points.

**Corollary 17.16** (Polynomial witnesses [C13]; also [C12]). *For  $\mathcal{C}$  as in [corollary 17.12](#), with real fiber  $C_{\mathbb{R}}$ , the following are equivalent:*

- (i)  $\mathcal{C}$  has a nonordinary omnific point;
- (ii)  $C_{\mathbb{R}} \cong \mathbb{A}_{\mathbb{R}}^1$  and  $\mathcal{C}(\mathbb{Z}) \neq \emptyset$ ;
- (iii)  $C_{\mathbb{R}}$  has a nonconstant real polynomial arc through an integer point;

(iv)  $\mathcal{C}$  has a nonordinary omnific point all of whose coordinates have finite support.

*Every nonordinary omnific point has the same constant term as some nonordinary point of finite support.*

*Proof.* (i) and (ii) are equivalent by [corollary 17.12](#). Given (ii) and  $\mathbf{n} = \phi(t_0)$ , the arc  $T \mapsto \phi(t_0 + T)$  gives (iii). Given (iii), translate the real parameter so that the integer point occurs at  $T = 0$ , then substitute  $T = \omega^\gamma$  with  $\gamma > 0$  ([proposition 14.1](#)); the coordinates have finite support and integer constants, and some coordinate has positive degree. Clearly (iv) implies (i). Given a nonordinary point, (i) gives the affine-line case; translating its isomorphism to the constant term of that point and making the same substitution proves the last assertion.  $\square$

For smooth affine curves this answers the existence part of [question 21.8](#): no nonordinary point exists only because of infinite supports ([section 21.4](#)). It is stronger than [theorem 14.7](#), which starts from a finite-support point, and it removes the finite-support hypothesis there for smooth curves [C12, C13]. It is not asserted for singular curves or in higher dimension, and it gives no algorithm for the integer points [C13]. Source C12 states the geometric part: over an algebraically closed field,  $C(\mathcal{A}_k(\Gamma)) \neq C(\mathbb{k})$  exactly when there is a nonconstant morphism  $\mathbb{A}^1 \rightarrow C$ , which is the last equivalence of [theorem 17.6](#); this classifies which curves acquire points, not the shape of an arbitrary infinite-support point. Status (batch 31): for singular curves source C16 proves the Gaussian form and the real form at integer points with a real preimage under the normalization ([corollary 17.43](#) and [proposition 17.42](#)); the remaining real case is [question 21.24](#).

In the setting of [corollary 17.16](#), the affine-line isomorphism also descends to  $\mathbb{Q}$ . Applying [lemma 17.14](#) at the same integer point therefore gives a stronger choice of witness: every coordinate can be taken in  $\mathbb{Z}[\omega^\gamma]$  for one  $\gamma > 0$ . The finite-support witness may have integer coefficients even when the original point has arbitrary real coefficients and infinite support. This does not put the original point on a polynomial arc with a finite-support parameter.

**Theorem 17.17** (Descent for separated models [C12]). *Let  $\mathcal{X}$  be a separated scheme of finite type over  $\mathbb{Z}$  and  $X = \mathcal{X}_{\mathbb{C}}$ . Suppose  $X(\mathcal{A}_{\mathbb{C}}(\Gamma)) = X(\mathbb{C})$  for every set-sized  $\Gamma \subseteq \mathbf{No}$ . Then inclusion and  $\text{ct}$  are inverse bijections  $\mathcal{X}(\mathbb{Z}) \cong \mathcal{X}(\mathbf{Oz}_{\Gamma})$  for every such  $\Gamma$ , and  $\mathcal{X}(\mathbf{Oz}) = \mathcal{X}(\mathbb{Z})$  in the class convention.*

*Proof.* Let  $p \in \mathcal{X}(\mathbf{Oz}_{\Gamma})$  and  $p_0 = \text{ct}(p) \in \mathcal{X}(\mathbb{Z})$ , and denote the constant extension of  $p_0$  to  $\mathbf{Oz}_{\Gamma}$  by  $\tilde{p}_0$ . The image of  $p$  in  $X(\mathcal{A}_{\mathbb{C}}(\Gamma))$  is constant, and since  $\mathbf{Oz}_{\Gamma} \rightarrow \mathbb{Z} \rightarrow \mathbb{C}$  and  $\mathbf{Oz}_{\Gamma} \rightarrow \mathcal{A}_{\mathbb{C}}(\Gamma) \rightarrow \mathbb{C}$  are compatible with  $\text{ct}$ , it is the image of  $p_0$ . So  $p$  and  $\tilde{p}_0$  agree over the field  $\mathbb{H}_{\mathbb{C}}(\Gamma)$ , in which the domain  $\mathbf{Oz}_{\Gamma}$  embeds, hence over  $\mathbf{Oz}_{\Gamma}$  by [lemma 16.13](#). Explicitly, their equalizer is a closed subscheme of  $\text{Spec } \mathbf{Oz}_{\Gamma}$  whose ideal maps to zero in that field; injectivity makes the ideal zero. Specialization of a constant point returns  $p_0$ , proving both inverse identities. In the class convention a point is represented in a set-sized workspace, so the same equality applies to each representative and remains true after enlarging the workspace.  $\square$

No smoothness of  $\mathcal{X}$  at any prime and no flatness of  $\mathbf{Oz}_{\Gamma} \rightarrow \mathcal{A}_{\mathbb{C}}(\Gamma)$  is used; smoothness enters only through the rigidity of the complex fiber, and separatedness recovers the integral point from the ambient field [C12]. The hypothesis holds when  $X$  is a smooth affine curve other than  $\mathbb{A}_{\mathbb{C}}^1$ , a smooth projective curve of positive genus, an abelian or semiabelian variety, a smooth proper variety with globally generated cotangent sheaf, or a separated scheme with a quasi-finite map to one of these ([theorems 17.2, 17.6, 18.2, 18.4 and 18.5](#) and [corollary 16.17](#)),

so all of these have only ordinary omnific points on every separated integral model [C12]. This is a reduction theorem, not an algorithm for the ordinary points, and it gives no height bound [C12].

## 17.4 Projective coordinates

A nonzero homogeneous tuple over a field determines a projective point. Over a domain  $S$ , a tuple  $\mathbf{a}$  generates the coordinate ideal  $J = \sum_i S a_i$ . If  $J$  is invertible, the surjection  $S^{N+1} \rightarrow J$  defines a projective  $S$ -point, as proved below. Here invertibility means that  $JJ' = S$  for a fractional ideal  $J' \subseteq \text{Frac } S$ ; it does not mean  $J = S$ . Every nonzero principal ideal is invertible, including a proper one. The special case  $J = S$  is a unimodular tuple [C10, C11, C12, C13, C14]. By [corollary 4.7](#), every point of  $\mathbb{P}^n(\mathbf{No})$  has coordinates in  $\Pi^{n+1}$ , and such coordinates generate a proper ideal; sources C10 and C12 reprove this clearing, which may fail when the clearing exponent is confined to a fixed workspace [C10]. So raw homogeneous solutions can be abundant while unimodular ones are constant; there is no conflict.

**Theorem 17.18** (Unimodular coordinates [C14]; for  $\mathfrak{o} = \mathbb{k}$  [C10, C13]). *Let  $\mathfrak{o} \subseteq \mathbb{k}$  be a subring,  $R = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ , and  $X \subseteq \mathbb{P}_{\mathbb{k}}^N$  a closed subscheme with  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$ . If a unimodular tuple  $\mathbf{a} \in R^{N+1}$  satisfies the homogeneous equations of  $X$ , then  $\mathbf{a} = \text{ct}(\mathbf{a}) \in \mathfrak{o}^{N+1}$ .*

*Proof.* Choose  $b_i \in R$  with  $\sum_i b_i a_i = 1$  and put  $c_i = \text{ct}(a_i)$ ,  $d_i = \text{ct}(b_i)$ , so  $\sum_i d_i c_i = 1$ . In particular  $\mathbf{c}$  is nonzero and unimodular over  $\mathfrak{o}$ . The original unit-ideal identity remains valid in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , so the tuple defines a point of  $X(\mathcal{A}_{\mathbb{k}}(\Gamma))$ , which is constant; applying  $\text{ct}$  identifies it with  $[c_0 : \cdots : c_N]$ . So  $a_i c_j = a_j c_i$  in  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  for all  $i, j$ . Put  $s = \sum_j d_j a_j \in R$ . Then  $s c_i = \sum_j d_j a_j c_i = a_i \sum_j d_j c_j = a_i$ , and  $1 = \sum_i b_i a_i = s \sum_i b_i c_i$ , so  $s \in R^\times = \mathfrak{o}^\times$  ([lemma 16.1](#)). As  $\text{ct}(s) = \sum_j d_j c_j = 1$  and  $s$  is constant,  $s = 1$  and  $a_i = c_i$ . This last step upgrades equality of projective points to equality of the given tuples: the residual scalar is forced to be the constant unit 1.  $\square$

**Lemma 17.19** (Invertible coordinate ideals [C12]). *Let  $S$  be a domain contained in a field  $L$ ,  $X \subseteq \mathbb{P}_S^N$  a closed subscheme, and  $\mathbf{a} \in S^{N+1}$  a tuple with  $[\mathbf{a}] \in X(L)$ . If  $\langle \mathbf{a} \rangle$  is invertible, the tuple defines a morphism  $\text{Spec } S \rightarrow X$  with that  $L$ -point.*

*Proof.* Put  $J = \langle \mathbf{a} \rangle$ . The surjection  $S^{N+1} \rightarrow J$ , sending the  $i$ th basis vector to  $a_i$ , is an invertible quotient. It defines a morphism  $f : \text{Spec } S \rightarrow \mathbb{P}_S^N$  [44, Tag 01ND]. Under the natural identification  $J \otimes_S L \cong L$ , its generating sections become precisely the coordinates  $a_i$ , so  $f_L = [\mathbf{a}]$ . The inverse image of  $X$  is a closed subscheme of  $\text{Spec } S$ , say with ideal  $I$ . Since  $f_L$  factors through  $X$ , every element of  $I$  maps to zero in  $L$ . Injectivity of  $S \hookrightarrow L$  gives  $I = 0$ , proving that  $f$  factors through  $X$ .  $\square$

Only this sufficient direction is used; no claim is made that every projective point over such a ring has a global unimodular coordinate tuple [C12].

**Theorem 17.20** (Coordinate ideals on rigid targets [C12]). *Let  $X \subseteq \mathbb{P}_{\mathbb{k}}^N$  be projective with  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$ , and let a nonzero  $\mathbf{a} \in \mathcal{A}_{\mathbb{k}}(\Gamma)^{N+1}$  represent  $p \in X(\mathbb{H}_{\mathbb{k}}(\Gamma))$ . Then  $\langle \mathbf{a} \rangle$  is invertible if and only if it is principal, if and only if  $p \in X(\mathbb{k})$ .*

*Proof.* If  $\langle \mathbf{a} \rangle$  is invertible, [lemma 17.19](#) gives an  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point, which is constant. If  $p = [c_0 : \cdots : c_N]$  with  $c_i \in \mathbb{k}$ , then  $a_i = \mu c_i$  with  $\mu = a_j/c_j \in \mathcal{A}_{\mathbb{k}}(\Gamma)$  for some  $c_j \neq 0$ , and the constants  $c_i$  generate the unit ideal, so  $\langle \mathbf{a} \rangle = \mu \mathcal{A}_{\mathbb{k}}(\Gamma)$ . Here  $\mu \neq 0$  because  $\mathbf{a} \neq 0$ . Its inverse fractional ideal is  $\mu^{-1} \mathcal{A}_{\mathbb{k}}(\Gamma)$ , proving invertibility without requiring  $\mu$  to be a unit.  $\square$

**Theorem 17.21** (Omnific coordinate ideals [C12]). *Let  $\mathcal{X} \subseteq \mathbb{P}_{\mathbb{Z}}^N$  be closed with complex fiber  $X$  satisfying  $X(\mathcal{A}_{\mathbb{C}}(\Gamma)) = X(\mathbb{C})$  for every set-sized  $\Gamma$ . Let  $\mathbf{a} \in \mathbf{Oz}^{N+1}$  be nonzero with  $[\mathbf{a}] \in \mathcal{X}(\mathbf{No})$ . Then  $\langle \mathbf{a} \rangle$  is invertible if and only if it is principal, if and only if  $[\mathbf{a}] \in \mathcal{X}(\mathbb{Q})$ . If  $\langle \mathbf{a} \rangle = \mathbf{Oz}$ , every  $a_i$  is an ordinary integer.*

*Proof.* Write  $J = \sum_i \mathbf{Oz} a_i$ . Invertibility gives finitely many  $b_i \in \mathbf{No}$  such that

$$\sum_i a_i b_i = 1, \quad a_j b_i \in \mathbf{Oz} \quad \text{for all } i, j.$$

Indeed, expand a finite expression of 1 in  $JJ'$  in terms of the given generators  $a_i$ , absorbing its coefficients into  $b_i \in J'$ . Write each  $b_i$  as a fraction of omnific integers. Put their numerators and nonzero denominators, all  $a_i$ , and all products  $a_j b_i$  into one workspace. Then  $b_i \in \text{Frac}(\mathbf{Oz}_{\Gamma})$  and  $J'_{\Gamma} = \sum_i \mathbf{Oz}_{\Gamma} b_i$  is a fractional ideal there (the product of the chosen denominators clears it). The displayed identities give  $J_{\Gamma} J'_{\Gamma} = \mathbf{Oz}_{\Gamma}$ . This uses finitely many witnesses, not a Noetherian hypothesis or a spectrum of a proper-class ring. By lemma 17.19 the tuple is a point of  $\mathcal{X}(\mathbf{Oz}_{\Gamma})$ , which is ordinary by theorem 17.17; so  $[\mathbf{a}]$  is rational. If  $[\mathbf{a}]$  is rational, choose a primitive integer representative  $\mathbf{c}$  and integers  $r_i$  with  $\sum r_i c_i = 1$ ; then  $\mathbf{a} = \mu \mathbf{c}$  with  $\mu = \sum r_i a_i \in \mathbf{Oz}$  and  $\langle \mathbf{a} \rangle = \mu \mathbf{Oz}$  (as after theorem 14.5). A nonzero principal ideal is invertible. If  $\langle \mathbf{a} \rangle = \mathbf{Oz}$ , then  $\mu \in \mathbf{Oz}^{\times} = \{\pm 1\}$ , giving the ordinary tuple  $\mathbf{a} = \pm \mathbf{c}$ .  $\square$

**Corollary 17.22** (Rational projective points [C14; also C10, C12, C13, C15]). *Let integer homogeneous equations define  $\mathcal{X} \subseteq \mathbb{P}_{\mathbb{Z}}^N$  whose complex fiber is smooth and proper with globally generated cotangent sheaf, for example a smooth projective curve of positive genus or a projectively embedded abelian variety. Every unimodular omnific solution tuple is an ordinary integer tuple, and likewise with  $\mathbf{Oz}[i]$  and  $\mathbb{Z}[i]$ . For a smooth projective curve of positive genus over  $\mathbb{Q}$  in such an embedding, the points with a unimodular omnific representative are exactly the points of  $\mathcal{X}(\mathbb{Q})$ , their unimodular representatives are  $\pm$  the primitive integer representatives, and every other point of  $\mathcal{X}(\mathbf{No})$  represented in  $\mathbf{Oz}^{N+1}$  has a coordinate ideal that is neither invertible nor principal.*

*Proof.* Put the tuple and a unit-ideal witness into one workspace and apply corollary 16.17 and theorem 17.18 with  $(\mathfrak{o}, \mathbb{k}) = (\mathbb{Z}, \mathbb{C})$  or  $(\mathbb{Z}[i], \mathbb{C})$ ; the constant tuple lies in  $\mathbb{Z}^{N+1}$  or  $\mathbb{Z}[i]^{N+1}$ . A rational point has a primitive integer representative with an ordinary Bézout witness, hence unimodular over  $\mathbf{Oz}$  [C14]. The last assertion is theorem 17.21 [C12].  $\square$

For the Gaussian analogue, a  $\mathbb{Q}(i)$ -point has a primitive  $\mathbb{Z}[i]$ -tuple because  $\mathbb{Z}[i]$  is a Euclidean domain. Its Bézout identity makes it unimodular over  $\mathbf{Oz}[i]$ . Conversely the first assertion makes any unimodular  $\mathbf{Oz}[i]$ -tuple Gaussian integral. Two primitive Gaussian representatives of the same point differ by one of  $1, -1, i, -i$ . Thus in the same rigid setting the Gaussian unimodular representatives are exactly the primitive ordinary Gaussian tuples, up to those units [C14].

An ordinary integer tuple is unimodular over  $\mathbf{Oz}$  exactly when its coordinates generate  $\mathbb{Z}$ : apply ct to an omnific witness, and use an ordinary Bézout identity for the converse [C13]. For the Fermat curve of exponent  $n \geq 3$ , theorem 17.18 with  $\mathfrak{o} = \mathbb{k} = \mathbb{C}$  gives a second route to theorem 7.9, without its nonvanishing hypothesis on the coordinates; excluding particular ordinary points still needs the theorem of Wiles and Taylor–Wiles (corollary 7.10) [C13, C14, C15]. This is a specialization, not a new Fermat result. The unimodular Pythagorean family of proposition 14.4 shows that positive genus cannot be dropped [C13]. The corollary uses actual unit-ideal witnesses; absence of a common nonunit divisor is a different condition in the non-Bézout ring  $\mathbf{Oz}$ , and question 21.2 is not answered [C10, C11, C12, C13, C14]. It is complementary to corollary 15.28: there, on rational curves, primitive coordinates exist exactly

at points of rational specialization; here, on rigid projective targets, only constant points have unimodular or invertible coordinates. The rationality conditions are different: the earlier result tests the constant-term specialization, whereas the rigid-target result tests the projective point itself. For example  $[\omega : 1] \in \mathbb{P}^1(\mathbf{No})$  has coordinate ideal  $\mathbf{Oz}$  and specialization  $[0 : 1] \in \mathbb{P}^1(\mathbb{Q})$ , but is not a rational point. Thus a principal coordinate ideal alone does not force rationality on a rational curve. For a constant projective direction, the two criteria agree by [theorem 14.5](#).

**Example 17.23** (A noninvertible coordinate ideal [C12]). Let  $\mathcal{E} : Y^2Z = X^3 - XZ^2 + Z^3$ , whose affine cubic has  $\Delta_0 = 23$ . It is squarefree, so the affine curve is smooth; its only point at infinity is  $[0 : 1 : 0]$ , where the derivative of the defining polynomial with respect to  $Z$  is 1. Thus the projective complex fiber is a smooth genus-one cubic. In  $\mathbf{No}$  put  $x = \omega^2$  and

$$y = \omega^3(1 - \omega^{-4} + \omega^{-6})^{1/2} = \omega^3 - \frac{1}{2}\omega^{-1} + \frac{1}{2}\omega^{-3} - \frac{1}{8}\omega^{-5} + \frac{1}{4}\omega^{-7} - \frac{3}{16}\omega^{-9} + \cdots,$$

so  $y^2 = x^3 - x + 1$  and  $y \notin \mathcal{B}_{\mathbb{R}}$ . With  $M = \omega^\omega$  the tuple  $(Mx, My, M)$  is purely infinite: the exponents of  $y$  are ordinary integers, and  $\omega + n > 0$  for every ordinary integer  $n$ . Multiplication by  $M$  shifts these exponents by  $\omega$ . The tuple represents the nonrational point  $[x : y : 1]$ . By [theorem 17.21](#) the ideal  $\langle Mx, My, M \rangle = \langle My, M \rangle$  is neither invertible nor principal. The monomial  $M$  is not a common divisor in  $\mathbf{Oz}$ , since  $My/M = y \notin \mathbf{Oz}$ , but  $\omega^{\omega/2}$  is: dividing each coordinate by it leaves only exponents  $\omega/2 + n > 0$ , with  $n$  ordinary. This divisor is a nonunit, so the tuple is not primitive and does not bear on the primitive question. The ideal equality uses  $Mx = xM$  and  $x = \omega^2 \in \mathbf{Oz}$ ; it does not assert that  $M$  divides  $My$ .

*Remark 17.24* (Field points outside the ring [C10; also C12, C13, C14, C15]). The theorems concern points over the support-restricted rings, not over  $\mathbf{No}$  or  $\mathbf{No}[i]$ . For real  $a, b$  with  $4a^3 + 27b^2 \neq 0$ , the point  $x = \omega$ ,  $y = \omega^{3/2}(1 + a\omega^{-2} + b\omega^{-3})^{1/2}$  of  $y^2 = x^3 + ax + b$  lies in  $\mathbf{No}$ ; its first correction term is  $\frac{a}{2}\omega^{-1/2}$  if  $a \neq 0$  and  $\frac{b}{2}\omega^{-3/2}$  otherwise, so  $y \notin \mathcal{B}_{\mathbb{R}}$  [C10]. Likewise  $x = \omega^2$ ,  $y = \omega^3(1 + \omega^{-4})^{1/2} = \omega^3 + \frac{1}{2}\omega^{-1} - \frac{1}{8}\omega^{-5} + \cdots$  on  $y^2 = x^3 + x$  [C14], the point of [example 17.23](#) [C12, C13], and, for real  $a, b$ , any positive infinite  $x$  with the positive square root of  $x^3 + ax + b$  in the real closed field  $\mathbf{No}$  [C15]. This last existence statement by itself does not locate a negative tail: for example  $a = b = 0$ ,  $x = \omega^2$  gives  $y = \omega^3$  in  $\mathbf{Oz}$ , on a singular cubic. For the explicit nonsingular examples above the displayed first negative coefficient proves failure of membership in  $\mathcal{B}_{\mathbb{R}}$ . The binomial expansions are support-controlled Hahn sums: powers of the infinitesimal inside the square root have reverse-well-ordered union of supports and finitely many contributions at each exponent. Truncating the tail changes the equation: replacing  $y$  by  $\omega^3$  in [example 17.23](#) turns  $\omega^6 - \omega^2 + 1$  into  $\omega^6$  [C12, C14]. Clearing such a point by [corollary 4.7](#) gives an omnific tuple with a proper coordinate ideal, not an affine omnific point [C10].

## 17.5 Singular curves and normalization

The classification of [theorem 17.6](#) is for smooth curves. The short Weierstrass cubics are settled completely ([corollary 16.9](#) and [proposition 16.11](#)) [C12]. The node and the cusp have normalizations isomorphic to  $\mathbb{A}^1$  but are not affine lines themselves, so smoothness cannot be replaced by integrality in [theorem 17.6](#) [C14]. In the other direction a singular curve that is a locally closed subscheme of an abelian variety ([theorem 18.2](#)) or has a quasi-finite map to a rigid target ([theorem 18.5](#)) is rigid; the embedding or map is a substantial extra hypothesis, and not every singular curve has one [C13, C14]. Sources C10–C15 leave reducible affine curves unclassified. A map from their coordinate ring to the domain  $\mathcal{A}_{\mathbb{K}}(\Gamma)$  kills the nilradical and has prime kernel, hence factors through an irreducible component; that component may be singular [C12]. Source C16 treats this case over algebraically closed fields in [theorem 17.36](#).

A normalization lift over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  cannot be assumed merely from integrality. For a nonconstant point of an integral constant curve the coordinate map is injective, as proved before [lemma 17.30](#), and identifies its function field with a subfield of  $\text{Frac } \mathcal{A}_{\mathbb{k}}(\Gamma)$ . The lift exists exactly when the image of the normalized coordinate ring lies in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ . Thus normality of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  in its own fraction field would suffice, by the universal property of normalization [[55](#), Lemma 29.55.5(4)]. Failure of normality removes that sufficient hypothesis; it does not by itself produce a constant curve point without a lift.

For the full class non-normality is [proposition 4.12](#), with the witness  $\sqrt{\omega^2 + 1}$  of sources 01, 02 and 05. Sources C10 and C13 give the same witness in a fixed workspace, and source C16 gives it again: for  $\Gamma = \mathbb{Q} + \mathbb{Q}\omega \subseteq \mathbf{No}$  (C13 allows any  $\mathbb{Q} + \mathbb{Q}H$  with  $H$  above every ordinary integer, and C16 any  $0 < e < H$  in  $\Gamma$  with  $ne < H$  for every ordinary  $n$ , with  $\omega^e$  in place of  $\omega$ ),  $q = \sqrt{\omega^2 + 1} = \omega + \frac{1}{2}\omega^{-1} - \frac{1}{8}\omega^{-3} + \dots$  is integral over  $\mathcal{A}_{\mathbb{R}}(\Gamma)$  and not in it, while  $\omega^\omega q \in \mathcal{A}_{\mathbb{R}}(\Gamma)$  because all its exponents  $\omega + 1 - 2n$  are positive for ordinary  $n \geq 0$ ; their decreasing sequence is a valid reverse-well-ordered support, so  $q \in \text{Frac } \mathcal{A}_{\mathbb{R}}(\Gamma) \setminus \mathcal{A}_{\mathbb{R}}(\Gamma)$ . Source C12 gives the same phenomenon for  $\Gamma = \mathbb{Z}^2$  ordered lexicographically, with  $g = (0, 1)$ ,  $h = (1, 0)$ ,  $z = \omega^g(1 + \omega^{-2g})^{1/2}$  and  $\omega^h z \in \mathcal{A}_{\mathbb{k}}(\Gamma)$ . Only these fixed-workspace forms are new.

The dominating scale matters: an integral element of the larger field  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  need not lie in  $\text{Frac } \mathcal{A}_{\mathbb{k}}(\Gamma)$  [C13]. For  $\Gamma = \mathbb{Z}$ ,  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}[\omega]$  is integrally closed, and  $z = \omega(1 + \omega^{-2})^{1/2}$  is integral over it but not in it; source C14 uses this only to show that integrality in the ambient field does not keep a lift in the ring, and explicitly does not claim non-normality. Source C15 gives the same element, as  $t^{-1}(1 + t^2)^{1/2}$  in the  $t$ -convention, and concludes that normality is unavailable even for  $\Gamma = \mathbb{Z}$ . That conclusion is not correct: for  $\Gamma = \mathbb{Z}$  the ring  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}[\omega]$  is a polynomial ring, hence normal, and the element, whose square  $\omega^2 + 1$  is not a square in  $\mathbb{k}(\omega)$ , lies outside  $\text{Frac } \mathcal{A}_{\mathbb{k}}(\Gamma)$ ; it shows only that  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is not integrally closed in the larger field  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ . It is not an obstruction to the normalization lift of a nonconstant curve point: for  $\Gamma = \mathbb{Z}$ , that lift does exist because  $\mathbb{k}[\omega]$  is normal in its own fraction field. Indeed  $\omega^2 + 1$  has simple roots over  $\bar{\mathbb{k}}$ , whereas the order of a zero or pole of a rational square is even; thus its square root is not in  $\mathbb{k}(\omega)$  ([appendix A](#)). Source C15's statement that  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is not assumed normal and that such a lift is not available as an unstated step is correct. Nor is a non-normality example by itself a point of a constant curve that fails to lift [C13].

For an arbitrary morphism  $f : X \rightarrow G$  to a rigid target, [proposition 16.21](#) gives only  $f(x) = f(\text{ct } x)$ ; applied to an Albanese map this confines nonconstant directions to its fibers, and positive-dimensional fibers need not be rigid [C10]. The remaining problem is [question 21.17](#). Status (batch 31): source C16 classifies the geometrically integral affine curves, singular or not, and the affine schemes of dimension at most one over an algebraically closed field ([theorems 17.25](#) and [17.36](#), [section 17.6](#)); what remains of the question is the lifting problem [question 21.23](#) and the real fibers [question 21.24](#).

## 17.6 Singular curves: normalization and conductor certificates

This subsection is built from a seventh manuscript on the subject of the part, *Singular Curves over Omnific Integers: Conductor differentials, a complete Hahn-rigidity criterion, and polynomial witnesses at every surreal scale*, tagged [C16] after its file prefix `16-singular-curves-` ([section 1.2](#)). It pins `bcac55a`. At that pin sources C10–C14 were placed, with their audits, but their material was not yet in this article, and source C15 was not yet placed; source C16 read the audits of sources C13 and C14, which leave singular curves outside the smooth classification, and saw none of the other curve manuscripts. It answers the first two clauses of [question 21.17](#): over every field of characteristic zero and for every  $\Gamma \neq 0$ , a geometrically integral affine curve, singular or not, has a nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point exactly when

its normalization is the affine line ([theorem 17.25](#)). The proof does not assume a normalization lift over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ : its existence cannot be inferred from integrality alone ([section 17.5](#)). A single conductor element of the curve clears every power of a contracted regular differential after a fixed finite multiplication (Seidenberg’s identity), and the discrete valuation at the one boundary point that a nonconstant point selects turns this into a contradiction after finitely many derivatives. Labels of this subsection carry the sub-prefix `odg:sg:`. The proof review in `RECONCILIATION.md` now covers the criterion, its conductor and boundary-place arguments, and the structural consequences through [corollary 17.37](#); the later applications remain outside that review. Throughout this subsection  $\Gamma \neq 0$  unless a statement explicitly treats the trivial case. If  $\Gamma = 0$ , then  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}$  and every point is constant; an existence criterion for nonconstant points must exclude this case.

Some statements of source C16 are already printed above and are not reprinted; source C16 is added to their credits. Its ring lemma (units, the retraction, transcendence of nonconstant elements, no nonzero element of positive valuation) is [lemma 16.1](#); its Euler derivations and their detection property are [lemmas 16.2](#) and [16.3](#), including the caution that one derivation need not detect every nonconstant series; its non-normality example is the fixed-workspace witness of [section 17.5](#); its descent from  $\bar{\mathbb{k}}$  to  $\mathbb{k}$  is the argument of [lemma 17.5](#); its smooth case is [theorem 17.6](#), for which its proof is a third route to the rigid direction ([remark 17.33](#)); and its elliptic Bézout certificate is ([16.5](#)) at  $(a, b) = (-1, 1)$ . Three short additions carry the tag `[merge]` and have complete proofs.

**Translation and letters.** Source C16 uses the  $t$ -convention  $t^\gamma = \omega^{-\gamma}$ . Its ring  $B_{k,\Gamma}$  of Hahn series with nonpositive  $t$ -exponents is  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  with exponents negated; it is not the valuation ring, and the italic  $B$  is a bilinear form here. Its field  $K = k((t^\Gamma))$ , valuation ring  $O$  and maximal ideal  $\mathfrak{m}$  are  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ ,  $\mathcal{O}_{\mathbb{k}}(\Gamma)$  and  $\mathfrak{m}_{\mathbb{k}}(\Gamma)$ . Its valuation  $v(f) = \min \text{supp}_t f$  is the valuation  $v = -\deg_\omega$  of [section 16.1](#), unchanged. Its purely infinite parts  $\mathcal{J}_{\mathbb{R}}$ ,  $\mathcal{J}_{\mathbb{C}}$  are  $\Pi$ ,  $\Pi_{\mathbb{C}}$ , and its  $\mathbf{Oz}[i]$  is the ring  $\mathbf{Oz}[i]$  of this report, unchanged. Its Euler derivation  $D_\lambda$  multiplies  $t^\gamma$  by  $\lambda(\gamma)$ ; it is  $\partial_{-\lambda}$  in the notation of ([16.1](#)), the same family. Its letter  $D$  also names a general derivation, written  $\partial$  here, and a coefficient subring, written  $\mathfrak{o}$  ( $D$  is the Pell parameter). The following letters are renamed because source C16 uses each twice or because the part reserves them: its coordinate ring  $A$  and normalization  $\tilde{A}$  are  $\mathbb{k}[C]$  and  $\mathbb{k}[\tilde{C}]$  ( $A$  is an abelian variety in this part); its smooth projective model  $X$  is  $\bar{C}$ , the smooth projective model of the function field  $F = \text{Frac } \mathbb{k}[C]$ , which completes  $\tilde{C}$ ; its boundary is  $\Sigma = \bar{C} \setminus \tilde{C}$  and its centre  $p$  is  $Q$ , as in [theorem 17.8](#); its conductor element  $d$  is  $\kappa$  ( $d$  is a degree and  $d$  the differential); its length  $m$  of a differential presentation is  $\ell$  ( $m$  is the superelliptic exponent); its contraction  $s = \alpha(D)$  is  $\alpha(\partial)$ , as  $\eta(\partial)$  in [theorem 17.8](#); its formal variable  $z$  and Taylor map  $E$  are  $w$  and  $\mathcal{T}$  ( $z$  is a coordinate and  $E$  an elliptic curve in [section 17.6.7](#)); in [section 17.6.6](#) its number  $r$  of distinct roots is  $k$  and  $r_\infty$  is  $n_\infty$  ( $r$  is the vanishing order of a differential in [section 17.6.2](#)), and its cofactors  $Q$ ,  $S$  are  $P_1$ ,  $P_{\text{odd}}$  ( $Q$  is the centre); its Bézout polynomial  $V$  is  $W$  ( $V$  is a valuation ring); its points  $P$ ,  $P_0$  are  $\mathbf{x}$ ,  $\mathbf{x}_0$ ; and its purely infinite parameter  $h$  and exponent  $a$  in [section 17.6.8](#) are  $s$  and  $\beta$ . In this subsection  $E$  is the affine elliptic curve of [section 17.6.7](#) and nothing else, and  $P_1$  is a polynomial, not a boundary point as in [lemma 17.7](#). The complete list is in [table 1](#).

Two readings are tempting and false. “The normalization of  $C$  is  $\mathbb{A}^1$ ” does not mean “ $C \cong \mathbb{A}^1$ ”: the node and the cusp are singular, have normalization  $\mathbb{A}^1$  and carry polynomial families ([proposition 16.11](#)). And “ $C$  has a nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point” does not mean that every nonconstant point lifts to the normalization, or is a polynomial family evaluated at a Hahn parameter; the criterion is about existence.

### 17.6.1 The criterion

**Theorem 17.25** (Singular curve criterion [C16]). *Let  $\mathbb{k}$  be a field of characteristic zero,  $\Gamma \neq 0$ , and  $C$  a geometrically integral affine curve of finite type over  $\mathbb{k}$ , with normalization  $\nu: \tilde{C} \rightarrow C$ . The following are equivalent:*

- (i)  $C(\mathcal{A}_{\mathbb{k}}(\Gamma))$  contains a point not in  $C(\mathbb{k})$ ;
- (ii)  $\tilde{C} \cong \mathbb{A}_{\mathbb{k}}^1$ ;
- (iii) there is a nonconstant  $\mathbb{k}$ -morphism  $\mathbb{A}_{\mathbb{k}}^1 \rightarrow C$ ;
- (iv) there is a nonconstant point in  $C(\mathbb{k}[T])$ .

When they hold, for every  $0 < \gamma \in \Gamma$  there is a nonconstant point whose coordinates have finite support in  $\{0, \gamma, 2\gamma, \dots, n\gamma\}$  for some ordinary  $n$ .

Here  $C(\mathbb{k}[T])$  is a polynomial family, not a single ordinary point: for a closed embedding  $C \hookrightarrow \mathbb{A}_{\mathbb{k}}^N$ , condition (iv) asks for polynomials  $p_1(T), \dots, p_N(T)$ , not all constant, that satisfy the equations of  $C$  identically. The theorem does not say that every nonconstant point is such a family evaluated at a Hahn parameter. For a smooth curve it is [theorem 17.6](#). The node, the cusp and the singular short Weierstrass cubics ([proposition 16.11](#)) are its first singular exceptional cases, and it removes every singularity restriction from the smooth classification while keeping arbitrary  $\Gamma$ . The proof, in [section 17.6.4](#), has four steps: a nonconstant point makes the function field a subfield of  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ ; the conductor gives a fixed denominator for all powers of a contracted differential ([theorem 17.28](#)); a regular differential has a positive valuation at the boundary place ([lemma 17.30](#)); and a fixed denominator against a positive discrete valuation is a contradiction ([corollary 17.29](#)).

The generalized-series setting and the normal forms of **Oz** are classical [[1](#), [2](#), [3](#)]; the finite identity used here is Seidenberg’s [[53](#)]; normalization, Riemann–Roch and Riemann–Hurwitz are standard [[55](#), [39](#), [41](#), [57](#)]. Source C16 names as its contribution the combination that removes all singularity restrictions from the affine-curve dichotomy at arbitrary rank, the finite-order certificate and the repeated-root consequences; it does not attribute Seidenberg’s identity, smooth rigidity or Riemann–Hurwitz to itself. “Complete criterion” refers to the class of curves and points of [theorem 17.25](#), not to a named problem of the literature. Its targeted comparison did not find the combined singular statement, and it is not a priority search. Van den Dries [[50](#)] concerns infinite points in *some* discretely ordered ring, not in the fixed ring **Oz**; source C16 checked only its record and abstract ([section 18.4](#)).

Source C16’s reason for the next step is the one recorded in [section 17.5](#): a point cannot be lifted through the normalization by an appeal to normality, because  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  need not be integrally closed even in its fraction field. Its witness is the fixed-workspace example printed there: for  $e, H \in \Gamma$  with  $0 < e$  and  $ne < H$  for every ordinary  $n$ , the element  $\omega^e(1 + \omega^{-2e})^{1/2}$  has exponents  $(1 - 2n)e$ , is integral over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  and not in it, and becomes an element of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  after multiplication by  $\omega^H$  [C10, C13, C16].

*Remark 17.26* (Why discreteness is needed [C16]). The same scales show why almost integrality alone does not force a positive-valued element to vanish. Put  $a = \omega^{-e}$ . Then  $v(a) = e > 0$ , but  $\omega^H a^n = \omega^{H-ne} \in \mathcal{A}_{\mathbb{k}}(\Gamma)$  for every ordinary  $n \geq 0$ . No finite multiple of  $e$  exceeds  $H$ . In the proof below the conductor lies in a one-variable function field whose restricted value group is  $\gamma\mathbb{Z}$  ([lemma 17.30](#)). The contracted differential may lie outside that function field: the Euler derivation need not preserve it. Instead, the choice of derivation gives  $v(\partial u) = v(u) = \gamma$ , and  $\alpha(\partial) = h \partial u$  with  $h \in F$  then gives  $v(\alpha(\partial)) \in \gamma\mathbb{Z}$ . These two valuations, rather than two field

memberships, justify the finite comparison. This is not an assumption that  $\Gamma$  has rank one. The hypothesis  $ne < H$  for all  $n$  is essential to the example, in accordance with [section 17.5](#): for  $\Gamma = \mathbb{Z}$  the ring is  $\mathbb{k}[\omega]$  and normal.

Every finite tuple of surreals lies in a set-sized workspace ([lemma 2.6](#)); scheme theory and derivations are used only there, and no spectrum of a proper-class ring is formed [C16]. In the surreal instances  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is the part of  $\mathcal{B}_{\mathbb{R}} = \mathbb{R} \oplus \Pi$  or  $\mathcal{B}_{\mathbb{C}} = \mathbb{C} \oplus \Pi_{\mathbb{C}}$  supported in  $\Gamma$ , and  $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$ ,  $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$  [1, 2, 3].

### 17.6.2 Seidenberg's identity and conductor certificates

Let  $C$  be an integral affine curve over a field. Its normalization  $\mathbb{k}[\tilde{C}]$  in  $F = \text{Frac } \mathbb{k}[C]$  is a finite  $\mathbb{k}[C]$ -module [56]; write finitely many module generators as fractions  $c_i/d_i$  with  $c_i, d_i \in \mathbb{k}[C]$  and  $d_i \neq 0$ . Their product  $\kappa = \prod_i d_i$  is nonzero and clears every module generator, hence every element of the normalization. It is a *conductor element*

$$0 \neq \kappa \in \mathbb{k}[C], \quad \kappa \mathbb{k}[\tilde{C}] \subseteq \mathbb{k}[C], \quad \text{hence} \quad \kappa b^j \in \mathbb{k}[C] \quad (b \in \mathbb{k}[\tilde{C}], j \geq 0). \quad (17.1)$$

No principal conductor or greatest common divisor is needed [C16].

**Lemma 17.27** (Seidenberg's power identity [C16]; after [53]). *Let  $R$  be a domain containing  $\mathbb{Q}$  in a field  $L$  with a derivation  $\partial$  such that  $\partial(R) \subseteq R$ . If  $\kappa, b \in L$  and  $\kappa b^j \in R$  for  $0 \leq j \leq n$ , then*

$$\kappa^2 (\partial b)^n = \frac{1}{n!} \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} (\kappa b^{n-j}) \partial^n (\kappa b^j) \in R. \quad (17.2)$$

*Proof.* Let  $w$  be an independent formal variable and  $\mathcal{T} : L \rightarrow L[[w]]$ ,  $\mathcal{T}(f) = \sum_{i \geq 0} \partial^i(f) w^i / i!$ ; the Leibniz rule makes  $\mathcal{T}$  a ring homomorphism: the coefficient of  $w^i$  in  $\mathcal{T}(fg)$  agrees with that in  $\mathcal{T}(f)\mathcal{T}(g)$  by  $\partial^i(fg) = \sum_{j=0}^i \binom{i}{j} \partial^j(f) \partial^{i-j}(g)$ . By the binomial theorem and multiplicativity,

$$\kappa \mathcal{T}(\kappa) (\mathcal{T}(b) - b)^n = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} (\kappa b^{n-j}) \mathcal{T}(\kappa b^j).$$

Since  $\mathcal{T}(b) - b = (\partial b)w + O(w^2)$  and  $\mathcal{T}(\kappa) = \kappa + O(w)$ , the coefficient of  $w^n$  on the left is  $\kappa^2 (\partial b)^n$ , and on the right it is the sum in (17.2). This includes  $n = 0$ , when both sides equal  $\kappa^2$ . Each  $\kappa b^{n-j}$  and each  $\partial^n (\kappa b^j)$  lies in  $R$ , and  $1/n! \in \mathbb{Q} \subseteq R$ .  $\square$

The variable  $w$  is an ordinary formal variable, separate from the monomials  $\omega^\gamma$ ; only one coefficient, involving finitely many derivatives, is used, and nothing is substituted for  $w$ . An element  $b \in \text{Frac } R$  is almost integral over  $R$  when  $\kappa b^n \in R$  for all  $n$  and a fixed  $0 \neq \kappa \in R$ ; the lemma is the finite form of Seidenberg's theorem that derivations preserve complete integral closure, also for non-Noetherian  $\mathbb{Q}$ -algebras [53, Section 3], [54, Section 2.5]. Here the actual multiplier  $\kappa^2$  is kept [C16].

**Theorem 17.28** (Conductor contraction certificate [C16]). *Let  $\mathbb{k}$  have characteristic zero, let  $\mathbb{k}[C] \subseteq R \subseteq L$  be  $\mathbb{k}$ -algebras in a field  $L$ , where  $\mathbb{k}[C]$  is the coordinate ring of an integral affine curve with normalization  $\mathbb{k}[\tilde{C}] \subseteq F \subseteq L$ , and let  $\partial$  be a  $\mathbb{k}$ -derivation of  $L$  with  $\partial(R) \subseteq R$ . Choose  $\kappa$  as in (17.1). For*

$$\alpha = \sum_{j=1}^{\ell} a_j db_j \in \Omega_{\mathbb{k}[\tilde{C}]/\mathbb{k}}^1, \quad a_j, b_j \in \mathbb{k}[\tilde{C}], \quad \ell \geq 1,$$

put  $\alpha(\partial) = \sum_{j=1}^{\ell} a_j \partial b_j \in L$ . Then for every ordinary  $n \geq 0$

$$\kappa^{2\ell+1} \alpha(\partial)^n \in R, \quad (17.3)$$

and for fixed  $n$  the proof uses derivatives of order at most  $n$  of finitely many elements of  $\mathbb{k}[C]$ .

*Proof.* Expand  $\alpha(\partial)^n$  by the multinomial theorem. Apart from an integer coefficient, a term is  $(\prod_j a_j^{e_j}) (\prod_j (\partial b_j)^{e_j})$  with  $e_j \geq 0$  and  $\sum_j e_j = n$ . The first factor lies in  $\mathbb{k}[\tilde{C}]$ , so  $\kappa$  times it lies in  $\mathbb{k}[C] \subseteq R$ . For each  $j$ , all  $\kappa b_j^i$  lie in  $\mathbb{k}[C]$ , so [lemma 17.27](#) with  $n = e_j$  puts  $\kappa^2 (\partial b_j)^{e_j}$  in  $R$ . Multiplying these  $\ell + 1$  memberships puts  $\kappa^{2\ell+1}$  times the term in  $R$ , and summing proves (17.3). The exponent  $2\ell + 1$  depends on the chosen presentation of  $\alpha$ , but is independent of  $n$ ; this uniformity is the point of multiplying the memberships before summing. Terms with  $e_j = 0$  cause no exception, since the identity at order zero gives  $\kappa^2 \in R$ . Only the derivatives  $\partial^{e_j}$ ,  $e_j \leq n$ , of the elements  $\kappa b_j^i$  occur.  $\square$

The ring  $R$  may be all of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ : no finite generation, normality or equality  $\text{Frac } R = L$  hidden in the statement. Only  $\mathbb{k}[\tilde{C}]$  is normalized, and its elements  $b_j$  may lie outside  $R$  [C16].

**Corollary 17.29** (A finite derivative-order bound [C16]). *In [theorem 17.28](#) take  $R = \mathcal{A}_{\mathbb{k}}(\Gamma) \subseteq L = \mathbb{H}_{\mathbb{k}}(\Gamma)$  and  $\partial$  an Euler derivation. Suppose that for some  $\gamma > 0$  and ordinary  $\delta, r \geq 0$*

$$v(\kappa) = -\delta\gamma, \quad v(\alpha(\partial)) = (r+1)\gamma.$$

*These data are impossible; the contradiction occurs at*

$$n = \left\lfloor \frac{(2\ell+1)\delta}{r+1} \right\rfloor + 1. \quad (17.4)$$

*Proof.* The element (17.3) is nonzero, lies in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  and has valuation  $(n(r+1) - (2\ell+1)\delta)\gamma > 0$ ; but  $\mathcal{A}_{\mathbb{k}}(\Gamma) \cap \mathfrak{m}_{\mathbb{k}}(\Gamma) = \{0\}$  ([lemma 16.1](#)).  $\square$

This is a finite certificate in differential algebra. The bound (17.4) is the least  $n$  for which that valuation is positive, for the given parameters; it is not claimed optimal over the choices of conductor, presentation and differential, and no algorithm is given for producing a conductor and a regular differential from an arbitrary presentation ([questions 21.25](#) and [21.26](#)) [C16].

### 17.6.3 The place at infinity

Let  $C = \text{Spec } \mathbb{k}[C]$  be an integral affine curve and  $\varphi : \mathbb{k}[C] \rightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$  a  $\mathbb{k}$ -algebra map. If its kernel is nonzero, then  $\mathbb{k}[C]/\ker \varphi$  is a field finite over  $\mathbb{k}$ , and its image in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is algebraic over  $\mathbb{k}$ , hence equal to  $\mathbb{k}$  ([lemma 16.1\(d\)](#)); the point is constant. So a nonconstant point gives

$$\mathbb{k}[C] \hookrightarrow \mathcal{A}_{\mathbb{k}}(\Gamma) \subseteq \mathbb{H}_{\mathbb{k}}(\Gamma), \quad F = \text{Frac } \mathbb{k}[C] \hookrightarrow \mathbb{H}_{\mathbb{k}}(\Gamma), \quad (17.5)$$

and some element of  $\mathbb{k}[C]$  has negative valuation [C16].

**Lemma 17.30** (The place at infinity [C16]). *Let  $\mathbb{k}$  be algebraically closed and let (17.5) come from a nonconstant point. There are a closed point  $Q \in \Sigma = \overline{C} \setminus \tilde{C}$ , a uniformizer  $u \in F$  at  $Q$  and  $\gamma = v(u) > 0$  such that*

$$v(f) = \text{ord}_Q(f) \gamma \quad (f \in F^\times). \quad (17.6)$$

*In particular  $v(F^\times) = \gamma\mathbb{Z}$ .*

*Proof.* The ring  $V = F \cap \mathcal{O}_{\mathbb{k}}(\Gamma)$  is a valuation ring of  $F$  containing  $\mathbb{k}$ , nontrivial because some element of  $\mathbb{k}[C]$  has negative valuation. By the valuative criterion of properness [38, Tag 0BX5], the generic point extends to  $\text{Spec } V \rightarrow \overline{C}$ . The centre  $Q$  is not the generic point, for then  $F \subseteq V$ ; so it is closed, and  $\mathcal{O}_{\overline{C},Q}$  is a discrete valuation ring [39]. The local inclusion  $\mathcal{O}_{\overline{C},Q} \rightarrow V$  sends a uniformizer  $u$  into the maximal ideal of  $V$ , so  $\gamma = v(u) > 0$ . Every  $f \in F^\times$  is  $u^n w$  with  $w \in \mathcal{O}_{\overline{C},Q}^\times$ ; both  $w$  and  $w^{-1}$  lie in  $V$ , so  $v(w) = 0$  and (17.6) holds. If  $Q$  were in  $\tilde{C}$ , every element of  $\mathbb{k}[C] \subseteq \mathbb{k}[\tilde{C}] \subseteq \mathcal{O}_{\overline{C},Q}$  would have nonnegative valuation, hence lie in  $\mathcal{A}_{\mathbb{k}}(\Gamma) \cap \mathcal{O}_{\mathbb{k}}(\Gamma) = \mathbb{k}$ , and the point would be constant.  $\square$

This is the centre of [theorem 17.8](#), with the discreteness of the induced valuation on  $F$  made explicit. The smooth projective model used here exists over any algebraically closed, hence perfect, field, and the lemma does not use the characteristic; the next proposition uses that [C16].

**Proposition 17.31** (Only one point may be missing [C16]). *Let  $\mathbb{k}$  be algebraically closed of any characteristic, with  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  defined by the same formula. If an integral affine curve  $C$  over  $\mathbb{k}$  has a nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point, then  $\Sigma = \overline{C} \setminus \tilde{C}$  consists of exactly one point.*

*Proof.* Let  $Q$  be the centre of [lemma 17.30](#), which is in  $\Sigma$ , and suppose that  $q \neq Q$  is also in  $\Sigma$ . For  $N \geq g(\overline{C}) + 1$  Riemann–Roch gives  $h^0(\overline{C}, \mathcal{O}(Nq - Q)) \geq N - g \geq 1$  [41]; a nonzero  $f$  in this space is regular on  $\tilde{C}$ , vanishes at  $Q$  and has its only possible pole at  $q$ . So  $f \in \mathbb{k}[\tilde{C}]$  and  $v(f) > 0$ . For a conductor element  $\kappa$  every  $\kappa f^n$  lies in  $\mathbb{k}[C] \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$ , while by (17.6) both  $v(\kappa) \leq 0$  and  $v(f) > 0$  are integer multiples of  $\gamma$ ; for large ordinary  $n$ ,  $v(\kappa f^n) > 0$ , contradicting  $\mathcal{A}_{\mathbb{k}}(\Gamma) \cap \mathfrak{m}_{\mathbb{k}}(\Gamma) = \{0\}$ . More explicitly, put  $\delta = -\text{ord}_Q(\kappa) \geq 0$  and  $r = \text{ord}_Q(f) \geq 1$ . Any integer  $n > \delta/r$  gives the contradiction, even if  $\Gamma$  has higher rank. Hence  $\Sigma = \{Q\}$ .  $\square$

The argument uses no derivation and no normality of the target ring; it is the part of the obstruction that survives in positive characteristic. In characteristic zero regular differentials add the restriction of genus zero [C16]. The two positive-characteristic examples of [section 18.3](#) satisfy it, as they must: the affine curve  $y^2 + y = x^3$  of [proposition 18.12](#) and the affine line  $y^p = x^p - x$  of [proposition 18.13](#) each have one point at infinity [merge]. So a characteristic-free statement can only be one-place, not genus zero.

#### 17.6.4 Proof of the criterion

**Theorem 17.32** (Positive genus across singularities [C16]). *Let  $\mathbb{k}$  be algebraically closed of characteristic zero. If the smooth projective model  $\overline{C}$  of an integral affine curve  $C$  over  $\mathbb{k}$  has genus at least one, then  $C(\mathcal{A}_{\mathbb{k}}(\Gamma)) = C(\mathbb{k})$ .*

*Proof.* Suppose a nonconstant point exists, and take  $Q$ ,  $u$  and  $\gamma$  from [lemma 17.30](#). As  $g \geq 1$ , there is a nonzero  $\alpha \in H^0(\overline{C}, \Omega_{\overline{C}/\mathbb{k}}^1)$ , a space of dimension  $g$  [41]. Its restriction to the smooth affine curve  $\tilde{C}$  lies in  $\Omega_{\mathbb{k}[\tilde{C}]/\mathbb{k}}^1$  and has a finite presentation

$$\alpha = \sum_{j=1}^{\ell} a_j db_j, \quad a_j, b_j \in \mathbb{k}[\tilde{C}]. \quad (17.7)$$

Since  $v(u) = \gamma$ , the leading exponent of  $u$  is  $-\gamma \neq 0$ ; choose  $\lambda$  with  $\lambda(-\gamma) = 1$  ([lemma 16.3](#)) and put  $\partial = \partial_\lambda$ . The leading term of  $u$  survives, so  $\partial u \neq 0$  and  $v(\partial u) = \gamma$ . At  $Q$  the differential  $du$  freely generates the module of differentials, since  $\overline{C}$  is smooth and  $\kappa(Q) = \mathbb{k}$  is separable over  $\mathbb{k}$  [57, Lemma 53.12.3]; write  $\alpha = h du$  with  $0 \neq h \in \mathcal{O}_{\overline{C},Q}$  and  $r = \text{ord}_Q(h) \geq 0$ . The

universal property of Kähler differentials, applied to the  $\mathbb{k}$ -derivation  $F \rightarrow \mathbb{H}_{\mathbb{k}}(\Gamma)$  obtained by restricting  $\partial$ , gives

$$\alpha(\partial) = \sum_{j=1}^{\ell} a_j \partial b_j = h \partial u,$$

which is nonzero with valuation  $(r+1)\gamma$  by (17.6). A conductor element  $\kappa \in \mathbb{k}[C] \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$  has  $v(\kappa) = -\delta\gamma$  with an ordinary  $\delta \geq 0$ . Theorem 17.28 and corollary 17.29 give the contradiction.  $\square$

The derivation is defined on  $F$  with values in  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ ; it need not preserve  $F$ , and it is never identified with an endomorphism of the function field. The equality  $\alpha(\partial) = h \partial u$  takes place in  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  and is justified by the relative module  $\Omega_{F/\mathbb{k}}^1$  [C16].

*Remark 17.33* (A third route in the smooth case [merge]). For smooth  $C$  one may take  $\kappa = 1$  and  $\delta = 0$ ; then (17.4) gives  $n = 1$ , that is,  $\alpha(\partial) \in \mathcal{A}_{\mathbb{k}}(\Gamma) \cap \mathfrak{m}_{\mathbb{k}}(\Gamma) = \{0\}$  directly, and proposition 17.31 replaces the unit argument of lemma 17.4 by Riemann–Roch. So the rigid direction of theorem 17.6 has a third proof, after the descent proof of source C14 and the logarithmic proof of source C12 (theorem 17.8): positive genus by one regular differential and its valuation at the centre, and genus zero with two or more boundary points by a function with a single pole.

**The algebraically closed case.** Suppose  $\mathbb{k}$  is algebraically closed and  $C$  has a nonconstant point. By theorem 17.32 the model  $\overline{C}$  has genus zero, and by proposition 17.31 the affine normalization omits exactly one point; a smooth projective genus-zero curve over an algebraically closed field is  $\mathbb{P}^1$  [42], so  $\tilde{C} \cong \mathbb{A}_{\mathbb{k}}^1$ . Conversely, if  $\tilde{C} \cong \mathbb{A}_{\mathbb{k}}^1$ , the normalization map in a closed embedding  $C \hookrightarrow \mathbb{A}_{\mathbb{k}}^N$  is a tuple  $(p_1(T), \dots, p_N(T))$  of polynomials, not all constant, and for  $0 < \gamma \in \Gamma$  the substitution  $T = \omega^\gamma$  gives a point of  $C(\mathcal{A}_{\mathbb{k}}(\Gamma))$  with support in  $\{0, \gamma, \dots, n\gamma\}$ ,  $n = \max \deg p_i$ . It is nonconstant because  $\omega^\gamma$  is transcendental over  $\mathbb{k}$  [C16].

**Descent to every field of characteristic zero.** Coefficientwise inclusion  $\mathcal{A}_{\mathbb{k}}(\Gamma) \subseteq \mathcal{A}_{\overline{\mathbb{k}}}(\Gamma)$  preserves nonconstancy, and  $C_{\overline{\mathbb{k}}}$  is integral because  $C$  is geometrically integral; so a nonconstant point over  $\mathbb{k}$  gives one over  $\overline{\mathbb{k}}$ , and the normalization of  $C_{\overline{\mathbb{k}}}$  is  $\mathbb{A}_{\overline{\mathbb{k}}}^1$ . Over the perfect field  $\mathbb{k}$  the normalization  $\tilde{C}$  is smooth and geometrically integral. To see the latter assertion, use the dense open where normalization is an isomorphism: after coefficient extension it is an integral dense open from  $C_{\overline{\mathbb{k}}}$ , while smoothness makes the whole base change normal and reduced. The base change of  $\tilde{C}$  is consequently the normalization of  $C_{\overline{\mathbb{k}}}$ : the morphism stays finite and birational and its source is normal. So  $\tilde{C}$  is a form of the affine line, and  $\tilde{C} \cong \mathbb{A}_{\mathbb{k}}^1$  by lemma 17.5: its completion has genus zero and one geometric boundary point, which is rational because in characteristic zero a closed point of higher degree splits into several geometric points [C16]. This proves (i) $\Rightarrow$ (ii) of theorem 17.25. The normalization map gives (ii) $\Rightarrow$ (iii); (iii) and (iv) are the same statement about the affine functor; and evaluating a nonconstant polynomial family at  $\omega^\gamma$  gives (iv) $\Rightarrow$ (i) with the stated finite support.

**What the proof does not use.** No analyticity, no countability of supports, no divisibility of  $\Gamma$ , no saturation, and no statement that an element integral over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  lies in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ ; the proof works for  $\Gamma = \mathbb{Z}$ , for dense rank-one groups and for every set-sized rank. The smooth normalization is a theorem about finite-type curves over a perfect field, not a hypothesis on  $C$ : nodes, cusps, several branches and non-planar singularities are allowed. The finite-type and constant-coefficient hypotheses are substantive [C16].

### 17.6.5 Structural consequences

**Corollary 17.34** (Independence of the scale [C16]). *For a geometrically integral affine curve  $C$  over a field of characteristic zero, whether  $C(\mathcal{A}_{\mathbb{k}}(\Gamma)) = C(\mathbb{k})$  does not depend on the nonzero ordered group  $\Gamma$ , and it holds exactly when  $C(\mathbb{k}[T]) = C(\mathbb{k})$ .*

*Proof.* Each statement is equivalent to  $\tilde{C} \not\cong \mathbb{A}_{\mathbb{k}}^1$  by [theorem 17.25](#).  $\square$

So transfinite supports never create the first nonconstant point on a constant-coefficient affine curve; they can create further points, and complicated fibers, on the exceptional curves. The corollary is an existence reduction, not a description of all points [C16].

**Corollary 17.35** (Finite birational invariance [C16]). *Let  $C' \rightarrow C$  be a finite birational morphism of geometrically integral affine curves over a field of characteristic zero. Then  $C'(\mathcal{A}_{\mathbb{k}}(\Gamma)) = C'(\mathbb{k})$  if and only if  $C(\mathcal{A}_{\mathbb{k}}(\Gamma)) = C(\mathbb{k})$ .*

*Proof.*  $\mathbb{k}[C']$  is integral over  $\mathbb{k}[C]$  with the same fraction field, so both have the same normalization; apply [theorem 17.25](#).  $\square$

The affine normalization, including which points at infinity are removed, must be retained: arbitrary birational equivalence is not enough, since  $\mathbb{A}^1$  is flexible and  $\mathbb{G}_m$  rigid with the same function field [C16].

**Theorem 17.36** (Affine schemes of dimension at most one [C16]). *Let  $\mathbb{k}$  be algebraically closed of characteristic zero, let  $\Gamma \neq 0$ , and let  $X$  be an affine  $\mathbb{k}$ -scheme of finite type and dimension at most one, possibly reducible or nonreduced. Then  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$  if and only if no one-dimensional irreducible component of  $X_{\text{red}}$  has affine normalization  $\mathbb{A}_{\mathbb{k}}^1$ .*

*Proof.* A map  $\mathbb{k}[X] \rightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$  kills nilpotents, since  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is a domain, and its kernel  $P$  is prime. If  $X$  is empty, both point sets are empty and the criterion is vacuous. For a point of nonempty  $X$ , if  $\dim \mathbb{k}[X]/P = 0$ , the image is a domain algebraic over  $\mathbb{k}$ , hence  $\mathbb{k}$  ([lemma 16.1\(d\)](#)), and the point is constant. Otherwise  $P$  is a minimal prime: a strict inclusion of a minimal prime in  $P$ , followed by a strict prime inclusion above  $P$ , would give a chain of length two in a ring of dimension at most one. Thus  $V(P)$  is a one-dimensional irreducible component of  $X_{\text{red}}$ , integral and, over the algebraically closed  $\mathbb{k}$ , geometrically integral, and the point is a nonconstant point of it; [theorem 17.25](#) applies. Conversely the normalization of an exceptional component gives a nonconstant polynomial map into  $X$ , which  $T = \omega^\gamma$  turns into a nonconstant point.  $\square$

The nonzero-group hypothesis is necessary: for  $\Gamma = 0$  and  $X = \mathbb{A}_{\mathbb{k}}^1$ , the left side is true because  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}$ , whereas the stated condition on the normalization is false.

This classifies the reducible affine curves over an algebraically closed field of characteristic zero, which [section 17.5](#) records as not classified by sources C10–C15; over other fields components need not be geometrically integral, and no statement is made there [C16].

**Corollary 17.37** (Polynomial witnesses in every fiber [C16]). *Let  $\mathbb{k}$  be algebraically closed of characteristic zero and  $\tilde{C} \cong \mathbb{A}_{\mathbb{k}}^1$ . For every  $\mathbf{x}_0 \in C(\mathbb{k})$  and every  $0 < \gamma \in \Gamma$  there is a nonconstant finite-support  $\mathbf{x} \in C(\mathcal{A}_{\mathbb{k}}(\Gamma))$  with  $\text{ct}(\mathbf{x}) = \mathbf{x}_0$ .*

*Proof.* The finite normalization map is surjective; choose  $c \in \mathbb{k}$  over  $\mathbf{x}_0$  and write the map as  $(p_i(T))_i$ . The coordinates  $p_i(c + \omega^\gamma)$  lie in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , have constant terms  $p_i(c)$ . At least one  $p_i$  is nonconstant, and translation by  $c$  preserves its degree. Evaluation at the transcendental monomial  $\omega^\gamma$  is injective on  $\mathbb{k}[T]$ , so that coordinate is not in  $\mathbb{k}$ . The resulting supports are finite subsets of  $\{0, \gamma, \dots, n\gamma\}$ , where  $n = \max_i \deg p_i$ .  $\square$

Over a field that is not algebraically closed the same construction works at every  $\mathbf{x}_0$  with a  $\mathbb{k}$ -rational preimage under the normalization, in particular at every nonsingular  $\mathbb{k}$ -point. It is not replaced by the existence of a  $\mathbb{k}$ -point on the singular curve; the distinction matters at real isolated singularities ([question 21.24](#)) [C16].

### 17.6.6 Repeated-root superelliptic equations

Let  $\mathbb{k}$  be algebraically closed of characteristic zero,  $m \geq 2$ , and

$$P(X) = c \prod_{i=1}^k (X - a_i)^{e_i}, \quad c \neq 0, \quad e_i \geq 1, \quad (17.8)$$

with distinct  $a_i$  and  $d = \deg P = \sum_i e_i$ . Assume throughout this theorem that  $Y^m - P(X)$  is irreducible, equivalently  $\gcd(m, e_1, \dots, e_k) = 1$ . Indeed,  $P$  is a  $p$ th power in  $\mathbb{k}(X)$  exactly when  $p$  divides every  $e_i$ : necessity follows from the orders at the  $a_i$ , and sufficiency from taking a  $p$ th root of  $c$ . For completeness, put  $K = \mathbb{k}(X)$  and  $b^m = P$ . Since  $K$  contains all  $m$ th roots of unity,  $K(b)/K$  is cyclic Galois of degree  $r \mid m$ , and  $b^r \in K$  [58]. If  $r < m$ , then  $P = (b^r)^{m/r}$  is a  $p$ th power for some prime  $p \mid m/r$ . Conversely, such a power makes  $Y^m - P$  reducible by a difference-of-powers factorization. Gauss's lemma transfers irreducibility between  $K[Y]$  and  $\mathbb{k}[X, Y]$ . Thus irreducibility means that for every prime  $p \mid m$ , some  $e_i$  is not divisible by  $p$  [C16].

**Theorem 17.38** (Superelliptic flexibility criterion [C16]). *Assume  $Y^m - P(X)$  is irreducible, in addition to the hypotheses above. Then  $Y^m = P(X)$  has a nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point if and only if exactly one index  $i$  has  $m \nmid e_i$ , and for that index  $\gcd(m, e_i) = 1$ . Equivalently, after extracting polynomial  $m$ th powers,*

$$P(X) = c P_1(X)^m (X - a)^e, \quad 1 \leq e < m, \quad \gcd(m, e) = 1, \quad (17.9)$$

and then a normalization parameter is

$$X = a + T^m, \quad Y = \eta P_1(a + T^m) T^e, \quad \eta^m = c. \quad (17.10)$$

In every other irreducible case all solutions in  $\mathcal{A}_{\mathbb{k}}(\Gamma)^2$  are constant.

*Proof.* Let  $\overline{C}$  be the smooth projective model of this degree- $m$  cover of the  $X$ -line. At  $a_i$  put  $s = X - a_i$ , and at infinity put  $s = 1/X$ . Locally  $P = s^a U(s)$  with  $U \in \mathbb{k}[[s]]^\times$ , where  $a = e_i$  at a finite root and  $a = -d$  at infinity. The unit  $U$  has an  $m$ th root: take a root of its constant coefficient and expand the remaining unit by the binomial series. After dividing  $Y$  by this root, the local equation is  $w^m = s^a$ . Put  $h = \gcd(m, |a|)$ . It has  $h$  branches, each with  $s = u^{m/h}$  and  $w = \zeta u^{a/h}$  for an appropriate constant  $\zeta$ . The relatively prime integers  $m/h$  and  $a/h$  generate  $u$  in the branch field, so its completed normalization is  $\mathbb{k}[[u]]$  and its ramification index is  $m/h$ . At finite nonroots  $a = 0$ , giving  $m$  unramified branches. Thus a root contributes  $m - \gcd(m, e_i)$  to the ramification sum, and infinity has  $n_\infty = \gcd(m, d)$  points and contributes  $m - n_\infty$ . Characteristic zero makes this ramification tame, so Riemann–Hurwitz [57] gives

$$2g - 2 = -2m + \sum_{i=1}^k (m - \gcd(m, e_i)) + m - \gcd(m, d). \quad (17.11)$$

The coordinate ring is integral over  $\mathbb{k}[X]$  because the equation is monic in  $Y$ . Its normalization is the integral closure of  $\mathbb{k}[X]$  in the same function field: integrality in either direction follows from transitivity and from inclusion of the base ring. It is finite over  $\mathbb{k}[X]$  and identifies with

the inverse image of the affine  $X$ -line in  $\overline{C}$ . Hence it omits exactly the  $n_\infty$  points over infinity. By [theorem 17.25](#) (with [theorem 17.32](#) and [proposition 17.31](#)), flexibility is equivalent to  $g = 0$  and  $n_\infty = 1$ , which by [\(17.11\)](#) give

$$\sum_{i=1}^k (m - \gcd(m, e_i)) = m - 1. \quad (17.12)$$

A nonzero summand is at least  $m/2$ , since a proper divisor of  $m$  is at most  $m/2$ ; so there is exactly one nonzero summand, and its gcd is one. Conversely that pattern gives  $d \equiv e \pmod{m}$ , so  $\gcd(m, d) = 1$ , and [\(17.12\)](#), hence  $g = 0$  and one point at infinity. Finally [\(17.10\)](#) satisfies the equation;  $T$  is integral over the coordinate ring since  $T^m = X - a$ ; and  $T^m$  and  $T^e = Y/(\eta P_1(X))$  generate  $T$  in the function field: choose integers  $a_0, b_0$  with  $a_0 m + b_0 e = 1$  and use  $T = (T^m)^{a_0} (T^e)^{b_0}$  there. Negative powers are allowed in the function field, not asserted to lie in the coordinate ring. So  $\mathbb{k}[T]$  is a finite birational normal extension of the coordinate ring, that is, its normalization.  $\square$

The irreducibility assumption is essential. Without it,  $Y^2 = X^2$  has the nonconstant polynomial families  $(T, \pm T)$  although no multiplicity is odd. The quadratic corollary below includes such reducible cases separately.

Repeated factors, and so arbitrarily singular affine models, are governed by the same test; the Riemann–Hurwitz computation itself is classical. Many repeated roots do not make a curve flexible: only the residues of the multiplicities modulo  $m$  matter [C16]. The theorem contains [theorem 16.6](#) over an algebraically closed field, where all  $e_i = 1$  and  $k = d \geq 2$  gives at least two indices with  $m \nmid e_i$ , and it agrees with [proposition 16.11](#): the singular cubic  $(X - r)^2(X + 2r)$ ,  $r \neq 0$ , and the cusp  $X^3$  have exactly one odd multiplicity. Over any field  $\mathbb{k}$  of characteristic zero, if  $Y^m - P(X)$  is irreducible over  $\overline{\mathbb{k}}$ , the test applied to the factorization over  $\mathbb{k}$  decides flexibility over  $\mathbb{k}$ : by [theorem 17.25](#) flexibility over  $\mathbb{k}$  means  $\tilde{C} \cong \mathbb{A}_{\mathbb{k}}^1$ , which holds exactly when  $\tilde{C}_{\overline{\mathbb{k}}} \cong \mathbb{A}_{\overline{\mathbb{k}}}^1$  by [lemma 17.5](#) and base change [merge].

For  $P \in \mathbb{C}[X] \setminus \{0\}$  write  $P = cP_1^2 P_{\text{odd}}$  with  $P_1, P_{\text{odd}}$  monic,  $P_{\text{odd}}$  squarefree and  $c \neq 0$ ;  $P_{\text{odd}}$  is the product of the irreducible factors of odd multiplicity. The same factorization is used over  $\mathbb{R}$ .

**Corollary 17.39** (The quadratic-root test [C16]). *For  $P \in \mathbb{C}[X] \setminus \{0\}$ , the equation  $Y^2 = P(X)$  has a nonconstant  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ -point if and only if  $\deg P_{\text{odd}} \leq 1$ ; if  $\deg P_{\text{odd}} \geq 2$ , every solution is constant. For  $P \in \mathbb{R}[X] \setminus \{0\}$  and the same factorization over  $\mathbb{R}$ , there is a nonconstant  $\mathcal{A}_{\mathbb{R}}(\Gamma)$ -point if and only if  $\deg P_{\text{odd}} = 1$ , or  $\deg P_{\text{odd}} = 0$  and  $c > 0$ .*

*Proof.* Over  $\mathbb{C}$ , a nonconstant  $P_{\text{odd}}$  makes the curve irreducible and [theorem 17.38](#) applies with  $m = 2$ ; if  $P_{\text{odd}} = 1$  the two components  $Y = \pm \sqrt{c} P_1(X)$  are polynomial graphs. Over  $\mathbb{R}$ ,  $\deg P_{\text{odd}} \geq 2$  is rigid after complexification. If  $P_{\text{odd}} = X - a$ , then  $X = a + cT^2$ ,  $Y = cP_1(a + cT^2)T$  is a real polynomial family for either sign of  $c$ . If  $P_{\text{odd}} = 1$  and  $c > 0$  the graphs are real. Evaluating any of these nonconstant polynomial families at  $T = \omega^\gamma$ ,  $\gamma > 0$ , gives a nonconstant finite-support Hahn point. If  $P_{\text{odd}} = 1$  and  $c < 0$ , the ordering of the real Hahn field forces  $Y = 0$  and  $P_1(X) = 0$ , and the roots of a real polynomial in an ordered extension of  $\mathbb{R}$  are real, so the points are constant.  $\square$

The excluded  $P = 0$  has the family  $(T, 0)$ . For example  $Y^2 = X^2(X - 1)$  is flexible and  $Y^2 = X^2(X - 1)(X - 2)$  is rigid, although both are singular at the origin; for  $\deg P_{\text{odd}} = 2$  the obstruction is two points at infinity, not positive genus, and for  $\deg P_{\text{odd}} \geq 3$  genus contributes as well [C16].

### 17.6.7 A pinched elliptic curve with a seventh-order certificate

Consider

$$Z^2 = X^2(X^3 - X + 1), \quad (17.13)$$

singular at the origin. Put  $f(X) = X^3 - X + 1$ , of discriminant  $-23$ , so the affine elliptic curve  $E : Y^2 = f(X)$  is smooth; it is the curve with  $\Delta_0 = 23$  of [corollary 16.9](#) and [example 17.23](#). The map  $(X, Y) \mapsto (X, Z = XY)$  is the normalization of (17.13) and identifies  $(0, 1)$  with  $(0, -1)$ . Indeed,  $y$  is integral by  $y^2 = f(x)$  and is the fraction  $z/x$ , while the smooth ring  $\mathbb{k}[x, y]/(y^2 - f(x))$  is normal. In ring terms  $\mathbb{k}[C] = \mathbb{k}[x, xy] \subseteq \mathbb{k}[\tilde{C}] = \mathbb{k}[x, y]$  with  $y^2 = f(x)$ , and  $x\mathbb{k}[\tilde{C}] \subseteq \mathbb{k}[C]$ : every element of the larger ring is  $A(x) + B(x)y$ , whose product with  $x$  is  $xA(x) + B(x)z$ . Thus  $\kappa = x$  is a conductor element [C16].

**Theorem 17.40** (A pinched elliptic curve [C16]). *Over every field of characteristic zero every solution of (17.13) in  $\mathcal{A}_{\mathbb{k}}(\Gamma)^2$  is constant. In particular*

$$\{(x, z) \in \mathbf{Oz}^2 : z^2 = x^2(x^3 - x + 1)\} = \{(x, z) \in \mathbb{Z}^2 : z^2 = x^2(x^3 - x + 1)\},$$

and likewise with  $\mathbf{Oz}[i]$  and  $\mathbb{Z}[i]$ . A nonconstant solution is excluded by derivatives of order at most seven.

*Proof.* Rigidity follows from [theorem 17.25](#), since the normalization  $E$  has genus one. Here is the explicit certificate. Suppose  $(x, z) \in \mathcal{A}_{\mathbb{k}}(\Gamma)^2$  is nonconstant. If  $x \in \mathbb{k}$ , then  $z$  is algebraic over  $\mathbb{k}$  and constant; so  $x \notin \mathbb{k}$ ,  $x \neq 0$ , and  $y = z/x \in \mathbb{H}_{\mathbb{k}}(\Gamma)$  satisfies  $y^2 = f(x)$ . Since a nonconstant element of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  has negative valuation,  $v(x) < 0$ . The term  $x^3$  has strictly smaller valuation than both  $-x$  and  $1$ , so it cannot cancel. Thus  $2v(y) = 3v(x)$ , and  $\gamma = v(x/y) \in \Gamma$  satisfies  $\gamma > 0$ ,  $v(x) = -2\gamma$  and  $v(y) = -3\gamma$ : these are the pole orders two and three at the point at infinity of  $E$ . Choose  $\lambda$  with  $\lambda(-\gamma) = 1$ , so  $\lambda(\deg_{\omega} x) = -2 \neq 0$ , and put  $\partial = \partial_{\lambda}$  and  $\alpha(\partial) = \partial x/(2y)$ ; the leading term of  $x$  survives, so  $v(\alpha(\partial)) = \gamma$ . The polynomials

$$U(X) = \frac{18X + 27}{23}, \quad W(X) = \frac{-6X^2 - 9X + 4}{23}, \quad Uf + Wf' = 1, \quad (17.14)$$

are those of (16.5) at  $(a, b) = (-1, 1)$ , divided by  $\Delta_0 = 23$ . With  $2y \partial y = f'(x) \partial x$  they give

$$\alpha(\partial) = \frac{U(x)y}{2} \partial x + W(x) \partial y. \quad (17.15)$$

For every  $j \geq 0$ ,  $xy^j \in \mathbb{k}[C]$ : explicitly,  $xy^{2a} = xf(x)^a$  and  $xy^{2a+1} = zf(x)^a$ . Applying [lemma 17.27](#) with conductor  $x$  and element  $y$  gives  $x^2(\partial y)^j \in \mathcal{A}_{\mathbb{k}}(\Gamma)$  for all  $j$ , including  $j = 0$ . In the binomial expansion of the  $n$ th power of (17.15), the summand indexed by  $i + j = n$ , after multiplication by  $x^3$ , is

$$\binom{n}{i} \frac{U(x)^i W(x)^j}{2^i} (\partial x)^i (xy^i) (x^2(\partial y)^j).$$

Every displayed factor lies in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ ; in particular, division by  $2^i$  is division by a nonzero constant. Hence

$$x^3 \alpha(\partial)^n \in \mathcal{A}_{\mathbb{k}}(\Gamma) \quad (n \geq 0). \quad (17.16)$$

At  $n = 7$  this nonzero element has valuation  $3(-2\gamma) + 7\gamma = \gamma > 0$ , a contradiction. For the term indexed by  $j \leq 7$ , Seidenberg's identity uses at most  $j$  derivatives of  $xy^a$  and powers of  $x$ , with  $0 \leq a \leq j$ . The latter derivatives are polynomial expressions in derivatives of  $x = xy^0$ . Thus only derivatives of order at most seven of the elements  $xy^a$ ,  $0 \leq a \leq 7$ , occur. No assertion that  $y$  itself belongs to  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is used. The omnific statements follow in a common workspace, as in [corollary 16.9](#).  $\square$

The certificate is not a squarefree equation in disguise:  $y = z/x$  is *not* assumed to be in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , and the fixed multiplier  $x^2$  for all powers of  $\partial y$  replaces that missing statement. The theorem is also [corollary 17.39](#) with  $P_{\text{odd}} = X^3 - X + 1$ . It classifies the additional omnific points by proving that there are none; it does not enumerate the ordinary integer points of the curve or of  $E$  [C16]. The general bound (17.4), with  $\ell = 2$ ,  $\delta = 2$  and  $r = 0$  for the differential  $dx/(2y)$ , which does not vanish at infinity, gives eleven; seven improves it, and no optimality is claimed ([question 21.25](#)).

### 17.6.8 Omnific and Gaussian omnific points

Let  $\mathcal{C} \subseteq \mathbb{A}_{\mathbb{Z}}^N$  be an affine scheme given by finitely many integer equations whose real fiber  $C_{\mathbb{R}}$  is a geometrically integral curve; the coordinate embedding is fixed.

**Theorem 17.41** (Omnific rigidity for singular curves [C16]). *If the affine normalization of  $C_{\mathbb{R}}$  is not  $\mathbb{A}_{\mathbb{R}}^1$ , then  $\mathcal{C}(\mathbf{Oz}) = \mathcal{C}(\mathbb{Z})$ . Likewise, for an affine  $\mathbb{Z}[i]$ -model with geometrically integral complex fiber  $C_{\mathbb{C}}$  whose affine normalization is not  $\mathbb{A}_{\mathbb{C}}^1$ ,  $\mathcal{C}(\mathbf{Oz}[i]) = \mathcal{C}(\mathbb{Z}[i])$ .*

*Proof.* A candidate tuple lies in a set-sized real or complex workspace ([lemma 2.6](#)); there the curve is rigid by [theorem 17.25](#), so the coordinates are constants, and  $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$ ,  $\mathbf{Oz}[i] \cap \mathbb{C} = \mathbb{Z}[i]$ . The reverse inclusions are clear.  $\square$

The argument works for any coefficient subring  $\mathfrak{o} \subseteq \mathbb{k}$  and the ring  $\mathfrak{o} \oplus \ker(\text{ct}) = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ ; it uses the constant-term condition, not an order on  $\mathfrak{o}$ . This is [theorem 17.11\(i\)](#) and [theorem 18.20](#) for singular curves. For reducible complex models [theorem 17.36](#) gives the componentwise test [C16].

If  $\nu(T) = (p_1(T), \dots, p_N(T))$ ,  $p_i \in \mathbb{k}[T]$ , parametrizes the normalization of an exceptional curve, then for  $c \in \mathbb{k}$  and every purely infinite  $s$  with coefficients in  $\mathbb{k}$ ,

$$\text{ct}(p_i(c + s)) = p_i(c), \quad (17.17)$$

since the positive powers of  $s$  are purely infinite [C16].

**Proposition 17.42** (Arithmetic families above liftable points [C16]). *Let  $\tilde{C} \cong \mathbb{A}_{\mathbb{k}}^1$  with  $\mathbb{k} = \mathbb{R}$  or  $\mathbb{C}$ , and  $\mathfrak{o} = \mathbb{Z}$  or  $\mathbb{Z}[i]$  respectively. If  $\mathbf{x}_0 \in \mathcal{C}(\mathfrak{o})$  has a  $\mathbb{k}$ -rational preimage  $c$  under the normalization, then  $\nu(c + \omega^\beta) \in \mathcal{C}(\mathfrak{o} \oplus \Pi_{\mathbb{k}})$  for every positive surreal  $\beta$ , where  $\Pi_{\mathbb{R}} = \Pi$ . These points are nonconstant, have finite support and constant term  $\mathbf{x}_0$ , and they form a proper class of distinct points.*

*Proof.* Write  $p_i(c + S) = p_i(c) + \sum_{j=1}^{d_i} b_{ij} S^j$ . At  $S = \omega^\beta$  the exponents  $j\beta$  are positive and distinct. The constant coefficient is  $p_i(c) \in \mathfrak{o}$ ; the other coefficients need only belong to  $\mathbb{k}$ , as allowed in the omnific ring. Every defining equation vanishes identically after substitution of the polynomial tuple  $\nu(T)$ , so it still vanishes at  $c + \omega^\beta$ . At least one coordinate polynomial has degree  $e \geq 1$ , because the normalization map is not constant. Its leading exponent is  $e\beta$  with nonzero coefficient, so the tuple is nonconstant. Multiplication by the positive integer  $e$  is injective on the ordered exponent group; distinct  $\beta$  therefore give distinct tuples. Each tuple has finite support. Finally the positive surreal exponents form a proper class, proving the claimed size of this family in the standing class convention.  $\square$

Over  $\mathbb{C}$  every ordinary point has a complex preimage under the finite surjective normalization: its fiber is a nonempty finite  $\mathbb{C}$ -scheme, and any closed point of that fiber has residue field a finite extension of  $\mathbb{C}$ , hence  $\mathbb{C}$ . Thus the existence dichotomy is complete for Gaussian points.

**Corollary 17.43** (The Gaussian existence dichotomy [C16]). *Let  $\mathcal{C}$  be an affine  $\mathbb{Z}[i]$ -model with geometrically integral complex fiber  $C_{\mathbb{C}}$ . Nonordinary  $\mathbf{Oz}[i]$ -points exist if and only if  $\tilde{C}_{\mathbb{C}} \cong \mathbb{A}_{\mathbb{C}}^1$  and  $\mathcal{C}(\mathbb{Z}[i]) \neq \emptyset$ . Then every ordinary Gaussian point is the constant term of a proper class of finite-support nonordinary points.*

*Proof.* A nonordinary tuple lies in a complex Hahn workspace and is nonconstant there, because  $\mathbf{Oz}[i] \cap \mathbb{C} = \mathbb{Z}[i]$ . It forces the normalization condition by [theorem 17.25](#). Constant extraction is a ring homomorphism fixing  $\mathbb{Z}[i]$ , so applying it to the defining equations gives an ordinary Gaussian point. The converse is [proposition 17.42](#).  $\square$

For real omnific points the converse holds at every nonsingular ordinary point and at every singular ordinary point with a real preimage under the normalization. A real point of a singular curve can have only nonreal preimages, so the complex argument does not transfer, and the real singular fibers are not classified ([question 21.24](#)) [C16]. For example,  $Y^2 = X^2(X - 1)$  has affine-line normalization  $(X, Y) = (T^2 + 1, T(T^2 + 1))$ , but the ordinary point  $(0, 0)$  has only the preimages  $T = \pm i$ . Thus the real polynomial-family construction does not apply at that point; this observation alone does not rule out all possible Hahn points in its constant-term fiber. For smooth curves the fibers are known exactly ([corollary 17.12](#)); for singular exceptional curves [proposition 17.42](#) gives families in the fibers, not the entire fibers: no lifting theorem for every point is proved ([question 21.23](#)).

The criterion decides whether a curve supports nonconstant points once its affine normalization is known, and for superelliptic equations [theorem 17.38](#) makes this a multiplicity computation. It does not decide whether ordinary integer points exist; the implication  $\mathcal{C}(\mathbf{Oz}) \neq \emptyset \Rightarrow \mathcal{C}(\mathbb{Z}) \neq \emptyset$  already follows from the retraction. What it decides is whether *additional* nonordinary solutions can occur [C16].

### 17.6.9 Boundaries of the criterion

**The field is different.** The Hahn field contains  $\omega^{\gamma}(1 + \omega^{-2\gamma})^{1/2}$  for every  $\gamma > 0$ , while  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  does not, and similar expansions give nonconstant field points on many curves of positive genus ([remark 17.24](#)). The criterion concerns  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -points, not  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ -points; the prohibition of negative  $\omega$ -exponents is indispensable [C16].

**Constant coefficients.** If a coefficient may be a chosen element of the Hahn ring, a prescribed tuple can be forced to solve a relation ([section 18.5](#)). The proof needs the Euler derivations to kill the coefficients, and the function field to have transcendence degree one over the constant field [C16].

**Characteristic zero is used twice.** Seidenberg's identity divides by  $n!$ , and an Euler derivation must detect a nonzero valuation. In characteristic  $p$  an additive map  $\Gamma \rightarrow \mathbb{k}$  kills  $p\Gamma$  and vanishes on a  $p$ -divisible group ([section 18.3](#)). Seidenberg also treats Hasse–Schmidt operators, but that alone does not supply a valuation-detecting operator. [Proposition 17.31](#) survives in every characteristic; the genus-zero conclusion is not asserted in positive characteristic [C16].

**Higher dimension.** For a curve a nontrivial valuation trivial on constants is controlled by one discrete place, and one nonzero regular differential has a scalar local expression  $h du$  whose contraction is nonzero with known positive valuation. In higher dimension valuations need not be discrete, the derivation can annihilate chosen global forms, and a centre may have several tangent directions. The conductor identity holds for finite birational extensions of

domains in any dimension, but the geometric step needs new hypotheses. The logarithmic and symmetric-differential criteria of [theorem 16.19](#) and [corollary 18.18](#) are not subsumed by this curve argument ([question 21.28](#)) [C16].

**Normalization lifting.** The criterion concerns the *existence* of a nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point and of a polynomial arc. It does not say that a given nonconstant point of a singular exceptional curve factors through an  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point of the normalization; [theorem 17.28](#) deliberately allows  $b_j \notin \mathcal{A}_{\mathbb{k}}(\Gamma)$ . So the complete existence dichotomy and the open lifting problem ([question 21.23](#)) are consistent [C16].

### 17.6.10 Source C16's own audit

Source C16 ends with a table of the pitfalls it checked. Support reversal:  $t^\gamma = \omega^{-\gamma}$  throughout, and its ring has nonpositive  $t$ -support. Normality of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ : never assumed, with an explicit counterexample in  $\text{Frac } \mathcal{A}_{\mathbb{k}}(\Gamma)$ . The conductor: it comes from the finite normalization of the finite-type curve, not from properties of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ . Infinite differential closure: only the finite identity ([17.2](#)) is used, with no Noetherian hypothesis on a differential hull. High rank: the conductor and the local parameter come from one function field, so their valuations are integer multiples of one  $\gamma$ . The derivation may leave  $F$ ; the contraction uses  $\Omega_{F/\mathbb{k}}^1$  and takes values in  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ . Nonzero contraction:  $\alpha = h du$  with  $h \neq 0$ , and the chosen derivation keeps the leading term of  $u$ . No hidden infinite sum: the only auxiliary series is formal in a new variable, and one coefficient is used. Arithmetic and geometry: ordinary integer points and real normalization fibers are kept separate from the geometric criterion. Proper classes: all algebraic geometry takes place in a set-sized workspace containing the finite tuple. Novelty: a proposed extension of the inspected baseline, with no exhaustive priority or peer-review claim.

Its script checks the finite Taylor identity, Euler derivations on finite rational supports, the multiplicity and genus arithmetic of [section 17.6.6](#), explicit normalization substitutions, the elliptic Bézout certificate and the derivative-order inequalities, with exact arithmetic ([appendix C](#)). These are regression checks for finite parts of the argument, not proofs of Hahn summability, normalization, Riemann–Roch or the universally quantified theorems; proofs, finite checks and a clean build are different kinds of evidence [C16]. Its summary: the smooth result extends through arbitrary affine-curve singularities, not by lifting a point into a normal ring but by keeping one conductor of the original curve to bound all powers of a contracted differential, which the one-dimensional place turns into a finite contradiction; the only escape is normalization by the affine line, and transfinite supports or larger scales cannot make a rigid constant curve flexible. What remains concerns the exceptional fibers, higher dimension and effective verification ([questions 21.23](#) to [21.30](#)).

## 18 Group varieties, coefficient algebras and sharpness

Until the explicit change of characteristic in [section 18.3](#),  $\mathbb{k}$  has characteristic zero. The exponent group  $\Gamma$  may be zero; then  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}$  and  $\Pi_{\mathbb{k}}(\Gamma) = 0$ . For a  $\mathbb{k}$ -scheme  $X$ , the notation  $\text{ct} : X(\mathcal{A}_{\mathbb{k}}(\Gamma)) \rightarrow X(\mathbb{k})$  denotes the map induced by the ring retraction. A constant point in  $X(\mathcal{A}_{\mathbb{k}}(\Gamma))$  is obtained by extending scalars along  $\mathbb{k} \hookrightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$ .

### 18.1 Abelian and semiabelian varieties

**Lemma 18.1** (Invariant differentials [C11, C12, C13, C14, C15]). *For a smooth algebraic group  $G$  over  $\mathbb{k}$  with identity  $e$ , translation gives an isomorphism  $\Omega_{G/\mathbb{k}}^1 \cong \mathcal{O}_G \otimes_{\mathbb{k}} e^* \Omega_{G/\mathbb{k}}^1$ ; in particular*

$\Omega_{G/\mathbb{k}}^1$  is generated by global invariant one-forms.

*Proof.* Differentiating the multiplication  $G \times G \rightarrow G$  in the second factor along  $G \times \{e\}$  gives a morphism of vector bundles  $G \times T_e G \rightarrow TG$ ,  $(g, v) \mapsto d(L_g)_e(v)$ , whose inverse comes from translation by  $g^{-1}$  and is algebraic because inversion is a morphism. Its fiber at each  $g$  is an isomorphism with inverse  $d(L_{g^{-1}})_g$ , so it trivializes  $TG$ ; dualizing gives the displayed cotangent isomorphism. A basis at the identity extends to global left-invariant forms spanning every fiber [C12, C13]. The statement is also [43, Tag 047I] and, for abelian varieties, [48, Chapter IV, Proposition 6.4(b)].  $\square$

**Theorem 18.2** (Abelian varieties [C10, C11, C13, C14, C15; also C12]). *For every abelian variety  $A$  over  $\mathbb{k}$ ,  $A(\mathcal{A}_{\mathbb{k}}(\Gamma)) = A(\mathbb{k})$  as groups, with  $\text{ct}$  inverse to the inclusion. Every locally closed subscheme of  $A$ , singular or not, has only constant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -points.*

*Proof.* An abelian variety is smooth and proper, and lemma 18.1 gives global generation; apply corollary 16.17 and proposition 16.21. The maps come from ring maps and preserve the group law.  $\square$

This concerns scheme-valued points over the ring, including points not given by a unimodular tuple in a fixed projective embedding; it is not the statement  $A(\mathbb{H}_{\mathbb{k}}(\Gamma)) = A(\mathbb{k})$ , which is false, and it is not restricted to torsion points [C10, C11, C13]. For  $\Gamma = \mathbb{Z}$  it is the familiar constancy of polynomially parametrized points of an abelian variety; for other groups the extra infinite supports create no new points [C10]. Source C12 proves the theorem over algebraically closed  $\mathbb{k}$  and for closed subvarieties.

**Lemma 18.3** (Tori [C14; also C10, C11, C12, C13, C15]). *For every  $\mathbb{k}$ -torus  $T$ ,  $T(\mathcal{A}_{\mathbb{k}}(\Gamma)) = T(\mathbb{k})$ .*

*Proof.* Over  $\bar{\mathbb{k}}$  the torus is  $\mathbb{G}_m^r$ , whose points over  $\mathcal{A}_{\bar{\mathbb{k}}}(\Gamma)$  are tuples of units, hence constants. Let  $t \in T(\mathcal{A}_{\mathbb{k}}(\Gamma))$  and  $t_0 = \text{ct}(t)$ . Coefficient extension commutes with constant extraction, so the image of  $t$  is the constant extension of the image of  $t_0$ . The two points agree after the injective coefficientwise map  $\mathcal{A}_{\mathbb{k}}(\Gamma) \hookrightarrow \mathcal{A}_{\bar{\mathbb{k}}}(\Gamma)$ . Since  $T$  is affine, equality of the extended coordinate homomorphisms and injectivity give  $t = t_0$  already over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ . Equivalently, the coordinate map takes values in  $\mathcal{A}_{\mathbb{k}}(\Gamma) \cap \bar{\mathbb{k}} = \mathbb{k}$  [C14].  $\square$

Source C10 descends instead along a finite splitting field  $L/\mathbb{k}$ : coefficientwise scalar extension gives  $\mathcal{A}_{\mathbb{k}}(\Gamma) \otimes_{\mathbb{k}} L \cong \mathcal{A}_L(\Gamma)$ , because a finite union of reverse-well-ordered supports is reverse well ordered. More explicitly, choose a finite  $\mathbb{k}$ -basis of  $L$  and split every coefficient in that basis. The resulting finitely many component series have supports contained in the original support; conversely their union is admissible. Linear independence of the basis proves injectivity, so this is the natural tensor-product isomorphism. The faithfully flat extension  $\mathcal{A}_{\mathbb{k}}(\Gamma) \rightarrow \mathcal{A}_{\mathbb{k}}(\Gamma) \otimes_{\mathbb{k}} L$  then reflects equality of points. Source C13 does the same from  $\mathbb{R}$  to  $\mathbb{C}$ , as in `ogl:alg:lem:torus`, and source C11 descends from  $\bar{\mathbb{k}}$  through lemma 16.13; source C12 treats split tori over an algebraically closed field. Source C15 descends along a finite splitting field  $L/\mathbb{k}$  using only the injectivity of the coefficientwise map  $\mathcal{A}_{\mathbb{k}}(\Gamma) \rightarrow \mathcal{A}_L(\Gamma)$  and the affineness of  $T$ .

**Theorem 18.4** (Semiabelian varieties [C10, C11, C12, C13, C14, C15]). *If  $1 \rightarrow T \rightarrow G \xrightarrow{\pi} A \rightarrow 0$  is an extension of an abelian variety by a torus over  $\mathbb{k}$ , then  $G(\mathcal{A}_{\mathbb{k}}(\Gamma)) = G(\mathbb{k})$ .*

*Proof.* For  $\sigma \in G(\mathcal{A}_{\mathbb{k}}(\Gamma))$  put  $\sigma_0 = \text{ct}(\sigma) \in G(\mathbb{k})$ . By theorem 18.2,  $\pi(\sigma)$  is constant with constant term  $\pi(\sigma_0)$ , so  $\pi(\sigma) = \pi(\sigma_0)$  and  $\sigma_0^{-1}\sigma$  lies in the scheme-theoretic kernel  $T(\mathcal{A}_{\mathbb{k}}(\Gamma))$ . By lemma 18.3 it is constant, and its constant term is the identity; so  $\sigma = \sigma_0$ .  $\square$

No surjectivity of  $G(\mathbb{k}) \rightarrow A(\mathbb{k})$  is used: the lift  $\sigma_0$  is the constant term of the given point [C10, C13]. Properness is not assumed for  $G$ ; the torus part is handled by the unit lemma, and a trivial cotangent sheaf alone is not enough, since  $\mathbb{G}_a(\mathcal{A}_{\mathbb{k}}(\Gamma)) = \mathcal{A}_{\mathbb{k}}(\Gamma)$  [C11, C14].

**Theorem 18.5** (Quasi-finite maps [C11, C15; also C10, C12, C14]). *Let  $f : X \rightarrow Y$  be a quasi-finite morphism of separated  $\mathbb{k}$ -schemes of finite type with  $Y(\mathcal{A}_{\mathbb{k}}(\Gamma)) = Y(\mathbb{k})$ . Then  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$ .*

*Proof.* A  $\mathbb{k}$ -scheme  $Z$  of finite type and dimension zero is affine and finite over  $\mathbb{k}$  [59, Lemma 33.20.2]. Write  $Z = \operatorname{Spec} E$ . For any  $\mathbb{k}$ -algebra map  $E \rightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$ , every image element is algebraic over  $\mathbb{k}$ , hence belongs to  $\mathbb{k}$  by lemma 16.1(d). Thus the map factors through  $\mathbb{k}$ , necessarily by its own constant term, and  $Z(\mathcal{A}_{\mathbb{k}}(\Gamma)) = Z(\mathbb{k})$ . This includes nonreduced  $Z$ : its nilpotents map to zero in the domain  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  [C11, C14]. For  $x \in X(\mathcal{A}_{\mathbb{k}}(\Gamma))$  write  $y = \operatorname{ct}(f(x)) \in Y(\mathbb{k})$ . Rigidity of  $Y$  says  $f(x)$  is the constant extension of  $y$ , so the fiber-product property factors  $x$  through  $X_y$ . Quasi-finiteness makes this a zero-dimensional finite-type scheme over  $\mathbb{k}$ , and the preceding argument makes  $x$  constant. No finiteness or surjectivity of  $f$  is needed.  $\square$

Thus locally closed subschemes and finite covers of semiabelian varieties, of positive-genus curves and of their finite products are rigid, and so are finite covers of products of positive-genus curves [C10, C11, C12, C14]. Source C14 states the finite case and source C10 the quasi-finite case over semiabelian and other rigid separated targets; source C10 proves the zero-dimensional step by corollary 16.4. Adjoining a free affine-line factor destroys rigidity: a product  $X \times \mathbb{A}^1$  has nonconstant points through each  $\mathbb{k}$ -point of  $X$  [C12].

**Theorem 18.6** (The nonconstant part of a commutative group [C13], for every  $\mathbb{k}$ ). *Let  $G$  be a smooth connected commutative algebraic group over  $\mathbb{k}$ , let  $U$  be its maximal connected unipotent affine subgroup, and fix an isomorphism  $U \cong \mathbb{G}_a^d$ . Then  $\operatorname{ct}$  gives a split exact sequence of abelian groups*

$$0 \longrightarrow \Pi_{\mathbb{k}}(\Gamma)^d \longrightarrow G(\mathcal{A}_{\mathbb{k}}(\Gamma)) \xrightarrow{\operatorname{ct}} G(\mathbb{k}) \longrightarrow 0, \quad \text{so} \quad G(\mathcal{A}_{\mathbb{k}}(\Gamma)) \cong G(\mathbb{k}) \oplus \Pi_{\mathbb{k}}(\Gamma)^d.$$

*The kernel is canonically  $\ker(U(\mathcal{A}_{\mathbb{k}}(\Gamma)) \rightarrow U(\mathbb{k}))$ ; its identification with  $\Pi_{\mathbb{k}}(\Gamma)^d$  depends on the chosen coordinates. For  $\Gamma = \mathbf{No}$  in the class convention the kernel is  $\Pi^d$  over  $\mathbb{R}$  and  $\Pi_{\mathbb{C}}^d$  over  $\mathbb{C}$ .*

*Proof.* By the Barsotti–Chevalley theorem there is a connected affine normal subgroup  $N \subseteq G$  with  $G/N$  an abelian variety; since  $G$  is commutative and  $\mathbb{k}$  is perfect,  $N \cong T \times U$  with  $T$  a torus and  $U$  unipotent, and in characteristic zero the commutative unipotent group  $U$  is a vector group [49, Theorem 8.27, Corollaries 14.33, 16.15]. Here the product decomposition is applied to the affine group  $N$ ; it is not a decomposition of  $G$  itself. The largest connected affine subgroup contains every connected unipotent affine subgroup of  $G$ , so the  $U$  in this decomposition is the subgroup in the statement. Thus  $G/U$  is an extension of  $G/N$  by  $N/U \cong T$ , a semiabelian variety. For  $\sigma \in G(\mathcal{A}_{\mathbb{k}}(\Gamma))$  with  $\sigma_0 = \operatorname{ct}(\sigma)$ , the image of  $\sigma$  in  $(G/U)(\mathcal{A}_{\mathbb{k}}(\Gamma))$  is constant by theorem 18.4 and equals that of  $\sigma_0$ ; so  $\sigma - \sigma_0 \in U(\mathcal{A}_{\mathbb{k}}(\Gamma))$  has constant term 0, that is, lies in  $\Pi_{\mathbb{k}}(\Gamma)^d$  in the chosen coordinates. Conversely  $\Pi_{\mathbb{k}}(\Gamma)^d \subseteq U(\mathcal{A}_{\mathbb{k}}(\Gamma))$  lies in the kernel of  $\operatorname{ct}$ . Let  $\iota : U \hookrightarrow G$  denote inclusion, using the chosen vector coordinates on  $U$ . The splitting and its inverse are explicitly

$$(a, s) \longmapsto a + \iota(s), \quad \sigma \longmapsto (\operatorname{ct}(\sigma), \iota^{-1}(\sigma - \operatorname{ct}(\sigma))).$$

The preceding kernel calculation justifies the second coordinate. These maps are mutually inverse group homomorphisms because  $G$  is commutative, proving the direct-sum assertion, also when  $d = 0$  or  $\Gamma = 0$ .  $\square$

Source C13 states and proves the theorem for  $\mathbb{k} = \mathbb{R}$  and  $\mathbb{C}$ . Its structure inputs are stated in its reference for perfect fields and for characteristic zero, and the torus and semiabelian steps hold over every field of characteristic zero (lemma 18.3 and theorem 18.4); so the statement is printed for every such  $\mathbb{k}$ . The cited statements have now been checked in the 2017 edition, on printed pages 154, 289 and 329; in the last, the affine-group convention is essential. The splitting is a splitting of groups of points supplied by the retraction, not an algebraic splitting of  $G$ : an extension of an abelian variety by a vector group can be nontrivial. It is not a formal-completion statement either:  $\Pi_{\mathbb{k}}(\Gamma)$  is not nilpotent, and its nonzero elements are infinite, not infinitesimal [C13]. For instance  $\mathbb{G}_a(\mathcal{A}_{\mathbb{k}}(\Gamma)) = \mathbb{k} \oplus \Pi_{\mathbb{k}}(\Gamma)$ ,  $\mathbb{G}_m(\mathcal{A}_{\mathbb{k}}(\Gamma)) = \mathbb{k}^{\times}$  and  $A(\mathcal{A}_{\mathbb{k}}(\Gamma)) = A(\mathbb{k})$ . No analogous claim is made for noncommutative groups:  $\mathrm{SL}_2(\mathcal{A}_{\mathbb{k}}(\Gamma))$  and  $\mathrm{GL}_2(\mathcal{A}_{\mathbb{k}}(\Gamma))$  contain the nonconstant unipotent matrices  $\begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}$ ,  $u \in \Pi_{\mathbb{k}}(\Gamma)$  [C12, C13].

*Remark 18.7* (Consistency with the omnific groups report [merge]). The report `docs/surreal/omnific-groups-and-lattices` decides rigidity for closed subgroup schemes of  $\mathrm{GL}_N$ : over  $\mathcal{B}_{\mathbb{R}}$  exactly when the real fiber has no nonidentity unipotent element, over  $\mathcal{B}_{\mathbb{C}}$  exactly when the identity component is a torus (`ogl:alg:thm:real`, `ogl:alg:thm:complex`), which answers question 21.1 for group schemes (`ogl:alg:rem:affine`). The two answers agree where they overlap. (1) In dimension one,  $\mathbb{G}_a$  is the affine line and is nonrigid in both;  $\mathbb{G}_m$ , which is  $\mathbb{P}^1$  minus two points, and the anisotropic torus  $\mathrm{SO}(2)$ , the conic  $x^2 + y^2 = 1$  with two conjugate points at infinity, are rigid in both. (2) For a commutative unipotent group the Baker–Campbell–Hausdorff product of `ogl:alg:prop:bch` is addition, and its kernel  $\exp(\mathfrak{u} \otimes \Pi_{\mathbb{k}})$  is the group  $\Pi_{\mathbb{k}}(\Gamma)^d$  of theorem 18.6 in vector coordinates; the latter theorem extends the description to commutative groups that are not affine. (3) Semiabelian rigidity extends the torus freezing of `ogl:alg:lem:torus` beyond linear groups. (4) For smooth curves real and complex rigidity coincide (corollary 17.12(a)), but for groups they do not:  $\mathrm{SO}_3$  is rigid over  $\mathcal{B}_{\mathbb{R}}$  because  $\mathrm{SO}_3(\mathbb{R})$  is compact, and not over  $\mathcal{B}_{\mathbb{C}}$  because  $\mathrm{SO}_{3,\mathbb{C}}$  is not a torus. There is no conflict, because sources C10–C14 transfer rigidity only from the complex fiber to the real one, never conversely.

## 18.2 Reduced coefficient algebras and dual numbers

For a commutative  $\mathbb{k}$ -algebra  $S$  let  $\mathcal{A}_S(\Gamma)$  be the formal sums  $\sum_{\gamma} a_{\gamma} \omega^{\gamma}$  with  $a_{\gamma} \in S$  and reverse-well-ordered support in  $\Gamma^{\geq 0}$ . The Hahn support lemma makes this a ring even when  $S$  has zero divisors, with maps  $S \hookrightarrow \mathcal{A}_S(\Gamma) \xrightarrow{\mathrm{ct}} S$  whose composite is the identity; for such  $S$  no leading coefficient is divided by [C11].

**Lemma 18.8** (Coefficientwise separation [C11]). *If  $S$  is reduced, the coefficientwise maps  $\mathcal{A}_S(\Gamma) \rightarrow \mathcal{A}_{F_{\mathfrak{p}}}(\Gamma)$ , with  $F_{\mathfrak{p}} = \mathrm{Frac}(S/\mathfrak{p})$  for the primes  $\mathfrak{p}$  of  $S$ , have kernels with zero intersection.*

*Proof.* For each prime  $\mathfrak{p}$ , the map  $S \rightarrow F_{\mathfrak{p}}$  has kernel  $\mathfrak{p}$ , so a series vanishes under that coefficientwise map exactly when all its coefficients lie in  $\mathfrak{p}$ . A series in every kernel therefore has each coefficient in the nilradical of  $S$ , which is zero. No uniform finite support or finite collection of primes is required.  $\square$

**Theorem 18.9** (Reduced coefficient algebras [C11]). *Let  $X$  be a separated  $\mathbb{k}$ -scheme with  $X(\mathcal{A}_F(\Gamma)) = X(F)$  for every field extension  $F/\mathbb{k}$ . Then for every reduced commutative  $\mathbb{k}$ -algebra  $S$  the inclusion and `ct` are inverse bijections  $X(S) \cong X(\mathcal{A}_S(\Gamma))$ . This applies to smooth proper varieties with globally generated cotangent sheaf, to semiabelian varieties, to smooth geometrically integral affine curves not isomorphic to the affine line, and to separated schemes of finite type with a quasi-finite map to one of these.*

*Proof.* Injectivity holds because  $\text{ct}$  is a retraction. For  $\sigma \in X(\mathcal{A}_S(\Gamma))$  let  $\sigma_0$  be the constant point of  $\text{ct}(\sigma)$ . For each prime  $\mathfrak{p}$  the image of  $\sigma$  over  $\mathcal{A}_{F_{\mathfrak{p}}}(\Gamma)$  is constant by hypothesis, and since all maps are coefficientwise its constant term is the image of  $\text{ct}(\sigma)$ ; so  $\sigma$  and  $\sigma_0$  agree over every  $\mathcal{A}_{F_{\mathfrak{p}}}(\Gamma)$ . To spell out the descent, separatedness makes their equalizer a closed subscheme of  $\text{Spec } \mathcal{A}_S(\Gamma)$ , defined by an ideal  $J$ . Equality after each coefficient map puts  $J$  in every kernel. Their zero intersection, proved in [lemma 18.8](#), gives  $J = 0$ , so the equalizer is the whole spectrum and  $\sigma = \sigma_0$ . If  $S$  is the zero ring both spectra are empty and the assertion is immediate. The listed classes satisfy the hypothesis over every  $F \supseteq \mathbb{k}$ , which has characteristic zero: properness, smoothness and global generation are stable under field extension ([corollary 16.17](#) and [theorem 18.4](#)), the affine-line condition is geometric ([theorem 17.6](#)), and [theorem 18.5](#) applies after base change.  $\square$

The proof uses joint injectivity of the coefficient maps and the closed diagonal, not a faithfully flat cover. Nor does it identify  $\mathcal{A}_S(\Gamma)$  with  $S \otimes_{\mathbb{k}} \mathcal{A}_{\mathbb{k}}(\Gamma)$  [C11]. For example, take  $S = \mathbb{k}[s]$  and  $\Gamma = \mathbb{Q}$ . The series

$$\sum_{n \geq 1} s^n \omega^{1/n} \in \mathcal{A}_{\mathbb{k}[s]}(\mathbb{Q})$$

has reverse-well-ordered positive support. Its coefficients span an infinite-dimensional  $\mathbb{k}$ -subspace of  $S$ . The coefficients of the image of any tensor  $\sum_{j=1}^r a_j \otimes f_j$  lie in the finite-dimensional span of  $a_1, \dots, a_r$ , so the natural tensor map is not surjective in this example. This does not assert failure of flatness. The principle uses all primes and needs no Jacobson or finite-type condition on  $S$ ; for  $S = \mathbb{k}[s_1, \dots, s_r]$  it says that a family of points with Hahn coordinates and polynomial coefficient functions already comes from a family over  $S$  [C11].

**Lemma 18.10** (Square-zero lifts [C11]). *Let  $X$  be locally of finite type over  $\mathbb{k}$ ,  $S$  a  $\mathbb{k}$ -algebra and  $p \in X(S)$ . Write  $S[\epsilon] := S[T]/(T^2)$ , with  $\epsilon$  the class of  $T$ . With the constant lift as origin, the lifts of  $p$  to  $S[\epsilon]$  correspond to  $\text{Hom}_S(p^* \Omega_{X/\mathbb{k}}^1, S)$ ; if  $p$  comes from a smooth point  $p_0 \in X(\mathbb{k})$ , this is  $T_{p_0} X \otimes_{\mathbb{k}} S$ .*

*Proof.* The square-zero closed immersion  $\text{Spec } S \hookrightarrow \text{Spec } S[\epsilon]$  is a homeomorphism on underlying spaces. Hence a lift has the same underlying map as  $p$ , and can be described on the same affine charts. Locally its ring map is uniquely  $\varphi + \epsilon\delta$ , with  $\delta$  additive and zero on  $\mathbb{k}$ . Multiplicativity is equivalent to  $\delta(ab) = \varphi(a)\delta(b) + \varphi(b)\delta(a)$ , because  $\epsilon^2 = 0$ . The universal property of Kähler differentials identifies these derivations with the displayed Hom module. This construction respects restrictions to smaller charts, so the local descriptions glue. For a constant smooth  $\mathbb{k}$ -point  $p_0$ , its cotangent space is finite dimensional, and taking its dual commutes with scalar extension to  $S$ . This gives  $T_{p_0} X \otimes_{\mathbb{k}} S$ .  $\square$

**Theorem 18.11** (The dual-number defect [C11]). *For an abelian variety  $A$  over  $\mathbb{k}$  there is a canonical split description*

$$A(\mathcal{A}_{\mathbb{k}}(\Gamma)[\epsilon]) \cong A(\mathbb{k}) \oplus (\text{Lie } A \otimes_{\mathbb{k}} \mathcal{A}_{\mathbb{k}}(\Gamma)),$$

*under which  $A(\mathbb{k}[\epsilon]) = A(\mathbb{k}) \oplus \text{Lie } A$ . Since  $\mathcal{A}_{\mathbb{k}[\epsilon]}(\Gamma) \cong \mathcal{A}_{\mathbb{k}}(\Gamma)[\epsilon]$  coefficientwise,*

$$A(\mathcal{A}_{\mathbb{k}[\epsilon]}(\Gamma))/A(\mathbb{k}[\epsilon]) \cong \text{Lie } A \otimes_{\mathbb{k}} \Pi_{\mathbb{k}}(\Gamma).$$

*Proof.* First split every coefficient  $a + \epsilon b$  into its two components. Their supports are subsets of the original support; conversely the union of two reverse-well-ordered supports is reverse well ordered. This proves the natural ring isomorphism  $\mathcal{A}_{\mathbb{k}[\epsilon]}(\Gamma) \cong \mathcal{A}_{\mathbb{k}}(\Gamma)[\epsilon]$  and its compatibility with reduction and constant extraction.

Reduction modulo  $\epsilon$  maps  $A(\mathcal{A}_{\mathbb{k}}(\Gamma)[\epsilon])$  onto  $A(\mathcal{A}_{\mathbb{k}}(\Gamma)) = A(\mathbb{k})$  (theorem 18.2), split by constant lifts. By lemma 18.10 the kernel over the identity is  $\text{Lie } A \otimes_{\mathbb{k}} \mathcal{A}_{\mathbb{k}}(\Gamma)$ , and on square-zero tangent vectors the group law is addition: the differential of multiplication at  $(e, e)$  sends  $(v, w)$  to  $v + w$ , and all products of the first-order errors vanish modulo  $\epsilon^2$ . Commutativity makes the split extension a direct sum. Applying the same natural construction over  $\mathbb{k}$  identifies the constant subgroup as  $A(\mathbb{k}) \oplus (\text{Lie } A \otimes_{\mathbb{k}} \mathbb{k})$ . Finally the additive decomposition  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k} \oplus \Pi_{\mathbb{k}}(\Gamma)$  splits  $\text{Lie } A \otimes_{\mathbb{k}} \mathcal{A}_{\mathbb{k}}(\Gamma)$  into the ordinary tangent space and  $\text{Lie } A \otimes_{\mathbb{k}} \Pi_{\mathbb{k}}(\Gamma)$ ; quotienting by the former gives the assertion.  $\square$

If  $\dim A > 0$  and  $\Gamma \neq 0$ , a nonzero  $v \in \text{Lie } A$  and  $h = \omega^\gamma$  with  $\gamma > 0$  give the new point  $\epsilon hv$ ; so reducedness cannot be dropped in theorem 18.9, even for elliptic curves. Concretely, on the smooth affine curve  $Y^2 = X^3 - X$  the pair  $(0, \epsilon h)$  with  $0 \neq h \in \Pi_{\mathbb{k}}(\Gamma)$  is a point over  $\mathcal{A}_{\mathbb{k}[\epsilon]}(\Gamma)$  not coming from  $\mathbb{k}[\epsilon]$ . Here  $\partial(Y^2 - X^3 + X)/\partial X = 1$  at  $(0, 0)$ , so the chosen point is smooth, and substitution gives zero because  $(\epsilon h)^2 = 0$ . Its  $Y$ -coordinate has a nonzero positive-growth coefficient and therefore is not in  $\mathbb{k}[\epsilon]$ . No omnific ring contains such a nilpotent; the example belongs to the coefficient-ring extension of the theory only [C11].

### 18.3 Positive characteristic

**Proposition 18.12** (An elliptic point in characteristic two [C11, C14, C15]). *Let  $\mathbb{k}$  have characteristic two. In  $\mathcal{A}_{\mathbb{k}}(\mathbb{Q})$  put*

$$x = \omega^{1/3}, \quad y = \sum_{n \geq 1} \omega^{1/2^n}.$$

*Then  $y^2 + y = x^3$ , and the projective closure  $Y^2Z + YZ^2 = X^3$  is a smooth cubic of genus one with the rational point  $[0 : 1 : 0]$ . The point  $(x, y)$  is nonconstant.*

*Proof.* The support  $\{2^{-n} : n \geq 1\}$  is positive and reverse well ordered: every nonempty subset has a largest exponent, indexed by its least  $n$ . The Hahn product has only finitely many contributions at each exponent. In characteristic two the off-diagonal terms pair as  $(i, j)$  and  $(j, i)$  and cancel, leaving precisely the diagonal terms. Consequently

$$y^2 = \sum_{n \geq 1} \omega^{2^{1-n}} = \omega + \sum_{n \geq 1} \omega^{2^{-n}} = x^3 + y.$$

This is an identity of Hahn coefficients, without a convergence claim for the sequence of partial sums. The partial derivatives of  $Y^2Z + YZ^2 - X^3$  are  $X^2$ ,  $Z^2$  and  $Y^2$  in characteristic two, with no common projective zero; a smooth plane cubic has genus one.  $\square$

Source C14 states this for every field of characteristic two with  $\Gamma = \mathbb{Q}$ ; source C11 states it for  $\mathbb{k} = \mathbb{F}_2$  and  $\Gamma = \mathbb{Z}[\frac{1}{2}]$  with the point  $x = \omega$ ,  $y = \sum_{n \geq 1} \omega^{3/2^n}$ , which is the same example with all exponents multiplied by three. Source C15 states it for every field of characteristic two with  $\Gamma = \mathbb{Z}[\frac{1}{2}]$  and the point  $x = \sum_{n \geq 1} \omega^{3/2^n}$ ,  $y = \omega$  of  $x^2 + x = y^3$ , which is source C11's example with  $x$  and  $y$  interchanged. The partial sums  $y_N$  satisfy  $y_N^2 + y_N + x^3 = \omega^{2^{-N}}$ , and this residual monomial does not tend to zero in the valuation topology; the proof is an exact coefficient identity, not a limit [C11, C14]. Every additive map from a 2-divisible group to a field of characteristic two is zero, since  $\lambda(\gamma) = 2\lambda(\gamma/2)$ ; so the separating Euler family disappears exactly where the characteristic-zero argument needs it. The exponent group has rank one, so neither rank nor non-Archimedean complexity explains the failure. The telescoping is the Artin–Schreier phenomenon in Hahn form and is not claimed as new [C11].

**Proposition 18.13** (Squarefree rigidity fails in characteristic  $p$  [C13]). *Over  $\mathbb{F}_p$  with  $\Gamma = \mathbb{Q}$ , the elements  $x = \omega$  and  $y = \omega - \omega^{1/p}$  of  $\mathcal{A}_{\mathbb{F}_p}(\mathbb{Q})$  satisfy  $y^p = x^p - x$ , although  $X^p - X$  is squarefree.*

*Proof.* The Frobenius is additive in characteristic  $p$ , so  $y^p = \omega^p - \omega = x^p - x$ ; and  $(X^p - X)' = -1$ .  $\square$

The curve  $y^p = x^p - x$  is itself an affine line in characteristic  $p$ : with  $w = y - x$  one has  $w^p = -x$ , so  $x = -w^p$  and  $y = w - w^p$  [merge]. Thus this example refutes the characteristic- $p$  analogue of [theorem 16.6](#), not of [theorem 17.6](#); the genus-one example of [proposition 18.12](#) refutes the latter. Characteristic zero is used twice: to separate exponents by rational linear functionals and to differentiate minimal polynomials of algebraic elements; ordinary derivations in characteristic  $p$  kill  $p$ -th powers, so the proofs give no positive-characteristic classification [C10, C12, C13].

## 18.4 Discretely ordered rings

The rigidity singles out the omnific integers; it does not extend to every discretely ordered subring of a surreal field. Let  $\mathbb{R}((\omega^{-1}))$  be the ordered field of Laurent series in the positive infinitesimal  $\omega^{-1}$ , a subfield of  $\mathbf{No}$ , and put

$$x = \omega^2, \quad y = \omega^3(1 - \omega^{-4})^{1/2} = \omega^3 - \frac{1}{2}\omega^{-1} - \frac{1}{8}\omega^{-5} - \frac{1}{16}\omega^{-9} - \dots,$$

so that  $y^2 = \omega^6 - \omega^2 = x^3 - x$ .

**Theorem 18.14** (A discretely ordered ring with no omnific copy [C11]). *The ring  $R_0 = \mathbb{Z}[x, y] \subseteq \mathbb{R}((\omega^{-1}))$  is isomorphic to  $\mathbb{Z}[X, Y]/(Y^2 - X^3 + X)$  and is discretely ordered by the order of  $\mathbf{No}$ . It admits no injective unital ring homomorphism into  $\mathbf{Oz}$ , although it is an ordered subring of  $\mathbf{No}$ .*

*Proof.* Division by the monic polynomial  $Y^2 - X^3 + X$  in  $\mathbb{Z}[X][Y]$  gives a unique remainder  $f(X) + Yg(X)$  with  $f, g \in \mathbb{Z}[X]$ . Substitution is well defined because the displayed binomial series squares to  $\omega^6 - \omega^2$ . If  $f \neq 0$ , the leading exponent of  $f(x)$  is  $2 \deg f$ , even; if  $g \neq 0$ , that of  $yg(x)$  is  $3 + 2 \deg g$ , odd, since the leading coefficient of  $y$  is 1. Thus the two leading terms cannot cancel. A nonzero remainder has nonzero image, proving the asserted ring isomorphism and excluding further polynomial relations.

If  $g \neq 0$  or  $\deg f \geq 1$ , the leading exponent of that image is positive and its absolute value exceeds every ordinary integer. Otherwise the image is an ordinary integer. Hence any positive element is at least 1, proving discrete order. If  $\psi : R_0 \rightarrow \mathbf{Oz}$  were an injective unital homomorphism, then  $\psi(x), \psi(y)$  would solve the nonsingular equation  $Y^2 = X^3 - X$ , so both would be integers by [corollary 16.9](#). Put  $n = \psi(x) \in \mathbb{Z}$ . A unital ring homomorphism fixes the ordinary integers, so  $\psi(x - n) = 0$ . But  $x - n = \omega^2 - n \neq 0$ , contradicting injectivity.  $\square$

The element  $y$  has an infinite leading term and a nonzero infinitesimal tail, so it is not omnific; discrete order says that no element lies strictly between 0 and 1, not that elements lack infinitesimal terms [C11]. So the canonical integer part  $\mathbf{Oz}$  of  $\mathbf{No}$  need not contain a copy of every discretely ordered ring inside  $\mathbf{No}$ . Van den Dries studied which curves over  $\mathbb{Z}$  have infinite points in *some* discretely ordered ring, in connection with open induction [50]; that quantifier over rings is not the fixed-ring question here. Sources C11, C14 and C15 consulted only the bibliographic record and the abstract of that paper, and no source uses it; the abstract was confirmed during placement, but the paper itself was not read for this report, and no

comparison or priority statement is made. The theorem is consistent with open induction in the nonnegative part of **Oz** (see [remark 10.9](#) and [theorem 10.10](#); the analogue for the definable omnific ring is `dsn:cor:openinduction`), which concerns those rings, not  $R_0$ .

## 18.5 Coefficients, rings and the scope of the method

The rigidity assertions above assume constant coefficients, which the Euler derivations kill. With varying coefficients this reasoning need not apply. One may choose a nonconstant point first and build it into the coefficients [C14]:  $y^2 = x^3 + (1 - \omega^3)$  has the omnific point  $(\omega, 1)$  [C10],  $y^2 = x^3 + x - u^2$  with  $u$  positive infinite has  $(u^2, u^3)$  although  $4 + 27u^4 \neq 0$  [C12], and  $y^2 = x^3 + x - \omega^2$  has  $(\omega^2, \omega^3)$  [C13]. No rigidity is claimed for equations whose coefficients vary at a surreal scale ([question 21.22](#)). Status (batch 33): in dimension zero source C17 proves such a theorem: a monic polynomial over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  whose discriminant is a nonzero constant is a purely infinite translate of its constant-term polynomial ([theorem 19.12](#)); the examples above are curves, where nothing of the kind is claimed.

The valuation ring is not rigid. For  $\gamma > 0$  and  $\varepsilon = \omega^{-\gamma} \in \mathfrak{m}_{\mathbb{k}}(\Gamma)$ , the nonsingular cubic  $y^2 = x^3 - x + 1$  has the nonconstant  $\mathcal{O}_{\mathbb{k}}(\Gamma)$ -point  $x = \varepsilon$ ,  $y = (1 - \varepsilon + \varepsilon^3)^{1/2}$  [C12]; over  $\mathbb{C}$ , at a point  $(x_0, y_0)$  of  $y^2 = x^3 + ax + b$  with  $y_0 \neq 0$ , the formal implicit solution  $x = x_0 + \varepsilon$ ,  $y = y_0 + \frac{3x_0^2 + a}{2y_0}\varepsilon + \cdots$  is another [C10]. For  $\Gamma = \mathbb{Z}$ , the curve  $y^2 = x^3 + 1$  has the  $\mathcal{O}_{\mathbb{k}}(\Gamma)$ -point  $x = \omega^{-1}$ ,  $y = (1 + \omega^{-3})^{1/2}$  [C15]. The two displayed square roots are formal binomial series with constant term 1, supported on nonpositive exponents. For example, the powers of  $-\varepsilon + \varepsilon^3$  have exponents tending downwards in the cyclic subgroup generated by  $\gamma$ ; each coefficient receives only finitely many contributions. The resulting Hahn sum squares to  $1 - \varepsilon + \varepsilon^3$ . This requires no topological convergence in the full value group. The discriminant factor of  $X^3 - X + 1$  is  $4(-1)^3 + 27 = 23 \neq 0$ , so that example is nonsingular. Confusing the two rings would reverse the main conclusion. Together with [remark 17.24](#), this shows that nothing is claimed over **No**, **No** $[i]$  or  $\mathcal{O}_{\text{fin}}$ .

The exponent group may be arbitrary: no order unit, rank, cofinality, cardinality, divisibility or finite-support condition is imposed, and well ordering with finite coefficient convolution suffices. For  $\Gamma = 0$  the ring is  $\mathbb{k}$  and everything is trivially rigid. Nontriviality is needed for nonconstant boundary examples and for the converse direction of [theorem 17.6](#) and [corollary 17.10](#) [C14]. Every field, module, scheme and space of functionals is set-sized; only the final collection of points ranges over a class [C10, C11, C12, C13, C14]. The results classify which smooth curves acquire nonordinary points; they give no enumeration, finiteness statement or height bound for the ordinary points, and they decide no Diophantine problem over  $\mathbb{Z}$  that was not already decided [C10, C11, C12, C13, C14]. What they leave open is recorded in [section 21.4](#).

## 18.6 Logarithmic symmetric differentials

This subsection is built from a sixth manuscript on the subject of the part, *Geometric Rigidity over Omnific Integer Rings*, tagged [C15] after its file prefix 15-logarithmic-rigidity- ([section 1.2](#)). Its pin precedes the placement of sources C10–C14, and it saw none of them. About three quarters of it proves results printed above; they are printed once, and source C15 is added to their credits: the two rings, the Euler derivations and their joint constants ([lemmas 16.1 to 16.3](#) and [corollary 16.4](#)), the squarefree theorem by the strict route, the cubic certificate and the Weierstrass corollary ([theorem 16.6](#), [proposition 16.8](#), and [corollary 16.9](#)), the singular families ([proposition 16.11](#)), symmetric differentials and ample cotangent bundles ([theorem 16.19](#) and [corollary 16.20](#)), inheritance ([proposition 16.21](#)), the curve classification by one logarithmic form over an algebraically closed field and its descent to every field of

characteristic zero (lemma 17.7, theorems 17.6 and 17.8, remark 17.3, and corollary 17.10), unimodular coordinates on curves of positive genus and the Fermat specialization (corollary 17.22), abelian varieties, tori, semiabelian varieties and quasi-finite maps (lemmas 18.1 and 18.3 and theorems 18.2, 18.4 and 18.5), the characteristic-two point (proposition 18.12), and the points over **No** and over the valuation ring (remark 17.24 and section 18.5). Its new material is printed here: logarithmic annihilation on a normal-crossings compactification (theorem 18.16), the tangent-separation criterion it gives (corollary 18.18), the logarithmic hypothesis of its abstract interface (remark 18.19), descent over an arbitrary coefficient ring (theorem 18.20) and closure under finite products (proposition 18.22). Labels of this subsection carry the sub-prefix `odg:log:`. Their manuscript proof review and source comparison are recorded in `RECONCILIATION.md`; Lean formalization remains pending.

**Translation and letters.** Source C15 uses the  $t$ -convention. Its ring  $\mathcal{A}$  of series supported in  $\Gamma_{\leq 0}$ , its ideal  $\mathcal{I}$  of series supported in  $\Gamma_{< 0}$ , its valuation ring  $\mathcal{V}$  and its field  $\mathcal{H} = \mathbb{k}((t^\Gamma))$  are  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ ,  $\Pi_{\mathbb{k}}(\Gamma)$ ,  $\mathcal{O}_{\mathbb{k}}(\Gamma)$  and  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  with exponents negated, and its ring  $R_D = D \oplus \mathcal{I}$  is  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  with  $D$  renamed  $\mathfrak{o}$  ( $D$  is the Pell parameter here). Its derivation  $D_\ell$ , for an additive map  $\ell : \Gamma \rightarrow (\mathbb{k}, +)$ , multiplies  $t^\gamma$  by  $\ell(\gamma)$ . In characteristic zero,  $\ell$  extends uniquely  $\mathbb{Q}$ -linearly to  $\Gamma_{\mathbb{Q}}$ ; using this extension, it is  $\partial_{-\ell}$  in the notation of (16.1), so the two families coincide. Its symmetric power  $q$  is  $r$  here, because  $q$  is the valuation-ring point, as in theorem 16.14. We return to characteristic zero throughout this subsection. Here  $\bar{X}$  is a smooth proper  $\mathbb{k}$ -scheme,  $E \subseteq \bar{X}$  a reduced divisor with simple normal crossings ( $E = \emptyset$  is allowed), and  $X = \bar{X} \setminus E$ ; the letter  $E$  has no other meaning in this subsection. For a field  $L \supseteq \mathbb{k}$  and  $x \in X(L)$  a tangent vector is an  $L$ -linear map  $\xi : x^* \Omega_{X/\mathbb{k}}^1 \rightarrow L$ , and a  $\mathbb{k}$ -derivation  $\partial$  of  $L$  gives the tangent vector  $\partial x$  with  $(\partial x)(df) = \partial(f(x))$ . For  $\sigma \in H^0(\bar{X}, \text{Sym}^r \Omega_{\bar{X}/\mathbb{k}}^1(\log E))$ , whose restriction to  $X$  is a regular symmetric differential,  $\sigma_x(\xi, \dots, \xi)$  is the value of  $x^* \sigma$  on  $\xi$ . Near every point  $z \in \bar{X}$  there are an affine open  $W \ni z$  and  $u_1, \dots, u_n \in \mathcal{O}_{\bar{X}}(W)$  such that  $du_1, \dots, du_n$  is a basis of  $\Omega_{W/\mathbb{k}}^1$ ,  $E \cap W$  is the zero set of  $u_1 \cdots u_s$  for some  $s \leq n$ , and

$$\Omega_{\bar{X}/\mathbb{k}}^1(\log E)|_W = \bigoplus_{i \leq s} \mathcal{O}_W \frac{du_i}{u_i} \oplus \bigoplus_{i > s} \mathcal{O}_W du_i, \quad (18.1)$$

the local description of logarithmic poles along a simple normal-crossings divisor [C15]. The simple normal-crossings hypothesis supplies the local boundary parameters [60, Tag 0BI9]; smoothness lets their differentials be completed to a coordinate basis after shrinking  $W$ . One can construct the displayed sheaf inside the regular differentials on  $X \cap W$ : adjoin  $du_i/u_i$  for the boundary components to  $\Omega_{W/\mathbb{k}}^1$ . Changing a component equation to  $a_i u_i$ , with  $a_i$  a unit, changes its logarithmic differential by the regular form  $da_i/a_i$ . Thus the modules agree on overlaps and glue. The displayed generators are independent because they become a rescaling of the ordinary coordinate basis on  $X \cap W$ . The construction for one smooth divisor is [47, Section 50.15]; that section does not itself state the full normal-crossings version used here.

**Lemma 18.15** (The valuation-ring side [C15]). *Let  $L/\mathbb{k}$  be a field extension,  $V$  a valuation ring of  $L$  containing  $\mathbb{k}$ , and  $\partial$  a  $\mathbb{k}$ -derivation of  $L$  with  $\partial(V) \subseteq V$  and  $\partial u/u \in V$  for every nonzero  $u \in V$ . For every  $x \in X(L)$ , every  $r \geq 1$  and every  $\sigma \in H^0(\bar{X}, \text{Sym}^r \Omega_{\bar{X}/\mathbb{k}}^1(\log E))$ ,*

$$\sigma_x(\partial x, \dots, \partial x) \in V.$$

*Proof.* By the valuative criterion of properness for arbitrary valuation rings [38, Tag 0BX5],  $x$ , viewed in  $\bar{X}(L)$ , extends to  $q \in \bar{X}(V)$ . Let  $z$  be the image of the closed point of  $\text{Spec } V$  and choose  $W \ni z$  and  $u_1, \dots, u_n$  as in (18.1). Every point of  $\text{Spec } V$  is a generization of the closed point and open sets are stable under generization, so  $q$  factors through  $W$ . This uses that the

point is closed in  $\text{Spec } V$ , not that its image  $z$  is closed in  $\overline{X}$ . Hence every regular function on  $W$  takes a value in  $V$  at  $q$ , and that value is its value at  $x$ . Write  $\sigma|_W$  as a finite sum

$$\sigma|_W = \sum_{|\alpha|=r} c_\alpha \prod_{i \leq s} \left( \frac{du_i}{u_i} \right)^{\alpha_i} \prod_{i > s} (du_i)^{\alpha_i}, \quad c_\alpha \in \mathcal{O}_{\overline{X}}(W).$$

For  $i \leq s$  the value  $u_i(x)$  is nonzero, since  $x$  lies in  $X$ , off the boundary; so  $\partial(u_i(x))/u_i(x) \in V$ . For  $i > s$ ,  $\partial(u_i(x)) \in V$ , and  $c_\alpha(x) \in V$ . The contraction

$$\sigma_x(\partial x, \dots, \partial x) = \sum_{|\alpha|=r} c_\alpha(x) \prod_{i \leq s} \left( \frac{\partial(u_i(x))}{u_i(x)} \right)^{\alpha_i} \prod_{i > s} (\partial(u_i(x)))^{\alpha_i}$$

therefore lies in  $V$ . No discreteness, rank or Noetherian hypothesis on  $V$  is used, and the point  $z$  may lie on  $E$  or not [C15].  $\square$

The logarithmic-derivative hypothesis cannot be replaced by stability of  $V$  alone. In  $L = \mathbb{k}((t))$ , the ordinary derivation  $d/dt$  preserves  $V = \mathbb{k}[[t]]$ . But for  $\overline{X} = \mathbb{P}^1$ ,  $E = \{0, \infty\}$  and  $x = t \in \mathbb{G}_m(L)$ , the global logarithmic form  $dz/z$  contracts to  $1/t \notin V$ . By contrast, an Euler derivation has  $\partial(t)/t \in \mathbb{k} \subseteq V$ .

**Theorem 18.16** (Logarithmic annihilation [C15]). *Let  $p \in X(\mathcal{A}_{\mathbb{k}}(\Gamma))$  and let  $p_{\mathbb{H}} \in X(\mathbb{H}_{\mathbb{k}}(\Gamma))$  be its image. For every  $\mathbb{Q}$ -linear  $\lambda : \Gamma_{\mathbb{Q}} \rightarrow \mathbb{k}$ , every  $r \geq 1$  and every  $\sigma \in H^0(\overline{X}, \text{Sym}^r \Omega_{\overline{X}/\mathbb{k}}^1(\log E))$ ,*

$$\sigma_{p_{\mathbb{H}}}(\partial_{\lambda} p_{\mathbb{H}}, \dots, \partial_{\lambda} p_{\mathbb{H}}) = 0. \quad (18.2)$$

*Proof.* Put  $\partial = \partial_{\lambda}$ . The restriction  $\sigma|_X$  is a regular symmetric differential. Pull it back along  $p : \text{Spec } \mathcal{A}_{\mathbb{k}}(\Gamma) \rightarrow X$  and apply the natural map on differentials. The image is an element of  $\text{Sym}_{\mathcal{A}_{\mathbb{k}}(\Gamma)}^r \Omega_{\mathcal{A}_{\mathbb{k}}(\Gamma)/\mathbb{k}}^1$ , a finite sum  $\sum_j a_j db_{j1} \cdots db_{jr}$  with  $a_j, b_{jh} \in \mathcal{A}_{\mathbb{k}}(\Gamma)$ ; no chart around  $p$  and no finite generation of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is needed, because  $\text{Spec } \mathcal{A}_{\mathbb{k}}(\Gamma)$  is affine. By functoriality, as in the proof of theorem 16.14, the left side of (18.2) is  $\sum_j a_j \partial b_{j1} \cdots \partial b_{jr}$ . The contraction on products is symmetric, so it descends directly to the symmetric power; a lift of a global symmetric section to a global tensor section is not required. Each  $\partial b_{jh}$  lies in  $\Pi_{\mathbb{k}}(\Gamma)$  (lemma 16.2), and  $\Pi_{\mathbb{k}}(\Gamma)$  is an ideal of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  (lemma 16.1), so the contraction lies in  $\Pi_{\mathbb{k}}(\Gamma)^r \subseteq \Pi_{\mathbb{k}}(\Gamma)$  because  $r \geq 1$ . On the other hand  $\mathcal{O}_{\mathbb{k}}(\Gamma)$  is a valuation ring of  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  containing  $\mathbb{k}$  (lemma 16.1),  $\partial(\mathcal{O}_{\mathbb{k}}(\Gamma)) \subseteq \mathfrak{m}_{\mathbb{k}}(\Gamma) \subseteq \mathcal{O}_{\mathbb{k}}(\Gamma)$ , and  $\partial u/u \in \mathcal{O}_{\mathbb{k}}(\Gamma)$  for every  $u \in \mathbb{H}_{\mathbb{k}}(\Gamma)^{\times}$  (lemma 16.2); so lemma 18.15 puts the same element in  $\mathcal{O}_{\mathbb{k}}(\Gamma)$ . The two constructions agree in  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ : both send a local differential  $df$  to  $\partial(f(p_{\mathbb{H}}))$ , and multiplication gives the same diagonal contraction on symmetric powers. They may use different affine charts. Since  $\Pi_{\mathbb{k}}(\Gamma) \cap \mathcal{O}_{\mathbb{k}}(\Gamma) = \{0\}$  (lemma 16.1), the contraction is zero.  $\square$

The proof uses the two inclusions into  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ , not a ring map between  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  and  $\mathcal{O}_{\mathbb{k}}(\Gamma)$  compatible with those inclusions. In fact, maps between these rings do exist: constant extraction followed by inclusion gives  $\mathcal{A}_{\mathbb{k}}(\Gamma) \rightarrow \mathbb{k} \hookrightarrow \mathcal{O}_{\mathbb{k}}(\Gamma)$  and  $\mathcal{O}_{\mathbb{k}}(\Gamma) \rightarrow \mathbb{k} \hookrightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$ . For  $\Gamma \neq 0$  they do not preserve the common-field images of nonconstant elements. The two points instead have a common image over  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ , where the scalars are compared. The derivation  $\partial_{\lambda}$  need not preserve the function field of the image of  $p$ ; its restriction there is a derivation into  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ , which is all that the universal property of differentials requires [C15]. For  $E = \emptyset$  the theorem is the diagonal symmetric case of theorem 16.14 for the Hahn data. The logarithmic case needs the support inclusion  $\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) \subseteq \Pi_{\mathbb{k}}(\Gamma)$ : a boundary factor contributes  $\partial u/u$ , which lies in  $\mathcal{O}_{\mathbb{k}}(\Gamma)$  but in general not in  $\mathfrak{m}_{\mathbb{k}}(\Gamma)$  (for  $u = \omega^{-\gamma}$  it is the constant  $\lambda(-\gamma)$ ), so the valuation-ring side gives only an element of  $\mathcal{O}_{\mathbb{k}}(\Gamma)$ , and  $\Pi_{\mathbb{k}}(\Gamma) \cap \mathcal{O}_{\mathbb{k}}(\Gamma) = \{0\}$  must replace

$\mathcal{A}_{\mathbb{k}}(\Gamma) \cap \mathfrak{m}_{\mathbb{k}}(\Gamma) = \{0\}$ . This is the mechanism of [theorem 17.8](#), now in every dimension. Degree zero would not work: the global section 1 contracts to 1, and  $\Pi_{\mathbb{k}}(\Gamma)^0 = \mathcal{A}_{\mathbb{k}}(\Gamma)$  supplies no annihilation.

**Definition 18.17** (Logarithmic tangent separation [C15]). The global logarithmic symmetric differentials of  $(\overline{X}, E)$  *separate tangent directions* on  $X$  if for every field extension  $L/\mathbb{k}$ , every  $x \in X(L)$  and every nonzero tangent vector  $\xi$  of  $X_L$  at  $x$  there are  $r \geq 1$  and  $\sigma \in H^0(\overline{X}, \text{Sym}^r \Omega_{\overline{X}/\mathbb{k}}^1(\log E))$  with  $\sigma_x(\xi, \dots, \xi) \neq 0$ .

For  $E = \emptyset$  this is detection of tangent directions ([definition 16.18](#)). As there, the quantification over extension fields is needed because the point  $p_{\mathbb{H}}$  is generally not classical.

**Corollary 18.18** (The logarithmic tangent criterion [C15]). *If the global logarithmic symmetric differentials of  $(\overline{X}, E)$  separate tangent directions on  $X$ , then  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$  for every  $\Gamma$ , and  $\text{ct}$  is inverse to the inclusion of constant points. This holds in particular if, for every field extension  $L/\mathbb{k}$  and every  $x \in X(L)$ , the global logarithmic one-forms span the cotangent space, that is,  $H^0(\overline{X}, \Omega_{\overline{X}/\mathbb{k}}^1(\log E)) \otimes_{\mathbb{k}} L \rightarrow x^* \Omega_{X/\mathbb{k}}^1$  is surjective. For  $\mathbb{k} = \mathbb{R}$  or  $\mathbb{C}$  the same holds over  $\mathcal{B}_{\mathbb{R}}$  or  $\mathcal{B}_{\mathbb{C}}$  in the class convention of [section 16.1](#).*

*Proof.* Let  $p \in X(\mathcal{A}_{\mathbb{k}}(\Gamma))$  and let  $\lambda$  be given. By [theorem 18.16](#) every global logarithmic symmetric differential vanishes on the diagonal of the tangent vector  $\partial_{\lambda} p_{\mathbb{H}}$  at  $p_{\mathbb{H}} \in X(\mathbb{H}_{\mathbb{k}}(\Gamma))$ , so tangent separation with  $L = \mathbb{H}_{\mathbb{k}}(\Gamma)$  gives  $\partial_{\lambda} p_{\mathbb{H}} = 0$ ; that is,  $\partial_{\lambda}(f(p_{\mathbb{H}})) = 0$  for every regular function  $f$  on an affine open neighbourhood of  $p_{\mathbb{H}}$ , and for every  $\lambda$ . The scheme  $X$  is open in the proper, hence separated, scheme  $\overline{X}$ , so it is separated and of finite type, and [lemma 16.15](#) applies to the separating data  $(\mathbb{H}_{\mathbb{k}}(\Gamma), \mathcal{A}_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \{\partial_{\lambda}\})$ . Concretely, the coordinate map of an affine neighbourhood of the image of  $p_{\mathbb{H}}$  has values in the joint constants  $\mathbb{k}$ . It defines a  $\mathbb{k}$ -point whose constant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -extension agrees with  $p$  over  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ ; separatedness reflects this equality back to  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ . The point  $p$  need not lie in that one affine chart before constancy is established. Applying  $\text{ct}$  identifies the resulting point with  $\text{ct}(p)$  and proves injectivity of constant inclusion. If the one-forms span, a nonzero tangent vector  $\xi$  at  $x$  is nonzero on the image of some global logarithmic one-form  $\eta$ , and  $r = 1$ ,  $\sigma = \eta$  separates it. A class point lies over some  $\mathcal{A}_{\mathbb{R}}(\Gamma)$  or  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ .  $\square$

For  $E = \emptyset$  the corollary is [theorem 16.19](#), which contains [corollaries 16.17](#) and [16.20](#). For curves a single form suffices without the criterion, and this is how source C15 proves the classification of smooth curves over an algebraically closed field: a nonconstant point has generic image, so the function field  $F$  embeds in  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ ; unless the completion has genus zero and at most one boundary point, there is a nonzero global logarithmic form  $\eta$  (by [lemma 17.7](#) for a nonempty boundary; without boundary the regular forms have dimension  $g$ ), which spans  $\Omega_{F/\mathbb{k}}^1$ ; by [theorem 18.16](#) its contraction with every  $\partial_{\lambda}$  vanishes, so the derivations kill  $F$ , contradicting  $\text{trdeg}_{\mathbb{k}} F = 1$ . This is [remark 17.3](#) and [theorem 17.8](#), with the computation at a local parameter replaced by [lemma 18.15](#), which works wherever the centre of the valuation-ring point lies; so this route needs no closed-centre step. For genus zero with two or more punctures the form  $dz/z$  replaces the unit argument of [lemma 17.4](#), as source C12 also notes. Source C15 proves semiabelian rigidity by the extension argument of [theorem 18.4](#) and the quasi-finite results by finite fibers, not through [corollary 18.18](#).

**Remark 18.19** (An abstract interface [C15]). Let  $\mathbb{k} \subseteq R \subseteq L$ , with  $L$  a field, let  $R \rightarrow \mathbb{k}$  be a ring retraction with kernel  $J$ , let  $V$  be a valuation ring of  $L$  containing  $\mathbb{k}$ , and let  $\mathcal{D}$  be a family of  $\mathbb{k}$ -derivations of  $L$  with

$$\partial(R) \subseteq J, \quad \partial(V) \subseteq V \quad (\partial \in \mathcal{D}), \quad J \cap V = \{0\}, \quad \bigcap_{\partial \in \mathcal{D}} \ker \partial = \mathbb{k}.$$

Then the proofs of [theorem 18.16](#) and [corollary 18.18](#) give  $X(R) = X(\mathbb{k})$  for every smooth proper  $X$  whose global symmetric differentials detect tangent directions and, if in addition  $\partial u/u \in V$  for every nonzero  $u \in V$  and every  $\partial \in \mathcal{D}$ , for every  $X = \overline{X} \setminus E$  as in [corollary 18.18](#). The constancy step uses only the joint constants and the injectivity of  $R \rightarrow L$ , as in the proof of [lemma 16.15](#): annihilation puts each contraction in  $J \cap V$ , tangent separation makes every  $\partial p_L$  zero, and the coordinate map of an affine chart at  $p_L$  then has image in  $\mathbb{k}$ . The resulting constant point agrees with  $p$  over  $R$  by the closed equalizer argument. No assertion that these data satisfy the different ideal conditions of [definition 16.12](#) is needed. This interface is dual to [definition 16.12](#): there the ring is only stable,  $\partial(R) \subseteq R$ , the valuation ring is pushed into its maximal ideal and  $R \cap \mathfrak{m}_V = \{0\}$ ; here the ring is pushed into the augmentation ideal, the valuation ring is only stable and  $J \cap V = \{0\}$ . The Hahn data  $(\mathbb{H}_{\mathbb{k}}(\Gamma), \mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma))$  satisfy both, and the logarithmic hypothesis by [lemma 16.2](#). Source C15 notes that properness alone is not enough, nor is a family of derivations without control of the augmentation ideal; these are what must be checked before the argument is used in another generalized-series setting.

**Theorem 18.20** (Descent over any coefficient ring [C15]). *Let  $\mathfrak{o} \subseteq \mathbb{k}$  be a subring,  $\mathcal{X}$  a separated  $\mathfrak{o}$ -scheme of finite type,  $X = \mathcal{X} \times_{\mathfrak{o}} \mathbb{k}$  and  $R = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ . If  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$ , then inclusion and  $\text{ct}$  are inverse bijections  $\mathcal{X}(\mathfrak{o}) \cong \mathcal{X}(R)$ .*

*Proof.* Let  $p \in \mathcal{X}(R)$  and  $p_0 = \text{ct}(p) \in \mathcal{X}(\mathfrak{o})$ . Composed with  $R \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$ ,  $p$  gives a point of  $\mathcal{X}(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathcal{A}_{\mathbb{k}}(\Gamma))$ , which is constant by hypothesis. Since  $\text{ct} : \mathcal{A}_{\mathbb{k}}(\Gamma) \rightarrow \mathbb{k}$  restricts to  $\text{ct} : R \rightarrow \mathfrak{o}$ , applying  $\text{ct}$  shows that this constant point is the image of  $p_0$ . So  $p$  and the constant  $R$ -point of  $p_0$  agree after the injection  $R \hookrightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$ , hence agree ([lemma 16.13](#)); faithful flatness is not needed. Injectivity of  $\mathcal{X}(\mathfrak{o}) \rightarrow \mathcal{X}(R)$  follows by applying  $\text{ct}$ . Explicitly, the equalizer of the two  $R$ -points is closed because  $\mathcal{X}$  is separated over  $\mathfrak{o}$ . Its defining ideal maps to zero in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , hence is zero by injectivity. This explains why neither flatness of  $R \hookrightarrow \mathcal{A}_{\mathbb{k}}(\Gamma)$  nor smoothness of the arithmetic model is required.  $\square$

With  $(\mathfrak{o}, \mathbb{k}) = (\mathbb{Z}, \mathbb{C})$ , the injection  $\mathbf{Oz}_{\Gamma} \subseteq \mathcal{R}_{\mathbb{Z}}(\mathbb{C}, \Gamma)$  and [lemma 16.13](#) give the conclusion of [theorem 17.17](#); with  $(\mathfrak{o}, \mathbb{k}) = (\mathbb{Z}[i], \mathbb{C})$  the ring  $R$  is the workspace form of  $\mathbf{Oz}[i]$ . For affine curves the theorem contains the rigid case [theorem 17.11\(i\)](#). No smoothness of  $\mathcal{X}$  over  $\mathfrak{o}$  is used: bad reduction at primes of  $\mathfrak{o}$  is allowed, since only the fiber over  $\mathbb{k}$  is assumed rigid [C15].

**Corollary 18.21** (Omnific points on logarithmic complements [C15]). *Let  $(\mathfrak{o}, \mathbb{k}) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , and let  $\mathcal{X}$  be a separated  $\mathfrak{o}$ -scheme of finite type whose fiber over  $\mathbb{k}$  is  $X = \overline{X} \setminus E$ , with  $\overline{X}$  smooth and proper over  $\mathbb{k}$ ,  $E$  a reduced divisor with simple normal crossings, and global logarithmic symmetric differentials separating tangent directions on  $X$ . Then  $\mathcal{X}(\mathbf{Oz}) = \mathcal{X}(\mathbb{Z})$  in the real case and  $\mathcal{X}(\mathbf{Oz}[i]) = \mathcal{X}(\mathbb{Z}[i])$  in the complex case, in the class convention of [section 16.1](#).*

*Proof.* In the class convention, one point is represented by finite presentation data in a set-sized ring. Include its affine-chart data, the finitely many denominators and unit-ideal witnesses defining their cover, and the identities gluing them. The union of the supports of these finitely many surreal numbers generates a set-sized ordered subgroup  $\Gamma$ . All the data therefore lie in one workspace  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  with  $\Gamma \subseteq \mathbf{No}$  set-sized ([lemma 2.6](#)). There  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$  by [corollary 18.18](#), and [theorem 18.20](#) makes the point ordinary. The ordinary point is its constant term, so it does not depend on the workspace.  $\square$

Source C15 states this corollary with two further cases, printed above: smooth geometrically connected curves whose complexification is neither  $\mathbb{A}^1$  nor  $\mathbb{P}^1$  ([corollary 17.12](#), and [theorem 17.17](#) with [theorem 17.2](#) for proper curves), and schemes with a quasi-finite map to a semiabelian

variety (theorems 17.17, 18.4 and 18.5). When  $X$  is affine, corollary 18.21 adds a class of rigid affine varieties in every dimension to the first part of question 21.1. It is an exact equality of point sets; it gives no enumeration, finiteness statement or height bound for the ordinary points [C15].

**Proposition 18.22** (Finite products [C15]). *For fixed  $\mathbb{k}$  and  $\Gamma$ , the class of separated  $\mathbb{k}$ -schemes  $X$  of finite type with  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$  is closed under finite products. By proposition 16.21 and theorem 18.5 it is also closed under locally closed subschemes and under quasi-finite morphisms into its members.*

*Proof.* For  $\mathbb{k}$ -schemes  $X, Y$  and every  $\mathbb{k}$ -algebra  $S$ ,  $(X \times_{\mathbb{k}} Y)(S) = X(S) \times Y(S)$  naturally in  $S$ , so the inclusion of constant points of the product is the product of two bijections; the empty product  $\text{Spec } \mathbb{k}$  is rigid trivially. Repeating the binary argument proves the assertion for every finite number of factors. Finite products of separated finite-type schemes remain separated and of finite type, so the construction stays within the stated class. A locally closed immersion is quasi-finite; the other closure assertions then follow from theorem 18.5 with their stated finiteness hypotheses.  $\square$

So products of curves of positive genus, locally closed subschemes of products of abelian varieties and tori, and their finite covers are rigid; the defining systems may be arbitrary finite systems, not only separated equations, and a finite cover may add algebraic dependencies [C15]. Quasi-finiteness cannot be dropped: the projection  $\mathbb{A}^1 \times G \rightarrow G$ , with  $G$  a semiabelian variety, has a rigid target. For  $\gamma > 0$  the point  $(\omega^\gamma, e)$  of its source is nonconstant and maps to the identity  $e$  of  $G$  [C12, C15]. Tangent separation is sufficient, not claimed necessary, and no conclusion is drawn from general type: positivity of the canonical bundle alone need not supply separating symmetric differentials [C15]. The criterion answers the first question of question 21.19 in sufficient form; what remains open is recorded in section 21.4.

## 19 Equations with omnific coefficients

For  $P \in \mathbf{Oz}[X_1, \dots, X_n]$  let  $\overline{P} \in \mathbb{Z}[X_1, \dots, X_n]$  apply ct to every coefficient. Then [02, 05]

$$\text{ct}(P(\mathbf{x})) = \overline{P}(\text{ct } x_1, \dots, \text{ct } x_n), \quad P(\mathbf{x}) = 0 \implies \overline{P}(\text{ct } \mathbf{x}) = 0.$$

The reverse implication need not hold, because the coefficients are no longer fixed by ct.

### 19.1 Quadratic equations

**Theorem 19.1** (Square-discriminant criterion [02]). *For  $b, c \in \mathbf{Oz}$ , the equation  $x^2 + bx + c = 0$  has a root in  $\mathbf{Oz}$  if and only if  $\Delta = b^2 - 4c$  is a square in  $\mathbf{Oz}$ . If  $y^2 = \Delta$  with  $y \in \mathbf{Oz}$ , its roots are  $(-b \pm y)/2$ , and both are omnific.*

*Proof.* A root gives  $(2x + b)^2 = \Delta$ . Conversely, suppose  $y^2 = \Delta$ . Specializing and reducing modulo 4 gives  $\text{ct}(y)^2 \equiv \text{ct}(b)^2 \pmod{4}$ . Integer squares are 0 or 1 modulo 4, so  $\text{ct}(y)$  and  $\text{ct}(b)$  have the same parity, and theorem 3.1 shows that  $-b \pm y$  are divisible by two in  $\mathbf{Oz}$ . The usual identity verifies the roots. Conversely every root determines one of these two square roots by  $y = 2x + b$ , so the list is exhaustive. When  $\Delta = 0$  the two expressions coincide; otherwise they are distinct.  $\square$

Monicity is essential to the criterion as stated. For a general quadratic  $ax^2 + bx + c$  with  $a \neq 0$  in  $\mathbf{Oz}$ , a root is equivalent to an element  $y \in \mathbf{Oz}$  satisfying both  $y^2 = b^2 - 4ac$  and

$y - b \in 2a\mathbf{Oz}$ . Indeed a root gives  $y = 2ax + b$ ; conversely write  $y - b = 2ax$  and use  $y^2 - (b^2 - 4ac) = 4a(ax^2 + bx + c)$ , cancelling  $4a \neq 0$ . A square discriminant alone does not ensure this divisibility: the polynomial  $6x^2 + x - 1 = (3x - 1)(2x + 1)$  has discriminant 25, but its two field roots  $1/3$  and  $-1/2$  are not omnific. For a monic quadratic, the parity argument above supplies exactly the missing divisibility by two.

Testing whether a nonnegative surreal has an omnific square root is not the same as testing whether it has a surreal square root: real closedness gives the latter, while nonnegative support and an integral constant must still be checked for the former.

**Example 19.2** (Residual and leading tests that do not lift [01, 02, 05]). The equation  $x^2 = \omega^2 + 1$  has no solution in  $\mathbf{Oz}$  (proposition 4.12). Nevertheless its residual equation  $\text{ct}(x)^2 = 1$  is solvable, and the leading terms allow degree 1 and leading coefficient  $\pm 1$ . The positive root in  $\mathbf{No}$  has the formal binomial expansion

$$\sqrt{\omega^2 + 1} = \omega + \frac{1}{2}\omega^{-1} - \frac{1}{8}\omega^{-3} + \frac{1}{16}\omega^{-5} - \dots,$$

whose negative exponents exclude it from  $\mathbf{Oz}$ . This is a support-controlled Hahn expansion, not a claim of convergence in the surreal order topology. In contrast,  $\omega^2 + 2\omega + 1 = (\omega + 1)^2$  is an omnific square.

The sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` prints a later manuscript's image-gap theorem at a fresh scale (its files carry the prefix `16-fresh-scale-`): if  $f$  has degree at least two and coefficients in the Hahn field  $K_G$  with exponents in a set-sized group  $G$ , and  $t = \omega^b$  with  $b > G$ , then  $f(\mathcal{B}_{\mathbb{R}}) \cap (f(t) + K_G) = \{f(t)\}$ , and likewise over  $\mathcal{B}_{\mathbb{C}}$ ; for ordinary  $f$  one may take  $G = \{0\}$  and any  $b > 0$ . The unsolvability of  $x^2 = \omega^2 + 1$  in example 19.2 is the case  $f = X^2$ ,  $t = \omega$ , and the two-term equations of proposition 19.4 are the cases  $f = aY^n$ ,  $t = \omega^{\gamma/n}$ , after the substitution  $X = a^{1/n}Y$  in  $\mathcal{B}_{\mathbb{R}}$ . The short proofs here are kept; the general statement is printed there.

## 19.2 Initial forms

**Proposition 19.3** (Initial forms of an omnific solution [01, 02]). *Let  $P = \sum_{\alpha} a_{\alpha} \mathbf{X}^{\alpha} \in \mathbf{Oz}[X_1, \dots, X_n]$  be nonzero, let  $\mathbf{x} \in \mathbf{Oz}^n$  have all coordinates nonzero, and write  $\gamma_i = \deg_{\omega}(x_i)$ ,  $r_i = \text{lc}(x_i)$ . Discarding zero coefficients, put*

$$W = \max_{\alpha} \left( \deg_{\omega}(a_{\alpha}) + \sum_i \alpha_i \gamma_i \right).$$

*If  $P(\mathbf{x}) = 0$ , then*

$$\sum_{\deg_{\omega}(a_{\alpha}) + \sum_i \alpha_i \gamma_i = W} \text{lc}(a_{\alpha}) \prod_i r_i^{\alpha_i} = 0, \quad (19.1)$$

*and in particular the maximum is attained by at least two monomials.*

*Proof.* By lemma 2.4, each evaluated monomial has degree  $\deg_{\omega}(a_{\alpha}) + \sum_i \alpha_i \gamma_i$  and leading coefficient  $\text{lc}(a_{\alpha}) \prod_i r_i^{\alpha_i}$ . The coefficient of  $P(\mathbf{x})$  at  $\omega^W$  is exactly the sum in (19.1), and it must vanish; a single maximizer would contribute a nonzero product.  $\square$

The maximum is taken over the finite nonempty polynomial support, so it exists even when the surreal weights have arbitrary rank. The exponent  $W$  is a candidate degree before cancellation, not an assertion that  $\deg_{\omega}(P(\mathbf{x})) = W$ . Several maximal terms can cancel; their leading coefficients, including their signs, must actually sum to zero. Merely having two terms

of maximal weight is insufficient: for  $P(X) = X^2 + \omega^2$  and a candidate with  $\deg_\omega(x) = 1$ , the initial equation is  $r^2 + 1 = 0$ , which has no real solution.

Variables assigned zero can first be substituted out. If this makes the polynomial identically zero, it imposes no initial-form condition; otherwise apply the proposition to the remaining nonzero coordinates. For ordinary coefficients this is a weighted initial polynomial over  $\mathbb{R}$  with surreal weights. The constant-coefficient equation and the initial equation constrain opposite ends of a normal form; they are complementary necessary tests, not a lifting criterion, as [example 19.2](#) shows: the missing information lies in the full tail of a potential root.

### 19.3 Two-term roots and failed lifting

**Proposition 19.4** (A two-term root obstruction [05]). *Let  $a \in \mathbb{R}_{>0}$ ,  $\gamma > 0$ ,  $b \in \mathbb{Z}$ , and let  $n \geq 2$  be ordinary. The equation  $X^n = a\omega^\gamma + b$  has a solution in  $\mathbf{Oz}$  if and only if  $b = 0$ . When  $b = 0$  its positive solution is  $a^{1/n}\omega^{\gamma/n}$ , together with its negative when  $n$  is even.*

*Proof.* When  $b = 0$ , put  $u = a^{1/n}\omega^{\gamma/n} \in \Pi$ . Then  $u^n = a\omega^\gamma$ , and strict monotonicity of positive integer powers on the positive cone gives the stated list of roots.

For  $b \neq 0$ , the right side is still positive and infinite. Suppose an omnific root exists, replacing it by its absolute value if  $n$  is even; the resulting root  $x$  is positive. With the same positive  $u$ , factor

$$b = x^n - u^n = (x - u)S, \quad S = \sum_{j=0}^{n-1} x^{n-1-j}u^j.$$

Both factors belong to  $\mathcal{B}_{\mathbb{R}}$ , so [lemma 6.1](#) would make  $S$  a nonzero real constant. But every summand is positive and  $S \geq u^{n-1}$  is infinite, a contradiction. This finite-algebra proof uses only the previously established unit computation.  $\square$

Source 05 instead proves the obstruction by displaying the positive field root. Its formal binomial expansion is

$$a^{1/n}\omega^{\gamma/n} \sum_{k=0}^{\infty} \binom{1/n}{k} \left(\frac{b}{a}\right)^k \omega^{-k\gamma}. \quad (19.2)$$

For every positive surreal  $\gamma$ , the exponents  $\gamma/n - k\gamma$  form a reverse-well-ordered set: the largest exponent of any nonempty subset corresponds to its least natural index. Each coefficient in an  $n$ -fold product involves finitely many index tuples with a fixed sum. Hence the formal power-series binomial identity evaluates coefficientwise to the  $n$ th-root identity in a set-sized Hahn workspace containing  $\gamma/n$ . The leading coefficient is positive, so this is the positive field root; the only other possible root is its negative for even  $n$ .

If  $b \neq 0$ , the  $k = 1$  term has nonzero coefficient  $(b/n)a^{1/n-1}$  at exponent  $\gamma/n - \gamma < 0$ , which no other term has. Thus the expansion identifies the forbidden term for both possible signs. No convergence of partial sums in the full surreal order topology is used, and no assumption that the multiples of  $\gamma$  are cofinal among positive surreals is needed.

**Example 19.5** (A simple residue root need not lift [05]). Let  $f(T) = T^2 - T - \omega$ . Modulo  $\Pi$  it becomes  $T(T - 1)$ , and the residue roots 0 and 1 are simple, with derivatives  $-1$  and  $1$ , respectively, both units in  $\mathbb{Z}$ . Nevertheless  $f$  has no root in  $\mathbf{Oz}$ : its discriminant  $1 + 4\omega$  is not an omnific square by [proposition 19.4](#), so [theorem 19.1](#) applies. Its roots in  $\mathbf{No}$  are

$$\frac{1}{2} \pm (\omega^{1/2} + \frac{1}{8}\omega^{-1/2} - \frac{1}{128}\omega^{-3/2} + \dots),$$

with constant coefficient  $\frac{1}{2}$  and negative exponents. Hensel-style intuition fails for  $\mathbf{Oz} \rightarrow \mathbb{Z}$ : the kernel  $\Pi$  is not contained in the Jacobson radical, since  $-\omega \in \Pi$  but  $1 + \omega$  is not a unit, and

$\Pi^2 = \Pi$ , so there is no decreasing nilpotent-error filtration. This is a failure of the simple-root lifting property itself.

This example concerns the constant-term map on the omnific ring. The standard-part map on the valuation ring of finite surreals has a different domain and an infinitesimal kernel. In particular  $\omega$  is not finite, so the displayed polynomial does not even have all its coefficients in that valuation ring. Simple-residue-root lifting there, as used in the trigonometry report (`trigonometry:lem:hensel`), makes no lifting claim for  $\mathbf{Oz} \rightarrow \mathbb{Z}$ . The two uses of residue must be kept separate. Status (batch 34): the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` studies what henselization adds, in set-sized rings. For every infinite cardinal  $\kappa$  it builds a real closed field  $\mathbb{E} \subseteq \mathbf{No}$  of cardinality  $\kappa$ , containing  $\omega$ , whose integer part  $\mathbb{E} \cap \mathbf{Oz}$  is localized at its idempotent constant-term ideal to a local domain  $\mathfrak{L}$  with residue field  $\mathbb{Q}$ ; the henselization  $\mathfrak{L}^h$ , which has roots of equations such as  $T^2 - T - \omega$  reducing to simple residue roots, has exactly  $2^\kappa$  minimal primes, each with fraction field  $\mathbb{E}[i]$ , and no  $\mathfrak{L}$ -algebra map to  $\mathbb{E}$  or to any ordered extension field of  $\mathbb{E}$  (`osq:hb:thm:main`). The complex branches are forced: if  $h > \frac{1}{4}$  lies in the maximal ideal of a local domain  $A$  with real closed fraction field  $\mathbb{F}$ , the root of  $U^2 - U + h$  in  $A^h$  reducing to 0 gives a square root of  $-1$  in  $\mathbb{F} \otimes_A A^h$  (`osq:hb:prop:complex`, applied with  $h = \omega$ ). This concerns set-sized subrings of  $\mathbf{Oz}$  and their henselizations; the example above, in  $\mathbf{Oz}$  itself, is unchanged.

For general coefficients, a viable approach must inspect actual normal-form supports. Matching constant terms is necessary, and occasionally useful, but is not a sufficient solvability criterion ([question 21.6](#)).

## 19.4 Constant discriminants and étale units

This subsection is built from a fifteenth manuscript, *Constant Discriminants and Étale Unit Rigidity over Omnific Hahn Rings: Translation classification, arithmetic descent, and normal spectral rigidity*, tagged [C17] after its file prefix `17-discriminant-rigidity-` ([section 1.2](#)); the letter C is kept as for sources C10–C16, because the sibling report has a manuscript 17 of its own. Source C17 works with equations whose coefficients are *not* constant: monic polynomials in one variable over the nonnegative-support ring. Its main theorem says that a nonzero *constant* discriminant forces such a polynomial to be a purely infinite translate of its constant-term polynomial ([theorem 19.12](#)). So a constant nondegeneracy invariant recovers a rigidity theorem for coefficients that vary at a surreal scale, in dimension zero. This is the first clause of [question 21.22](#) for univariate equations, and for monic polynomials over  $\mathbf{Oz}$  with discriminant in  $\mathbb{Z} \setminus \{0\}$  it answers [question 21.6](#) ([corollary 19.16](#)). The same mechanism shows that every unit of a finite étale algebra over the ring is algebraic over the coefficient field ([theorem 19.26](#)), which contains [theorem 6.15](#) in characteristic zero ([remark 19.28](#)). Labels of this subsection carry the sub-prefix `odg:disc:`. It was appended at the end of [section 19](#), so no existing section, statement or equation number changed; its ten questions follow the last question of [section 21](#). The elementary lemmas, root-velocity certificate and translation classification, its factor/Galois/resultant consequences and the finite étale unit theorem, monogenic descent, critical-point and matrix applications, boundary examples and the second proof have been reviewed. The later question/status notes and complete parallel-source reconciliation remain pending, as recorded in `RECONCILIATION.md`.

**Conventions and renamed symbols.** As in the curve part ([section 16.1](#)),  $\mathbb{k}$  is a field of characteristic zero,  $\bar{\mathbb{k}}$  an algebraic closure,  $\Gamma$  a set-sized ordered abelian group, not assumed divisible, and  $\Gamma_{\mathbb{Q}}$  its divisible hull. We use the  $\omega$ -convention:  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is the ring of series with support in  $\Gamma^{\geq 0}$ ,  $\Pi_{\mathbb{k}}(\Gamma)$  its ideal of series with support in  $\Gamma^{>0}$ ,  $\mathcal{O}_{\mathbb{k}}(\Gamma)$  the valuation ring of

series with support in  $\Gamma^{\leq 0}$ , and  $v = -\deg_\omega$ . Source C17 uses the  $t$ -convention  $t^\gamma = \omega^{-\gamma}$ : its  $\mathcal{A} = k((t^{\Gamma^{\leq 0}}))$ ,  $\mathcal{I}$ ,  $\mathcal{O}_K$  and  $K = k((t^\Gamma))$  are  $\mathcal{A}_k(\Gamma)$ ,  $\Pi_k(\Gamma)$ ,  $\mathcal{O}_k(\Gamma)$  and  $\mathbb{H}_k(\Gamma)$  with exponents negated, and its algebraically closed workspace  $\widehat{K} = \widehat{k}((t^{\Gamma_\mathbb{Q}}))$  with valuation ring  $\widehat{\mathcal{O}}$  is here the Hahn field  $\mathbb{H}_{\widehat{k}}(\Gamma_\mathbb{Q})$  with coefficients in  $\widehat{k}$  and exponents in  $\Gamma_\mathbb{Q}$ , with valuation ring  $\mathcal{O}_{\widehat{k}}(\Gamma_\mathbb{Q})$ . Its coefficient subring  $D \subseteq k$  is  $\mathfrak{o} \subseteq \mathbb{k}$  (here  $D$  is the Pell parameter), and its pullback  $\mathcal{A}_D = D \oplus \mathcal{I}$  is  $\mathcal{R}_\mathfrak{o} = \mathcal{R}_\mathfrak{o}(\mathbb{k}, \Gamma) = \mathfrak{o} + \Pi_k(\Gamma)$ . For  $P \in \mathcal{A}_k(\Gamma)[X]$ ,  $\overline{P} = \text{ct } P$  applies  $\text{ct}$  to every coefficient, as at the start of [section 19](#) (the source writes  $P_0$ ); the bar over a polynomial or a matrix is this constant-term map, while  $\widehat{k}$  is an algebraic closure. Its Euler derivations  $D_\chi$ ,  $t^\gamma \mapsto \chi(\gamma)t^\gamma$  for  $\chi \in \text{Hom}_\mathbb{Q}(\Gamma_\mathbb{Q}, \mathbb{Q})$ , are the  $\partial_{-\chi}$  of [\(16.1\)](#), the same family; we write  $\partial_\lambda$  with  $\lambda \in \text{Hom}_\mathbb{Q}(\Gamma_\mathbb{Q}, \mathbb{Q})$ , acting coefficientwise on  $\mathbb{H}_{\widehat{k}}(\Gamma_\mathbb{Q})$  and on the coefficients of polynomials. They are not  $d/dX$ . The source's finite étale algebra  $B$ , its constant subalgebra  $C_B$  and its splitting algebra  $S$  are  $B$ ,  $C_B$  and  $B_P$  (here  $B$  is a bilinear form,  $\mathcal{B}_\mathbb{R}, \mathcal{B}_\mathbb{C}$  are rings and  $S$  a target ring); the rank  $N$  of  $B$  is  $r$  (here  $N$  is a norm form, and  $r$  is the dimension of the étale algebra  $E$  of [theorem 6.15](#), which is the rank in the constant case). The velocity polynomial  $Q$  is  $\dot{P}$ , the divided difference  $S(X, Y)$  is  $W(X, Y)$ , the root velocities  $u_i$  are  $\dot{r}_i$ , the certificate polynomials  $F_{n,s}$  are  $\mathcal{F}_{n,s}$ , the characteristic polynomial  $H_y$  is  $\text{ch}_y$ , the spectral idempotents  $E_j$  are  $P_j$  (here  $E$  is an étale algebra), the target translation  $c$  of the critical-point theorem is  $\beta$  (the letter of the sibling report, [remark 19.35](#)), the number  $r$  of polynomials in the resultant theorem is  $\ell$  and its graph  $G$  is  $\mathcal{G}$  ( $G$  is the Gram matrix), the source's real and complex workspaces  $H$ ,  $H[i]$  are  $\mathbb{H}_\mathbb{R}(\Gamma_\mathbb{Q})$  and  $\mathbb{H}_\mathbb{C}(\Gamma_\mathbb{Q})$ , and its  $\mathcal{A}_\mathbb{C}$ ,  $\mathcal{I}_\mathbb{C}$  are  $\mathcal{A}_\mathbb{C}(\Gamma)$ ,  $\Pi_\mathbb{C}(\Gamma)$ . The letters  $h$  (the translation),  $\Delta$  (a discriminant),  $m = \binom{n}{2}$ ,  $w_{ij}$ ,  $e_s$ ,  $F = \text{Frac } \mathcal{A}_k(\Gamma)$  and  $L_0$  are the source's. [Table 1](#) lists these renamings and a few local ones (for example the quadratic velocity coefficients  $\alpha, \beta$  become  $b_1, b_2$ , and the infinitesimal root  $r$  of [example 19.41](#) becomes  $\rho$ ).

Four readings must be kept apart. A *constant* discriminant is one in  $\mathbb{k}$ , that is, supported at the exponent 0; it is not “independent of  $X$ ”, which every discriminant is. The *constant coefficient in  $X$*  of  $P$  is  $a_n$ , whereas the constant term of a series is its coefficient at  $\omega^0$ . A *nonzero* constant discriminant in  $\mathfrak{o}$  need not be a *unit* of  $\mathfrak{o}$ : the translation theorem needs only the former, étaleness the latter. And  $\mathcal{A}_k(\Gamma)$  is not the set of series of nonpositive valuation ([section 1.4](#)): the theorems below are false when the support condition is weakened to a condition on the leading exponent ([example 19.41](#)).

Here a polynomial over  $\mathfrak{o}$  is called *squarefree* when it has no repeated root over  $\widehat{k}$ , equivalently when it is squarefree over the field  $\mathbb{k}$ . This is a field-theoretic condition, not a statement about square divisors in the possibly nonnormal ring  $\mathfrak{o}[X]$ . The translation classification and its coefficient test concern positive degree: for  $P = 1$  every translation gives the same polynomial, so there is no unique  $h$  when  $\Pi_k(\Gamma) \neq 0$ .

### 19.4.1 The splitting field and the Euler family

Source C17 reproves the ring facts it uses:  $\mathcal{A}_k(\Gamma) = \mathbb{k} \oplus \Pi_k(\Gamma)$  with the retraction  $\text{ct}$ ,  $\mathcal{A}_k(\Gamma) \cap \mathcal{O}_k(\Gamma) = \mathbb{k}$ ,  $\Pi_k(\Gamma) \cap \mathcal{O}_k(\Gamma) = 0$ ,  $\mathcal{A}_k(\Gamma)^\times = \mathbb{k}^\times$ , and for the pullback  $\mathcal{R}_\mathfrak{o} \cap \mathbb{k} = \mathfrak{o}$  and  $\mathcal{R}_\mathfrak{o}^\times = \mathfrak{o}^\times$  (its Lemma 2.1, here [lemma 16.1](#)); the Euler derivations with  $\partial(\mathcal{A}_k(\Gamma)) \subseteq \Pi_k(\Gamma)$ ,  $\partial(\Pi_k(\Gamma)) \subseteq \Pi_k(\Gamma)$ ,  $v(\partial x) \geq v(x)$ ,  $\partial x/x \in \mathcal{O}_k(\Gamma)$  and joint constants  $\mathbb{k}$  (its Lemma 3.1, here [lemmas 16.2](#) and [16.3](#)); and that an element of  $\mathbb{H}_k(\Gamma)$  algebraic over  $\mathbb{k}$  lies in  $\mathbb{k}$ , so that every nonzero element of  $\Pi_k(\Gamma)$  is transcendental over  $\mathbb{k}$  (its Lemma 6.3, here [corollary 16.4](#) and [lemma 16.1\(d\)](#); over  $\mathbf{Oz}$  this is [proposition 2.8](#)). These are printed once, with C17 named in their credits. Its example  $\sum_{m \geq 1} \omega^{1/m} \in \Pi_k(\Gamma)$ , whose support has no least element, is the one after [lemma 16.1](#); no proof below assumes that  $\Pi_k(\Gamma)$  is generated by monomials. Both proofs of the source use these facts in the larger workspace  $\mathbb{H}_{\widehat{k}}(\Gamma_\mathbb{Q})$ , and the next lemma records what is used there.

**Lemma 19.6** (The Hahn splitting field [C17]). *The field  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$  is algebraically closed, and it contains  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  and  $F = \text{Frac } \mathcal{A}_{\mathbb{k}}(\Gamma)$ . The coefficientwise Euler derivations  $\partial_{\lambda}$ ,  $\lambda \in \text{Hom}_{\mathbb{Q}}(\Gamma_{\mathbb{Q}}, \mathbb{Q})$ , are  $\bar{\mathbb{k}}$ -linear derivations of  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$  extending those of  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ , with*

$$\mathcal{A}_{\mathbb{k}}(\Gamma) \cap \mathcal{O}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}}) = \mathbb{k}, \quad \Pi_{\mathbb{k}}(\Gamma) \cap \mathcal{O}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}}) = 0, \quad \frac{\partial_{\lambda} x}{x} \in \mathcal{O}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}}) \ (x \neq 0), \quad \bigcap_{\lambda} \ker \partial_{\lambda} = \bar{\mathbb{k}}. \quad (19.3)$$

*Proof.* Algebraic closedness of a Hahn field with algebraically closed coefficient field of characteristic zero and divisible exponent group is the standard criterion [3, Fact 2.1.1]. Extending coefficients along  $\mathbb{k} \subseteq \bar{\mathbb{k}}$  and exponents along  $\Gamma \subseteq \Gamma_{\mathbb{Q}}$  preserves supports and coefficient convolutions, so  $\mathbb{H}_{\mathbb{k}}(\Gamma) \subseteq \mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$ , hence  $F \subseteq \mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$ ; the field  $F$  need not be  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  (for  $\Gamma = \mathbb{Z}$ ,  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}[\omega]$  and  $F = \mathbb{k}(\omega)$ ). The intersections compare supports, as in lemma 16.1(b). The Leibniz rule follows from additivity of  $\lambda$  on each finite coefficient convolution, as in lemma 16.2; linearity over  $\bar{\mathbb{k}}$  follows directly from the coefficient formula. Coefficientwise differentiation can only remove support terms, so  $v(\partial_{\lambda} x) \geq v(x)$  and  $v(\partial_{\lambda} x/x) \geq 0$  when the derivative is nonzero; the zero derivative also lies in  $\mathcal{O}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$ . Finally, for any nonzero  $\gamma \in \Gamma_{\mathbb{Q}}$ , extend  $\gamma$  to a  $\mathbb{Q}$ -basis and choose a rational linear functional with  $\lambda(\gamma) = 1$ . It detects any nonzero coefficient at  $\gamma$ . Thus a series killed by every  $\partial_{\lambda}$  is supported only at 0, giving the joint kernel in (19.3), with no restriction on the size of its support.  $\square$

These intersections do the work that a convergence or compactness argument might otherwise be asked to do [C17].

**Lemma 19.7** (Compatibility with algebraic embeddings [C17]). *Let  $L/F$  be a finite separable extension and  $\sigma : L \hookrightarrow \mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$  an  $F$ -embedding. Each  $\partial_{\lambda}$  on  $F$  has a unique extension to a derivation of  $L$ , and  $\sigma$  intertwines it with the coefficientwise  $\partial_{\lambda}$  on  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$ .*

*Proof.* The derivation  $\partial = \partial_{\lambda}$  preserves  $F$ , since it preserves  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  and obeys the quotient rule. For  $z \in L$  let  $R \in F[X]$  be its minimal polynomial. Differentiating  $\sigma(R)(\sigma z) = 0$  in  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$  gives

$$\partial(\sigma z) = -\frac{\sigma(R^{\partial}(z))}{\sigma(R'(z))} = \sigma\left(-\frac{R^{\partial}(z)}{R'(z)}\right),$$

where  $R^{\partial}$  applies  $\partial$  to the coefficients and  $R'(z) \neq 0$  by separability. Hence the ambient derivation preserves  $\sigma(L)$ ; pulling its restriction back through  $\sigma$  gives a derivation on  $L$  extending the one on  $F$  and proves the required compatibility. Any extension must satisfy the same formula for every  $z$ , so it is unique.  $\square$

**Lemma 19.8** (Constant coefficients remain independent [C17]). *Let  $L/\mathbb{k}$  be a field extension. If  $c_1, \dots, c_k \in L$  are  $\mathbb{k}$ -linearly independent, they are linearly independent over  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ , hence over every subfield of  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ , in the Hahn field  $\mathbb{H}_L(\Gamma_{\mathbb{Q}})$  (for  $L \subseteq \bar{\mathbb{k}}$ , in  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$ ). The same holds for vectors with entries in  $L$ .*

*Proof.* Expand a relation  $\sum_j f_j c_j = 0$  with  $f_j \in \mathbb{H}_{\mathbb{k}}(\Gamma)$  coefficientwise. At each exponent  $\gamma$  it gives the  $\mathbb{k}$ -linear relation  $\sum_j (f_j)_{\gamma} c_j = 0$ , so every coefficient of every  $f_j$  vanishes. A finite union of supports is reverse well ordered, so the comparison is legitimate. The vector case is the same.  $\square$

### 19.4.2 The finite certificate for root velocities

If  $r_1, \dots, r_n$  are distinct roots of a monic polynomial and  $\partial$  is a derivation, put  $w_{ij} = (\partial r_i - \partial r_j)/(r_i - r_j)$  for  $i < j$ . Differentiating the discriminant gives only

$$\frac{\partial \text{Disc}(P)}{\text{Disc}(P)} = 2 \sum_{i < j} w_{ij}, \quad (19.4)$$

and a vanishing sum does not force the individual  $w_{ij}$  to vanish. The missing information is supplied by *every* elementary symmetric function of the  $w_{ij}$  [C17]. We use  $\text{Disc}(P) = \prod_{i < j} (r_i - r_j)^2 = (-1)^{n(n-1)/2} \text{Res}(P, P')$  for monic  $P$ , with the empty-product convention  $\text{Disc}(P) = 1$  in degrees zero and one; the sign disappears under logarithmic differentiation, so (19.4) has no extra sign.

**Lemma 19.9** (Universal denominator certificate [C17]). *Let  $P = X^n + a_1 X^{n-1} + \dots + a_n$  be monic and separable over a field, with roots  $r_1, \dots, r_n$ , let  $\dot{P} = b_1 X^{n-1} + \dots + b_n$ , and put*

$$\dot{r}_i = -\dot{P}(r_i)/P'(r_i), \quad w_{ij} = (\dot{r}_i - \dot{r}_j)/(r_i - r_j), \quad m = \binom{n}{2}, \quad \Delta = \text{Disc}(P).$$

*For  $1 \leq s \leq m$  there is a universal polynomial  $\mathcal{F}_{n,s} \in \mathbb{Z}[a_1, \dots, a_n, b_1, \dots, b_n]$ , homogeneous of degree  $s$  in  $b_1, \dots, b_n$ , with*

$$\Delta^{2s} e_s((w_{ij})_{i < j}) = \mathcal{F}_{n,s}(a_1, \dots, a_n, b_1, \dots, b_n), \quad (19.5)$$

*where  $e_s$  is the  $s$ th elementary symmetric polynomial in the  $m$  edge variables. No optimality of the exponent  $2s$  is asserted.*

*Proof.* The numerator of

$$W(X, Y) = \frac{\dot{P}(Y)P'(X) - \dot{P}(X)P'(Y)}{X - Y} \quad (19.6)$$

vanishes at  $X = Y$ , so  $W$  is a polynomial over  $\mathbb{Z}[a, b]$ , symmetric in  $X, Y$  and linear in  $b$ . A direct calculation gives

$$w_{ij} = \frac{W(r_i, r_j)}{P'(r_i)P'(r_j)}. \quad (19.7)$$

By the root product identity  $\prod_{i=1}^n P'(r_i) = (-1)^m \Delta$  we have  $\Delta/P'(r_i) = (-1)^m \prod_{u \neq i} P'(r_u)$ . In particular the two signs cancel in

$$\Delta^2 w_{ij} = W(r_i, r_j) \prod_{u \neq i} P'(r_u) \prod_{v \neq j} P'(r_v).$$

This is a polynomial in the roots and  $b$ , linear in  $b$ . Its  $s$ th elementary symmetric function is  $\Delta^{2s} e_s(w)$ , a polynomial homogeneous of degree  $s$  in  $b$ . Permuting the roots permutes the unordered edges, and both  $w_{ij}$  and the polynomial just built respect this action, so the latter is symmetric in the roots. Treat the roots as indeterminates, with  $a_j = (-1)^j e_j(r)$  and  $b$  independent. The fundamental theorem on symmetric polynomials over  $\mathbb{Z}[b]$  expresses the polynomial in the  $e_j(r)$ ; substituting  $e_j(r) = (-1)^j a_j$  gives  $\mathcal{F}_{n,s}$  over  $\mathbb{Z}$ . The elimination can be done in each  $b$ -homogeneous component separately, so the homogeneity is kept.  $\square$

The lemma is an identity in universal polynomial rings; it does not assert that integral elements of  $\text{Frac } \mathcal{A}_{\mathbb{k}}(\Gamma)$  lie in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , and no integral closedness of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is used [C17];  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is not integrally closed in general (section 17.5).

*Remark 19.10* (The quadratic certificate [C17]). For  $P = X^2 + aX + b$  and  $\dot{P} = b_1X + b_2$  there is one relative velocity, and (19.4) gives  $w = (ab_1 - 2b_2)/\Delta$  with  $\Delta = a^2 - 4b$ . So the general certificate is  $\mathcal{F}_{2,1} = \Delta(ab_1 - 2b_2)$ , although the exponent 1 already suffices. In the application  $b_1, b_2 \in \Pi_{\mathbb{k}}(\Gamma)$  and  $\Delta \in \mathbb{k}^\times$ , so  $w \in \Pi_{\mathbb{k}}(\Gamma)$ , while  $w \in \mathcal{O}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$ ; hence  $w = 0$ . This is completing the square, and the lemma explains why the same method works in every degree.

**Proposition 19.11** (All relative Euler velocities vanish [C17]). *Let  $P \in \mathcal{A}_{\mathbb{k}}(\Gamma)[X]$  be monic with  $\Delta = \text{Disc}(P) \in \mathbb{k}^\times$ . For every Euler derivation  $\partial = \partial_\lambda$  of  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$  and any two roots  $r_i, r_j \in \mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$  of  $P$ ,  $\partial(r_i - r_j) = 0$ .*

*Proof.* For  $n = 1$  there is nothing to prove. Otherwise differentiating  $P(r_i) = 0$  gives  $\partial r_i = -\dot{P}(r_i)/P'(r_i)$ , where  $\dot{P} = P^\partial$  has the coefficients  $b_j = \partial a_j \in \Pi_{\mathbb{k}}(\Gamma)$  (lemma 16.2). By (19.3) each  $w_{ij} = \partial(r_i - r_j)/(r_i - r_j)$  lies in  $\mathcal{O}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$ , so every  $e_s(w)$  lies in  $\mathcal{O}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$ . On the other hand  $\mathcal{F}_{n,s}$  has positive degree in  $b$  and coefficients in  $\mathbb{Z}[a] \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$ , so  $\mathcal{F}_{n,s}(a, b) \in \Pi_{\mathbb{k}}(\Gamma)$ , and with  $\Delta \in \mathbb{k}^\times$ , (19.5) gives  $e_s(w) \in \Pi_{\mathbb{k}}(\Gamma)$ . By (19.3),  $e_s(w) = 0$  for  $1 \leq s \leq m$ . Thus  $\prod_{i < j} (T - w_{ij}) = T^m$ ; evaluating at an edge velocity gives  $w_{ij}^m = 0$ . Since  $m \geq 1$  and  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$  is a field, every  $w_{ij}$  is zero.  $\square$

### 19.4.3 Translation classification

**Theorem 19.12** (Translation classification [C17]). *Let  $\mathfrak{o} \subseteq \mathbb{k}$  be a unital subring and  $P = X^n + a_1X^{n-1} + \dots + a_n \in \mathcal{R}_{\mathfrak{o}}[X]$ ,  $n \geq 1$ . The following are equivalent:*

- (i)  $\text{Disc}(P)$  is a nonzero constant, that is,  $\text{Disc}(P) \in \mathfrak{o} \setminus \{0\}$ ;
- (ii)  $\bar{P}$  is squarefree and there is  $h \in \Pi_{\mathbb{k}}(\Gamma)$  with

$$P(X) = \bar{P}(X - h). \quad (19.8)$$

The translation is unique and equals

$$h = -\frac{a_1 - \text{ct } a_1}{n}. \quad (19.9)$$

The discriminant need not be a unit of  $\mathfrak{o}$ , and  $1/n$  need not lie in  $\mathfrak{o}$ .

*Proof.* First let  $\mathfrak{o} = \mathbb{k}$ . A nonzero discriminant gives distinct roots  $r_1, \dots, r_n$  in  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$ . Proposition 19.11 for every  $\lambda$  and (19.3) give  $r_i - r_j \in \bar{\mathbb{k}}$  for all  $i, j$ . Let  $\mu = -a_1/n$  be the mean of the roots. Every  $r_i - \mu = n^{-1} \sum_j (r_i - r_j)$  lies in  $\bar{\mathbb{k}}$ , so  $R(X) = P(X + \mu)$  has its coefficients in  $\bar{\mathbb{k}}$ . They also lie in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , because  $\mu \in \mathcal{A}_{\mathbb{k}}(\Gamma)$ , hence in  $\mathcal{A}_{\mathbb{k}}(\Gamma) \cap \bar{\mathbb{k}} = \mathbb{k}$ . Write  $\mu = \mu_0 + h$  with  $\mu_0 = \text{ct } \mu$  and  $h \in \Pi_{\mathbb{k}}(\Gamma)$ . Then  $P(X) = R(X - \mu_0 - h)$ , and applying the ring homomorphism  $\text{ct}$  to the coefficients gives  $\bar{P}(X) = R(X - \mu_0)$ . So  $P(X) = \bar{P}(X - h)$  and  $h = -(a_1 - \text{ct } a_1)/n$ . Translation preserves root differences, so  $\text{Disc}(\bar{P}) = \text{Disc}(P) \neq 0$ ; hence  $\bar{P}$  is squarefree over  $\mathbb{k}$ .

For general  $\mathfrak{o}$ , regard  $P$  as a polynomial over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ . A nonzero constant discriminant lies in  $\mathcal{R}_{\mathfrak{o}} \cap \mathbb{k} = \mathfrak{o}$  and is a unit of  $\mathbb{k}$ , so the first case applies. Since  $a_1 - \text{ct } a_1 \in \Pi_{\mathbb{k}}(\Gamma)$  and  $\Pi_{\mathbb{k}}(\Gamma)$  is a  $\mathbb{k}$ -vector space,  $h \in \Pi_{\mathbb{k}}(\Gamma) \subseteq \mathcal{R}_{\mathfrak{o}}$  even when  $1/n \notin \mathfrak{o}$ , and  $\bar{P} \in \mathfrak{o}[X]$ . Conversely a translation does not change root differences, so  $\bar{P}(X - h)$  has the discriminant of the squarefree  $\bar{P}$ . Uniqueness follows from the coefficient of  $X^{n-1}$ ; for  $n = 1$  the formula is immediate.  $\square$

**Corollary 19.13** (Complete parameterization [C17]). *For  $n \geq 1$  the map  $(\bar{P}, h) \mapsto \bar{P}(X - h)$  is a bijection from pairs of a squarefree monic  $\bar{P} \in \mathfrak{o}[X]$  of degree  $n$  and  $h \in \Pi_{\mathbb{k}}(\Gamma)$  onto the monic polynomials of degree  $n$  in  $\mathcal{R}_{\mathfrak{o}}[X]$  with nonzero constant discriminant. Every fiber of  $P \mapsto \bar{P}$  on the latter set is a free transitive orbit of the additive group  $\Pi_{\mathbb{k}}(\Gamma)$ .*

*Proof.* [Theorem 19.12](#) gives existence and uniqueness. Translation by an element of  $\Pi_{\mathbf{k}}(\Gamma)$  preserves  $\bar{P}$ , and a nonzero translation changes the coefficient of  $X^{n-1}$  in characteristic zero.  $\square$

An equivalent finite coefficient test avoids computing a large discriminant [C17]: given monic  $P$  of positive degree, compute  $h$  by (19.9) and expand  $P(X + h)$ ; then  $P$  has a nonzero constant discriminant if and only if

$$P(X + h) = \bar{P}(X) \quad \text{and} \quad \text{Disc}(\bar{P}) \neq 0. \quad (19.10)$$

This is an algebraic criterion, not an algorithm for arbitrary presentations of infinite Hahn series; it becomes effective only in a representation with exact arithmetic, constant extraction and equality [C17].

**Example 19.14** (An infinite-support family [C17]). Let  $h = \sum_{m \geq 1} \omega^{1/m} \in \Pi$ , whose exponents decrease to 0. The cubic

$$P(X) = (X - h)^3 - 2(X - h) + 1 = X^3 - 3hX^2 + (3h^2 - 2)X - h^3 + 2h + 1$$

lies in  $\mathbf{Oz}[X]$  and has discriminant  $-4(-2)^3 - 27 = 5$ . Its roots are  $h + c_i$ , where  $c_1 = 1$  and  $c_{2,3} = (-1 \pm \sqrt{5})/2$  are the roots of  $X^3 - 2X + 1$ . The translation has infinite support accumulating at the exponent 0; the theorem is not confined to finitely many monomials.

**Corollary 19.15** (Omnific and Gaussian omnific form [C17]). *Let  $P \in \mathbf{Oz}[X]$  be monic of positive degree. Then  $\text{Disc}(P) \in \mathbb{Z} \setminus \{0\}$  if and only if  $P = \bar{P}(X - h)$  with  $\bar{P} \in \mathbb{Z}[X]$  squarefree and  $h \in \Pi$ , and  $h$  is unique. Its roots in  $\mathbf{No}[i]$  are  $h + c_i$  with  $c_i$  the complex roots of  $\bar{P}$ . The same holds over  $\mathbf{Oz}[i]$ , with  $\bar{P} \in \mathbb{Z}[i][X]$  and  $h \in \Pi_{\mathbb{C}}$ .*

*Proof.* The finitely many coefficients lie in a workspace  $\mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \Gamma)$  or  $\mathcal{R}_{\mathbb{Z}[i]}(\mathbb{C}, \Gamma)$  over a set-sized divisible group  $\Gamma \subseteq \mathbf{No}$  ([lemma 2.6](#)). Apply [theorem 19.12](#) with  $(\mathfrak{o}, \mathbf{k}) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , and read the identity in the full class. The roots of  $\bar{P}$  are algebraic over  $\mathbb{Q}$  or  $\mathbb{Q}(i)$ .  $\square$

**Corollary 19.16** (Omnific roots at a constant discriminant [merge]). *Let  $P \in \mathbf{Oz}[X]$  be monic of positive degree with  $\text{Disc}(P) \in \mathbb{Z} \setminus \{0\}$ , and let  $h$  be given by (19.9). The roots of  $P$  in  $\mathbf{Oz}$  are exactly the elements  $h + c$  with  $c$  an integer root of  $\bar{P}$ ; in particular  $P$  has a root in  $\mathbf{Oz}$  if and only if  $\bar{P}$  has one in  $\mathbb{Z}$ , and then every integer root of  $\bar{P}$  lifts to exactly one omnific root. The same holds over  $\mathbf{Oz}[i]$  with  $\mathbb{Z}[i]$ .*

*Proof.* By [corollary 19.15](#) every root in  $\mathbf{No}[i]$  is  $h + c$  with  $c$  a root of  $\bar{P}$ . If  $h + c \in \mathbf{Oz}$ , then  $c = (h + c) - h \in \mathbf{Oz} \cap \mathbb{C} = \mathbb{Z}$  ([proposition 2.5](#)). Conversely  $h + c \in \mathbf{Oz}$  for  $c \in \mathbb{Z}$ . The Gaussian case is the same with  $\mathbf{Oz}[i] \cap \mathbb{C} = \mathbb{Z}[i]$ .  $\square$

**Remark 19.17** (Comparison with the quadratic and lifting results [merge]). For a monic quadratic  $x^2 + bx + c$  over  $\mathbf{Oz}$  with  $\Delta = b^2 - 4c \in \mathbb{Z} \setminus \{0\}$ , [theorem 19.1](#) asks whether  $\Delta$  is a square in  $\mathbf{Oz}$ , and by [proposition 2.8](#) a nonzero integer is an omnific square only if it is an integer square; the roots  $(-b \pm y)/2$  with  $y \in \mathbb{Z}$  are then  $h + c$  for the integer roots  $c$  of  $\bar{P}$ , as [corollary 19.16](#) says. For such polynomials the residue roots do lift: every root of  $\bar{P}$  is simple, and each integer one lifts to exactly one omnific root. The polynomial  $T^2 - T - \omega$  of [example 19.5](#), whose simple residue roots 0 and 1 do not lift, has the nonconstant discriminant  $1 + 4\omega$ ; its translation  $h$  from (19.9) is 0, and  $\bar{P}(X - h) = X^2 - X \neq P$ . So the constant discriminant is exactly the hypothesis that separates the two behaviours in these examples. Monicity cannot simply be dropped:  $P(X) = \omega X^2 + X$  has discriminant 1 and  $\bar{P} = X$ , but no translate of  $\bar{P}$  has degree two. In contrast, the ordinary polynomial  $6x^2 + x - 1$  after [theorem 19.1](#) illustrates a different obstruction: its discriminant is a square but its roots are not omnific integers, because the denominator in the nonmonic quadratic formula is not harmless.

**The polynomial-parameter case [C17].** For  $\Gamma = \mathbb{Z}$  a reverse-well-ordered subset of  $\mathbb{Z}^{\geq 0}$  is finite and  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}[\omega]$ . Writing  $Y = \omega$ , the theorem says that a monic  $P(X, Y) \in \mathbb{k}[Y][X]$  with nonzero constant discriminant satisfies

$$P(X, Y) = P(X - h(Y), 0), \quad h(Y) = -\frac{a_1(Y) - a_1(0)}{n}.$$

This is a uniform answer in all positive degrees and characteristic zero to a question posed on Mathematics Stack Exchange by the user Oblomov [64], and it shows why the familiar quadratic and cubic computations are not isolated. For several parameters, embed  $\mathbb{k}[Y_1, \dots, Y_k]$  in a Hahn ring with  $Y_i = \omega^{\varepsilon_i}$  for the standard basis  $\varepsilon_i$  of lexicographically ordered  $\mathbb{Z}^k$ : every nonconstant monomial has positive exponent, the embedding is injective, and  $\text{ct}$  is evaluation at the origin, so a monic polynomial over  $\mathbb{k}[Y_1, \dots, Y_k]$  with nonzero constant discriminant is a translate of its value at the origin. Source C17 does not present these finite-support cases as new or as a deep theorem about the affine line; its proposed contribution is the arbitrary-rank, arbitrary-support theorem and its consequences.

#### 19.4.4 Factors, Galois groups, marks and resultants

**Proposition 19.18** (Monic factors are translated constant factors [C17]). *Let  $P = \overline{P}(X - h)$  be as in theorem 19.12. Translation by  $h$  is a bijection between the monic divisors of  $\overline{P}$  in  $\mathfrak{o}[X]$  and the monic divisors of  $P$  in  $\mathcal{R}_{\mathfrak{o}}[X]$ ; in particular their monic factorization patterns agree.*

*Proof.* Substitution  $X \mapsto X - h$  carries a monic factorization of  $\overline{P}$  over  $\mathfrak{o}$  to one of  $P$  over  $\mathcal{R}_{\mathfrak{o}}$ . Conversely write  $P = QT$  with  $Q$  monic. Since  $P$  is monic and the coefficient ring is a domain,  $T$  is monic too. Translating gives  $\overline{P} = Q(X + h)T(X + h)$ . In  $\mathbb{H}_{\mathbb{k}}(\Gamma_{\mathbb{Q}})[X]$  the roots of each factor are among the roots of  $\overline{P}$ , hence lie in  $\mathbb{k}$ . Monicity then puts every coefficient of both factors in  $\mathbb{k}$ , including when one factor is 1. These coefficients also lie in  $\mathcal{R}_{\mathfrak{o}}$ , so they belong to  $\mathcal{R}_{\mathfrak{o}} \cap \mathbb{k} = \mathfrak{o}$ . Thus the translated divisor and its quotient both descend. The two substitutions are inverse, giving the bijection.  $\square$

The word “monic” matters over a ring  $\mathfrak{o}$  that is not a field: an assertion about arbitrary factorizations must also account for units and leading coefficients [C17].

**Theorem 19.19** (Galois groups are unchanged [C17]). *Let  $P \in \mathcal{A}_{\mathbb{k}}(\Gamma)[X]$  be monic of positive degree with  $\text{Disc}(P) \in \mathbb{k}^{\times}$ , let  $F = \text{Frac } \mathcal{A}_{\mathbb{k}}(\Gamma)$ , and let  $L_0 \subseteq \mathbb{k}$  be the splitting field of  $\overline{P}$  over  $\mathbb{k}$ . The splitting field of  $P$  over  $F$  in  $\mathbb{H}_{\mathbb{k}}(\Gamma_{\mathbb{Q}})$  is the compositum  $FL_0$ , and restriction is an isomorphism  $\text{Gal}(FL_0/F) \cong \text{Gal}(L_0/\mathbb{k})$ . An irreducible factor of  $\overline{P}$  over  $\mathbb{k}$  stays irreducible over  $F$ , and its translate has the same degree.*

*Proof.* By theorem 19.12 the roots of  $P$  are  $h + c_i$  with  $h \in \Pi_{\mathbb{k}}(\Gamma) \subset F$  and  $c_i$  the roots of  $\overline{P}$ . Thus adjoining them to  $F$  adjoins the  $c_i$ . By lemma 19.8 a  $\mathbb{k}$ -basis of  $L_0$  remains  $F$ -linearly independent. Hence the multiplication map  $F \otimes_{\mathbb{k}} L_0 \rightarrow \mathbb{H}_{\mathbb{k}}(\Gamma_{\mathbb{Q}})$  is injective. Its image is a finite-dimensional domain over  $F$ , therefore a field, and is precisely  $FL_0$ , of degree  $[L_0 : \mathbb{k}]$  over  $F$ .

Every  $\mathbb{k}$ -automorphism of  $L_0$  extends by  $\text{id}_F \otimes \sigma$  through this isomorphism. Conversely an  $F$ -automorphism of  $FL_0$  permutes the roots of  $\overline{P} \in \mathbb{k}[X]$ , so preserves the field  $L_0$  they generate over  $\mathbb{k}$ . Its restriction determines it because  $F$  and  $L_0$  generate the compositum. These operations are inverse. For a root  $c_i$ , the same basis argument for  $\mathbb{k}(c_i)$  gives  $[F(c_i) : F] = [\mathbb{k}(c_i) : \mathbb{k}]$ , proving that its minimal polynomial remains irreducible over  $F$ . Translation is an automorphism of  $F[X]$  preserving degree, which proves the last assertion.  $\square$

For omnific polynomials the coefficient theorem descends to  $\mathbb{Z}$ , but the field of this Galois statement must be specified: with  $\mathbb{k} = \mathbb{R}$  the finer rational Galois group of  $\bar{P}$  is lost. To keep it one uses a Hahn workspace with coefficient field  $\mathbb{Q}$  when the coefficients admit such a presentation, or applies ordinary splitting theory over  $\mathbb{Q}$  to  $\bar{P}$  separately. There is no claim that the rational Galois group survives after extending constants to  $\mathbb{R}$ : for example  $X^2 - 2$  already splits over  $\mathbb{R}$ , although its splitting field over  $\mathbb{Q}$  has degree two [C17].

**Corollary 19.20** (Marked-value rigidity [C17]). *Let  $P = \bar{P}(X - h)$  be as in [theorem 19.12](#), with  $h \in \Pi_{\mathbb{k}}(\Gamma)$ . If  $a \in \mathcal{A}_{\mathbb{k}}(\Gamma)$  and  $P(a) \in \mathbb{k}$ , then  $a - h \in \mathbb{k}$ , that is,  $a = h + \text{ct } a$ . Consequently an ordinary coefficient of  $X^{n-1}$ , or an ordinary value  $P(c)$  at one  $c \in \mathbb{k}$ , forces  $h = 0$  and  $P = \bar{P}$ .*

*Proof.*  $\bar{P}(a - h) = P(a) \in \mathbb{k}$  makes  $a - h \in \mathcal{A}_{\mathbb{k}}(\Gamma)$  a root of  $\bar{P}(T) - P(a) \in \mathbb{k}[T]$ . This polynomial is nonzero because  $\bar{P}$  has positive degree. Thus  $a - h \in \mathbb{k}$  by [lemma 16.1\(d\)](#), and  $\text{ct}(a - h) = \text{ct } a$  identifies this constant, since  $\text{ct } h = 0$ . For  $a = c \in \mathbb{k}$  this gives  $h \in \mathbb{k} \cap \Pi_{\mathbb{k}}(\Gamma) = 0$ ; the coefficient case is [\(19.9\)](#).  $\square$

The marked value may be zero: a monic polynomial with nonzero constant discriminant whose coefficient  $a_n$  (its constant coefficient in  $X$ ) is constant is entirely constant. The constant coefficient in  $X$  must not be confused with the constant term of a series [C17].

**Theorem 19.21** (Coherent translations [C17]). *Let  $\ell \geq 1$  and let  $P_1, \dots, P_\ell \in \mathcal{R}_0[X]$  be monic of positive degree, each with nonzero constant discriminant, and let  $\mathcal{G}$  be a connected graph on  $\{1, \dots, \ell\}$  such that  $\text{Res}(P_i, P_j)$  is a nonzero constant for every edge  $ij$ . Then there is a unique  $h \in \Pi_{\mathbb{k}}(\Gamma)$  with  $P_i(X) = \bar{P}_i(X - h)$  for all  $i$ .*

*Proof.* Write  $P_i = \bar{P}_i(X - h_i)$ . For an edge  $ij$  let  $c_1, \dots, c_a$  and  $d_1, \dots, d_b$  be the roots of  $\bar{P}_i$  and  $\bar{P}_j$ . Then

$$\text{Res}(P_i, P_j) = \prod_{u=1}^a \prod_{v=1}^b (h_i - h_j + c_u - d_v),$$

the value at  $h_i - h_j$  of  $G_{ij}(T) = \text{Res}_X(\bar{P}_i(X - T), \bar{P}_j(X)) \in \mathbb{k}[T]$ . The displayed product shows that  $G_{ij}$  is monic of degree  $ab \geq 1$ . If the resultant equals  $c \in \mathbb{k}$ , then  $h_i - h_j$  is a root of the nonzero polynomial  $G_{ij}(T) - c$ . If  $h_i \neq h_j$ , then  $h_i - h_j$  is a nonzero element of  $\Pi_{\mathbb{k}}(\Gamma)$ , transcendental over  $\mathbb{k}$  ([lemma 16.1\(d\)](#)), which contradicts a constant resultant. So  $h_i = h_j$  along every edge, and connectedness gives a common  $h$ , unique by any one  $P_i$ . When  $\ell = 1$  this is the original translation theorem, with no edge condition. Nonemptiness is needed for uniqueness: an empty family places no restriction on  $h$ .  $\square$

Only the edges of a spanning tree are needed. The nonzero condition says that the marked configurations do not meet along the chosen edges; the proof of equal translations would also work with a zero constant resultant. For the complete graph with nonzero resultants,  $\text{Disc}(\prod_i P_i) = \prod_i \text{Disc}(P_i) \prod_{i < j} \text{Res}(P_i, P_j)^2$  gives a second proof by applying [theorem 19.12](#) to the product; the connected-graph version records the weaker hypothesis [C17].

#### 19.4.5 Finite étale algebras and their units

Throughout,  $B$  is a nonzero finite étale  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -algebra, with the usual commutative-algebra definition of étaleness [61]. Finiteness, flatness and finite presentation give a finite locally free, hence finite projective,  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -module [65]. Its rank is constant, say  $r \geq 1$ , because  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is a domain and so  $\text{Spec } \mathcal{A}_{\mathbb{k}}(\Gamma)$  is connected. The standard inputs are preservation under base change, formal étaleness [62], and the description of a finite étale algebra over a field as a finite

product of finite separable field extensions [52]. The generic algebra  $B_F = F \otimes_{\mathcal{A}_k(\Gamma)} B$  is such a product. A projective module over a domain is a direct summand of a free module, hence torsion free, so  $B \hookrightarrow B_F$  and  $B$  is reduced. The rank cannot be zero: local freeness would then give  $B = 0$ . These statements do not require  $B$  to be a domain; its generic algebra may have several field factors. For  $y \in B$ , multiplication by  $y$  has a characteristic polynomial

$$\text{ch}_y(T) = T^r + c_1(y)T^{r-1} + \cdots + c_r(y) \in \mathcal{A}_k(\Gamma)[T], \quad (19.11)$$

defined in local bases of the projective module; the local coefficients glue by invariance under change of basis, commute with base change and satisfy Cayley–Hamilton, both being local matrix identities. In a splitting field of  $B_F$  it is the product of  $T - \sigma(y)$  over the  $r$  generic embeddings  $\sigma$  of all the field factors. No global basis of  $B$  is assumed [C17].

**Lemma 19.22** (Characteristic polynomials after reduction [C17]). *Let  $M$  be a finite projective module of constant rank  $r$  over a commutative ring  $R$ , let  $J \subseteq R$  be an ideal, and let  $\tau \in \text{End}_R(M)$  have image in  $JM$ . Then the nonleading coefficients of the characteristic polynomial of  $\tau$  lie in  $J$ , and Cayley–Hamilton holds for  $\tau$ .*

*Proof.* On a localization where  $M$  is free, the reduction of the matrix of  $\tau$  modulo  $J$  is zero, so the characteristic polynomial reduces to  $X^r$ . The images of the coefficients in  $R/J$  therefore vanish after localization at every prime, hence vanish. Substituting  $\tau$  into its characteristic polynomial gives an endomorphism that is zero on every free localization, hence zero.  $\square$

**Lemma 19.23** (The extended derivation vanishes in the augmentation fiber [C17]). *Each  $\partial = \partial_\lambda$  extends uniquely to a derivation  $\tilde{\partial}$  of  $B$ , and*

$$\tilde{\partial}(B) \subseteq \Pi_k(\Gamma)B. \quad (19.12)$$

*Proof.* Give  $B[\epsilon]/(\epsilon^2)$  the  $\mathcal{A}_k(\Gamma)$ -algebra structure  $a \mapsto a + \partial(a)\epsilon$ . The identity of its quotient  $B$  lifts uniquely along the square-zero ideal  $(\epsilon)$  by formal étaleness [62]; the lift has the form  $b \mapsto b + \tilde{\partial}(b)\epsilon$ , and its multiplicativity is the Leibniz rule. Since  $\partial(\Pi_k(\Gamma)) \subseteq \Pi_k(\Gamma)$ ,  $\tilde{\partial}$  preserves  $\Pi_k(\Gamma)B$ , and its reduction is a derivation of  $B_0 = B/\Pi_k(\Gamma)B$  that vanishes on  $\mathcal{A}_k(\Gamma)/\Pi_k(\Gamma) = \mathbb{k}$ . But  $B_0$  is finite étale over  $\mathbb{k}$ , a product of finite separable extensions. A derivation  $D$  kills every idempotent  $e$ :  $D(e) = 2eD(e)$ , and  $1 - 2e$  is a unit with square 1. It therefore preserves each field factor. In a separable field factor, differentiating a minimal polynomial  $g$  over  $\mathbb{k}$  gives  $g'(z)D(z) = 0$ , hence  $D(z) = 0$ . This proves (19.12).  $\square$

The quotient rule also extends  $\tilde{\partial}$  to  $B_F$ . The same idempotent calculation shows that it preserves each generic field factor, and lemma 19.7 makes every embedding of that factor into  $\mathbb{H}_{\mathbb{k}}(\Gamma_{\mathbb{Q}})$  compatible with the ambient Euler derivation. Thus the generic images used below are compatible with differentiation even when  $B$  is disconnected.

**Proposition 19.24** (No nonconstant logarithmic derivative of a unit [C17]). *For  $u \in B^\times$  and every  $\lambda$ ,  $\tilde{\partial}_\lambda u = 0$ .*

*Proof.* Put  $y = u^{-1}\tilde{\partial}u$ , which lies in  $\Pi_k(\Gamma)B$  by lemma 19.23. Multiplication by  $y$  has image in  $\Pi_k(\Gamma)B$ , so  $c_j(y) \in \Pi_k(\Gamma)$  for  $j \geq 1$  (lemma 19.22). Realize every generic embedding  $\sigma$  of every field factor of  $B_F$  in  $\mathbb{H}_{\mathbb{k}}(\Gamma_{\mathbb{Q}})$ . By lemma 19.7,  $\sigma(y) = \partial_\lambda \sigma(u)/\sigma(u)$ , with nonzero denominator because  $u$  is a unit, so  $\sigma(y) \in \mathcal{O}_{\mathbb{k}}(\Gamma_{\mathbb{Q}})$  by (19.3). The  $c_j(y)$  are up to sign elementary symmetric functions of these  $r$  values, so  $c_j(y) \in \Pi_k(\Gamma) \cap \mathcal{O}_{\mathbb{k}}(\Gamma_{\mathbb{Q}}) = 0$ . Cayley–Hamilton gives  $m_y^r = 0$  for multiplication  $m_y$  by  $y$ ; evaluating this endomorphism at 1 gives  $y^r = 0$ . As  $B$  is reduced,  $y = 0$ , and so  $\tilde{\partial}u = uy = 0$ .  $\square$

This is the finite-projective analogue of the root-velocity argument: the ideal information comes from the reduction of a multiplication operator instead of a universal discriminant denominator, and disconnected algebras are allowed [C17].

**Proposition 19.25** (Constants, their dimension and their fiber [C17]). *The set  $C_B$  of elements of  $B$  algebraic over  $\mathbb{k}$  is a finite étale  $\mathbb{k}$ -subalgebra, and*

$$C_B = \bigcap_{\lambda} \ker \tilde{\partial}_{\lambda}. \quad (19.13)$$

*The map  $C_B \rightarrow B/\Pi_{\mathbb{k}}(\Gamma)B$  is injective, so  $\dim_{\mathbb{k}} C_B \leq r$ , and the natural map  $\mathcal{A}_{\mathbb{k}}(\Gamma) \otimes_{\mathbb{k}} C_B \rightarrow B$  is injective.*

*Proof.* Elements algebraic over a field form a subalgebra, since finitely many of them generate a finite-dimensional algebra. For an algebraic  $b$ , choose an annihilating polynomial and let  $g \in \mathbb{k}[T]$  be its squarefree part. Some power of  $g$  is divisible by the original polynomial, so  $g(b)$  is nilpotent and hence zero in the reduced algebra  $B$ . In characteristic zero,  $\gcd(g, g') = 1$ ; a Bézout identity evaluated at  $b$  makes  $g'(b)$  invertible in  $\mathbb{k}[b]$ . Since  $\tilde{\partial}$  kills  $\mathbb{k}$ ,  $0 = \tilde{\partial}g(b) = g'(b)\tilde{\partial}b$  gives  $\tilde{\partial}b = 0$ . Conversely, if every  $\tilde{\partial}_{\lambda}$  kills  $b$ , then each of its finitely many generic images is killed by every  $\partial_{\lambda}$  and lies in  $\bar{\mathbb{k}}$  by (19.3); the product of their minimal polynomials over  $\mathbb{k}$  vanishes in every generic embedding. Those embeddings jointly inject  $B_F$  into  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})^r$ , so this polynomial annihilates  $b$  in  $B_F$ , hence in  $B$ .

For  $b \in C_B$  all generic images lie in  $\bar{\mathbb{k}}$ , so the coefficients of  $\text{ch}_b$  lie in  $\mathcal{A}_{\mathbb{k}}(\Gamma) \cap \bar{\mathbb{k}} = \mathbb{k}$ . If moreover  $b \in \Pi_{\mathbb{k}}(\Gamma)B$ , these constant nonleading coefficients lie in  $\Pi_{\mathbb{k}}(\Gamma)$  (lemma 19.22), so  $\text{ch}_b = T^r$ , and Cayley–Hamilton and reducedness give  $b = 0$ . Hence  $C_B$  embeds in the  $r$ -dimensional fiber  $B/\Pi_{\mathbb{k}}(\Gamma)B$ . A finite-dimensional reduced algebra over a field of characteristic zero is a product of finite separable extensions, hence finite étale. Finally embed  $B$  in the product of the generic splitting fields, and then in  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})^r$ . The images of a  $\mathbb{k}$ -basis of  $C_B$  are  $\mathbb{k}$ -linearly independent vectors with entries in  $\bar{\mathbb{k}}$ , so they are  $F$ -linearly independent by the vector form of lemma 19.8, hence  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -linearly independent. Since these basis elements also form an  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -basis of  $\mathcal{A}_{\mathbb{k}}(\Gamma) \otimes_{\mathbb{k}} C_B$ , this is exactly the asserted injectivity; it does not assert surjectivity.  $\square$

**Theorem 19.26** (Units of finite étale algebras [C17]). *Let  $B$  be a nonzero finite étale  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -algebra of rank  $r$ . Then  $C_B$  is finite étale over  $\mathbb{k}$  with  $\dim_{\mathbb{k}} C_B \leq r$ , and*

$$B^{\times} = C_B^{\times}. \quad (19.14)$$

*The map  $\mathcal{A}_{\mathbb{k}}(\Gamma) \otimes_{\mathbb{k}} C_B \rightarrow B$  is injective, and it is an isomorphism if and only if  $B$  is generated as an  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -algebra by its units.*

*Proof.* Proposition 19.24 and (19.13) give  $B^{\times} \subseteq C_B$ , and the inverse of a unit is killed by the same derivations; so  $B^{\times} = C_B^{\times}$ . The other structural assertions are proposition 19.25. If  $B$  is generated by its units, it is generated by elements of  $C_B$ , so the injective map is onto. Conversely a finite étale  $\mathbb{k}$ -algebra  $C = \prod_j L_j$  is spanned by its units: for  $c \in C$  choose  $0 \neq \lambda_0 \in \mathbb{k}$  different from every component  $c_j$  that lies in  $\mathbb{k}$ , which is possible because  $\mathbb{k}$  is infinite; then  $\lambda_0$  and  $c - \lambda_0$  are units with sum  $c$ . So  $\mathcal{A}_{\mathbb{k}}(\Gamma) \otimes_{\mathbb{k}} C_B$ , when it equals  $B$ , is generated by units.  $\square$

**Corollary 19.27** (Geometrically constant coefficients [C17]). *If  $\mathbb{k}$  is algebraically closed, then  $C_B \cong \mathbb{k}^j$  for some  $1 \leq j \leq r$  and  $B^{\times} \cong (\mathbb{k}^{\times})^j$ ; if moreover  $B$  is connected,  $C_B = \mathbb{k}$  and  $B^{\times} = \mathbb{k}^{\times}$ .*

*Proof.* A finite étale algebra over an algebraically closed field is a product of copies of the field, and two or more factors give nontrivial idempotents of  $B$ . More generally, every idempotent of  $B$  already belongs to  $C_B$ , because it satisfies  $T^2 - T = 0$ . Thus the constant subalgebra records all idempotents, even before assuming that  $\mathbb{k}$  is algebraically closed.  $\square$

**Why this is not yet a covering classification [C17].** A finite étale algebra might contain elements that are not combinations of its units, and [theorem 19.26](#) does not exclude them. The generic primitive element theorem does not give a global monogenic presentation over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ : a generic generator may need localization, and its discriminant need not be a unit before localization. The injection  $C_B \rightarrow B/\Pi_{\mathbb{k}}(\Gamma)B$  isolates the issue: every algebraic constant is visible in the augmentation fiber, but it is not shown that every element of the fiber lifts to one. Such a surjectivity, with a global descent argument, would be a substantive further step, and it is not used below ([question 21.31](#)).

*Remark 19.28* (The étale norm theorem as a constant case [merge]). [Theorem 19.26](#) contains [theorem 6.15](#) when  $\mathbb{k}$  has characteristic zero. The coordinate conclusion is vacuous when  $E = 0$ , so assume  $E \neq 0$ . With  $K \subseteq \mathbb{k}$ ,  $E$  and  $e_1, \dots, e_r$  as there, the algebra  $B = \mathcal{A}_{\mathbb{k}}(\Gamma) \otimes_K E$  is finite étale over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  of rank  $r$  by base change, and  $z = \sum_j e_j x_j$  with  $x_j \in A \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$  has norm  $N(\mathbf{x}) = a \in K^\times$ , so multiplication by  $z$  has invertible determinant in the induced  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -basis. It is therefore an invertible endomorphism; applying its inverse to 1 gives the inverse of  $z$ . Thus  $z \in B^\times \subseteq C_B$ . Each of the  $r$  maps  $\sigma_\ell : E \rightarrow \bar{\mathbb{k}}$  extends to  $B \rightarrow \mathcal{A}_{\bar{\mathbb{k}}}(\Gamma)$ , and  $\sigma_\ell(z) = \sum_j \sigma_\ell(e_j) x_j$  is algebraic over  $\mathbb{k}$ , hence constant ([corollary 16.4](#) over  $\bar{\mathbb{k}}$ ). For every  $\gamma \neq 0$  the coefficient vector  $\mathbf{x}_\gamma$  therefore satisfies  $M\mathbf{x}_\gamma = 0$  with the invertible matrix  $M = (\sigma_\ell(e_j))$ , so  $x_j \in \mathbb{k} \cap A = \mathfrak{o}$ . The last step is the one of the printed proof; the unit theorem replaces its factorization of the norm. It does not replace the printed proof, which works in every characteristic (paragraph *Boundary examples and characteristic* after [theorem 6.15](#)), whereas source C17 uses characteristic zero throughout. [Theorem 19.26](#) goes further:  $B$  need not be a base change, and all its units are algebraic. The dual-number example after [theorem 6.15](#), a unit with a nonconstant nilpotent coordinate, and the finite free reduced algebra of [example 19.45](#), a nonconstant unit without nilpotents, are the two ways the étale hypothesis can fail.

#### 19.4.6 Monogenic and arithmetic descent

**Theorem 19.29** (Monogenic finite étale algebras are constant [C17]). *If a finite étale  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -algebra  $B$  is generated by one element  $b$ , then  $B \cong \mathcal{A}_{\mathbb{k}}(\Gamma) \otimes_{\mathbb{k}} C$  for a finite étale  $\mathbb{k}$ -algebra  $C$ . More precisely, if  $B$  has rank  $r$  and  $P = \text{ch}_b$ , then*

$$B \cong \mathcal{A}_{\mathbb{k}}(\Gamma)[X]/(P) \cong \mathcal{A}_{\mathbb{k}}(\Gamma)[X]/(\bar{P}),$$

*the second isomorphism being induced by a translation by an element of  $\Pi_{\mathbb{k}}(\Gamma)$ .*

*Proof.* If  $B = 0$ , take  $C = 0$  and  $P = 1$ ; both displayed quotients are zero. Otherwise  $r \geq 1$ . Evaluation at the generator  $b$  is surjective, and Cayley–Hamilton makes it factor through  $\mathcal{A}_{\mathbb{k}}(\Gamma)[X]/(P)$ . Since  $P$  is monic of degree  $r$ , its quotient has basis  $1, X, \dots, X^{r-1}$ . On any localization where  $B$  is free, the resulting surjection is a square matrix with a right inverse, so its determinant is a unit and the matrix is invertible. Thus the map is an isomorphism locally everywhere, hence globally. In the induced power basis of  $B$ , the trace-pairing determinant is  $\text{Disc}(P)$ : after splitting over a field it is the determinant of  $V^T V$  for the Vandermonde matrix of roots. This is a universal polynomial identity in the coefficients, so it holds over the original ring. Étaleness makes this a unit of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  by the discriminant criterion [63], hence an element of  $\mathbb{k}^\times$  ([lemma 16.1\(c\)](#)). [Theorem 19.12](#) gives  $P = \bar{P}(X - h)$  with  $h \in \Pi_{\mathbb{k}}(\Gamma)$ , and  $Y = X - h$

gives the second isomorphism explicitly by  $Y \mapsto X - h$ , with inverse  $X \mapsto Y + h$ . Now  $C = \mathbb{k}[Y]/(\overline{P})$  is finite étale because  $\overline{P}$  is squarefree, and base change identifies  $\mathcal{A}_{\mathbb{k}}(\Gamma) \otimes_{\mathbb{k}} C$  with  $\mathcal{A}_{\mathbb{k}}(\Gamma)[Y]/(\overline{P})$ .  $\square$

**Corollary 19.30** (The split complex case [C17]). *Every monogenic finite étale  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ -algebra is a finite product of copies of  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ .*

*Proof.* Over  $\mathbb{C}$  the squarefree  $\overline{P}$  splits into distinct linear factors whose pairwise differences are units of  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ , so the Chinese remainder theorem applies over  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ .  $\square$

**Theorem 19.31** (Arithmetic monic descent [C17]). *Let  $\mathfrak{o} \subseteq \mathbb{k}$  be a unital subring and  $P \in \mathcal{R}_{\mathfrak{o}}[X]$  monic with  $\text{Disc}(P) \in \mathfrak{o} \setminus \{0\}$ . Then there is an  $\mathcal{R}_{\mathfrak{o}}$ -algebra isomorphism*

$$\mathcal{R}_{\mathfrak{o}}[X]/(P) \cong \mathcal{R}_{\mathfrak{o}} \otimes_{\mathfrak{o}} \mathfrak{o}[X]/(\overline{P}). \quad (19.15)$$

*If  $\text{Disc}(P) \in \mathfrak{o}^{\times}$ , both sides are finite étale over  $\mathcal{R}_{\mathfrak{o}}$  and  $\mathfrak{o}[X]/(\overline{P})$  is finite étale over  $\mathfrak{o}$ . Every monogenic finite étale  $\mathcal{R}_{\mathfrak{o}}$ -algebra descends in this way.*

*Proof.* For the degree-zero polynomial  $P = 1$ , both sides of (19.15) are zero and the assertion is immediate. Otherwise translation by  $h \in \Pi_{\mathbb{k}}(\Gamma) \subseteq \mathcal{R}_{\mathfrak{o}}$  is an automorphism of the polynomial ring, so [theorem 19.12](#) gives  $\mathcal{R}_{\mathfrak{o}}[X]/(P) \cong \mathcal{R}_{\mathfrak{o}}[Y]/(\overline{P})$ . The latter is the displayed tensor product by base change of the presentation; no flatness of  $\mathcal{R}_{\mathfrak{o}}$  over  $\mathfrak{o}$  is needed for this identity. For a unit discriminant the finite free trace pairing is perfect, and the discriminant criterion gives étaleness over each coefficient ring. For a monogenic nonzero finite étale algebra the characteristic-polynomial argument of [theorem 19.29](#) works over the domain  $\mathcal{R}_{\mathfrak{o}}$ , and its discriminant is a unit of  $\mathcal{R}_{\mathfrak{o}}$ , hence of  $\mathfrak{o}$  ([lemma 16.1\(c\)](#)). The zero algebra descends from the zero  $\mathfrak{o}$ -algebra separately.  $\square$

For example  $P = (X - h)^2 - 2 \in \mathbf{Oz}[X]$ ,  $h \in \Pi$ , has discriminant 8 and is a translate of an ordinary polynomial, although its algebra is not étale at the prime 2. The theorem classifies the polynomial when the stronger étale hypothesis fails; confusing “nonzero integer” with “unit integer” would erase the distinction [C17].

*Remark 19.32* (The arithmetic scope [C17]). The descent for  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  concerns finite monic presentations, read in set-sized workspaces as before. [Theorem 19.26](#) is proved for  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k} \oplus \Pi_{\mathbb{k}}(\Gamma)$ , not for  $\mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma)$  by replacing  $\mathbb{k}$  with  $\mathfrak{o}$ : finite étale algebras over  $\mathfrak{o}$  may have a richer arithmetic unit group, and reduction modulo  $\Pi_{\mathbb{k}}(\Gamma)$  need not be a product of fields. No extension of the unit theorem to that case is made ([question 21.32](#)).

### 19.4.7 Critical-point configurations

A translation theorem for the derivative gives a consequence about polynomials as algebraic maps, not about iteration or analytic continuation [C17].

**Theorem 19.33** (Two-translation classification [C17]). *Let  $f = X^d + a_1 X^{d-1} + \cdots + a_d \in \mathcal{R}_{\mathfrak{o}}[X]$  with  $d \geq 2$ , and suppose that the monic critical polynomial  $f'/d \in \mathcal{A}_{\mathbb{k}}(\Gamma)[X]$  has nonzero constant discriminant. Then there are unique  $h, \beta \in \Pi_{\mathbb{k}}(\Gamma)$  with*

$$f(X) = \overline{f}(X - h) + \beta, \quad h = -\frac{a_1 - \text{ct } a_1}{d}, \quad \beta = f(h) - \overline{f}(0). \quad (19.16)$$

*Conversely, a polynomial of the form (19.16) has nonzero constant critical discriminant exactly when  $\overline{f}'/d$  is squarefree.*

*Proof.* The coefficient of  $X^{d-2}$  in  $f'/d$  is  $(d-1)a_1/d$ , so [theorem 19.12](#) over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  gives  $f'(X) = \bar{f}'(X-h)$  with  $h$  as stated. We apply the theorem over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ : division by  $d$  need not preserve  $\mathcal{R}_{\mathfrak{o}}$ . Constant extraction commutes with formal differentiation, so the constant-term polynomial of  $f'/d$  is  $\bar{f}'/d$ . Then  $f(X) - \bar{f}(X-h)$  has zero derivative. In characteristic zero every positive degree coefficient of this difference vanishes, so it is a constant  $\beta$  with respect to  $X$ . It belongs to  $\mathcal{R}_{\mathfrak{o}}$ , and applying  $\text{ct}$  to the identity gives  $\bar{f}(X) - \bar{f}(X) = \text{ct } \beta = 0$ . Thus  $\beta \in \Pi_{\mathbb{k}}(\Gamma)$ . Evaluation at  $X = h$  gives the formula. Uniqueness of  $h$  follows from the coefficient of  $X^{d-1}$ , and that of  $\beta$  is then immediate. The converse follows by differentiating, since translation does not change the discriminant.  $\square$

If  $\xi_1, \dots, \xi_{d-1} \in \bar{\mathbb{k}}$  are the critical points of  $\bar{f}$ , those of  $f$  are  $h + \xi_j$ , with critical values  $\beta + \bar{f}(\xi_j)$ ; all gaps between critical points and between critical values lie in  $\bar{\mathbb{k}}$  (not necessarily in  $\mathbb{k}$ ), and distinct critical values of  $\bar{f}$  remain distinct. This is a classification up to independent source and target translations. It is not in general a conjugacy  $f = T_h \circ \bar{f} \circ T_{-h}$ , where  $T_h(X) = X + h$ ; that would need  $\beta = h$  [C17].

**Example 19.34** (A repeated critical value [C17]). For  $s, t \in \Pi_{\mathbb{k}}(\Gamma)$ ,  $f(X) = (X-s)^4 - 2(X-s)^2 + t$  has critical polynomial  $(X-s)^3 - (X-s)$ , of discriminant 4, critical points  $s-1, s, s+1$  and critical values  $t-1, t, t+1$ , and  $\bar{f} = X^4 - 2X^2$ . The critical points must be simple, but the theorem allows repeated critical values.

*Remark 19.35* (The same normal forms in the sibling report [merge]). The sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` reaches the same normal forms from a different hypothesis. For a positive-degree integer-valued polynomial  $f$  over  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  with more than  $d = \deg f$  inputs of ordinary output, its canonical translation (`osq:or:thm:canonical`, `osq:or:thm:eta`) gives  $f = \text{ct}_*(f)(X-\eta)$  with  $\eta = -\hat{a}_{d-1}/(\hat{a}_d)$ , where  $\text{ct}_*(f) = \bar{f}$  and  $\hat{a} = a - \text{ct } a$ ; for monic  $f$  this is (19.9). Its rich-target theorem (`osq:or:thm:targets`), for  $d \geq 2$ , gives  $f = \bar{f}(X-\eta) + \beta$  with  $\beta = f(\eta) - \text{ct } f(\eta)$ , the form (19.16). For monic  $P \in \mathbf{Oz}[X]$  with  $\text{Disc}(P) \in \mathbb{Z} \setminus \{0\}$ , [corollary 19.15](#) gives  $P(h+c) = \bar{P}(c) \in \mathbb{Z}$  for every  $c \in \mathbb{Z}$ , so the sibling hypothesis holds and the two translations coincide. Likewise, for  $f \in \mathbf{Oz}[X]$ , [theorem 19.33](#) makes  $\beta + \mathbb{Z}$  a rich target, since  $f(h+c) = \bar{f}(c) + \beta$  for all  $c \in \mathbb{Z}$ . For  $f \in \mathbf{Oz}[i][X]$  the same assertion uses  $\beta + \mathbb{Z}[i]$  and  $c \in \mathbb{Z}[i]$ . The converse fails:  $(X-h)^2$  with  $0 \neq h \in \Pi$  has ordinary outputs on all of  $h + \mathbb{Z}$  and discriminant zero. The sibling theorems concern integer-valued polynomials and the specified ordinary-output sets  $\mathbb{Z}$  or  $\mathbb{Z}[i]$ , not discreteness alone, whereas [theorem 19.12](#) holds over every  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  and  $\mathcal{R}_{\mathfrak{o}}$ . Source C17 does not cite the sibling report, of which it read only the guide.

### 19.4.8 Normal and Hermitian matrices

Work in  $\mathbb{H}_{\mathbb{R}}(\Gamma_{\mathbb{Q}})$ , ordered by the sign of the coefficient of the dominant monomial (the one of largest exponent), and in  $\mathbb{H}_{\mathbb{C}}(\Gamma_{\mathbb{Q}})$ , with conjugation on coefficients. For a vector  $v$  the form  $\langle v, v \rangle = \sum_j v_j \bar{v}_j$  is positive for  $v \neq 0$ ; the arguments are finite algebra over this ordered field, not Hilbert-space arguments [C17]. For a matrix,  $M^*$  is the conjugate transpose and  $\text{ct } M$  applies  $\text{ct}$  entrywise.

**Lemma 19.36** (A normal idempotent is an orthogonal projection [C17]). *Let  $P \in M_n(\mathbb{H}_{\mathbb{C}}(\Gamma_{\mathbb{Q}}))$  satisfy  $P^2 = P$  and  $PP^* = P^*P$ . Then  $P = P^*$ ,  $0 \leq P_{jj} \leq 1$  and  $|P_{jk}|^2 \leq 1$ . If its entries lie in  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ , they lie in  $\mathbb{C}$ .*

*Proof.* Normality gives  $\langle Pv, Pv \rangle = \langle P^*v, P^*v \rangle$ , so positive definiteness gives  $\ker P = \ker P^*$ . Idempotence decomposes each vector as  $v = Pv + (v - Pv)$ , in  $\text{im } P \oplus \ker P$ . The summands are orthogonal: if  $z \in \ker P$ , then  $\langle Px, z \rangle = \langle x, P^*z \rangle = 0$ . Applying these decompositions to two

vectors gives  $\langle Pv, w \rangle = \langle Pv, Pw \rangle = \langle v, Pw \rangle$ , hence  $P = P^*$ . Its diagonal entries are therefore real. The  $jj$  entry of  $P = PP^*$  is  $P_{jj} = \sum_k P_{jk} \overline{P_{jk}}$ , so  $P_{jj} \geq 0$  and  $P_{jj} \geq P_{jj}^2$ , whence  $P_{jj} \leq 1$ , and every summand is at most 1. An element of positive degree has squared modulus of positive degree with positive dominant coefficient, so it is infinite and exceeds 1. Hence every entry lies in  $\mathcal{O}_{\mathbb{C}}(\Gamma_{\mathbb{Q}})$ , and an entry in  $\mathcal{A}_{\mathbb{C}}(\Gamma)$  lies in  $\mathbb{C}$  by (19.3).  $\square$

**Theorem 19.37** (Normal matrices [C17]). *Let  $n \geq 1$  and let  $M \in M_n(\mathcal{A}_{\mathbb{C}}(\Gamma))$  satisfy  $MM^* = M^*M$ , and suppose that its characteristic polynomial has nonzero constant discriminant. Then*

$$M = \text{ct } M + hI_n, \quad h = \frac{\text{tr } M - \text{ct}(\text{tr } M)}{n} \in \Pi_{\mathbb{C}}(\Gamma). \quad (19.17)$$

*For Hermitian  $M$ ,  $h$  has real coefficients. In particular this applies to normal matrices over  $\mathbf{Oz}[i]$  and real symmetric matrices over  $\mathbf{Oz}$ , read in a workspace.*

*Proof.* Since the determinant is a polynomial in the matrix entries,  $\text{ct}(\det(XI_n - M)) = \det(XI_n - \text{ct } M)$ , where  $\text{ct}$  acts on coefficients. Thus theorem 19.12 applied to the characteristic polynomial says that the eigenvalues of  $M$  are the distinct elements  $h + c_j$ , where  $c_1, \dots, c_n \in \mathbb{C}$  are the eigenvalues of  $\text{ct } M$  and  $h$  is as stated. The Lagrange spectral idempotents

$$P_j = \prod_{u \neq j} \frac{M - (h + c_u)I_n}{c_j - c_u} \quad (19.18)$$

have entries in  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ , since the denominators are nonzero complex numbers. Interpolation modulo the squarefree characteristic polynomial and Cayley–Hamilton give  $P_j^2 = P_j$ ,  $P_j P_u = 0$  for  $j \neq u$ ,  $\sum_j P_j = I_n$  and  $M = \sum_j (h + c_j)P_j$ . These identities follow by evaluating the corresponding scalar polynomials at each of the distinct roots, then using divisibility by the characteristic polynomial. For  $n = 1$  the empty product in (19.18) is  $I_1$ . The adjoint of a polynomial in  $M$  is a polynomial in  $M^*$  with conjugated scalar coefficients. Since  $M$  and  $M^*$  commute, polynomials in them commute as well, including with these possibly nonconstant scalar coefficients. Thus every  $P_j$  is normal, and lemma 19.36 gives  $P_j \in M_n(\mathbb{C})$ . Thus  $M = hI_n + \sum_j c_j P_j$ , and taking constant terms identifies the sum as  $\text{ct } M$ . For Hermitian  $M$  the trace is real, hence so is  $h$ .  $\square$

**Corollary 19.38** (Traceless rigidity and fixed eigenspaces [C17]). *Under the hypotheses of theorem 19.37,  $M - \frac{\text{tr } M}{n}I_n = \text{ct } M - \frac{\text{tr}(\text{ct } M)}{n}I_n$ . An ordinary trace forces  $M$  to be an ordinary matrix, all spectral projections are ordinary, and an ordinary unitary change of basis diagonalizes  $\text{ct } M$  and  $M$  simultaneously.*

*Proof.* Subtract the traces in (19.17). The proof of the theorem makes every spectral projection ordinary. Constant extraction commutes with conjugation, transpose and matrix multiplication, so  $\text{ct } M$  is a normal complex matrix. The ordinary finite-dimensional spectral theorem supplies a unitary matrix over  $\mathbb{C}$  diagonalizing  $\text{ct } M$ ; it also diagonalizes  $M$ , since adding  $hI_n$  leaves the eigenspaces unchanged.  $\square$

**Remark 19.39** (What positivity contributes [C17]). The characteristic-polynomial theorem controls eigenvalues but not entries or eigenspaces. Positivity turns normal idempotents into bounded projections, and one-sided support turns bounded entries into constants; without this step the matrix conclusion is false (example 19.44). No physical prediction about infinitesimal observables follows from the algebraic statement.

**Remark 19.40** (Symmetric omnific matrices [merge]). For real symmetric matrices over  $\mathbf{Oz}$  there is a second route through corollary 9.5. If the characteristic polynomial of  $M$  has discriminant in  $\mathbb{Z} \setminus \{0\}$ , corollary 19.15 writes it as  $\overline{P}(X - h)$  with  $h = (\text{tr } M - \text{ct } \text{tr } M)/n \in \Pi$ . Then

$M - hI_n$  is a symmetric matrix over  $\mathbf{Oz}$  with the integer characteristic polynomial  $\bar{P}$ , so all its entries are ordinary integers by [corollary 9.5](#), and  $M - hI_n = \text{ct } M$ . That corollary needs no discriminant hypothesis when the characteristic polynomial is already constant, while [theorem 19.37](#) allows a purely infinite trace and complex normal matrices. The nilpotent example after [corollary 9.5](#) and [example 19.44](#) show the same failure without symmetry or normality.

#### 19.4.9 Counterexamples and the boundary

**Example 19.41** (Two-sided coefficients [C17]). In a real closed Hahn field, or in  $\mathbf{No}$ , let  $x = \omega$  and

$$\rho = \frac{x - \sqrt{x^2 - 4/x}}{2}. \quad (19.19)$$

The square root exists since  $x^2 - 4/x > 0$ . Rationalizing gives  $\rho = 2/[x(x + \sqrt{x^2 - 4/x})]$ ; its denominator has dominant term  $2x^2$ , so  $\rho$  is positive infinitesimal with dominant term  $x^{-2}$ . The elements  $\rho$  and  $x - \rho$  are the roots of  $Y^2 - xY + 1/x$ , so  $\rho(x - \rho) = x^{-1}$  and  $0 < \rho < x$ . The monic cubic  $P = X(X - x)(X - \rho)$  has distinct roots and  $\text{Disc}(P) = x^2\rho^2(x - \rho)^2 = 1$ , but its root gap  $x$  is infinite, so it is not a translate of a constant polynomial. Its coefficient  $x\rho$  is a nonzero infinitesimal, so  $P$  does not have coefficients in  $\mathcal{B}_{\mathbb{R}}$ . The real diagonal matrix  $\text{diag}(0, x, \rho)$  is normal with the same discriminant and violates [theorem 19.37](#) once the support hypothesis is dropped. So “over surreal numbers” cannot replace “with coefficients in the one-sided ring”. (Here  $\rho$  replaces the source’s  $r$ .)

**Example 19.42** (Characteristic  $p$  [C17]). Let  $\mathbb{k}$  have characteristic  $p > 0$ ,  $\gamma > 0$  and  $u = \omega^\gamma$  in the nonnegative-support ring over  $\mathbb{k}$ . The polynomial

$$P(X) = X^{p^2} + uX^p + X \quad (19.20)$$

has derivative 1, so it is separable with the constant discriminant  $(-1)^{p^2(p^2-1)/2}$ , and  $\bar{P} = X^{p^2} + X$ . But  $\bar{P}(X - h) = X^{p^2} + X - h^{p^2} - h$  has no term in  $X^p$  for any  $h$ , so  $P \neq \bar{P}(X - h)$ . This works in every positive characteristic and avoids an Artin–Schreier example whose right side could be absorbed by an infinite translation. It is consistent with the failure of squarefree rigidity in characteristic  $p$  ([section 18.3](#)).

**Example 19.43** (Zero discriminant [C17]). For  $0 \neq s \in \Pi_{\mathbb{k}}(\Gamma)$ ,  $X^2(X - s)$  has discriminant zero and  $\bar{P} = X^3$ , but it is not  $(X - h)^3$ , which has one distinct root. Repeated roots allow relative motion that the logarithmic-gap argument cannot constrain, because its denominators vanish.

**Example 19.44** (Nonnormal matrices [C17]). For distinct  $c_1, c_2 \in \mathbb{C}$  and  $0 \neq s \in \Pi_{\mathbb{C}}(\Gamma)$ , the matrix  $M = \begin{pmatrix} c_1 & s \\ 0 & c_2 \end{pmatrix}$  has constant characteristic polynomial  $(X - c_1)(X - c_2)$  with nonzero constant discriminant, but it is not  $\text{ct } M + hI_2$ ; its corner entry moves an eigenspace without moving an eigenvalue. For  $c_1 = 0$ ,  $c_2 = 1$  and real  $s$ ,  $MM^* - M^*M$  has the diagonal entry  $s^2 \neq 0$ .

**Example 19.45** (Finite free and reduced is not enough [C17]). Take  $\Gamma = \mathbb{Z}$ , so  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}[\omega]$ , and  $\mathbf{B} = \mathcal{A}_{\mathbb{k}}(\Gamma)[Y]/(Y^2 - \omega^2 - 1)$ . It is finite free of rank two and reduced, since its generic polynomial is separable and it injects into its generic fiber. It has the unit  $u = \omega + Y$  with inverse  $Y - \omega$ , and  $u$  is not algebraic over  $\mathbb{k}$ : otherwise  $u^{-1}$  would be algebraic too, and so would  $2\omega = u - u^{-1}$ . Its discriminant  $4(\omega^2 + 1)$  is not a unit of  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , so  $\mathbf{B}$  is not étale, and replacing “finite étale” by “finite free and reduced” makes [theorem 19.26](#) false. In  $\mathbb{H}_{\mathbb{k}}(\Gamma)$  the generic image of  $Y$  is  $\pm\sqrt{\omega^2 + 1}$ , the element of [example 19.2](#) and [proposition 4.12](#); the unit  $u$  becomes  $\omega + \sqrt{\omega^2 + 1}$ , infinite, with inverse  $\sqrt{\omega^2 + 1} - \omega = \frac{1}{2}\omega^{-1} - \dots$  [merge].

Source C17's theorems do not classify the prime elements of  $\mathbf{Oz}$ , prove a unique factorization or refinement conjecture, determine birthdays of products, or construct a preferred derivation of the surreal field. No set-theoretic strength beyond set-sized Hahn and vector-space constructions is used; a request for a broad research direction was answered by a focused algebraic theorem package, not by a claim about large cardinals [C17].

#### 19.4.10 A second proof and the common mechanism

The unit theorem gives a second route to [theorem 19.12](#) that does not use the universal certificate [C17].

**Lemma 19.46** (Finite étale splitting algebra [C17]). *If  $R$  is a commutative ring and  $P \in R[X]$  is monic of degree  $n$  with unit discriminant, there is a finite free étale  $R$ -algebra  $B_P$  of rank  $n!$  in which  $P = \prod_{j=1}^n (X - r_j)$  and every difference  $r_i - r_j$ ,  $i \neq j$ , is a unit.*

*Proof.* For  $n = 0$ , take  $B_P = R$ , using the empty product  $P = 1$  and  $0! = 1$ . For  $n = 1$ , its root already lies in  $R$ , so again take  $B_P = R$ . For  $n \geq 2$ , adjoin a root  $r_1$  by  $R_1 = R[X]/(P)$ , finite free of rank  $n$  and étale by the discriminant criterion. Divide  $P$  by  $X - r_1$  to get a monic  $Q_1 \in R_1[X]$  of degree  $n - 1$ . The identity  $\text{Disc}(P) = \text{Disc}(Q_1)Q_1(r_1)^2$  shows that  $\text{Disc}(Q_1)$  is a unit, since factors of a unit are units. Apply induction to  $Q_1$  over  $R_1$ . The composite has an  $R$ -basis obtained by multiplying the two free bases, so its rank is  $n(n - 1)! = n!$ , and compositions of finite étale maps are finite étale. In the result the product of the squared differences is  $\text{Disc}(P)$ , so each difference is a unit.  $\square$

*Second proof of [theorem 19.12](#) [C17].* Over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ , the splitting algebra in [lemma 19.46](#) is nonzero because it is free of positive rank  $n!$ . Apply [theorem 19.26](#): the root differences in  $B_P$  are units, hence algebraic over  $\mathbb{k}$ . Map a field factor of the generic algebra of  $B_P$  to  $\mathbb{H}_{\bar{\mathbb{k}}}(\Gamma_{\mathbb{Q}})$ . The images of the roots split  $P$ ; their differences remain nonzero because units map to units. Thus the roots remain distinct, and every gap is algebraic over  $\mathbb{k}$ , so it lies in  $\bar{\mathbb{k}}$  (by [corollary 16.4](#) for the coefficient field  $\bar{\mathbb{k}}$ ). Subtracting the mean and taking constant terms proves  $P = \bar{P}(X - h)$  as in the first proof, and the arithmetic pullback follows as there.  $\square$

There is no circularity: the proof of [theorem 19.26](#) uses the Euler derivations, formal étale lifting, characteristic polynomials and opposite supports, not [theorem 19.12](#). The first proof remains useful because its certificate is finite and uniform in the degree and avoids the splitting algebra of rank  $n!$  [C17].

Both proofs follow one pattern. A derivation sends the coefficient ring  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  into the strict ideal; logarithmic differentiation produces quantities in the opposite valuation ring; a symmetric or characteristic polynomial puts all their invariants into the strict ideal as well; the intersection is zero, so the polynomial is a pure power and reducedness kills the quantities; a separating family of derivations then identifies the residual constants. The point is not merely that “infinite and finite cannot agree”: individual roots or elements of a finite algebra need not lie in the ring, and the invariant polynomial carries the information back to the ring, where the intersection argument is legitimate. So the trace of a logarithmic derivative alone, or the product of the root gaps alone, does not suffice [C17]. This is the two-ring mechanism of [section 16.5](#), with the pair  $(\mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma))$  and the valuation ring, applied to roots and to finite algebras instead of points of varieties [merge].

### 19.4.11 Source C17’s verification and formalization notes

Source C17’s script (`code/17-discriminant-rigidity-verify.py`) performs exact finite computations over rational polynomial rings and finite fields: translation formulas, discriminants, resultants, the quadratic certificate, root-velocity identities, critical configurations, projection examples and the counterexamples. Its recorded run (`data/17-discriminant-rigidity-verification.json`) passed 1,126 assertions in 13 groups ([appendix C](#)). Such checks can catch sign errors, wrong normalizations, false examples and failures of a finite identity; they cannot certify the theorem for arbitrary supports, the formal étale argument, or the existence of the certificate in every degree. No Lean implementation accompanies the source, and the Lean coverage of the repository does not by itself cover these statements [C17].

| Claim                         | Decisive dependencies and exclusions [C17]                                                                                                                                                                                                    |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Theorem 19.12</a> | Hahn splitting field; separating Euler family; universal symmetric certificate; $\Pi_{\mathbb{k}}(\Gamma) \cap \mathcal{O}_{\mathbb{k}}(\Gamma_{\mathbb{Q}}) = 0$ . No rank-one norm and no normality of $\mathcal{A}_{\mathbb{k}}(\Gamma)$ . |
| <a href="#">Theorem 19.26</a> | Formal étale extension; reduction modulo $\Pi_{\mathbb{k}}(\Gamma)$ ; characteristic polynomials on a finite projective module; reduced generic fiber. No global primitive element.                                                           |
| <a href="#">Theorem 19.31</a> | The translation lies in $\Pi_{\mathbb{k}}(\Gamma)$ , so division by the degree does not need the degree to be a unit of $\mathfrak{o}$ .                                                                                                      |
| <a href="#">Theorem 19.37</a> | Polynomial projectors; normal idempotents; positive Hermitian form; bounded one-sided entries are constants.                                                                                                                                  |
| Full surreal cases            | Finite input supports lie in a set-sized divisible exponent group; no proper-class summation, no large-cardinal axiom.                                                                                                                        |

A formalization should keep these divisions: in particular it should not replace the strict ideal  $\Pi_{\mathbb{k}}(\Gamma)$  by an ideal of support bounded away from 0, nor the separating set of Euler functionals by one alleged faithful real-valued functional. Source C17 proposes an order: first a generic interface (a ring  $A$  and an ideal  $I$  inside a valued field, derivations carrying  $A$  into  $I$ , and  $I \cap \mathcal{O} = 0$ ); then the certificate, organized through the permutation action on unordered pairs, and the vanishing lemma for characteristic polynomials, first for free algebras and then descended from local bases; then the instantiation with Hahn series, whose joint-constants lemma needs coefficient extensionality and separation of exponents by linear functionals; the surreal specialization through the repository’s normal-form and omnific-ring identifications, with an exact theorem mapping rather than a claim that an abstract theorem has been proved for the constructed surreals; and, largely independently, the projectors and the normal-idempotent lemma. This is a proposal; no stage is claimed done ([question 21.40](#)). The interface is the abstract one of [remark 18.19](#) with the roles of the ring and the valuation ring as in [section 16.5](#).

### 19.4.12 What source C17 does not claim

The limitations stated by source C17 are listed in [appendix B](#). In short: it does not show that every finite étale  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -algebra is constant; it does not extend the unit theorem to  $\mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma)$ ; it proves no factorization, prime or birthday result and uses no large cardinal; it constructs no preferred surreal derivation; it does not claim the exponent  $2s$  optimal; its Galois statement depends on the chosen coefficient field; its critical-point theorem is not a conjugacy; its main theorems fail without one-sided support, characteristic zero or a nonzero discriminant, and the matrix theorem without normality; the polynomial-parameter case is not claimed new; its

finite checks do not prove the arbitrary-support theorems; it has no Lean formalization and certifies no novelty.

## 20 Fixed Hahn workspaces and a formalization route

### 20.1 Which results are local in the exponent group?

For a set-sized ordered abelian group  $\Gamma$ , work in a Hahn field whose monomials are written  $\omega^\gamma$  with reverse-well-ordered supports, and define [01, 02]

$$\mathbf{Oz}_\Gamma = \mathbb{Z} + \left\{ \sum_{\gamma > 0} r_\gamma \omega^\gamma \right\}, \quad \mathcal{B}_\Gamma = \mathbb{R} + \left\{ \sum_{\gamma > 0} r_\gamma \omega^\gamma \right\}.$$

The displayed sums are elements of the ambient Hahn field with the specified support restrictions; this is a convention, not an identification of every abstract group element with a surreal exponent. When an ordered embedding of  $\Gamma$  into  $\mathbf{No}$  is fixed, it gives the corresponding surreal workspace.

**Local results.** The ring decompositions, the unit lemma, the retraction, finite quotients, equational transfer, the Smith criterion, constant-product and decomposable rigidity, binary, Pell and norm rigidity, the quadratic constructions, the quartic guards, and all explicit polynomial identities apply to  $\mathbf{Oz}_\Gamma$  [01, 02]. Their proofs use finite algebra, leading exponents and real coefficients, not the availability of a new exponent outside  $\Gamma$ . The derivation arguments of [section 7](#) take place in the workspace of the rational span of the relevant supports. A functional on that rational span restricts to  $\Gamma$ , and its Euler derivation changes coefficients without changing exponents. Thus it preserves the original workspace; passing to the divisible hull to choose the functional does not require the point or its derivatives to acquire new exponents [05]. If  $\Gamma \neq 0$ , its positive cone provides nonconstant parameters for the infinite-family statements; the case  $\Gamma = 0$  is  $\mathbf{Oz}_\Gamma = \mathbb{Z}$ . The Lorentz family uses only  $t, t^2$  and real scalars, so one positive monomial already produces examples. If the coefficient field is restricted below  $\mathbb{R}$ , the general indefinite lifting additionally needs the chosen real vectors  $e, u$  to be available, or an explicit scalar extension [02]. The definability results of sources 06 and 07 ([sections 11](#) and [12](#)) and the multiplier identity are proved for an arbitrary ordered exponent group, so they apply to  $\mathbf{Oz}_\Gamma = \mathcal{R}_\mathbb{Z}(\mathbb{R}, \Gamma)$  as stated; the multiplier identity needs  $\Gamma \neq 0$ . Only  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  and the reconstruction of  $\mathbf{No}$  ([theorem 13.4](#)) use the full class, and the fraction field of  $\mathbf{Oz}_\Gamma$  need not be the whole Hahn field [06].

**Full-class results.** The common-divisor theorem for an arbitrary set of purely infinite elements, the monomial clearing theorem,  $\Pi^2 = \Pi$ , the non-generation of  $\Pi$  by a set,  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ , the general pair without a gcd ([theorem 4.10](#)) and the set-sized quotient theorem ([cited theorem 3.6](#)) require bounds that may not exist in  $\Gamma$ . For instance, in  $\Gamma = \mathbb{R}$  the set  $\{1/k : k \geq 1\}$  has no strictly positive real lower bound, whereas in  $\mathbf{No}$  it has one. A support in  $\Gamma$  may also be unbounded below in  $\Gamma$ . These conclusions are not asserted for  $\mathbf{Oz}_\Gamma$ ; in particular one must not infer  $\Pi_\Gamma^2 = \Pi_\Gamma$  by copying the proof for  $\mathbf{No}$  [01, 02, 05]. Proofs that stay inside a workspace must be distinguished from proofs that enlarge it. The conclusion of [theorem 4.10](#), though not the proof printed with it, does hold in every  $\mathbf{Oz}_\Gamma$  for each  $0 \neq t \in \Pi_\Gamma$  and irrational real  $r$ , by the halving route of sources 08 and 09 ([remark 15.8](#)). Nontriviality of  $\Gamma$  is needed for such a pair to exist; when  $\Gamma = 0$  the ring is  $\mathbb{Z}$ , a GCD domain. For a concrete failure of another full-class conclusion, take  $\Gamma = \mathbb{Z}$ . A reverse-well-ordered subset of the positive integers is finite, so  $\Pi_\Gamma = \omega\mathbb{R}[\omega]$  and  $\Pi_\Gamma^2 = \omega^2\mathbb{R}[\omega] \neq \Pi_\Gamma$ . In particular,  $\omega$  is not a sum of products of two purely infinite elements in that workspace.

**Fractions.** In [section 15](#) the multiplier theorem is ring-theoretic: it uses a domain containing  $\mathbb{R}$ , a retraction to  $\mathbb{R}$ , its integer pullback, and a unimodular tuple in that domain ([theorem 15.11](#)). These hypotheses, especially unimodularity, are retained; an arbitrary ring map alone does not give the conclusion. The polynomial models, the scale defect and the  $\text{Tor}_1$  computation concern set-sized rings; and the classification of representations, primitivity and least denominators uses only Bézout identities and constant coefficients [08]. The non-generation of the denominator ideals  $q(t)\Pi$ , the absence of a set of denominators for all surreals, the orbit density and the focusing theorem use the full class [08, 09]. Sources 08 and 09 show in one workspace that the fraction field of its omnific ring is smaller than the Hahn field ([example 15.52](#)).

**Curve and differential rigidity.** The characteristic-zero rigidity criteria in [sections 16](#) to [18](#) are first proved for set-sized Hahn rings. No divisibility or rank restriction on the ordered exponent group is imposed. Their own hypotheses on the coefficient field, geometry and coefficient ring are still needed: a theorem about  $\mathcal{A}_k(\Gamma)$ -points is not automatically a theorem about  $\mathbf{Oz}_\Gamma$ -points. The arithmetic conclusions use the descent statements [theorems 17.11](#), [17.17](#) and [18.20](#). The positive-characteristic counterexamples and reduced- or dual-number coefficient extensions have their separately stated domains.

For the actual rings  $\mathcal{B}_\mathbb{R}$ ,  $\mathcal{B}_\mathbb{C}$ ,  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , each point and its finite presentation witnesses lie in one set-sized workspace [C10, C11, C12, C13, C14, C15, C16]. This proves the asserted class-level equalities one point at a time. Conversely, nonconstant points on an affine line require  $\Gamma \neq 0$ ; at  $\Gamma = 0$  every point is constant, regardless of the target. In the affine-line case the arithmetic constant-term fibers are copies of  $\Pi_\mathbb{R}(\Gamma)$ , whereas the rigid curve cases have singleton fibers over their ordinary points. For singular curves, the arithmetic converse needs the normalization-preimage condition of [proposition 17.42](#); it must not be replaced by the smooth fiber formula.

The full-class projective clearing in [corollary 4.7](#), quoted in [section 17.4](#), is a separate step. So is the statement that nonempty affine-line fibers are proper classes ([corollary 17.12](#)): fibers inside a fixed workspace are sets.

**Constant discriminants.** Every result of [section 19.4](#) is proved over  $\mathcal{A}_k(\Gamma)$  or  $\mathcal{R}_o(k, \Gamma)$  for an arbitrary set-sized ordered group  $\Gamma$ , with the splitting field  $\mathbb{H}_k(\Gamma_\mathbb{Q})$  over its divisible hull, and the statements about  $\mathbf{Oz}$ ,  $\mathbf{Oz}[i]$  and omnific matrices are read in a workspace containing the finitely many coefficients [C17]. They therefore hold for  $\mathbf{Oz}_\Gamma$  as stated, and no statement of that subsection uses the full class.

## 20.2 A formalization route

The implementation ledger `docs/FORMALIZATION.md` distinguishes proved Lean declarations from manuscript proofs and proposed modules. At commit `38425db`, the implemented prerequisites concern the actual surreal and surcomplex carriers, using their proved normal-form equivalence. They are not merely assertions about one fixed Hahn field. The dependency order is as follows.

First come the real and complex nonnegative-support rings, constant-term retractions, purely infinite ideals and omnific integer ring ([proposition 2.2](#)); then degree, units, discreteness and the exact floor ([lemma 2.4](#), [proposition 2.5](#), and [theorem 2.7](#)). The floor includes the subtraction of one at an integer constant coefficient with a negative infinitesimal tail. The files are `OmnificIntegers.lean`, `OmnificDegree.lean`, `OmnificUnits.lean` and `OmnificFloor.lean` under `Surreal/Foundations/`, with the complex support-ring and unit counterparts under `Surreal/Surcomplex/`.

Next, ordinary polynomial-root rigidity, residue arithmetic, integer divisors, positive-characteristic ideals and the canonical retraction are proved. The separate support-bound arguments prove common monomial clearing,  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ , idempotence of  $\Pi$ , and the stated failure of GCD, Noetherian and integral-closure properties. These full-class arguments use the normal-form bridge’s explicit smallness hypotheses; they are not consequences of the elementary fixed-workspace ring package alone.

The equation-retraction theorem and positive-existential transfer are proved in Lean, while the separate undecidability claims remain pending. For linear systems, [theorem 5.4](#) is implemented relative to a chosen integer Smith reduction; [theorem 5.5](#) has its full ordinary-plus-purely-infinite decomposition and unique rational-kernel parameters in `Surreal/Foundation s/OmnificLinearSystems.lean`. A zero transformed row requires its whole right-hand entry to vanish, not just its constant term.

Finally, [lemma 6.1](#) and [theorem 6.2](#) are implemented in the `DecomposableFibers.lean` modules and the omnific specializations under `Surreal/Surcomplex/`. The real kernel of complex forms on real omnific variables is kept distinct from full complex column rank for complex support-ring variables. The coefficientwise statement uses actual Conway coefficients; no finite-support restriction is introduced. The Gaussian fiber theorem and its coefficientwise complex-kernel description are also proved in commit 4037368; commit ae0210c proves the converse existence criterion, using a nonzero purely infinite monomial. The files `GaussianOmnificIntegers.lean`, `GaussianDecomposableFibers.lean` and `DecomposableKernelCriterion.lean` under `Surreal/Surcomplex/` retain the ordinary-point and nonzero-level hypotheses. Differential and geometric rigidity and the remaining unmapped Diophantine results are pending. Exact theorem names and any newer coverage belong in the ledger.

The sources proposed overlapping routes; the following merges them. The module names in the table are proposals, not the names of the implemented files.

| Proposed layer                             | Main obligations                                                                                                                                                                                                                       |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>Omnific.Basic</code> [01, 02, 05]    | In a real Hahn field over a set-sized ordered divisible group: the nonnegative-support subring, the integer-constant subring, the retraction $\text{ct}$ from the zero-exponent convolution formula, and the finite-element criterion. |
| <code>Omnific.Residue</code> [01, 02]      | $n\mathbf{Oz} = \Pi + n\mathbb{Z}$ , finite quotients, preservation of $\Pi$ ; source 02 suggests proving these once for any ring retraction onto $\mathbb{Z}$ with an integer-divisible kernel.                                       |
| <code>Omnific.Rigidity</code> [01, 02, 05] | Additive leading degree, units of the real and complex support rings, the exact decomposable fibers (finite linear algebra plus coefficient extensionality), binary and norm rigidity.                                                 |
| <code>Omnific.Derivation</code> [05]       | Euler derivations from a $\mathbb{Q}$ -linear functional (Leibniz from finite convolution fibers), the quotient step for separated powers, comaximality of powers, and Wronskian degree bounds.                                        |
| <code>Omnific.Quadratic</code> [01, 02]    | The finite bilinear identities for $U_t$ over any field of characteristic zero, the isotropic-vector choices, and the level classification.                                                                                            |

| Proposed layer                                               | Main obligations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>Omnific.IntegralityGuard</code> [01, 05]               | Pell rigidity with <code>Nat.sum_four_squares</code> (source 05's six-auxiliary form); source 01's five-auxiliary form needs the three-square input separately. The manuscript's imported number theory is not supplied by the ring package.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>Omnific.Definability</code> [06, 07]                   | Units of $\mathcal{A}_{\mathbb{k}}(\Gamma)$ on the large-monomial side (a small-monomial Hahn implementation must reverse the exponent sign), Pell rigidity by coefficient extension, the finite Pell-divisibility argument, intersectivity of $\Lambda$ with every prime power including 2, 13 and 17 (a finite modular computation is no substitute), the order-free system, the augmentation root detector (a statement about augmented commutative algebras, independent of surreals), the multiplier identity, and then the transport to $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ .                                                                                                                             |
| <code>Omnific.Global</code> [01, 02, 05]                     | The small-cut theorem for support bounds, common monomial divisibility and global clearing, kept separate and with explicit smallness (universe) hypotheses; the upper bound for clearing and the positive lower bound for idempotence are distinct lemmas.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>Omnific.Fractions</code> [09]                          | Pullback algebra for a domain with a split map to a field (the multiplier theorem and the Bézout lift); ordered denominator ideals in a discretely ordered domain with integer-part division (principal ideals and least members, covariance); then the normal-form instantiation; last, the matrices $F_{x,b}$ and focusing, with an explicit smallness hypothesis on the input family.                                                                                                                                                                                                                                                                                                                       |
| <code>Omnific.CurveRigidity</code> [C12, C13; also C10, C15] | First the elementary chain in a set-sized Hahn field: the one-sided rings, the image condition $\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) \subseteq \Pi_{\mathbb{k}}(\Gamma)$ , the certificate (16.3) and the ordered-group degree contradiction; the cubic certificate is a ring-normalization identity. Then, separately, the geometric chain in three layers: a differential support algebra with $\Pi_{\mathbb{k}}(\Gamma) \cap \mathcal{O}_{\mathbb{k}}(\Gamma) = 0$ ; a one-variable function-field statement with a logarithmic differential and a valuation centre; and the compactification and Riemann–Roch input producing that differential. No proper-class scheme; the surreal instances last. |

Source 05 points to the normal-form bridge (`docs/NORMAL_FORM_BRIDGE.md`) and the modules `SmallNormalFormFieldEquiv.lean`, `SmallNormalFormHahnEmbedding.lean`, `SmallNormalFormWorkspace.lean` and `SignSequenceWorkspace.lean` under `Surreal1/Foundations/` as the transfer route to the actual surreal carrier; all four exist at the placement commit (with `SignSequenceWorkspaceExponents.lean`), and their signatures must be checked before use. The imported logical theorems (Tarski's quantifier elimination, MRDP) remain separate dependencies; formalizing the retraction does not formalize them. The transvection identity is finite bilinear algebra with no dependence on birthday recursion,

and the only hard number-theoretic input to the guards is the three- or four-square theorem. The order-free system needs only elementary local arithmetic of  $\Lambda$  and the finite Pell argument; the number-field extension also needs Dirichlet's theorem and quadratic reciprocity, and the reconstruction of  $\mathbf{No}$  the real closedness of  $\mathbf{No}$ . Sources 06 and 07 propose the order followed in the row `Omnific.Ddefinability`, keeping the class-sized instantiation separate from the set-sized Hahn results; this is a proposal, not a report of completed formalization. Sources C12 and C13 propose the order of the row `Omnific.CurveRigidity`, and source C10 notes that finite coordinate certificates for arbitrary smooth curves would make the geometric chain checkable in the same way as the cubic certificate ([question 21.22](#)); these too are proposals. Source C15 proposes the same two stages, with the local description of logarithmic forms in the geometric one, and notes that a conditional Lean theorem would not by itself verify the instantiations. Source C16 proposes, for its own results, finite differential algebra first, then the abstract valuation contradiction, then normalization of curves and Riemann–Roch, and the proper-class application last ([question 21.30](#)); this too is a proposal. Source C17 proposes a generic interface of a ring, a strict ideal and derivations with  $I \cap \mathcal{O} = 0$ , then its certificate and the characteristic-polynomial lemma as finite algebra, then the Hahn instantiation and the surreal specialization ([section 19.4.11](#) and [question 21.40](#)); this is a proposal as well.

## 21 Questions and synthesis

The sources list continuation problems that their proofs do not decide. None of them claims that these problems are open in the literature, or that the list is exhaustive. Two were settled inside the batch of manuscripts from which sources 01, 02 and 05 come. In the merge of sources 06 and 07 one more was answered and two were answered in part. Source 08 answers [question 21.3](#) in part, and sources 08 and 09 add five questions ([section 21.3](#)). Sources C10–C14 answer [question 21.1](#) for smooth curves, answer [questions 21.3](#) and [21.8](#) in part, and add six questions ([section 21.4](#)); the three questions stay open, re-scoped. Source C15 answers the first question of [question 21.19](#) in sufficient form and adds a class of rigid varieties to [question 21.1](#); both stay open, re-scoped. Source C16 answers the first two clauses of [question 21.17](#) and the curve part of [question 21.1](#), answers the existence part of [question 21.8](#) for singular curves over  $\mathbf{Oz}[i]$  and in part over  $\mathbf{Oz}$ , and adds eight questions; the questions stay open, re-scoped. Source C17 answers [question 21.6](#) for monic polynomials over  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  with a nonzero constant discriminant, answers the first clause of [question 21.22](#) in dimension zero, and adds ten questions; both questions stay open, re-scoped.

### 21.1 Questions settled by source 05

**Coprime exponents and  $Y^2 = X^3 + 1$  (source 01).** Source 01 asked which curves such as  $Y^2 = X^3 + 1$  have genuinely infinite omnific points, and proved rigidity of  $x^m - y^n = c \neq 0$  for positive  $m, n$  with  $\gcd(m, n) > 1$  ([remark 6.13](#)). [Theorem 7.4](#) of source 05 answers this for every separated equation  $ax^m + by^n = c$  with  $abc \neq 0$  and  $m, n \geq 2$ ; in particular  $Y^2 = X^3 + 1$  and  $x^2 - y^3 = 1$  have only ordinary omnific points ([corollary 7.6](#)). The general part of source 01's question, a criterion for arbitrary affine curves, was left open as [question 21.1](#); sources C10–C14 answer it for smooth curves ([section 21.4](#)), and it remains open for singular curves and in higher dimension. Status (batch 31): source C16 settles the geometrically integral singular curves ([section 21.4](#)); higher dimension remains open.

**Unimodular Fermat (sources 01 and 02).** Source 01 asked for the classification of primitive or unimodular omnific solutions of  $X^n + Y^n = Z^n$  with nonzero coordinates,  $n \geq 3$ ;

source 02 asked which homogeneous systems admit nonzero omnific solutions generating  $\mathbf{Oz}$ . [Theorem 7.9](#) and [corollary 7.10](#) of source 05 settle the unimodular Fermat case: there are none. The primitive case (no common nonunit divisor) remains open as [question 21.2](#), and source 02's question for other homogeneous systems remains open as [question 21.3](#).

## 21.2 Questions answered in the merge of sources 06 and 07

**An existential definition of  $\Pi$  (source 01; source 07).** Source 01 asked for an existential definition of  $\Pi$  over  $\mathbf{Oz}$  ([question 21.4](#)), and source 07 closed by asking whether the purely infinite ideal, or the graph of  $\text{ct}$ , has a positive-existential definition. Source 06 answers both:  $\exists y (x^2 = 2y^2)$  defines  $\Pi$ , with one witness and no parameters, in  $\mathbf{Oz}$ , in  $\mathbf{Oz}[i]$  and in every ring  $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$  with  $\sqrt{2} \in \mathbb{k}$  and  $\sqrt{2} \notin \text{Frac}(\mathfrak{o})$  ([theorem 11.2](#)), and the graph of  $\text{ct}$  is Diophantine ([theorem 12.8](#)). Source 07's detector makes the complement  $\{\text{ct} \neq 0\}$  Diophantine too ([theorem 12.2](#)). The second part of [question 21.4](#), which fibers or subideals of  $\Pi$  are Diophantine, remains open.

**The least degree of a definition of  $\mathbb{Z}$  (source 06).** Source 06 asked for the least total degree, and the best trade-off between degree and witnesses, of a single integer polynomial defining  $\mathbb{Z}$  in  $\mathbf{Oz}$ , offering a quintic with seven witnesses. The guards of sources 01 and 05 ([theorems 10.3](#) and [10.4](#)), which source 06 did not see, reach degree four with five and six witnesses. This answers the question in part; the least degree and the trade-off remain open and are merged into [question 21.5](#), together with source 06's question on a single Gaussian polynomial and source 07's on better quantifier and degree bounds.

## 21.3 Questions in the merge of sources 08 and 09

**Primitive homogeneous solutions (source 02).** Source 02 asked which homogeneous systems admit a nonzero omnific solution whose coordinates generate  $\mathbf{Oz}$  ([question 21.3](#)). Source 08's rational-curve criterion answers it in part ([corollaries 15.28](#) and [15.30](#)). If the projective zero locus contains a curve  $[p_0 : \cdots : p_n]$  given by real polynomials with gcd one and rational specialization at  $T = 0$ , then after a real normalization every  $0 \neq t \in \Pi$  gives a unimodular omnific solution  $(p_0(t), \dots, p_n(t))$ . Along such a curve, the points with primitive coordinates are exactly those of rational specialization. Points off rational curves and the dependence on the integral model remain open, so the question stays open, re-scoped. Neither source addresses [question 21.2](#): for  $n \geq 3$  the Fermat curve has positive genus and contains no nonconstant rational curve, so the criterion does not apply to it.

**New questions (sources 08 and 09).** Sources 08 and 09 end with questions about denominator ideals beyond one parameter, the extent of the unimodular region, orbits outside it, several parameters and other scale extensions. They are [questions 21.12](#) to [21.16](#) below; source 08's first question and source 09's second are merged into [question 21.12](#). The sources present them as questions of their articles, not as certified open problems.

## 21.4 Questions in the merge of sources C10–C15

**Rigid affine varieties and curves (sources 01, 02, 05).** All five sources answer the second part of [question 21.1](#) for smooth geometrically integral curves: over any field of characteristic zero and any nontrivial ordered exponent group, a smooth affine curve has nonconstant points over the one-sided Hahn ring exactly when it is the affine line ([theorem 17.6](#)). The trivial

exponent group gives only the coefficient field and is rigid for every target. For an affine integral model with smooth geometrically integral real fiber,  $\mathcal{C}(\mathbf{Oz}) = \mathcal{C}(\mathbb{Z})$  unless that real fiber is an affine line, in which case every fiber of  $\text{ct}$  over an integer point is a copy of  $\Pi$  ([corollary 17.12](#)). The named test  $Y^2 = X^3 + aX + b$  is settled completely: only ordinary solutions when  $4a^3 + 27b^2 \neq 0$  ([corollary 16.9](#), also for general Weierstrass equations, [corollary 16.10](#)) and injective polynomial families otherwise ([proposition 16.11](#)). For the first part, [theorem 17.17](#) reduces the question for every separated model to the rigidity of its complex fiber over the rings  $\mathcal{A}_{\mathbb{C}}(\Gamma)$ , and the part supplies rigid affine varieties in every dimension: locally closed subschemes and quasi-finite covers of abelian and semiabelian varieties ([theorems 18.2 and 18.5](#)). Source C15 adds descent over every coefficient ring ([theorem 18.20](#)) and the complements of simple normal-crossings divisors whose logarithmic symmetric differentials separate tangent directions ([corollaries 18.18 and 18.21](#)). The question stays open for singular curves and for general affine varieties of higher dimension; its refined forms are [questions 21.17 and 21.19](#). Status (batch 31): source C16 settles the curve part; see the paragraph on singular curves below.

**Effective finite-support discovery (source 05).** For smooth affine curves the existence part of [question 21.8](#) is answered: a nonordinary omnific point exists if and only if one with finite supports exists, and then one with the same constant term ([corollary 17.16](#)). In the affine-line case integer polynomial arcs through every integer point and a finite congruence test describe the points once an affine-line isomorphism and its inverse are supplied ([lemma 17.14](#) and [proposition 17.15](#)). The question stays open beyond smooth curves, and [corollary 17.16](#) is not an algorithm for the ordinary points. Status (batch 31): source C16 extends the answer to singular curves in part (paragraph on singular curves below).

**Primitive homogeneous solutions (source 02).** For projective varieties whose complex fiber is rigid, in particular smooth projective curves of positive genus and abelian varieties, every unimodular omnific solution is an ordinary integer tuple, and a coordinate ideal is invertible exactly when it is principal, exactly when the point is rational ([theorems 17.18 and 17.21](#) and [corollary 17.22](#)). For these targets the answer does not depend on the integral model beyond the fixed projective embedding: the unimodular points are the rational points. With [corollary 15.30](#) for rational curves, this answers [question 21.3](#) for two opposite classes of varieties; other varieties, and primitive tuples that are not unimodular, remain open ([questions 21.2 and 21.20](#)).

**Logarithmic forms in higher dimension (source C12; source C15).** Source C12 asked which hypotheses on logarithmic one-forms force all  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -points of a smooth affine variety with a smooth normal-crossings compactification to be constant, and noted that in higher dimension one form need not span the cotangent space and that the centre of an arbitrary-rank valuation must be handled. Source C15, which did not see source C12, answers this in sufficient form in every dimension, without assuming the variety affine: if the global logarithmic symmetric differentials separate tangent directions after every field extension, in particular if the global logarithmic one-forms span the cotangent space at every point, then  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$  ([theorem 18.16](#) and [corollary 18.18](#)); its local computation handles the centre of an arbitrary valuation ring and the boundary parameters ([lemma 18.15](#)). Which hypotheses are necessary, a classification of the rigid complements, the converse from the absence of polynomial curves and the question of source C10 on surviving tangent directions remain open, so [question 21.19](#) stays open, re-scoped.

**New questions (sources C10–C14).** The sources' continuation questions are merged into six. Singular curves (all five sources) are [question 21.17](#); rigidity beyond cotangent generation (C11, C13, C14) is [question 21.18](#); higher-dimensional varieties (C10's discussion, C12's logarithmic question and C13's polynomial-arc question) are [question 21.19](#); primitive versus unimodular tuples (C12, C13, C14, and the corresponding remarks of C10 and C11) are [question 21.20](#); positive characteristic (C11, and C10's suggestion of higher derivations) is [question 21.21](#); omnific coefficients, effective certificates and other coefficient systems (C12, and C10's remarks) are [question 21.22](#). Source C15's four closing questions fall under these: singular targets under [question 21.17](#); rigidity beyond differential positivity, and whether the absence of nonconstant maps from  $\mathbb{A}^1$  implies rigidity, under [question 21.19](#) (for proper varieties see also [question 21.18](#)); positive characteristic under [question 21.21](#); and minimal certificates for curves given by several equations under [question 21.22](#). None of the sources presents these as known open problems of the literature.

**Singular curves (source C16).** Source C16 answers the first two clauses of [question 21.17](#). Uniformly in  $\Gamma \neq 0$  and over every field of characteristic zero, a geometrically integral affine curve has only constant  $\mathcal{A}_k(\Gamma)$ -points unless its normalization is  $\mathbb{A}^1$  ([theorem 17.25](#)); the answer is expressed through the normalization and its points at infinity (genus zero and exactly one boundary point), and the conductor supplies finite certificates ([theorem 17.28](#) and [corollary 17.29](#)). Over an algebraically closed field this extends to all affine schemes of dimension at most one, reducible or not ([theorem 17.36](#)). The third clause, which finite normalization maps admit nonconstant points without a lift, stays open as [question 21.23](#), so [question 21.17](#) stays open, re-scoped. For [question 21.1](#) this settles the curve part: every omnific point of a model whose real fiber is geometrically integral is ordinary unless the normalization of that fiber is  $\mathbb{A}_{\mathbb{R}}^1$ , and every Gaussian omnific point of a model whose complex fiber is geometrically integral is ordinary unless that normalization is  $\mathbb{A}_{\mathbb{C}}^1$  ([theorem 17.41](#)); conversely nonordinary Gaussian points exist exactly when in addition there is a Gaussian integer point ([corollary 17.43](#)), and nonordinary omnific points exist when some integer point has a real preimage under the normalization ([proposition 17.42](#)), in particular when it is nonsingular. For [question 21.8](#) it answers the existence part for singular curves over  $\mathbf{Oz}[i]$ , and over  $\mathbf{Oz}$  at integer points with a real normalization preimage; the case of integer points whose preimages are all nonreal, such as an isolated real node, stays open ([question 21.24](#)). For [question 21.21](#) it adds a necessary condition in every characteristic, one boundary point ([proposition 17.31](#)); for [question 21.22](#) it gives finite derivative-order certificates for every rigid curve, without optimality or an algorithm ([questions 21.25](#) and [21.26](#)).

**New questions (source C16).** Its eight questions are printed as stated: lifting through the normalization ([question 21.23](#), the third clause of [question 21.17](#)), real singular fibers ([question 21.24](#)), optimal certificates and a certificate-producing procedure ([questions 21.25](#) and [21.26](#), refining [question 21.22](#)), positive characteristic ([question 21.27](#), refining [question 21.21](#)), normalization invariance in higher dimension ([question 21.28](#)), images of the constant term ([question 21.29](#)) and formalization ([question 21.30](#)). Source C16 does not present them as known open problems of the literature.

**Roots and coefficients (source C17).** For a monic  $P \in \mathbf{Oz}[T]$  with  $\text{Disc}(P) \in \mathbb{Z} \setminus \{0\}$ , source C17 answers [question 21.6](#):  $P = \overline{P}(T - h)$  with  $h = -(a_1 - ct a_1)/n \in \Pi$ , so the omnific roots of  $P$  are  $h + c$  for the integer roots  $c$  of  $\overline{P}$ , and the finite coefficient test ([19.10](#)) decides whether the hypothesis holds, relative to exact coefficient arithmetic ([theorem 19.12](#) and [corollary 19.16](#)); the same holds over  $\mathbf{Oz}[i]$ . This complements [theorem 19.1](#) and explains

**example 19.5**, whose discriminant  $1 + 4\omega$  is not constant (**remark 19.17**). Nonconstant discriminants, zero discriminants and nonmonic polynomials stay open, and so does the decidability clause for general support classes. For the first clause of **question 21.22**, a nonzero constant discriminant is a nondegeneracy condition that recovers rigidity for coefficients varying at a surreal scale in dimension zero, over every  $\mathcal{A}_k(\Gamma)$  and  $\mathcal{R}_o(k, \Gamma)$ ; units of finite étale algebras and normal matrices follow the same pattern (**theorems 19.26** and **19.37**). Curves and higher-dimensional varieties with nonconstant coefficients stay open (**section 18.5**).

**New questions (source C17).** Its ten questions are printed as stated: full and arithmetic étale descent (**questions 21.31** and **21.32**), projective modules and nonnormal spectral data (**question 21.33**), sharp certificates (**question 21.34**), nonconstant discriminants (**question 21.35**, refining **question 21.6**), positive characteristic (**question 21.36**), restricted support rings (**question 21.37**), several configurations (**question 21.38**), polynomial dynamics (**question 21.39**) and formalization (**question 21.40**). Source C17 does not present them as known open problems of the literature.

## 21.5 Open questions

**Question 21.1** (Rigid affine varieties and curves [01, 02, 05]; smooth curves answered by [C10–C14]). For which affine varieties  $X$  over  $\mathbb{Z}$  does  $X(\mathbf{Oz}) = X(\mathbb{Z})$  hold? For which affine curves does every omnific point have ordinary coordinates? A natural next test is  $Y^2 = X^3 + aX + b$  with  $a \neq 0$ , outside **theorem 7.4**.

**Theorems 6.2, 6.7** and **7.4** answer this for nonzero decomposable levels with trivial common kernel and for separated equations, **theorems 9.2** and **9.6** for bounded loci and systems with no real point at infinity, and **theorem 8.1** and polynomial arcs give large opposing classes. A satisfactory criterion should separate polynomial parametrization, torus rigidity and genuinely algebraic branches [01]; it needs either a new degree inequality or a structural analysis of possible supports [05]. No claim is made that all other fibers are rigid, or that all nontrivial fibers come from real polynomial curves [02]. The report `docs/surreal/omnific-groups-and-lattices` answers the question for closed subgroup schemes of  $GL_N$  over  $\mathbb{Z}$  (and over  $\mathbb{Z}[i]$  for  $\mathbf{Oz}[i]$ ). Sources C10–C14 answer the second part for smooth geometrically integral curves and settle the test equation (**section 21.4**). An affine model with smooth geometrically integral real fiber can have nonordinary omnific points only when that real fiber is the affine line; in that case such points exist precisely when the model has an ordinary integer point. This is a condition on the real fiber, not an assertion that the model is  $\mathbb{A}_{\mathbb{Z}}^1$ . Nonsingular Weierstrass equations have only ordinary solutions, while the singular cases have the displayed polynomial families. For smooth curves the criterion is exactly the separation into polynomial parametrization, torus rigidity and genuinely algebraic branches asked for above, and source 02’s caution is resolved: every nontrivial fiber of a smooth affine curve comes from a real polynomial curve (**corollaries 17.12** and **17.16**). Source C15 adds a class in every dimension: complements of normal-crossings divisors whose logarithmic symmetric differentials separate tangent directions (**corollary 18.21**). Singular curves and general affine varieties of higher dimension remain open (**questions 21.17** and **21.19**). Status (batch 31): source C16 settles the curve part for geometrically integral curves, and over the Gaussian ring completely (**section 21.4**); over  $\mathbf{Oz}$  the case of an exceptional curve whose integer points all lack a real normalization preimage remains (**question 21.24**). General affine varieties of higher dimension remain open.

**Question 21.2** (Primitive Fermat triples [01, 05]). For  $n \geq 3$ , can the unit-ideal hypothesis of **theorem 7.9** be weakened to “no common nonunit divisor”? That is, are there primitive, non-unimodular omnific solutions of  $X^n + Y^n = Z^n$  with nonzero coordinates?

The answer cannot be read off from ordinary integer intuition; it needs an explicit relation between common divisors, ideal generation and the applicable factorization theory. [Theorem 7.9](#) deliberately avoids any such relation, and the binary nonzero-fiber theorem does not directly answer a three-variable homogeneous question [01].

**Question 21.3** (Primitive homogeneous solutions [02]). Which homogeneous systems admit a nonzero omnific solution whose coordinates generate  $\mathbf{Oz}$ , and how does this depend on the chosen integral model of the projective variety?

The real-solvability criterion [theorem 14.2](#) omits unimodularity, and [theorem 14.5](#) exhibits the gap. A gcd normalization cannot be assumed, by [theorem 4.10](#). Fermat equations of exponent at least three have none with all coordinates nonzero ([corollary 7.10](#)), although  $(1, 0, 1)$  is a unimodular solution with a zero coordinate; the exponent-two Pythagorean and indefinite quadratic cases have many ([proposition 14.4](#), (8.4)). Source 08 answers the question in part ([section 21.3](#)): a rational curve of real polynomials with gcd one in the zero locus gives unimodular omnific solutions exactly when its specialization at  $T = 0$  is rational ([corollary 15.30](#)). Sources C10–C14 answer it in part at the opposite extreme ([section 21.4](#)): on a projective variety whose complex fiber is rigid, such as a smooth curve of positive genus or an abelian variety, the unimodular omnific solutions are exactly the ordinary unimodular ones, and in a fixed embedding the points with such solutions are the rational points ([theorem 17.18](#) and [corollary 17.22](#)). Other varieties, and the dependence on the integral model in general, remain open.

**Question 21.4** (An existential definition of  $\Pi$  [01]; first part answered by [06]). Is there an existential definition of  $\Pi$  over  $\mathbf{Oz}$  with ordinary integer parameters? More generally, which fibers or subideals of  $\Pi$  are Diophantine? Retraction stability is necessary but far from sufficient.

The first question is answered:  $\exists y (x^2 = 2y^2)$  defines  $\Pi$  ([theorem 11.2](#), [section 21.2](#)). The second remains open. Retraction stability is still necessary ([theorem 13.13](#)): a Diophantine set containing a nonzero element of  $\Pi$  contains 0.

**Question 21.5** (Complexity of the integrality guard [01, 06, 07]). Can the number of auxiliary variables be reduced below five, with a single quartic equation, or can a meaningful lower bound be proved? What is the least total degree of a single integer polynomial defining  $\mathbb{Z}$  in  $\mathbf{Oz}$ , and what is the best trade-off between degree and witnesses [06]? Can the Gaussian three-equation system of [theorem 11.9](#) be replaced by a single polynomial of controlled degree [06], and can its quantifier and degree bounds be improved without an order [07]? The present constructions give explicit upper bounds only (five in [theorem 10.3](#), six in [theorem 10.4](#), seven in the quintic (11.5)); other rigid arithmetic families might serve as finite bounds.

Any Gaussian sum-of-squares replacement must address isotropy ( $1^2 + i^2 = 0$ ) rather than assume it away [06]. The quantifier-free obstruction [proposition 13.14](#) is only a first lower bound on logical complexity; it does not make any of the present counts optimal.

**Question 21.6** (Roots with omnific coefficients [01, 02, 05]). For polynomials with finitely represented omnific coefficients, which root questions in a chosen support class are decidable? Can one give tractable necessary and sufficient support criteria for a root of a polynomial in  $\mathbf{Oz}[T]$  to be omnific?

[Theorem 19.1](#) reduces the monic quadratic case to square membership, [proposition 19.4](#) settles two-term binomials, and [example 19.5](#) rules out a naive lifting theorem; multiterm cancellation needs separate analysis. Any effective claim must specify the coefficients, the

exponent representation, and whether workspace enlargement is permitted. An unrestricted algorithm for ordinary-coefficient affine existence would contradict [corollary 5.2](#). Status (batch 33): for monic polynomials with a discriminant in  $\mathbb{Z} \setminus \{0\}$  (in  $\mathbb{Z}[i] \setminus \{0\}$  over  $\mathbf{Oz}[i]$ ) source C17 gives a complete answer: the omnific roots are  $h + c$  for the integer roots  $c$  of  $\bar{P}$ , with  $h$  read off the second coefficient ([corollary 19.16](#) and [equation \(19.10\)](#)). The question stays open for other polynomials ([section 21.4](#)).

**Question 21.7** (The exact size boundary [05]). Which set-sized exponent groups or universe-relative constructions admit analogues of the common-monomial and set-sized quotient theorems?

The sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` contains cardinal-threshold results on this question for specific set-sized integer parts.

**Question 21.8** (Effective finite-support discovery [05]). How far can a search for polynomial arcs and exact Bézout certificates, suggested by [theorem 14.7](#), go before transfinite support constructions are needed? It can find witnesses and expose obstructions, but it cannot become a complete decision procedure for integer-coefficient systems, by [corollary 5.2](#).

For smooth affine curves the existence part is answered by sources C12 and C13: a nonordinary omnific point exists exactly when one of finite support with the same constant term exists ([corollary 17.16](#)), and source C14 makes the affine-line case constructive once a parametrization is supplied ([lemma 17.14](#) and [proposition 17.15](#)). Beyond smooth curves the question stays open ([section 21.4](#)). Status (batch 31): source C16 answers the existence part for geometrically integral singular curves over  $\mathbf{Oz}[i]$  ([corollary 17.43](#)) and over  $\mathbf{Oz}$  at integer points with a real normalization preimage ([proposition 17.42](#)); the remaining real case is [question 21.24](#), and higher dimension stays open.

**Question 21.9** (Coefficient recovery of lower complexity [06]). Can the canonical  $\mathbb{R}$  or  $\mathbb{C}$  in the interpreted fraction field be recovered with weaker quantifier complexity than the multiplier formula of [corollary 13.3](#), keeping the integer-part ring as the only named structure?

The pure field and the field with its integer-part predicate must be distinguished in formulating this. The present formulas are first order in the interpreted ring structure; they give no Diophantine definition of  $\mathbb{Z}$  in  $\mathbb{Q}$  and no solution of Hilbert’s tenth problem over  $\mathbb{Q}$  [06].

**Question 21.10** (Which real structure must be added? [06]). What additional natural structure on  $\mathbf{Oz}[i]$  rules out the coefficient-fixing twists of [lemma 13.9](#) and determines the real omnific subring, without simply naming conjugation or  $\mathbf{Oz}$ ?

Naming all complex constants does not suffice ([theorem 13.10](#)); naming a suitable involution does, but gives the real structure explicitly rather than reconstructing it [06]. Status (batch 32): further negative information, not an answer. By `saut:fs:thm:main-count` of `docs/surcomplex/surcomplex-field-automorphisms`, the involutions of  $\mathbf{No}[i]$  that preserve the valuation, Hahn summation and  $\mathbf{Oz}[i]$  fall into  $2^{2^{\aleph_0}}$  conjugacy types, with representatives whose fixed fields are pairwise nonisomorphic and whose fixed rings are integer parts; so these structures alone do not single out the real omnific subring, even up to isomorphism. Status (batch 34): ordinary arithmetic does not separate these rings either. Each integer-part ring  $R_F$  of that report, for a set-sized real closed field  $F$  (with  $\mathbf{Oz} = R_{\mathbb{R}}$ ), has  $R_F/nR_F \cong \mathbb{Z}/n\mathbb{Z}$  for every positive ordinary integer  $n$ , with every finite quotient factoring through `ct`, and a system of integer polynomial equations is solvable in  $R_F$  exactly when it is solvable in  $\mathbb{Z}$  (`saut:gr:prop:finite`, `saut:gr:prop:dio`). This is further negative information, not an answer, and not elementary equivalence.

**Question 21.11** (Infinite algebraic extensions [07]). Which coefficient subrings of algebraic extensions of  $\mathbb{Q}$  of infinite degree admit a root-free intersective polynomial, or another positive-existential certificate of a nonzero constant, so that [theorem 12.14](#) extends to them?

No polynomial of the shape used in [lemma 12.13](#) is root-free in a field containing all algebraic numbers, so a different certificate would be needed there [07].

**Question 21.12** (Denominator ideals beyond one parameter [08, 09]). Which surreal tuples have denominator ideals generated by a set? Must such an ideal be principal, or can a finitely or set-generated nonprincipal denominator ideal occur [08]? For arbitrary  $x \in \mathbf{No}$ , does a principal  $\mathfrak{D}(x)$  force a unimodular presentation; which nonprincipal denominator ideals occur outside  $\mathcal{V}$ ; and can a primitive presentation exist there without a Bézout identity [09]?

[Theorems 15.15](#) and [15.18](#) answer this for tuples in  $\mathbb{R}(t)$  and on  $\mathcal{V}$  only; they do not classify  $\mathfrak{D}(x)$  for all  $x \in \mathbf{No}$ .

**Question 21.13** (The unimodular region [09]). Which points of  $\mathbb{P}^1(\mathbf{No})$  lie in  $\mathcal{V}$ ? Give a support-theoretic criterion for a fraction to admit a unimodular presentation over  $\mathcal{B}_{\mathbb{R}}$ , or an explicit obstruction.

Every rational function of one parameter lies in  $\mathcal{V}$ , even when its normal form has infinitely many terms. The dense fibers do not imply  $\mathcal{V} = \mathbb{P}^1(\mathbf{No})$  [09].

**Question 21.14** (Orbits outside the region [09]). By [corollary 15.36](#), every  $K$ -orbit is dense, including any orbit outside  $\mathcal{V}$ . What complete invariants distinguish such orbits, if they exist?

Within  $\mathcal{V}$  the invariant is exactly one real projective point ([theorem 15.32](#)).

**Question 21.15** (Several parameters [08]). Is there a similarly explicit classification for fractions in  $\mathbb{R}(t_1, \dots, t_k)$  with suitably independent purely infinite parameters? What local or projective data replace the single specialization at zero?

The proof does not repeat directly: a multivariate gcd of one does not give a Bézout identity, as  $T_1, T_2$  in  $\mathbb{R}[T_1, T_2]$  show, and [theorem 15.11](#) needs unimodularity [08].

**Question 21.16** (Other scale extensions [08]). For inclusions of more general set-sized omnific Hahn models, which extensions commute with denominator ideals, and can their relation defects be expressed in terms of the newly admitted support scales?

[Theorem 15.46](#) and [proposition 15.48](#) settle one explicit case; they give no general flatness or normalization theorem for Hahn integer parts [08].

**Question 21.17** (Singular curves [C10, C11, C12, C13, C14, C15]). Which integral singular affine curves over  $\mathbb{k}$  have only constant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -points, uniformly in  $\Gamma$ ? Can the answer be expressed through the normalization, the conductor and the points at infinity, and which finite normalization maps admit nonconstant points downstairs without a lift over the same ring?

The node and the cusp show that “only the affine line” is no longer the right statement. A normalization lift cannot simply be assumed from integrality, and  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  need not be normal in its own fraction field ([section 17.5](#)) [C10, C11, C13, C14]. Finite or quasi-finite maps to rigid targets are covered ([theorem 18.5](#)), but not every singular curve has one [C14]. Source C12 proposes a route through logarithmic differentials on the normalization whose pullbacks are regular on the singular curve, with the conductor controlling the descent; this is a proposal, not a proved criterion. Source C15 asks the same question for every exponent group, naming

the identifications of branches among the data; the quasi-finite criterion settles some singular targets but gives no classification [C15]. Status (batch 31): source C16 answers the first two clauses for geometrically integral curves: uniformly in  $\Gamma \neq 0$ , only constant points exist unless the normalization is  $\mathbb{A}^1$ , a condition on the genus and the points at infinity of the normalization, and a conductor gives finite certificates ([theorems 17.25](#) and [17.28](#)). The third clause stays open as [question 21.23](#); the question stays open, re-scoped.

**Question 21.18** (Beyond cotangent generation [C11, C13, C14]). For which smooth proper varieties does  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$  hold for every  $\Gamma$ ? Must a smooth proper complex variety containing no rational curves satisfy  $X(\mathcal{B}_{\mathbb{C}}) = X(\mathbb{C})$  [C11, C13]? Which positivity conditions on the cotangent bundle suffice, and what obstruction is encoded by the locus where global forms fail to generate [C11, C14]?

Global generation of  $\Omega^1$ , or tangent detection by symmetric differentials, is sufficient ([theorems 16.16](#) and [16.19](#)) and not claimed necessary; no implication from the absence of rational curves, from nefness, bigness or general type is proved, and varieties without global one-forms are not classified [C11, C13, C14].

**Question 21.19** (Affine varieties of higher dimension [C10, C12, C13, C15]; logarithmic sufficient condition [C15]). For a smooth affine variety with a smooth normal-crossings compactification, which hypotheses on logarithmic one-forms force all  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -points to be constant [C12]? For an affine complex variety  $X$ , does every nonconstant  $\mathcal{B}_{\mathbb{C}}$ -point force a nonconstant polynomial arc  $\mathbb{A}^1 \rightarrow X$  [C13]? Which of the Euler tangent directions allowed by the global forms, and by the fibers of an Albanese map, arise from Hahn points [C10]? For which smooth varieties does the absence of nonconstant morphisms  $\mathbb{A}^1 \rightarrow X$  imply  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$ , and which additional hypothesis bridges that condition and tangent separation by logarithmic symmetric differentials [C15]?

For smooth curves the first two questions have affirmative answers ([theorems 17.6](#) and [17.8](#)). In every dimension source C15 answers the first question in sufficient form: tangent separation by global logarithmic symmetric differentials, in particular spanning by global logarithmic one-forms, forces every  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point to be constant ([corollary 18.18](#)); necessary conditions and a classification remain open ([section 21.4](#)). In higher dimension one nonzero logarithmic form need not span the cotangent space, the centre of an arbitrary-rank valuation must be handled, and no equivalence with uniruledness, rational connectedness or the existence of polynomial curves is asserted [C10, C12]. Source C15's local computation handles the centre of an arbitrary valuation ring ([lemma 18.15](#)); it asserts no equivalence between rigidity and the absence of nonconstant maps from  $\mathbb{A}^1$ , and such an equivalence may not be inferred from the classification of curves [C15]. If  $\Gamma \neq 0$  and  $X(\mathbb{k}) \neq \emptyset$ , then  $X \times \mathbb{A}^1$  is not rigid: for  $x_0 \in X(\mathbb{k})$  and  $\gamma > 0$ ,  $(x_0, \omega^\gamma)$  is a nonconstant point [C12]. The point hypothesis matters: constant extraction gives  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) \rightarrow X(\mathbb{k})$ , so if the latter is empty, both the source and the product's point set are empty and the equality holds vacuously.

**Question 21.20** (Primitive versus unimodular [C12, C13, C14; also C10, C11]). For positive-genus projective curves over  $\mathbb{Z}$ , can the unit-ideal hypothesis of [corollary 17.22](#) be weakened to the absence of a common nonunit divisor in  $\mathbf{Oz}$ ? Can a nonrational point of such a curve be represented by a primitive omnific tuple whose coordinate ideal is not invertible [C12]?

This contains [question 21.2](#). [Theorem 17.21](#) decides invertibility and principality, not the existence of a common nonunit divisor in a noninvertible ideal; the tuple of [example 17.23](#) has one and does not answer the question [C12]. No gcd or Bézout property of  $\mathbf{Oz}$  may be substituted for an explicit unit-ideal identity [C10, C11, C13, C14].

**Question 21.21** (Positive characteristic [C11; also C10, C15]). For which ordered groups  $\Gamma$ , fields  $\mathbb{k}$  of positive characteristic and smooth curves do nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -points exist? Can the Artin–Schreier mechanism of [proposition 18.12](#) be characterized by ramification and the  $p$ -divisible part of  $\Gamma$  [C11], and can iterative higher derivations replace the Euler derivations [C10]?

[Propositions 18.12](#) and [18.13](#) show that both the curve classification and the squarefree theorem fail as stated in positive characteristic ([section 18.3](#)): a characteristic-free version of the theorems is false, not merely unproved. Source C15 asks the same question, how the  $p$ -divisibility of  $\Gamma$  and the inseparable geometry of the target control the points, and whether higher or iterative differential operators can replace the Euler family [C15]. Status (batch 31): source C16 proves in every characteristic that a curve with a nonconstant point has exactly one boundary point in its normalization ([proposition 17.31](#)); its refinement is [question 21.27](#).

**Question 21.22** (Omnific coefficients and effective certificates [C12; also C10, C15]). For equations with genuinely omnific coefficients, which support and nondegeneracy conditions recover a rigidity theorem [C12]? For an arbitrary smooth curve, can the global-form argument be turned into a finite coordinate certificate like [proposition 16.8](#) with useful complexity bounds [C10], and can the support obstruction be combined with effective methods for integral points without losing explicit bounds [C12]? Which other pairs of a ring and a valuation ring in one field satisfy ([16.10](#)) [C10]?

With nonconstant coefficients even nonsingular cubics have nonordinary omnific points ([section 18.5](#)); these are separate from the proved constant-coefficient statements [C12]. A finite certificate would formalize the mechanism; it would not by itself enumerate integer points [C10]. Source C15 asks for such a certificate of the logarithmic obstruction, extracted effectively, for a smooth affine curve presented by several equations, so that a checker need not reconstruct the geometry [C15]. Status (batch 31): source C16 gives finite certificates for every rigid curve, smooth or singular, with an explicit derivative-order bound ([theorems 17.28](#) and [17.40](#) and [corollary 17.29](#)); optimal bounds and a certificate-producing procedure are [questions 21.25](#) and [21.26](#). Status (batch 33): for the first clause source C17 gives a rigidity theorem in dimension zero: a monic univariate polynomial over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  or  $\mathcal{R}_{\circ}(\mathbb{k}, \Gamma)$  whose discriminant is a nonzero constant is a purely infinite translate of its constant-term polynomial ([theorem 19.12](#)), with a universal finite certificate ([lemma 19.9](#)); the batch-31 note above concerns only the certificate clause. Curves and varieties with nonconstant coefficients stay open.

The next eight questions are those of source C16, printed as it states them; it presents them as continuations of its article, not as recognized open problems of the literature.

**Question 21.23** (Exact normalization lifting [C16]). If  $C$  has normalization  $\mathbb{A}_{\mathbb{k}}^1$ , does every nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point of  $C$  lift to an  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -point of the normalization? If not, what are the smallest explicit singular curve and ordered group exhibiting the failure?

This is the third clause of [question 21.17](#), which [theorem 17.25](#) leaves open. The conductor proof controls powers only after multiplication. A first test is a rational curve with two distinct parameter values identified at a node; a positive answer must deal with the non-normality of [section 17.5](#) rather than assume it away [C16].

**Question 21.24** (Real singular arithmetic fibers [C16]). For a real curve with normalization  $\mathbb{A}_{\mathbb{R}}^1$ , which ordinary integer points are constant terms of nonordinary omnific points? Is a real preimage under the normalization necessary, as well as sufficient?

This is the remaining real–Gaussian asymmetry of [corollary 17.43](#). A concrete test is the isolated real node of  $Y^2 = X^2(X-1)$  at  $(0, 0)$ : the normalization is  $X = T^2 + 1$ ,  $Y = T(T^2 + 1)$ , the preimages of the origin are  $T = \pm i$ , and the real polynomial families of [proposition 17.42](#) do not reach that fiber, while the complex ones do [C16].

**Question 21.25** (Optimal certificate complexity [C16]). How small can the derivative order of (17.4) be made by optimizing the conductor, the presentation of  $\alpha$  and the regular differential? Can the least order be bounded by the genus, the conductor length and the pole semigroup at infinity alone?

The seventh-order certificate of [theorem 17.40](#) improves the direct bound eleven; no optimality is claimed. The identity is an effective target for symbolic algebra and formal verification [C16]. This refines the certificate part of [question 21.22](#).

**Question 21.26** (A constructive curve-rigidity procedure [C16]). Can normalization, conductor computation and the construction of a regular differential be combined into a certificate-producing procedure for arbitrary affine curve presentations over effective fields of characteristic zero, with proofs checked independently of the computer algebra system?

For superelliptic input the multiplicities already decide ([theorem 17.38](#)). For general input a certificate should specify a normalization presentation, conductor witnesses, a differential presentation and the finite derivative identity, not only a yes-or-no answer [C16]. This refines the effective-certificate part of [question 21.22](#).

**Question 21.27** (Positive-characteristic replacement [C16]). Among one-place affine curves in characteristic  $p$ , which admit nonconstant  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -points? Can support-preserving Hasse–Schmidt operators separate subclasses according to the  $p$ -divisibility of  $\Gamma$ ?

The first step is to separate the failure of a derivation to detect a scale from the genuine existence of a Hahn point; Seidenberg’s higher-derivation theorem is relevant but not sufficient by itself [C16]. [Proposition 17.31](#) gives the one-place condition in every characteristic; the question refines [question 21.21](#).

**Question 21.28** (Higher-dimensional normalization invariance [C16]). For which affine varieties is the property  $X(\mathcal{A}_{\mathbb{k}}(\Gamma)) = X(\mathbb{k})$  invariant under finite birational morphisms? Can conductor certificates be combined with enough differential tensors to remove singularity assumptions from the higher-dimensional criteria?

A promising restricted setting has a controlled rank-one valuation at the centre and differential tensors detecting every relevant tangent direction; the curve theorem is the one-dimensional model, not a proof there [C16]. The criteria in question are [theorem 16.19](#) and [corollary 18.18](#) (see [question 21.19](#)).

**Question 21.29** (Images, not just existence [C16]). For an exceptional curve, describe the image of  $C(\mathcal{A}_{\mathbb{k}}(\Gamma))$  under  $\text{ct}$ , the possible support order types in each fiber, and their behaviour under embeddings of ordered groups. How much of this is encoded by the conductor?

[Corollary 17.34](#) concerns existence only; fibers may keep rank and support information that the dichotomy forgets [C16]. For smooth curves, [theorem 17.11](#) gives singleton fibers in the rigid case and copies of  $\Pi_{\mathbb{k}}(\Gamma)$  in the affine-line case, over each ordinary point. It does not give the affine-line fiber formula for singular curves.

**Question 21.30** (Formalization across the normal-form bridge [C16]). Can the finite power identity, the conductor certificate and the one-variable valuation argument be formalized in Lean and transported to the repository’s surreal normal-form interface?

Source C16 proposes the order: finite differential algebra, then the abstract valuation contradiction, then normalization of curves and Riemann–Roch, and the proper-class application last; none of these stages is claimed done [C16].

The next ten questions are those of source C17 (section 19.4), printed as it states them, in the notation of section 19.4. Source C17 calls them questions left open by its article, not a catalogue certified by a literature search, and names the first two as the most substantial.

**Question 21.31** (Full finite étale descent [C17]). Is base change an equivalence from finite étale  $\mathbb{k}$ -algebras to finite étale  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -algebras, for every field  $\mathbb{k}$  of characteristic zero and every ordered abelian group  $\Gamma$ ?

The monogenic and unit-generated cases are settled (theorems 19.26 and 19.29). A first objective is to prove or disprove that the injection  $\mathbf{C}_{\mathbf{B}} \rightarrow \mathbf{B}/\Pi_{\mathbb{k}}(\Gamma)\mathbf{B}$  is always surjective. A proof must bridge a global issue, which generic primitive elements and local monogenic charts do not settle; a counterexample would have to hide its nonconstant structure outside its unit group [C17].

**Question 21.32** (Arithmetic étale descent [C17]). For a number ring  $\mathfrak{o} \subseteq \mathbb{k}$ , does every finite étale  $\mathcal{R}_{\mathfrak{o}}$ -algebra descend from  $\mathfrak{o}$ ? Which additional hypotheses on  $\mathbb{k}$  are necessary?

Theorem 19.31 handles monogenic presentations. A next step compares base change to  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  with reduction to  $\mathfrak{o}$ , keeping track of flatness and arithmetic fibers; the field-coefficient unit theorem must not be applied over  $\mathfrak{o}$  (remark 19.32). The cases  $\mathfrak{o} = \mathbb{Z}, \mathbb{Z}[i]$  concern  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  [C17].

**Question 21.33** (Projective modules and nonnormal spectral data [C17]). What is  $\mathrm{Pic}(\mathcal{A}_{\mathbb{k}}(\Gamma))$ , and which projective modules occur as eigenmodules of a matrix with nonzero constant characteristic discriminant?

The translation theorem fixes eigenvalues; the rest of the matrix is a system of projective summands attached to the spectral idempotents. Normality makes them constant by positivity, while example 19.44 shows freedom without it. A classification up to similarity should separate changes of basis from nontrivial projective summands [C17].

**Question 21.34** (Sharp denominator certificates [C17]). What is the least exponent  $d(n, s)$  for which  $\Delta^{d(n,s)}e_s(w)$  is a universal polynomial in the coefficients and their derivatives? Can the certificates be generated with practical size?

Lemma 19.9 gives  $d(n, s) \leq 2s$ , and for  $n = 2, s = 1$  one power of  $\Delta$  suffices (remark 19.10). The problem is finite algebra, independent of surreal foundations [C17].

**Question 21.35** (Nonconstant discriminants and quantitative defects [C17]). Can the support or valuation of  $\mathrm{Disc}(P)$  bound the possible nonconstant motion of the root gaps when the discriminant is not constant?

The identities (19.5) remain valid, and their valuations give inequalities between derivative coefficients and powers of  $\Delta$ , but the zero-intersection argument no longer closes. A Newton-polygon theory of the velocity polynomial could measure how far a polynomial is from a constant configuration plus a translation [C17]. The example  $T^2 - T - \omega$  of example 19.5, with discriminant  $1 + 4\omega$ , is a test case.

**Question 21.36** (Positive characteristic [C17]). Which restrictions on degree, ramification or additive-polynomial structure restore a useful translation theorem in characteristic  $p$ ?

[Example 19.42](#) shows that separability and a unit discriminant do not suffice. A substitute must account for Frobenius and for derivations that do not see  $p$ -divisible exponents; degrees below  $p$ , then tame splitting extensions, are natural first cases, neither asserted sufficient [C17]. Compare [questions 21.21](#) and [21.27](#).

**Question 21.37** (Restricted support rings [C17]). For which proper subrings of a Hahn ring does the discriminant theorem hold, with the translation in the subring?

A candidate should be closed under the Euler derivations, contain the purely nonconstant part divided by the degree, and keep the opposite-support intersection; effective left-finite, bounded-denominator and definability-restricted representations are natural tests [C17].

**Question 21.38** (Several polynomial configurations [C17]). What are the minimal invariant conditions forcing a family of finite root configurations to share one translation, or one affine transformation?

Constant discriminants and a spanning tree of constant resultants suffice for a translation ([theorem 19.21](#)). A controlled common scaling would need a normalization of the weights of discriminants and resultants, distinguishing labelled configurations from their symmetric quotients and handling arithmetic descent [C17].

**Question 21.39** (Polynomial dynamics [C17]). Under which extra conditions does  $f = \bar{f}(X - h) + \beta$  become conjugate, rather than left-right equivalent, to an ordinary polynomial?

[Theorem 19.33](#) reduces this to a relation between  $h$ ,  $\beta$  and the chosen dynamical normalization; fixed points, critical orbits or an ordinary marked value are possible extra data, and iterates must be kept separate from the fixed-degree theorem [C17].

**Question 21.40** (Formal verification and constructive extraction [C17]). Can the universal certificate and the finite étale unit theorem be formalized for the repository's actual Hahn and surreal constructions, with an executable translation extractor for supported notation systems?

The extractor is (19.9); the difficulty is certified equality of the resulting infinite coefficients and a proof that the input denotes a legal Hahn series. A verified finite-support implementation, then the generic support interface, is proposed as a first milestone ([section 19.4.11](#)) [C17].

## 21.6 Synthesis

The arithmetic of  $\mathbf{Oz}$  is governed by three simultaneous facts:

$$\mathbf{Oz}/\Pi = \mathbb{Z}, \quad \mathbf{Oz}^\times = \{\pm 1\}, \quad \text{Frac}(\mathbf{Oz}) = \mathbf{No}.$$

The first makes ordinary congruences and equation-existence questions look classical. The second imposes powerful rigidity on nonzero product fibers. The third makes projective representation flexible while saying little about primitive representatives. Growth degree, support and chosen Euler derivations then govern whether individual solutions escape the ordinary integers [05].

For Diophantine equations the key divide is not only between finite and infinite numbers. It is between *equation existence*, which the retraction controls; *multiplicative, differential or bounded rigidity*, which can force every coordinate to be ordinary; and *polynomial flexibility*, which creates infinite points without any infinitesimal tail [02]. The difference between a Pell conic and a three-variable indefinite quadric is structural: binary factorization forces constants,

while an additional indefinite dimension supports finite polynomial isometries. The guards use the rigid side of this dichotomy to recover the ordinary integers inside  $\mathbf{Oz}$ . Omnific integers preserve much more of ordinary equational existence than their size suggests, and much less of ordinary quantified arithmetic than their discreteness suggests [05]. None of this requires unique factorization, a Euclidean termination principle, or Conway’s refinement property.

Sources 06 and 07 show that the decomposition itself is visible from inside the ring. The ideal  $\Pi$  is defined by  $\exists y (x^2 = 2y^2)$ , its complement is the set of divisors of values of  $\Lambda$ , and  $\mathbb{Z}$  is cut out by a Pell equation, a divisibility and an intersective certificate; so the retraction  $\text{ct}$ , although not a symbol of the language, has a Diophantine graph. The multiplier ring of  $\Pi$  then recovers the real coefficients, and with them the valued field  $\mathbf{No}$ , from the pure ring  $\mathbf{Oz}$ , while the Gaussian ring  $\mathbf{Oz}[i]$  recovers  $\mathbb{C}$  but not its real form. Arithmetic recovery and recovery of a real form are different tasks [06].

Sources 08 and 09 say how much the third fact leaves of ordinary fractions. Wherever a point has a unimodular presentation over  $\mathcal{B}_{\mathbb{R}}$ , and in particular for all rational functions of one purely infinite parameter, the retraction decides lowest terms: a rational constant vector gives a principal denominator ideal and a primitive presentation unique up to an overall sign, and an irrational one gives a scalar copy of  $\Pi$  and no gcd. This arithmetic is invisible to the order: both kinds of point occur in every interval, and the specialization is continuous nowhere.

Sources C10–C14 turn the differential side of the second divide into geometry. Euler derivations preserve the one-sided ring and push the opposite valuation ring into its maximal ideal. Properness lets the same field point be seen from both rings, so each positive-degree global symmetric differential has zero contraction against the induced Euler tangent. The differential itself need not vanish at that point. Tangent separation then forces constancy. For smooth curves the three mechanisms are then exhaustive: positive genus gives differential rigidity, two or more punctures of a rational curve give unit rigidity, and the single remaining case, the affine line, gives polynomial flexibility with every fiber a copy of  $\Pi$ . The elliptic test equation of [question 21.1](#) is the first case with a one-line certificate, and its singular members are the polynomial case. Source C15 carries the differential side to logarithmic forms: a form with logarithmic poles along the boundary still contracts into the valuation ring, so with the image condition  $\partial(\mathcal{A}_{\mathbb{K}}(\Gamma)) \subseteq \Pi_{\mathbb{K}}(\Gamma)$  its Euler contraction vanishes as well, and complements whose logarithmic forms separate tangent directions are rigid in every dimension.

Source C16 removes smoothness from the curve case without assuming that points lift to the normalization. One conductor element of the curve clears every power of a contracted regular differential after a fixed multiplication, and at the one boundary place selected by a nonconstant point the valuation is discrete; so a regular differential forces a contradiction after finitely many derivatives. The only escape is normalization by the affine line, and larger scales never make a rigid constant curve flexible.

## 21.7 Relations to neighbouring reports

- `docs/foundations-and-computation/foundations`: defines  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  (`found:eq:omnific`, subsection `found:sub:omnific`) with the same letter  $\Pi$ ; this report develops the arithmetic of that ring.
- `docs/surcomplex/trigonometry`: writes the purely infinite part with  $t$ -exponents of the opposite sign (`trigonometry:eq:split`; translation in [section 1.4](#)); its integer-part construction agrees with [theorem 2.7](#), and its identity  $2\pi\mathbf{Oz} = \Pi + 2\pi\mathbb{Z}$  (`trigonometry:eq:periodclass`) rests on the same fact  $r\Pi = \Pi$  for real  $r \neq 0$  as [theorem 3.1](#).
- `docs/surcomplex/gamma-and-zeta-functions`: the parity statements of `gz:cor:`

**canonical-zeros** ( $-\omega/2 \in \mathbf{Oz}$ ,  $-\omega - 1 \notin 2\mathbf{Oz}$ ) are instances of [theorem 3.1](#); its statement that the internal integers of an ultrapower are not  $\mathbf{Oz}$  is consistent with [theorem 10.10](#) and [remark 10.11](#).

- **docs/surreal/transcendence-over-bounded-support**: its hypothesis `bst:hyp:gcd` concerns the rank-one bounded-support ring over an algebraically closed coefficient field, whose units are the nonzero monomials. Its support is bounded in both directions and may contain either sign of exponent. By contrast, [theorem 4.10](#) concerns  $\mathbf{Oz}$ , whose exponents are all nonnegative and whose units are  $\pm 1$ . The conditional GCD assertion and the no-GCD theorem apply to different rings and do not imply or contradict each other.
- **docs/surreal/set-sized-quotients-of-omnific-integers**: prints the universal set-sized quotient theorem ([cited theorem 3.6](#)) with source 05's proof and further threshold results; its preliminaries overlap [sections 2 to 4](#). Its manuscript 13 proves the Gaussian fiber theorem, the kernel converse and étale norm rigidity printed here once in [section 6](#); its manuscript 16 proves the fresh-scale image-gap theorem of which the equations of [example 19.2](#) and [proposition 19.4](#) are special cases. Its manuscript 14 proves the constant-direction criterion over  $\mathbb{Z}$  and  $\mathbb{Z}[i]$  printed here as [corollary 15.55](#); its coefficient-lattice results contain the rank-two case of [theorem 15.7](#) (`osq:tn:thm:duals`), the intersection formula of [example 15.27](#) for monomial  $t$  (`osq:tn:thm:syzygy`), and the scalar step of [theorem 15.11](#) (`osq:tn:thm:hom`). Localization at the positive monomials gives  $\mathbf{No}$  (`osq:prop:fractions`), and the residue field  $\mathbb{Q}$  of  $\mathbf{Oz}_\Pi$  agrees with its list of small primes (`osq:cor:smallprimes`). Its class-valued derivations  $\mathbf{Oz} \rightarrow \Pi$  (`osq:prop:classder`) contain the support inclusion  $\partial(\mathcal{A}_k(\Gamma)) \subseteq \Pi_k(\Gamma)$  used in [section 16](#); its theorem that derivations of the class ring into set-sized modules vanish (`osq:thm:derivations`) is no conflict, since the Euler derivations here act on set-sized workspaces and take values in them. A derivation on a workspace is not a derivation from the full class  $\mathbf{Oz}$  into a set-sized  $\mathbf{Oz}$ -module; the latter module structure is an additional hypothesis. Its manuscript 18's canonical translation and rich-target theorems (`osq:or:thm:canonical`, `osq:or:thm:targets`) give the same translation normal forms as [theorems 19.12](#) and [19.33](#), for integer-valued polynomials with more inputs of ordinary output, or of output in one ordinary coset, than their degree. These are different hypotheses from a constant discriminant ([remark 19.35](#)); discreteness alone does not force a polynomial into translation form.
- **docs/surreal/omnific-preserving-automorphisms**: its support-preserving Euler derivations map  $\mathcal{R}_o(k, \Gamma)$  into  $\Pi_k(\Gamma)$  (`opa:par:prop:Euler`), the support inclusion of [lemma 16.2](#) on the arithmetic rings; sources C10–C14 did not cite it.
- **docs/surcomplex/hahn-tate-uniformization**: its Tate points concern field points of curves with nonconstant  $j$ -invariant, outside the constant-coefficient hypothesis of [section 17](#); like [remark 17.24](#) and [section 18.5](#), they show that field points and points with varying coefficients are not constrained by the ring rigidity here.
- **docs/surreal/omnific-groups-and-lattices**: its non-elementary unipotents and cusp-residue obstruction (`ogl:el:thm:none1`, `ogl:el:prop:cuspresidue`) show that the focusing matrices of [lemma 15.34](#) are not products of elementary matrices at irrational centres, and that the density of [corollary 15.36](#) needs such matrices ([remark 15.39](#)). That report also cites [theorems 4.10](#) and [14.5](#), and it answers [question 21.1](#) for closed subgroup schemes of  $\mathrm{GL}_N$  over  $\mathbb{Z}$  (and over  $\mathbb{Z}[i]$  for  $\mathbf{Oz}[i]$ ), not for general affine varieties or curves; its updated remark `ogl:alg:rem:affine` now records the smooth classification and the normalization criterion for geometrically integral singular curves. The latter is a geometric criterion for nonconstant  $\mathcal{A}_k(\Gamma)$ -points, not a complete arithmetic existence test over  $\mathbf{Oz}$ : the case of an affine-line normalization with integer points but no real normalization preimage

of any of them remains open here ([question 21.24](#)). Over  $\mathbf{Oz}[i]$ , an ordinary Gaussian point and an affine-line normalization do suffice ([corollary 17.43](#)). The two answers agree where they overlap ([remark 18.7](#)):  $\mathbb{G}_a$  is nonrigid and the one-dimensional tori are rigid; the vector-group kernel of [theorem 18.6](#) is its unipotent kernel `ogl:alg:prop:bch` for commutative groups; semiabelian rigidity extends its torus freezing. For a smooth geometrically integral real affine curve, being the affine line is equivalent to its complexification being the affine line. Thus rigidity over the real and complex support rings coincides; arithmetic existence still requires the relevant ordinary point. Such geometric rigidity does not coincide for all groups ( $\mathrm{SO}_3$ ). [Theorem 12.12](#) concerns homomorphisms into class-sized Hahn rings as well and is consistent with the quotient theorem there.

- `docs/surcomplex/surcomplex-field-automorphisms`: constructs the coefficient-fixing phase twists and the dilations  $S_a$ ; source 06 credits it and adds that the twist of [lemma 13.9](#) preserves  $\mathbf{Oz}[i]$ , whence [theorem 13.10](#).
- `docs/surcomplex/single-dilation-hahn-support`: shows that the single exponent dilation  $S_2$  of the full Hahn field  $k((t^\Gamma))$ , with  $k$  of characteristic zero and  $\Gamma$  nonzero and 2-divisible, defines its coefficient field, constant coefficient and valuation ring in the language of fields with one named automorphism; [theorems 13.2](#) and [13.4](#) recover the coefficient field, order and valuation ring from the pure ring  $\mathbf{Oz}$  and its Diophantine ideal instead.
- `docs/surreal/euclidean-three-space`: its theorem that class homomorphisms from  $\mathrm{SO}(3, \mathbf{No})$  to set-sized groups factor through the standard part (`e3:cut:thm:smallquotient`) is an analogous size phenomenon for groups, not the same theorem; by [corollary 9.4](#),  $\mathrm{SO}(3, \mathbf{Oz})$  is finite.

## A Provenance and merge record

### A.1 Sources

The report is built from fifteen manuscripts. Sources 01, 02 and 05 are three of a batch of eleven placed in commit `be06fc8`, and their numbers are the batch item numbers; they are dated September 22, 2026, and the repository commit they pin is `2cb9c0200af749fbbd27796c97d017e7f16fbf33`. Sources 06 and 07 are items 01 and 07 of a batch of eight placed in commit `cf350b1`, numbered here after the earlier sources, as their shipped file prefixes are; they are dated September 23, 2026, and the commit they pin is `71e9606d297c9b98c732070dc462682cc6948981` (“Prove Prony’s main theorem, holonomic ODE rigidity and eight more results”), which precedes `be06fc8`. None of the five was aware of the others. Item 03 of the second batch is manuscript 13 of the sibling report; its Gaussian and étale fiber results (its Sections 8 and 9) are printed here, tagged [13], and its remaining results there. The manuscripts themselves are not shipped. Their code and data are shipped under the prefixes `01-diophantine-geometry-`, `02-diophantine-`, `05-diophantine-rigidity-`, `06-definability-reconstruction-` and `07-defining-arithmetic-`; source 02’s provenance note, source 05’s build note and source 06’s source audit are shipped as `02-diophantine-PROVENANCE.md`, `05-diophantine-rigidity-BUILD.md` and `06-definability-reconstruction-SOURCE_AUDIT.md`. Source 01 delivered a checksum manifest, which is not shipped because it describes the delivered file names; the delivered descriptions of sources 06 and 07 are not shipped either. Source 13’s files are shipped with the sibling report.

Sources 08 and 09 are items 01 and 02 of a batch of nine placed in commit `a4dcb91`, numbered here by their shipped file prefixes. Both are dated September 23, 2026, and both pin

9a385d3957bdf9ea79f9a524751c90bd2c894 (“Merge the rewritten Diophantine report and refresh its pending source index”), 35 commits before that placement. Neither was aware of the other. Their code and data are shipped under the prefixes `08-fractions-` and `09-fraction-fibres-`, and source 08’s source audit as `08-fractions-SOURCE_AUDIT.md`. Their texts, PDFs and delivered READMEs are not shipped, nor is source 09’s checksum manifest, which lists the delivered file names. Manuscript 14 of the sibling report is item 04 of the batch placed in `cf350b1`; its files are shipped there with the prefix `14-arithmetic-tensors-`.

Sources C10–C14 are items 02, 03, 04, 07 and 08 of a batch of nine placed in commit `c6359e4`, whose archives arrived in `5610500` (C10, C11, C12) and `21016dc` (C13, C14); they are tagged by their shipped file prefixes `10-curve-rigidity-`, `11-curve-abelian-rigidity-`, `12-curve-logarithmic-`, `13-differential-rigidity-` and `14-hahn-differential-rigidity-`, with the letter C to keep them apart from the sibling report’s manuscripts 13 and 14. All are dated September 23, 2026. Sources C10, C13 and C14 pin `4cdeaec43ff1692ed2ad9f5bdf761a41872975a5` (“Record merged validation for the quadratic geometry review”), 23 commits before `5610500`; source C11 pins `f6e031a8ebe78203f7d7b449d7e97cb7a540c949` and source C12 `89bec380b218c0a5bb865f53a86f5abf7f8fb5ad`, 26 and 27 commits before it. None of the five was aware of the others. Their code, data and audits are shipped under those prefixes: the source audits of all five, source C11’s build audit, and the build records of sources C10 and C14 (in `data/`). Their texts, PDFs and delivered READMEs are not shipped, nor are the checksum manifests of sources C10 and C12, which list the delivered file names.

Source C15 is item 02 of a batch of nine placed in commit `66d7e55`, whose archives arrived in `65775fb`; it is tagged by its shipped file prefix `15-logarithmic-rigidity-`, with the letter C as for sources C10–C14. It is dated September 23, 2026, and pins `934810abdde4c6d74c08777b069445de43602934` (“Complete sharp angular root stability with native fine derivatives”), 18 commits before `c6359e4` and 23 before `66d7e55`; so it saw none of sources C10–C14. Its proof audit, code and data are shipped under that prefix (`15-logarithmic-rigidity-PROOF_AUDIT.md`, the verification script, its two build scripts, the recorded run, the build report of its own article and the requirements file). Its text, PDF and delivered README are not shipped, nor is its checksum manifest, which lists the delivered file names.

Source C16 is item 04 of a batch of nine placed in commit `9d28e28`, whose archives arrived in `39fe674`; it is tagged by its shipped file prefix `16-singular-curves-`, with the letter C as for sources C10–C15 (the sibling report’s manuscript 16, files `16-fresh-scale-`, is not a source of this report). It is dated September 23, 2026, and pins `bcac55ae6fd3f2b354e568ba1d2e94d496a2cc59` (“Construct Fejer-Riesz factors on the full affine Cayley chart”), 87 commits before `9d28e28`. At that pin sources C10–C14 had been placed (`c6359e4`), but their material was not yet written into this article (`ac54217` is not an ancestor of the pin), and source C15 had not been placed. Its source audit, code and data are shipped under that prefix (`16-singular-curves-SOURCE_AUDIT.md`, the verification script, its build script, the recorded run, the requirements file and the build report of its own article). Its text, PDF and delivered README are not shipped, nor is its checksum manifest, which lists the delivered file names.

Source C17 is manuscript 07 of the batch of nine placed in commit `aa9c891` (batch 33), whose archives arrived in `73043eb`; it is the archive `omnific_discriminant_rigidity`, local number 17 of this report, tagged by its shipped file prefix `17-discriminant-rigidity-` with the letter C as for sources C10–C16 (the sibling report’s manuscript 17, files `17-relations-arithmetic-`, is not a source of this report). Its title is *Constant Discriminants and Étale Unit Rigidity over Omnific Hahn Rings: Translation classification, arithmetic descent, and normal spectral rigidity*; it is dated September 23, 2026, has 27 pages, and pins `efc5446229dbf61a97bdd5edae91dba20af9d131` (“Prove the arithmetic homeomorphisms

of omnific separation quotients”; tree [aba698290ee2cd54f97db3a8b4038c010fcb47d8](#)), 70 commits before [aa9c891](#) and 31 before [73043eb](#). At that pin sources C10–C15 were written into this article ([ac54217](#) is an ancestor), source C16 was placed ([9d28e28](#)) but not yet written ([1ad4ad8](#) is not an ancestor), and the questions cited below were [questions 21.6](#) and [21.22](#) under the same labels. Its source and proof audits, code and data are shipped under that prefix (`17-discriminant-rigidity-SOURCE_AUDIT.md`, `17-discriminant-rigidity-PROOF_AUDIT.md`, the verification script, its build script, the recorded run, the requirements file and the build report of its own article). Its text, PDF and delivered README are not shipped; it delivered no checksum manifest.

## A.2 Section map

| Section                                         | Sources                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">1</a> Scope and conventions         | 01 base; 02 and 05 results; new notation and provenance text                                                                                                                                                                                                                                             |
| <a href="#">2</a> Rings and integer part        | 01 base; 05 degree of sums, workspaces, univariate roots; 02 examples                                                                                                                                                                                                                                    |
| <a href="#">3</a> Residues and quotients        | 01 base; 02 mixed gcds and completions; 05 $p$ -adic orders, divisors of integers, CRT; 05’s set-sized quotient theorem cited                                                                                                                                                                            |
| <a href="#">4</a> Full-class support            | 01 base; 02 nilpotent tests, common multiples, general no-gcd pair; 05 witness $\sqrt{2}$ and prime examples                                                                                                                                                                                             |
| <a href="#">5</a> Transfer and linear equations | 01 base; 02 fibers and examples; 05 rational-basis form                                                                                                                                                                                                                                                  |
| <a href="#">6</a> Constant products             | 02 exact fibers; 05 binary form over $\mathbb{C}$ , $D = 2$ classification, conics; 01 zero Pell fiber, $\gcd > 1$ powers, norm forms with any $\mathbb{Q}$ -basis; 13 Gaussian fibers, converse, kernel remark, rank-deficient example, étale norms; 07 norms over any base field; merged étale theorem |
| <a href="#">7</a> Derivations                   | 05, extended to $\mathbb{C}$ by the review; merge remark <a href="#">7.5</a> ; pointers to the squarefree theorem of C10–C14                                                                                                                                                                             |
| <a href="#">8</a> Quadratic dichotomy           | 01 base; 02 skewness, determinant, Lorentz matrix, unipotent subgroup                                                                                                                                                                                                                                    |
| <a href="#">9</a> Bounded geometry              | 02 semialgebraic theorem; 01 positive bounds, symmetric matrices, no real point at infinity; 05 even powers                                                                                                                                                                                              |
| <a href="#">10</a> Defining $\mathbb{Z}$        | 01 three-square guard; 05 four-square definition and induction; merge remark <a href="#">10.11</a> ; 07 quartic variant                                                                                                                                                                                  |

| Section                                     | Sources                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">11</a> Diophantine arithmetic   | 06 base: setting, ideal, Pell, $\Lambda$ , quintic remark; 07 intermediate rings, integer multiples, the system in its general form (with 06)                                                                                                                                                                                                                                                                                                      |
| <a href="#">12</a> Constant term            | 07 detector, support bounds, ideal test, universal forms, number fields; 06 graph of ct, splitting, homomorphisms; merge quartic graph and number-field remark                                                                                                                                                                                                                                                                                     |
| <a href="#">13</a> Reconstruction and logic | 06 multiplier, reconstruction, automorphisms, twist, c.e. sets, QF bound; 07 recursive saturation and arithmetic; 06 and 07 collapse obstruction                                                                                                                                                                                                                                                                                                   |
| <a href="#">14</a> Families and primitivity | 01 base; 02 sign conditions and decidability; 05 witness and finite-support specialization; pointers to C12–C14                                                                                                                                                                                                                                                                                                                                    |
| <a href="#">15</a> Fractions                | 08 base: multiplier theorem, rational-function classification, real constants, curves, localization, models, scale defect, $\text{Tor}_1$ , monomials, Gaussian dichotomy; 09 region $\mathcal{V}$ , denominator calculus, least denominators, orbits, focusing, density, nowhere continuity, examples; 14's Gaussian direction criterion credited; merge remarks <a href="#">15.8</a> , <a href="#">15.39</a> and <a href="#">corollary 15.30</a> |

| Section                                  | Sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">16</a> Differential rigidity | C14 base: two rings, two-ring contraction, proper rigidity; C12 and C13 strict certificate; C13 tensors and symmetric differentials, general Weierstrass; C10, C11, C14 unit route; all five cubic certificate and singular families (node of C11)                                                                                                                                                                                                                                                                                                                                                          |
| <a href="#">17</a> Smooth curves         | C14 base: classification over every field, forms of $\mathbb{A}^1$ , integer arcs, congruences, unimodular coordinates over $\mathcal{R}_o$ ; C12 logarithmic route (closed-centre step [merge]), descent, coordinate ideals; C11 fibers over any $o$ , class size, one-form remark; C13 all curves, polynomial witnesses; C10 fiber bijection, field points; C16 singular-curve criterion, conductor certificates, one-place obstruction, curves of dimension at most one, superelliptic test, pinched elliptic curve, Gaussian dichotomy ( <a href="#">17.6</a> ), and credits on the results it reproves |
| <a href="#">18</a> Groups and boundaries | all five abelian and semiabelian; C11 quasi-finite, reduced algebras, dual numbers, discretely ordered ring; C13 group kernel, characteristic $p$ ; C11 and C14 characteristic two; C10, C12, C13, C14 coefficient and valuation-ring boundaries; merge remark <a href="#">18.7</a> ; C15 logarithmic annihilation, tangent criterion, interface, descent over any coefficient ring and finite products ( <a href="#">18.6</a> ), and credits on the results it reproves in Sections <a href="#">16–18</a>                                                                                                  |
| <a href="#">19</a> Omnific coefficients  | 02 quadratic criterion and initial forms (also 01); 05 binomials and failed lifting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <a href="#">19</a> (continued)           | pointer to the image-gap theorem of the sibling report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <a href="#">19</a> (continued)           | C17 translation classification, certificate, factors, Galois groups, resultants, étale units, monogenic and arithmetic descent, critical configurations, normal matrices, boundary examples and second proof ( <a href="#">19.4</a> ); merge corollary <a href="#">19.16</a> and remarks <a href="#">19.17</a> , <a href="#">19.28</a> , <a href="#">19.35</a> , <a href="#">19.40</a>                                                                                                                                                                                                                      |
| <a href="#">20</a> , <a href="#">21</a>  | All fifteen; settled questions recorded in <a href="#">sections 21.1 to 21.4</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

### A.3 Where the merge chose

1. *Base text.* Source 01, which has the weakest hypotheses on most shared statements (Pell with  $D \neq 0$ , norm forms for any  $\mathbb{Q}$ -basis and  $c \in \mathbb{Q}^\times$ , the only exhaustive quadratic dichotomy, the guard uniform in  $n$ , the QE-free homogeneous criterion, the localization discussion).
2. *Decomposable fibers.* Source 02's exact-fiber theorem is printed, with source 01's rank- $n$  theorem as case (b) of [theorem 6.2](#).

3. *Binary forms*. Source 05's complex form is printed; sources 01 and 02 proved the integer case.
4. *Power equations*. Source 05's separated-power theorem replaces source 01's  $\gcd > 1$  corollary, which is kept as [remark 6.13](#).
5. *Isotropic vectors*. Source 02's hypothesis  $a \neq 0$  replaces source 01's  $q(a) \neq 0$  in [lemma 8.3](#).
6. *Pairs without gcd*. Source 02's general theorem is printed; source 01's degree proof is kept as a second route ([remark 4.11](#)) because it localizes to divisible workspaces.
7. *Homogeneous systems*. Source 01's elementary proof and source 02's Tarski proof are both kept, since only the latter handles sign conditions and decidability.
8. *Uniqueness of ct*. Source 01's divisibility proof is printed; source 05 derives it from the set-sized quotient theorem, which is quoted and whose proof belongs to the sibling report.
9. *Guards*. Both are printed: source 01's (five auxiliaries, three squares) and source 05's (six auxiliaries, four squares).
10. *Additions*. Two remarks carry the tag [merge]: [remark 7.5](#) (a direct identity giving the divisibility in the separated-power proof) and [remark 10.11](#) (existential induction on the nonnegative cone fails in the pure ring language). Both have complete proofs.
11. *Definability base*. Source 06 is the base of [sections 11 to 13](#): its existential definitions of  $\Pi$  and of the graph of ct supersede source 07's universal and prenex ones, which are kept as [corollary 12.7](#) and [remark 12.11](#), and its homomorphism theorem covers all unital homomorphisms, where source 07's covers automorphisms.
12. *The order-free system*. The two sources' systems are identical; source 07's statement (both pairs, intermediate rings) is printed, with source 06 credited ([theorem 11.9](#)).
13. *Real single polynomials*. The quartics of sources 01 and 05 stay the printed versions. Source 07's quartic is source 05's up to the substitution of [remark 10.5](#) and is printed there once; source 06's quintic, dominated by both, is a remark with the comparison ([remark 11.10](#)).
14. *Pell divisibility*. Source 07's permutation argument is printed with source 06's finite-ring argument as a second route ([lemma 11.6](#)).
15. *Norms*. Source 07's theorem (any base field, intermediate rings) and source 13's (étale algebras, Gaussian variables) are combined in [theorem 6.15](#); [theorem 6.14](#) of sources 01 and 02 is its special case.
16. *Fibers*. Source 13's real fiber theorem is [theorem 6.2\(a\)](#), printed once with source 02 as base; its Gaussian case, converse, kernel remark and rank-deficient example are added.
17. *Transfer and obstructions*. Sources 06, 07 and 13 re-prove the transfer theorem, printed once ([theorem 5.1](#)); the collapse obstructions of sources 06 and 07 generalize [proposition 5.3](#) and are printed as [theorem 13.13](#). Source 07's recursive-saturation statement is printed, with source 06's countable type as the simpler instance.
18. *Fraction fields*. Source 06's common-denominator lemma is [theorem 4.6](#), printed once; the Gaussian consequence is stated in [section 13](#). Source 06's remark distinguishing ct from st joins [remark 2.3](#).
19. *Four squares*. Source 06's appendix proof of Lagrange's theorem is not reprinted; the theorem is cited ([14] for Mathlib, source 07's [27]).

20. *Further additions.* [Corollary 12.9](#) (a quartic with six witnesses for the graph of  $\text{ct}$ , improving source 06's degree-ten polynomial), [remark 12.15](#) (existential definitions over number-field coefficients), the Gaussian example in [remark 6.5](#), the collapse of the quintic over  $\mathbf{Oz}[i]$  in [remark 11.10](#), and the combined [theorem 6.15](#) carry the tag `[merge]` and have complete proofs.
21. *Fractions base.* Source 08 is the base of [section 15](#). Its multiplier theorem has the weaker hypotheses: any domain with a retraction onto  $\mathbb{R}$ , any number of coordinates, and any  $t \in \Pi \setminus \{0\}$ . Source 09 proves the pair version over  $\mathcal{B}_{\mathbb{R}}$  and the rational functions of one monomial  $t = \omega^\alpha$ . So source 09's multiplier theorem and one-scale classification are special cases of [theorems 15.11](#) and [15.18](#) and are not printed separately (credited in [sections 15.3](#) and [15.5](#)). Source 09's proofs use only  $\mathcal{B}_{\mathbb{R}}$ -unimodularity and full-class clearing, so they carry over unchanged to source 08's setting.
22. *The region  $\mathcal{V}$ .* Source 09's formulation on the region  $\mathcal{V}$ , with  $\mathcal{U} = \text{sp}^{-1}(\mathbb{P}^1(\mathbb{Q}))$ , is kept as [theorem 15.15](#), proved from [theorem 15.11](#).
23. *Real constants and ordinary denominators.* Source 08's five characterizations of rationality and source 09's example are printed once ([theorem 15.7](#)). Source 08's localization identity and source 09's statements on units, subfields and Bézout pairs are combined in [proposition 15.5](#).
24. *Second routes.* The two Bézout lifts are kept: source 08's ([15.4](#)) and source 09's adjustment of constants ([remark 15.12](#)). The halving argument of both sources for the absence of a gcd is kept next to the proofs of [theorem 4.10](#) and [remark 4.11](#) ([remark 15.8](#)). The two non-rational elements of one workspace (source 08's exponential series and source 09's factorial-gap series) and the monomial examples of both sources are all printed ([examples 15.51](#) and [15.52](#)).
25. *Tables.* The two sources' example tables are merged into one ([example 15.23](#)); the rows  $m/n$ ,  $r$  and  $(t+r)/(t+1)$  occur in both.
26. *Questions.* Source 08's question on set-generated denominator ideals and source 09's on principal ideals outside  $\mathcal{V}$  are merged into [question 21.12](#); the other three are printed as stated.
27. *Credits added.* Source 08's conic example is [proposition 14.4](#) with source 01's witness and is not reprinted. Its example without an lcm is, for monomial  $t$ , the sibling report's `osq:tn:thm:syzygy(iii)`, and its real-constant ideal is, for irrational  $r$ , `osq:tn:thm:duals`. Source 09's localization  $\mathcal{M}^{-1}\mathbf{Oz} = \mathbf{No}$  is `osq:prop:fractions(iii)`. Source 08's Gaussian dichotomy contains manuscript 14's direction criterion, which is printed once, crediting both ([corollary 15.55](#)).
28. *Further additions.* [Remark 15.8](#) (the conclusion of [theorem 4.10](#) in every  $\mathbf{Oz}_\Gamma$ ), [corollary 15.30](#) (the formulation of source 08's curve criterion for homogeneous systems) and [remark 15.39](#) (comparison with the elementary group of the omnific groups report) carry the tag `[merge]` and have complete proofs.
29. *Curve base.* Source C14 is the base of [sections 16 to 18](#): on the shared theorems it has the weakest hypotheses (any field of characteristic zero, any exponent group, the curve classification over the field itself, the arithmetic dichotomy with the generic fiber over  $\mathbb{Q}$ , unimodular coordinates over  $\mathcal{R}_o(\mathbb{k}, \Gamma)$ ). Source C11 has the most additional material but states its classification over an algebraically closed field or  $\mathbb{R}$ ; source C12 assumes an algebraically closed field throughout; source C13 assumes a divisible exponent group and the fields  $\mathbb{R}$  and  $\mathbb{C}$ ; source C10 assumes an algebraically closed field for curves.

30. *Hidden hypotheses.* Proofs taken from sources with stronger hypotheses were re-read before being attached to weaker ones. Source C12's logarithmic proof uses closed points with residue field  $\mathbb{k}$ , so it is printed over an algebraically closed field ([theorem 17.8](#)), and the general case descends as in source C14. Source C13's tensor theorem, general Weierstrass corollary, curve corollary and group kernel use only a functional  $\Gamma_{\mathbb{Q}} \rightarrow \mathbb{k}$  and characteristic zero, and are printed for every  $\mathbb{k}$  and  $\Gamma$  ([theorems 16.14](#) and [18.6](#) and [corollaries 16.10](#) and [17.10](#)); source C11's arithmetic fibers extend to every  $\mathbb{k}$  by the classification ([theorem 17.11](#)). These extensions use the sources' own proofs.
31. *Squarefree equations.* The strict-ideal proof of sources C12 and C13 is printed ([theorem 16.6](#)); the unit argument at the corner  $(2, 2)$  of sources C10, C11 and C14 is kept as a second route ([remark 16.7](#)). The support inclusion  $\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) \subseteq \Pi_{\mathbb{k}}(\Gamma)$  is credited to `osq:prop:classder` and `opa:par:prop:Euler`, which none of the five cites.
32. *Two rings.* Source C13's version for global tensors and symmetric differentials is printed ([theorems 16.14](#) and [16.19](#)); the one-form versions of the other four are its case  $r = 1$ . Source C14's bijection form and source C11's equalizer lemma for families of maps are used ([theorem 16.16](#) and [lemma 16.13](#)).
33. *Curves.* Source C14's proof of the classification is printed; source C12's logarithmic proof is a second route; source C11's one-form remark and source C12's projective-curve proof are combined in [remark 17.3](#); the real descent of sources C10, C11 and C13 is the case  $\mathbb{k} = \mathbb{R}$  of [lemma 17.5](#).
34. *Arithmetic.* Source C11's fibers over any coefficient ring are printed ([theorem 17.11](#)), with the fiber bijections of sources C10 and C13 and the generic-fiber dichotomy of source C14 as [corollary 17.12](#). Source C14's integer arcs and congruence criterion, source C13's polynomial witnesses and source C12's descent and coordinate ideals are added. The coordinate clearing reproved by sources C10 and C12 is [corollary 4.7](#) and is not reprinted.
35. *Groups.* Source C14's torus proof is printed, with the faithful-flatness descent of source C10 and the complex descent of source C13 recorded; source C11's quasi-finite theorem for any rigid target is printed, with the finite and semiabelian cases of the others.
36. *Further additions.* The closed-centre step in [theorem 17.8](#), the comparison with the omnific groups report ([remark 18.7](#)) and the observation that source C13's characteristic- $p$  curve is an affine line (after [proposition 18.13](#)) carry the tag [merge] and have complete proofs.
37. *Logarithmic addition.* Source C15 arrived after sources C10–C14 had been integrated. Its new results are appended as [section 18.6](#), at the end of [section 18](#), so that no existing number moves; its other results are not reprinted, and C15 is added to the credits of the statements it proves.
38. *Routes of source C15.* Its squarefree proof is the strict route of sources C12 and C13, with the element in both  $\Pi_{\mathbb{k}}(\Gamma)$  and  $\mathcal{O}_{\mathbb{k}}(\Gamma)$ , and is not reprinted. Its curve classification by one logarithmic form is described after [corollary 18.18](#); it needs no closed-centre step. Its torus descent along a finite splitting field is recorded after [lemma 18.3](#), and its unimodular proof, through the principal ideal domains  $\mathbb{Z}$  and  $\mathbb{Z}[i]$ , is that of [theorem 17.18](#). Its descent theorem, for every subring of every field of characteristic zero, is printed as [theorem 18.20](#); [theorem 17.17](#) of source C12 is its case over  $\mathbb{Z}$ .
39. *Hidden hypotheses in source C15.* Its curve classification is stated over an algebraically closed field and its descent corollary over any field of characteristic zero; both are covered by the statements above over every field. Its abstract interface asks for a valuation ring

$V \subseteq K$ ; the valuative criterion needs  $V$  to be a valuation ring of  $K$ , and [remark 18.19](#) says so. No result is added by the merge in [section 18.6](#).

40. *Singular addition.* Source C16 arrived after source C15. It answers a question of this part and shares its spine, so its new results are appended as [section 17.6](#), at the end of [section 17](#), directly after the subsection that states the problem; appending at the end of a section moves no existing number. Its rings, Euler derivations, non-normality example, descent of forms of the affine line and smooth case are not reprinted, and C16 is added to the credits of [lemmas 16.1 to 16.3](#) and [17.5](#) and [theorem 17.6](#) and of the witness in [section 17.5](#).
41. *Routes of source C16.* Its proof of the smooth case is recorded as a third route ([remark 17.33](#)). For the pinched elliptic curve both of its proofs are printed, by the general theorem and by the explicit seventh-order certificate ([theorem 17.40](#)); its Bézout polynomials are those of ([16.5](#)).
42. *Stated generality of source C16.* Its results are printed at the generality it states: the criterion over every field of characteristic zero, the one-place proposition over an algebraically closed field of any characteristic, and the dimension-one theorem and superelliptic test over an algebraically closed field of characteristic zero. Three items carry the tag [merge] and have complete proofs: the consistency of the one-place obstruction with the positive-characteristic examples (after [proposition 17.31](#)), the third route ([remark 17.33](#)) and the superelliptic test over an arbitrary field of characteristic zero (after [theorem 17.38](#)).
43. *Questions of source C16.* Its eight questions are printed as stated ([questions 21.23 to 21.30](#)); the first is the third clause of [question 21.17](#), and the third, fourth and fifth refine [questions 21.21](#) and [21.22](#). They are not merged into those questions, so that each is printed as its source states it.
44. *Discriminant addition.* Source C17 arrived after source C16. It answers [question 21.6](#) for monic polynomials with a nonzero constant discriminant and the first clause of [question 21.22](#) in dimension zero, and it shares the Euler and two-ring mechanism of the curve part, but its subject is equations with omnific coefficients. So it is appended as [section 19.4](#), at the end of [section 19](#), after the questions' own pointer to it; appending at the end of a section moves no existing number, whereas a new section before [section 20](#) would renumber Sections 20 and 21. Its ring, Euler and algebraic-constant lemmas are not reprinted, and C17 is added to the credits of [lemmas 16.1 to 16.3](#), [corollary 16.4](#), and [proposition 2.8](#).
45. *Routes of source C17.* Both of its proofs of the translation theorem are printed, the root-velocity certificate and the route through the unit theorem ([section 19.4.10](#)). Its appendix on characteristic polynomials after reduction is printed as [lemma 19.22](#), its quadratic normalization as [remark 19.10](#), and its sign check in the text before [lemma 19.9](#).
46. *Stated generality of source C17.* Its results are printed at the generality it states, for every field of characteristic zero and every set-sized ordered group. Five items carry the tag [merge] and have complete proofs: the omnific roots at a constant discriminant ([corollary 19.16](#)); the comparison with the quadratic criterion and the lifting example ([remark 19.17](#)); the étale norm theorem as a constant case ([remark 19.28](#)); the comparison with the sibling report's canonical translation ([remark 19.35](#)); and the second route for symmetric omnific matrices ([remark 19.40](#)), with two sentences identifying the generic image in [example 19.45](#) and the two-ring pattern in [section 19.4.10](#).
47. *Questions of source C17.* Its ten questions are printed as stated ([questions 21.31 to 21.40](#)); the fifth refines [question 21.6](#) and the sixth sits beside [questions 21.21](#) and [21.27](#). They are not merged into those questions.

**Printed once.** The floor formula, retraction, congruences, finite quotients and completions, transfer, Hilbert’s tenth problem, the Smith criterion, units, the Euclid example, non-atomicity, the ACC chain,  $\Pi^2 = \Pi$  and non-generation,  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ , the Pythagorean family, the Fermat scaling,  $x^2 = \omega^2 + 1$ , the orthogonal group, the nilpotent isometry, the torus remark, and the failure of four squares for  $\omega - 1$  appear in two or three of sources 01, 02 and 05 and are printed once, with all sources named. From sources 06, 07 and 13: the units lemma, Pell rigidity (with the Gaussian solution  $(i, i)$ ), the Pell divisibility lemma, the intersective polynomial and its proof, the order-free system and its Gaussian witness for  $1 + i$ , the transfer theorem and Hilbert’s tenth problem, the collapse obstruction,  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ , the zero Pell fiber, the cubic norm, the negative-exponent Pell solutions and the omitted types are each printed once, with every proving source named. Sources 08 and 09 both reprove the rings and their units, the common monomial divisor, the non-generation of  $\Pi$  and  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  (credited by both to Conway and to L’Innocente–Mantova [3, Proposition 2.4.5]); source 09 also reproves the floor, ordered division and  $\Pi^2 = \Pi$ . Both recover the pair  $(t, rt)$  without a gcd and the rational-direction theorem as special cases and credit them. These are printed once, in sections 2, 4 and 14. Within section 15, the unimodular-fraction lemma, the real constants, ordinary denominators, the one-function criterion, the monomial criterion and the Gaussian fraction field are printed once with both sources named. All five of sources C10–C14 reprove the rings, the retraction, the units, the degree rules, the Euler derivations and their detection property, which are printed once (sections 2 and 7 and lemma 11.1, extended in lemmas 16.1 to 16.3), and all five credit them to this report. Within sections 16 to 18, the following are printed once with every proving source named: the squarefree theorem and the cubic certificate (all five), the singular Weierstrass families (C10, C11, C12, C14), the two-ring principle and proper rigidity (all five), the canonical-bundle lemma (all five), the puncture lemma (C10, C11, C13, C14), the curve classification (all five, at different generality), the arithmetic fibers (C10, C11, C13, C14), unimodular coordinates (C10, C13, C14), abelian and semiabelian rigidity (all five), the quasi-finite theorem (C10, C11, C12, C14), the characteristic-two point (C11 and C14, the same example with exponents rescaled by three), the fixed-workspace non-normality examples (C10, C12, C13, with proposition 4.12), the field points outside the ring (C10, C12, C13, C14) and the examples with nonconstant coefficients (C10, C12, C13, C14). Source C15 reproves the rings, units, Euler derivations and joint constants, the squarefree theorem, the cubic certificate with the example  $\Delta_0 = 23$ , the singular families, abelian, torus and semiabelian rigidity, the quasi-finite theorem and inheritance, the curve classification with  $\mathbb{P}^1$ , symmetric differentials and ample cotangent bundles, unimodular coordinates on curves of positive genus, the characteristic-two point (C11’s example with the coordinates interchanged), the  $\Gamma = \mathbb{Z}$  integrality example of C14, and the field and valuation-ring points; these are printed once, with C15 named in their credits, and the workspace lemma it uses is lemma 2.6. Source C16 reproves the rings, the retraction and the units, the transcendence of nonconstant elements, the Euler derivations and their detection property, the fixed-workspace non-normality example (for any  $0 < e < H$  with  $ne < H$ ), the descent of forms of the affine line and, as its smooth case, the smooth classification; these are printed once, with C16 named in their credits (lemmas 16.1 to 16.3 and 17.5 and theorem 17.6, section 17.5). Source C17 reproves the rings, the retraction, the units and the pullback  $\mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma)$  (its Lemma 2.1), the Euler derivations, logarithmic boundedness and joint constants (its Lemma 3.1), the algebraic constants of the Hahn field (its Lemma 6.3) and the example  $\sum_m \omega^{1/m}$ ; these are printed once, with C17 named in their credits (lemmas 16.1 to 16.3, corollary 16.4, and proposition 2.8).

#### A.4 Renamed symbols

Table 1: Symbols renamed from each source. No normalization changed except where stated.

| Source | Source symbol                                                                                   | This report                                                                                                               |
|--------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 01     | $A$ (omnific integers)                                                                          | $\mathbf{Oz}$                                                                                                             |
| 01     | $\mathcal{I}$ (purely infinite ideal)                                                           | $\Pi$                                                                                                                     |
| 01     | $\text{Int}, \text{Int}(x)$                                                                     | $\text{Std}_n, \text{Std}$                                                                                                |
| 01     | $G_n$ (guard polynomial)                                                                        | $\Psi_n$ ( $G$ is the Lorentz Gram matrix)                                                                                |
| 01     | $A_\Gamma, B_\Gamma$                                                                            | $\mathbf{Oz}_\Gamma, \mathcal{B}_\Gamma$                                                                                  |
| 01     | $P$ (purely infinite part in the floor)                                                         | $x_{>0}$                                                                                                                  |
| 01     | $\epsilon$ (common-divisor exponent)                                                            | $\delta$                                                                                                                  |
| 01     | $\alpha = \sqrt{2} - 1$                                                                         | $\rho$                                                                                                                    |
| 01     | $\delta = \deg_\omega d$ (gcd proof)                                                            | $\theta$                                                                                                                  |
| 01     | $E$ (hyperboloid coefficient)                                                                   | $D'$                                                                                                                      |
| 01     | $T_t$ (transvection)                                                                            | $U_t$                                                                                                                     |
| 01     | $\lambda_\alpha, c_\alpha, M$ (initial forms)                                                   | $\deg_\omega a_\alpha, \text{lc } a_\alpha, W$                                                                            |
| 01     | $\lambda$ (scalar in decomposable forms)                                                        | divided into the nonzero level                                                                                            |
| 01     | $f$ (endomorphism)                                                                              | $\psi$                                                                                                                    |
| 02     | $\mathcal{I}, \mathcal{I}_\mathbb{C}$                                                           | $\Pi, \Pi_\mathbb{C}$                                                                                                     |
| 02     | $\mathcal{A}_\mathbb{R}, \mathcal{A}_\mathbb{C}$                                                | $\mathcal{B}_\mathbb{R}, \mathcal{B}_\mathbb{C}$                                                                          |
| 02     | $d(f)$                                                                                          | $\deg_\omega f$                                                                                                           |
| 02     | $Q, B$ (quadratic and bilinear form)                                                            | $q, B$                                                                                                                    |
| 02     | $a, q = Q(a)$ (lemma vectors)                                                                   | $u, k = q(u)$                                                                                                             |
| 02     | $u$ (integer point)                                                                             | $\mathbf{a}$                                                                                                              |
| 02     | $U_T, T, S$                                                                                     | $U_t, t, s$                                                                                                               |
| 02     | $J = \text{diag}(1, -1, -1)$                                                                    | $G$                                                                                                                       |
| 02     | $J$ (index set)                                                                                 | sets $S$ of elements                                                                                                      |
| 02     | $J$ (ideal)                                                                                     | $\mathbf{a}$ or $\mathbf{n}$                                                                                              |
| 02     | $A$ (integer matrix)                                                                            | $M$                                                                                                                       |
| 02     | $K$ (common real kernel)                                                                        | $\mathcal{K}$                                                                                                             |
| 02     | $\eta$ (infinitesimal part), $P$ (purely infinite part)                                         | $\varepsilon, x_{>0}$                                                                                                     |
| 02     | $s = \sqrt{2} - 1$                                                                              | $\rho$                                                                                                                    |
| 02     | $T$ (in the no-gcd theorem)                                                                     | $t$                                                                                                                       |
| 05     | $\mathcal{I}$                                                                                   | $\Pi$                                                                                                                     |
| 05     | $\mathcal{B}, \mathcal{B}_H, \mathcal{B}_{\mathbb{C},H}$                                        | $\mathcal{B}_\mathbb{R}, \mathcal{B}_{\mathbb{R},H}, \mathcal{B}_{\mathbb{C},H}$                                          |
| 05     | $A$ (generic ring; set-sized target)                                                            | $R; S$                                                                                                                    |
| 05     | $D_\lambda, W_D$                                                                                | $\partial_\lambda, W_\partial$                                                                                            |
| 05     | $D, N$ (Pell)                                                                                   | $D, c$                                                                                                                    |
| 05     | $p$ (Fermat exponent)                                                                           | $n$ ( $p$ is reserved for primes)                                                                                         |
| 05     | $a, b$ (ratio constants in the Fermat proof)                                                    | $\mu, \nu$                                                                                                                |
| 05     | $\deg_\omega, \lfloor \cdot \rfloor_{\mathbf{Oz}}, X_k, Y_k, \rho, \text{Std}, \Phi, x_{>0}$    | unchanged                                                                                                                 |
| 06     | $G, k, D$ (exponent group, coefficient field and ring)                                          | $\Gamma, \mathbb{k}, \mathbf{o}$ ( $G$ is the Gram matrix, $k$ an index, $D$ the Pell parameter)                          |
| 06     | $X^g; d(f); \mathbf{K} = \mathbf{No}(i)$                                                        | $\omega^g; \deg_\omega f; \mathbf{No}[i]$                                                                                 |
| 06     | $\mathcal{A}_k(G), \mathcal{I}_k(G), B_D(k, G); \mathcal{I}_\mathbb{R}, \mathcal{I}_\mathbb{C}$ | $\mathcal{A}_\mathbb{k}(\Gamma), \Pi_\mathbb{k}(\Gamma), \mathcal{R}_\mathbf{o}(\mathbb{k}, \Gamma); \Pi, \Pi_\mathbb{C}$ |

| Source | Source symbol                                                                                                                   | This report                                                                                                                                                            |
|--------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 06     | $H$ (intersective polynomial); witnesses $h, r$                                                                                 | $\Lambda$ ( $H$ is a workspace group); $s, t$                                                                                                                          |
| 06     | $P$ and its parts $A, B, C$ (quintic); $\text{Std}_D$                                                                           | the display (11.5); $\Xi$ or any definition of $\mathbb{Z}$                                                                                                            |
| 06     | $(u_m, v_m)$ ; $F = \text{Frac } B$ ; $\mathcal{O}$                                                                             | $(X_k, Y_k)$ ; $\mathcal{F}$ ; $\mathcal{O}_{\text{fin}}$                                                                                                              |
| 06     | $T, T_2$ (twist, dilation); $\ell$ ( $\omega^0$ -coefficient on $\mathbf{No}$ )                                                 | $\tau, S_2$ ; $\eta$ ( $\ell_j$ are linear forms)                                                                                                                      |
| 06     | $\text{Inf}, \text{CT}, \text{Mult}, \text{Coeff}, \text{ST}, \chi, \text{st}, \mathbf{m}$                                      | unchanged                                                                                                                                                              |
| 07     | $\Gamma, k, D, \Omega$ (large monomial), $c_0$                                                                                  | $\Gamma, \mathbb{k}, \mathbf{o}, \omega, \text{ct}$                                                                                                                    |
| 07     | $H_k^+(\Gamma), I_k^+(\Gamma), R_D(\Gamma)$                                                                                     | $\mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma), \mathcal{R}_{\mathbf{o}}(\mathbb{k}, \Gamma)$                                                             |
| 07     | $P, P_K$ (intersective polynomials)                                                                                             | $\Lambda, \Lambda_K$ ( $P$ names polynomials in general)                                                                                                               |
| 07     | $\Phi, \Phi_K$ (constant definitions); $\mathcal{I}, \mathcal{C}$                                                               | $\Xi, \Xi_K$ ( $\Phi$ is the induction formula of theorem 10.10); the formula of corollary 12.7, $\text{CT}^\vee$                                                      |
| 07     | $B, M, F$ (augmented algebra, kernel, polynomial)                                                                               | $\mathcal{H}, \ker \varepsilon, f$                                                                                                                                     |
| 07     | $(u_n, v_n)$ ; $L/K$ (norm)                                                                                                     | $(X_k, Y_k)$ ; $E/K$                                                                                                                                                   |
| 07     | $\mathcal{E}, \Theta, F_4, \varepsilon$                                                                                         | unchanged                                                                                                                                                              |
| 13     | $J_k, J_{\mathbb{R}}, J_{\mathbb{C}}; P_k; A_D$                                                                                 | $\Pi_{\mathbb{k}}(\mathbf{No}), \Pi, \Pi_{\mathbb{C}}; \mathcal{A}_{\mathbb{k}}(\mathbf{No}); \mathcal{R}_{\mathbf{o}}(\mathbb{k}, \mathbf{No})$ (its $P_k$ is a ring) |
| 13     | $\ell_j$ (affine forms), $A$ (matrix of linear parts), $W, J(W)$                                                                | $L_j$ , the linear parts $\ell_j, \mathcal{K}$ or $\mathcal{K}_{\mathbb{C}}$ , the coefficient-vector description                                                      |
| 13     | $d, m$ (Pell); $n, r$ (norm)                                                                                                    | $D, c; r, a$                                                                                                                                                           |
| 08     | $\mathcal{B}, \text{Deg}$                                                                                                       | $\mathcal{B}_{\mathbb{R}}, \text{deg}_{\omega}$                                                                                                                        |
| 08     | $\tau$ (purely infinite parameter)                                                                                              | $t$ ( $\tau$ is the phase twist)                                                                                                                                       |
| 08     | $B, c, I, A = c^{-1}(\mathbb{Z}), F$ (abstract pullback)                                                                        | $R, c, I_R, R_{\mathbb{Z}}, K$ ( $B$ is a bilinear form, $A$ an intermediate ring)                                                                                     |
| 08     | $L_{\mathbf{v}}; P$ (projective specialization)                                                                                 | $\mathcal{L}_{\mathbf{v}}; \text{sp}(\mathbf{x})$                                                                                                                      |
| 08     | $[q : p_1 : \cdots : p_n]$ (denominator first)                                                                                  | $[p_1 : \cdots : p_n : q]$ (denominator last)                                                                                                                          |
| 08     | $\Delta = \text{Deg } q(\tau)$                                                                                                  | $\text{deg}_{\omega} q(t)$ ( $\Delta$ is a discriminant)                                                                                                               |
| 08     | $\mathcal{A} = \mathbf{Oz}_{\Pi}, S = \mathbf{Oz} \setminus \Pi, \rho, \mathcal{F} = \mathcal{A} \cap \mathcal{O}, \mathcal{O}$ | $\mathbf{Oz}_{\Pi}, \mathbf{Oz} \setminus \Pi, \text{res}, \mathbf{Oz}_{\Pi}^{\text{fin}}, \mathcal{O}_{\text{fin}}$                                                   |
| 08     | $A_{\tau}, I_{\tau}; A_0, A_m, I_0, I_m$                                                                                        | $\mathcal{P}_t, t\mathbb{R}[t]; \mathcal{P}[T], \mathcal{P}[U], I_T, I_U$                                                                                              |
| 08     | $C_0, M(x), \mathbf{c}$                                                                                                         | $C, \text{Ex}(x), \mathbf{c}$                                                                                                                                          |
| 08     | $\mathbf{Oz}_{\mathbb{C}}, \mathcal{B}_{\mathbb{C}}$                                                                            | $\mathbf{Oz}[i], \mathcal{B}_{\mathbb{C}}$                                                                                                                             |
| 08     | $\Pi, \mathfrak{D}, \text{ct}, \text{st}$                                                                                       | unchanged                                                                                                                                                              |
| 09     | $\mathcal{B}$                                                                                                                   | $\mathcal{B}_{\mathbb{R}}$                                                                                                                                             |
| 09     | $T = \omega^{\alpha}$                                                                                                           | $t$ (any $t \in \Pi \setminus \{0\}$ in the general statements)                                                                                                        |
| 09     | $D(x)$                                                                                                                          | $\mathfrak{D}(x)$ ( $D$ is the Pell parameter)                                                                                                                         |
| 09     | $\mathcal{A}_{\mathbb{Q}} = S^{-1}\mathbf{Oz}$                                                                                  | $\mathbb{Q} \oplus \Pi$                                                                                                                                                |
| 09     | $\sigma$                                                                                                                        | $\text{sp}$ ( $\sigma$ is an automorphism)                                                                                                                             |
| 09     | $\Pi, \mathcal{V}, \mathcal{U}, \text{deg}_{\omega}, \text{ct}, \text{st}, \mathcal{O}_{\text{fin}}$                            | unchanged                                                                                                                                                              |
| 09     | $\Gamma$ (congruence kernel)                                                                                                    | $\mathbf{K}$ ( $\Gamma$ is an exponent group)                                                                                                                          |
| 09     | $G_{x,b}, N_x$                                                                                                                  | $\mathbf{F}_{x,b}, \mathbf{N}_x$ ( $G$ is the Gram matrix, $U_t$ the quadratic transvection)                                                                           |
| 09     | $\Lambda(p, q)$                                                                                                                 | the set (15.3) ( $\Lambda$ is the intersective polynomial)                                                                                                             |

| Source | Source symbol                                                                                       | This report                                                                                                                                                                                                                |
|--------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09     | $E(x), O_{\mathbb{Z}}, M$ (monomials)                                                               | $\text{Ex}(x), \mathcal{P}_{\omega}, \mathcal{M}$                                                                                                                                                                          |
| 09     | “finite” point; “divisor-primitive”; “Bézout fraction”                                              | affine point; primitive; unimodular presentation                                                                                                                                                                           |
| 09     | fibre                                                                                               | fiber                                                                                                                                                                                                                      |
| C10    | $k, H; K_{k,H}, B_{k,H}, O_{k,H}, \mathfrak{m}_{k,H}$                                               | $\mathbb{k}, \Gamma; \mathbb{H}_{\mathbb{k}}(\Gamma), \mathcal{A}_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \mathfrak{m}_{\mathbb{k}}(\Gamma)$                                                               |
| C10    | $D_{\lambda}$ , script $D$ (family); $\backslash \text{dd}$ ( $\deg_{\omega}$ )                     | $\partial_{\lambda}, \mathcal{D}; \deg_{\omega}$                                                                                                                                                                           |
| C10    | $f$ (squarefree polynomial), $U, V; U_0, V_0$                                                       | $P, U, W$ ; the polynomials of (16.5)                                                                                                                                                                                      |
| C10    | $\Phi_p, c_p, q$ (inverse)                                                                          | $s \mapsto \phi(c_{\mathbf{n}} + s), c_{\mathbf{n}}, q$                                                                                                                                                                    |
| C10    | $\eta_x(D); X(\mathcal{B}_k)$                                                                       | $\eta(\partial)$ ; the class convention of section 16.1                                                                                                                                                                    |
| C11    | $k; B_{\Gamma}(k), P_{\Gamma}(k), K_{\Gamma}(k), \mathcal{O}, \mathfrak{m}$ ( $t$ -convention)      | $\mathbb{k}; \mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma), \mathbb{H}_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \mathfrak{m}_{\mathbb{k}}(\Gamma)$ (exponents negated)                         |
| C11    | $D \subseteq k, R_{\Gamma}(D, k); A$ (reduced algebra); $B_{\Gamma}(A)$                             | $\mathfrak{o}, \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma); S; \mathcal{A}_S(\Gamma)$                                                                                                                                   |
| C11    | $D_{\lambda}$ , script $D$ (family), $\backslash \text{dd}$ ( $d$ ); $A_0, C_0$                     | $\partial_{\lambda}, \mathcal{D}, d$ ; the polynomials of (16.5)                                                                                                                                                           |
| C11    | $U$ (curve), $a_P, \mathcal{U}; h$ (fiber parameter)                                                | $U$ or $C, c_{\mathbf{n}}, \mathcal{U}; s$                                                                                                                                                                                 |
| C12    | $k; B, I, V, \mathfrak{m}, K$ ( $t$ -convention)                                                    | $\mathbb{k}; \mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \mathfrak{m}_{\mathbb{k}}(\Gamma), \mathbb{H}_{\mathbb{k}}(\Gamma)$                                             |
| C12    | $\deg_H; D_{\lambda}; U, W; D_{\infty}, r$                                                          | $\deg_{\omega}; \partial_{\lambda}; U, W; \Sigma, r$                                                                                                                                                                       |
| C12    | $J(\mathbf{a}); \mathbf{Oz}_{\Gamma} = \mathbb{Z} \oplus I_{\mathbb{R}, \Gamma}; \Pi_{\mathbb{R}}$  | $\langle \mathbf{a} \rangle; \mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \Gamma); \Pi$                                                                                                                                            |
| C13    | $\mathcal{B}_k, \mathcal{B}_{k, \Gamma}, I, O = \mathcal{O}_{k, \Gamma}, \mathfrak{m}, K$           | $\mathcal{B}_{\mathbb{R}}$ or $\mathcal{B}_{\mathbb{C}}, \mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \mathfrak{m}_{\mathbb{k}}(\Gamma), \mathbb{H}_{\mathbb{k}}(\Gamma)$ |
| C13    | $D_{\lambda}$ with $\lambda : \Gamma \rightarrow \mathbb{Q}; \deg_{\omega}; F$ (squarefree), $U, V$ | $\partial_{\lambda}$ with $\lambda : \Gamma_{\mathbb{Q}} \rightarrow \mathbb{k}; \deg_{\omega}; P, U, W$                                                                                                                   |
| C13    | $R, V, \mathfrak{n}$ (abstract rings)                                                               | $R, V, \mathfrak{m}_V$                                                                                                                                                                                                     |
| C13    | $L = \mathcal{O}_{\mathbb{P}(\Omega)}(1); P_R, P_K$                                                 | $\mathcal{L}; p, p_L$                                                                                                                                                                                                      |
| C14    | $k, K; \mathcal{B}_{\Gamma}(k), \mathcal{B}_{<0}, \mathcal{O}, \mathfrak{m}$ ( $t$ -convention)     | $\mathbb{k}, \mathbb{H}_{\mathbb{k}}(\Gamma); \mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \mathfrak{m}_{\mathbb{k}}(\Gamma)$                                             |
| C14    | $R_{\Gamma}(D; k)$                                                                                  | $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ ( $D$ is the Pell parameter)                                                                                                                                              |
| C14    | $D_{\lambda}, \mathcal{D}; A, C$ (Bézout)                                                           | $\partial_{\lambda}, \mathcal{D}; U, W$                                                                                                                                                                                    |
| C14    | $B, O, \mathfrak{n}$ (abstract data); $\langle \eta, D \rangle_p$                                   | $R, V, \mathfrak{m}_V$ ; the contraction $\eta(\partial)$ at $p$                                                                                                                                                           |
| C14    | $M$ (arcs); $D, L$ (congruence criterion)                                                           | $N; N_1, N_2$                                                                                                                                                                                                              |
| C14    | $S$ (boundary); script $C$ (model)                                                                  | $\Sigma; \mathcal{C}$                                                                                                                                                                                                      |
| C10–   | $\backslash \mathbb{A}(\mathbb{A}), \backslash \text{OO}(\mathcal{O})$                              | $\mathbb{A}, \mathcal{O}_{\mathbb{k}}(\Gamma)$ (this report’s $\backslash \text{OO}$ is the orthogonal group)                                                                                                              |
| C14    |                                                                                                     |                                                                                                                                                                                                                            |
| C15    | $k; \mathcal{A}, \mathcal{I}, \mathcal{V}, \mathcal{H} = k((t^{\Gamma}))$ ( $t$ -convention)        | $\mathbb{k}; \mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \mathbb{H}_{\mathbb{k}}(\Gamma)$ (exponents negated)                                                            |
| C15    | $D \subseteq k, R_D = D \oplus \mathcal{I}; \mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}$     | $\mathfrak{o}, \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma); \mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}$                                                                                                         |
| C15    | $D_{\ell}, \ell : \Gamma \rightarrow (k, +)$                                                        | $\partial_{-\ell}$ (the same family), $\lambda$                                                                                                                                                                            |
| C15    | $f$ (squarefree), $U, V; w_D; Q = 4a^3 + 27b^2$                                                     | $P, U, W; h/m; \Delta_0$                                                                                                                                                                                                   |
| C15    | $q$ (symmetric power); $q$ (valuation-ring centre); $p_{\mathcal{H}}$                               | $r; q; p_{\mathbb{H}}$                                                                                                                                                                                                     |
| C15    | $\backslash \mathbb{A}, \backslash V, \backslash \text{OO}(\mathcal{O}, \text{structure sheaf})$    | $\mathcal{A}_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \mathcal{O}_{\overline{X}}$ (here $\backslash V$ is a zero set)                                                                                       |
| C15    | “tangent-separating”                                                                                | separating tangent directions (definition 18.17)                                                                                                                                                                           |

| Source | Source symbol                                                                                                                                        | This report                                                                                                                                                                               |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C16    | $k; B_{k,\Gamma}, K = k((t^\Gamma)), O, \mathfrak{m}$<br>( $t$ -convention); $v$                                                                     | $\mathbb{k}; \mathcal{A}_{\mathbb{k}}(\Gamma), \mathbb{H}_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \mathfrak{m}_{\mathbb{k}}(\Gamma)$ (exponents negated); $v$ (unchanged) |
| C16    | $\mathcal{J}_{\mathbb{R}}, \mathcal{J}_{\mathbb{C}}; B_{\mathbb{R}}^{\text{No}}, B_{\mathbb{C}}^{\text{No}}$                                         | $\Pi, \Pi_{\mathbb{C}}; \mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}$                                                                                                               |
| C16    | $D_{\lambda} (t^\gamma \mapsto \lambda(\gamma)t^\gamma); D$ (a derivation); $D \subseteq k$ (coefficient ring)                                       | $\partial_{-\lambda}$ (the same family); $\partial; \mathfrak{o}$                                                                                                                         |
| C16    | $A, \tilde{A}; X$ (smooth projective model); $p$ (centre); $d$ (conductor element)                                                                   | $\mathbb{k}[C], \mathbb{k}[\tilde{C}]; \overline{C}; Q; \kappa$                                                                                                                           |
| C16    | $m$ (terms of $\alpha$ ); $s = \alpha(D); z, E$ (formal variable, Taylor map)                                                                        | $\ell; \alpha(\partial); w, \mathcal{T}$                                                                                                                                                  |
| C16    | $r$ (number of roots), $r_{\infty}; Q, S$ (cofactors); $d$ (support bound)                                                                           | $k, n_{\infty}; P_1, P_{\text{odd}}; n$                                                                                                                                                   |
| C16    | $V$ (Bézout); $P, P_0$ (points); $h, a$ (arithmetic families)                                                                                        | $W; \mathbf{x}, \mathbf{x}_0; s, \beta$                                                                                                                                                   |
| C16    | $m$ (superelliptic exponent),<br>$d = \sum e_i, r$ (vanishing order), $\delta, \gamma, \lambda, E$ (elliptic curve), $F, U, \nu$                     | unchanged                                                                                                                                                                                 |
| C17    | $k; \mathcal{A} = k((t^{\Gamma \leq 0})), \mathcal{I}, \mathcal{O}_K, K = k((t^\Gamma))$ ( $t$ -convention)                                          | $\mathbb{k}; \mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma), \mathcal{O}_{\mathbb{k}}(\Gamma), \mathbb{H}_{\mathbb{k}}(\Gamma)$ (exponents negated)                           |
| C17    | $\widehat{K} = \widehat{k}((t^{\Gamma_{\mathbb{Q}}}))$ , $\widehat{\mathcal{O}}; D \subseteq k, \mathcal{A}_D = D \oplus \mathcal{I}$                | $\mathbb{H}_{\mathbb{k}}(\Gamma_{\mathbb{Q}}), \mathcal{O}_{\mathbb{k}}(\Gamma_{\mathbb{Q}}); \mathfrak{o}, \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$                               |
| C17    | $D_{\chi}, \chi \in \text{Hom}(\Gamma_{\mathbb{Q}}, \mathbb{Q}); D$ (a derivation); $P_0 = \text{ct } P$                                             | $\partial_{-\chi}$ (the same family), written $\partial_{\lambda}; \partial; \overline{P}$                                                                                                |
| C17    | $B, C_B, N$ (rank), $H_y; S$ (splitting algebra)                                                                                                     | $B, C_B, r, \text{ch}_y; B_P$                                                                                                                                                             |
| C17    | $Q, b_j, u_i; S(X, Y); F_{n,s}; \alpha, \beta$ (quadratic velocity)                                                                                  | $\dot{P}, b_j, \dot{r}_i; W(X, Y); \mathcal{F}_{n,s}; b_1, b_2$                                                                                                                           |
| C17    | $E_j$ (spectral idempotents), $H, H[i], \mathcal{A}_{\mathbb{C}}, \mathcal{I}_{\mathbb{C}}$                                                          | $P_j, \mathbb{H}_{\mathbb{R}}(\Gamma_{\mathbb{Q}}), \mathbb{H}_{\mathbb{C}}(\Gamma_{\mathbb{Q}}), \mathcal{A}_{\mathbb{C}}(\Gamma), \Pi_{\mathbb{C}}(\Gamma)$                             |
| C17    | $c$ (target shift); $r, G$ (resultant theorem); $r$ (two-sided root); $\lambda$ (unit proof); $c_1, \dots, c_r, k^r$ (independence, split constants) | $\beta; \ell, \mathcal{G}; \rho; \lambda_0; c_1, \dots, c_k, \mathbb{k}^j$                                                                                                                |
| C17    | $h, \Delta, m, w_{ij}, e_s, F, L_0, \Pi, \Pi_{\mathbb{C}}, \mathbf{Oz}[i]$                                                                           | unchanged                                                                                                                                                                                 |

## A.5 Corrections to the sources

1. Source 02 states Pell rigidity for  $D > 0$  (in its abstract, its introduction, its catalogue table and its Pell corollary); its proof uses only  $D \neq 0$ , as in sources 01 and 05. [Corollary 6.9](#) states  $D \neq 0$ .
2. Source 01's abstract says that nonzero fibers of “nondegenerate binary forms” are rigid. Its theorem requires, and proves, the weaker hypothesis of at least two distinct projective linear factors, which in degree at least three is weaker than squarefreeness ( $X^2Y$  qualifies). The abstract of this report states the theorem's hypothesis.
3. Source 02 cites Matiyasevich's translation as *Soviet Math. Dokl.* 11 (1970), 354–358, and

- source 05 as 354–357. The range 354–357 is the one given, with MathSciNet number MR 258744, in the bibliography of Bhatt and Poonen’s notes *Diophantine sets*, and in the title of the review of the paper in the *Journal of Symbolic Logic*; it is used here.
4. Source 05’s proof of the set-sized quotient theorem invokes global choice to pick a cardinal; the Hartogs ordinal suffices ([remark 3.7](#)).
  5. Source 01’s isotropic-vector lemma assumes  $q(a) \neq 0$  where  $a \neq 0$  suffices ([lemma 8.3](#)).
  6. Source 05’s verification script checks the separated-power identity only for  $a = 1$ ,  $b = -1$ ; the general identity of [remark 7.5](#) was checked separately (symbolically, for  $2 \leq m, n \leq 7$ ) when this report was assembled. The shipped script is unchanged.
  7. Source 01’s questions on  $Y^2 = X^3 + 1$  and on unimodular Fermat triples, and the Fermat case of source 02’s question on unimodular homogeneous solutions, are recorded as settled by source 05 ([section 21.1](#)); the remaining parts are re-scoped as open questions.
  8. Source 02’s attribution of the nonzero omnific Fermat failure to MathOverflow comments could not be verified; it is printed as source 02’s attribution, marked unverified ([example 14.3](#)).
  9. Source 06 presents its quintic (degree five, seven witnesses) as its principal real Diophantine formula. The guards of sources 01 and 05, which it did not see, have degree four with five and six witnesses; the quintic is printed as a remark with that comparison ([remark 11.10](#)), and source 06’s question on the least degree is recorded as answered in part ([section 21.2](#)).
  10. Source 06’s single real polynomial for the graph of  $\text{ct}$  has degree ten and eight witnesses; [corollary 12.9](#) gives degree four and six witnesses.
  11. Source 07’s universal definition of  $\Pi$  and its  $\exists^5 \forall^2$  and  $\forall^2 \exists^5$  definitions of the graph of  $\text{ct}$  are superseded as definitions by source 06’s existential ones; they are kept as the negation of the detector and as a second route ([corollary 12.7](#) and [remark 12.11](#)), and source 07’s closing question on positive-existential definitions is recorded as answered ([section 21.2](#)).
  12. Source 13 states Pell rigidity for positive nonsquare  $d$ ; its own fiber theorem covers every  $D \neq 0$ , which [corollary 6.9](#) states.
  13. Sources 06 and 07 credit the interseptive polynomial to Lê–Spencer (and 07 also to Berend–Bilu). Lê and Spencer attribute it to Borevich and Shafarevich [[22](#), p. 3], which is added as the classical source.
  14. Source 06 cites Matiyasevich’s translation with pages 354–358, as source 02 does; the range 354–357 is used here, as explained above.
  15. The formula  $\exists y (x^2 = 2y^2)$  fails over number-field coefficient rings containing  $\sqrt{2}$  in their fraction field; source 06’s remark on other radicands supplies the replacement used in [remark 12.15](#).
  16. Source 09 says “for finite  $x$ ” for points  $x \neq \infty$  of  $\mathbb{P}^1(\mathbf{No})$ ; “finite” means bounded here, so such points are called affine ([section 15.1](#)).
  17. Source 09’s local density proposition concludes “unless  $g = f$  exactly”; the clause is unnecessary, since  $g = f$  gives degree  $-\infty$  ([proposition 15.26](#)).
  18. Source 09’s introduction speaks of “a unique positive least denominator”; a least positive denominator is unique by definition, and the statement is [theorem 15.15\(i\)](#).

19. Source 08 writes  $A = c^{-1}(\mathbb{Z})$  for its abstract pullback and  $\mathcal{A}$  for the localization at  $\Pi$ , and orders projective coordinates with the denominator first, source 09 with the numerator first. One order is fixed here and both symbols are renamed ([table 1](#)).
20. Source 09’s lemma on the matrices  $F_{x,b}$  is called “lem:transvection” in its source, the name of a different lemma here ([lemma 8.2](#)); it is [lemma 15.34](#).
21. Source 08 does not name this report’s Pythagorean witness ([14.2](#)), which was in the report at its pin, although it disclaims novelty for the conic. The sibling report’s coefficient-lattice results and its localization  $\mathcal{M}^{-1}\mathbf{Oz} = \mathbf{No}$  were written after the pin of sources 08 and 09. These credits are added (merge choices above).
22. Source C12’s logarithmic proof of the curve classification takes “the centre”  $Q$  of the valuation-ring point without showing that  $Q$  is a closed point. If it were the generic point, the function field would lie in the valuation ring and the residue map would embed it into  $\mathbb{k}$ , which is impossible for transcendence degree one. This step is added ([theorem 17.8](#)).
23. Source C13 assumes a divisible exponent group, with  $\lambda : \Gamma \rightarrow \mathbb{Q}$ , and, for curves and groups, the fields  $\mathbb{R}$  and  $\mathbb{C}$ . Its proofs need neither; its results are printed for every ordered exponent group and every field of characteristic zero, with  $\lambda : \Gamma_{\mathbb{Q}} \rightarrow \mathbb{k}$  as in sources C10 and C14. Sources C10 and C12 state their curve theorems over algebraically closed fields and source C11 over such fields or  $\mathbb{R}$ ; the report states them over every field of characteristic zero by source C14’s descent.
24. Sources C10, C11 and C14 treat the corner  $m = d = 2$  of the squarefree theorem by units. The support inclusion  $\partial(\mathcal{A}_{\mathbb{k}}(\Gamma)) \subseteq \Pi_{\mathbb{k}}(\Gamma)$  used by sources C12 and C13 removes it; it is already in `opa:par:prop:Euler` and `osq:prop:classder`, which none of the five cites. The strict proof is printed first and the credit is added.
25. Source C13 cites Stacks Project Tag 02AT for Kähler differentials; that tag is Section 111.50 of the chapter of exercises. The citation is replaced by Tag 00RM (Section 10.131), which sources C10 and C14 cite. Source C13’s Tag 01KF for the valuative criterion was not checked; Tag 0BX5 (Lemma 29.43.1), cited by sources C12 and C14, is cited instead. Sources C10 and C11 cite Tag 0BX4, the section containing that lemma, and Riemann–Roch is cited as Section 53.5 (Tag 0B5B) or as its Lemma 53.5.2 (Tag 0BS6); these are consistent. Milne’s *Abelian Varieties*, Chapter IV, 6.7 is called a proposition by sources C10 and C11 and a corollary by source C14; this was not checked.
26. The non-integral-closure examples of sources C10, C12 and C13 repeat, in fixed workspaces, the witness  $\sqrt{\omega^2 + 1}$  of [proposition 4.12](#) (sources 01, 02, 05), which was in this report at their pins; only the fixed-workspace forms are new, and the credit is added ([section 17.5](#)).
27. Source C13’s characteristic- $p$  example is correct as stated, a counterexample to squarefree rigidity; its curve is, however, an affine line in characteristic  $p$ , which is recorded after [proposition 18.13](#).
28. Source C15 says that its example  $t^{-1}(1 + t^2)^{1/2}$  for  $\Gamma = \mathbb{Z}$ , integral over its ring but not in it, shows that “normality really is unavailable, even in this simple workspace”. For  $\Gamma = \mathbb{Z}$  the ring  $\mathcal{A}_{\mathbb{k}}(\Gamma) = \mathbb{k}[\omega]$  is a polynomial ring and normal; the element shows only failure of integral closedness in the larger field  $\mathbb{H}_{\mathbb{k}}(\Gamma)$ . It is outside  $\text{Frac } \mathcal{A}_{\mathbb{k}}(\Gamma)$  and cannot obstruct the lift of a nonconstant curve point over this normal ring. The corrected distinction is printed ([section 17.5](#)); it is the example of source C14, which disclaims non-normality. Source C15’s proof audit words the point correctly.

29. Source C15’s abstract says that “consequently” every semiabelian variety is rigid, after the logarithmic annihilation theorem. Its proof goes through the abelian quotient and the torus kernel, as in [theorem 18.4](#), not through the logarithmic criterion; no claim is made here that semiabelian rigidity follows from [corollary 18.18](#).
30. Source C15’s abstract interface asks for a valuation ring  $V \subseteq K$  containing  $k$ ; the proof needs  $V$  to be a valuation ring of  $K$ , so that the valuative criterion applies ([remark 18.19](#)).
31. Source C16 introduces the element  $t^{-1}(1 + t^2)^{1/2}$  of the Hahn field “over a sufficiently divisible group”. The element lies in  $\mathbb{k}((t^{\mathbb{Z}}))$ , and for every  $\Gamma \neq 0$  and  $0 < \gamma \in \Gamma$  the element  $\omega^\gamma(1 + \omega^{-2\gamma})^{1/2}$  lies in  $\mathbb{H}_k(\Gamma)$  and not in  $\mathcal{A}_k(\Gamma)$ ; no divisibility is needed, and the statement is printed in that form ([section 17.6.9](#)).

No mathematical error was found in any main proof of the seven sources, or in the printed results of sources 13 and 14. In sources C10–C14 the only gap found is the closed-centre step of source C12, supplied above; there is no false theorem and no contradiction among the five, with the collection, or with the other sources of this report. In source C15 no gap was found in the proofs; its remark on normality and the logical route stated in its abstract are corrected above, and it contradicts no other source. In source C16 no gap was found in the proofs; its one remark with an unnecessary hypothesis is noted above, its one-place proposition is consistent with the positive-characteristic examples of [section 18.3](#), and it contradicts no other source. In source C17 no gap was found in the proofs, and it contradicts no other source. The later proof review of [sections 5 to 7](#) (recorded in `RECONCILIATION.md`) corrects the coefficient-derivation hypothesis in the merge’s own [remark 7.5](#): the derivation must kill  $a$  and  $b$ , and its use at level  $c$  needs  $\partial c = 0$ ; the Euler derivations used in the proof satisfy both. The complex extensions of one-variable and separated-power rigidity ([proposition 2.8](#) and [theorem 7.4](#)) are explained after their proofs. A further review of [sections 8 to 10](#) expands the polarization argument and the transfer in [theorem 9.2](#), which now covers quantified ordered-ring formulas explicitly; it adds that the quadratic orbits preserve the coordinate ideal, and it makes the nonnegative induction domain and the order-free semiring formula of [remark 10.11](#) explicit. It did not cover [remark 10.5](#).

## A.6 Stale repository statements

Sources 01, 02 and 05 inspected the repository through the GitHub connector, without a checkout or build, at the commit `2cb9c0200af749fbbd27796c97d017e7f16fbf33` named above; sources 06 and 07 inspected it at `71e9606` in the same way, and source 06 also read the opening of the surcomplex automorphism report. Sources 08 and 09 inspected it at `9a385d3`, also through the connector.

1. Source 01 says that the repository’s trigonometry material already uses the omnific integer part. At the pin this was true but incomplete: the foundations report already defined  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  (`found:eq:omnific`). The collection now also contains this report and the sibling report on set-sized quotients.
2. All three sources correctly found no report devoted to  $\mathbf{Oz}$  and no Lean declaration about it at the pin. Both statements are now historical: this report exists, and [proposition 2.2](#) has a Lean implementation.
3. Sources 02 and 05 describe `docs/NORMAL_FORM_BRIDGE.md` and the workspace modules; these descriptions are still accurate at the placement commit.

4. The status of Conway’s refinement conjecture is recorded as of September 22, 2026 ([appendix D](#)); the transcendence-over-bounded-support report takes the same stance.
5. Sources 06 and 07 say that the explicit Diophantine definitions of  $\mathbb{Z}$  in **Oz** they give were not found in the repository. At their pin 71e9606 this was true; since the placement be06fc8 the collection contains the quartic guards of this report ([theorems 10.3](#) and [10.4](#)). Source 13’s statement that the exact fiber formulation was not found is stale in the same way ([theorem 6.2](#)). None of the three saw this report or its sibling.
6. Source 06’s descriptions of the surcomplex automorphism report (phase twists that move the real axis while fixing  $\mathbb{C}$ ) and of the catalogue entry on coefficient recovery by a monomial dilation, and source 07’s description of the formalization ledger as separating source claims from Lean declarations, are accurate at the placement commit cf350b1. The absence of omnific Lean code was true then; [section 20](#) records the implemented ring package.
7. Sources 08 and 09 inspected the repository through the GitHub connector at 9a385d3, without a checkout or build. Both cite the rational-direction criterion as “Theorem 11.5” of this report; it was 11.5 at their pin and is now [theorem 14.5](#), since sources 06 and 07 were inserted as [sections 11](#) to [13](#). Likewise source 09’s “§§4,11” for the absence of lowest terms are now [sections 4](#) and [14](#). Their other citations ([theorems 4.2](#), [4.6](#) and [4.10](#) and [corollary 4.3](#), and the floor of [section 2](#)) are accurate.
8. Source 09 describes this report as merged from manuscripts dated September 22, 2026. That was true at its pin, when the report had three sources; it now has fourteen, dated September 22 and 23. Its fraction results remain pending in Lean; the basic ring package is now implemented.
9. Source 08’s bibliography gives the sibling report the title *The Universal Set-Sized Quotient of the Omnific Integers*. That was its title at the pin; it is now *Set-Sized Quotients of the Omnific Integers*. Source 08 uses that report only for comparison, not as a proof dependency.
10. The sibling report states that manuscript 14’s Gaussian and principal-ideal form of the direction criterion is printed in this report with [theorem 14.5](#). Before this merge it was not; it is now [corollary 15.55](#).
11. Sources C10–C14 inspected the repository through the GitHub connector at 4cdeaec (C10, C13, C14), f6e031a (C11) and 89bec38 (C12), without a checkout or build. At all three pins this report had only sources 01, 02 and 05 and fourteen sections. All five cite [question 21.1](#) as “Question 14.1”; it is now [question 21.1](#), and source C14’s “Question 14.2” is [question 21.2](#). Source C13’s “Section 11” (projective tuples and finite-support specialization) is now [section 14](#), and the sources’ “Section 14” is now [section 21](#); their Sections 6 and 7, and Theorem 7.4 cited by source C12, keep their numbers. Their descriptions of the Euler derivations, the separated-power proof and its stated boundary for  $y^2 = x^3 + ax + b$ , the finite-support specialization (now [theorem 14.7](#)) and the Pythagorean family were accurate at the pins.
12. The blob hashes recorded by the sources are correct: the report file was 4c26f4e4bee62f76428d7c69adf4aabe5af5ec53 at 4cdeaec (C10, C13, C14) and 47af413cd5f5d186a8c1391514d69e9933034498 at 89bec38 (C12), the same blob as at f6e031a (C11). Source C14 describes the report as a three-source assembly and source C11 says that its guide separates the reviewed Sections 1–7 from later material; both were true at their pins. The report now has fourteen sources, and the reviews recorded in RECONCILIATION.md extend further, but do not cover [sections 16](#) to [18](#).

13. Source C12 describes the collection as having 51 reports, true at [89bec38](#); the catalogue has since grown. The sources' statements that no Lean declaration concerns omnific integers were true at their pins; the ring package is now implemented ([section 20](#)).
14. Source C15 inspected the repository through the GitHub connector at [934810a](#), without a build. At that pin this report had sources 01, 02, 05, 06 and 07, and its guide stated that  $y^2 = x^3 + ax + b$  with  $a \neq 0$  is outside the method; source C15 quotes this correctly and presents its elliptic theorem as settling it. Since [c6359e4](#) the collection has settled the equation independently ([corollary 16.9](#), sources C10–C14), and source C15's proof is printed there once. Its statements that the unimodular Fermat case is already in this report in a more specific form ([theorem 7.9](#)) and that the ledger separates source assertions, proof review and Lean coverage are accurate. Its absence-of-Lean statement is historical; [section 20](#) records the current coverage.
15. Source C16 inspected the repository through the GitHub connector at [bcac55a](#), without a checkout or build, and did not read the assembled article in full. It says that this report “and later companion audits” contain smooth curve rigidity and the smooth classification, citing the audits of sources C13 and C14. At its pin the classification was in those staged audits and their manuscripts, not in this article; since [ac54217](#) it is [theorem 17.6](#). Its statement that these audits leave arbitrary singular curves outside their scope is accurate, and the report said the same until this integration ([section 17.5](#) and [question 21.17](#)). It saw neither the written curve part nor source C15, so it does not cite [theorem 17.8](#) and [proposition 16.11](#) or [section 18.6](#); the comparisons are made in [section 17.6](#). Its statement that its results are not Lean-verified remains accurate.
16. Source C17 inspected the repository at [efc5446](#), through the GitHub interface and in part through excerpts and a truncated catalogue response, without a checkout or build; it read this report's guide, not its article. Its statement that the repository already develops the constant/purely infinite decomposition, coefficient reconstruction and the Euler and two-ring methods, and credits its curve and logarithmic results to integrated manuscripts, is accurate at the pin and now. It saw neither source C16's integration nor the questions of this report that it answers, and it does not cite [questions 21.6](#) and [21.22](#), [theorem 6.15](#), [theorem 19.1](#), [corollary 9.5](#) or the sibling report's canonical translation; these comparisons are made in [section 19.4](#). Its statement that the repository's Lean coverage does not cover its theorems remains accurate.

## B Non-claims by source

Every limitation, disclaimer and priority caveat stated by a source is kept. Most also appear at their point of use in the text; this appendix is the per-source record. The non-claims of sources C10–C15 about singular curves record what those sources claim; the singular-curve criterion is now supplied by source C16 ([section 17.6](#)), within its own non-claims below. Likewise their statements that no rigidity is asserted for coefficients varying at a surreal scale record what they claim; source C17 supplies such a theorem in dimension zero, within its own non-claims.

**Common to sources 01, 02 and 05.** No independent peer review; no Lean or other proof-assistant verification; the finite scripts check finite identities and examples, not the theorems; the repository was not checked out or built; the Conway refinement announcement of September 18, 2026 was not audited, is not used, and is neither assumed nor claimed settled; no result assumes refinement, GCD or UFD properties; no priority claim is made.

**Source 01 (24 items).**

1. Proving a statement is not a claim that nobody proved it before; no exhaustive historical-priority claim; the quadratic classification and the guard are named as its most distinctive constructions, found in the inspected sources in no such form.
2. Familiar mechanisms (coefficient retractions, unit arguments, polynomial parametrizations, nilpotent quadratic isometries) are not claimed as new; a focused search does not establish priority.
3. The script is a supplement, not a proof; it does not implement the surreal numbers.
4. Transfer does not include inequations or order conditions.
5. Transfer concerns existence, not cardinalities or complete solution sets.
6. The Hilbert's-tenth-problem statement concerns ordinary integer coefficients; omnific coefficients are a different question.
7. Thue levels: absence of additional omnific solutions only; no new proof of finiteness and no algorithm for ordinary solutions.
8. The  $\gcd > 1$  power argument does not classify coprime cases such as  $x^2 - y^3 = 1$  (now covered by source 05).
9. The torus remark does not say that every affine variety is constant.
10. The quadratic classification concerns represented nonzero levels; it does not decide representability or classify all infinite points.
11. The guard's five auxiliary variables are not claimed optimal.
12. The first-order definition of  $\Pi$  is not an existential definition.
13. Polynomial lifting does not license evaluating infinite power series. (Status (batch 32): entire series over a fixed workspace are treated in `docs/surcomplex/entire-functions-at-arbitrary-rank`; by `ent:as:thm:sampling` and `ent:as:thm:integer` those preserving the omnific part are polynomials.)
14. Fermat scaling solutions are not primitive and resolve no primitive Fermat problem.
15. The rational-direction theorem concerns real projective points only.
16. Initial equations are necessary, not sufficient.
17. The converse of the no-real-point-at-infinity theorem is false.
18. Common divisors, clearing,  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  and non-generation of  $\Pi$  are full-class and not asserted for fixed workspaces; clearing is representability, not primitivity.
19. The three-square theorem, MRDP and the normal-form foundations are imported, not reproved.
20. L'Innocente–Mantova's Theorem B is not used in any proof.
21. None of its arguments refutes the refinement property.
22. Proposed module names are proposals; no Lean files; not checked against a built repository.
23. A missing repository search match is not proof of absence.
24. Its questions are not all claimed open in the literature.

**Source 02 (22 items).**

1. The definition, integer-part property and pathological factorization belong to established theory; the retraction and many consequences should not be advertised as new.
2. The literature search does not establish priority for its principal formulations (exact decomposable fibers, uniform quadratic lifting); independent rediscovery is likely; absence of an identical formulation is no evidence of priority.
3. AI-assisted; no peer review; no complete Lean verification; no Lean build.
4. The checks do not prove transfinite-support results, the Diophantine claims, or the imported quantifier-elimination and undecidability theorems.
5. Transfer says nothing about how many points lie above an integer point.
6. The common-multiple theorem is an existence theorem, not an algorithm; no nonzero element is divisible by every nonzero omnific integer.
7.  $\Pi/\Pi^2 = 0$  does not mean the infinite part is small.
8. The L’Innocente–Mantova prime is not reproved.
9. No effective enumeration of surreals is assumed; arbitrary class coefficients have no implicit finite encoding.
10. Nonvanishing and order do not transfer; the retraction is not an elementary embedding; residual solvability need not lift for non-integer coefficients.
11. The failure of four squares does not contradict open induction.
12. Residual and initial tests are not a complete lifting criterion.
13. The homogeneous theorem does not decide affine omnific equations.
14. The Fermat example is not a refutation of Fermat’s Last Theorem and not a novelty claim.
15. Clearing and homogeneous solvability do not answer the primitive-representative question.
16. Polynomial maps are not assumed to exhaust all constructions.
17. Binary and norm rigidity give no effective bounds for ordinary points.
18. Not every omnific linear group is finite.
19. The quadratic classification concerns the appearance of infinite points, conditional on ordinary solvability; not every point lies on the orbit; degenerate forms and the zero level are excluded from its statement.
20. Do not infer  $\Pi_G^2 = \Pi_G$  for fixed workspaces; restricted coefficient fields need the vectors  $e, u$  or a scalar extension.
21. Its questions: no claim that other fibers are rigid or come from curves; no general root-membership algorithm (one would contradict H10).
22. Formalization stages are proposals; formalizing the retraction does not formalize Tarski or MRDP; the repository review was targeted and the normal form theorems are used as mathematical inputs, not as an audit of the implementation.

**Source 05 (26 items).**

1. No general solution of omnific Diophantine equations and no effective presentation of every surreal.

2. No new proof of the theorems of Lagrange, Matiyasevich, Shepherdson or Wiles.
3. No historical-priority claim for the derived results.
4. Not peer reviewed; not formally verified; finite checks are not substitutes for proofs and do not check arbitrary supports or the class-theoretic pigeonhole argument.
5. The normal-form bridge implementation is not independently certified; the Lean implementation was not built or audited.
6. The substantial factorization theory of L’Innocente–Mantova is a separate achievement, not replaced by elementary arguments.
7. Transfer does not make fibers singletons.
8. The computational statement does not cover omnific coefficients.
9. Projective representability does not give unimodular representatives.
10. The Euler derivations are local and not canonical, and are not the Berarducci–Mantova derivation.
11. Full-class results are not asserted for a preassigned workspace.
12. The set-sized quotient theorem does not apply to class targets such as  $\mathbf{Oz}$ , and it does not say that endomorphisms are the identity.
13. The size conclusion for  $\mathbf{Oz}/(f)$  concerns the class ring, not a set-sized quotient.
14. There is no Pell solution at index  $\omega$ .
15. No Thue finiteness proof is contained in the reduction.
16. The conic statement covers central conics, not every quadratic equation.
17. Separated-power rigidity gives no bound or complete list of integer solutions; the script certifies no completeness.
18.  $y^2 = x^3 + ax + b$  with  $a \neq 0$  is outside the method, a genuine boundary.
19. Only fixed ordinary exponents are treated, not variable omnific exponents.
20. The Fermat theorem says nothing about triples that merely lack a common nonunit divisor, and assumes no gcd equivalence.
21. The first-order definition of  $\text{ct}$  is not existential.
22. The four-square theorem is used only on ordinary nonnegative integers.
23. Transcendence of infinite omnific integers does not imply algebraic independence of tuples.
24. Finite-support specialization does not turn infinite-support solutions into arcs; finite-support search is not a complete decision procedure; matching constant terms is not a sufficient solvability criterion.
25. Its questions are not an exhaustive list of established open problems.
26. Its formalization route is a proposal; declarations were not invented; the size-sensitive layer must not be generalized to fixed Hahn groups.

**Common to sources 06 and 07.** Unrefereed research drafts with written proofs; no Lean files and no Lean-verified result; the repository was not built; the finite checks are exact but do not certify quantified statements over Hahn supports or the class interpretation; the literature and repository comparisons were focused, not exhaustive, and a missing search hit is not evidence of absence; no named classical factorization conjecture, and no named longstanding

conjecture, is claimed to be resolved; historical priority is not asserted. Their pin 71e9606 predates both omnific reports.

### Source 06 (25 items).

1. First-order definability of  $\mathbb{Z}$  in  $\mathbf{Oz}$  and the nonsaturation of  $\mathbf{Oz}$  are not new (*nombre*, 2018); the ideal equation belongs to the same obstruction and is not presented as an unrelated discovery.
2. Undecidability of integer-coefficient solvability follows from the retraction and is not a new consequence; the logical limitations of  $\mathbf{Oz}$  were discussed by Jeřábek in 2011.
3. The intersective polynomial and its local-root property are classical; Pell methods in definability have a long history [20, 21]; phase twists are an existing repository construction, reproved with attribution.
4. No optimality assertion for degree five, seven witnesses or the coefficient sizes of the quintic, nor for the Gaussian system.
5. The Gaussian predicate is a system; no single-polynomial Gaussian claim is made, since a sum of squares does not preserve zero sets over a ring containing  $i$ .
6. The certificate  $vs = \Lambda(t)$  is not a global definition of nonzeroness.
7. The multiplier and coefficient-field formulas are first order, not positive existential; the finite valuation ring is defined in the interpreted fraction field, not as a subset of  $\mathbf{Oz}$ .
8. The value group is recovered as an interpreted quotient, not a monomial cross-section; no definability of the  $\omega$ -map, the simplicity order, birthdays or the exponential is claimed.
9. Pointwise fixation of  $\mathbb{R}$  is proved for automorphisms only, not for nonsurjective embeddings.
10. The homomorphism theorem does not say that homomorphisms kill the ideal; preservation, annihilation and pointwise fixation are different assertions.
11.  $\Gamma \neq 0$  is necessary for coefficient reconstruction, and the fraction field of a fixed Hahn ring need not be the whole Hahn field ( $\Gamma = \mathbb{Z}$  gives  $\mathbb{k}(\omega)$ ).
12. Full-field reconstruction uses surreal bounds for sets of exponents and the real closedness of  $\mathbf{No}$ ; no common denominator for a proper class is asserted; pointwise fixation uses the density of  $\mathbb{Q}$  in  $\mathbb{R}$ , and a general ordered coefficient field can have nontrivial automorphisms.
13. The c.e. classification concerns standard-supported sets and integer coefficients; arbitrary surreal parameters are not finite inputs to an algorithm.
14. The quantifier-free obstruction does not prove a lower bound on degree.
15. A formula defining the standard integers in this ring need not select a standard copy in an elementary extension.
16. No convergence, analytic continuation, canonical complex logarithm or infinite-height trigonometry is used; the only exponential is the ordinary complex exponential of a real multiple of  $i$ .
17. Class statements are formula by formula: no uniform truth predicate, no set of all surreals, no class of all class automorphisms.
18. Recovering  $\mathbb{C}$  in the Gaussian case does not recover the real form or conjugation; naming an involution gives the real structure explicitly rather than reconstructing it.
19. The coefficient-recovery formulas give no Diophantine definition of  $\mathbb{Z}$  in  $\mathbb{Q}$  and no solution of Hilbert's tenth problem over  $\mathbb{Q}$ .

20. No named classical factorization conjecture is addressed.
21. No Lean files, no kernel-checked theorem, no repository rebuild; the checks guard against mistakes and do not certify quantified statements.
22. Qualified novelty only; the comparison was focused, and a missing search hit was not used as evidence of mathematical nonexistence.
23. Its formalization order is a proposal, not a report of completed work.
24. Its questions are not claimed to be recognized open problems.
25. Lagrange's four-square theorem is classical; its appendix proof is not a new theorem (and is not reprinted here).

**Source 07 (19 items).**

1. The normal-form machinery, the Pell method, the intersective polynomial, the four-square theorem and the undecidability background are established and credited; the literature check is targeted, not exhaustive.
2. No prime-element or factorization theorem is rediscovered, and the historical conjecture list is not treated as a list of current open problems; the ideal characterization proves no factorization, unique factorization or normality property.
3. No single-equation claim in the Gaussian case; the real quartic is not asserted over  $\mathbf{Oz}[i]$ , where it accepts every  $x$ .
4. Two universal variables, five existential witnesses and degree four are upper bounds only.
5. The  $\exists^5\forall^2$  and  $\forall^2\exists^5$  bounds are not claimed minimal, and no claim is made that the graph of  $ct$  has no existential definition.
6. The detector needs a root of  $\Lambda$  in the coefficient field; it fails for  $\mathbb{Z} + \omega\mathbb{Q}[\omega]$  and is not asserted for all intermediate rings.
7. The number-field theorem is not uniform in  $K$  and does not apply as stated to infinite algebraic extensions.
8. Its interpretation of  $\mathbb{Z}[i]$  is not identified with a Diophantine definition of  $\mathbb{Z}$  inside  $\mathbb{Z}[i]$ ; no such result is needed.
9. Undecidability and non-axiomatizability are consequences of established results (Jeřábek 2011, DPRM), not new undecidability theorems about  $\mathbb{Q}$  or number fields.
10. It does not show that  $\Pi$  lacks a positive-existential definition (and source 06 shows that it has one).
11. Its automorphism corollary concerns ring automorphisms of the coefficient-restricted ring, not field automorphisms of  $\mathbf{No}$  or  $\mathbf{No}[i]$ , and does not say that an arbitrary field automorphism preserves  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$ .
12. Norm rigidity needs a nonzero right side and separability; it does not say that every variety has only constant points; coefficients outside  $A$  are read in the ambient ring.
13.  $I = \bigcap nA$  must not be transferred to coefficient rings containing  $\mathbb{Q}$ .
14. The results are not claimed for the full surreal or surcomplex fields without the omnific restriction.
15. No efficient algorithm; witness existence, decision complexity and computability of representations are different questions, and the witness recipe assumes access to coefficient data.

16. Saturation and  $\text{Th}(A)$  are asserted for set-sized rings; class statements are formula by formula, without a global satisfaction predicate.
17. The checks do not verify arbitrary Hahn supports, the all-moduli proof, prime existence for number fields, the model-theoretic claims or the class interpretation; the script must not be run with `python -O`.
18. Its formalization route is a proposal; no Lean module is supplied.
19. Its questions are not claimed to be previously catalogued open problems.

**Source 13, for the results printed here (7 items).** Its other non-claims are recorded in the sibling report.

1. The fiber theorem classifies the infinite points above ordinary points only; it gives no Thue or Baker finiteness and no enumeration or bound of the ordinary solutions, including those of the cubic norm equation.
2. The zero level is excluded, and the exclusion is sharp.
3. The fiber is a disjoint union of constant-coefficient fibers, with no assertion that the total solution class is a set.
4. The norm is evaluation of a finite polynomial, not a new norm on a class; nonordinary surreal solutions with negative exponents exist, so the theorem is about the nonnegative-support ring.
5. Equational transfer and the Hilbert’s-tenth-problem statement transfer the classical theorem and do not reprove it; inputs are finite integer polynomials, no encoding of surreal parameters is postulated, and no complexity claim is made for Gaussian or number-field problems.
6. The exact-fiber formulation is a proposed contribution, not a certified priority claim; the search was bounded.
7. Unrefereed, not checked in Lean; the finite script does not verify arbitrary Hahn sums; no claim is made about the status of Conway’s refinement conjecture.

**Common to sources 08 and 09.** AI-assisted research drafts with written proofs; not refereed; no Lean files and no Lean-verified result; the repository was inspected through the GitHub connector at 9a385d3 and not checked out or built; the SymPy checks are exact but finite and verify identities and examples, not the theorems; historical priority is not asserted, and each names its candidate contributions without certifying them.

**Source 08 (25 items).**

1.  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  is classical (Conway; L’Innocente–Mantova, Proposition 2.4.5) and not claimed as new.
2. Hamkins’s observations on the failure of lowest terms and on irrational reals as omnific fractions are not new.
3. The repository’s constant-term arithmetic, set-uniform clearing and common divisors, the pair  $(t, rt)$  without a gcd and the constant-direction criterion are credited; the needed arguments are reproved, not imported.
4. The pullback construction and the use of coefficient retractions are classical; L’Innocente–Mantova’s Lemma 9.2.1 is a methodological precedent.

5. Integrality, nonflatness and non-finiteness of such extensions are classical phenomena; the equational flatness criterion (Stacks 00HK) is standard.
6. The novelty assessment is qualified: the search was targeted, not exhaustive; no priority is claimed; a result may be a specialization of known algebra.
7. No named longstanding conjecture is claimed to be solved.
8. The parameter  $t$  must be nonzero and purely infinite; merely infinite elements with a nonintegral constant are excluded.
9. The polynomial vector needs gcd one in one variable; a multivariate gcd of one does not give a Bézout identity, and the several-parameter case is open.
10. Rational residues require a common real normalization to a primitive integer vector before the principal generator is stated.
11. Primitivity and unimodularity are shown to coincide only for the classified representations, not over all of  $\mathbf{Oz}$ .
12. Denominator ideals contain 0; representations need a nonzero multiplier.
13. The scale-defect dimension is a real dimension, not a module length.
14. Tensor products and Tor are taken only over set-sized rings.
15.  $\text{res}$  and  $\text{st}$  have different domains, and neither is a retraction of  $\mathbf{No}$  onto a smaller field.
16. There is no homomorphism  $\mathbf{No} \rightarrow \mathbb{R}$  that sets  $\omega = 0$ ;  $f(0)$  is a substitution in a rational function.
17. Monic monomial denominators are more restrictive than arbitrary ones ( $1/(\omega - 1)$ ).
18. The class of surreals with primitive fractions is not a subring; this concerns primitive representations, not the existence of fractions.
19. The classification covers tuples in  $\mathbb{R}(t)$  and  $\mathbb{C}(t)$  only; it does not classify all  $\mathfrak{D}(x)$ .
20. There is no general flatness or normalization theorem for Hahn integer parts.
21. The certificate procedure is effective only over exact coefficient fields; arbitrary reals are not assumed computable.
22. The conic example illustrates the classification; it is not a new construction of Pythagorean triples.
23. The checks do not verify the normal-form theorem, arbitrary supports, class-size claims or universal quantifiers, and they are not a proof assistant.
24. The repository audit read catalogues and guides through a connector whose responses were sometimes truncated; theorem numbers are cited from the catalogue, and an empty code search was not taken as proof of absence.
25. Its questions are not assertions that the problems are unstudied.

#### Source 09 (20 items).

1.  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  is Conway's theorem and not new.
2. The repository's constant-term arithmetic and floor, small common divisors,  $\Pi^2 = \Pi$  and non-generation, the fraction field, the pair without a gcd and the rational-direction theorem are not relabelled as new.

3. It does not claim that every surreal lies in  $\mathcal{V}$ , that divisor-coprime pairs are unimodular, or that  $\mathbf{Oz}$  is a GCD or Bézout domain.
4. The five-fold equivalence is proved on  $\mathcal{V}$  only.
5. It does not claim that a fixed Hahn field has the full-class clearing properties; in a workspace a formal cancellation does not give a unimodular pair.
6. No theorem needs Conway’s refinement conjecture or any announced proof of it.
7.  $\text{sp}$  is not a field homomorphism, does not extend  $\text{ct}$ , and is defined only on  $\mathcal{V}$ .
8. The dense fibers are not open and not separated; focusing fails for proper classes.
9. The one-scale test at  $T = 0$  is a formal specialization, not a surreal value and not a valuation residue.
10. The topology at infinity on  $\mathbb{R}(T)$  is not the surreal order topology.
11. The certificate procedure needs exact coefficients and is not uniform for reals given by oracles; the checks are not a surreal computer-algebra system.
12. It gives no complex lowest-term classification (source 08 does).
13. “Reduced” in L’Innocente–Mantova is a support notion and is not used for lowest terms.
14. The pullback and matrix-completion methods are ordinary algebra; Klawia is cited for background only; no invention of pullbacks, projective coordinates or unipotent matrices is claimed.
15. The candidate contributions are its multiplier and dichotomy theorems, the one-scale classification, and the orbit and focusing theorems; no priority is established.
16. The literature search was keyword-based; a failed match is not evidence of absence.
17. The checks do not prove well-ordering of supports, proper-class separation, non-generation of ideals, the theorems or novelty.
18. Its formalization guide is a proposed order, not code; writing **Set** in a statement does not discharge a universe condition.
19. The repository was not cloned or built, and only selected sources were reviewed, not every manuscript in its history.
20. Its questions are the article’s own, not certified open problems.

**Source 14, for the result printed here (2 items).** Its other non-claims are recorded in the sibling report.

1. The unimodular-direction criterion concerns coordinate representatives, not all points of a projective scheme over a ring.
2. The example  $[1 : \sqrt{2}]$  is not new; it predates manuscript 14.

**Common to sources C10–C14.** AI-assisted research drafts with written proofs; not refereed or independently reviewed; no Lean files and no Lean-verified result, and the repository’s build and axiom audit are not a verification of these results; the repository was inspected through the GitHub connector and not built; the finite checks verify identities and finite-support examples, not arbitrary infinite supports, properness, Riemann–Roch, global generation or the class transfer, and a count of passing checks is not a formal proof; historical priority is not established, the searches were targeted, and a missing search hit is not evidence of absence;

the question answered is a continuation question of this report, not a literature conjecture, and no named long-standing conjecture, in particular none of Conway's factorization or refinement questions, is claimed to be solved; nothing is claimed about singular curves in general, about general affine varieties of higher dimension, or about primitive, non-unimodular Fermat triples; no enumeration, finiteness statement, height bound or algorithm for the ordinary integer points is given; no rigidity over  $\mathbf{No}$ ,  $\mathbf{No}[i]$  or the valuation ring is claimed, and field points outside the rings exist; coefficients must be constants; the Euler derivations are local, not canonical and not the Berarducci–Mantova derivation; no spectrum of a proper-class ring is formed; the standard inputs (Hahn arithmetic, Kähler differentials, the valuative criterion, smooth completion, Riemann–Roch, invariant differentials) are imported and credited, and the constant term, units, Euler derivations, separated-power and unimodular Fermat rigidity and finite-support specialization of this report are credited, not claimed.

### Source C10 (14 items).

1. The finite-polynomial special case, that maps from an affine line to a curve of positive genus are constant, is standard and not claimed as new.
2. The weaker verticality conclusion  $f(x) = f(\text{ct } x)$  is not a claim of rigidity for positive-dimensional fibers.
3. No equivalence with uniruledness, rational connectedness or the existence of polynomial curves is asserted in higher dimension.
4. The clearing lemma may fail when the clearing exponent is confined to a fixed workspace.
5.  $\text{Frac}(\mathcal{A}_{\mathbb{k}}(\Gamma)) = \mathbb{H}_{\mathbb{k}}(\Gamma)$  is not assumed; normality and unique factorization are not assumed; ambient roots are never truncated into the ring.
6. The two-ring argument does not assert a ring map from the ring to the valuation ring; properness supplies a second point, not an inclusion of rings.
7. Failure of global generation identifies the possible surviving directions; it does not prove that they integrate to nonconstant Hahn points.
8. The arithmetic criterion is not an enumeration algorithm, decides no arbitrary Diophantine system and uses no finiteness theorem for integral points.
9. Replacing a singular curve by its normalization changes the point problem unless lifting is justified; that issue is not hidden in the smooth completion.
10. The argument is not a positive-characteristic theorem.
11. No rigidity is asserted for equations whose coefficients vary at a surreal scale.
12. Its suggestions of other coefficient systems and of iterative higher derivations are suggestions, not results.
13. Its questions are boundaries of its arguments, not open problems certified in the literature.
14. A native code search that returned no match is not evidence of absence.

### Source C11 (15 items).

1. No integer-part assertion is made for an arbitrary subring  $\mathfrak{o}$  or an arbitrary  $\Gamma$ .
2.  $\mathcal{A}_S(\Gamma)$  is in general not  $S \otimes_{\mathbb{k}} \mathcal{A}_{\mathbb{k}}(\Gamma)$ ; the reduced-base principle is not faithful flatness.

3. Van den Dries (1981) was compared through its bibliographic record and abstract only; no theorem-by-theorem comparison is made, and its discretely ordered ring theorem uses nothing from that paper.
4. The Artin–Schreier telescoping identity is not claimed as new.
5. No implication from the absence of rational curves to rigidity is proved.
6. The pointwise criterion is an obstruction, not a classification of varieties whose global forms fail to generate somewhere.
7. The curve-subalgebra corollary needs a smooth algebra, does not say that every finitely generated subalgebra is a polynomial ring, and does not cover higher transcendence degree.
8. The surcomplex conclusion does not say that the curves have no nonconstant points in  $\mathbf{No}[i]$ ; such points exist.
9. The projective theorem does not settle the primitive, non-unimodular Fermat question, and projective points over a ring are not arbitrary homogeneous tuples over its fraction field.
10. Integer coefficients cannot be replaced by omnific coefficients without qualification.
11. The canonical integer part need not contain a copy of every discretely ordered ring inside  $\mathbf{No}$ .
12. Conway’s and Gonshor’s texts were not newly read; for L’Innocente–Mantova only the abstract and version information were checked, and their factorization theorems are not used.
13. Enlarging  $\Gamma$  to its divisible hull is permitted but not needed, and no cofinal set in the class of surreal exponents is used.
14. Its build audit concerns its own artifact, not a mathematical proof.
15. Its questions are proposed continuation questions, not assertions of open status in the literature.

**Source C12 (13 items).**

1. The definition of  $\mathbf{Oz}$ , the Hahn construction, the retractions, the degree estimates and the Euler derivations are not claimed as new.
2. Only the sufficient direction of the coordinate test is used; no claim is made that every projective point over the ring has a global unimodular tuple.
3. The ideal of its elliptic example has an explicit common nonunit divisor and does not answer the primitive question.
4. Reducible affine curves are not classified.
5. Its formalization route is a proposal; no formalization was done.
6. The descent theorem is a reduction, not an algorithm for integer points; the squarefree theorem does not decide the remaining ordinary questions; no effective enumeration or height bound is given.
7. The Euler contractions are not claimed to separate all elements of  $\Omega_{\mathbb{H}_k(\Gamma)/\mathbb{k}}^1$ ; they separate functions.
8. No characteristic- $p$  classification is given.
9. In higher dimension one logarithmic form does not suffice, and no assertion is made that it does.

10. Group structure alone does not give rigidity; the result is not about every algebraic group ( $\mathbb{G}_a$ ,  $\mathrm{SL}_2$ ).
11. Uniform normality of the ring across exponent groups is not assumed.
12. Nonduplication is not certified against every supplementary manuscript, historical commit or paper; the 51-report collection of its pin was not audited exhaustively.
13. Its symbolic checks guard against transcription errors; they do not prove the arbitrary-support lemmas, the compactification argument or the coordinate-ideal theorem.

**Source C13 (13 items).**

1. No conclusion is drawn from nefness, bigness or general type; “ample” means the literal tautological-bundle condition, and conventions for semiample bundles vary.
2. The splitting of group points is not an algebraic splitting of the group, and not a formal-completion approximation.
3. Its non-normality example is not itself a failed lift of a point of a constant curve, and a normalization lift may not be declared to exist.
4. Embedding a singular curve in an abelian variety is a substantial extra hypothesis; the singular-subscheme consequence does not classify singular curves.
5. The abelian theorem is not  $A(\mathbb{H}_{\mathbb{k}}(\Gamma)) = A(\mathbb{k})$  and not a statement about torsion points.
6. The polynomial-witness principle is not asserted for singular curves or in higher dimension, and gives no uniform algorithm for integer points.
7. No resolution of Conway’s factorization or refinement questions, or of any named conjecture, is claimed.
8. Its 12,874 assertions are not independently verified theorems; the checks do not verify the valuative criterion, the sheaf-theoretic contraction, Riemann–Roch, group structure, the class assembly, correctness of every proof or originality.
9. Division by 2 takes place in  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ ; 2 is not claimed to be a unit of  $\mathbf{Oz}$ .
10. The Fermat case is a specialization of an earlier result, not a new Fermat theorem, and nothing is concluded about primitive, non-unimodular tuples.
11.  $\mathcal{C}(\mathbf{Oz}) = \mathcal{C}(\mathbb{Z})$  is neither an algorithm nor a finiteness statement, and no undecidability problem is solved.
12. No theorem is asserted to have a matching Lean declaration; its formalization order is a proposal.
13. Its example equations come with no claimed complete list of integer points.

**Source C14 (12 items).**

1. The integral-closure remark for  $\Gamma = \mathbb{Z}$  does not assert that  $\mathbb{k}[\omega]$  is not normal.
2. The congruence criterion is an algorithm only after a rational parametrization and its inverse are supplied, not for arbitrary curves.
3. Varieties with no global one-forms are not classified; global generation is sufficient, not claimed necessary.

4. The supplementary definability manuscripts were not reviewed line by line, and the repository was not audited exhaustively.
5. The singular families are polynomial families; not every point of a singular curve is claimed to have a parameter in the same ring.
6. The arithmetic dichotomy permits bad reduction but not a singular generic fiber, and gives no algorithm for integer points on curves of positive genus.
7. The factorization results of Berarducci and of L’Innocente–Mantova, and Conway’s refinement problem, are neither claimed nor solved.
8. Van den Dries (1981) was consulted through its record and abstract only; no exhaustive comparison with its consequences, or priority over them, is claimed.
9. For a Fermat curve, excluding particular ordinary points needs separate arithmetic input.
10. A nonzero exponent group is needed only for the converse of the curve classification.
11. No assertion is made that field-valued surreal points are constant.
12. A finite check count is not a formal proof of the paper.

**Source C15 (18 items).**

1. Its geometric package is a candidate-original contribution; priority is not certified, targeted searches found no published statement of the whole package, and negative search results do not prove novelty; not every repository companion was read.
2. Van den Dries (1981) was inspected through its bibliographic record and abstract only, and overlap beyond the abstract is not excluded; its statements concern the support-defined Hahn ring and its omnific subring, not a new criterion for all discretely ordered rings.
3. Not refereed; no Lean verification; the repository was not built in its investigation.
4. It settles no named long-standing conjecture, only a stated limitation of this report, and supersedes neither the theory of integer parts nor the classical algebraic-geometric ingredients.
5. No computation, enumeration, finiteness statement or height bound for the ordinary integral points; it does not replace Siegel’s theorem or methods for integral points, and its example equations come with no computation of their ordinary solutions.
6. No classification of singular curves.
7. It decides no arbitrary omnific Diophantine system and says nothing about Hilbert’s tenth problem.
8. Normal forms, Hahn multiplication, the constant-term retraction, the classical geometry of curves and algebraic groups and the Euler derivations are not claimed as new; the Fermat case is prior and not claimed.
9. No statement about general type: positivity of the canonical bundle alone need not supply tangent-separating symmetric differentials; no equivalence between rigidity and the absence of nonconstant maps from  $\mathbb{A}^1$  is established, and none may be inferred from the curve classification.
10. No spectrum of a proper-class ring is formed;  $\text{Frac}(\mathcal{A}_{\mathbb{k}}(\Gamma)) = \mathbb{H}_{\mathbb{k}}(\Gamma)$  is not assumed; no morphism between  $\text{Spec } \mathcal{A}_{\mathbb{k}}(\Gamma)$  and  $\text{Spec } \mathcal{O}_{\mathbb{k}}(\Gamma)$  is asserted.

11. The theorems concern points over the support-restricted rings, not over  $\mathbf{No}$  or  $\mathbf{No}[i]$ , where elliptic curves have nonconstant points; field solvability and integer-part solvability are different problems.
12. Affine integer points are not projective integer points, and clearing field denominators by a monomial does not create unimodularity.
13. Quasi-finiteness is essential ( $\mathbb{A}^1 \times G \rightarrow G$ ).
14.  $\mathcal{A}_{\mathbb{k}}(\Gamma)$  is not assumed normal, and lifting a point to a normalization is not an available unstated step.
15. It is not asserted that every nontrivial form of  $\mathbb{A}^1$  or  $\mathbb{P}^1$  over an arbitrary field has a nonconstant point; faithful flatness of the coefficient extension in the descent is neither asserted nor needed.
16. The finite checks (2,820 cases, most of them valuation-margin samples) do not verify the infinite-support or scheme-theoretic assertions.
17. A conditional Lean theorem using its geometric hypotheses would not by itself verify the instantiations; its formalization plan is a proposal.
18. Its proof audit is an author-side audit, not an independent referee report or a machine verification, and supplies neither priority nor independent proof checking.

**Source C16 (19 items).**

1. Characteristic zero is required for the criterion. In positive characteristic only the one-place obstruction is proved; the genus-zero conclusion is not asserted, and Seidenberg's Hasse–Schmidt operators are relevant but not claimed to supply a valuation-detecting operator.
2. Constant coefficients are required; nothing is claimed for coefficients that vary at a surreal scale.
3. Geometric integrality, dimension one and finite type are required; the theorem on affine schemes of dimension at most one is stated over an algebraically closed field only.
4. The criterion is an existence and classification statement: it does not claim that every nonconstant point lifts to the normalization, nor that every point is a polynomial family evaluated at a Hahn parameter.
5. The real singular arithmetic fibers are not classified; a real normalization preimage is shown sufficient, not necessary, and the isolated real node is left open.
6. No description is given of the fibers or the image of  $ct$  on exceptional curves, or of the support order types in them.
7. The derivative-order bound is not claimed optimal, and no algorithm is given for normalizations, conductors or regular differentials of an arbitrary presentation.
8. Ordinary integer-point existence is not decided, and no integer points are enumerated, for the pinched curve or otherwise.
9. Nothing is claimed about points over the Hahn field,  $\mathbf{No}$  or  $\mathbf{No}[i]$ ; it is not a claim that all field-valued surreal points are constant.
10. No higher-dimensional statement is made, and the curve argument does not subsume the existing higher-dimensional criteria.

11. Full surreal statements are made only after reduction to set-sized workspaces; no spectrum of a proper-class ring is formed.
12. Not refereed and not formalized in Lean; its formalization order is a proposal, and no stage of it is claimed done.
13. Historical priority is not asserted: the targeted comparison is not exhaustive, and “complete criterion” refers only to the stated class of curves and points, not to a named long-standing problem.
14. Van den Dries (1981) was compared through its bibliographic record and abstract only; no comparison with its proofs is claimed.
15. Seidenberg’s identity, the smooth classification, normalization, Riemann–Roch, Riemann–Hurwitz, Hahn normal forms, the retraction, the units, the Euler derivations and the non-normality of the ring are credited, not claimed.
16. The 113,940 finite assertions are regression checks, not counts of verified theorems; they do not verify infinite Hahn series, normalization of arbitrary curves, Riemann–Roch, Riemann–Hurwitz, the universal conclusions or novelty.
17. A clean compilation and the checksum manifest are not substitutes for mathematical review or formal verification.
18. The repository comparison was not exhaustive: the assembled article was not read in full, and no report, historical source, unmerged manuscript or Lean module was reviewed line by line.
19. Its questions are proposed continuations, not recognized open problems of the literature.

**Source C17 (22 items).**

1. It does not show that every finite étale  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ -algebra is constant: the injection  $\mathbb{C}_{\mathbb{B}} \rightarrow \mathbb{B}/\Pi_{\mathbb{k}}(\Gamma)\mathbb{B}$  is not shown surjective, and full descent is left open.
2. It does not assume or produce a global primitive element; a generic primitive element does not give a monogenic presentation over  $\mathcal{A}_{\mathbb{k}}(\Gamma)$ .
3. The unit theorem is not extended to  $\mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma)$ ; arithmetic finite étale algebras may have richer unit groups, and reduction modulo  $\Pi_{\mathbb{k}}(\Gamma)$  need not be a product of fields.
4. A nonzero constant discriminant is distinguished from a unit discriminant throughout; the translation theorem does not assert étaleness when the discriminant is not a unit.
5. No factorization, prime-element, unique factorization or refinement result, and no birthday result, is claimed; L’Innocente–Mantova’s factorization theorems are neither reproved nor claimed.
6. No large cardinal or other set-theoretic strength is used or claimed, and no proper-class finite-projective formalism is introduced.
7. No preferred derivation of the surreal field is constructed; the Euler derivations act on coefficients, are not  $d/dX$  and are not the Berarducci–Mantova derivation.
8. The exponent  $2s$  in the certificate is not claimed optimal.
9. The Galois statement depends on the chosen coefficient field; the rational Galois group is not claimed to survive extension of constants to  $\mathbb{R}$ .
10. The critical-point theorem is a classification up to independent source and target translations, not a conjugacy, and says nothing about iteration.

11. The main polynomial and matrix theorems are false without one-sided support, characteristic zero or a nonzero discriminant, and the matrix theorem without normality; finite free and reduced does not suffice for the unit theorem.
12. The characteristic-polynomial theorem alone controls eigenvalues, not entries or eigenspaces.
13. The polynomial-parameter case  $\mathbb{K}[Y]$  and its multivariable form are not claimed new and not presented as a deep theorem about the affine line; they answer the cited Stack Exchange question as special cases only.
14. The coefficient test is an algebraic criterion, not an algorithm for arbitrary presentations of infinite Hahn series.
15. No physical prediction about infinitesimal observables follows from the matrix theorem.
16. The 1,126 finite assertions check identities, normalizations and examples; they do not verify arbitrary supports, all degrees, the existence of the certificate in every degree, the formal étale argument, or priority.
17. A clean build and the build report are production checks, not mathematical review.
18. Not refereed and not formalized in Lean; the repository's Lean coverage does not cover these statements, and the formalization order is a proposal.
19. No novelty or priority is certified: the comparison was focused, not exhaustive, and a prior equivalent result may exist under other names.
20. The repository comparison read some material through excerpts or a truncated catalogue response; a search without useful matches is not evidence of absence.
21. The Hahn and omnific background, the Stacks foundations and the repository's support, Euler and two-ring antecedents are credited, not claimed; the elementary support facts are reproved independently.
22. Its questions are questions left open by its article, not a certified catalogue of previously unknown open problems.

## C Dependencies and verification

### C.1 Imported inputs

The foundation imported from surreal theory is the existence, uniqueness, order and multiplication of set-supported Conway normal forms, the set-cut property, and the real closedness of **No** [1, 2]. The other imported theorems are: MRDP (corollary 5.2); Tarski's quantifier elimination (theorem 9.2, theorem 14.2(b)); the three-square theorem (lemma 10.2); Lagrange's four-square theorem (theorem 10.4, corollary 5.2); Shepherdson's theorem (open induction); the Fermat theorem of Wiles and Taylor–Wiles (corollary 7.10 only, after the rigidity theorem has reduced to ordinary integers); the algebraic closedness of  $\mathbb{C}$  (to factor binary forms; the algebraic closedness of a class-sized surcomplex field is not used); and, as context only, the primes of [3] and [4, Lemma 11.1]. The material of sources 06 and 07 imports in addition MRDP (theorem 13.11), Lagrange's theorem (corollary 11.11 and remarks 10.5 and 11.10), Dirichlet's theorem on primes in progressions and quadratic reciprocity (lemma 12.13 only), the existence of algebraic closures and the discriminant criterion for étale algebras (theorem 6.15), and the real closedness of **No** (theorem 13.4); the local arithmetic of  $\Lambda$ , Hensel lifting and the Chinese remainder theorem are proved in the text. The material of sources 08 and 09 imports only the normal-form foundations, the set-cut property, ordinary polynomial and integer Bézout identities, the Euclidean property of  $\mathbb{Z}[i]$ , the equational criterion of flatness

[34, Tag 00HK] ([theorem 15.47](#)) and standard tensor exact sequences ([proposition 15.48](#)), and it uses [remark 15.39](#)’s citation of the omnific groups report. The material of sources C10–C14 imports the universal property of Kähler differentials [35, Tag 00RM]; the valuative criterion of properness for arbitrary valuation rings [38, Tag 0BX5]; smooth projective completion of curves, Riemann–Roch with duality, and the genus-zero classification [39, 41, 42]; invariant differentials on algebraic groups [43, 48]; the functorial description of projective space, the projective bundle of a locally free sheaf and globally generated powers of ample bundles [44, 45, 46] ([lemma 17.19](#) and [corollary 16.20](#)); the Barsotti–Chevalley theorem and the structure of commutative algebraic groups [49] ([theorem 18.6](#) only); and the Nullstellensatz, the square-zero description of tangent vectors, the smoothness of normal curves over a perfect field and the genus of a smooth plane cubic. It also uses [remark 18.7](#)’s citations of the omnific groups report. The material of source C15 uses in addition differentials with logarithmic poles along a simple normal-crossings divisor ([lemma 18.15](#)). The local construction and coordinate-independence are spelled out before that lemma; [47, Tag 0FMU] treats the one-smooth-divisor construction, and [60, Tag 0BI9] records the local normal-crossings condition. Its other inputs are among those just listed. The material of source C16 imports in addition the finiteness of normalization for curves [55, 39], Seidenberg’s identity [53] (reproved in [lemma 17.27](#)), the free generation of local differentials by a uniformizer and the Riemann–Hurwitz formula [57], and Kummer theory over a field containing the roots of unity ([section 17.6.6](#)). The material of source C17 imports in addition the algebraic closedness of Hahn fields with divisible exponent group and algebraically closed coefficients [3, Fact 2.1.1], the fundamental theorem on symmetric polynomials, the definition and base change of étale ring maps and the structure of finite étale algebras over a field [61, 52], formal étale lifting [62], the discriminant criterion for finite locally free algebras [63], and local characteristic polynomials of finite projective modules ([lemma 19.22](#)). All other deductions are proved here from these and finite algebra.

The principal dependency chains are:

normal forms  $\implies$  ct is a retraction  $\implies$  transfer and finite quotients;  
normal forms  $\implies$  constant units in  $\mathcal{B}_{\mathbb{C}}$   $\implies$  decomposable, binary and norm rigidity;  
local Euler derivations + binary rigidity  $\implies$  separated powers and unimodular Fermat;  
Pell rigidity + three- or four-square theorem  $\implies$  quartic definitions of  $\mathbb{Z}$ ;  
real quadratic linear algebra + support-ring algebra  $\implies$  quadratic-level dichotomy;  
normal forms + set-cut bounds  $\implies$  common divisors and clearing;  
Pell rigidity and divisibility + intersective  $\Lambda$   $\implies$  order-free definition of  $\mathbb{Z}$ ,  $\mathbb{Z}[i]$ ;  
 $\text{ct} + \sqrt{2} \notin \text{Frac}(\mathfrak{o}) \implies$  the Diophantine ideal  $\implies$  graph of ct and homomorphisms;  
augmentation + root of  $\Lambda$  in  $\mathbb{k} \implies$  detector and ideal test;  
Diophantine ideal + one negative exponent  $\implies$  multiplier identity;  
multiplier identity + real closedness of  $\mathbf{No}$   $\implies$  reconstruction of  $\mathbb{R}$  and  $\mathbf{No}$ ;  
retraction + Bézout identities  $\implies$  multiplier theorem  $\implies$  denominator dichotomy;  
matrix completion + set-cut bounds  $\implies$  congruence orbits, focusing, dense fibers;  
Euler degrees + image condition +  $\gcd(P, P') = 1 \implies$  squarefree rigidity;  
two opposite rings + properness + global forms  $\implies$  proper rigidity;  
positive genus + units + forms of  $\mathbb{A}^1 \implies$  smooth affine curves;  
smooth affine curves + ct + separatedness  $\implies$  arithmetic fibers and descent;  
image condition + properness + logarithmic poles  $\implies$  logarithmic tangent criterion;  
conductor + Seidenberg's identity + discrete boundary place  $\implies$  singular affine curves;  
Euler family + velocity certificate  $\implies$  discriminant translation.

The source-05 proofs that most deserve independent review are, by its own account, the separated-power theorem (divisibility in a quotient ring without unique factorization), the unimodular Fermat theorem (strict leading-term cancellation in the equal-degree Wronskian), and the set-sized quotient theorem (positivity of every exponent in the geometric quotient; printed in the sibling report). Sources C10 and C13 ask for specialist review of the scheme-valued two-ring argument at the level of its functorial maps, and of the exact arithmetic fibers; the elementary chain of the squarefree theorem is independent of algebraic geometry, and the Weierstrass result has both proofs. Source C15's proof audit asks first for review of the functorial contraction and of the passage from logarithmic annihilation to constancy, then of the set-sized formulation of the class applications. Source C16's audit lists fifteen critical checks; it singles out the commensurability of the conductor and the contracted differential at the boundary place as the step that would fail without discreteness ([remark 17.26](#)). Source C17's proof audit singles out the ideal intersection that kills the coefficients of the velocity polynomial, the characteristic polynomial of a logarithmic derivative on a finite projective module, and the passage from fixed eigenvalues to fixed eigenspaces by positivity ([propositions 19.11](#) and [19.24](#) and [lemma 19.36](#)).

## C.2 Finite checks

Fifteen scripts are shipped in `code/`, each with its recorded output in `data/`. The first three need Python 3.9 or later and SymPy (tested with SymPy 1.14.0); source 06's needs Python 3.10 or later and SymPy, and source 07's Python 3.10 or later and the standard library only; source 08's needs Python 3.10 or later and source 09's Python 3.9 or later, both with SymPy. Of the scripts of sources C10–C14, source C11's needs Python 3.10 or later and the standard library

only, source C12's Python 3.9 or later with SymPy, and the others Python 3.10 or later with SymPy; source C15's needs Python 3.9 or later and SymPy, and source C16's Python 3.10 or later and SymPy, and source C17's Python 3.10 or later and SymPy 1.14.0. They check finite identities and examples only.

- **01-diophantine-geometry-verification.py**: scalar identities (Pythagorean and hyperboloid identities and their Bézout certificates, the cubic norm by resultant, the Euclid ratio), 20 quadratic-isometry cases in dimensions 3 to 7, Pell indices 0 to 30, and 256 guard witnesses. It writes a JSON report, by default `verification_report.json` in the working directory, so it should be run with `-output` pointing elsewhere or on a copy; the shipped record is `01-diophantine-geometry-verification_report.json`.
- **02-diophantine-verify\_examples.py**: 13 exact identity checks (Pythagorean, cubic norm two ways,  $N^3 = 0$ , skewness, Lorentz preservation, determinant, group law and inverse, the hyperboloid family, binomial coefficients, the Euclid recurrence) and 49 finite-support products; output only to the terminal, recorded in `02-diophantine-verification.txt`.
- **05-diophantine-rigidity-verify\_examples.py**: nine groups (Pell recurrence, Pythagorean witness, the  $\rho$  identity, 64 separated-power identities, Fermat Wronskian identities for  $2 \leq p \leq 10$ , 16 finite telescoping identities, binomial root coefficients, quartic-definition certificates for  $|t| \leq 40$  by default, and five points on  $y^2 = x^3 + 1$ ); output to the terminal, recorded in `05-diophantine-rigidity-verification.txt`.
- **06-definability-reconstruction-verify.py**: the degrees of the quintic (five, seven witnesses) and of source 06's degree-ten polynomial for the graph of `ct`, the displayed real witnesses, six Gaussian examples of the three-equation system, the Pell unit order for moduli 1 to 500, roots of  $\Lambda$  modulo 1 to 2000, the four-square product identity, and finite-support ideal and twist identities. It writes a JSON report only to the path given with `-output` and otherwise prints it; the shipped record is `06-definability-reconstruction-verification.json`.
- **07-defining-arithmetic-verify.py**: 13 groups (the expansion of  $\Lambda$ , the universal witness identity of [theorem 12.1](#) symbolically, the witness degrees, the certificate for  $1 + \omega$ , roots of  $\Lambda$  modulo 1 to 4096, the 2-adic lifting step through  $2^{33}$ , Pell identities to index 300, the modular return for moduli 1 to 500, witnesses of the system for  $-6 \leq x \leq 6$  and six Gaussian samples, quartic witnesses for  $-100 \leq x \leq 100$ , the degree of  $F_4$  and its Gaussian collapse); output to the terminal only, recorded in `07-defining-arithmetic-verification.txt`. It uses assertions and must not be run with `python -O`.
- **08-fractions-verify.py**: 833 exact assertions in 13 categories, among them 40 random polynomial vectors (seed 23092026) with polynomial and integer Bézout identities, the lifted certificates ([15.4](#)), the rational or irrational classification over  $\mathbb{Q}(\sqrt{2})$ , sample scalar memberships, the examples, the substitution  $T = U^m$  and the defect layers, the nonflat relation, integrality equations, the residue identities and falling-factorial independence. It writes a JSON report, by default `verification.json` next to the script, so it should be run with `-output` pointing elsewhere or on a copy; the shipped record is `08-fractions-verification.json`.
- **09-fraction-fibres-verify\_examples.py**: 92 exact assertions (the matrices  $F_{x,b}$ : nilpotence, determinant, composition, inverse, action, fixed centre and image of  $\infty$ ; the covariance ([15.1](#)); Bézout certificates such as ([15.5](#)); nine one-scale examples with coprimality, specialization, direction type and lifted certificates; the local density construction; five finite focusing samples over  $\mathbb{Q}$  and the image of  $\infty$ ). It always writes `verification_report.json`

next to the script, so it should be run on a copy; the shipped record is `09-fraction-fibres-verification_report.json`.

- `10-curve-rigidity-verify.py`: 3,991 exact assertions in eight groups (the cubic certificate and its differential form, squarefree certificates for eleven polynomials and  $2 \leq m \leq 8$ , the degree obstruction on the grid  $2 \leq d, m \leq 30$ , the nodal family, binomial coefficients, the constant-term boundary, finite-support Euler laws in lexicographic ranks one to three, and lexicographic clearing). It writes `verification.json` next to itself unless `-output` is given; the shipped record is `10-curve-rigidity-verification.json`, and `10-curve-rigidity-BUILD_AUDIT.json` records source C10's build of its own article.
- `11-curve-abelian-rigidity-verify.py`: 13,049 exact assertions in 19 categories over  $\mathbb{Q}$  and  $\mathbb{F}_2$  (the universal cubic certificate, the divisibility identity, finite-support Euler laws, the characteristic-two tail and Frobenius squares, the discretely ordered ring's parity and leading terms, binomial coefficients, the dual-number example). It writes `verification.json` next to itself unless `-output` is given, and prints the report; the shipped record is `11-curve-abelian-rigidity-verification.json`.
- `12-curve-logarithmic-verification.py`: 49 recorded checks, among them 35 differential-ideal certificates for five squarefree polynomials and  $2 \leq m \leq 8$ , the cubic certificate and discriminant, the singular parametrization, the nonconstant-coefficient example, the integer threshold  $(m-1)d \geq m$ , the coefficients of the elliptic root in [example 17.23](#), and 400 finite-support Euler trials over lexicographic  $\mathbb{Z}^2$ . It writes `verification_report.json` next to itself unless `-output` is given; the shipped record is `12-curve-logarithmic-verification_report.json`.
- `13-differential-rigidity-verify_certificates.py`: 12,874 exact assertions in 17 checks (the universal cubic certificate, the general Weierstrass completion, squarefree and differential certificates, degree inequalities, finite Hahn Leibniz, support, degree and detection tests over lexicographic pairs, the two-ring inclusions, the single-derivation caution, sharpness and affine-line fiber identities). It prints its report and writes a file only to the path given with `-output`; the shipped record is `13-differential-rigidity-verification_report.txt`.
- `14-hahn-differential-rigidity-verify.py`: 3,440 exact assertions in five groups (symbolic certificates, finite series, degrees, arithmetic including the congruence example, and characteristic-two telescoping). It always writes `verification.json` next to itself, so it should be run on a copy; the shipped record is `14-hahn-differential-rigidity-verification.json`, and `14-hahn-differential-rigidity-BUILD_AUDIT.json` records source C14's build of its own article.
- `15-logarithmic-rigidity-verify.py`: 2,820 cases in seven groups (the universal cubic identity and its case  $\Delta_0 = 23$ , 49 differentiated superelliptic certificates for seven squarefree polynomials and  $2 \leq m \leq 8$ , 2,401 valuation-margin samples for  $2 \leq d, m \leq 50$ , the cusp and node families, 300 finite rank-two Hahn examples checking the Leibniz rule, supports, valuations and constant terms, 64 characteristic-two telescoping checks with the three partial derivatives, and a truncated binomial series for the valuation-ring point). It prints its report and writes a file only to the path given with `-output`; the shipped record is `15-logarithmic-rigidity-verification.json`, and `15-logarithmic-rigidity-build_report.json` records source C15's build of its own article.
- `16-singular-curves-verify.py`: 113,940 exact assertions in 28 groups: 264 rational and 8 symbolic checks of Seidenberg's identity; 1,493 checks of finite-support Hahn arithmetic over  $\mathbb{Q}$  (Leibniz rule, constant terms, valuations, iterated supports and leading-term detection);

200 lexicographic rank-two comparisons for the non-normality example; the superelliptic test for all 36,071 multiplicity patterns with  $2 \leq m \leq 20$  and one to three nonzero residues (genus integrality, the equivalence of [theorem 17.38](#), invariance under full  $m$ th powers) and 21 normalization identities; the elliptic Bézout and differential identities, the discriminant  $-23$  and the normalization of the pinched curve; the conductor memberships  $xy^j \in \mathbb{k}[C]$  for  $j < 15$ ; the node, cusp, acnode and real signed parametrizations; and 3,696 checks of the derivative-order bound for  $1 \leq \ell \leq 8$ ,  $0 \leq \delta \leq 20$ ,  $0 \leq r \leq 10$ . It prints its report and writes a file only to the path given with `-output`; the shipped record is `16-singular-curves-verification.json`, and `16-singular-curves-build_report.json` records source C16's build of its own article.

- `17-discriminant-rigidity-verify.py`: 1,126 exact assertions in 13 groups, with the fixed seed 20260923: 96 translation checks (extracted shift, constant extraction, inverse substitution, marked value) for random integer roots in degrees 1 to 6, 20 discriminants and 20 translated factorizations; 420 checks of (19.7) and of the clearing by  $\Delta^2$ , 510 of homogeneity, edge-permutation invariance and the velocity characteristic polynomial, and 6 of (19.4), in degrees 2 to 6; 2 quadratic-certificate checks; 8 resultant and 18 critical-configuration checks; 9 matrix checks; 8 characteristic- $p$  checks for  $p = 2, 3, 5$ ; 7 counterexample checks; and the discriminants 5 and 4 of [examples 19.14](#) and [19.34](#). It writes its report to the path given with `-output`, by default `verification.json` in the current directory, and prints it; the shipped record is `17-discriminant-rigidity-verification.json`, and `17-discriminant-rigidity-BUILD_REPORT.json` records source C17's build of its own article.

When this report was assembled the first three passed with Python 3.14.4 and SymPy 1.14.0, and source 05's script also passed with `-standard-radius 200`. The outputs of sources 01 and 02 differ from the shipped records only in the recorded Python version (3.13.5 there); the output of source 05 is identical to its record. The scripts of sources 06 and 07 also passed with Python 3.14.4 (and SymPy 1.14.0 for source 06), during placement on copies and again when this report was assembled, with source 06's report written outside this directory; that report differs from its shipped record only in the recorded Python version, and source 07's output is identical to its record. The degrees in [corollary 12.9](#) and the collapse of the quintic over  $\mathbf{Oz}[i]$  in [remark 11.10](#) were checked separately when this report was assembled; the shipped scripts are unchanged. The scripts of sources 08 and 09 passed with Python 3.14.4 and SymPy 1.14.0, on copies, during placement and again when [section 15](#) was written. Source 08's report (833 assertions) has the same content as its shipped record, which carries no timestamp; source 09's (92 assertions) differs from its record only in the generation time and the recorded Python version (3.13.5 there). The certificates of [remarks 15.12](#) and [15.39](#) and the table of [example 15.23](#) were checked separately with SymPy when the section was written; the shipped scripts are unchanged. The five scripts of sources C10–C14 passed with Python 3.14.4 and SymPy 1.14.0 when [sections 16](#) to [18](#) were written, each run on a copy in a scratch directory: 3,991, 13,049, 49, 12,874 and 3,440 assertions or checks. Their reports differ from the shipped records only in the recorded Python version (3.13.5 there) and, for source C10, the run time and timestamp; source C11's record contains neither and its report is identical. The identities printed in [propositions 16.8](#) and [16.11](#), [lemma 16.5](#), and [corollary 16.10](#), the expansions in [example 17.23](#), [remark 17.24](#), and [section 18.4](#), the Frobenius identity of [proposition 18.13](#), the arcs of the congruence example and the vanishing of  $g + r - 1$  only at  $(0, 1)$  were checked separately with SymPy when the part was written; the shipped scripts are unchanged. Source C15's script passed with Python 3.14.4 and SymPy 1.14.0 when [section 18.6](#) was written, run on a copy in a scratch directory: seven groups and 2,820 cases. The report it wrote with `-output` differs from the shipped record only in the recorded Python version (3.13.5 there),

and without `-output` it only prints. The universal cubic identity and the squarefreeness of the polynomials named for source C15 in [section 16.4](#) were checked separately with SymPy; the shipped script is unchanged. Source C16’s script passed with Python 3.14.4 and SymPy 1.14.0 when [section 17.6](#) was written, run on a copy in a scratch directory: 113,940 assertions in 28 groups, with the same group counts as the shipped record. The report it wrote with `-output` differs from that record only in the recorded Python version (3.13.5 there) and the run time; without `-output` it only prints. The parametrizations and identities printed in [sections 17.6.6](#) and [17.6.7](#) and [corollary 17.39](#) were checked separately with SymPy; the shipped script is unchanged. Source C17’s script passed with Python 3.14.4 and SymPy 1.14.0 when [section 19.4](#) was written, run on a copy in a scratch directory with `-output` pointing there: 1,126 assertions in 13 groups, with the same group counts as the shipped record, from which the report differs only in the recorded Python version (3.13.5 there).

## D Repository and literature audit

### D.1 Repository snapshots

Sources 01, 02 and 05 pin `2cb9c0200af749fbbd27796c97d017e7f16fbf33` (“Merge origin/main after the inverse-sine series formalization”). Source 01 read the tree, the root README, the documentation index, the trigonometry listing and its period-arithmetic source audit; source 02 read the README, the documentation index and the opening of the normal-form bridge note; source 05 read the documentation index and the bridge note. Each source says that this was a focused inspection through the GitHub connector, not a checkout, build, or review of every manuscript and declaration. Sources 06 and 07 pin `71e9606`. Source 06 read the root README, the documentation index, the opening of `docs/FORMALIZATION.md` and the first 235 source lines of the surcomplex automorphism report; source 07 read the root README, the documentation index and the opening of `docs/FORMALIZATION.md`. Both call this a focused comparison, not an exhaustive audit, and neither modified or built the repository. Sources 08 and 09 pin `9a385d3`. Source 08 read the root README, the tree, the documentation catalogue, the guides of this report and of the sibling report, and source 06’s source audit; source 09 read this report’s article and README, the root README and the collection guide. Both call this a selective inspection through the connector, without a checkout or build. Sources C10, C13 and C14 pin `4cdeaec` and read the report’s article at blob `4c26f4e`: source C10 its lines 1–130, 1350–1690 and 2910–3100, the README and the catalogues; source C13 lines 1–230, 1000–1200, 1460–1670, 2650–2860 and 2980–3125, the README and the opening of `docs/FORMALIZATION.md`; source C14 lines 1–180, 1000–1185, 1550–1785 and 2920–3080, the README, the catalogue and other reports’ READMEs. Source C11 pins `f6e031a` and compared against the report’s README and the catalogue. Source C12 pins `89bec38` and read Sections 6–7 of the article (blob `47af413`), its README, the root README and the catalogue. All five call this a targeted inspection through the connector, without a checkout or build. Source C15 pins `934810a` and read, at that pin, the first 180 lines of `docs/FORMALIZATION.md`, lines 130–280 of this report’s article (blob `cfc25a5`), lines 1–150 and 360–540 of its README and the first 210 lines of the sibling report’s article, through the connector and without a build. Source C16 pins `bcac55a` and read, through the connector, the repository metadata and recursive tree, the root and documentation READMEs, the directory and README of this report, and the audits of sources C13 and C14 in full; it did not read the assembled article in full and did not build the repository. Source C17 pins `efc5446` (tree `aba6982`) and read the root README, the documentation catalogue, the guides of this report, of the sibling report, of the omnific automorphism report and of the independent-copies report, and the opening of the holonomic-rigidity article, some through excerpts or a truncated

catalogue response; it did not read this report’s article and did not build the repository. The checks of all these repository statements against the placement commits are in [appendix A](#).

## D.2 Literature checks

- L’Innocente–Mantova, arXiv:1710.07304v5: Theorem B states that  $\omega^{\sqrt{2}} + \omega + 1$  is prime in  $\mathbf{Oz}$ , answering a question of Gonshor; their Section 1.3 reviews the earlier results on  $1 + \sum_k \omega^{1/k}$ ; their introduction records, citing Shepherdson, that the nonnegative part of  $\mathbf{Oz}$  is a model of open induction. The citations of all three sources agree with this.
- Shiu (*Math. Gazette* 104 (2020)) and the Archive of Formal Proofs entry of Danilkin and Chevalier (2023) for the three-square theorem, and Ehrlich–Kaplan Lemma 11.1 (as also cited in the trigonometry report), were confirmed during placement.
- Jeřábek’s MLQ article on rigid models of Presburger arithmetic (source 01) is cited for background; it was not re-read for this report.
- Source 02’s MathOverflow attribution could not be checked ([example 14.3](#)).
- Matiyasevich’s page range is corrected as described in [appendix A](#).
- Source 06’s attribution to *nombre* was confirmed during placement through the Stack Exchange interface: a comment of 6 July 2018 on MathOverflow question 304290 defines the ordinary integers in  $\mathbf{Oz}$  by excluding nonzero solutions of  $x^2 = 2y^2$  between  $-|n|$  and  $|n|$ , and observes that  $\mathbf{Oz}$  is not saturated [18].
- Jeřábek’s answer of 15 August 2011 to MathOverflow question 72691, cited by sources 06 and 07, was confirmed: it notes that the theory of  $\mathbf{Oz}$  is a complete extension of Robinson’s arithmetic and hence not recursively axiomatizable [19].
- Lê–Spencer, arXiv:1309.2259v1, Section 2, Example 4 lists  $\Lambda$  and attributes it to Borevich–Shafarevich, page 3; Berend–Bilu is *Proc. Amer. Math. Soc.* 124 (1996), 1663–1671. Miller–Shlapentokh, arXiv:1706.03302, cited by source 07 as related Pell-method precedent for function rings, has the stated title [25]. These were confirmed during placement; Borevich–Shafarevich itself was not re-read.
- The following were confirmed during placement for sources 08 and 09. In the arXiv version 1710.07304v5 of L’Innocente–Mantova, Proposition 2.4.5 states a cofinality criterion for the fraction field of  $\mathbb{Z} + K((G^{<0}))$  to be  $K((G))$ , and records  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  as Conway’s Theorem 32; their Lemma 9.2.1 is a primality transfer along an algebra retraction, and “reduced” is a support notion in their paper. Hamkins’s post of 4 November 2025 [32] says in its public part that lowest terms fail and that  $\sqrt{2}$  and  $\pi$  are omnific fractions; it does not discuss gcds there. Kława’s arXiv:2211.13412v2 [33] exists with the stated title and is cited by source 09 for background only. Stacks Tag 00HK is the equational criterion of flatness [34].
- For sources C10–C14 the following were checked during placement. Stacks Tag 0BX5 is Lemma 29.43.1, the valuative criterion of properness for arbitrary valuation rings, inside the section Tag 0BX4 cited by sources C10 and C11; Tag 0BS6 is Lemma 53.5.2 (Riemann–Roch) inside Section 53.5, Tag 0B5B. Tag 02AT, cited by source C13 for differentials, is Section 111.50 of the chapter of exercises, so Tag 00RM, cited by sources C10 and C14, is used instead ([appendix A](#)). The other Stacks tags and the statement numbers in Milne’s *Abelian Varieties* and *Algebraic Groups* are cited as the sources give them and were not rechecked for this report.

- Van den Dries’s paper [50] is cited by sources C11 and C14 from its bibliographic record and the abstract reproduced in the author’s bibliography on Celebratio Mathematica, where the abstract says that a criterion is given for curves over  $\mathbb{Z}$  “to have an infinite point in a discrete ordered ring”. The placement check confirmed that abstract; neither source nor this report read the paper itself, no result here depends on it, and no comparison or priority statement is made (section 18.4). Source C15 cites it in the same way, from its record and abstract, under the publisher’s DOI 10.1090/S0002-9947-1981-0597875-5.
- For source C15 the following were checked during placement: Stacks Tag 0FMU is Section 50.15, *Log poles along a divisor*; Tag 0BY6 is Section 53.8, the genus of a curve; Tag 0BYA is Section 53.9, plane curves, with the genus  $(d-1)(d-2)/2$ ; and Tag 0A22 is Section 33.43, curves, with the remark on compactification. Its citations of Milne’s *Abelian Varieties* (Chapter III, Proposition 2.3; Chapter IV, Proposition 6.4) were not rechecked.
- For source C16 the following were checked when section 17.6 was written. Seidenberg’s *Derivations and integral closure* is Pacific J. Math. 16 (1966), 167–173 [53]. Ciupercă’s *Integral closure of strongly Golod ideals* is Nagoya Math. J. 241, 204–216, first published online on 18 July 2019 [54]. Stacks Tag 035E is Section 29.55, *Normalization*, and Tag 0C1B is Section 53.12, *Riemann–Hurwitz*, whose Lemma 53.12.3 says that the differential of a uniformizer freely generates the module of differentials at a closed point with separable residue field. Its other Stacks tags (0BXX, 0BY3, 0BX5, 0BS6, 0C6L) are cited above. Seidenberg’s Section 3 and Ciupercă’s Section 2.5 were not re-read; the identity is reproved in lemma 17.27. Source C16 cites van den Dries from its record and abstract, as sources C11, C14 and C15 do.
- For source C17 the following were checked when section 19.4 was written. Stacks Tag 00U0 is Section 10.143, *Étale ring maps*; Tag 00UP is Section 10.150, *Formally étale maps*, containing Definition 10.150.1 and Lemma 10.150.3; Tag 0BJF is Lemma 49.3.1, in Section 49.3, *Discriminant of a finite locally free morphism*. The Mathematics Stack Exchange question 1626152 of the user Oblomov (25 January 2016) could not be retrieved from this environment and is cited as source C17 gives it [64]; nothing here depends on it. Its citation of L’Innocente–Mantova, Fact 2.1.1, for the algebraic closedness of Hahn fields was not rechecked.

### D.3 A dated factorization announcement

On September 18, 2026 Dan Abramov announced a Lean proof of Conway’s refinement conjecture [29, 30], noting explicitly the absence of independent mathematical verification. None of the sources audited that code or used the announced result. Accordingly this report does not repeat older descriptions of the refinement conjecture as a verified *currently open* problem, does not claim to settle it, and none of its proofs depends on its status.

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